Krishna's

TEXT BOOK on

LINEAR ALGEBRA & LINEAR PROGRAMMING

(For B.A. and B.Sc. IIIrd year students of Garhwal University)

THIRD YEAR–FIRST PAPER

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Preface

This book on **Linear Algebra & Linear Programming** has been specially written according to the latest **Syllabus** to meet the requirements of the **B.A. and B.Sc. Part-III Students** of all colleges affiliated to Garhwal University & Kumaun University.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. The proofs of various theorems and examples have been given with minute details. Each chapter of this book contains complete theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. At the end of each chapter an exercise containing objective questions has been given.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors, in general, hope that the present book will be warmly received by the students and teachers. We shall **indeed** be very thankful to our colleagues for their recommending this book to their students.

The authors wish to express their thanks to **Mr. S.K. Rastogi**, *M.D.,* **Mr. Sugam Rastogi**, *Executive Director,* **Mrs. Kanupriya Rastogi**, *Director* and **entire team** of **KRISHNA Prakashan Media (P) Ltd.**, **Meerut** for bringing out this book in the present nice form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. Suggestions for the improvement of the book are always welcome.

September, 2013 ***—Authors***

Syllabus

Linear Algebra & Linear Programming

B.A./B.Sc. III Year

Garhwal & Kumaun Universities *w.e.f.* (2006-07)

THIRD YEAR – FIRST PAPER

B.A./B.Sc. Paper-I M.M.-50

Note: Five questions are to be attempted selecting at least one from section (B).

(A) Linear Algebra (Eight questions)

Vector spaces-vector space, Sub spaces, Linear combinations, Linear spans, Sums and direct sums, linear dependence, Bases and dimensions, Dimensions and subspaces, coordinates and change of bases. Linear transformations-Linear transformations, rank and nullity, Operations with linear transformations, Linear Operators, algebra of linear operators, Invertible linear operators, Matrix of linear transformation, Matrices and linear transformation, Matrix of a linear operator, Change of basis, similarity.

Linear functionals-linear functional, Dual space and dual basis, Double dual space, Annihilators, Transpose of linear transformation. Bilinear, Quadratic and Hermitian forms -bilinear forms, Quadratic forms.

(B) Linear Programming (Two questions)

Linear programming Graphical method, Simplex method, The dual of linear programming problem.

Brief Contents

Section- A

Section-B

SECTION

A

LINEAR ALGEBRA

Chapters

Chapter 1

Vector Spaces

So far we have studied groups and rings. Now we shall study another important algebraic structure known as **vector space**. Before giving the definition of a vector space we shall make a distinction between **internal** and **external compositions**.

Let A be any set. If $a * b \in A \; \forall \; a, b \in A$, and $a * b$ is unique then $*$ is said to be an **internal composition** in the set A. Here a and b are both elements of the set A.

Let V and F be any two sets. If $a \circ \alpha \in V$ for all $a \in F$ and for all $\alpha \in V$ and $a \circ \alpha$ is unique, then $\circ$ is said to be an **external composition** in V over F. Here a is an element of the set F and α is an element of the set V and the resulting element $a \circ \alpha$ is an element of the set V.

1.1 Vector Space

(Kumaon 2002; Garhwal 10)

Definition : *Let* $(F, +, .)$ *be a field. The elements of F will be called* **scalars**. *Let V be a non-empty set whose elements will be called* **vectors**. *Then V is a vector space over the field F, if* *denoted by '+'. Also for this composition V is an abelian group.*

2. *There is an external composition in V over F called* **scalar multiplication** *and denoted multiplicatively i.e., $a\,\alpha \in V$ for all $a \in F$ and for all $\alpha \in V$. In other words V is closed with respect to scalar multiplication.*
3. *The two compositions i.e., scalar multiplication and addition of vectors satisfy the following postulates :*
   (i) $a\,(\alpha + \beta) = a\,\alpha + a\,\beta\ \forall\ a \in F$ *and* $\forall\ \alpha, \beta \in V$.
   (ii) $(a + b)\,\alpha = a\alpha + b\alpha\ \forall\ a, b \in F$ *and* $\forall\ \alpha \in V$.
   (iii) $(ab)\,\alpha = a\,(b\alpha)\ \forall\ a, b \in F$ *and* $\forall\ \alpha \in V$.
   (iv) $1\,\alpha = \alpha\ \forall\ \alpha \in V$ *and* 1 *is the unity element of the field* F.

When V is a vector space over the field F, we shall say that $V\ (F)$ is a vector space. If the field F is understood we can simply say that V is a vector space. If F is the field $\mathbf{R}$ of real numbers, V is called a *real vector space* ; similarly if F is $\mathbf{Q}$ or F is $\mathbf{C}$, we speak of *rational vector spaces or complex vector spaces.*

In the above definition of a vector space V over the field F, we have denoted the addition of vectors by the symbol '+'. This symbol also denotes the addition composition of the field F, *i.e.*, addition of scalars. There should be no confusion about the two compositions though we have used the same symbol to denote each of them. If $\alpha, \beta \in V$, then $\alpha + \beta$ represents addition in V *i.e.*, addition of vectors. If $a, b \in F$ then $a + b$ represents addition of scalars *i.e.*, addition in the field F. Similarly there should be no confusion in multiplication of scalars *i.e.*, multiplication of the elements of F and in scalar multiplication *i.e.*, multiplication of an element of V by an element of F. If $a, b \in F$, then ab represents multiplication of F and $ab \in F$. If $a \in F$, and $\alpha \in V$, then $a\,\alpha$ represents scalar multiplication and $a\,\alpha \in V$. Since $1 \in F$ and $\alpha \in V$, therefore 1α represents scalar multiplication. Again $a\,\alpha \in V$, $a\,\beta \in V$, therefore $a\alpha + a\beta$ represents addition of vectors and thus $a\alpha + a\beta$ is an element of V. Further $a \in F$ and $\alpha + \beta \in V$, therefore $a\,(\alpha + \beta)$ represents scalar multiplication and we have $a\,(\alpha + \beta) \in V$.

Note 1: For V to be an abelian group with respect to addition of vectors, we must have the following conditions satisfied :

(i) $\alpha + \beta \in V$ for all $\alpha, \beta \in V$.

(ii) $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in V$.

(iii) $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$ for all $\alpha, \beta, \gamma \in V$.

(iv) There exists an element $\mathbf{0} \in V$ such that $\mathbf{0} + \alpha = \alpha\ \forall\ \alpha \in V$.
The element $\mathbf{0} \in V$ will be called the **zero vector**. It is the additive identity in V.

(v) To every vector $\alpha \in V$ there exists a vector $-\alpha \in V$ such that $-\alpha + \alpha = \mathbf{0}$. Thus each vector should possess additive inverse. The vector $-\alpha$ is called the negative of the vector α.

Note 2: Since $(V, +)$ is an abelian group therefore all the properties of an abelian group will hold in V. A few of them are as follows :

(ii) $\beta + \alpha = \gamma + \alpha \Rightarrow \beta = \gamma$ (right cancellation law)

(iii) $\alpha + \beta = \mathbf{0} \Rightarrow \alpha = -\beta$ and $\beta = -\alpha$.

(iv) $-(\alpha + \beta) = -\alpha - \beta$ where by $\alpha - \beta$ we mean $\alpha + (-\beta)$.

(v) The additive identity $\mathbf{0}$ will be unique.

(vi) The additive inverse of each vector will be unique.

(vii) If $\alpha + \beta = \gamma$, then $\alpha + \beta - \gamma = \mathbf{0}$.

Note 3: There should also be no confusion about the use of the word vector. Here by vector we do not mean the vector quantity which we have defined in vector algebra as a directed line segment. Here we shall call the elements of the set V as vectors.

Note 4: In a vector space we shall be dealing with two types of zero elements. One is the zero vector and the other is the zero element of the field F *i.e.*, the 0 scalar. To distinguish between the two, we shall use the zero letter in bold type to represent the zero vector. However the students may use the same symbol 0 to denote the zero vector as well as the zero scalar. There will be no confusion in this use. The use of 0 will itself tell whether it stands for zero vector or for zero scalar.

Note 5: We shall usually use the lower case Greek letters α, β, γ etc. to denote vectors *i.e.*, the elements of V and the lower case Latin letters, a, b, c etc. to denote the scalars *i.e.*, the elements of the field F.

Illustrative Examples

Example 1: *A field K can be regarded as a vector space over any subfield F of K.*
(Garhwal 2006)

Solution: Here K is the set of vectors. Addition of vectors is the addition composition in the field K. Since K is a field, therefore $(K, +)$ is an abelian group. Further the elements of the subfield F constitute the set of scalars. The composition of scalar multiplication is the multiplication composition in the field K. K is a field, therefore $a\,\alpha \in K \;\forall\, a \in F$ and $\forall\, \alpha \in K$ because both a and α are elements of K. If 1 is the unity element of K, then 1 is also the unity element of the subfield F. We make the following observations.

(i) $a\,(\alpha + \beta) = a\,\alpha + a\beta \;\forall\, a \in F$ and $\forall\, \alpha, \beta \in K$. This result follows from the left distributive law in K.

(ii) $(a + b)\,\alpha = a\alpha + b\alpha \;\forall\, a, b \in F$ and $\forall\, \alpha \in K$. This result is a consequence of the right distributive law in K.

(iii) $(ab)\,\alpha = a\,(b\alpha) \;\forall\, a, b \in F$ and $\forall\, \alpha \in K$. This result is a consequence of associativity of multiplication in K.

(iv) $1\alpha = \alpha \;\forall\, \alpha \in K$ and 1 is the unity element of the subfield F. Since 1 is also the unity element of the field K, therefore $1\,\alpha = \alpha \;\forall\, \alpha \in K$.

Hence $K\,(F)$ is a vector space.

Note 2: If **C** is the field of complex numbers and **R** is the field of real numbers, then **C** is a vector space over **R** because **R** is a subfield of **C.** But **R** is not a vector space over **C**. Here **R** is not closed with respect to scalar multiplication. For example, $2 \in$ **R** and $3 + 4\,i \in$ **C** and $(3 + 4\,i)\,2 \notin$ **R.**

Example 2: *The set V of all m × n matrices with their elements as real numbers is a vector space over the field F of real numbers with respect to addition of matrices as addition of vectors and multiplication of a matrix by a scalar as scalar multiplication.*

Solution: As in groups, we can easily prove that V is an abelian group with respect to addition of matrices. The null matrix O of the type $m \times n$ is the additive identity of this abelian group.

If $a \in F$ and $\alpha \in V$ (*i.e.*, α is a matrix of the type $m \times n$ with elements as real numbers), then $a\alpha \in V$ because $a\alpha$ is also a matrix of the type $m \times n$ with elements as real numbers. Therefore V is closed with respect to scalar multiplication. Also from our study of matrices we observe that

(i) $a\,(\alpha + \beta) = a\alpha + a\beta \ \forall \ a \in F$ and $\forall \ \alpha, \beta \in V$.

(ii) $(a + b)\,\alpha = a\alpha + b\alpha \ \forall \ a, b \in F$ and $\forall \ \alpha \in V$.

(iii) $(ab)\,\alpha = a\,(b\alpha) \ \forall \ a, b \in F$ and $\forall \ \alpha \in V$.

(iv) $1\alpha = \alpha \ \forall \ \alpha \in V$ where 1 is the unity element of the field F of real numbers.

Hence $V\,(F)$ is a vector space.

Note: If V is the set of all $m \times n$ matrices with their elements as rational numbers and F is the field of real numbers, then V will not be closed with respect to scalar multiplication. For $\sqrt{7} \in F$ and if $\alpha \in V$, then $\sqrt{7}\alpha \notin V$ because the elements of the matrix $\sqrt{7}\alpha$ will not be rational numbers. Therefore $V\,(F)$ will not be a vector space.

Example 3: *The vector space of all ordered n-tuples over a field F.*

Solution: Let F be a field. An ordered set $\alpha = (a_1, a_2, a_3, \ldots, a_n)$ of n elements of F is called an *n-tuple* over F. Let V be the totality of all ordered *n-tuples over F i.e.,* let

$$V = \{(a_1, a_2, \ldots, a_n) : a_1, a_2, a_3, \ldots, a_n \in F\}.$$

Now we shall give a vector space structure to V over the field F. For this we define equality of two n-tuples, addition of two n-tuples and multiplication of an n-tuple by a scalar as follows :

Equality of two n-tuples. Two elements

$$\alpha = (a_1, a_2, \ldots, a_n)$$

and

$$\beta = (b_1, b_2, \ldots, b_n)$$

of V are said to be equal if and only if $a_i = b_i$ for each $i = 1, 2, \ldots, n$.

Addition composition in V. We define

$$\alpha + \beta = (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n)$$

$$\forall \ \alpha = (a_1, a_2, \ldots, a_n), \beta = (b_1, b_2, \ldots, b_n) \in V.$$

thus V is closed with respect to addition of n-tuples.

Scalar multiplication in V over F. We define

$$a\,\alpha = (aa_1, aa_2, \ldots, aa_n)\ \forall\ a \in F, \forall\ \alpha = (a_1, a_2, \ldots, a_n) \in V.$$

Since $aa_1, aa_2, \ldots, aa_n$ are all elements of F, therefore $a\,\alpha \in V$ and thus V is closed with respect to scalar multiplication.

Now we shall see that V is a vector space for these two compositions.

Associativity of addition in V. We have

$$\begin{aligned}
&(a_1, a_2, \ldots, a_n) + [(b_1, b_2, \ldots, b_n) + (c_1, c_2, \ldots, c_n)] \\
&= (a_1, a_2, \ldots, a_n) + (b_1 + c_1, b_2 + c_2, \ldots, b_n + c_n) \\
&= (a_1 + [b_1 + c_1], a_2 + [b_2 + c_2], \ldots, a_n + [b_n + c_n]) \\
&= ([a_1 + b_1] + c_1, [a_2 + b_2] + c_2, \ldots, [a_n + b_n] + c_n) \\
&= (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n) + (c_1, c_2, \ldots, c_n) \\
&= [(a_1, a_2, \ldots, a_n) + (b_1, b_2, \ldots, b_n)] + (c_1, c_2, \ldots, c_n).
\end{aligned}$$

Commutativity of addition in V. We have

$$\begin{aligned}
&(a_1, a_2, \ldots, a_n) + (b_1, b_2, \ldots, b_n) = (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n) \\
&= (b_1 + a_1, b_2 + a_2, \ldots, b_n + a_n) = (b_1, b_2, \ldots, b_n) + (a_1, a_2, \ldots, a_n)
\end{aligned}$$

.

Existence of additive identity in V. We have

$(0, 0, \ldots, 0) \in V$. Also if $(a_1, a_2, \ldots, a_n) \in V$, then

$$\begin{aligned}
(a_1, a_2, \ldots, a_n) + (0, 0, \ldots, 0) &= (a_1 + 0, a_2 + 0, \ldots, a_n + 0) \\
&= (a_1, a_2, \ldots, a_n).
\end{aligned}$$

$\therefore$ $(0, 0, \ldots, 0)$ is the additive identity in V.

Existence of additive inverse of each element of V. If

$$(a_1, a_2, \ldots, a_n) \in V, \text{ then } (-a_1, -a_2, \ldots, -a_n) \in V.$$

Also we have $(-a_1, -a_2, \ldots, -a_n) + (a_1, a_2, \ldots, a_n)$

$$= (-a_1 + a_1, -a_2 + a_2, \ldots, -a_n + a_n) = (0, 0, \ldots, 0).$$

$\therefore$ $(-a_1, -a_2, \ldots, -a_n)$ is the additive inverse of $(a_1, a_2, \ldots, a_n)$.

Thus V is an abelian group with respect to addition. Further we observe that

1. If $a \in F$ and $\alpha = (a_1, a_2, \ldots, a_n)$, $\beta = (b_1, b_2, \ldots, b_n) \in V$, then

$$\begin{aligned}
a\,(\alpha + \beta) &= a\,(a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n) \\
&= (a\,[a_1 + b_1], a\,[a_2 + b_2], \ldots, a\,[a_n + b_n]) \\
&= (aa_1 + ab_1, aa_2 + ab_2, \ldots, aa_n + ab_n) \\
&= (aa_1, aa_2, \ldots, aa_n) + (ab_1, ab_2, \ldots, ab_n) \\
&= a\,(a_1, a_2, \ldots, a_n) + a\,(b_1, b_2, \ldots, b_n) = a\alpha + a\beta.
\end{aligned}$$

2. If $a, b \in F$ and $\alpha = (a_1, a_2, \ldots, a_n) \in V$, then

$$(a + b)\,\alpha = ([a + b]\,a_1, [a + b]\,a_2, \ldots, [a + b]\,a_n)$$

$= (aa_1, aa_2, \ldots, aa_n) + (ba_1, ba_2, \ldots, ba_n)$
$= a\,(a_1, a_2, \ldots, a_n) + b\,(a_1, a_2, \ldots, a_n) = a\,\alpha + b\,\alpha.$

3. If $a, b \in F$ and $\alpha = (a_1, a_2, \ldots, a_n) \in V$, then

$$(ab)\,\alpha = ([ab]\,a_1, [ab]\,a_2, \ldots, [ab]\,a_n)$$
$$= (a\,[ba_1], a\,[ba_2], \ldots, a\,[ba_n])$$
$$= a\,(ba_1, ba_2, \ldots, ba_n) = a\,[b\,(a_1, a_2, \ldots, a_n)] = a\,(b\alpha).$$

4. If 1 is the unity element of F and $\alpha = (a_1, a_2, \ldots, a_n) \in V$, then

$$1\,\alpha = (1a_1, 1a_2, \ldots, 1\,a_n) = (a_1, a_2, \ldots, a_n) = \alpha.$$

Hence V is a vector space over F. *The vector space of all ordered n-tuples over F will be denoted by* $V_n\,(F)$. *Sometimes we also denote it by* F^n *or* $F^{(n)}$ *or* $F^n\,(F)$. Here the zero vector *i.e.*, **0** is the n-tuple $(0, 0, \ldots, 0)$.

Note: $V_2\,(F) = \{(a_1, a_2) : a_1, a_2 \in F\}$ is the vector space of all **ordered pairs** over F. Similarly $V_3\,(F) = \{(a_1, a_2, a_3) : a_1, a_2, a_3 \in F\}$ is the vector space of all **ordered triads** over F.

Example 4: *The vector space of all polynomials over a field F.*

Solution: Let $F\,[x]$ denote the set of all polynomials in an indeterminate x over a field F. Then $F\,[x]$ is a vector space over the field F with respect to addition of two polynomials as addition of vectors and the product of a polynomial by a constant polynomial (*i.e.*, by an element of F) as scalar multiplication.

Let $\quad f\,(x) = \Sigma\, a_i\, x^i = a_0 + a_1\, x + a_2\, x^2 + a_3\, x^3 + \ldots,$

$\quad g\,(x) = \Sigma\, b_i\, x^i = b_0 + b_1\, x + b_2\, x^2 + b_3\, x^3 + \ldots$

and $\quad h\,(x) = \Sigma\, c_i\, x^i = c_0 + c_1\, x + c_2\, x^2 + c_3\, x^3 + \ldots$

be any arbitrary members of $F\,[x]$.

Equality of two polynomials. We define $f\,(x) = g\,(x)$ if and only if $a_i = b_i$ for each $i = 0, 1, 2, \ldots$.

Addition composition in $F\,[x]$. We define

$$f\,(x) + g\,(x) = (a_0 + b_0) + (a_1 + b_1)\,x + (a_2 + b_2)\,x^2 + \ldots$$
$$= \Sigma\,(a_i + b_i)\,x^i.$$

Since $a_0 + b_0, a_1 + b_1, a_2 + b_2, \ldots$ are all elements of F, therefore

$$f\,(x) + g\,(x) \in F\,[x]$$

and thus $F\,[x]$ is closed with respect to addition of polynomials.

Scalar multiplication in $F\,[x]$ over F. If k is any scalar *i.e.*, $k \in F$, we define

$$k f\,(x) = ka_0 + (ka_1)\,x + (ka_2)\,x^2 + (ka_3)\,x^3 + \ldots$$
$$= \Sigma\,(ka_i)\,x^i.$$

is closed with respect to scalar multiplication.

Now we shall show that $F[x]$ is a vector space for these two compositions.

Commutativity of addition in $F[x]$. We have

$$f(x) + g(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \ldots$$

$$= (b_0 + a_0) + (b_1 + a_1)x + (b_2 + a_2)x^2 + \ldots$$

[$\because$ addition in the field F is commutative]

$$= g(x) + f(x).$$

Associativity of addition in $F[x]$. We have

$$[f(x) + g(x)] + h(x) = \Sigma (a_i + b_i) x^i + \Sigma c_i x^i$$

$$= \Sigma [(a_i + b_i) + c_i] x^i = \Sigma [a_i + (b_i + c_i)] x^i$$

$$= \Sigma a_i x^i + \Sigma (b_i + c_i) x^i$$

$$= f(x) + [g(x) + h(x)].$$

Existence of additive identity in $F[x]$. Let 0 denote the zero polynomial over the field F *i.e.*, $0 = 0 + 0x + 0x^2 + 0x^3 + \ldots$

Then $0 \in F[x]$ and $0 + f(x) = f(x)$.

$\therefore$ the zero polynomial 0 is the additive identity.

Existence of additive inverse of each member of $F[x]$. Let $-f(x)$ be the polynomial over the field F defined as

$$-f(x) = -a_0 + (-a_1)x + (-a_2)x^2 + \ldots$$

$$= -a_0 - a_1 x - a_2 x^2 - \ldots.$$

Then $-f(x) \in F[x]$ and we have

$$-f(x) + f(x) = 0 \text{ i.e., the zero polynomial.}$$

$\therefore$ $-f(x)$ is the additive inverse of $f(x)$.

Thus $F[x]$ is an abelian group with respect to addition of polynomials.

Now for the operation of scalar multiplication we make the following observations.

1. If $k \in F$, then

$$k[f(x) + g(x)] = k[(a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \ldots]$$

$$= k(a_0 + b_0) + k(a_1 + b_1)x + k(a_2 + b_2)x^2 + \ldots$$

$$= (ka_0 + kb_0) + (ka_1 + kb_1)x + (ka_2 + kb_2)x^2 + \ldots$$

$$= [ka_0 + (ka_1)x + (ka_2)x^2 + \ldots] + [kb_0 + (kb_1)x + (kb_2)x^2 + \ldots]$$

$$= k(a_0 + a_1 x + a_2 x^2 + \ldots) + k(b_0 + b_1 x + b_2 x^2 + \ldots)$$

$$= kf(x) + kg(x).$$

$$(k_1 + k_2) f(x) = (k_1 + k_2) a_0 + [(k_1 + k_2) a_1] x + [(k_1 + k_2) a_2] x^2 + \ldots$$
$$= (k_1 a_0 + k_2 a_0) + (k_1 a_1 + k_2 a_1) x + (k_1 a_2 + k_2 a_2) x^2 + \ldots$$
$$= [k_1 a_0 + (k_1 a_1) x + (k_1 a_2) x^2 + \ldots] + [k_2 a_0 + (k_2 a_1) x + (k_2 a_2) x^2 + \ldots]$$
$$= k_1 (a_0 + a_1 x + a_2 x^2 + \ldots) + k_2 (a_0 + a_1 x + a_2 x^2 + \ldots)$$
$$= k_1 f(x) + k_2 f(x).$$

3. If $k_1, k_2 \in F$, then

$$(k_1 k_2) f(x) = (k_1 k_2) a_0 + [(k_1 k_2) a_1] x + [(k_1 k_2) a_2] x^2 + \ldots$$
$$= k_1 (k_2 a_0) + [k_1 (k_2 a_1)] x + [k_1 (k_2 a_2)] x^2 + \ldots$$
$$= k_1 [k_2 a_0 + (k_2 a_1) x + (k_2 a_2) x^2 + \ldots] = k_1 [k_2 f(x)].$$

4. If 1 is the unity element of the field F, then

$$1 f(x) = (1a_0) + (1a_1) x + (1a_2) x^2 + \ldots$$
$$= a_0 + a_1 x + a_2 x^2 + \ldots = f(x).$$

Hence $F[x]$ is a vector space over the field F.

Example 5: *The set of all convergent real sequences is a vector space over the field of real numbers.*

Solution: Let V denote the set of all convergent sequences over the field of real numbers.

Let $\alpha = <\alpha_1, \alpha_2, \ldots, \alpha_n, \ldots> = <\alpha_n>$, $\beta = <\beta_1, \beta_2, \ldots, \beta_n, \ldots> = <\beta_n>$

and $\gamma = <\gamma_1, \gamma_2, \ldots, \gamma_n, \ldots> = <\gamma_n>$

be any three convergent sequences.

1. (V, +) is an abelian group.

(i) We have $\alpha + \beta = <\alpha_n> + <\beta_n> = <\alpha_n + \beta_n>$ which is also a convergent sequence. Therefore V is closed for addition of sequences.

(ii) Commutativity of addition. We have

$$\alpha + \beta = <\alpha_n> + <\beta_n> = <\alpha_n + \beta_n> = <\beta_n + \alpha_n>$$
$$= <\beta_n> + <\alpha_n> = \beta + \alpha$$

(iii) Associativity of addition. We have

$$\alpha + (\beta + \gamma) = <\alpha_n> + [<\beta_n> + <\gamma_n>] = <\alpha_n> + <\beta_n + \gamma_n>$$
$$= <\alpha_n + (\beta_n + \gamma_n)> = <(\alpha_n + \beta_n) + \gamma_n> = <\alpha_n + \beta_n> + <\gamma_n>$$
$$= [<\alpha_n> + <\beta_n>] + <\gamma_n> = (\alpha + \beta) + \gamma.$$

(iv) Existence of additive identity. The zero sequence $<0> = <0, 0, 0, \ldots>$ is the additive identity.

sequence $< -\alpha_n >$ such that

$$< \alpha_n > + < -\alpha_n > = < \alpha_n - \alpha_n > = < 0 > = \text{ the additive identity.}$$

$\therefore$ $(V, +)$ is an abelian group.

2. ***V* is closed for scalar multiplication.**

Let a be any scalar *i.e.*, a be any real number. Then

$a\alpha = a < \alpha_n > = < a\alpha_n >$ which is also a convergent sequence because

$$\lim_{n \to \infty} a\alpha_n = a \lim_{n \to \infty} \alpha_n.$$

Thus V is closed for scalar multiplication.

3. **Laws of scalar multiplication.** Let $a, b \in \mathbf{R}$. We have

(i) $a(\alpha + \beta) = a[< \alpha_n > + < \beta_n >] = a < \alpha_n + \beta_n >$

$$= < a(\alpha_n + \beta_n) > = < a\alpha_n + a\beta_n > = < a\alpha_n > + < a\beta_n >$$

$$= a < \alpha_n > + a < \beta_n > = a\alpha + a\beta.$$

(ii) $(a + b)\alpha = (a + b) < \alpha_n > = < (a + b)\alpha_n >$

$$= < a\alpha_n + b\alpha_n > = < a\alpha_n > + < b\alpha_n >$$

$$= a < \alpha_n > + b < \alpha_n > = a\alpha + b\alpha.$$

(iii) $(ab)\alpha = (ab) < \alpha_n > = < (ab)\alpha_n > = < a(b\alpha_n) >$

$$= a < b\alpha_n > = a[b < \alpha_n >] = a(b\alpha).$$

(iv) $1\alpha = 1 < \alpha_n > = < 1\alpha_n > = < \alpha_n > = \alpha.$

Thus all the postulates of a vector space are satisfied. Hence V is a vector space over the field of real numbers.

Example 6: *Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define*

$$(x, y) + (x_1, y_1) = (x + x_1, 0)$$

$$c(x, y) = (cx, 0).$$

Is V, with these operations, a vector space over the field of real numbers ?

Solution: If any of the postulates of a vector space is not satisfied, then V will not be a vector space. We shall show that for the operation of addition of vectors as defined in this problem the identity element does not exist. Suppose the ordered pair (x_1, y_1) is to be the identity element for the operation of addition of vectors. Then we must have

$$(x, y) + (x_1, y_1) = (x, y) \;\forall\; x, y \in \mathbf{R}$$

$$\Rightarrow \quad (x + x_1, 0) = (x, y) \;\forall\; x, y \in \mathbf{R}.$$

But if $y \neq 0$, then we cannot have $(x + x_1, 0) = (x, y)$. Thus there exists no element (x_1, y_1) of V such that

$$(x, y) + (x_1, y_1) = (x, y) \;\forall\; (x, y) \in V.$$

Therefore the identity element does not exist and V is not a vector space over the field $\mathbf{R}$.

numbers. Define

$$(x, y) + (x_1, y_1) = (3y + 3y_1, -x - x_1)$$
$$c\,(x, y) = (3cy, -cx).$$

Verify that V, with these operations, is not a vector space over the field of real numbers.

Solution: Take $\alpha = (1, 2), \beta = (2, 3), \gamma = (1, 3)$

Then $(\alpha + \beta) + \gamma = \{(1, 2) + (2, 3)\} + (1, 3)$
$$= (15, -3) + (1, 3) = (0, -16)$$

and $\alpha + (\beta + \gamma) = (1, 2) + \{(2,3) + (1,3)\}$
$$= (1, 2) + (18, -3) = (-3, -19).$$

Thus we conclude that in general $(\alpha + \beta) + \gamma \neq \alpha + (\beta + \gamma)$. Hence V is not a vector space over the field of real numbers.

Example 8: *How many elements are there in the vector space of polynomials of degree at most n in which the coefficients are the elements of the field* **I** *(p) over the field* **I** *(p), p being a prime number ?*

Solution: The field **I** (p) is the field $(\{0, 1, 2, \ldots, p-1\}, +_p, \times_p)$.
The number of distinct elements in the field **I** (p) is p.
If $f(x)$ is a polynomial of degree at most n over the field **I** (p), then

$$f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n, \text{ where}$$

$a_0, a_1, a_2, \ldots, a_n \in \mathbf{I}(p)$.

Now in the polynomial $f(x)$, the coefficient of each of the $n + 1$ terms $a_0, a_1 x, a_2 x^2, \ldots, a_n x^n$, can be filled in p ways because any of the p elements of the field **I** (p) can be filled there.

Thus we can have $p \times p \times p \times \ldots$upto $(n + 1)$ times *i.e.*, p^{n+1} distinct polynomials of degree at most n over the field **I** (p). Hence if P_n is the vector space of polynomials of degree at most n in which the coefficients are the elements of the field **I** (p) over the field **I** (p), then P_n has p^{n+1} distinct elements.

Comprehensive Exercise 1

1. Show that a field F may be considered as a vector space over F if scalar multiplication is identified with field multiplication.
2. Show that the complex field C is a vector space over the real field **R**.

(Garhwal 2007)

3. Let V be the set of ordered pairs (z_1, z_2) of complex numbers. Show that V is a vector space over the real field **R** with addition in V and scalar multiplication on V defined by $(z_1, z_2) + (w_1, w_2) = (z_1 + w_1, z_2 + w_2)$ and $a\,(z_1, z_2) = (az_1, az_2)$ where $z_1, z_2, w_1, w_2 \in \mathbf{C}$ and $a \in \mathbf{R}$.

space. How many vectors are there in this vector space ?

5. Let S be any non-empty set and let F be any field. Let V be the set of all functions from S to F *i.e.*, let

$$V = \{f : f : S \to F\}$$

Let us define sum of two elements f and g in V as follows :

$$(f + g)(x) = f(x) + g(x) \forall x \in S.$$

Also let us define scalar multiplication of an element f in V by an element c in F as follows :

$$(cf)(x) = cf(x) \forall x \in S.$$

Then $V(F)$ is a vector space.

6. The vector space of all real valued continuous functions defined in some interval $[0, 1]$.

7. Prove that the set of all vectors in a plane over the field of real numbers is a vector space.

8. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Examine in each of the following cases whether V is a vector space over the field of real numbers or not :

(i) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$
$c(x, y) = (|c|x, |c|y)$

(ii) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$
$c(x, y) = (0, cy)$

(iii) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$
$c(x, y) = (c^2 x, c^2 y)$

(iv) $(x, y) + (x_1, y_1) = (x + y_1, y + x_1)$
$c(x, y) = (cx, cy)$

9. Show that the axiom $1\alpha = \alpha \forall \alpha \in V$ and $1 \in F$, defining the vector space is not a consequence of the remaining axioms.

10. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$$
$$c(x, y) = (cx, y).$$

Show that with these operations V is not a vector space over the field of real numbers.

11. Let V be the set of ordered pairs (x, y) of real numbers, Show that V is not a vector space over **R** with addition in V and scalar multiplication on V defined by :

(i) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$ and $c(x, y) = (x, y)$;

(ii) $(x, y) + (x_1, y_1) = (x, y)$ and $c(x, y) = (cx, cy)$;

(iii) $(x, y) + (x_1, y_1) = (0, 0)$ and $c(x, y) = (cx, cy)$;

(iv) $(x, y) + (x_1, y_1) = (xx_1, yy_1)$ and $c(x, y) = (cx, cy)$.

numbers. Define

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$$
$$c\,(x, y) = (cx, 0).$$

Is V a vector space over the field of real numbers with these operations ?

13. Let $\mathbf{R}$ be the field of real numbers and let P_n be the set of all polynomials (of degree at most n) over the field $\mathbf{R}$. Prove that P_n is a vector space over the field $\mathbf{R}$.

14. Let U and W be vector spaces over a field F. Let V be the set of ordered pairs *i.e.*,

$$V = \{(u, w) : u \in U, w \in W\}.$$

Show that V is a vector space over F with addition in V and scalar multiplication on V defined by

$$(u, w) + (u_1, w_1) = (u + u_1, w + w_1)$$

and $\quad k\,(u, w) = (ku, kw)$ where $u, u_1 \in U$ and $w, w_1 \in W$ and $k \in F$.

(Kumaon 2003)

Answers 1

4. p^n

8. (i) not a vector space (ii) not a vector space
(iii) not a vector space (iv) not a vector space

12. not a vector space

1.2 General Properties of Vector Spaces

Theorem 1. *Let $V(F)$ be a vector space and $\mathbf{0}$ be the zero vector of V. Then*

(i) $a\,\mathbf{0} = \mathbf{0}\ \forall\ a \in F.$

(ii) $0\,\alpha = \mathbf{0}\ \forall\ \alpha \in V.$

(iii) $a(-\alpha) = -(a\alpha)\ \forall\ a \in F, \forall\ \alpha \in V.$ **(Garhwal 2007)**

(iv) $(-a)\,\alpha = -(a\alpha)\ \forall\ a \in F, \forall\ \alpha \in V.$ **(Garhwal 2007)**

(v) $a(\alpha - \beta) = a\alpha - a\beta\ \forall\ a \in F$ *and* $\forall\ \alpha, \beta \in V.$

(vi) $a\alpha = \mathbf{0} \Rightarrow a = 0$ or $\alpha = \mathbf{0}.$

Proof. (i) We have $a\mathbf{0} = a(\mathbf{0} + \mathbf{0})$ $\quad [\because\ \mathbf{0} = \mathbf{0} + \mathbf{0}]$

$$= a\mathbf{0} + a\mathbf{0}.$$

$\therefore \quad \mathbf{0} + a\mathbf{0} = a\mathbf{0} + a\mathbf{0}$ $\quad [\because\ a\mathbf{0} \in V$ and $\mathbf{0} + a\mathbf{0} = a\mathbf{0}]$

Now V is an abelian group with respect to addition.

Therefore by right cancellation law in V, we get $\mathbf{0} = a\mathbf{0}$.

(ii) We have $0\,\alpha = (0 + 0)\,\alpha$ $\quad [\because\ 0 = 0 + 0]$

$\therefore \qquad \mathbf{0} + 0\alpha = 0\alpha + 0\alpha. \qquad [\because \; 0\,\alpha \in V \text{ and } \mathbf{0} + 0\alpha = 0\alpha]$

Now V is an abelian group with respect to addition of vectors. Therefore by right cancellation law in V, we get $\mathbf{0} = 0\,\alpha$.

(iii) We have $a\,[\alpha + (-\alpha)] = a\,\alpha + a\,(-\alpha) \Rightarrow a\mathbf{0} = a\alpha + a(-\alpha)$

$\Rightarrow \qquad \mathbf{0} = a\alpha + a\,(-\alpha) \qquad [\because \; a\mathbf{0} = \mathbf{0}]$

$\Rightarrow \qquad a\,(-\alpha)$ is the additive inverse of $a\,\alpha \;\Rightarrow a\,(-\alpha) = -\,(a\,\alpha)$.

(iv) We have $[a + (-a)]\,\alpha = a\alpha + (-a)\,\alpha \Rightarrow 0\,\alpha = a\alpha + (-a)\,\alpha$

$\Rightarrow \qquad \mathbf{0} = a\,\alpha + (-a)\,\alpha \Rightarrow (-a)\,\alpha$ is the additive inverse of $a\alpha$

$\Rightarrow \qquad (-a)\,\alpha = -\,(a\,\alpha)$.

(v) We have $a\,(\alpha - \beta) = a\,[\alpha + (-\beta)] = a\alpha + a\,(-\beta)$

$= a\alpha + [-\,(a\beta)] \qquad [\because \; a\,(-\beta) = -\,(a\beta)]$

$= a\alpha - a\beta$.

(vi) Let $a\alpha = \mathbf{0}$ and $a \neq 0$. Then a^{-1} exists because a is a non-zero element of the field F.

$\therefore \qquad a\,\alpha = \mathbf{0} \Rightarrow a^{-1}\,(a\alpha) = a^{-1}\,\mathbf{0} \Rightarrow (a^{-1}\,a)\,\alpha = \mathbf{0} \Rightarrow 1\,\alpha = \mathbf{0} \Rightarrow \alpha = \mathbf{0}$.

Again let $a\alpha = \mathbf{0}$ and $\alpha \neq \mathbf{0}$. Then to prove that $a = 0$. Suppose $a \neq 0$. Then a^{-1} exists.

$\therefore \qquad a\alpha = \mathbf{0} \Rightarrow a^{-1}\,(a\alpha) = a^{-1}\,\mathbf{0} \Rightarrow (a^{-1}\,a)\,\alpha = \mathbf{0} \Rightarrow 1\alpha = \mathbf{0} \Rightarrow \alpha = \mathbf{0}$.

Thus we get a contradiction that α must be a zero vector. Therefore a must be equal to 0. Hence $\alpha \neq \mathbf{0}$ and $a\,\alpha = \mathbf{0} \Rightarrow a = 0$.

Theorem 2. *Let $V\,(F)$ be a vector space. Then*

(*i*) *If a, $b \in F$ and α is a non-zero element of V, we have $a\alpha = b\,\alpha \Rightarrow a = b$.*

(*ii*) *If α, $\beta \in V$ and a is a non-zero element of F, we have $a\alpha = a\,\beta \Rightarrow \alpha = \beta$.*

Proof: (i) We have $a\alpha = b\alpha \Rightarrow a\alpha - b\alpha = \mathbf{0} \Rightarrow (a - b)\,\alpha = \mathbf{0}$.

But $\alpha \neq \mathbf{0}$. Therefore $(a - b)\,\alpha = \mathbf{0} \Rightarrow a - b = 0 \Rightarrow a = b$.

(ii) We have $a\alpha = a\beta \Rightarrow a\alpha - a\beta = \mathbf{0} \Rightarrow a\,(\alpha - \beta) = \mathbf{0}$.

But $a \neq 0$. Therefore $a\,(\alpha - \beta) = \mathbf{0} \Rightarrow \alpha - \beta = \mathbf{0} \Rightarrow \alpha = \beta$.

1.3 Vector Subspaces

Definition: *Let V be a vector space over the field F and $W \subseteq V$. Then W is called a subspace of V if W itself is a vector space over F with respect to the operations of vector addition and scalar multiplication in V.* **(Garhwal 2010)**

Remark: Let $V\,(F)$ be any vector space. Then V itself and the subset of V consisting of zero vector only are always subspaces of V. These two are called **improper subspaces**. If V has any other subspace, then it is called a **proper subspace**. The subspace of V consisting of zero vector only is called the **zero subspace**.

space $V(F)$ to be a subspace of V is that W is closed under vector addition and scalar multiplication in V. **(Kumaon 2003)**

Proof: If W itself is a vector space over F with respect to vector addition and scalar multiplication in V, then W must be closed with respect to these two compositions. Hence the condition is necessary.

The condition is sufficient. Now suppose that W is a non-empty subset of V and W is closed under vector addition and scalar multiplication in V.

Let $\alpha \in W$. If 1 is the unity element of F, then $-1 \in F$. Now W is closed under scalar multiplication. Therefore

$$-1 \in F, \alpha \in W \Rightarrow (-1)\,\alpha \in W \Rightarrow -(1\alpha) \in W \Rightarrow -\alpha \in W.$$

$$[\because \quad \alpha \in W \Rightarrow \alpha \in V \text{ and } 1\,\alpha = \alpha \text{ in } V.]$$

Thus the additive inverse of each element of W is also in W.

Now W is closed under vector addition.

Therefore $\alpha \in W, -\alpha \in W \Rightarrow \alpha + (-\alpha) \in W \quad \Rightarrow \quad \mathbf{0} \in W$,

where $\mathbf{0}$ is the zero vector of V.

Hence the zero vector of V is also the zero vector of W. Since the elements of W are also the elements of V, therefore vector addition will be commutative as well as associative in W. Hence W is an abelian group with respect to vector addition. Also it is given that W is closed under scalar multiplication. The remaining postulates of a vector space will hold in W since they hold in V of which W is a subset.

Hence W itself is a vector space for the two compositions.

$\therefore$ W is a subspace of V.

Theorem 2: *The necessary and sufficient conditions for a non-empty subset W of a vector space $V(F)$ to be a subspace of V are*

(i) $\alpha \in W, \beta \in W \Rightarrow \alpha - \beta \in W$.

(ii) $a \in F, \alpha \in W \Rightarrow a\alpha \in W$.

Proof: The conditions are necessary. If W is a subspace of V, then W is an abelian group with respect to vector addition. Therefore $\alpha \in W$, $\beta \in W \Rightarrow \alpha - \beta \in W$. Also W must be closed under scalar multiplication. Therefore the condition (ii) is also necessary.

The conditions are sufficient: Now suppose W is a non-empty subset of V satisfying the two given conditions. From condition (i), we have $\alpha \in W$, $\alpha \in W \Rightarrow \alpha - \alpha \in W \Rightarrow \mathbf{0} \in W$.

Thus the zero vector of V belongs to W and it will also be the zero vector of W.

Now $\qquad \mathbf{0} \in W, \alpha \in W \Rightarrow \mathbf{0} - \alpha \in W \Rightarrow -\alpha \in W.$

Thus the additive inverse of each element of W is also in W.

Again $\qquad \alpha \in W, \beta \in W \Rightarrow \alpha \in W, -\beta \in W$

$\Rightarrow \qquad \alpha - (-\beta) \in W \Rightarrow \alpha + \beta \in W.$

Since the elements of W are also the elements of V, therefore vector addition will be commutative as well as associative in W. Hence W is an abelian group under vector addition. Also from condition (ii), W is closed under scalar multiplication. The remaining postulates of a vector space will hold in W since they hold in V which is a superset of W. Hence W is a subspace of V.

Theorem 3: *The necessary and sufficient condition for a non-empty subset W of a vector space V (F) to be a subspace of V is*

$$a, b \in F \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + b\,\beta \in W.$$ **(Kumaon 2004; Garhwal 10)**

Proof: The condition is necessary. If W is a subspace of V, then W must be closed under scalar multiplication and vector addition.

Therefore $\quad a \in F,\ \alpha \in W \Rightarrow a\alpha \in W$

and $\quad b \in F, \beta \in W \Rightarrow b\beta \in W.$

Now $\quad a\alpha \in W, b\beta \in W \Rightarrow a\alpha + b\beta \in W$. Hence the condition is necessary.

The condition is sufficient. Now suppose W is a non-empty subset of V satisfying the given condition *i.e.*,

$$a, b \in F \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W.$$

Taking $a = 1, b = 1$, we see that if $\alpha, \beta \in W$ then $1\,\alpha + 1\,\beta \in W$

$\Rightarrow \quad \alpha + \beta \in W. \qquad [\because \ \alpha \in W \Rightarrow \alpha \in V \text{ and } 1\,\alpha = \alpha \text{ in } V]$

Thus W is closed under vector addition.

Now taking $a = -1, b = 0$, we see that if $\alpha \in W$, then

$(-1)\,\alpha + 0\,\alpha \in W \qquad$ [in place of β we have taken α]

$\Rightarrow \quad -(1\,\alpha) + \mathbf{0} \in W \Rightarrow -\alpha \in W.$

Thus the additive inverse of each element of W is also in W.

Taking $a = 0, b = 0$, we see that if $\alpha \in W$, then

$$0\,\alpha + 0\,\alpha \in W \Rightarrow \mathbf{0} + \mathbf{0} \in W \Rightarrow \mathbf{0} \in W.$$

Thus the zero vector of V belongs to W. It will also be the zero vector of W.

Since the elements of W are also the elements of V, therefore vector addition will be associative as well as commutative in W. Thus W is an abelian group with respect to vector addition.

Now taking $\beta = \mathbf{0}$, we see that if $a, b \in F$

and $\quad \alpha \in W$, then

$$a\alpha + b\,\mathbf{0} \in W \ \ i.e.,\ a\alpha + \mathbf{0} \in W \ i.e.,\ a\,\alpha \in W.$$

Thus W is closed under scalar multiplication.

The remaining postulates of a vector space will hold in W since they hold in V of which W is a subset. Hence $W\,(F)$ is a subspace of $V\,(F)$.

Theorem 4: *A non-empty subset W of a vector space V (F) is a subspace of V if and only if for each pair of vectors* α, β *in W and each scalar a in F the vector* $a\alpha + \beta$ *is again in W.*

closed with respect to scalar multiplication and as well as with respect to vector addition. Therefore

$$a \in F, \alpha \in W \Rightarrow a\alpha \in W.$$

Further $\qquad a\alpha \in W,\ \beta \in W \Rightarrow a\alpha + \beta \in W.$

Hence the condition is necessary.

The condition is sufficient. It is given that W is a non-empty subset of V and $a \in F, \alpha, \beta \in W \Rightarrow a\alpha + \beta \in W$. We are to prove that W is a subspace of V.

(i) Since W is non-empty, therefore there is at least one vector in W, say γ. Now $1 \in F \Rightarrow -1 \in F$. Therefore taking $a = -1, \alpha = \gamma, \beta = \gamma$, we get from the given condition that

$$(-1)\gamma + \gamma = -(1\gamma) + \gamma = -\gamma + \gamma = \mathbf{0} \text{ is in } W.$$

(ii) Now let $a \in F, \alpha \in W$. Since $\mathbf{0}$ is in W, therefore taking $\beta = \mathbf{0}$ in the given condition, we get

$$a\alpha + \mathbf{0} = a\alpha \text{ is in } W.$$

Thus W is closed with respect to scalar multiplication.

(iii) Let $\alpha \in W$. Since $-1 \in F$ and W is closed with respect to scalar multiplication, therefore $(-1)\alpha = -(1\alpha) = -\alpha$ is in W.

(iv) We have $1 \in F$. If $\alpha, \beta \in W$, then $1\alpha + \beta = \alpha + \beta$ is in W. Thus W is closed with respect to vector addition.

The remaining postulates of a vector space will hold in W since they hold in V of which W is a subset.

Hence W is a subspace of V.

Note: If we are to prove that a non-empty subset W of a vector space V is a subspace of V, then either it is sufficient to prove that

$$a, b \in F$$

and $\qquad \alpha, \beta \in W$

$\Rightarrow \qquad a\alpha + b\beta \in W$

or it is sufficient to prove that

$$a \in F, \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + \beta \in W.$$

Illustrative Examples

Example 9: *The set W of ordered triads $(a_1, a_2, 0)$ where $a_1, a_2 \in F$ is a subspace of $V_3(F)$.*

Solution: W is non-empty since $\mathbf{0} = (0, 0, 0) \in W$. Let $\alpha = (a_1, a_2, 0)$ and $\beta = (b_1, b_2, 0)$ be any two elements of W. Then $a_1, a_2, b_1, b_2 \in F$. If a, b be any two elements of F, we have

$$\begin{aligned} a\alpha + b\beta &= a(a_1, a_2, 0) + b(b_1, b_2, 0) \\ &= (aa_1, aa_2, 0) + (bb_1, bb_2, 0) \end{aligned}$$

since $aa_1 + bb_1, aa_2 + bb_2 \in F$ and the last co-ordinate of this triad is zero. Hence W is a subspace of $V_3(F)$.

Example 10: *Let V be the vector space of all polynomials in an indeterminate x over a field F. Let W be a subset of V consisting of all polynomials of degree $\leq n$. Then W is a subspace of V.*

Solution: Let α and β be any two elements of W. Then α, β are polynomials over F of degree $\leq n$. If a, b are any two elements of F, then $a\alpha + b\beta$ will also be a polynomial of degree $\leq n$.

Therefore $a\alpha + b\beta \in W$. Hence W is a subspace of V.

Example 11: *If a_1, a_2, a_3 are fixed elements of a field F, then the set W of all ordered triads (x_1, x_2, x_3) of elements of F, such that $a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$, is a subspace of $V_3(F)$.*

Solution: W is non-empty since $(0, 0, 0) \in W$. Let $\alpha = (x_1, x_2, x_3)$ and $\beta = (y_1, y_2, y_3)$ be any two elements of W. Then $x_1, x_2, x_3, y_1, y_2, y_3$ are elements of F and are such that

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0 \qquad \text{...(1)}$$

and

$$a_1 y_1 + a_2 y_2 + a_3 y_3 = 0. \qquad \text{...(2)}$$

If a, b be any two elements of F, we have

$$\begin{aligned} a\alpha + b\beta &= a(x_1, x_2, x_3) + b(y_1, y_2, y_3) \\ &= (ax_1, ax_2, ax_3) + (by_1, by_2, by_3) \\ &= (ax_1 + by_1, ax_2 + by_2, ax_3 + by_3). \end{aligned}$$

Now $a_1(ax_1 + by_1) + a_2(ax_2 + by_2) + a_3(ax_3 + by_3)$

$$\begin{aligned} &= a(a_1 x_1 + a_2 x_2 + a_3 x_3) + b(a_1 y_1 + a_2 y_2 + a_3 y_3) \\ &= a0 + b0 \qquad \text{[by (1) and (2)]} \\ &= 0. \end{aligned}$$

$\therefore \qquad a\alpha + b\beta = (ax_1 + by_1, ax_2 + by_2, ax_3 + by_3) \in W.$

Hence W is a subspace of $V_3(F)$.

Example 12: *Which of the following sets of vectors $\alpha = (a_1, a_2, \ldots, a_n)$ in $\mathbf{R}^n$ are subspaces of $\mathbf{R}^n$ $(n \geq 3)$?*

(i) all α such that $a_1 \leq 0$;

(ii) all α such that a_3 is an integer;

(iii) all α such that $a_2 + 4a_3 = 0$;

(iv) all α such that $a_1 + a_2 + \ldots + a_n = k$ (a given constant).

Solution: (i) Let $W = \{\alpha : \alpha \in \mathbf{R}^n \text{ and } a_1 \leq 0\}$.

If we take $a_1 = -3$, then $a_1 < 0$ and so

$$\alpha = (-3, a_2, \ldots, a_n) \in W.$$

Now if we take $a = -2$, then

$$a\alpha = (6, -2a_2, \ldots, -2a_n).$$

Thus $\alpha \in W$, $a \in \mathbf{R}$ but $a\alpha \notin W$. Therefore W is not closed for scalar multiplication and so W is not a subspace of $\mathbf{R}^n$.

(ii) Let $W = \{\alpha : \alpha \in \mathbf{R}^n$ and a_3 is an integer$\}$.

If we take $a_3 = 5$, then a_3 is an integer and so

$$\alpha = (a_1, a_2, 5, \ldots, a_n) \in W.$$

Now if we take $a = \frac{1}{2}$, then

$$a\alpha = \left(\frac{1}{2}a_1, \frac{1}{2}a_2, \frac{5}{2}, \ldots, \frac{1}{2}a_n\right).$$

Since the third coordinate of $a\alpha$ is $5/2$ which is not an integer therefore $a\alpha \notin W$.

Thus $\alpha \in W, a \in \mathbf{R}$ but $a\alpha \notin W$. Therefore W is not closed for scalar multiplication and so W is not a subspace of $\mathbf{R}^n$.

(iii) Let $W = \{\alpha : \alpha \in \mathbf{R}^n$ and $a_2 + 4a_3 = 0\}$.

W is non-empty since $\mathbf{0} = (0, 0, \ldots, 0) \in W$.

Let $\alpha = (a_1, \ldots, a_n)$ and $\beta = (b_1, \ldots, b_n)$ be any two members of W.

Then $a_2 + 4a_3 = 0$ and $b_2 + 4b_3 = 0$.

If $a, b \in \mathbf{R}$, then $a\alpha + b\beta = (aa_1 + bb_1, \ldots, aa_n + bb_n)$.

We have $(aa_2 + bb_2) + 4(aa_3 + bb_3)$

$= a(a_2 + 4a_3) + b(b_2 + 4b_3) = a\,.\,0 + b\,.\,0 = 0.$

Thus according to the definition of W, $a\alpha + b\beta \in W$.

In this way $\alpha, \beta \in W$ and $a, b \in \mathbf{R} \Rightarrow a\alpha + b\beta \in W$. Hence W is a subspace of $\mathbf{R}^n$.

(iv) A subspace if $k = 0$ and not a subspace if $k \neq 0$.

Example 13: *Let* $\mathbf{R}$ *be the field of real numbers. Which of the following are subspaces of* $V_3(\mathbf{R})$ *?*

(i) $\{(x, 2y, 3z) : x, y, z \in \mathbf{R}\}$.

(ii) $\{(x, x, x) : x \in \mathbf{R}\}$.

(iii) $\{(x, y, z) : x, y, z$ *are rational numbers*$\}$.

Solution: (i) Let $W = \{(x, 2y, 3z) : x, y, z \in \mathbf{R}\}$.

Let $\alpha = (x_1, 2y_1, 3z_1)$ and $\beta = (x_2, 2y_2, 3z_2)$ be any two elements of W. Then $x_1, y_1, z_1, x_2, y_2, z_2$ are real numbers. If a, b are any two real numbers, then

$$\begin{aligned} a\alpha + b\beta &= a(x_1, 2y_1, 3z_1) + b(x_2, 2y_2, 3z_2) \\ &= (ax_1 + bx_2, 2ay_1 + 2by_2, 3az_1 + 3bz_2) \\ &= (ax_1 + bx_2, 2[ay_1 + by_2], 3[az_1 + bz_2]) \\ &\in W, \text{since } ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2 \end{aligned}$$

are real numbers.

Thus $a, b \in \mathbf{R}$ and $\alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W$.

(ii) Let $W = \{(x, x, x) : x \in \mathbf{R}\}$.

Let $\alpha = (x_1, x_1, x_1)$ and $\beta = (x_2, x_2, x_2)$ be any two elements of W. Then x_1, x_2 are real numbers. If a, b are any real numbers, then

$$a\alpha + b\beta = a(x_1, x_1, x_1) + b(x_2, x_2, x_2)$$
$$= (ax_1 + bx_2, ax_1 + bx_2, ax_1 + bx_2) \in W, \text{ since}$$

$ax_1 + bx_2 \in \mathbf{R}$.

Thus W is a subspace of $V_3(\mathbf{R})$.

(iii) Let $W = \{(x, y, z) : x, y, z$ are rational numbers$\}$.

Now $\alpha = (3, 4, 5)$ is an element of W. Also $a = \sqrt{7}$ is an element of $\mathbf{R}$. But

$$a\alpha = \sqrt{7}(3, 4, 5) = (3\sqrt{7}, 4\sqrt{7}, 5\sqrt{7}) \notin W \text{ since } 3\sqrt{7}, 4\sqrt{7}, 5\sqrt{7}$$

are not rational numbers.

Therefore W is not closed under scalar multiplication. Hence W is not a subspace of $V_3(\mathbf{R})$.

Example 14: *Let V be the vector space of all 2×2 matrices over the field $\mathbf{R}$. Show that W is not a subspace of V, where W contains all 2×2 matrices with zero determinants.*

Solution: Let $A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & b \end{bmatrix}$, where $a, b \in \mathbf{R}$ and $a \neq 0, b \neq 0$.

We have $|A| = 0, |B| = 0$ so $A, B \in W$.

Now $A + B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$. We have $|A + B| = ab \neq 0$.

$\therefore \quad A + B \notin W$.

Thus $\quad A \in W, B \in W$ but $A + B \notin W$.

Hence W is not a subspace of V.

Example 15: *If a vector space V is the set of all real valued continuous functions over the field of real numbers $\mathbf{R}$, then show that the set W of solutions of the differential equation.*

$$2\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 0$$

is a subspace of V.

Solution: We have $W = \left\{ y : 2\dfrac{d^2y}{dx^2} - 9\dfrac{dy}{dx} + 2y = 0 \right\}$, where $y = f(x)$.

Obviously $y = 0$ satisfies the given differential equation and as such it belongs to W and thus $W \neq \varnothing$.

Now let $y_1, y_2 \in W$. Then

$$2\frac{d^2y_1}{dx^2} - 9\frac{dy_1}{dx} + 2y_1 = 0 \qquad \ldots(1)$$

and

$$2\frac{d^2y_2}{dx^2} - 9\frac{dy_2}{dx} + 2y_2 = 0. \qquad \ldots(2)$$

belongs to W *i.e.*, it is a solution of the given differential equation.

We have $\quad 2 \dfrac{d^2}{dx^2}(ay_1 + by_2) - 9 \dfrac{d}{dx}(ay_1 + by_2) + 2(ay_1 + by_2)$

$$= 2a \frac{d^2 y_1}{dx^2} + 2b \frac{d^2 y_2}{dx^2} - 9a \frac{dy_1}{dx} - 9b \frac{dy_2}{dx} + 2ay_1 + 2by_2$$

$$= a\left(2 \frac{d^2 y_1}{dx^2} - 9 \frac{dy_1}{dx} + 2y_1\right) + b\left(2 \frac{d^2 y_2}{dx^2} - 9 \frac{dy_2}{dx} + 2y_2\right)$$

$$= a \cdot 0 + b \cdot 0, \text{ by (1) and (2)}$$

$$= 0.$$

Thus $ay_1 + by_2$ is a solution of the given differential equation and so it belongs to W.

Hence W is a subspace of V.

Example 16: *Find the solution of system of homogeneous linear equations. Let $V(F)$ be the vector space of all $n \times 1$ matrices over the field F. Let A be an $m \times n$ matrix over F. Then the set W of all $n \times 1$ matrices X over F such that $AX = O$ is a subspace of V. Here O is a null matrix of the type $m \times 1$.*

Solution: W is non-empty since $O_{n \times 1} \in W$. Let $X, Y \in W$. Then X and Y are $n \times 1$ matrices over F such that $AX = O$, $AY = O$.

Let $a \in F$. Then $aX + Y$ is also an $n \times 1$ matrix over F.

We have $\;A(aX + Y) = A(aX) + AY = a(AX) + AY = aO + O = O + O = O$

Therefore $aX + Y \in W$. Thus

$$a \in F,\ X, Y \in W \Rightarrow aX + Y \in W.$$

Hence W is a subspace of V.

Comprehensive Exercise 2

1. Show that the subset $\{(a, b, c) : a + b + c = 0\}$ of $\mathbf{R}^3$ is a subspace of $\mathbf{R}^3$.
2. Prove that the set of all solutions (a, b, c) of the equation $a + b + 2c = 0$ is a subspace of the vector space V_3 **(R)**.
3. Show that the set W of the elements of the vector space V_3 **(R)** of the form $(x + 2y, y, -x + 3y)$ where $x, y \in \mathbf{R}$ is a subspace of V_3 **(R)**.
4. Let V be the vector space of all polynomials over the field $\mathbf{R}$. Determine whether or not W is a subspace of V where (i) W contains all polynomials with integral coefficients (ii) W consists of all polynomials $b_0 + b_1 x^2 + b_2 x^4 + \ldots + b_n x^{2n}$ *i.e.*, polynomials with only even powers of x.
5. Which of the following sets of vectors $\alpha = (a_1, \ldots, a_n)$ in $\mathbf{R}^n$ are subspaces of $\mathbf{R}^n$? $(n \geq 3)$.

   (i) all α such that $a_1 \geq 0$;

(iii) all α such that $a_2 = a_1^2$; (iv) all α such that $a_1 a_2 = 0$;

(v) all α such that a_2 is rational.

6. Let F be the field of integers modulo 2, V be the set of all 2×2 matrices over F. Show that $V(F)$ is a finite vector space. Give two non-trivial subspaces of this vector space.

7. Let V be the vector space of all $n \times n$ square matrices over a field F. Show that W is a subspace of V if W consists of all matrices which are

(i) symmetric *i.e.*, all matrices $A = [a_{ij}]$ for which $a_{ji} = a_{ij}$.

(ii) skew-symmetric.

(iii) upper triangular.

(iv) diagonal.

(v) scalar.

8. Let V be a vector space of all real $n \times n$ matrices. Prove that the set W consisting of all $n \times n$ real matrices which commute with a given matrix T of V forms a subspace of V.

9. Let V be the vector space of all 2×2 matrices over the real field $\mathbf{R}$. Show that the subset of V consisting of all matrices A for which $A^2 = A$ is not a subspace of V.

10. Determine whether or not W is a subspace of $\mathbf{R}^3$ if W consists of those vectors $(a, b, c) \in \mathbf{R}^3$ for which :

(i) $a = 2b$ (ii) $ab = 0$

(iii) $a \leq b \leq c$ (iv) $a = b^2$

(v) $a^2 + b^2 + c^2 \leq 1$ (vi) $k_1 a + k_2 b + k_3 c = 0, k_i \in \mathbf{R}$.

11. Let V be the (real) vector space of all functions f from $\mathbf{R}$ into $\mathbf{R}$. Which of the following sets of functions are sub-spaces of V ?

(i) all f such that $f(x^2) = [f(x)]^2$;

(ii) all f such that $f(0) = f(1)$;

(iii) all f such that $f(3) = 1 + f(-5)$;

(iv) all f such that $f(-1) = 0$;

(v) all f which are continuous.

12. Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Check whether W is a subspace of V where

(i) $W = \{f : f(7) = 2 + f(1)\}$;

(ii) W consists of all bounded functions;

(iii) W consists of all integrable functions in the interval $[0, 1]$.

13. Let V be the vector space of infinite sequences $< a_1, a_2, \ldots a_n, \ldots >$ in a field F. Show that W is a subspace of V, if

(i) W consists of all sequences with 0 as the first component ;

(ii) W consists of all sequences with only a finite number of non-zero components.

unknowns over a field F. Show that the solution set W of the system is not a subspace of F^n.

Answers 2

4.	(i) not a subspace	(ii) a subspace	
5.	(i) not a subspace	(ii) a subspace	(iii) not a subspace;
	(iv) not a subspace	(v) not a subspace	
10.	(i) a subspace	(ii) not a subspace	
	(iii) not a subspace	(iv) not a subspace	
	(v) not a subspace	(vi) a subspace	
11.	(i) not a subspace	(ii) a subspace	(iii) not a subspace
	(iv) a subspace	(v) a subspace	
12.	(i) not a subspace	(ii) a subspace	(iii) a subspace

1.4 Algebra of Subspaces

Theorem 1: *The intersection of any two subspaces W_1 and W_2 of a vector space $V(F)$ is also a subspace of $V(F)$.* **(Kumaon 2002; Garhwal 07, 12)**

Proof: Since $\mathbf{0} \in W_1$ and W_2 both, therefore $W_1 \cap W_2$ is not empty.

Let $\alpha, \beta \in W_1 \cap W_2$ and $a, b \in F$.

Now $\alpha \in W_1 \cap W_2 \Rightarrow \alpha \in W_1$ and $\alpha \in W_2$

and $\beta \in W_1 \cap W_2 \Rightarrow \beta \in W_1$ and $\beta \in W_2$.

Since W_1 is a subspace, therefore $a, b \in F$ and $\alpha, \beta \in W_1 \Rightarrow a\alpha + b\beta \in W_1$.

Similarly $a, b \in F$ and $\alpha, \beta \in W_2 \Rightarrow a\alpha + b\beta \in W_2$.

Now $a\alpha + b\beta \in W_1, a\alpha + b\beta \in W_2 \Rightarrow a\alpha + b\beta \in W_1 \cap W_2$.

Thus $a, b \in F$ and $\alpha, \beta \in W_1 \cap W_2 \Rightarrow a\alpha + b\beta \in W_1 \cap W_2$.

Hence $W_1 \cap W_2$ is a subspace of $V(F)$.

Note: The union of two subspaces of $V(F)$ may not be a subspace of $V(F)$. For example if $\mathbf{R}$ be the field of real numbers, then $W_1 = \{(0, 0, z) : z \in \mathbf{R}\}$ and $W_2 = \{(0, y, 0) : y \in \mathbf{R}\}$ are two subspaces of $V_3(\mathbf{R})$.

We have $(0, 0, 3) \in W_1$ and $(0, 5, 0) \in W_2$.

But $(0, 0, 3) + (0, 5, 0) = (0, 5, 3) \notin W_1 \cup W_2$ since neither $(0, 5, 3) \in W_1$ nor $(0, 5, 3) \in W_2$. Thus $W_1 \cup W_2$ is not closed under vector addition. Hence $W_1 \cup W_2$ is not a subspace of $V_3(\mathbf{R})$.

Theorem 2: *The union of two subspaces is a subspace if and only if one is contained in the other.* **(Kumaon 2003; Garhwal 06)**

Proof. Suppose W_1 and W_2 are two subspaces of a vector space V.

subspaces and therefore, $W_1 \cup W_2$ is also a subspace.

Conversely, suppose $W_1 \cup W_2$ is a subspace.

To prove that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Let us assume that W_1 is not a subset of W_2 and W_2 is also not a subset of W_1.

Now W_1 is not a subset of $W_2 \Rightarrow \exists\, \alpha \in W_1$ and $\alpha \notin W_2$ ...(1)

and W_2 is not a subset of $W_1 \Rightarrow \exists\, \beta \in W_2$ and $\beta \notin W_1$. ...(2)

From (1) and (2), we have $\alpha \in W_1 \cup W_2$ and $\beta \in W_1 \cup W_2$.

Since $W_1 \cup W_2$ is a subspace, therefore $\alpha + \beta$ is also in $W_1 \cup W_2$.

But $\alpha + \beta \in W_1 \cup W_2 \Rightarrow \alpha + \beta \in W_1$ or W_2.

Suppose $\alpha + \beta \in W_1$. Since $\alpha \in W_1$ and W_1 is a subspace, therefore $(\alpha + \beta) - \alpha = \beta$ is in W_1.

But from (2), we have $\beta \notin W_1$. Thus we get a contradiction. Again suppose that $\alpha + \beta \in W_2$. Since $\beta \in W_2$ and W_2 is a subspace, therefore $(\alpha + \beta) - \beta = \alpha$ is in W_2. But from (1), we have $\alpha \notin W_2$. Thus here also we get a contradiction. Hence either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Theorem 3. *Arbitrary intersection of subspaces i.e., the intersection of any family of subspaces of a vector space is a subspace.*

Proof. Let $V(F)$ be a vector space and let $\{W_t : t \in T\}$ be any family of subspaces of V. Here T is an index set and is such that $\forall\, t \in T$, W_t is a subspace of V.

$$\text{Let} \qquad U = \bigcap_{t \in T} W_t = \{x \in V : x \in W_t \ \forall\, t \in T\}$$

be the intersection of this family of subspaces of V. Then to prove that U is also a subspace of V.

Obviously $U \neq \emptyset$, since at least the zero vector $\mathbf{0}$ of V is in $W_t \ \forall\, t \in T$.

Now let $a, b \in F$ and α, β be any two elements of $\bigcap_{t \in T} W_t$.

Then $\alpha, \beta \in W_t \ \forall\, t \in T$. Since each W_t is a subspace of V, therefore

$$a\alpha + b\beta \in W_t \ \forall\, t \in T.$$

Thus $a\alpha + b\beta \in \bigcap_{t \in T} W_t$.

Thus $a, b \in F$ and $\alpha, \beta \in \bigcap_{t \in T} W_t \Rightarrow a\alpha + b\beta \in \bigcap_{t \in T} W_t$.

Hence $\bigcap_{t \in T} W_t$ is a subspace of $V(F)$.

Note: In the above theorem we cannot replace intersection by union.

Smallest subspace containing any subset of $V(F)$. Let $V(F)$ be a vector space and S be any subset of V. If U is a subspace of V containing S and is itself contained in every subspace of V containing S, then U is called the **smallest subspace** of V containing S. The smallest subspace of V containing S is also called the subspace of V generated or spanned by S and we shall denote it by the subspaces of $V(F)$ containing S is the subspace of $V(F)$ generated by S. If $\{S\} = V$, then we say that V is spanned by S.

1.5 Linear Combination of Vectors and Linear Span of a Set

Linear combination. Definition. *Let $V(F)$ be a vector space. If $\alpha_1, \alpha_2, \ldots, \alpha_n \in V$, then any vector*

$$\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n \text{ where } a_1, a_2, \ldots, a_n \in F$$

is called a linear combination of the vectors $\alpha_1, \alpha_2, \ldots, \alpha_n$.

Linear span. Definition: *Let $V(F)$ be a vector space and S be any non-empty subset of V. Then the linear span of S is the set of all linear combinations of finite sets of elements of S and is denoted by $L(S)$.* Thus we have

$$L(S) = \{a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n : \alpha_1, \alpha_2, \ldots, \alpha_n \text{ is any arbitrary finite subset of } S \text{ and } a_1, a_2, \ldots, a_n \text{ is any arbitrary finite subset of } F\}.$$

Theorem 1: *The linear span $L(S)$ of any subset S of a vector space $V(F)$ is a subspace of V generated by S i.e., $L(S) = \{S\}$.*

Proof: Let α, β be any two elements of $L(S)$.

Then $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_m \alpha_m$

and $\beta = b_1 \beta_1 + b_2 \beta_2 + \ldots + b_n \beta_n$

where the a's and b's are elements of F and the α's and β's are elements of S.

If a, b be any two elements of F, then

$$a\alpha + b\beta = a(a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_m \alpha_m) + b(b_1 \beta_1 + b_2 \beta_2 + \ldots + b_n \beta_n)$$

$$= a(a_1 \alpha_1) + a(a_2 \alpha_2) + \ldots + a(a_m \alpha_m) + b(b_1 \beta_1) + b(b_2 \beta_2) + \ldots + b(b_n \beta_n)$$

$$= (aa_1)\alpha_1 + (aa_2)\alpha_2 + \ldots + (aa_m)\alpha_m + (bb_1)\beta_1 + (bb_2)\beta_2 + \ldots + (bb_n)\beta_n.$$

Thus $a\alpha + b\beta$ has been expressed as a linear combination of a finite set $\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_n$ of the elements of S. Consequently $a\alpha + b\beta \in L(S)$.

Thus $a, b \in F$ and $\alpha, \beta \in L(S) \Rightarrow a\alpha + b\beta \in L(S)$. Hence $L(S)$ is a subspace of $V(F)$.

Also each element of S belongs to $L(S)$ because if $\alpha_r \in S$, then $\alpha_r = 1\alpha_r$ and this implies that $\alpha_r \in L(S)$. Thus $L(S)$ is a subspace of V and S is contained in $L(S)$.

Now if W is any subspace of V containing S, then each element of $L(S)$ must be in W because W is to be closed under vector addition and scalar multiplication. Therefore $L(S)$ will be contained in W.

Illustration 1. *The subset containing a single element* $(1, 0, 0)$ *of the vector space* $V_3(F)$ *generates the subspace which is the totality of elements of the form* $(a, 0, 0)$.

Illustration 2. *The subset* $\{(1, 0, 0), (0, 1, 0)\}$ *of* $V_3(F)$ *generates the subspace which is the totality of the elements of the form* $(a, b, 0)$.

Illustration 3. *The subset* $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ *of* $V_3(F)$ *generates or spans the entire vector space* $V_3(F)$ *i.e.,* $L(S) = V$.

If (a, b, c) be any element of V, then

$$(a, b, c) = a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1).$$

Thus $$(a, b, c) \in L(S).$$

Hence $V \subseteq L(S)$. Also $L(S) \subseteq V$. Hence $L(S) = V$.

Illustration 4. *Let V be the vector space of all polynomials over the field F. Let S be the subset of V consisting of the polynomials* $f_0, f_1, f_2, \ldots,$ *defined by* $f_n = x^n, n = 0, 1, 2, \ldots$

Then $V = L(S)$.

Note 1: **Important.** Suppose S is a non-empty subset of a vector space $V(F)$. Then a vector $\alpha \in V$ will be in the subspace of V generated by S if it can be expressed as a linear combination over F of a finite number of vectors belonging to S.

Note 2: If in any case we are to prove that $L(S) = V$, then we should prove that $V \subseteq L(S)$ because $L(S) \subseteq V$ since $L(S)$ is a subspace of V. In order to prove that $V \subseteq L(S)$, we should prove that each element of V can be expressed as a linear combination of a finite number of elements of S. Then each element of V will also be an element of $L(S)$ and we shall have $V \subseteq L(S)$.

Finally $V \subseteq L(S)$ and $L(S) \subseteq V \Rightarrow L(S) = V$.

1.6 Linear Sum of Two Subspaces

Definition: *Let* W_1 *and* W_2 *be two subspaces of the vector space* $V(F)$. *Then the linear sum of the subspaces* W_1 *and* W_2, *denoted by* $W_1 + W_2$, *is the set of all sums* $\alpha_1 + \alpha_2$ *such that* $\alpha_1 \in W_1, \alpha_2 \in W_2$.

Thus $$W_1 + W_2 = \{\alpha_1 + \alpha_2 : \alpha_1 \in W_1, \alpha_2 \in W_2\}.$$

For any subspace W of a vector space V, we have $W + W = W$.

Theorem 1. *If* W_1 *and* W_2 *are subspaces of the vector space* $V(F)$, *then*

(i) $W_1 + W_2$ *is a subspace of* $V(F)$. **(Kumaon 2004; Garhwal 10)**

(ii) $W_1 + W_2 = \{W_1 \cup W_2\}$ *i.e.,* $L(W_1 \cup W_2) = W_1 + W_2$. **(Kumaon 2002)**

Then $\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$ where $\alpha_1, \beta_1 \in W_1$ and $\alpha_2, \beta_2 \in W_2$. If $a, b \in F$, we have

$$a\alpha + b\beta = a(\alpha_1 + \alpha_2) + b(\beta_1 + \beta_2) = (a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2).$$

Since W_1 is a subspace of V, therefore $a, b \in F$

and $\qquad \alpha_1, \beta_1 \in W_1 \Rightarrow a\alpha_1 + b\beta_1 \in W_1$.

Similarly $a\alpha_2 + b\beta_2 \in W_2$.

Consequently $a\alpha + b\beta = (a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2) \in W_1 + W_2$.

Thus $a, b \in F$ and $\alpha, \beta \in W_1 + W_2 \Rightarrow a\alpha + b\beta \in W_1 + W_2$.

Hence $W_1 + W_2$ is a subspace of $V(F)$.

(ii) Since W_2 contains the zero vector, therefore if $\alpha_1 \in W_1$, then we can write

$$\alpha_1 = \alpha_1 + \mathbf{0} \in W_1 + W_2.$$

Thus $\qquad W_1 \subseteq W_1 + W_2$.

Similarly $\qquad W_2 \subseteq W_1 + W_2$.

Hence $\qquad W_1 \cup W_2 \subseteq W_1 + W_2$.

Therefore $\qquad W_1 + W_2$ is a subspace of $V(F)$ containing $W_1 \cup W_2$.

Now to prove that $W_1 + W_2 = \{W_1 \cup W_2\}$ we should prove that

$$W_1 + W_2 \subseteq L(W_1 \cup W_2) \text{ and } L(W_1 \cup W_2) \subseteq W_1 + W_2.$$

Let $\alpha = \alpha_1 + \beta_1$ be any element of $W_1 + W_2$. Then $\alpha_1 \in W_1$ and $\beta_1 \in W_2$. Therefore $\alpha_1, \beta_1 \in W_1 \in W_2$. We can write

$$\alpha_1 + \beta_1 = 1\alpha_1 + 1\beta_1.$$

Thus $\alpha_1 + \beta_1$ is a linear combination of a finite number of elements

$$\alpha_1, \beta_1 \in W_1 \cup W_2.$$

Therefore $\alpha_1 + \beta_1 \in L(W_1 \cup W_2)$.

$\therefore \qquad W_1 + W_2 \subseteq L(W_1 \cup W_2)$.

Also $L(W_1 \cup W_2)$ is the smallest subspace containing $W_1 \cup W_2$ and $W_1 + W_2$ is a subspace containing $W_1 \cup W_2$. Therefore $L(W_1 \cup W_2)$ must be contained in $W_1 + W_2$. Consequently

$$L(W_1 \cup W_2) \subseteq W_1 + W_2.$$

Hence $\qquad W_1 + W_2 = L(W_1 \cup W_2) = \{W_1 \cup W_2\}$.

Note: If $W_1, W_2, \ldots, W_k$ are subspaces of the vector space V, then their linear sum, denoted by $W_1 + W_2 + \ldots + W_k$, is the set of all sums $\alpha_1 + \alpha_2 + \ldots + \alpha_k$ such that $\alpha_i \in W_i$. It can be proved that $W_1 + W_2 + \ldots + W_k$ is a subspace of V which contains each of the subspaces W_i and which is spanned by the union of $W_1, W_2, \ldots, W_k$.

Illustration: *Let V be the vector space of all 2×2 matrices over $\mathbf{R}$. If W_1 consists of those matrices in V whose second row is zero and W_2 consists of those matrices in V whose second column is zero i.e.,*

$$W_1 = \left\{\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} : a, b \in \mathbf{R}\right\}, W_2 = \left\{\begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} : a, c \in \mathbf{R}\right\}$$

then W_1 and W_2 are subspaces of V. We have

$$W_1 + W_2 = \left\{\begin{bmatrix} a & b \\ c & 0 \end{bmatrix} : a, b, c \in \mathbf{R}\right\}, W_1 \cap W_2 = \left\{\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} : a \in \mathbf{R}\right\}.$$

Illustrative Examples

Example 17: *If S, T are subsets of $V(F)$, then*

(i) $S \subseteq T \Rightarrow L(S) \subseteq L(T)$.

(ii) $L(S \cup T) = L(S) + L(T)$.

(iii) S is a subspace of $V \Leftrightarrow L(S) = S$.

(iv) $L(L(S)) = L(S)$.

Solution: (i) Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n \alpha_n \in L(S)$ where $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is a finite subset of S. Since $S \subseteq T$, therefore $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is also a finite subset of T. So $\alpha \in L(T)$.

Thus $\qquad \alpha \in L(S) \Rightarrow \alpha \in L(T)$.

$\therefore \qquad L(S) \subseteq L(T)$.

(ii) Let α be any element of $L(S \cup T)$. Then

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + \ldots + b_p\beta_p$$

where $\{\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_p\}$ is a finite subset of $S \cup T$ such that

$$\{\alpha_1, \alpha_2, \ldots, \alpha_m\} \subseteq S \text{ and } \{\beta_1, \beta_2, \ldots, \beta_p\} \subseteq T.$$

Now $\qquad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m \in L(S)$

and $\qquad b_1\beta_1 + b_2\beta_2 + \ldots + b_p\beta_p \in L(T)$.

Therefore $\qquad \alpha \in L(S) + L(T)$.

Consequently $L(S \cup T) \subseteq L(S) + L(T)$.

Now let γ be any element of $L(S) + L(T)$. Then $\gamma = \beta + \delta$ where $\beta \in L(S)$ and $\delta \in L(T)$. Now β will be a linear combination of a finite number of elements of S and δ will be a linear combination of a finite number of elements of T. Therefore $\beta + \delta$ will be a linear combination of a finite number of elements of $S \cup T$.

Thus $\beta + \delta \in L(S \cup T)$. Consequently

$$L(S) + L(T) \subseteq L(S \cup T).$$

Hence $\qquad L(S \cup T) = L(S) + L(T)$.

(iii) Suppose S is a subspace of V. Then we are to prove that $L(S) = S$.

Let $\alpha \in L(S)$. Then $\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$ where $a_1, \ldots, a_n \in F$ and $\alpha_1, \ldots, \alpha_n \in S$. But S is a subspace of V. Therefore it is closed with respect to scalar multiplication and vector addition. Hence $\alpha = a_1\alpha_1 + \ldots + a_n\alpha_n \in S$. Thus

$$\alpha \in L(S) \Rightarrow \alpha \in S.$$

Therefore $L(S) \subseteq S$. Also $S \subseteq L(S)$. Therefore $L(S) = S$.

know that $L(S)$ is a subspace of V. Since $S = L(S)$, therefore S is also a subspace of V.

(iv) $L(L(S))$ is the smallest subspace of V containing $L(S)$. But $L(S)$ is a subspace of V. Therefore the smallest subspace of V containing $L(S)$ is $L(S)$ itself.

Hence $L(L(S)) = L(S)$.

Example 18: *In the vector space* $\mathbf{R}^3$ *express the vector* $(1, -2, 5)$ *as a linear combination of the vectors* $(1, 1, 1), (1, 2, 3)$ *and* $(2, -1, 1)$.

Solution: Let $\alpha = (1, -2, 5), \alpha_1 = (1, 1, 1), \alpha_2 = (1, 2, 3), \alpha_3 = (2, -1, 1)$.

We wish to express α as a linear combination of $\alpha_1, \alpha_2, \alpha_3$.

Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$ where $a_1, a_2, a_3 \in \mathbf{R}$.

Then $(1, -2, 5) = a_1(1, 1, 1) + a_2(1, 2, 3) + a_3(2, -1, 1)$

$$= (a_1, a_1, a_1) + (a_2, 2a_2, 3a_2) + (2a_3, -a_3, a_3)$$

$$= (a_1 + a_2 + 2a_3, a_1 + 2a_2 - a_3, a_1 + 3a_2 + a_3)$$

$\therefore$

$$a_1 + a_2 + 2a_3 = 1 \qquad \ldots(1)$$

$$a_1 + 2a_2 - a_3 = -2 \qquad \ldots(2)$$

$$a_1 + 3a_2 + a_3 = 5 \qquad \ldots(3)$$

Method 1. The augmented matrix of the system of equations (1), (2) and (3) is

$$[A : B] = \begin{bmatrix} 1 & 1 & 2 & \vdots & 1 \\ 1 & 2 & -1 & \vdots & -2 \\ 1 & 3 & 1 & \vdots & 5 \end{bmatrix}.$$

We shall reduce the matrix $[A : B]$ to Echelon form by applying the row transformations only.

We have
$$[A : B] \sim \begin{bmatrix} 1 & 1 & 2 & \vdots & 1 \\ 0 & 1 & -3 & \vdots & -3 \\ 0 & 2 & -1 & \vdots & 4 \end{bmatrix}.$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & \vdots & 1 \\ 0 & 1 & -3 & \vdots & -3 \\ 0 & 0 & 5 & \vdots & 10 \end{bmatrix}.$$

Thus we have rank A = rank $[A : B]$ *i.e.*, the system of equations (1), (2), (3) is consistent and so has a solution *i.e.*, α can be expressed as a linear combination of $\alpha_1, \alpha_2, \alpha_3$. The equations (1), (2) and (3) are equivalent to equations

$$a_1 + a_2 + 2a_3 = 1, \; a_2 - 3a_3 = -3, \; 5a_3 = 10.$$

Solving for unknowns, we get $a_3 = 2, \; a_2 = 3, \; a_1 = -6$.

Hence $(1, -2, 5) = -6(1, 1, 1) + 3(1, 2, 3) + 2(2, -1, 1)$.

Method 2. Eliminating a_3 between (1) and (2), we get

$$3a_1 + 5a_2 = -3 \qquad \ldots(4)$$

$$2a_1 + 5a_2 = 3 \qquad \ldots(5)$$

Solving (4) and (5), we get $a_1 = -6,\ a_2 = 3$.

Putting $a_1 = -6,\ a_2 = 3$ in (1), we get $a_3 = 2$.

Hence $(1, -2, 5) = -6\,(1, 1, 1) + 3\,(1, 2, 3) + 2\,(2, -1, 1)$.

Example 19: *For what value of m, the vector $(m, 3, 1)$ is a linear combination of the vectors $(3, 2, 1)$ and $(2, 1, 0)$?*

Solution: Let $\alpha_1 = (3, 2, 1)$ and $\alpha_2 = (2, 1, 0)$.

Let $(m, 3, 1) = a\alpha_1 + b\alpha_2$ where $a, b \in \mathbf{R}$.

Then $(m, 3, 1) = a\,(3, 2, 1) + b\,(2, 1, 0) = (3a, 2a, a) + (2b, b, 0)$

$$= (3a + 2b, 2a + b, a)$$

$\therefore \quad 3a + 2b = m,\ 2a + b = 3,\ a = 1.$

These give $a = 1, b = 1, m = 5$.

Hence the required value of m is 5.

Example 20: *In the vector space $\mathbf{R}^4$ determine whether or not the vector $(3, 9, -4, -2)$ is a linear combination of the vectors $(1, -2, 0, 3), (2, 3, 0, -1)$ and $(2, -1, 2, 1)$.*

Solution: Let $\alpha = (3, 9, -4, -2), \alpha_1 = (1, -2, 0, 3)$,

$$\alpha_2 = (2, 3, 0, -1), \alpha_3 = (2, -1, 2, 1).$$

Let $\quad \alpha = a\alpha_1 + b\alpha_2 + c\alpha_3$ where $a, b, c \in \mathbf{R}$.

Then $\quad (3, 9, -4, -2) = a\,(1, -2, 0, 3) + b\,(2, 3, 0, -1) + c\,(2, -1, 2, 1)$

or $\quad (3, 9, -4, -2) = (a + 2b + 2c, -2a + 3b - c, 2c, 3a - b + c)$

$\therefore$
$$a + 2b + 2c = 3 \qquad \ldots(1)$$
$$-2a + 3b - c = 9 \qquad \ldots(2)$$
$$2c = -4 \qquad \ldots(3)$$
$$3a - b + c = -2 \qquad \ldots(4)$$

Solving (1), (2) and (3), we get $a = 1, b = 3, c = -2$. These values satisfy (4). Thus the system of equations (1), (2), (3) and (4) has a solution *i.e.*, α can be written as a linear combination of α_1, α_2 and α_3.

Also we have $\alpha = \alpha_1 + 3\alpha_2 - 2\alpha_3$.

Example 21: *In the vector space $\mathbf{R}^3$, let $\alpha = (1, 2, 1), \beta = (3, 1, 5), \gamma = (3, -4, 7)$. Show that the subspaces spanned by $S = \{\alpha, \beta\}$ and $T = \{\alpha, \beta, \gamma\}$ are the same.*

Solution: First we shall show that the vector γ can be expressed as a linear combination of the vectors α and β. Let

$$(3, -4, 7) = a\,(1, 2, 1) + b\,(3, 1, 5).$$

Then $a + 3b = 3, 2a + b = -4, a + 5b = 7$. Solving the first two equations, we get $a = -3, b = 2$ and these values satisfy the third equation also. Therefore we can write $\gamma = -3\alpha + 2\beta$.

Now $\quad S \subseteq T \Rightarrow L(S) \subseteq L(T)$.

vectors α, β and γ. In this linear combination the vector γ can be replaced by $-3\alpha + 2\beta$. Thus δ can be expressed as a linear combination of the vectors α and β. Therefore $\delta \in L(S)$. Thus $\delta \in L(T) \Rightarrow \delta \in L(S)$. Therefore $L(T) \subseteq L(S)$.

Hence $\quad L(T) = L(S)$.

Example 22: *Show that* $(1, 1, 1)$, $(0, 1, 1)$ *and* $(0, 1, -1)$ *generate* $\mathbf{R}^3$.

Solution: We shall show that any vector $(a, b, c) \in \mathbf{R}^3$ is a linear combination of the given vectors.

Let $\quad (a, b, c) = x(1, 1, 1) + y(0, 1, 1) + z(0, 1, -1)$, where $x, y, z \in \mathbf{R}$.

Then $\quad (a, b, c) = (x, x + y + z, x + y - z)$.

$\therefore \quad x = a, \ x + y + z = b, \ x + y - z = c$.

These give $x = a, \ y = \dfrac{b + c - 2a}{2}, \ z = \dfrac{b - c}{2}$ *i.e.*, the above system has a solution.

Thus the given vectors generate $\mathbf{R}^3$.

Example 23: *Let V be the vector space of all functions from $\mathbf{R}$ into $\mathbf{R}$; let V_e be the subset of even functions, $f(-x) = f(x)$; let V_o be the subset of odd functions, $f(-x) = -f(x)$.*

(i) Prove that V_e and V_o are subspaces of V.

(ii) Prove that $V_e + V_o = V$.

(iii) Prove that $V_e \cap V_o = \{\mathbf{0}\}$.

Solution: (i) Suppose f_e and $g_e \in V_e$ and a is any scalar *i.e.*, $a \in \mathbf{R}$. Then

$$(af_e + g_e)(-x) = af_e(-x) + g_e(-x)$$
$$= af_e(x) + g_e(x) = (af_e + g_e)(x).$$

Therefore $af_e + g_e$ is an even function.

Thus $a \in \mathbf{R}$ and $f_e, g_e \in V_e \Rightarrow af_e + g_e \in V_e$.

Hence V_e is a subspace of V.

Again suppose f_o and $g_o \in V_o$ and a is any scalar. Then

$$(af_o + g_o)(-x) = af_o(-x) + g_o(-x) = a[-f_o(x)] - g_o(x)$$
$$= -[af_o(x) + g_o(x)] = -(af_o + g_o)(x).$$

Therefore $af_o + g_o$ is an odd function.

Thus $a \in \mathbf{R}$ and $f_o, g_o \in V_o \Rightarrow af_o + g_o \in V_o$.

Hence V_o is a subspace of V.

(ii) Since V_e and V_o are subspaces of V, therefore $V_e + V_o$ is also a subspace of V and consequently $V_e + V_o \subseteq V$.

Now let $f \in V$. We shall show that f can be expressed as the sum of an even and an odd function.

Let $\quad f_{e_1}(x) = \dfrac{1}{2}[f(x) + f(-x)]$ and $f_{o_1}(x) = \dfrac{1}{2}[f(x) - f(-x)]$.

Then obviously f_{e_1} is an even function and f_{o_1} is an odd function. We can easily see that $f_{e_1}(-x) = f_{e_1}(x)$ and $f_{o_1}(-x) = -f_{o_1}(x)$.

$$= f_{e_1}(x) + f_{o_1}(x) = (f_{e_1} + f_{o_1})(x).$$

Therefore $\quad f = f_{e_1} + f_{o_1}$ where $f_{e_1} \in V_e,\ f_{o_1} \in V_o$.

Thus $\quad f \in V \Rightarrow f \in V_e + V_o$. Therefore $V \subseteq V_e + V_o$.

Hence $\quad V = V_e + V_o$.

(iii) Let $\mathbf{0}$ denote the zero function *i.e.*, $\mathbf{0}(x) = 0 \ \forall \ x \in \mathbf{R}$.

Then $\quad \mathbf{0} \in V_e$ and also $\mathbf{0} \in V_o$.

Let $\quad f \in V_e$ and also $f \in V_o$.

Then $\quad f(-x) = f(x) = -f(x)$.

$\therefore \quad 2f(x) = 0 \quad \Rightarrow \quad f(x) = 0 = \mathbf{0}(x)$.

Therefore $f = \mathbf{0}$ (zero function).

Thus $\quad f \in V_e \cap V_o \Leftrightarrow f = \mathbf{0}$.

Hence $\quad V_e \cap V_o = \{\mathbf{0}\}$.

Comprehensive Exercise 3

1. For which value of k will the vector $(1, -2, k)$ in $\mathbf{R}^3$ be a linear combination of the vectors $(3, 0, -2)$ and $(2, -1, -5)$?
2. In the vector space $\mathbf{R}^4$ determine whether or not the vector $(3, 9, -4, 2)$ is a linear combination of the vectors $(1, -2, 0, 3)$, $(2, 3, 0, -1)$ and $(2, -1, 2, 1)$.
3. Is the vector $(3, -1, 0, -1)$ in the subspace of $\mathbf{R}^4$ spanned by the vectors $(2, -1, 3, 2)$, $(-1, 1, 1, -3)$ and $(1, 1, 9, -5)$?
4. Is the vector $(2, -5, 3)$ in the subspace of $\mathbf{R}^3$ spanned by the vectors $(1, -3, 2), (2, -4, -1), (1, -5, 7)$?
5. Express the matrix $E = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}$ as a linear combination of the matrices
$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix}.$$
6. Write E as a linear combination of
$$A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$
where (i) $E = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ (ii) $E = \begin{bmatrix} 2 & 1 \\ -1 & -2 \end{bmatrix}$.
7. Express the polynomial $f = x^2 + 4x - 3$ over $\mathbf{R}$ as a linear combination of the polynomials
$$f_1 = x^2 - 2x + 5,\ f_2 = 2x^2 - 3x \text{ and } f_3 = x + 3.$$

$$f_1 = 2x^2 + 3x - 4,\ f_2 = x^2 - 2x - 3$$

where (i) $f = 3x^2 + 8x - 5$ (ii) $f = 4x^2 - 6x - 1$.

9. Find the condition on a, b and c so that $(a, b, c) \in \mathbf{R}^3$ is in the space generated by $(2, 1, 0), (1, -1, 2)$ and $(0, 3, -4)$.

10. Find one vector in $\mathbf{R}^3$ which generates the intersection of W_1 and W_2 where $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and W_2 is the space generated by the vectors $(1, 2, 3)$ and $(1, -1, 1)$.

11. If α, β and γ are vectors such that $\alpha + \beta + \gamma = \mathbf{0}$, then α and β span the same subspace as β and γ.

12. Give an example of a subset W of a vector space V which is not a subspace of V but for which

    (i) $W + W = W$. (ii) $W + W \subset W$.

13. Let V be the vector space of $n \times n$ matrices over a field F. Let W_1 and W_2 be the subspaces of upper triangular matrices and lower triangular matrices respectively. Find

    (i) $W_1 + W_2$ (ii) $W_1 \cap W_2$.

14. Show that the space U generated by the vectors $u_1 = (1, 2, -1, 3)$, $u_2 = (2, 4, 1, -2)$ and $u_3 = (3, 6, 3, -7)$ and the space V generated by the vectors $v_1 = (1, 2, -4, 11)$ and $v_2 = (2, 4, -5, 14)$ are equal.

Answers 3

1. $k = -8$
2. No
3. No
4. No
5. $E = 3A - 2B - C$
6. (i) $E = 2A - B + 2C$ (ii) Not possible
7. $f = -3f_1 + 2f_2 + 4f_3$
8. (i) $f = 2f_1 - f_2$ (ii) Not possible
9. $2a - 4b - 3c = 0$
10. $(-2, 5, 0)$
13. (i) $W_1 + W_2 = V$ (ii) space of diagonal matrices

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. The zero vector in the vector space $\mathbf{R}^4$ is

    (a) (0, 0) (b) (0, 0, 0)

    (c) (0, 0, 0, 0) (d) None of these.

n in which the coefficients are the elements of the field $\mathbf{I}\,(p)$ over the field $\mathbf{I}\,(p)$, p being a prime number, is

(a) p^{n+1} (b) p^{n}

(c) p^{n-1} (d) None of these.

3. If $(m, 3, 1)$ is a linear combination of vectors $(3, 2, 1)$ and $(2, 1, 0)$ in $\mathbf{R}^3$, then the value of m is

(a) 2 (b) 3

(c) 5 (d) None of these.

4. A subspace of a vector space V over a field F other than V is called.

(a) zero subspace of V (b) trivial subspace of V

(c) proper subspace of V (d) none of these **(Garhwal 2007)**

5. Let V be a vector space over the field of F. The intersection of any collection of subspaces of V is

(a) zero subspace of V (b) trivial subspace of V

(c) a subspace of V (d) none of these **(Garhwal 2008)**

6. If $\alpha_1 = (1, 1, 1), \alpha_2 = (1, 2, 3)$ and $\alpha_3 = (2, -1, 1)$ be three vectors in R^3, the vector $(1, -2, 5) \in R^3$ as a linear combination of $\alpha_1, \alpha_2, \alpha_3$ will be

(a) $-6\alpha_1 + 3\alpha_2 + 2\alpha_3$ (b) $6\alpha_1 - 3\alpha_2 + 2\alpha_3$

(c) $6\alpha_1 + 3\alpha_2 - 2\alpha_3$ (d) $6\alpha_1 + 3\alpha_2 + 2\alpha_3$

7. If W_1 and W_2 be two subspaces of the vector space V over a field F, then which one will always be the subspace of V

(a) $W_1 \cup W_2$ (b) $W_1 \cap W_2$

(c) $W_1 - W_2$ (d) none of these

Fill in the Blank(s)

Fill in the blanks "......" so that the following statements are complete and correct.

1. The necessary and sufficient condition for a non-empty subset W of a vector space $V\,(F)$ to be a subspace of V is

$$a, b \in F \text{ and } \alpha, \beta \in W \Rightarrow \ldots\ldots \in W.$$

2. The subset containing a single element $(1, 0, 0)$ of the vector space $V_3\,(F)$ generates the subspace which is the totality of the elements of the form ....

3. If W_1 and W_2 be two subspaces of the vector space $V\,(F)$, then $L\,(W_1 \cup W_2) = \ldots.$ **(Garhwal 2006)**

True or False

Write 'T' for true and 'F' for false statement.

1. A subspace of $V_3\,(\mathbf{R})$, where $\mathbf{R}$ is the real field, must always contain the origin.
2. The set of vectors $\alpha = (x, y) \in V_2\,(\mathbf{R})$ for which $x^2 = y^2$ is a subspace of $V_2\,(\mathbf{R})$.
3. The set of ordered triads (x, y, z) of real numbers with $x > 0$ is a subspace of $V_3\,(\mathbf{R})$.

of V_3 (**R**).

5. The union of any two subspaces W_1 and W_2 of a vector space $V(F)$ is always a subspace of $V(F)$.
6. The intersection of two subspaces of a vector space V is also a subspace of V.
7. If A and B are subsets of a vector space, then $A \neq B \Rightarrow L(A) \neq L(B)$.

Answers

Multiple Choice Questions

1. (c)	2. (a)	3. (c)	4. (c)	5. (c)
6. (a)	7. (b)			

Fill in the Blank(s)

1. $a\alpha + b\beta$	2. $(a, 0, 0)$	3. $W_1 + W_2$

True or False

1. T	2. F	3. F	4. T
5. F	6. T	7. F	

□

Chapter 2

Linear Dependence of Vectors

2.1 Linear Dependence and Linear Independence of Vectors

Linear dependence. Definition. *Let $V(F)$ be a vector space. A finite set $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ of vectors of V is said to be linearly dependent if there exist scalars $a_1, a_2, \ldots, a_n \in F$ not all of them 0 (some of them may be zero) such that*

$$a_1 \alpha_1 + a_2 \alpha_2 + a_3 \alpha_3 + \ldots + a_n \alpha_n = \mathbf{0}.$$

Linear independence. Definition. *Let $V(F)$ be a vector space. A finite set $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ of vectors of V is said to be linearly independent if every relation of the form*

$$a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n = \mathbf{0}, \; a_i \in F, 1 \le i \le n$$

$$\Rightarrow \quad a_i = 0 \text{ for each } 1 \le i \le n.$$

Any **infinite set** *of vectors of V is said to be linearly independent if its every finite subset is linearly independent, otherwise it is linearly dependent.*

Note: The empty set $\varnothing$ is defined as a linearly independent set.

2.2 Some Theorems on Linear Dependence and Linear Independence

Theorem 1: *Let $V(F)$ be a vector space. If $\alpha_1, \alpha_2, \ldots, \alpha_n$ are non-zero vectors $\in V$ then either they are linearly independent or some α_k, $2 \le k \le n$, is a linear combination of the preceding ones $\alpha_1, \alpha_2, \ldots, \alpha_{k-1}$.* **(Garhwal 2004)**

Proof: If $\alpha_1, \alpha_2, \ldots, \alpha_n$ are linearly independent we are nothing to prove. So let $\alpha_1, \alpha_2, \ldots, \alpha_n$ are linearly dependent. Then there exists a relation of the form

$$a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n = \mathbf{0}, \qquad \ldots(1)$$

where, not all the scalar coefficients $a_1, a_2, \ldots, a_n$ are 0. Let k be the largest integer for which $a_k \neq 0$, *i.e.*, $a_{k+1} = 0, a_{k+2} = 0, \ldots, a_n = 0$ and $a_k \neq 0$. There is no harm in this assumption because at the most if $a_n \neq 0$ then $k = n$.
Also $2 \le k$. Because if $a_2 = 0, a_3 = 0, \ldots, a_n = 0$ then $a_1\alpha_1 = \mathbf{0}$ and $\alpha_1 \neq \mathbf{0} \Rightarrow a_1 = 0$. This contradicts the fact that not all the a's are 0.
Now the relation (1) reduces to

$$a_1\alpha_1 + a_2\alpha_2 + \ldots + a_k\alpha_k = \mathbf{0}, \text{ where } a_k \neq 0$$

or $$a_k\alpha_k = -a_1\alpha_1 - a_2\alpha_2 - \ldots - a_{k-1}\alpha_{k-1}$$

or $$a_k^{-1}(a_k\alpha_k) = a_k^{-1}(-a_1\alpha_1 - a_2\alpha_2 - \ldots - a_{k-1}\alpha_{k-1})$$

or $$\alpha_k = (-a_k^{-1}a_1)\alpha_1 + (-a_k^{-1}a_2)\alpha_2 + \ldots + (-a_k^{-1}a_{k-1})\alpha_{k-1}.$$

Thus α_k is a linear combination of its preceding vectors.

Theorem 2: *The set of non-zero vectors $\alpha_1, \alpha_2, \ldots, \alpha_n$ of $V(F)$ is linearly dependent if some α_k, $2 \le k \le n$, is a linear combination of the preceding ones.*

Proof: If some α_k, $2 \le k \le n$, is a linear combination of the preceding ones $\alpha_1, \alpha_2, \ldots, \alpha_{k-1}$ then $\exists$ scalars $a_1, a_2, \ldots, a_{k-1}$ such that

$$\alpha_k = a_1\alpha_1 + \ldots + a_{k-1}\alpha_{k-1}$$

$\Rightarrow$ $$1\alpha_k - a_1\alpha_1 - a_2\alpha_2 - \ldots - a_{k-1}\alpha_{k-1} = \mathbf{0}$$

$\Rightarrow$ the set $\{\alpha_1, \alpha_2, \ldots, \alpha_k\}$ is linearly dependent.
Hence the set $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ of which $\{\alpha_1, \alpha_2, \ldots, \alpha_k\}$ is a subset, must be linearly dependent.

Theorem 3: *If in a vector space $V(F)$, a vector β is a linear combination of the set of vectors $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n$, then the set of vectors $\beta, \alpha_1, \alpha_2, \ldots, \alpha_n$ is linearly dependent.*

Proof: Since β is a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_n$, therefore there exist scalars $a_1, a_2, \ldots, a_n$ such that

$$\beta = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$$

$\Rightarrow$ $$1\beta - a_1\alpha_1 - a_2\alpha_2 - \ldots - a_n\alpha_n = \mathbf{0}. \qquad \ldots(1)$$

In the relation (1) the scalar coefficient of β is 1 which is $\neq 0$. Hence in the relation (1) not all the scalar coefficients are 0. Therefore the set $\beta, \alpha_1, \ldots, \alpha_n$ is linearly dependent.

Theorem 4: *Let S be a linearly independent subset of a vector space V. Suppose β is a*

S is linearly independent.

Proof: Suppose $\alpha_1, \alpha_2, \ldots, \alpha_m$ are distinct vectors in S. Let

$$c_1 \alpha_1 + c_2 \alpha_2 + \ldots + c_m \alpha_m + b\beta = \mathbf{0}. \qquad \ldots(1)$$

Then b must be zero, for otherwise

$$\beta = \left(-\frac{c_1}{b}\right)\alpha_1 + \ldots + \left(-\frac{c_m}{b}\right)\alpha_m$$

and consequently β is in the subspace spanned by S which is a contradiction.

Putting $b = 0$ in (1), we get

$$c_1 \alpha_1 + c_2 \alpha_2 + \ldots + c_m \alpha_m = \mathbf{0} \Rightarrow c_1 = 0, c_2 = 0, \ldots, c_m = 0$$ because

the set $\{\alpha_1, \alpha_2, \ldots, \alpha_m\}$ is linearly independent since it is a subset of a linearly independent set S.

Thus the relation (1) implies $c_1 = 0, c_2 = 0, \ldots, c_m = 0, b = 0$.

Therefore the set $\{\alpha_1, \alpha_2, \ldots, \alpha_m, \beta\}$ is linearly independent. If S' is the set obtained by adjoining β to S, then we have proved that every finite subset of S' is linearly independent. Hence S' is linearly independent.

Theorem 5: *The set of non-zero vectors $\alpha_1, \alpha_2, \ldots, \alpha_n$ of $V(F)$ is linearly dependent iff one of these vectors is a linear combination of the remaining $(n-1)$ vectors.*

Proof: This theorem can be easily proved.

Illustrative Examples

Example 1: *Prove that if two vectors are linearly dependent, one of them is a scalar multiple of the other.*

Solution: Let α, β be two linearly dependent vectors of the vector space V. Then $\exists$ scalars a, b not both zero, such that

$$a\alpha + b\beta = \mathbf{0}.$$

If $a \neq 0$, then we get $a\alpha = -b\beta$

$$\Rightarrow \quad \alpha = \left(-\frac{b}{a}\right)\beta \Rightarrow \alpha \text{ is a scalar multiple of } \beta.$$

If $b \neq 0$, then we get $b\beta = -a\alpha$

$$\Rightarrow \quad \beta = \left(-\frac{a}{b}\right)\alpha \Rightarrow \beta \text{ is a scalar multiple of } \alpha.$$

Thus one of the vectors α and β is a scalar multiple of the other.

Example 2: *In the vector space $V_n(F)$, the system of n vectors*

$$e_1 = (1, 0, 0, \ldots, 0), e_2 = (0, 1, 0, \ldots, 0, 0), \ldots, e_n = (0, 0, \ldots, 0, 1)$$

is linearly independent where 1 denotes the unity of the field F.

Solution: If $a_1, a_2, a_3, \ldots, a_n$ be any scalars, then

$$a_1 e_1 + a_2 e_2 + \ldots + a_n e_n = \mathbf{0}$$

$$\Rightarrow \quad a_1 (1, 0, 0, \ldots, 0) + a_2 (0, 1, 0, \ldots, 0) + \ldots + a_n (0, 0, \ldots, 0, 1) = \mathbf{0}$$

Therefore the given set of n vectors is linearly independent.

In particular $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ *is a linearly independent subset of* $V_3\ (F)$.

Example 3: *If the set* $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ *of vectors of* $V\ (F)$ *is linearly independent, then none of the vectors* $\alpha_1, \alpha_2, \ldots, \alpha_n$ *can be zero vector.*

Solution: Let α_r be equal to zero vector where $1 \le r \le n$.

Then $0\ \alpha_1 + 0\ \alpha_2 + \ldots + a\alpha_r + 0\ \alpha_{r+1} + \ldots + 0\ \alpha_n = \mathbf{0}$ for any $a \ne 0$ in F.
Since $a \ne 0$, therefore from this relation we conclude that S is linearly dependent. Thus we get a contradiction because it is given that S is linearly independent. Hence none of the vectors $\alpha_1, \alpha_2, \ldots, \alpha_n$ can be zero vector. We also conclude that *a set of vectors which contains the zero vector is necessarily linearly dependent.*

Example 4: *Every superset of a linearly dependent set of vectors is linearly dependent.*

Solution: Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a linearly dependent set of vectors. Then there exist scalars $a_1, a_2, \ldots, a_n$ not all zero such that

$$a_1\ \alpha_1 + a_2\ \alpha_2 + \ldots + a_n\ \alpha_n = \mathbf{0}. \qquad \ldots(1)$$

Now let $S' = \{\alpha_1, \alpha_2, \ldots, \alpha_n, \beta_1, \beta_2, \ldots, \beta_m\}$ be a superset of S.
Then we have from (1)

$$a_1\ \alpha_1 + a_2\ \alpha_2 + \ldots + a_n\ \alpha_n + 0\ \beta_1 + 0\ \beta_2 + \ldots + 0\ \beta_m = \mathbf{0}. \qquad \ldots(2)$$

Since in the relation (2) the scalar coefficients are not all 0, therefore S' is linearly dependent.

From this we also conclude that *any subset of a linearly independent set of vectors is also linearly independent.*

Example 5: *A system consisting of a single non-zero vector is always linearly independent.*

Solution: Let $S = \{\alpha\}$ be a subset of a vector space V and let α be not equal to zero vector. If a is any scalar, then

$$a\ \alpha = \mathbf{0}$$

$\Rightarrow$ $\quad a = 0.$ $\qquad$ [Since α is not-zero vector]

$\therefore$ $\quad$ the set S is linearly independent.

Example 6: *Show that* $S = \{(1, 2, 4), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ *is a linearly dependent subset of the vector space* $V_3\ (\mathbf{R})$ *where* $\mathbf{R}$ *is the field of real numbers.*

Solution: We have

$$1\ (1, 2, 4) + (-1)\ (1, 0, 0) + (-2)\ (0, 1, 0) + (-4)\ (0, 0, 1)$$
$$= (1, 2, 4) + (-1, 0, 0) + (0, -2, 0) + (0, 0, -4) = (0, 0, 0)$$

i.e., the zero vector.

Since in this relation the scalar coefficients $1, -1, -2, -4$ are not all zero, therefore the given system S is linearly dependent.

Example 7: *Show that the system of three vectors* $(1, 3, 2), (1, -7, -8), (2, 1, -1)$ *of* $V_3\ (\mathbf{R})$ *is linearly dependent.* **(Garhwal 2010)**

Solution: Let a, b, c be scalars *i.e.*, real numbers such that

$$a\ (1, 3, 2) + b\ (1, -7, -8) + c\ (2, 1, -1) = (0, 0, 0)$$

i.e., $\quad (a + b + 2c,\ 3a - 7b + c,\ 2a - 8b - c) = (0, 0, 0).$

$$3a - 7b + c = 0, \qquad \ldots(2)$$

and $$2a - 8b - c = 0. \qquad \ldots(3)$$

If the equations (1), (2), (3) possess a non-zero solution, then the given vectors are linearly dependent.

Adding (2) and (3), we get

$$5a - 15b = 0 \quad \text{or} \quad a - 3b = 0. \qquad \ldots(4)$$

Multiplying (3) by 2 and adding to (1), we get

$$5a - 15b = 0 \quad \text{or} \quad a - 3b = 0. \qquad \ldots(5)$$

The equations (4) and (5) are the same and give $a = 3b$.

Putting $a = 3b$ in (1), we get $2c + 4b = 0$ or $c = -2b$.

If we take $b = 1$, we get $a = 3, c = -2$.

Thus $a = 3, b = 1, c = -2$ is a non-zero solution of the equations (1), (2) and (3). Hence the given set of vectors is linearly dependent.

Alternative method. The coefficient matrix A of the system of equations (1), (2) and (3) is

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & -7 & 1 \\ 2 & -8 & -1 \end{bmatrix}.$$

We have $$\det A = \begin{vmatrix} 1 & 1 & 2 \\ 3 & -7 & 1 \\ 2 & -8 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 3 & -10 & -5 \\ 2 & -10 & -5 \end{vmatrix} \text{ by } C_2 - C_1 \text{ and } C_3 - 2C_1$$

$$= 50 - 50 = 0.$$

$\therefore$ rank $A < 3$ *i.e.*, rank $A <$ the number of unknowns a, b, c in the equations (1), (2) and (3).

Therefore the equations (1), (2) and (3) must possess a non-zero solution. Hence the given vectors are linearly dependent.

Example 8: *In V_3 (**R**), where **R** is the field of real numbers, examine each of the following sets of vectors for linear dependence :*

(i) $\{(2, 1, 2), (8, 4, 8)\}$

(ii) $\{(1, 2, 0), (0, 3, 1), (-1, 0, 1)\}$

(iii) $\{(-1, 2, 1), (3, 0, -1), (-5, 4, 3)\}$

(iv) $\{(2, 3, 5), (4, 9, 25)\}$

(v) $\{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$.

Solution: (i) We have $4\,(2, 1, 2) + (-1)\,(8, 4, 8) = (8, 4, 8) + (-8, -4, -8)$
$= (0, 0, 0)$ *i.e.*, the zero vector.

Since in this relation the scalar coefficients $4, -1$ are not both zero, therefore the given set is linearly dependent.

(ii) Let a, b, c be scalars *i.e.*, real numbers such that

$$a\,(1, 2, 0) + b\,(0, 3, 1) + c\,(-1, 0, 1) = (0, 0, 0)$$

i.e., $$(a - c, 2a + 3b, b + c) = (0, 0, 0)$$

i.e., $$a + 0b - c = 0, 2a + 3b + 0c = 0, 0a + b + c = 0.$$

not all zero if the rank of the coefficient matrix is less than three *i.e.*, the number of unknowns a, b, c. If the rank is 3, then the zero solution $a = 0, b = 0, c = 0$ will be the only solution.

Coefficient matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

We have $|A| = 1(3 - 0) - 2(0 + 1) = 1 \neq 0$.

$\therefore$ Rank $A = 3$. Hence $a = 0, b = 0, c = 0$ is the only solution. Therefore the given system is linearly independent.

(iii) Let a, b, c be scalars such that

$$a(-1, 2, 1) + b(3, 0, -1) + c(-5, 4, 3) = (0, 0, 0)$$

i.e., $$(-a + 3b - 5c, 2a + 0b + 4c, a - b + 3c) = (0, 0, 0)$$

i.e., $$-a + 3b - 5c = 0, 2a + 0b + 4c = 0, a - b + 3c = 0.$$

The coefficient matrix of these equations is

$$A = \begin{bmatrix} -1 & 3 & -5 \\ 2 & 0 & 4 \\ 1 & -1 & 3 \end{bmatrix}.$$

We have $|A| = -1(0 + 4) - 2(9 - 5) + 1(12 - 0) = 0$.

$\therefore$ Rank A is < 3 *i.e.*, the number of unknowns a, b, c. Therefore the given system of equations will possess a non-zero solution. For example $a = -2, b = 1, c = 1$ is a non-zero solution. Hence the given system of vectors is linearly dependent.

(iv) Let a, b be scalars *i.e.*, real numbers such that

$$a(2, 3, 5) + b(4, 9, 25) = (0, 0, 0)$$

i.e., $$(2a + 4b, 3a + 9b, 5a + 25b) = (0, 0, 0)$$

i.e., $$2a + 4b = 0, 3a + 9b = 0, 5a + 25b = 0.$$

The coefficient matrix of these equations is $A = \begin{bmatrix} 2 & 4 \\ 3 & 9 \\ 5 & 25 \end{bmatrix}$.

Obviously rank $A = 2$ *i.e.*, equal to the number of unknowns a and b.

Therefore these equations have the only solution $a = 0, b = 0$.

Hence the given set of vectors is linearly independent.

(v) Let a, b, c be scalars *i.e.*, real numbers such that

$$a(1, 2, 1) + b(3, 1, 5) + c(3, -4, 7) = (0, 0, 0)$$

i.e., $$(a + 3b + 3c, 2a + b - 4c, a + 5b + 7c) = (0, 0, 0)$$

i.e., $$a + 3b + 3c = 0, \qquad \text{...(1)}$$

$$2a + b - 4c = 0, \qquad \text{...(2)}$$

$$a + 5b + 7c = 0. \qquad \text{...(3)}$$

Multiplying (1) by 2, we get

$$2a + 6b + 6c = 0. \qquad \text{...(4)}$$

Subtracting (4) from (2), we get

or $\quad b + 2c = 0.$ $\quad \ldots(5)$

Again subtracting (3) from (1), we get

$$-2b - 4c = 0 \quad \text{or} \quad b + 2c = 0. \qquad \ldots(6)$$

The equations (5) and (6) are the same and give $b = -2c$.

Putting $b = -2c$ in (1), we get $a = 3c$. If we take $c = 1$, we get $b = -2$ and $a = 3$. Thus $a = 3, b = -2, c = 1$ is a non-zero solution of the equations (1), (2) and (3). Hence the given set of vectors is linearly dependent.

Example 9: *If F is the field of complex numbers, prove that the vectors* (a_1, a_2) *and* (b_1, b_2) *in* $V_2(F)$ *are linearly dependent iff* $a_1 b_2 - a_2 b_1 = 0$.

Solution: Let $x, y \in F$. Then

$$x(a_1, a_2) + y(b_1, b_2) = (0, 0) \Rightarrow (xa_1 + yb_1, xa_2 + yb_2) = (0, 0).$$

Therefore, and

$$\left.\begin{aligned} a_1 x + b_1 y &= 0 \\ a_2 x + b_2 y &= 0 \end{aligned}\right\}.$$

The necessary and sufficient condition for these equations to possess a non-zero solution is that

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0 \text{ i.e., } a_1 b_2 - a_2 b_1 = 0.$$

Hence the given system is linearly dependent iff $a_1 b_2 - a_2 b_1 = 0$.

Example 10: *If* α_1 *and* α_2 *are vectors of* $V(F)$, *and* $a, b \in F$, *show that the set* $\{\alpha_1, \alpha_2, a\alpha_1 + b\alpha_2\}$ *is linearly dependent.*

Solution: We have $(-a)\alpha_1 + (-b)\alpha_2 + 1(a\alpha_1 + b\alpha_2)$

$$= (-a + a)\alpha_1 + (-b + b)\alpha_2$$

$$= 0\alpha_1 + 0\alpha_2 = \mathbf{0} \text{ i.e., zero vector.}$$

In the above linear combination the scalar coefficient $1 \neq 0$. Therefore whatever may be the scalars $-a$ and $-b$, the given set of vectors is linearly dependent.

Example 11: *If* α, β, γ *are linearly independent vectors of* $V(F)$ *where F is any subfield of the field of complex numbers, then so also are* $\alpha + \beta, \beta + \gamma, \gamma + \alpha$.

Solution: Let a, b, c be scalars such that

$$a(\alpha + \beta) + b(\beta + \gamma) + c(\gamma + \alpha) = \mathbf{0}$$

i.e.,

$$(a + c)\alpha + (a + b)\beta + (b + c)\gamma = \mathbf{0}. \qquad \ldots(1)$$

But α, β, γ are linearly independent. Therefore (1) implies

$$a + 0b + c = 0, a + b + 0c = 0, 0a + b + c = 0.$$

The coefficient matrix A of these equations is

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

We have rank $A = 3$ *i.e.,* the number of unknowns a, b, c. Therefore $a = 0, b = 0, c = 0$ is the only solution of the given equations.

Hence $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ are also linearly independent.

vector space of all polynomials over the real number field.

Solution: Let a, b, c be scalars (real numbers) such that

$$a\,(1) + bx + c\,(1 + x + x^2) = 0.$$

We have $\quad a\,(1) + bx + c\,(1 + x + x^2) = 0$

$\Rightarrow \quad (a + c) + (b + c)\,x + cx^2 = 0$

$\Rightarrow \quad a + c = 0,\ b + c = 0,\ c = 0$

$\Rightarrow \quad c = 0, b = 0, a = 0.$

$\therefore$ the vectors $1, x, 1 + x + x^2$ are linearly independent over the field of real numbers.

Example 13: *In the vector space $F\,[x]$ of all polynomials over the field F the infinite set $S = \{1, x, x^2, x^3, \ldots\}$ is linearly independent.*

Solution: Let $S' = \{x^{m_1}, x^{m_2}, \ldots, x^{m_n}\}$ be any finite subset of S having n vectors. Here $m_1, m_2, \ldots, m_n$ are some non-negative integers. Let $a_1, a_2, \ldots, a_n$ be scalars such that

$$a_1 x^{m_1} + a_2 x^{m_2} + \ldots + a_n x^{m_n} = \mathbf{0}. \quad (\textit{i.e.}, \text{ zero polynomial}) \quad \ldots(1)$$

By the definition of equality of two polynomials we have from (1)

$$a_1 = 0, a_2 = 0, \ldots, a_n = 0.$$

Thus every finite subset of S is linearly independent.

Therefore S is linearly independent.

Example 14: *Show that the three vectors $(1, 1, -1), (2, -3, 5)$ and $(-2, 1, 4)$ of $\mathbf{R}^3$ are linearly independent.*

Solution: Let a, b, c be scalars *i.e.*, real numbers such that

$$a\,(1, 1, -1) + b\,(2, -3, 5) + c\,(-2, 1, 4) = (0, 0, 0)$$

i.e., $\quad (a + 2b - 2c, a - 3b + c, -a + 5b + 4c) = (0, 0, 0)$

i.e., $\quad a + 2b - 2c = 0 \quad \ldots(1)$

$a - 3b + c = 0 \quad \ldots(2)$

$-a + 5b + 4c = 0 \quad \ldots(3)$

Now we shall solve the simultaneous equations (1), (2) and (3).

Multiplying (2) by 2 and adding to (1), we get

$$3a - 4b = 0. \quad \ldots(4)$$

Again multiplying (1) by 2 and adding to (3), we get

$$a + 9b = 0. \quad \ldots(5)$$

Multiplying (5) by 3 and subtracting from (4), we get

$$-31b = 0 \quad \text{or} \quad b = 0.$$

Putting $b = 0$ in (5), we get $a = 0$.

Now putting $a = 0, b = 0$ in (1), we get $c = 0$.

Thus $a = 0, b = 0, c = 0$ is the only solution of the equations (1), (2) and (3).

$\therefore \quad a\,(1, 1, -1) + b\,(2, -3, 5) + c\,(-2, 1, 4) = (0, 0, 0)$

$\Rightarrow \quad a = 0, b = 0, c = 0.$

Example 15: *Find a maximal linearly independent subsystem of the system of vectors*

$$\alpha_1 = (2, -2, -4), \alpha_2 = (1, 9, 3), \alpha_3 = (-2, -4, 1)$$

and $\alpha_4 = (3, 7, -1)$.

Solution: Let A denote the matrix $\begin{bmatrix} 2 & -2 & -4 \\ 1 & 9 & 3 \\ -2 & -4 & 1 \\ 3 & 7 & -1 \end{bmatrix}$ whose rows consist of the vectors $\alpha_1, \alpha_2, \alpha_3$ and α_4.

We shall reduce the matrix A to Echelon form by applying the row transformations. We have

$$A \sim \begin{bmatrix} 1 & -1 & -2 \\ 1 & 9 & 3 \\ -2 & -4 & 1 \\ 3 & 7 & -1 \end{bmatrix}, \text{ applying } R_1 \to \frac{1}{2} R_1$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & 10 & 5 \\ 0 & -6 & -3 \\ 0 & 10 & 5 \end{bmatrix}, \quad \begin{aligned} \text{by } R_2 &\to R_2 - R_1 \\ R_3 &\to R_3 + 2R_1 \\ R_4 &\to R_4 - 3R_1 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \end{bmatrix}, \quad \begin{aligned} \text{by } R_2 &\to \frac{1}{10} R_2 \\ R_3 &\to -\frac{1}{6} R_3 \\ \mathrm{R}_4 &\to \frac{1}{10} R_4 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{aligned} \text{by } R_3 &\to R_3 - R_2 \\ R_4 &\to R_4 - R_2 \end{aligned}$$

which is in Echelon form.

We have rank A = the number of non-zero rows in its Echelon form = 2.

$\therefore$ the maximum number of linearly independent row vectors in the matrix A

= the rank $A = 2$.

The vectors α_1 and α_2 are linearly independent and so $\{\alpha_1, \alpha_2\}$ is a maximal linearly independent subsystem of the given system of vectors. We observe that none of the given four vectors is a scalar multiple of any of the remaining three vectors. So any two of the given four vectors form a maximal linearly independent subsystem of the given system of vectors.

Comprehensive Exercise 1

1. Show that the vectors $(1, 1, 2, 4), (2, -1, -5, 2), (1, -1, -4, 0)$ and $(2, 1, 1, 6)$ are linearly dependent in $\mathbf{R}^4$.
2. Show that the vectors $(1, 1, 0, 0), (0, 1, -1, 0), (0, 0, 0, 3)$ in $\mathbf{R}^4$ are linearly independent.
3. Determine whether the following set of vectors in $V_3(\mathbf{Q})$ is linearly dependent or independent, $\mathbf{Q}$ being the field of rational numbers :

   $\{(-1, 2, 1), (3, 1, -2)\}$.
4. Find a linearly independent subset T of the set $S = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ where $\alpha_1 = (1, 2, -1), \alpha_2 = (-3, -6, 3), \alpha_3 = (2, 1, 3), \alpha_4 = (8, 7, 7) \in \mathbf{R}^3$ which spans the same space as S.
5. If the vectors $(0, 1, x), (x, 1, 0), (1, x, 1)$ of the vector space $\mathbf{R}^3$ are linearly dependent, then find the value of x.
6. Show that the vectors $(0, 2, -4)$, $(1, -2, -1), (1, -4, 3)$ in $\mathbf{R}^3$ are linearly dependent. Also express $(0, 2, -4)$ as a linear combination of $(1, -2, -1)$ and $(1, -4, 3)$.
7. Determine if the vectors $(1, -2, 1), (2, 1, -1), (7, -4, 1)$ in $\mathbf{R}^3$ are linearly independent or linearly dependent.
8. Let $\alpha_1, \alpha_2, \alpha_3$ be the vectors of $V(F)$, $a, b \in F$. Show that the set $\{\alpha_1, \alpha_2, \alpha_3\}$ is linearly dependent if the set $\{\alpha_1 + a\alpha_2 + b\alpha_3, \alpha_2, \alpha_3\}$ is linearly dependent.
9. If α, β, γ are linearly independent vectors of $V(F)$ where F is the field of complex numbers, then so also are $\alpha + \beta, \alpha - \beta, \alpha - 2\beta + \gamma$.
10. Show that the set $\{1, x, x(1 - x)\}$ is a linearly independent set of vectors in the space of all polynomials over the real number field.
11. Find whether the vectors

    $$2x^3 + x^2 + x + 1, x^3 + 3x^2 + x - 2 \text{ and } x^3 + 2x^2 - x + 3 \text{ of } \mathbf{R}[x],$$

    the vector space of all polynomials over the real number field, are linearly independent or not.
12. Prove that a set of vectors which contains the zero vector is linearly dependent.
13. Let V be the vector space of 2×2 matrices over $\mathbf{R}$. Determine whether the matrices $A, B, C \in V$ are linearly dependent, where

    $$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 3 & -1 \\ 2 & 2 \end{bmatrix}, C = \begin{bmatrix} 1 & -5 \\ -4 & 0 \end{bmatrix}.$$
14. Prove that the three non-zero non-coplanar vectors are linearly independent.
15. Show that any three non-zero coplanar vectors are linearly dependent in $\mathbf{R}^3$.
16. Show that the set of functions $\{x, |x|\}$ is L.I. in vector space of continuous functions defined in $(-1, 1)$.

(**R**) are L.I.

18. If $V(F)$ is a vector space and $\alpha_1, \alpha_2, \ldots, \alpha_n \in V$, $\lambda_2, \lambda_3, \ldots, \lambda_n \in F$ such that $\{\alpha_2 + \lambda_2 \alpha_1, \alpha_3 + \lambda_3 \alpha_1, \ldots, \alpha_n + \lambda_n \alpha_1\}$ is L.D. then show that $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is L.D.

19. Let V be the vector space of functions from **R** into **R**. Show that $f, g, h \in V$ are linearly independent where

(i) $f(x) = e^{2x}$, $g(x) = x^2$, $h(x) = x$

(ii) $f(x) = \sin x$, $g(x) = \cos x$, $h(x) = x$.

20. Test the vectors $(0, 1, 0, 1, 1), (1, 0, 1, 0, 1), (0, 1, 0, 1, 1), (1, 1, 1, 1, 1)$ in V_5 over the field of rational numbers for linear independence.

21. Prove that the four vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$ in V_3 **(C)** form a linearly dependent set but any three of them are linearly independent.

Answers 1

3. linearly independent **4.** $\{\alpha_1, \alpha_3\}$ **5.** $0, \pm\sqrt{2}$

6. $(0, 2, -4) = (1, -2, -1) - (1, -4, 3)$ **7.** linearly dependent

11. linearly independent **13.** Yes **20.** linearly dependent

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. The singleton set $\{x\}$ is linearly independent if and only if

(a) $x = 0$ (b) $x \neq 0$

(c) may or may not be zero (d) none of these

(Garhwal 2006, 10)

2. The set of vectors $\{(1, 2, 0), (0, 3, 1), (-1, 0, 1)\}$ are

(a) linearly independent (b) linearly dependent

(c) neither linearly independent nor dependent

(d) none of these

3. The vectors (a_1, a_2) and (b_1, b_2) are linearly dependent if

(a) $a_1 b_1 + a_2 b_2 = 0$ (b) $a_1 b_2 + a_2 b_1 = 0$

(c) $a_1 b_2 - a_2 b_1 = 0$ (d) $a_1 b_1 - a_2 b_2 = 0$

4. Every superset of a linearly dependent set of vectors is

(a) linearly dependent (b) linearly independent

(c) neither linearly dependent nor independent

(d) none of these

(a) linearly independent
(b) linearly dependent
(c) neither dependent nor independent
(d) none of these

Fill in the Blank(s)

Fill in the blanks "……" so that the following statements are complete and correct.

1. Any set of vectors containing the zero vector as a member is linearly …… .
2. Intersection of two linearly independent subsets of a vector space will be linearly …… .
3. The subset {(1, 0, 0), (0, 1, 0), (0, 0, 1)} of the vector space $\mathbf{R}^3$ is linearly ……
4. A system consisting of a single non-zero vector is always linearly …… .
5. In the vector space $\mathbf{R}^3$ the vectors (1, 0, 1), (3, 7, 0) and (– 1, 0, – 1) are linearly …… .

True or False

Write 'T' for true and 'F' for false statement.

1. A set containing a linearly independent set of vectors is itself linearly independent.
2. Union of two linearly independent subsets of a vector space is linearly independent.
3. The vectors (2, – 3, 7), (0, 0, 0), (3, – 1, – 4) are linearly independent in $\mathbf{R}^3$.
4. The vectors (1, – 3, 7), (2, 0, – 6), (3, – 1, – 1), (2, 4, – 5) are linearly dependent in $\mathbf{R}^3$.

Answers

Multiple Choice Questions

1. (b) 2. (a) 3. (c) 4. (a) 5. (a)

Fill in the Blank(s)

1. dependent 2. independent 3. independent
4. independent 5. dependent

True or False

1. F 2. *F* 3. *F* 4. *T*

□

Chapter

3

Basis and Dimensions

3.1 Basis of A Vector Space

Definition: *A subset S of a vector space V (F) is said to be a basis of V (F), if*

(i) S consists of linearly independent vectors,

(ii) S generates V (F) i.e., L (S) = V i.e., each vector in V is a linear combination of a finite number of elements of S. **(Garhwal 2010)**

Illustrative Examples

Example 1: *A system S consisting of n vectors,*

$$e_1 = (1, 0, 0, \ldots, 0),\ e_2 = (0, 1, 0, \ldots, 0), \ldots,\ e_n = (0, 0, \ldots, 0, 1)$$

is a basis of V_n (F).

Solution: First we should show that S is a linearly independent set of vectors. We have proved it in one of the previous examples.

Now we should prove that $L(S) = V_n(F)$. We have always $L(S) \subseteq V_n(F)$. So we should prove that $V_n(F) \subseteq L(S)$ *i.e.*, each vector in $V_n(F)$ is a linear combination of elements of S.

We can write

$$(a_1, a_2, \ldots, a_n) = a_1\ (1, 0, \ldots, 0) + a_2\ (0, 1, \ldots, 0) + \ldots\ + a_n\ (0, 0, \ldots, 0, 1)$$

i.e., $\alpha = a_1\ e_1\ + a_2\ e_2\ + \ldots + a_n\ e_n$.

Hence S is a basis of $V_n\ (F)$. We shall call this particular basis the **standard basis** of $V_n\ (F)$.

Note: The set $\{(1, 0), (0, 1)\}$ is a basis of $V_2\ (F)$. The set $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a basis of $V_3\ (F)$. As a particular case a basis of $F\ (F)$ is the set consisting of only the unity element of F.

Example 2: *Show that the infinite set $S = \{1, x, x^2, \ldots, x^n, \ldots\}$ is a basis of the vector space $F\ [x]$ of polynomials over the field F.*

Solution: First we should prove that S is a linearly independent set of vectors. For proof refer some previous example.

Now we should show that S spans $F\ [x]$ *i.e.*, each polynomial in $F\ [x]$ can be expressed as a linear combination of a finite number of elements of S.

Let $f\ (x) = a_0\ + a_1\ x + a_2\ x^2\ + \ldots + a_t\ x^t$ be a polynomial of degree t.

Then $f\ (x) = (a_0)\ 1 + a_1\ x + a_2\ x^2\ + \ldots + a_t\ x^t$. Hence S is a basis of $F\ [x]$.

Note: The vector space $F\ [x]$ has no finite basis. If we take any finite set W of polynomials, we can find a polynomial of degree greater than that of each of them. Such a polynomial cannot at any cost be expressed as a linear combination of the elements of W.

3.2 Finite Dimensional Vector Space

(Garhwal 2003, 07)

Definition: *The vector space $V\ (F)$ is said to be* **finite dimensional** *or* **finitely generated** *if there exists a* **finite** *subset S of V such that $V = L\ (S)$.*

The vector space $V_n\ (F)$ of n-tuples is a finite dimensional vector space.

The vector space $F\ [x]$ of all polynomials over a field F is not finite dimensional. There exists no finite subset S of $F\ [x]$ which spans $F\ [x]$. *A vector space which is not finitely generated may be referred to as an infinite dimensional space.* Thus the vector space $F\ [x]$ of all polynomials over a field F is infinite dimensional.

Note: The dimension of the zero space $\{\mathbf{0}\}$ is defined to be zero. It is generated by $\varnothing$.

Existence of basis of a finite dimensional vector space.

Theorem: *There exists a basis for each finite dimensional vector space.*

(Garhwal 2006, 10, 12)

Proof: Let $V\ (F)$ be a finitely generated vector space.

Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_m\}$

be a finite subset of V such that $L\ (S) = V$. We may suppose that no member of S is $\mathbf{0}$.

If S is linearly dependent, then there exists a vector $\alpha_i \in S$ which can be expressed as a linear combination of the preceding vectors $\alpha_1, \alpha_2, \ldots, \alpha_{i-1}$.
If we omit this vector α_i from S, then the remaining set S' of $m-1$ vectors

$$\alpha_1, \alpha_2, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_m$$

also generates V *i.e.*, $V = L(S')$. For if α is any element of V, then $L(S) = V$ implies that α can be written as a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_m$. Let

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_{i-1}\alpha_{i-1} + a_i\alpha_i + a_{i+1}\alpha_{i+1} + \ldots + a_m\alpha_m.$$

But α_i can be expressed as a linear combination of $\alpha_1, \ldots, \alpha_{i-1}$. Let

$$\alpha_i = b_1\alpha_1 + \ldots + b_{i-1}\alpha_{i-1}.$$

Putting this value of α_i in the expression for α, we get

$$\alpha = a_1\alpha_1 + \ldots + a_{i-1}\alpha_{i-1} + a_i(b_1\alpha_1 + \ldots + b_{i-1}\alpha_{i-1}) + a_{i+1}\alpha_{i+1} + \ldots + a_m\alpha_m.$$

Thus α has been expressed as a linear combination of the vectors

$$\alpha_1, \alpha_2, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_m.$$

In this way $\alpha \in V \Rightarrow \alpha$ can be expressed as a linear combination of the vectors belonging to the set S'.
Thus S' generates V *i.e.*, $L(S') = V$.
If S' is linearly independent, then S' will be a basis of V. If S' is linearly dependent, then proceeding as above we shall get a new set of $n-2$ vectors which generates V. Continuing this process, we shall, after finite number of steps, obtain a linearly independent subset of S which generates V and which is therefore a basis of V.
At the most it may happen that we shall be left with a subset of S which contains only one non-zero vector and which spans V. We know that a set containing a single non-zero vector is definitely linearly independent and so it will form a basis of V.

Note: The above theorem may also be stated as below :
If a finite set S of vectors spans a finite dimensional vector space $V(F)$, there exists a subset of S which forms a basis of V.

Invariance of number of elements in the basis of a finite dimensional vector space.

Dimension theorem for vector spaces. *If $V(F)$ is a finite dimensional vector space, then any two bases of V have the same number of elements.* **(Garhwal 2004)**

Proof. Suppose $V(F)$ is a finite dimensional vector space. Then V definitely possesses a basis. Let

$$S_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_m\}$$

and $$S_2 = \{\beta_1, \beta_2, \ldots, \beta_n\}$$

be two bases of V. We shall prove that $m = n$.
Since $V = L(S_1)$ and $\beta_1 \in V$, therefore β_1 can be expressed as a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_m$. Consequently the set $S_3 = \{\beta_1, \alpha_1, \alpha_2, \ldots, \alpha_m\}$ which also obviously generates $V(F)$ is linearly dependent. Therefore there exists a member $\alpha_i \neq \beta_1$ of this set S_3 such that α_i is a linear combination of the preceding vectors

If we omit the vector α_i from S_3 then V is also generated by the remaining set

$$S_4 = \{\beta_1, \alpha_1, \alpha_2, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_m\}.$$

Since $V = L(S_4)$ and $\beta_2 \in V$, therefore β_2 can be expressed as a linear combination of the vectors belonging to S_4. Consequently the set

$$S_5 = \{\beta_2, \beta_1, \alpha_1, \alpha_2, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_m\}$$

is linearly dependent. Therefore there exists a member α_j of this set S_5 such that α_j is a linear combination of the preceding vectors. Obviously α_j will be different from β_1 and β_2 since $\{\beta_1, \beta_2\}$ is a linearly independent set. If we exclude the vector α_j from S_5, then the remaining set will generate $V(F)$.

We may continue to proceed in this manner. Here each step consists in the exclusion of an α and the inclusion of a β in the set S_1.

Obviously the set S_1 of α's cannot be exhausted before the set S_2 of β's otherwise $V(F)$ will be a linear span of a proper subset of S_2 and thus S_2 will become linearly dependent. Therefore we must have

$$m \not< n.$$

Interchanging the roles of S_1 and S_2, we shall get that

$$n \not< m.$$

Hence $\qquad n = m.$

Illustration: For the vector space $V_3(F)$, the sets

$$S_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ and } S_2 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$$

are bases as can easily be seen. Both these bases contain the same number of elements *i.e.*, 3.

Dimension of a finitely generated vector space. Definition. *The number of elements in any basis of a finite dimensional vector space $V(F)$ is called the dimension of the vector space $V(F)$ and will be denoted by dim V.* **(Garhwal 2010)**

The vector space $V_n(F)$ is of dimension n. The vector space $V_3(F)$ is of dimension 3. If a field F is regarded as a vector space over F, then F will be of dimension 1 and the set $S = \{1\}$ consisting of unity element of F alone is a basis of F. In fact every non-zero element of F will form a basis of F.

3.3 Some Properties of Finite Dimensional Vector Spaces

Theorem 1. Extension theorem. *Every linearly independent subset of a finitely generated vector space $V(F)$ forms a part of a basis of V.*

Or

Every linearly independent subset of a finitely generated vector space $V(F)$ is either a basis of V or can be extended to form a basis of V.

Proof: Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_m\}$ be a linearly independent subset of a finite dimensional vector space $V(F)$. If dim $V = n$, then V has a finite basis, say $\{\beta_1, \beta_2, \ldots, \beta_n\}$. Consider the set

$$S_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_n\}.$$

β's therefore the set S_1 is linearly dependent.

Therefore there is some vector of S_1 which is a linear combination of its preceding vectors. This vector cannot be any of the α's since the α's are linearly independent. Therefore this vector must be some β, say β_i. Now omit the vector β_i from S_1 and consider the set

$$S_2 = \{\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_{i-1}, \beta_{i+1}, \ldots, \beta_n\}.$$

Obviously $L(S_2) = V$. If S_2 is linearly independent, then S_2 will be a basis of V and it is the required extended set which is a basis of V. If S_2 is not linearly independent, then repeating the above process a finite number of times, we shall get a linearly independent set containing $\alpha_1, \alpha_2, \ldots, \alpha_m$ and spanning V. This set will be a basis of V and it will contain S. Since each basis of V contains the same number of elements, therefore exactly $n - m$ elements of set of β's will be adjoined to S so as to form a basis of V.

Theorem 2: *Each set of $(n + 1)$ or more vectors of a finite dimensional vector space $V(F)$ of dimension n is linearly dependent.*

Proof: Let $V(F)$ be a finite dimensional vector space of dimension n. Let S be a linearly independent subset of V containing $(n + 1)$ or more vectors. Then S will form a part of basis of V. Thus we shall get a basis of V containing more than n vectors. But every basis of V will contain exactly n vectors. Hence our assumption is wrong. Therefore if S contains $(n + 1)$ or more vectors, then S must be linearly dependent.

Theorem 3: *Let V be a vector space which is spanned by a finite set of vectors $\beta_1, \beta_2, \ldots, \beta_m$. Then any linearly independent set of vectors in V is finite and contains no more than m vectors.*

Proof: Let $S = \{\beta_1, \beta_2, \ldots, \beta_m\}$.

Since $L(S) = V$, therefore V has a finite basis and $\dim V \le m$. Hence every subset S' of V which contains more than m vectors is linearly dependent. This proves the theorem.

Theorem 4: *If $V(F)$ is a finite dimensional vector space of dimension n, then any set of n linearly independent vectors in V forms a basis of V.*

Proof: Let $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a linearly independent subset of a finite dimensional vector space $V(F)$ of dimension n. If S is not a basis of V, then it can be extended to form a basis of V. Thus we shall get a basis of V containing more than n vectors.

But every basis of V must contain exactly n vectors. Therefore our assumption is wrong and S must be a basis of V.

Theorem 5: *If a set S of n vectors of a finite dimensional vector space $V(F)$ of dimension n generates $V(F)$, then S is a basis of V.*

Proof: Let $V(F)$ be a finite dimensional vector space of dimension n. Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a subset of V such that $L(S) = V$. If S is linearly independent, then S will form a basis of V. If S is not linearly independent, then there will exist a proper subset of S which will form a basis of V. Thus we shall get a exactly n elements. Hence S cannot be linearly dependent and so S must be a basis of V.

Note: *If V is a finite dimensional vector space of dimension n, then V cannot be generated by fewer than n vectors.*

Theorem 6: Dimension of a subspace. *Each subspace W of a finite dimensional vector space $V(F)$ of dimension n is a finite dimensional space with dim $m \leq n$.*

Also $V = W$ iff dim V = dim W. **(Garhwal 2008)**

Proof: Let $V(F)$ be a finite dimensional vector space of dim n. Let W be a subspace of V. Any subset of W containing $(n+1)$ or more vectors is also a subset of V and any $(n+1)$ vectors in V are linearly dependent. Therefore any linearly independent set of vectors in W can contain, at the most n vectors. Let

$$S = \{\alpha_1, \alpha_2, \ldots, \alpha_m\}$$

be a linearly independent subset of W with maximum number of elements. We claim that S is a basis of W. The proof is as follows :

(i) S is a linearly independent subset of W.

(ii) $L(S) = W$. Let α be any element of W.

Then the $(m+1)$ vectors $\alpha, \alpha_1, \alpha_2, \ldots, \alpha_m$ belonging to W are linearly dependent because we have supposed that the largest independent subset of W contains m vectors.

Now $\{\alpha_1, \alpha_2, \ldots, \alpha_m, \alpha\}$ is a linearly dependent set. Therefore there exists a vector belonging to it which can be expressed as a linear combination of the preceding vectors. Since $\alpha_1, \alpha_2, \ldots, \alpha_m$ are linearly independent, therefore this vector cannot be any of these m vectors. So it must be α itself. Thus α can be expressed as a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_m$. Hence $L(S) = W$.

$\therefore$ S is a basis of W.

$\therefore$ dim $W = m$ and $m \leq n$.

Now if $V = W$, then every basis of V is also a basis of W.

Hence dim V = dim $W = n$.

Conversely let dim W = dim $V = n$. Then to prove that $W = V$.

Let S be a basis of W. Then $L(S) = W$ and S contains n vectors. Since S is also a subset of V and S contains n linearly independent vectors, therefore S will also be a basis of V. Therefore $L(S) = V$. Hence $W = V$. We thus conclude :

If W is a proper subspace of a finite-dimensional vector space V, then W is finite dimensional and dim $W <$ dim V. **(Garhwal 2008)**

Theorem 7: *If W is a subspace of a finite-dimensional vector space V, every linearly independent subset of W is finite and is part of a (finite) basis for W.*

Proof: Let dim $V = n$. Let W be a subspace of V. Let S_0 be a linearly independent subset of W. Let S be a linearly independent subset of W which contains S_0 and which has maximum number of elements. Then S is also a linearly independent subset of V. So S will have at the most n elements. Therefore S is finite and consequently S_0 is finite.

(i) S is a linearly independent subset of W.

(ii) $L(S) = W$. Because if $\beta \in W$, then β must be in the linear span of S. If β is not in the linear span of S, then the subset of W obtained by adjoining β to S will be linearly independent. Thus S will not remain the maximum linearly independent subset of W containing S_0. Hence $\beta \in W \Rightarrow \beta \in L(S)$. Thus $W \subseteq L(S)$. Since S is a subset of the subspace W, therefore $L(S) \subseteq W$.

Hence $\quad L(S) = W$.

Thus S is a finite basis of W and $S_0 \subseteq S$.

Theorem 8: *Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a basis of a finite dimensional vector space $V(F)$ of dimension n. Then every element α of V can be* **uniquely** *expressed as*

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n \quad \textit{where} \quad a_1, a_2, \ldots, a_n \in F.$$

Proof: Since S is a basis of V, therefore $L(S) = V$. Therefore any vector $\alpha \in V$ can be expressed as

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n.$$

To show uniqueness let us suppose that

$$\alpha = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n.$$

Then we must show that $a_1 = b_1, a_2 = b_2, \ldots, a_n = b_n$.

We have $\quad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n$

$\Rightarrow \quad (a_1 - b_1)\alpha_1 + (a_2 - b_2)\alpha_2 + \ldots + (a_n - b_n)\alpha_n = \mathbf{0}$

$\Rightarrow \quad a_1 - b_1 = 0, a_2 - b_2 = 0, \ldots, a_n - b_n = 0,$

since $\alpha_1, \alpha_2, \ldots, \alpha_n$ are linearly independent

$\Rightarrow \quad a_1 = b_1, a_2 = b_2, \ldots, a_n = b_n$.

Hence the theorem.

Theorem 9: *If W_1, W_2 are two subspaces of a finite dimensional vector space $V(F)$, then*

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

(Kumaon 2004; Garhwal 08)

Proof: Let $\dim(W_1 \cap W_2) = k$ and let the set

$$S = \{\gamma_1, \gamma_2, \gamma_3, \ldots, \gamma_k\}$$

be a basis of $W_1 \cap W_2$. Then $S \subseteq W_1$ and $S \subseteq W_2$.

Since S is linearly independent and $S \subseteq W_1$, therefore S can be extended to form a basis of W_1. Let

$$\{\gamma_1, \gamma_2, \ldots, \gamma_k, \alpha_1, \alpha_2, \ldots, \alpha_m\}$$

be a basis of W_1. Then $\dim W_1 = k + m$. Similarly let

$$\{\gamma_1, \gamma_2, \ldots, \gamma_k, \beta_1, \beta_2, \ldots, \beta_t\}$$

be a basis of W_2. Then $\dim W_2 = k + t$.

$\therefore \quad \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2) = (m + k) + (k + t) - k$

$$= k + m + t.$$

$$\dim (W_1 + W_2) = k + m + t.$$

We claim that the set

$$S_1 = \{\gamma_1, \gamma_2, \ldots, \gamma_k, \alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_t\}$$

is a basis of $W_1 + W_2$.

First we show that S_1 is linearly independent. Let

$$c_1\gamma_1 + c_2\gamma_2 + \ldots + c_k\gamma_k + a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + \ldots + b_t\beta_t = \mathbf{0} \qquad \ldots(1)$$

$$\Rightarrow \quad b_1\beta_1 + b_2\beta_2 + \ldots + b_t\beta_t = -(c_1\gamma_1 + \ldots + c_k\gamma_k + a_1\alpha_1 + \ldots + a_m\alpha_m) \qquad \ldots(2)$$

Now $-(c_1\gamma_1 + \ldots + c_k\gamma_k + a_1\alpha_1 + \ldots + a_m\alpha_m) \in W_1$ since it is a linear combination of elements of a basis of W_1. Again

$$b_1\beta_1 + b_2\beta_2 + \ldots + b_t\beta_t \in W_2$$

since it is a linear combination of elements belonging to a basis of W_2.

Also by virtue of the equality (2), $b_1\beta_1 + \ldots + b_t\beta_t \in W_1$. Therefore

$$b_1\beta_1 + b_2\beta_2 + \ldots + b_t\beta_t \in W_1 \cap W_2.$$

Therefore it can be expressed as a linear combination of elements of the basis of $W_1 \cap W_2$. Thus we have a relation of the form

$$b_1\beta_1 + b_2\beta_2 + \ldots + b_t\beta_t = d_1\gamma_1 + \ldots + d_k\gamma_k$$

$$\Rightarrow \quad b_1\beta_1 + b_2\beta_2 + \ldots + b_t\beta_t - d_1\gamma_1 - d_2\gamma_2 - \ldots - d_k\gamma_k = \mathbf{0}.$$

But $\beta_1, \beta_2, \ldots, \beta_t, \gamma_1, \ldots, \gamma_k$ are linearly independent vectors.

Therefore we must have

$$b_1 = 0, b_2 = 0, \ldots, b_t = 0.$$

Putting these values of b's in (1), it reduces to

$$c_1\gamma_1 + c_2\gamma_2 + \ldots + c_k\gamma_k + a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m = \mathbf{0}$$

$$\Rightarrow \quad c_1 = 0, c_2 = 0, \ldots, c_k = 0, a_1 = 0, a_2 = 0, \ldots, a_m = 0$$

since the vectors $\gamma_1, \gamma_2, \ldots, \gamma_k, \alpha_1, \alpha_2, \ldots, \alpha_m$ are linearly independent.

Thus the relation (1) implies that

$$c_1 = 0, c_2 = 0, \ldots, c_k = 0, a_1 = 0, \ldots, a_m = 0, b_1 = 0, \ldots, b_t = 0.$$

Therefore the set S_1 of vectors

$$\gamma_1, \ldots, \gamma_k, \alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_t$$

is linearly independent.

Now to show that $L(S_1) = W_1 + W_2$.

Since $W_1 + W_2$ is a subspace of V and each element of S_1 belongs to $W_1 + W_2$, therefore $L(S_1) \subseteq W_1 + W_2$.

Again let α be any element of $W_1 + W_2$. Then

$$\begin{aligned}\alpha &= \text{some element of } W_1 + \text{some element of } W_2\\ &= \text{a linear combination of elements of basis of } W_1\\ &\qquad + \text{a linear combination of elements of basis of } W_2\end{aligned}$$

$\therefore \quad \alpha \in L(S_1)$. Hence $W_1 + W_2 \subseteq L(S_1)$.

$\therefore \quad L(S_1) = W_1 + W_2$.

$\therefore \quad S_1$ is a basis of $W_1 + W_2$ and consequently

$$\dim(W_1 + W_2) = k + m + t.$$

Hence the theorem.

Illustrative Examples

Example 3: *Let V be the vector space of all 2×2 matrices over the field F. Prove that V has dimension 4 by exhibiting a basis for V which has 4 elements.* **(Garhwal 2006)**

Solution: Let $\alpha = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \beta = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \gamma = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ and $\delta = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

be four elements of V.

The subset $S = \{\alpha, \beta, \gamma, \delta\}$ of V is linearly independent because

$$a\alpha + b\beta + c\,\gamma + d\,\delta = \mathbf{0}$$

$$\Rightarrow \quad a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \quad a = 0, b = 0, c = 0, d = 0.$$

Also $L(S) = V$ because if $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is any vector in V, then we can write

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a\alpha + b\beta + c\gamma + d\delta.$$

Therefore S is a basis of V. Since the number of elements in S is 4, therefore

$$\dim V = 4.$$

Example 4: *Show that if $S = \{\alpha, \beta, \gamma\}$ is a basis of* $\mathbf{C}^3$ **(C)**, *then the set*

$$S' = \{\alpha + \beta, \beta + \gamma, \gamma + \alpha\}$$

is also a basis of $\mathbf{C}^3$ **(C).**

Solution: The vector space $\mathbf{C}^3$ **(C)** is of dimension 3. Any subset of $\mathbf{C}^3$ having three linearly independent vectors will form a basis of $\mathbf{C}^3$. We have shown in one of the previous examples that if S is a linearly independent subset of $\mathbf{C}^3$, then S' is also a linearly independent subset of $\mathbf{C}^3$.

Therefore S' is also a basis of $\mathbf{C}^3$.

Example 5: *Let V be the vector space of ordered pairs of complex numbers over the real field* **R** *i.e., let V be the vector space* $\mathbf{C}^2$ **(R).** *Show that the set* *is a basis for V.*

Solution: **S is linearly independent.** We have

$$a\,(1,0)+b\,(i,0)+c\,(0,1)+d\,(0,i)=(0,0) \text{ where } a,b,c,d\in\mathbf{R}$$

$$\Rightarrow \quad (a+ib,\, c+id)=(0,0)$$

$$\Rightarrow \quad a+ib=0,\, c+id=0$$

$$\Rightarrow \quad a=0,\, b=0,\, c=0,\, d=0.$$

Therefore S is linearly independent.

Now we shall show that $L(S)=V$. Let any ordered pair $(a+ib,\, c+id)\in V$ where $a,b,c,d\in\mathbf{R}$. Then as shown above we can write

$$(a+ib,\, c+id)=a\,(1,0)+b\,(i,0)+c\,(0,1)+d\,(0,i).$$

Thus any vector in V is expressible as a linear combination of elements of S. Therefore $L(S)=V$ and so S is a basis for V.

Example 6: *Show that the vectors* $(1,2,1),(2,1,0),(1,-1,2)$ *form a basis of* $\mathbf{R}^3$.

(Garhwal 2010)

Solution: We know that the set $\{(1,0,0),(0,1,0),(0,0,1)\}$ forms a basis for $\mathbf{R}^3$. Therefore $\dim \mathbf{R}^3=3$. If we show that the set $S=\{(1,2,1),(2,1,0),(1,-1,2)\}$ is linearly independent, then this set will also form a basis for $\mathbf{R}^3$.

[See theorem 4 of article 3.3]

We have $\quad a_1\,(1,2,1)+a_2\,(2,1,0)+a_3\,(1,-1,2)=(0,0,0)$

$$\Rightarrow \quad (a_1+2a_2+a_3,\, 2a_1+a_2-a_3,\, a_1+2a_3)=(0,0,0).$$

$$\therefore \quad a_1+2a_2+a_3=0 \qquad \ldots(1)$$

$$2a_1+a_2-a_3=0 \qquad \ldots(2)$$

$$a_1+2a_3=0. \qquad \ldots(3)$$

Now we shall solve these equations to get the values of a_1, a_2, a_3. Multiplying the equation (2) by 2, we get

$$4a_1+2a_2-2a_3=0. \qquad \ldots(4)$$

Subtracting (4) from (1), we get

$$-3a_1+3a_3=0$$

or

$$-a_1+a_3=0. \qquad \ldots(5)$$

Adding (3) and (5), we get $3a_3=0$ or $a_3=0$. Putting $a_3=0$ in (3), we get $a_1=0$. Now putting $a_3=0$ and $a_1=0$ in (1), we get $a_2=0$.

Thus solving the equations (1), (2) and (3), we get $a_1=0,\ a_2=0,\ a_3=0$. Therefore the set S is linearly independent. Hence it forms a basis for $\mathbf{R}^3$.

Example 7: *Determine whether or not the following vectors form a basis of* $\mathbf{R}^3$:

$$(1,1,2),(1,2,5),(5,3,4).$$

Solution: We know that $\dim \mathbf{R}^3=3$. If the given set of vectors is linearly independent, it will form a basis of $\mathbf{R}^3$ otherwise not. We have

$\Rightarrow \quad (a_1 + a_2 + 5a_3, a_1 + 2a_2 + 3a_3, 2a_1 + 5a_2 + 4a_3) = (0, 0, 0).$

$$\therefore \quad \left.\begin{array}{r} a_1 + a_2 + 5a_3 = 0 \\ a_1 + 2a_2 + 3a_3 = 0 \\ 2a_1 + 5a_2 + 4a_3 = 0 \end{array}\right\}. \quad \begin{array}{r} \ldots(1) \\ \ldots(2) \\ \ldots(3) \end{array}$$

Now we shall solve these equations to get the values of a_1, a_2, a_3. Subtracting (2) from (1), we get

$$-a_2 + 2a_3 = 0. \qquad \ldots(4)$$

Multiplying (1) by 2, we get

$$2a_1 + 2a_2 + 10a_3 = 0. \qquad \ldots(5)$$

Subtracting (5) from (3), we get

$$3a_2 - 6a_3 = 0$$

or $\quad a_2 - 2a_3 = 0. \qquad \ldots(6)$

We see that the equations (4) and (6) are the same and give $a_2 = 2a_3$. Putting $a_2 = 2a_3$ in (1), we get $a_1 = -7a_3$. If we put $a_3 = 1$, we get $a_2 = 2$ and $a_1 = -7$. Thus $a_1 = -7, a_2 = 2, a_3 = 1$ is a non-zero solution of the equations (1), (2) and (3). Hence the given set is linearly dependent so it does not form a basis of $\mathbf{R}^3$.

Example 8: *Show that the vectors* $\alpha_1 = (1, 0, -1), \alpha_2 = (1, 2, 1), \alpha_3 = (0, -3, 2)$ *form a basis for* $\mathbf{R}^3$. *Express each of the standard basis vectors as a linear combination of* $\alpha_1, \alpha_2, \alpha_3$.

Solution: Let $S = \{\alpha_1, \alpha_2, \alpha_3\}$.

First we shall show that the set S is linearly independent. Let a, b, c be scalars *i.e.*, real numbers such that

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \mathbf{0}$$

i.e., $\quad a(1, 0, -1) + b(1, 2, 1) + c(0, -3, 2) = (0, 0, 0)$

i.e., $\quad (a + b + 0c, 0a + 2b - 3c, -a + b + 2c) = (0, 0, 0)$

i.e., $\quad a + b = 0 \qquad \ldots(1)$

$$2b - 3c = 0 \qquad \ldots(2)$$

$$-a + b + 2c = 0. \qquad \ldots(3)$$

Adding both sides of the equations (1) and (3), we get

$$2b + 2c = 0 \qquad \ldots(4)$$

Subtracting (2) from (4), we get

$$5c = 0 \quad \text{or} \quad c = 0.$$

Putting $c = 0$ in (2), we get $b = 0$ and then putting $b = 0$ in (1), we get $a = 0$. Thus $a = 0, b = 0, c = 0$ is the only solution of the equations (1), (2) and (3) and so

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \mathbf{0} \Rightarrow a = 0, b = 0, c = 0.$$

$\therefore$ The vectors $\alpha_1, \alpha_2, \alpha_3$ are linearly independent.

We have dim $\mathbf{R}^3 = 3$. Therefore the vectors $\alpha_1, \alpha_2, \alpha_3$ form a basis of $\mathbf{R}^3$.

$$\gamma = (p, q, r) = x\alpha_1 + y\alpha_2 + z\alpha_3, \; x, y, z \in \mathbf{R}$$
$$= x(1, 0, -1) + y(1, 2, 1) + z(0, -3, 2). \qquad \ldots(5)$$

Then $x + y = p \qquad \ldots(6)$

$2y - 3z = q \qquad \ldots(7)$

$-x + y + 2z = r \qquad \ldots(8)$

Adding both sides of equations (6) and (8), we get

$$2y + 2z = p + r. \qquad \ldots(9)$$

Subtracting (7) from (9), we get

$$5z = p + r - q \quad \text{or} \quad z = \frac{1}{5}p - \frac{1}{5}q + \frac{1}{5}r.$$

Then from (9), we get

$$y = -z + \frac{1}{2}p + \frac{1}{2}r = \frac{3}{10}p + \frac{1}{5}q + \frac{3}{10}r$$

and from (6), we get

$$x = p - y = p - \frac{3}{10}p - \frac{1}{5}q - \frac{3}{10}r = \frac{7}{10}p - \frac{1}{5}q - \frac{3}{10}r.$$

Thus, the relation (5) expresses the vector $\gamma = (p, q, r)$ as a linear combination of α_1, α_2 and α_3 where $x, y, z \in \mathbf{R}$ are as found above.

The standard basis vectors are

$$e_1 = (1, 0, 0), e_2 = (0, 1, 0) \quad \text{and} \quad e_3 = (0, 0, 1).$$

If $\gamma = e_1$, then $p = 1, q = 0, r = 0$ and so

$$x = \frac{7}{10}, \; y = \frac{3}{10} \text{ and } z = \frac{1}{5}.$$

$$\therefore \quad e_1 = \frac{7}{10}\alpha_1 + \frac{3}{10}\alpha_2 + \frac{1}{5}\alpha_3.$$

If $\gamma = e_2$, then $p = 0, q = 1, r = 0$ and so

$$x = -\frac{1}{5}, \; y = \frac{1}{5} \text{ and } z = -\frac{1}{5}.$$

$$\therefore \quad e_2 = -\frac{1}{5}\alpha_1 + \frac{1}{5}\alpha_2 - \frac{1}{5}\alpha_3.$$

Finally if $\gamma = e_3$, then $p = 0, q = 0, r = 1$ and so

$$x = -\frac{3}{10}, \quad y = \frac{3}{10} \quad \text{and} \quad z = \frac{1}{5}.$$

$$\therefore \quad e_3 = -\frac{3}{10}\alpha_1 + \frac{3}{10}\alpha_2 + \frac{1}{5}\alpha_3.$$

Example 9: *Show that a system X consisting of the vectors*

$$\alpha_1 = (1, 0, 0, 0), \alpha_2 = (0, 1, 0, 0), \alpha_3 = (0, 0, 1, 0) \textit{ and } \alpha_4 = (0, 0, 0, 1)$$

is a basis set of $\mathbf{R}^4$ $(\mathbf{R})$.

Solution: First we show that the set X is a linearly independent set of vectors. If a_1, a_2, a_3, a_4 be any scalars *i.e.*, elements of the field $\mathbf{R}$, then

$$a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3 + a_4\alpha_4 = \text{zero vector}$$

$\Rightarrow \qquad (a_1, a_2, a_3, a_4) = (0, 0, 0, 0) \quad \Rightarrow \quad a_1 = 0, a_2 = 0, a_3 = 0, a_4 = 0.$

Therefore the given set X of four vectors is linearly independent.

Now we shall show that X generates $\mathbf{R}^4$ *i.e.*, each vector of $\mathbf{R}^4$ can be expressed as a linear combination of the vectors of X.

Let (a, b, c, d) be any vector in $\mathbf{R}^4$. We can write

$$\begin{aligned}(a, b, c, d) &= a\ (1, 0, 0, 0) + b\ (0, 1, 0, 0) + c\ (0, 0, 1, 0) + d\ (0, 0, 0, 1)\\ &= a\alpha_1 + b\alpha_2 + c\alpha_3 + d\alpha_4.\end{aligned}$$

Thus (a, b, c, d) has been expressed as a linear combination of the vectors of X and so X generates $\mathbf{R}^4$.

Since X is a linearly independent subset of $\mathbf{R}^4$ and it also generates $\mathbf{R}^4$, therefore it is a basis of $\mathbf{R}^4$.

Example 10: *Show that the set $S = \{1, x, x^2, \ldots, x^n\}$ of $n + 1$ polynomials in x is a basis of the vector space P_n $(\mathbf{R})$, of all polynomials in x (of degree at most n) over the field of real numbers.*

Solution: P_n $(\mathbf{R})$ is the vector space of all polynomials in x (of degree at most n) over the field $\mathbf{R}$ of real numbers.

$S = \{1, x, x^2, \ldots, x^n\}$ is a subset of P_n consisting of $n + 1$ polynomials. To prove that S is a basis of the vector space P_n $(\mathbf{R})$.

First we show that the vectors in the set S are linearly independent over the field $\mathbf{R}$.

The zero vector of the vector space P_n $(\mathbf{R})$ is the zero polynomial. Let $a_0, a_1, a_2, \ldots, a_n \in \mathbf{R}$ be such that

$$a_0\ (1) + a_1\ x + a_2\ x^2 + \ldots + a_n\ x^n = 0 \quad \textit{i.e.,} \text{ zero polynomial.}$$

Now by the definition of the equality of two polynomials, we have

$$a_0 + a_1\ x + a_2\ x^2 + \ldots + a_n\ x^n = 0$$

$$\Rightarrow \qquad a_0 = 0,\ a_1 = 0,\ a_2 = 0, \ldots, a_n = 0.$$

$\therefore$ the vectors $1, x, x^2, \ldots, x^n$ of the vector space P_n $(\mathbf{R})$ are linearly independent.

Now we shall show that the set S generates the whole vector space P_n $(\mathbf{R})$.

Let $\alpha = a_0 + a_1\ x + a_2\ x^2 + \ldots + a_n\ x^n$ be any arbitrary member of P_n, where $a_0, a_1, \ldots, a_n \in \mathbf{R}$. Then α is a linear combination of the polynomials $1, x, x^2, \ldots, x^n$ over the field $\mathbf{R}$. Therefore S generates P_n $(\mathbf{R})$.

Since S is a linearly independent subset of P_n $(\mathbf{R})$ and it also generates P_n $(\mathbf{R})$, therefore it is a basis of P_n $(\mathbf{R})$.

Example 11: *Show that a finite subset W of a vector space V (F) is linearly dependent if and only if some element of W can be expressed as a linear combination of the others.*

Solution: Let $W = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a finite subset of a vector space V (F).

First suppose that W is linearly dependent. Then to show that some vector of W can be expressed as a linear combination of the others.

scalars $a_1, a_2, \dots, a_n$ not all zero such that

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = \mathbf{0}. \qquad \dots(1)$$

Suppose $a_r \neq 0, 1 \le r \le n$.

Then from (1), we have

$$a_r \alpha_r = -a_1\alpha_1 - a_2\alpha_2 - \dots - a_{r-1}\alpha_{r-1} - a_{r+1}\alpha_{r+1} - \dots - a_n\alpha_n$$

or $$a_r^{-1}(a_r\alpha_r) = a_r^{-1}(-a_1\alpha_1 - a_2\alpha_2 - \dots - a_{r-1}\alpha_{r-1} - a_{r+1}\alpha_{r+1} - \dots - a_n\alpha_n)$$

or $$\alpha_r = (-a_r^{-1}a_1)\alpha_1 + (-a_r^{-1}a_2)\alpha_2 + \dots + (-a_r^{-1}a_{r-1})\alpha_{r-1} + (-a_r^{-1}a_{r+1})\alpha_{r+1} + \dots + (-a_r^{-1}a_n)\alpha_n.$$

Thus α_r is a linear combination of the other vectors of the set W.

Conversely suppose that some vector of W is a linear combination of the others. Then to prove that W is linearly dependent.

Without loss of generality suppose α_1 is a linear combination of the vectors $\alpha_2, \dots, \alpha_n$.

Let $\alpha_1 = b_2\alpha_2 + b_3\alpha_3 + \dots + b_n\alpha_n$, where $b_2, b_3, \dots, b_n \in F$.

Then $$\alpha_1 - b_2\alpha_2 - b_3\alpha_3 - \dots - b_n\alpha_n = 0. \qquad \dots(2)$$

Since in the linear relation (2) among the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$, the scalar coefficient of α_1 is 1 which is $\neq 0$, therefore the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ are linearly dependent.

Example 12: *If n vectors span a vector space V containing r linearly independent vectors, then show that $n \ge r$.*

Solution: Suppose a subset S of V containing n vectors spans V. Then there exists a linearly independent subset T of S which also spans V. This subset T of S will form a basis of V. Suppose the number of vectors in T is m. Then $\dim V = m \le n$.

Since $\dim V = m$, therefore any subset of V containing more than m vectors will be linearly dependent.

Hence if S_1 is a linearly independent subset of V and S_1 contains r vectors, we must have

$$r \le m \Rightarrow r \le n. \qquad [\because m \le n]$$

Example 13: *Let W_1 and W_2 be distinct subspaces of V and that $\dim W_1 = 4$, $\dim W_2 = 4$ and $\dim V = 6$. Find the possible dimensions of $W_1 \cap W_2$.*

Solution: Since W_1 and W_2 are distinct, so $W_1 + W_2$ properly contains W_1 and W_2 *i.e.*, $\dim(W_1 + W_2) > 4$. Also $\dim V = 6 \Rightarrow \dim(W_1 + W_2)$ cannot be greater than 6. Thus we have two possibilities *i.e.*, dimension of $(W_1 + W_2)$ must be either 5 or 6.

Using theorem $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$, we have

(i) $5 = 4 + 4 - \dim(W_1 \cap W_2)$ or $\dim W_1 \cap W_2 = 3$

(ii) $6 = 4 + 4 - \dim(W_1 \cap W_2)$ or $\dim W_1 \cap W_2 = 2$.

Hence the possible dimensions of $W_1 \cap W_2$ are 2, 3.

Comprehensive Exercise 1

1. Give a basis for each of the following vector spaces over the indicated fields.
   (i) $\mathbf{R}(\sqrt{2})$ over $\mathbf{R}$ (ii) $\mathbf{C}$ over $\mathbf{R}$ (iii) $\mathbf{Q}(2^{1/4})$ over $\mathbf{Q}$
   where $\mathbf{Q}, \mathbf{R}, \mathbf{C}$ are fields of rational, real and complex numbers respectively.
2. Find the dimension of $\mathbf{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbf{Q}$.
3. Find basis for the subspaces
   $$W_1 = \{(a, b, 0): a, b, \in \mathbf{R}\}, W_2 = \{(0, b, c): b, c \in \mathbf{R}\} \text{ of } \mathbf{R}^3.$$
   Find the dimensions of W_1, W_2 and $W_1 \cap W_2$.
4. Show that the real field $\mathbf{R}$ is a vector space of infinite dimension over the rational field $\mathbf{Q}$.
5. Let V be the vector space of all 2×2 symmetric matrices over F. Show that $\dim V = 3$.
6. Prove that the space of all $m \times n$ matrices over the field F has dimension mn, by exhibiting a basis for this space.
7. If $\{\alpha_1, \alpha_2, \alpha_3\}$ is a basis of $V_3(\mathbf{R})$, show that $\{\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1\}$ is also a basis of $V_3(\mathbf{R})$.
8. If W_1 and W_2 are finite-dimenisonal subspaces with the same dimension, and if $W_1 \subseteq W_2$, then $W_1 = W_2$.
9. In the vector space $\mathbf{R}^3$, let $\alpha = (1, 2, 1), \beta = (3, 1, 5), \gamma = (3, -4, 7)$. Show that there exists more than one basis for the subspace spanned by the set $S = \{\alpha, \beta, \gamma\}$.
10. For the 3-dimensional space $\mathbf{R}^3$ over the field of real numbers $\mathbf{R}$, determine if the set $\{(2, -1, 0), (3, 5, 1), (1, 1, 2)\}$ is a basis.
11. Show that the set $\{(1, i, 0), (2i, 1, 1), (0, 1 + i, 1 - i)\}$ is a basis for $V_3(\mathbf{C})$.
12. Find three vectors in $\mathbf{R}^3$ which are linearly dependent, and are such that any two of them are linearly independent.
13. Select a basis, if any, of $\mathbf{R}^3(\mathbf{R})$ from the set $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, where
    $$\alpha_1 = (1, -3, 2), \alpha_2 = (2, 4, 1), \alpha_3 = (3, 1, 3), \alpha_4 = (1, 1, 1).$$
14. If W is a subspace of a finite dimensional vector space V, prove that any basis of W can be extended to form a basis of V.
15. Prove that any finite set S of vectors, not all the zero vectors, contains a linearly independent subset T which spans the same space as S.
16. Let V be a vector space. Let W be a subspace of V generated by the vectors $\alpha_1, \ldots, \alpha_s$. Prove that W is spanned by a linearly independent subset of $\alpha_1, \ldots, \alpha_s$.
17. If a vector space V is spanned by a finite set of m vectors, then show that any linearly independent set of vectors in V has at most m elements.

the following two statements are equivalent :

(i) B is linearly independent

(ii) If $\alpha \in V$, then the expression $\alpha = \sum_{i=1}^{n} a_i \alpha_i$ with $a_i \in F$ is unique.

19. Determine a basis of the subspace spanned by the vectors

$$\alpha_1 = (1, 2, 3), \alpha_2 = (2, 1, -1), \alpha_3 = (1, -1, -4), \alpha_4 = (4, 2, -2).$$

20. Let W be the subspace of $\mathbf{R}^4$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$ and $(3, 8, -3, -5)$. Find a basis and the dimension of W. Extend the basis of W to a basis of the whole space $\mathbf{R}^4$.

21. Let E be a subfield of a field F and F a subfield of a field K *i.e.*, $E \subset F \subset K$. Let K be of dimension n over F and F is of dimension m over E. Show that K is of dimension mn over E.

22. Let W_1 and W_2 be the subspaces of $\mathbf{R}^4$:

$$W_1 = \{(a, b, c, d) : b + c + d = 0\}, W_2 = \{(a, b, c, d) : a + b = 0, c = 2d\}.$$

Find the dimension of $W_1 \cap W_2$.

23. Let W_1 and W_2 be subspaces of V and that $\dim W_1 = 4$, $\dim W_2 = 5$ and $\dim V = 7$. Find the possible dimensions of $W_1 \cap W_2$.

24. Let W_1 and W_2 be the subspaces of $\mathbf{R}^4$ generated by the sets

$$B_1 = \{(1, 1, 0, -1), (1, 2, 3, 0), (2, 3, 3, -1)\}$$

and $B_2 = \{(1, 2, 2, -2), (2, 3, 2, -3), (1, 3, 4, -3)\}$ respectively.

Find (i) $\dim (W_1 + W_2)$ (ii) $\dim (W_1 \cap W_2)$.

Answers 1

1. (i) $\{1, \sqrt{2}\}$ (ii) $\{1, i\}$ (iii) $\{1, 2^{1/4}\}$ **2.** 4

3. Basis for $W_1 = \{(1, 0, 0), (0, 1, 0)\}$; $\dim W_1 = 2$

Basis for $W_2 = \{(0, 1, 0), (0, 0, 1)\}$; $\dim W_2 = 2$; $\dim (W_1 \cap W_2) = 1$

10. Yes **12.** $(1, 0, 0), (0, 1, 0), (1, 1, 0)$

13. $\{\alpha_1, \alpha_2, \alpha_4\}$ **19.** $\{\alpha_1, \alpha_2\}$

20. A basis of $W = \{(1, -2, 5, -3), (0, 7, -9, 2)\}$; $\dim W = 2$

A basis of $\mathbf{R}^4$ is $\{(1, -2, 5, -3), (0, 7, -9, 2), (0, 0, 1, 0), (0, 0, 0, 1)\}$

22. 1 **23.** 2, 3 or 4

24. (i) 3 (ii) 1

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. The vector space V_3 (**R**) is of dimension

(a) 1 (b) 3
(c) 2 (d) None of these.

2. Let V be the vector space of all 2×2 matrices over the field F. The dimension of V is

(a) 2 (b) 3
(c) 4 (d) None of these. **(Garhwal 2007)**

3. The dimension of the vector space $\mathbf{C}^2$ (**R**) is

(a) 2 (b) 4
(c) 1 (d) None of these.

4. The dimension of the vector space **C** (**R**) is

(a) 1 (b) 2
(c) 4 (d) None of these.

5. Let W_1 and W_2 be finite dimensional subspaces of a vector space V. If dim $W_1 = 2$, dim $W_2 = 2$, dim $(W_1 + W_2) = 3$ then dim $(W_1 \cap W_2)$ is

(a) 1 (b) 2
(c) 3 (d) 4.

6. The dimension of the vector space spanned by $(1, -2, 3, -1)$ and $(1, 1, -2, 3)$ is

(a) 1 (b) 2
(c) 4 (d) None of these.

7. The dimension of the vector space spanned by $(3, -6, 3, -9)$ and $(-2, 4, -2, 6)$ is

(a) 1 (b) 2
(c) 4 (d) None of these.

8. In the vector space $\mathbf{R}^3$, the coordinate vector of $(3, 5, -2)$ relative to the ordered basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is

(a) $(5, 3, -2)$ (b) $(3, 5, -2)$
(c) $(-2, 5, 3)$ (d) $(-2, 3, 5)$.

Fill in the Blank(s)

Fill in the blanks "......" so that the following statements are complete and correct.

1. Each set of $(n + 1)$ or more vectors of a finite dimensional vector space $V(F)$ of dimension ...... is linearly dependent.

$\dim (W_1 + W_2) + \dim (W_1 \cap W_2) = \ldots\ldots$

3. The set $S = \ldots\ldots$ of $n+1$ polynomials in x is a basis of the vector space P_n (**R**), of all polynomials in x (of degree at most n) over the field of real numbers.
4. Let V be the vector space of $m \times n$ matrices over a field F. Then dim $V = \ldots\ldots$
5. If W is a proper subspace of a finite dimensional vector space V, then W is finite dimensional and .... **(Garhwal 2008)**
6. The number of members in any basis of a vector space is called .... . **(Garhwal 2010, 12)**

True or False

Write 'T' for true and 'F' for false statement.

1. If a subset of an n-dimensional vector space V consists of n non-zero vectors, then it will be a basis of V.
2. If A and B are subspaces of a vector space, then $\dim A < \dim B \Rightarrow A \subset B$.
3. If A and B are subspaces of a vector space, then

$$A \neq B \Rightarrow \dim A \neq \dim B.$$

4. If M and N are finite dimensional subspaces with the same dimension, and if $M \subset N$, then $M = N$.
5. In an n-dimensional space any subset consisting of n linearly independent vectors will form a basis.

Answers

Multiple Choice Questions

1. (b) 2. (c) 3. (b) 4. (b) 5. (a)
6. (b) 7. (a) 8. (b)

Fill in the Blank(s)

1. n 2. $\dim W_1 + \dim W_2$ 3. $\{1, x, x^2, \ldots, x^n\}$
4. mn 5. $\dim W < \dim V$ 6. dimension

True or False

1. F 2. F 3. F 4. T
5. T

□

Chapter

4

Linear Transformations

4.1 Homomorphism of Vector Spaces or Linear Transformation

Definition. *Let* $U(F)$ *and* $V(F)$ *be two vector spaces. Then a mapping*

$$f : U \to V$$

is called a homomorphism or a linear transformation of U **into** V, *if*

(i) $f(\alpha + \beta) = f(\alpha) + f(\beta), \forall\, \alpha, \beta \in U$

and (ii) $f(a\alpha) = af(\alpha)\ \forall\, a \in F,\ \forall\, \alpha \in U$.

The conditions (i) and (ii) can be combined into a single condition

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta)\ \forall\, a, b \in F \text{ and } \forall\, \alpha, \beta \in U.$$

If f is a homomorphism of U **onto** V, then V is called a homomorphic image of U.

Theorem: *If* f *is a homomorphism of* $U(F)$ *into* $V(F)$, *then*

(i) $f(\mathbf{0}) = \mathbf{0}'$ *where* $\mathbf{0}$ *and* $\mathbf{0}'$ *are the zero vectors of* U *and* V *respectively*.

(ii) $f(-\alpha) = -f(\alpha)\ \forall\, \alpha \in U$.

$$f(\alpha) + \mathbf{0}' = f(\alpha) = f(\alpha + \mathbf{0}) = f(\alpha) + f(\mathbf{0}).$$

Now V is an abelian group with respect to addition of vectors.

$\therefore \quad f(\alpha) + \mathbf{0}' = f(\alpha) + f(\mathbf{0})$

$\Rightarrow \quad \mathbf{0}' = f(\mathbf{0})$ by left cancellation law.

(ii) If $\alpha \in U$, then $-\alpha \in U$. Also we have

$$\mathbf{0}' = f(\mathbf{0}) = f[\alpha + (-\alpha)] = f(\alpha) + f(-\alpha).$$

Now $f(\alpha) + f(-\alpha) = \mathbf{0}' \Rightarrow f(-\alpha) =$ the additive inverse of $f(\alpha)$

$\Rightarrow \quad f(-\alpha) = -f(\alpha).$

4.2 Isomorphism of Vector Spaces

(Garhwal 2010)

Definition: *Let $U(F)$ and $V(F)$ be two Vector spaces. Then a mapping*

$$f : U \to V$$

is called an isomorphism of U onto V, if

(i) f is one-one,

(ii) f is onto,

(iii) $f(a\alpha + b\beta) = a f(\alpha) + b f(\beta) \; \forall \; a, b \in F, \alpha, \beta \in U.$

Also then the two vector spaces U and V are said to be isomorphic and symbolically we write $U(F) \cong V(F)$.

The vector space $V(F)$ is also called the isomorphic image of the vector space $U(F)$.

If f is a homomorphism of $U(F)$ into $V(F)$, then f will become an isomorphism of U into V if f is one-one. Also in addition if f is onto V, then f will become an isomorphism of U onto V.

4.3 Isomorphism of Finite Dimensional Vector Spaces

Theorem 1: *Two finite dimensional vector spaces over the same field are isomorphic if and only if they are of the same dimension.* **(Garhwal 2006, 10)**

Proof: First suppose that $U(F)$ and $V(F)$ are two finite dimensional vector spaces each of dimension n. Then to prove that

$$U(F) \cong V(F).$$

Let the sets of vectors

$$\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

and $\quad \{\beta_1, \beta_2, \ldots, \beta_n\}$

be the bases of U and V respectively.

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n .$$

Let $f : U \in V$ be defined by

$$f(\alpha) = a_1\beta_1 + a_2\beta_2 + \ldots + a_n\beta_n .$$

Since in the expression of α as a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_n$ the scalars $a_1, a_2, \ldots, a_n$ are unique, therefore the mapping f is well defined *i.e.*, $f(\alpha)$ is a unique element of V.

***f* is one-one.** We have

$$f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) = f(b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n)$$

$\Rightarrow$ $a_1\beta_1 + a_2\beta_2 + \ldots + a_n\beta_n = b_1\beta_1 + b_2\beta_2 + \ldots + b_n\beta_n$

$\Rightarrow$ $(a_1 - b_1)\beta_1 + (a_2 - b_2)\beta_2 + \ldots + (a_n - b_n)\beta_n = \mathbf{0}'$ (zero vector of V)

$\Rightarrow$ $a_1 - b_1 = 0, a_2 - b_2 = 0, \ldots, a_n - b_n = 0$ because $\beta_1, \beta_2, \ldots, \beta_n$ are linearly independent

$\Rightarrow$ $a_1 = b_1, a_2 = b_2, \ldots, a_n = b_n$

$\Rightarrow$ $a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n$.

$\therefore$ f is one-one.

***f* is onto *V*.** If $a_1\beta_1 + a_2\beta_2 + \ldots + a_n\beta_n$ is any element of V, then $\exists$ an element $a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n \in U$ such that

$$f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) = a_1\beta_1 + a_2\beta_2 + \ldots + a_n\beta_n .$$

$\therefore$ f is onto V.

***f* is a linear transformation.** We have

$$\begin{aligned} f\,[a(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) + b(b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n)] \\ = f\,[(aa_1 + bb_1)\alpha_1 + (aa_2 + bb_2)\alpha_2 + \ldots + (aa_n + bb_n)\alpha_n] \\ = (aa_1 + bb_1)\beta_1 + (aa_2 + bb_2)\beta_2 + \ldots + (aa_n + bb_n)\beta_n \\ = a(a_1\beta_1 + a_2\beta_2 + \ldots + a_n\beta_n) + b(b_1\beta_1 + b_2\beta_2 + \ldots + b_n\beta_n) \\ = a f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) + b f(b_1\alpha_1 + b_2\alpha_2 \\ + \ldots + b_n\alpha_n). \end{aligned}$$

$\therefore$ f is a linear transformation.

Hence f is an isomorphism of U onto V.

$\therefore$ $U \cong V$.

Conversely, let $U(F)$ and $V(F)$ be two isomorphic finite dimensional vector spaces.

Then to prove that $\dim U = \dim V$.

Let $\dim U = n$. Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a basis of U. If f is an isomorphism of U onto V, we shall show that

$$S' = \{f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_n)\}$$

is a basis of V. Then V will also be of dimension n.

Let $a_1 \ f(\alpha_1) + a_2 \ f(\alpha_2) + \ldots + a_n \ f(\alpha_n) = \mathbf{0}'$ (zero vector of V)

$\Rightarrow \quad f(a_1\alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n) = \mathbf{0}'$

[$\because f$ is a linear transformation]

$\Rightarrow \quad a_1\alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n = \mathbf{0}$

[$\because f$ is one-one and $f(\mathbf{0}) = \mathbf{0}'$ where $\mathbf{0}$ is zero vector of U]

$\Rightarrow \quad a_1 = 0, a_2 = 0, \ldots, a_n = 0$ since $\alpha_1, \alpha_2, \ldots, \alpha_n$ are linearly independent.

$\therefore \quad S'$ is linearly independent.

Now to prove that $L(S') = V$. For this we shall prove that any vector $\beta \in V$ can be expressed as a linear combination of the vectors of the set S'. Since f is onto V, therefore $\beta \in V \Rightarrow$ there exists $\alpha \in U$ such that $f(\alpha) = \beta$.

Let $\quad \alpha = c_1 \ \alpha_1 + c_2 \ \alpha_2 + \ldots + c_n \ \alpha_n$.

Then $\quad \beta = f(\alpha) = f(c_1 \ \alpha_1 + c_2 \ \alpha_2 + \ldots + c_n \ \alpha_n)$

$\quad = c_1 \ f(\alpha_1) + c_2 \ f(\alpha_2) + \ldots + c_n \ f(\alpha_n)$.

Thus β is a linear combination of the vectors of S'.

Hence $\quad V = L(S')$.

$\therefore \quad S'$ is a basis of V. Since S' contains n vectors, therefore $\dim V = n$.

Note: While proving the converse, we have proved that if *f is an isomorphism of U onto V, then f maps a basis of U onto a basis of V.*

Theorem 2: *Every n-dimensional vector space $V(F)$ is isomorphic to $V_n(F)$.*

(Garhwal 2007)

Proof: Let $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be any basis of $V(F)$. Then every vector $\alpha \in V$ can be uniquely expressed as

$$\alpha = a_1 \ \alpha_1 + a_2 \ \alpha_2 + \ldots + a_n \ \alpha_n, a_i \in F.$$

The ordered n-tuple $(a_1, a_2, \ldots, a_n) \in V_n(F)$.

Let $\ f : V(F) \to V_n(F)$ be defined by $f(\alpha) = (a_1, a_2, \ldots, a_n)$.

Since in the expression of α as a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_n$ the scalars $a_1, a_2, \ldots, a_n$ are unique, therefore $f(\alpha)$ is a unique element of $V_n(F)$ and thus the mapping f is well defined.

***f* is one-one.** Let $\alpha = a_1\alpha_1 + a_2 \ \alpha_2 + \ldots + a_n \ \alpha_n$

and $\quad \beta = b_1\alpha_1 + b_2 \ \alpha_2 + \ldots + b_n \ \alpha_n$

be any two elements of V. We have

$$f(\alpha) = f(\beta)$$

$\Rightarrow \quad f(a_1 \ \alpha_1 + a_2 \ \alpha_2 + \ldots + a_n \ \alpha_n) = f(b_1 \ \alpha_1 + b_2 \ \alpha_2 + \ldots + b_n \ \alpha_n)$

$\Rightarrow \quad (a_1, a_2, \ldots, a_n) = (b_1, b_2, \ldots, b_n)$

$\Rightarrow \quad a_1 = b_1, a_2 = b_2, \ldots, a_n = b_n$

$\therefore \quad f$ is one-one.

f is onto $V_n(F)$. Let $(a_1, a_2, \ldots, a_n)$ be any element of $V_n(F)$. Then there exists an element $a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n \in V(F)$ such that

$$f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) = (a_1, a_2, \ldots, a_n).$$

$\therefore \quad f$ is onto $V_n(F)$.

f is a linear transformation. If $a, b \in F$ and $\alpha, \beta \in V(F)$, we have

$$\begin{aligned}
f(a\alpha + b\beta) &= f[a(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) + b(b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n)] \\
&= f[(aa_1 + bb_1)\alpha_1 + (aa_2 + bb_2)\alpha_2 + \ldots + (aa_n + bb_n)\alpha_n] \\
&= (aa_1 + bb_1, aa_2 + bb_2, \ldots, aa_n + bb_n) \\
&= (aa_1, aa_2, \ldots, aa_n) + (bb_1, bb_2, \ldots, bb_n) \\
&= a(a_1, a_2, \ldots, a_n) + b(b_1, b_2, \ldots, b_n) \\
&= af(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) + bf(b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n) \\
&= af(\alpha) + bf(\beta).
\end{aligned}$$

$\therefore \quad f$ is a linear transformation.

$\therefore \quad f$ is an isomorphism of $V(F)$ onto $V_n(F)$.

Hence $V(F) \cong V_n(F)$.

Illustrative Examples

Example 1: *Show that the mapping $f : V_3(F) \to V_2(F)$ defined by*

$$f(a_1, a_2, a_3) = (a_1, a_2)$$

is a homomorphism of $V_3(F)$ onto $V_2(F)$.

Solution: Let $\alpha = (a_1, a_2, a_3)$ and $\beta = (b_1, b_2, b_3)$ be any two elements of $V_3(F)$. Also let a, b be any two elements of F. We have

$$\begin{aligned}
f(a\alpha + b\beta) &= f[a(a_1, a_2, a_3) + b(b_1, b_2, b_3)] \\
&= f[(aa_1 + bb_1, aa_2 + bb_2, aa_3 + bb_3)] \\
&= (aa_1 + bb_1, aa_2 + bb_2) \\
&= a(a_1, a_2) + b(b_1, b_2) \\
&= af(a_1, a_2, a_3) + bf(b_1, b_2, b_3) \\
&= af(\alpha) + bf(\beta).
\end{aligned}$$

$\therefore \quad f$ is a linear transformation.

To show that f is onto $V_2(F)$. Let (a_1, a_2) be any element of $V_2(F)$. Then $(a_1, a_2, 0) \in V_3(F)$ and we have $f(a_1, a_2, 0) = (a_1, a_2)$. Therefore f is onto $V_2(F)$.

Therefore f is a homomorphism of $V_3(F)$ onto $V_2(F)$.

R *and let T be a mapping from* $V(\mathbf{R})$ *to* $V_2(\mathbf{R})$ *defined as*

$$T(a+ib)=(a,b).$$

Show that T is an isomorphism.

Solution: **T is one-one.** Let

$$\alpha=a+ib,\beta=c+id$$

be any two members of $V(\mathbf{R})$. Then $a,b,c,d\in\mathbf{R}$.

We have $\quad T(\alpha)=T(\beta)\Rightarrow(a,b)=(c,d) \qquad [\because\ T(\alpha)=(a,b)]$

$\Rightarrow \qquad a=c,b=d \Rightarrow a+ib=c+id$

$\Rightarrow \qquad \alpha=\beta.$

$\therefore \quad T$ is one-one.

***T* is onto.** Let (a,b) be an arbitrary member of $V_2(\mathbf{R})$. Then $\exists$ a vector $a+ib\in V(\mathbf{R})$ such that $T(a+ib)=(a,b)$. Hence T is onto.

***T* is a linear transformation.** Let $\alpha=a+ib,\beta=c+id$ be any two members of $V(\mathbf{R})$ and k_1,k_2 be any two elements of the field **R**. Then

$$k_1\alpha+k_2\beta=k_1(a+ib)+k_2(c+id)=(k_1a+k_2c)+i(k_1b+k_2d).$$

We have $\quad T(k_1\alpha+k_2\beta)=(k_1a+k_2c,k_1b+k_2d)$, by definition of T

$$=(k_1a,k_1b)+(k_2c,k_2d)=k_1(a,b)+k_2(c,d)$$

$$=k_1T(a+ib)+k_2T(c+id) \qquad \text{[by definition of } T\text{]}$$

$$=k_1T(\alpha)+k_2T(\beta).$$

Hence T is a linear transformation.

Hence T is an isomorphism.

Example 3: *If V is a finite dimensional vector space and f is an isomorphism of V into V, prove that f must map V onto V.*

Solution: Let $V(F)$ be a finite dimensional vector space of dimension n. Let f be an isomorphism of V into V *i.e.*, f is a linear transformation and f is one-one. To prove that f is onto V.

Let $S=\{\alpha_1,\alpha_2,\ldots,\alpha_n\}$ be a basis of V. We shall first prove that

$$S'=\{f(\alpha_1),f(\alpha_2),\ldots,f(\alpha_n)\}$$

is also a basis of V. We claim that S' is linearly independent. The proof is as follows :

Let $a_1f(\alpha_1)+a_2f(\alpha_2)+\ldots+a_nf(\alpha_n)=\mathbf{0}$ (zero vector of V)

$\Rightarrow \qquad f(a_1\alpha_1+a_2\alpha_2+\ldots+a_n\alpha_n)=\mathbf{0} \qquad [\because\ f$ is linear transformation]

$\Rightarrow \qquad a_1\alpha_1+a_2\alpha_2+\ldots+a_n\alpha_n=\mathbf{0} \qquad [\because\ f$ is one-one and $f(\mathbf{0})=\mathbf{0}]$

$\Rightarrow \qquad a_1=0,a_2=0,\ldots,a_n=0$

since $\alpha_1,\alpha_2,\ldots,\alpha_n$ are linearly independent.

$\therefore \quad S'$ is linearly independent.

vectors. Therefore S' must be a basis of V. Therefore each vector in V can be expressed as a linear combination of the vectors belonging to S'.

Now we shall show that f is onto V. Let α be any element of V. Then there exist scalars $c_1, c_2, \ldots, c_n$ such that

$$\alpha = c_1 \, f(\alpha_1) + c_2 \, f(\alpha_2) + \ldots + c_n \, f(\alpha_n)$$
$$= f(c_1 \alpha_1 + c_2 \alpha_2 + \ldots + c_n \alpha_n).$$

Now $c_1 \alpha_1 + c_2 \alpha_2 + \ldots + c_n \alpha_n \in V$ and the f-image of this element is α. Therefore f is onto V. Hence f is an isomorphism of V onto V.

Example 4: *If V is finite dimensional and f is a homomorphism of V onto V prove that f must be one-one, and so, an isomorphism.*

Solution: Let $V(F)$ be a finite dimensional vector space of dimension n. Let f be a homomorphism of V onto V *i.e.,* f is a linear transformation and f is onto V. To prove that f is one-one.

Let $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a basis of V. We shall first prove that

$$S' = \{f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_n)\}$$

is also a basis of V. We claim that $L(S') = V$. The proof is as follows :

Let α be any element of V. We shall show that α can be expressed as a linear combination of $f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_n)$. Since f is onto V, therefore $\alpha \in V$ implies that there exists $\beta \in V$ such that $f(\beta) = \alpha$. Now β can be expressed as a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_n$. Let

$$\beta = a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n .$$

Then
$$\alpha = f(\beta) = f(a_1 \alpha_1 + a_2 \alpha_2 + \ldots + a_n \alpha_n)$$
$$= a_1 \, f(\alpha_1) + a_2 \, f(\alpha_2) + \ldots + a_n \, f(\alpha_n).$$

Thus α has been expressed as a linear combination of

$$f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_n).$$

Therefore $L(S') = V$.

Since V is of dimension n and S' is a subset of V containing n vectors and $L(S') = V$, therefore S' must be a basis of V. Therefore each vector in V can be expressed as a linear combination of the vectors belonging to S' and S' is linearly independent.

Now we shall show that f is one-one. Let γ and δ be any two elements of V such that

$$\gamma = c_1 \alpha_1 + c_2 \alpha_2 + \ldots + c_n \alpha_n , \quad \delta = d_1 \alpha_1 + d_2 \alpha_2 + \ldots + d_n \alpha_n .$$

We have $f(\gamma) = f(\delta)$

$$\Rightarrow \quad f(c_1 \alpha_1 + c_2 \alpha_2 + \ldots + c_n \alpha_n) = f(d_1 \alpha_1 + d_2 \alpha_2 + \ldots + d_n \alpha_n)$$

$$\Rightarrow \quad c_1 \, f(\alpha_1) + c_2 \, f(\alpha_2) + \ldots + c_n \, f(\alpha_n) = d_1 \, f(\alpha_1) + d_2 \, f(\alpha_2) + \ldots + d_n \, f(\alpha_n)$$

$$\Rightarrow \quad (c_1 - d_1) \, f(\alpha_1) + (c_2 - d_2) \, f(\alpha_2) + \ldots + (c_n - d_n) \, f(\alpha_n) = \mathbf{0}$$

$f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_n)$ are linearly independent

$\Rightarrow \quad c_1 = d_1, c_2 = d_2, \ldots, c_n = d_n$

$\Rightarrow \quad \gamma = \delta.$

$\therefore \quad f$ is one-one.

$\therefore \quad f$ is an isomorphism of V onto V.

Example 5: *If $f : U \to V$ is an isomorphism of the vector space U into the vector space V, then a set of vectors $\{f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_r)\}$ is linearly independent if and only if the set $\{\alpha_1, \alpha_2, \ldots, \alpha_r\}$ is linearly independent.*

Solution: $U(F)$ and $V(F)$ are two vector spaces over the same field F and f is an isomorphism of U into V *i.e.*, $f : U \to V$ such that f is 1-1 and

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) \ \forall \ a, b \in F \text{ and } \forall \ \alpha, \beta \in U.$$

Let $\{\alpha_1, \alpha_2, \ldots, \alpha_r\}$ be a subset of U. First suppose that the vectors $\alpha_1, \alpha_2, \ldots, \alpha_r$ are linearly independent. Then to show that the vectors

$$f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_r)$$

are also linearly independent.

We have $a_1 f(\alpha_1) + a_2 f(\alpha_2) + \ldots + a_r f(\alpha_r) = \mathbf{0}'$,

where $\quad a_1, a_2, \ldots, a_r \in F$

$\Rightarrow \quad f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_r\alpha_r) = \mathbf{0}'$ [$\because f$ is a linear transformation]

$\Rightarrow \quad f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_r\alpha_r) = f(\mathbf{0})$ $\quad$ [$\because f(\mathbf{0}) = \mathbf{0}'$]

$\Rightarrow \quad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_r\alpha_r = \mathbf{0}$ $\quad$ [$\because f$ is 1-1]

$\Rightarrow \quad a_1 = 0, a_2 = 0, \ldots, a_r = 0$

since the vectors $\alpha_1, \alpha_2, \ldots, \alpha_r$ are linearly independent.

Hence the vectors $f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_r)$ are also linearly independent.

Conversely suppose that the vectors

$$f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_r)$$

are linearly independent. Then to show that the vectors $\alpha_1, \alpha_2, \ldots, \alpha_r$ are also linearly independent.

We have $a_1\alpha_1 + a_2\alpha_2 + \ldots + a_r\alpha_r = \mathbf{0}$, where $a_1, a_2, \ldots, a_r \in F$

$\Rightarrow \quad f(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_r\alpha_r) = f(\mathbf{0})$

$\Rightarrow \quad a_1 f(\alpha_1) + a_2 f(\alpha_2) + \ldots + a_r f(\alpha_r) = \mathbf{0}'$

[$\because f$ is a linear transformation]

$\Rightarrow \quad a_1 = 0, a_2 = 0, \ldots, a_r = 0$

Since the vectors $f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_r)$ are linearly independent.

Hence the vectors $\alpha_1, \alpha_2, \ldots, \alpha_r$ are also linearly independent.

1. Let $T : V_2(\mathbf{R}) \to V_2(\mathbf{R})$ be defined as

$$T(a_1, b_1) = (b_1, a_1).$$

Show that T is an isomorphism.

2. Define linear transformation of a vector space $V(F)$ into a vector space $W(F)$. Show that the mapping

$$T : (a, b) \to (a + 2, b + 3)$$

of $V_2(\mathbf{R})$ into itself is not a linear transformation.

3. $V(F)$ and $W(F)$ are two finite dimensional vector spaces such that dim $V =$ dim W. If f is an isomorphism of V into W, prove that f must map V into W.

4. Let f be a linear transformation from a vector space U into a vector space V. If S is a subspace of U, prove that $f(S)$ will be a subspace of V.

5. If V is finite dimensional and f is a homomorphism of V into itself which is not onto prove that there is some $\alpha \neq \mathbf{0}$ in V such that $f(\alpha) = \mathbf{0}$.

6. If $f : (U \to V)$ is an isomorphism of the vector space U into the vector space V, then a set of vectors $f(\alpha_1), f(\alpha_2), \ldots, f(\alpha_r)$ is linearly dependent in V if and only if the set $\alpha_1, \alpha_2, \ldots \alpha_r$ is linearly dependent in U.

7. Give an example of a one-one linear transformation of an infinite dimensional vector space which is not an isomorphism.

8. If T be a linear operator on a finite dimensional vector space V, show that T is one-one if and only if T is onto.

9. If f is an isomorphism of a vector space V onto a vector space W, prove that f maps a basis of V onto a basis of W.

10. Prove that a finite dimensional vector space $V(\mathbf{R})$ with dimension $V = n$ is isomorphic to $\mathbf{R}^n$.

4.4 Quotient Space

(Garhwal 2007, 10, 12)

Right and Left Cosets.

Let W be any subspace of a vector space $V(F)$. Let α be any element of V. Then the set

$$W + \alpha = \{\gamma + \alpha : \gamma \in W\}$$

is called a **right coset** of W in V generated by α. Similarly the set

$$\alpha + W = \{\alpha + \gamma : \gamma \in W\}$$

is called a **left coset** of W in V generated by α.

commutative, therefore we have $W + \alpha = \alpha + W$. Hence we shall call $W + \alpha$ as simply a **coset** of W in V generated by α.

The following results about cosets are both to be remembered :

(i) *We have* $\mathbf{0} \in V$ and $W + \mathbf{0} = W$. *Therefore W itself is a coset of W in V.*

(ii) $\alpha \in W \Rightarrow W + \alpha = W$.

Proof. First we shall prove that $W + \alpha \subseteq W$.

Let $\gamma + \alpha$ be any arbitrary element of $W + \alpha$. Then $\gamma \in W$.

Now W is a subspace of V. Therefore

$$\gamma \in W, \alpha \in W \Rightarrow \gamma + \alpha \in W.$$

Thus every element of $W + \alpha$ is also an element of W. Hence

$$W + \alpha \subseteq W.$$

Now we shall prove that $W \subseteq W + \alpha$.

Let $\beta \in W$. Since W is a subspace, therefore

$$\alpha \in W \Rightarrow -\alpha \in W.$$

Consequently $\beta \in W, -\alpha \in W \Rightarrow \beta - \alpha \in W$. Now we can write

$$\beta = (\beta - \alpha) + \alpha \in W + \alpha \text{ since } \beta - \alpha \in W.$$

Thus $\qquad \beta \in W \Rightarrow \beta \in W + \alpha$. Therefore $W \subseteq W + \alpha$.

Hence $\qquad W = W + \alpha$.

(iii) *If $W + \alpha$ and $W + \beta$ are two cosets of W in V, then*

$$W + \alpha = W + \beta \Leftrightarrow \alpha - \beta \in W.$$

Proof. Since $\mathbf{0} \in W$, therefore $\mathbf{0} + \alpha \in W + \alpha$. Thus

$$\alpha \in W + \alpha.$$

Now $\qquad W + \alpha = W + \beta \Rightarrow \alpha \in W + \beta$

$\Rightarrow \qquad \alpha - \beta \in W + (\beta - \beta)$

$\Rightarrow \qquad \alpha - \beta \in W + \mathbf{0} \Rightarrow \alpha - \beta \in W$.

Conversely,

$$\alpha - \beta \in W \Rightarrow W + (\alpha - \beta) = W$$

$\Rightarrow \qquad W + [(\alpha - \beta) + \beta] = W + \beta$

$\Rightarrow \qquad W + \alpha = W + \beta$.

Let V/W denote the set of all cosets of W in V *i.e.*, let

$$V/W = \{W + \alpha : \alpha \in V\}.$$

We have just seen that if $\alpha - \beta \in W$, then $W + \alpha = W + \beta$. Thus a coset of W in V can have more than one representation.

Now if $V(F)$ is a vector space, then we shall give a vector space structure to the set V/W over the same field F. For this we shall have to define addition in V/W *i.e.*, addition of cosets of W in V and multiplication of a coset by an element of F *i.e.*, scalar multiplication.

$W + \alpha$ where α is any arbitrary element of V, is a vector space over F for the addition and scalar multiplication compositions defined as follows :

$$(W + \alpha) + (W + \beta) = W + (\alpha + \beta) \ \forall \ \alpha, \beta \in V$$

and $$a\,(W + \alpha) = W + a\alpha \ ; a \in F, \alpha \in V.$$

Proof: We have $\alpha, \beta \in V \ \Rightarrow \ \alpha + \beta \in V$.

Also $a \in F, \alpha \in V \ \Rightarrow \ a\alpha \in V$.

Therefore $W + (\alpha + \beta) \in V / W$ and also $W + a\alpha \in V / W$. Thus V / W is closed with respect to addition of cosets and scalar multiplication as defined above. Now first of all we shall show that these two compositions are well defined *i.e.*, are independent of the particular representation chosen to denote a coset.

Let $W + \alpha = W + \alpha', \alpha, \alpha' \in V$

and $W + \beta = W + \beta', \beta, \beta' \in V$.

We have $W + \alpha = W + \alpha' \ \Rightarrow \ \alpha - \alpha' \in W$

and $W + \beta = W + \beta' \ \Rightarrow \ \beta - \beta' \in W$.

Now W is a subspace, therefore

$$\alpha - \alpha' \in W, \beta - \beta' \in W$$

$$\Rightarrow \quad (\alpha - \alpha') + (\beta - \beta') \in W$$

$$\Rightarrow \quad (\alpha + \beta) - (\alpha' + \beta') \in W$$

$$\Rightarrow \quad W + (\alpha + \beta) = W + (\alpha' + \beta')$$

$$\Rightarrow \quad (W + \alpha) + (W + \beta) = (W + \alpha') + (W + \beta').$$

Therefore addition in V/W is well defined.

Again $a \in F, \alpha - \alpha' \in W \ \Rightarrow \ a\,(\alpha - \alpha') \in W$

$\Rightarrow$ $a\alpha - a\alpha' \in W \ \Rightarrow \ W + a\alpha = W + a\alpha'$.

$\therefore$ scalar multiplication in V/W is also well defined.

Commutativity of addition. Let $W + \alpha$, $W + \beta$ be any two elements of V/W. Then

$$(W + \alpha) + (W + \beta) = W + (\alpha + \beta) = W + (\beta + \alpha)$$
$$= (W + \beta) + (W + \alpha).$$

Associativity of addition. Let $W + \alpha$, $W + \beta$, $W + \gamma$ be any three elements of V/W. Then

$$\begin{aligned}(W + \alpha) + [(W + \beta) + (W + \gamma)] &= (W + \alpha) + [W + (\beta + \gamma)] \\ &= W + [\alpha + (\beta + \gamma)] \\ &= W + [(\alpha + \beta) + \gamma] \\ &= [W + (\alpha + \beta)] + (W + \gamma) \\ &= [(W + \alpha) + (W + \beta)] + (W + \gamma).\end{aligned}$$

Existence of additive identity. If $\mathbf{0}$ is the zero vector of V, then

$$W + \mathbf{0} = W \in V/W.$$

If $W + \alpha$ is any element of V/W, then

$\therefore \qquad W + \mathbf{0} = W$ is the additive identity.

Existence of additive inverse. If $W + \alpha$ is any element of V / W, then

$$W + (-\alpha) = W - \alpha \in V/W.$$

Also we have

$$(W + \alpha) + (W - \alpha) = W + (\alpha - \alpha) = W + \mathbf{0} = W.$$

$\therefore \quad W - \alpha$ is the additive inverse of $W + \alpha$.

Thus V/W is an abelian group with respect to addition composition. Further we observe that if

$a, b \in F$ and $W + \alpha, W + \beta \in V / W$, then

1. $a\,[(W + \alpha) + (W + \beta)] = a\,[W + (\alpha + \beta)]$

$$= W + a\,(\alpha + \beta) = W + (a\alpha + a\beta)$$
$$= (W + a\alpha) + (W + a\beta)$$
$$= a\,(W + \alpha) + a\,(W + \beta).$$

2. $(a + b)\,(W + \alpha) = W + (a + b)\,\alpha$

$$= W + (a\alpha + b\alpha)$$
$$= (W + a\alpha) + (W + b\alpha)$$
$$= a\,(W + \alpha) + b\,(W + \alpha).$$

3. $(ab)\,(W + \alpha) = W + (ab)\,\alpha = W + a\,(b\alpha)$

$$= a\,(W + b\alpha) = a\,[b\,(W + \alpha)].$$

4. $1\,(W + \alpha) = W + 1\alpha = W + \alpha.$

$\therefore \; V/W$ is a vector space over F for these two compositions. The vector space V/W is called the **Quotient Space** of V relative to W. The coset W is the zero vector of this vector space.

4.5 Dimension of a Quotient Space

Theorem: *If W be a subspace of a finite dimensional vector space $V\,(F)$, then*

$$\dim V/W = \dim V - \dim W. \qquad \textbf{(Garhwal 2006, 07, 08, 10)}$$

Proof: Let m be the dimension of the subspace W of the vector space $V\,(F)$.

Let $\qquad S = \{\alpha_1, \alpha_2, \ldots, \alpha_m\}$

be a basis of W. Since S is a linearly independent subset of V, therefore it can be extended to form a basis of V. Let

$$S' = \{\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_l\}$$

be a basis of V. Then $\dim V = m + l$.

$\therefore \qquad \dim V - \dim W = (m + l) - m = l.$

So we should prove that $\dim V / W = l$.

$$S_1 = \{W + \beta_1, W + \beta_2, \dots, W + \beta_l\}$$
is a basis of V/W.

First we show that S_1 is linearly independent. The zero vector of V/W is W.

Let $\quad a_1 (W + \beta_1) + a_2 (W + \beta_2) + \dots + a_l (W + \beta_l) = W$

$\Rightarrow \quad (W + a_1 \beta_1) + (W + a_2 \beta_2) + \dots + (W + a_l \beta_l) = W$

$\Rightarrow \quad W + (a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l) = W + \mathbf{0}$

$\Rightarrow \quad a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l \in W$

$\Rightarrow \quad a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l = b_1 \alpha_1 + b_2 \alpha_2 + \dots + b_m \alpha_m$

[$\because$ any vector in W can be expressed as a linear combination of its basis vectors]

$\Rightarrow \quad a_1 \beta_1 + a_2 \beta_2 + \dots + a_l \beta_l - b_1 \alpha_1 - b_2 \alpha_2 - \dots - b_m \alpha_m = \mathbf{0}$

$\Rightarrow \quad a_1 = 0, a_2 = 0, \dots, a_l = 0$ since the vectors $\beta_1, \beta_2, \dots \beta_l, \alpha_1, \alpha_2, \dots, \alpha_m$ are linearly independent.

$\therefore$ The set S_1 is linearly independent.

Now to show that $L(S_1) = V/W$. Let $W + \alpha$ be any element of V/W. The vector $\alpha \in V$ can be expressed as

$$\alpha = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l$$
$$= \gamma + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l \quad \text{where}$$
$$\gamma = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_m \alpha_m \in W.$$

So
$$\begin{aligned} W + \alpha &= W + (\gamma + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l) \\ &= (W + \gamma) + d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l \\ &= W + (d_1 \beta_1 + d_2 \beta_2 + \dots + d_l \beta_l) \quad [\because \gamma \in W \Rightarrow W + \gamma = W] \\ &= (W + d_1 \beta_1) + (W + d_2 \beta_2) + \dots + (W + d_l \beta_l) \\ &= d_1 (W + \beta_1) + d_2 (W + \beta_2) + \dots + d_l (W + \beta_l). \end{aligned}$$

Thus any element $W + \alpha$ of V/W can be expressed as a linear combination of elements of S_1.

$\therefore \quad V/W = L(S_1). \quad \therefore \quad S_1$ is a basis of V/W.

$\therefore \quad \dim V/W = l.$ Hence the theorem.

4.6 Direct Sum of Spaces

(Garhwal 2006, 10)

Vector space as a direct sum of subspaces.

Definition: *Let $V(F)$ be a vector space and let $W_1, W_2, \dots, W_m$ be subspaces of V. Then V is said to be the direct sum of $W_1, W_2, \dots, W_m$ if every element $\alpha \in V$ can be written in one and only one way as $\alpha = \alpha_1 + \alpha_2 + \dots + \alpha_m$ where*
$$\alpha_1 \in W_1, \alpha_2 \in W_2, \dots, \alpha_m \in W_m.$$

If a vector space $V(F)$ is a direct sum of its two subspaces W_1 and W_2 then we should have not only $V = W_1 + W_2$ but also that each vector of V can be uniquely direct sum is represented by the notation $V = W_1 \oplus W_2$.

Illustration 1. *Let $V_2(F)$ be the vector space of all ordered pairs of F. Then $W_1 = \{(a, 0) : a \in F\}$ and $W_2 = \{(0, b) : b \in F\}$ are two subspaces of $V_2(F)$. Obviously any element $(x, y) \in V_2(F)$ can be uniquely expressed as the sum of an element of W_1 and an element of W_2. The unique expression is $(x, y) = (x, 0) + (0, y)$. Thus $V_2(F)$ is the direct sum of W_1 and W_2. Also we observe that the only element common to both W_1 and W_2 is the zero vector* $(0, 0)$.

Illustration 2. *Let $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and $W_2 = \{(0, 0, c) : c \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$.*

Now any vector $(a, b, c) \in \mathbf{R}^3$ can be uniquely expressed as the sum of a vector in W_1 and a vector in W_2 : $(a, b, c) = (a, b, 0) + (0, 0, c)$.

Thus $\mathbf{R}^3$ is the direct sum of W_1 and W_2.

Also we observe that the only element common to both W_1 and W_2 is the zero vector $(0, 0, 0)$.

Illustration 3. *Let $W_1 = \{(a, b, 0) : a, b \in \mathbf{R}\}$ and $W_2 = \{(0, b, c) : b, c \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$. Then $\mathbf{R}^3 = W_1 + W_2$ since every vector in $\mathbf{R}^3$ is the sum of a vector in W_1 and a vector in W_2. However, $\mathbf{R}^3$ is not the direct sum of W_1 and W_2 since such sums are not unique.*

For example, $(4, 6, 8) = (4, 5, 0) + (0, 1, 8)$.

Also $\qquad (4, 6, 8) = (4, -1, 0) + (0, 7, 8)$.

Here we observe that $W_1 \cap W_2 \neq \{\mathbf{0}\}$.

4.7 Disjoint Subspaces

Definition. *Two subspaces W_1 and W_2 of the vector space $V(F)$ are said to be disjoint if their intersection is the zero subspace i.e., if $W_1 \cap W_2 = \{\mathbf{0}\}$.*

Theorem. *The necessary and sufficient conditions for a vector space $V(F)$ to be a direct sum of its two subspaces W_1 and W_2 are that*

(i) $V = W_1 + W_2$

(ii) $W_1 \cap W_2 = \{\mathbf{0}\}$ *i.e., W_1 and W_2 are disjoint.* **(Garhwal 2010)**

Proof: The conditions are necessary.

Let V be direct sum of its two subspaces W_1 and W_2. Then each element of V is expressible uniquely as sum of an element of W_1 and an element of W_2. Therefore we have

$$V = W_1 + W_2.$$

Let, if possible $\mathbf{0} \neq \alpha \in W_1 \cap W_2$. Then $\alpha \in W_1, \alpha \in W_2$. Also $\alpha \in V$ and we can write and $\qquad \alpha = \alpha + \mathbf{0}$ where $\alpha \in W_1, \mathbf{0} \in W_2$.

Thus $\alpha \in V$ can be expressed in at least two different ways as sum of an element of W_1 and an element of W_2. This contradicts the fact that V is direct sum of W_1 and W_2. Hence $\mathbf{0}$ is the only vector common to both W_1 and W_2 *i.e.*, $W_1 \cap W_2 = \{\mathbf{0}\}$. Thus the conditions are necessary.

The conditions are sufficient.

Let $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$. Then to show that V is direct sum of W_1 and W_2.

$V = W_1 + W_2 \Rightarrow$ that each element of V can be expressed as sum of an element of W_1 and an element of W_2. Now to show that this expression is unique.

Let, if possible,

$$\alpha = \alpha_1 + \alpha_2, \alpha \in V, \alpha_1 \in W_1, \alpha_2 \in W_2,$$

and $\qquad \alpha = \beta_1 + \beta_2, \beta_1 \in W_1, \beta_2 \in W_2$.

Then to show that $\alpha_1 = \beta_1$ and $\alpha_2 = \beta_2$.

We have $\qquad \alpha_1 + \alpha_2 = \beta_1 + \beta_2$

$\Rightarrow \qquad \alpha_1 - \beta_1 = \beta_2 - \alpha_2$.

Since W_1 is a subspace, therefore

$$\alpha_1 \in W_1, \beta_1 \in W_1 \Rightarrow \alpha_1 - \beta_1 \in W_1.$$

Similarly, $\qquad \beta_2 - \alpha_2 \in W_2$.

$\therefore \qquad \alpha_1 - \beta_1 = \beta_2 - \alpha_2 \in W_1 \cap W_2$.

But $\mathbf{0}$ is the only vector which belongs to $W_1 \cap W_2$. Therefore

$$\alpha_1 - \beta_1 = \mathbf{0} \Rightarrow \alpha_1 = \beta_1.$$

Also $\qquad \beta_2 - \alpha_2 = \mathbf{0} \Rightarrow \alpha_2 = \beta_2$.

Thus each vector $\alpha \in V$ is uniquely expressible as sum of an element of W_1 and an element of W_2. Hence $V = W_1 \oplus W_2$.

4.8 Dimension of a Direct Sum

Theorem: *If a finite dimensional vector space $V(F)$ is a direct sum of two subspaces W_1 and W_2, then*

$$dim\ V = dim\ W_1 + dim\ W_2.$$

Proof: Let dim $W_1 = m$ and dim $W_2 = l$. Also let the sets of vectors

$$S_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_m\} \quad \text{and} \quad S_2 = \{\beta_1, \beta_2, \ldots, \beta_l\}$$

be the bases of W_1 and W_2 respectively.

We have dim W_1 + dim $W_2 = m + l$.

In order to prove that dim V = dim W_1 + dim W_2, we should therefore prove that dim $V = m + l$. We claim that the set

$$S = S_1 \cup S_2 = \{\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_l\}$$

First we show that the set S is linearly independent. Let

$$a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l = \mathbf{0}.$$

$\Rightarrow \quad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m = -(b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l).$

Now $\quad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m \in W_1$

and $\quad -(b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l) \in W_2.$

Therefore $\quad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m \in W_1 \cap W_2$

and $\quad -(b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l) \in W_1 \cap W_2.$

But V is the direct sum of W_1 and W_2. Therefore $\mathbf{0}$ is the only vector belonging to $W_1 \cap W_2$. Then we have

$$a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m = \mathbf{0},\ b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l = \mathbf{0}.$$

Since both the sets $\{\alpha_1, \alpha_2, \ldots, \alpha_m\}$ and $\{\beta_1, \beta_2, \ldots, \beta_l\}$ are linearly independent, therefore we have

$$a_1 = 0, a_2 = 0, \ldots, a_m = 0, b_1 = 0, b_2 = 0, \ldots, b_l = 0.$$

Therefore S is linearly independent.

Now we shall show that $L(S) = V$. Let α be any element of V. Then

$\alpha =$ an element of W_1 + an element of W_2.

$\quad =$ a linear combination of S_1 + a linear combination of S_2

$\quad =$ a linear combination of elements of S.

$\therefore \quad L(S) = V$

$\therefore \quad S$ is a basis of V. Therefore dim $V = m + l$.

Hence the theorem.

A sort of converse of this theorem is true. It has been proved in the following theorem.

Theorem: *Let V be a finite dimensional vector space and let W_1, W_2 be subspaces of V such that* $\quad V = W_1 + W_2$ *and* $\dim V = \dim W_1 + \dim W_2$.

Then $\quad V = W_1 \oplus W_2$.

Proof: Let dim $W_1 = l$ and dim $W_2 = m$. Then

$$\dim V = l + m.$$

Let $\quad S_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_l\}$

be a basis of W_1 and

$$S_2 = \{\beta_1, \beta_2, \ldots, \beta_m\}$$

be a basis of W_2. We shall show that $S_1 \cup S_2$ is a basis of V.

Let $\quad \alpha \in V$. Since $V = W_1 + W_2$, therefore we can write

$$\alpha = \gamma + \delta \text{ where } \gamma \in W_1, \delta \in W_2.$$

Now $\gamma \in W_1$ can be expressed as a linear combination of the elements of S_1 and $\delta \in W_2$ can be expressed as a linear combination of the elements of S_2. Therefore $\alpha \in V$ can be expressed as a linear combination of the elements of $S_1 \cup S_2$. Therefore $V = L(S_1 \cup S_2)$. Since dim $V = l + m$ and $L(S_1 \cup S_2) = V$, therefore

$S_1 \cup S_2$ has $l+m$ distinct elements and therefore $S_1 \cup S_2$ is a basis of V. Therefore the set

$$\{\alpha_1, \alpha_2, \dots, \alpha_l, \beta_1, \beta_2, \dots, \beta_m\}$$

is linearly independent.

Now we shall show that

$$W_1 \cap W_2 = \{\mathbf{0}\}.$$

Let $\alpha \in W_1 \cap W_2$.

Then $\quad \alpha \in W_1, \alpha \in W_2$.

Therefore $\quad \alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_l\alpha_l$

and $\quad \alpha = b_1\beta_1 + b_2\beta_2 + \dots + b_m\beta_m$

for some a's and b's $\in F$.

$\therefore \quad a_1\alpha_1 + a_2\alpha_2 + \dots + a_l\alpha_l = b_1\beta_1 + b_2\beta_2 + \dots + b_m\beta_m$

$\Rightarrow \quad a_1\alpha_1 + a_2\alpha_2 + \dots + a_l\alpha_l - b_1\beta_1 - b_2\beta_2 - \dots - b_m\beta_m = \mathbf{0}$

$\Rightarrow \quad a_1 = 0, a_2 = 0, \dots, a_l = 0, b_1 = 0, b_2 = 0, \dots, b_m = 0$

$\Rightarrow \quad \alpha = \mathbf{0}$.

$\therefore \quad W_1 \cap W_2 = \{\mathbf{0}\}$.

4.9 Complementary Subspaces

Definition. *Let $V(F)$ be a vector space and W_1, W_2 be two subspaces of V. Then the subspace W_2 is called the complement of W_1 in V if V is the direct sum of W_1 and W_2.*

Existence of complementary subspaces.

Theorem: *Corresponding to each subspace W_1 of a finite dimensional vector space $V(F)$, there exists a subspace W_2 such that V is the direct sum of W_1 and W_2.*

Proof. Let dim $W_1 = m$. Let the set

$$S_1 = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$$

be a basis of W_1.

Since S_1 is a linearly independent subset of V, therefore S_1 can be extended to form a basis of V. Let the set

$$S = \{\alpha_1, \alpha_2, \dots, \alpha_m, \beta_1, \beta_2, \dots, \beta_l\}$$

be a basis of V.

Let W_2 be the subspace of V generated by the set

$$S_2 = \{\beta_1, \beta_2, \dots, \beta_l\}.$$

We shall prove that V is the direct sum of W_1 and W_2. For this we shall prove that $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$.

Let α be any element of V. Then we can express

$$\alpha = \text{a linear combination of elements of } S$$

$$= \text{an element of } W_1 + \text{an element of } W_2.$$

$\therefore \quad V = W_1 + W_2.$

Again let $\beta \in W_1 \cap W_2$. Then β can be expressed as a linear combination of S_1 and also as a linear combination of S_2. So we have

$$\beta = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m = b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l.$$

$\Rightarrow \quad a_1\alpha_1 + a_2\alpha_2 + \ldots + a_m\alpha_m - b_1\beta_1 - b_2\beta_2 - \ldots - b_l\beta_l = \mathbf{0}$

$\Rightarrow \quad a_1 = 0, a_2 = 0, \ldots, a_m = 0, b_1 = 0, b_2 = 0, \ldots, b_l = 0$

since $\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_l$ are linearly independent.

$\therefore \quad \beta = \mathbf{0}$ (zero vector).

Thus $\quad W_1 \cap W_2 = \{\mathbf{0}\}$.

Hence V is the direct sum of W_1 and W_2.

Dimension of a Quotient space. Alternative method.

Theorem: *If W_1 and W_2 are complementary subspaces of a vector space V, then the mapping f which assigns to each vector β in W_2 the coset $W_1 + \beta$ is an isomorphism between* W_2 and V/W_1.

Proof. It is given that

$$V = W_1 \oplus W_2$$

and $\quad f : W_2 \to V/W_1$ such that

$$f(\beta) = W_1 + \beta \;\forall\; \beta \in W_2.$$

We shall show that f is an isomorphism of W_2 onto V/W_1.

(i) ***f* is one-one.**

If $\beta_1, \beta_2 \in W_2$, then

$f(\beta_1) = f(\beta_2) \Rightarrow W_1 + \beta_1 = W_1 + \beta_2$ [by def. of f]

$\Rightarrow \quad \beta_1 - \beta_2 \in W_1$

$\Rightarrow \quad \beta_1 - \beta_2 \in W_1 \cap W_2$ [$\because$ $\beta_1 - \beta_2 \in W_2$ because W_2 is a subspace]

$\Rightarrow \quad \beta_1 - \beta_2 = \mathbf{0}$ [$\because$ $W_1 \cap W_2 = \{\mathbf{0}\}$]

$\Rightarrow \quad \beta_1 = \beta_2.$

$\therefore \quad f$ is one-one.

(ii) ***f* is onto.**

Let $W_1 + \alpha$ be any coset in V/W_1, where $\alpha \in V$. Since V is direct sum of W_1 and W_2, therefore we can write

$$\alpha = \gamma + \beta \text{ where } \gamma \in W_1, \beta \in W_2.$$

This gives $\gamma = \alpha - \beta \in W_1$.

Since $\quad \alpha - \beta \in W_1$, therefore $W_1 + \alpha = W_1 + \beta$.

Now $\quad f(\beta) = W_1 + \beta$ [by def. of f]

$$= W_1 + \alpha.$$

Thus $W_1 + \alpha \in V/W_1 \Rightarrow \exists\; \beta \in W_2$ such that

Therefore f is onto.

(iii) **f is a linear transformation.**

Let $a, b \in F$ and $\beta_1, \beta_2 \in W_2$.

Then $$f(a\beta_1 + b\beta_2) = W_1 + (a\beta_1 + b\beta_2)$$
$$= (W_1 + a\beta_1) + (W_1 + b\beta_2)$$
$$= a(W_1 + \beta_1) + b(W_1 + \beta_2)$$
$$= af(\beta_1) + bf(\beta_2).$$

Therefore f is a linear transformation.

Hence f is an isomorphism between W_2 and V/W_1.

Corollary: Dimension of a quotient space. *If W is an m-dimensional subspace of an n-dimensional vector space V, then the dimension of the quotient space V/W is $n - m$.*

(Garhwal 2006)

Proof: Since W is a subspace of a finite dimensional vector space V, therefore there exists a subspace W_1 of V such that

$$V = W \oplus W_1.$$

Also $$\dim V = \dim W + \dim W_1$$

or $$\dim W_1 = \dim V - \dim W = n - m.$$

Now by the above theorem, we have

$$V/W \cong W_1.$$

$$\therefore \quad \dim V/W = \dim W_1 = n - m.$$

4.10 Direct Sum of Several Subspaces

Now we shall have some discussion on the direct sum of several subspaces. For this purpose we shall first define the concept of independence of subspaces analogous to the disjointness condition on two subspaces.

Definition: *Suppose $W_1, W_2, \ldots, W_k$ are subspaces of the vector space V. We shall say that $W_1, W_2, \ldots, W_k$ are* ***independent****, if*

$$\alpha_1 + \alpha_2 + \ldots + \alpha_k = \mathbf{0}, \alpha_i \in W_i \text{ implies that each } \alpha_i = \mathbf{0}.$$

Theorem 1: *Suppose $V(F)$ is a vector space. Let $W_1, W_2, \ldots, W_k$ be subspaces of V and let $W = W_1 + W_2 + \ldots + W_k$. The following are equivalent :*

(i) $W_1, W_2, \ldots, W_k$ are independent.

(ii) Each vector α in W can be uniquely expressed in the form

$$\alpha = \alpha_1 + \alpha_2 + \ldots + \alpha_k \text{ with } \alpha_i \text{ in } W_i, i = 1, \ldots, k.$$

(iii) For each $j, 2 \le j \le k$, the subspace W_j is disjoint from the sum

$$(W_1 + \ldots + W_{j-1}).$$

Proof: In order to prove the equivalence of the three statements we shall prove that (i) $\Rightarrow$ (ii), (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (i).

Since $W = W_1 + \ldots + W_k$, therefore we can write

$$\alpha = \alpha_1 + \ldots + \alpha_k \text{ with } \alpha_i \text{ in } W_i .$$

Suppose that also $\alpha = \beta_1 + \ldots + \beta_k$ with β_i in W_i.

Then $\alpha_1 + \ldots + \alpha_k = \beta_1 + \ldots + \beta_k$

$\Rightarrow \quad (\alpha_1 - \beta_1) + \ldots + (\alpha_k - \beta_k) = \mathbf{0}$ with $\alpha_i - \beta_i$ in W_i as W_i is a subspace

$\Rightarrow \quad \alpha_i - \beta_i = \mathbf{0}$ [$\because$ $W_1, \ldots, W_k$ are independent]

$\Rightarrow \quad \alpha_i = \beta_i$, $i = 1, \ldots, k$.

Therefore the α_i's are uniquely determined by α.

(ii) $\Rightarrow$ (iii). Let $\alpha \in W_j \cap (W_1 + \ldots + W_{j-1})$.

Then $\alpha \in W_j$ and $\alpha \in W_1 + \ldots + W_{j-1}$.

Now $\alpha \in W_1 + \ldots + W_{j-1}$ implies that there exist vectors $\alpha_1, \ldots, \alpha_{j-1}$ with α_i in W_i such that

$$\alpha = \alpha_1 + \ldots + \alpha_{j-1} .$$

Also $\qquad \alpha \in W_j$.

Therefore we get two expressions for α as a sum of vectors, one in each W_i. These are

$$\alpha = \alpha_1 + \ldots + \alpha_{j-1} + \mathbf{0} + \ldots + \mathbf{0}$$

in which the vector belonging to W_j is $\mathbf{0}$

and $\qquad \alpha = \mathbf{0} + \ldots + \mathbf{0} + \alpha + \ldots + \mathbf{0}$

in which the vector belonging to W_j is α.

Since the expression for α is given to be unique, therefore we must have

$$\alpha_1 = \ldots = \alpha_{j-1} = \mathbf{0} = \alpha.$$

Thus $\qquad W_j \cap (W_1 + \ldots + W_{j-1}) = \{\mathbf{0}\}$.

(iii) $\Rightarrow$ (i).

Let $\alpha_1 + \ldots + \alpha_k = \mathbf{0}$ where $\alpha_i \in W_i$, $i = 1, \ldots, k$. ...(1)

Then we are to prove that each $\alpha_i = \mathbf{0}$.

Suppose that for some i we have $\alpha_i \neq \mathbf{0}$.

Let j be the largest integer i between 1 and k such that $\alpha_i \neq \mathbf{0}$. Obviously j must be ≥ 2 and at the most j can be equal to k. Then (1) reduces to

$$\alpha_1 + \ldots + \alpha_j = \mathbf{0}, \alpha_j \neq \mathbf{0}$$

$\Rightarrow \quad \alpha_j = -\alpha_1 - \ldots - \alpha_{j-1}$

$\Rightarrow \quad \alpha_j \in W_1 + \ldots + W_{j-1}$ [$\because$ $-\alpha_1 - \ldots - \alpha_{j-1} \in W_1 + \ldots + W_{j-1}$]

$\Rightarrow \quad \alpha_j \in W_j \cap (W_1 + \ldots + W_{j-1})$ [$\Rightarrow$ $\alpha_j \in W_j$]

$\Rightarrow \quad \alpha_j = \mathbf{0}$.

Thus we get a contradiction. Hence each $\alpha_i = \mathbf{0}$.

Note: If any (and hence all) of the three conditions of theorem 1 hold for $W_1, \ldots, W_k$, then we shall say that W is the **direct sum** of $W_1, \ldots, W_k$ and we write

$$W = W_1 \oplus \ldots \oplus W_k .$$

that

$$V = W_1 + \ldots + W_n$$

and that $\quad W_i \cap (W_1 + \ldots + W_{i-1} + W_{i+1} + \ldots + W_n) = \{\mathbf{0}\}$

for every $i = 1, 2, \ldots, n$. Prove that V is the direct sum of $W_1, \ldots, W_n$.

Proof: In order to prove that V is the direct sum of $W_1, \ldots, W_n$, we should prove that each vector $\alpha \in V$ can be uniquely expressed as

$$\alpha = \alpha_1 + \ldots + \alpha_n \text{ where } \alpha_i \in W_i, \; i = 1, \ldots, n.$$

Since $V = W_1 + \ldots + W_n$, therefore any vector α in V can be written as

$$\alpha = \alpha_1 + \ldots + \alpha_n \text{ where } \alpha_i \in W_i. \qquad \ldots(1)$$

To show that $\alpha_1, \ldots, \alpha_n$ are unique.

Let $\quad \alpha = \beta_1 + \ldots + \beta_n$ where $\beta_i \in W_i$. $\qquad \ldots(2)$

From (1) and (2), we get

$$\alpha_1 + \ldots + \alpha_n = \beta_1 + \ldots + \beta_n$$

$$(\alpha_1 - \beta_1) + \ldots + (\alpha_{i-1} - \beta_{i-1}) + (\alpha_i - \beta_i) + (\alpha_{i+1} - \beta_{i+1}) + \ldots + (\alpha_n - \beta_n) = \mathbf{0}. \qquad \ldots(3)$$

Now each W_i is a subspace of V. Therefoe $\alpha_i - \beta_i$ and also its additive inverse

$$\beta_i - \alpha_i \in W_i, \; i = 1, \ldots, n.$$

From (3), we get

$$(\alpha_i - \beta_i) = (\beta_1 - \alpha_1) + \ldots + (\beta_{i-1} - \alpha_{i-1}) + (\beta_{i+1} - \alpha_{i+1}) + \ldots + (\beta_n - \alpha_n). \qquad \ldots(4)$$

Now the vector on the right hand side of (4) and consequently the vector $\alpha_i - \beta_i$ is in $W_1 + \ldots + W_{i-1} + W_{i+1} + \ldots + W_n$.

Also $\quad \alpha_i - \beta_i \in W_i$.

$\therefore \quad \alpha_i - \beta_i \in W_i \cap (W_1 + \ldots + W_{i-1} + W_{i+1} + \ldots + W_n)$.

But for every $i = 1, \ldots, n$, it is given that

$$W_i \cap (W_1 + \ldots + W_{i-1} + W_{i+1} + \ldots + W_n) = \{\mathbf{0}\}.$$

Therefore $\quad \alpha_i - \beta_i = \mathbf{0}, i = 1, \ldots, n$

$\Rightarrow \quad \alpha_i = \beta_i, \; i = 1, \ldots, n$

$\Rightarrow$ the expression (1) for α is unique.

Hence V is the direct sum of $W_1, \ldots, W_n$.

Theorem 3: *Let $V(F)$ be a finite dimensional vector space and let $W_1, \ldots, W_k$ be subspaces of V. Then the following two statements are equivalent.*

(i) V is the direct sum of $W_1, \ldots, W_k$.

(ii) If B_i is a basis of W_i, $i = 1, \ldots, k$, then the union

$$B = \bigcup_{i=1}^{k} B_i \textit{ is also a basis for } V.$$

Proof: Let $B_i = \{\alpha_1^{\,i}, \alpha_2^{\,i}, \ldots, \alpha_{n_i}^{\,i}\}$ be a basis for W_i.

Here $n_i = \dim W_i =$ number of vectors in B_i. Also let B be the union of the basis B_i.

we can write

$$\alpha = \alpha_1 + \ldots + \alpha_k \text{ for } \alpha_i \in W_i\ , i = 1, \ldots, k.$$

Now α_i can be expressed as a linear combination of the vectors in B_i which is a basis of W_i. Therefore α can be expressed as a linear combination of the elements of $B = \bigcup_{i=1}^{k} B_i$. Therefore $L(B) = V$ *i.e.*, B spans V.

Now to show that B is linearly independent. Let

$$\sum_{i=1}^{k} (a_1{}^i \alpha_1{}^i + a_2{}^i \alpha_2{}^i + \ldots + a_{n_i}{}^i \alpha_{n_i}{}^i) = \mathbf{0}. \qquad \ldots(1)$$

Since V is the direct sum of $W_1, \ldots, W_k$, therefore $\mathbf{0} \in V$ can be uniquely expressed as a sum of vectors one in each W_i. This unique expression is

$$\mathbf{0} = \mathbf{0} + \ldots + \mathbf{0} \text{ where } \mathbf{0} \in W_i\ , i = 1, \ldots, k.$$

Now $a_1{}^i \alpha_1{}^i + \ldots + a_{n_i}{}^i \alpha_{n_i}{}^i \in W_i$. Therefore from (1) which is an expression for $\mathbf{0} \in V$ as a sum of vectors one in each W_i, we get

$$a_1{}^i \alpha_1{}^i + \ldots + a_{n_i}{}^i \alpha_{n_i}{}^i = \mathbf{0}, i = 1, \ldots, k$$

$$a_1{}^i = \ldots = a_{n_i}{}^i = 0$$

since $\{\alpha_1{}^i, \ldots, \alpha_{n_i}{}^i\}$ is linearly independent being a basis for W_i. Therefore $B = \bigcup_{i=1}^{k} B_i$ is linearly independent. Therefore B is a basis of V.

(ii) $\Rightarrow$ (i). It is given that $B = \bigcup_{i=1}^{k} B_i$ is a basis of V.

Therefore for any $\alpha \in V$, we can write

$$\alpha = \sum_{i=1}^{k} (a_1{}^i \alpha_1{}^i + a_2{}^i \alpha_2{}^i + \ldots + a_{n_i}{}^i \alpha_{n_i}{}^i)$$

$$= \alpha_1 + \alpha_2 + \ldots + \alpha_k \qquad \ldots(2)$$

where $\qquad \alpha_i = a_1{}^i \alpha_1{}^i + \ldots + a_{n_i}{}^i \alpha_{n_i}{}^i \in W_i$.

Thus each vector in V can be expressed as a sum of vectors one in each W_i.

Now V will be the direct sum of $W_1, \ldots, W_k$ if the expression (2) for α is unique. Let

$$\alpha = \beta_1 + \beta_2 + \ldots + \beta_k \qquad \ldots(3)$$

where $\qquad \beta_i = b_1{}^i \alpha_1{}^i + \ldots + b_{n_i}{}^i \alpha_{n_i}{}^i \in W_i$.

From (2) and (3), we get

$$\alpha_1 + \ldots + \alpha_k = \beta_1 + \ldots + \beta_k$$

$$\Rightarrow \quad (\alpha_1 - \beta_1) + \ldots + (\alpha_i - \beta_i) + \ldots + (\alpha_k - \beta_k) = \mathbf{0}$$

$$\Rightarrow \quad \sum_{i=1}^{k} (\alpha_i - \beta_i) = \mathbf{0}$$

$$\Rightarrow \quad \sum_{i=1}^{k} [(a_1{}^i - b_1{}^i)\alpha_1{}^i + \ldots + (a_{n_i}{}^i - b_{n_i}{}^i)\alpha_{n_i}{}^i] = \mathbf{0}$$

$[\because \bigcup_{i=1}^{k} B_i$ is linearly independent being a basis of V]

$\Rightarrow \qquad a_1{}^i = b_1{}^i, \dots, a_{n_i}{}^i = b_{n_i}{}^i, i = 1, \dots, k$

$\Rightarrow \qquad \alpha_i = \beta_i, \ i = 1, \dots, k$

$\Rightarrow$ the expression (2) for α is unique.

Hence V is the direct sum of $W_1, \dots, W_k$.

Note: While proving this theorem we have proved that *if a finite dimensional vector space $V(F)$ is the direct sum of its subspaces $W_1, \dots, W_k$, then*

$$\dim V = \dim W_1 + \dots + \dim W_k .$$

Illustrative Examples

Example 6: *Find dimension of V/W where $V = \mathbf{R}^3$, $W = \{(a, 0, 0) : a \in \mathbf{R}\}$.*

Solution: Here $V = \mathbf{R}^3$. $\dim V = 3$.

$$W = \{(a, 0, 0) : a \in \mathbf{R}\}. \{(1, 0, 0)\} \text{ is a basis of } W \text{ so } \dim W = 1.$$

Now $\dim V/W = \dim V - \dim W = 3 - 1 = 2$.

Example 7: *If $W_1 = \{(x, 0, z) : x, z \in \mathbf{R}\}$, $W_2 = \{(0, y, z) : y, z \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$ then show that $\mathbf{R}^3 = W_1 + W_2$ but $\mathbf{R}^3 \neq W_1 \oplus W_2$.*

Solution: Any $(x, y, z) \in \mathbf{R}^3$ can be written as a sum of a vector of W_1 and a vector of W_2 as shown below.

We have $(x, y, z) = \left(x, 0, \frac{1}{2} z\right) + \left(0, y, \frac{1}{2} z\right)$, where $\left(x, 0, \frac{1}{2} z\right) \in W_1$

and $\qquad \left(0, y, \frac{1}{2} z\right) \in W_2$.

Thus $\mathbf{R}^3 = W_1 + W_2$.

Also we have $(x, y, z) = \left(x, 0, \frac{1}{4} z\right) + \left(0, y, \frac{3}{4} z\right)$ where $\left(x, 0, \frac{1}{4} z\right) \in W_1$ and $\left(0, y, \frac{3}{4} z\right) \in W_2$.

It follows that representation of (x, y, z) as a sum of a vector of W_1 and a vector of W_2 is not unique.

Hence $\mathbf{R}^3 \neq W_1 \oplus W_2$.

Example 8: *If $W_1 = \{(a, b, c) : a = b = c\}$, $W_2 = \{(a, b, c) : a = 0\}$ are two subspaces of $\mathbf{R}^3$, show that $\mathbf{R}^3 = W_1 \oplus W_2$.*

Solution: We claim that $\mathbf{R}^3 = W_1 + W_2$.

For if $\alpha = (a, b, c) \in \mathbf{R}^3$, then $\alpha = (a, a, a) + (0, b - a, c - a)$ where $(a, a, a) \in W_1$ and $(0, b - a, c - a) \in W_2$.

For $\alpha = (a, b, c) \in W_1 \cap W_2 \Rightarrow a = b = c$ and $a = 0$

$\Rightarrow$ $a = 0, b = 0, c = 0$ *i.e.*, $\alpha = \mathbf{0}$.

Thus $\mathbf{R}^3 = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$.

Hence $\mathbf{R}^3 = W_1 \oplus W_2$.

Example 9: *Construct three subspaces W_1, W_2, W_3 of a vector space V so that $V = W_1 \oplus W_2 = W_1 \oplus W_3$ but $W_2 \neq W_3$.*

Solution: Take the vector space $V = \mathbf{R}^2$.

Obviously $W_1 = \{(a, 0) : a \in \mathbf{R}\}$,

$W_2 = \{(0, a) : a \in \mathbf{R}\}$, and $W_3 = \{(a, a) : a \in \mathbf{R}\}$,

are three subspaces of $\mathbf{R}^2$.

We have $V = W_1 + W_2$ and $W_1 \cap W_2 = \{(0, 0)\}$. $\therefore$ $V = W_1 \oplus W_2$.

Also it can be easily shown that

$$V = W_1 + W_3 \text{ and } W_1 \cap W_3 = \{(0, 0)\}. \quad \therefore \quad V = W_1 \oplus W_3.$$

Thus $V = W_1 \oplus W_2 = W_1 \oplus W_3$ but $W_2 \neq W_3$.

Comprehensive Exercise 2

1. Let V be the vector space of square matrices of order n over the field $\mathbf{R}$. Let W_1 and W_2 be the subspaces of symmetric and antisymmetric matrices respectively. Show that $V = W_1 \oplus W_2$.
2. Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Let U be the subspace of even functions and W the subspace of odd functions. Show that $V = U \oplus W$.
3. Let W_1, W_2 and W_3 be the following subspaces of $\mathbf{R}^3$:

   $$W_1 = \{(a, b, c) : a + b + c = 0\}, W_2 = \{(a, b, c) : a = c\},$$
   $$W_3 = \{(0, 0, c) : c \in \mathbf{R}\}.$$

   Show that (i) $\mathbf{R}^3 = W_1 + W_2$; (ii) $\mathbf{R}^3 = W_1 + W_3$; (iii) $\mathbf{R}^3 = W_2 + W_3$.

   When is the sum direct ?
4. Let W be a subspace of a vector space V over a field F. Show that $\alpha \in \beta + W$ iff $\alpha - \beta \in W$.
5. Let W_1 and W_2 be two subspaces of a finite dimensional vector space V. If $\dim V = \dim W_1 + \dim W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$, prove that $V = W_1 \oplus W_2$.

Answers 2

3. The sum is direct in (ii) and (iii)

4.11 Co-ordinates

Let $V(F)$ be a finite dimensional vector space. Let

$$B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$$

be an ordered basis for V. By an ordered basis we mean that the vectors of B have been enumerated in some well-defined way *i.e.*, the vectors occupying the first, second, ..., nth places in the set B are fixed.

Let $\alpha \in V$. Then there exists a unique n-tuple $(x_1, x_2, \ldots, x_n)$ of scalars such that

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \ldots + x_n\alpha_n = \sum_{i=1}^{n} x_i \alpha_i .$$

The n-tuple $(x_1, x_2, \ldots, x_n)$ is called the n-tuple of **co-ordinates of** α **relative to the ordered basis *B*.** The scalar x_i is called the ith **coordinate** of α **relative to the ordered basis *B*.** The $n \times 1$ matrix

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ x_n \end{bmatrix}$$

is called the **coordinate matrix** of α **relative to the ordered basis B.** We shall use the symbol $[\alpha]_B$ for the coordinate matrix of the vector α relative to the ordered basis B. **(Garhwal 2008)**

It should be noted that for the same basis set B, the coordinates of the vector α are unique only with respect to a particular ordering of B. The basis set B can be ordered in several ways. The coordinates of α may change with a change in the ordering of B.

Note: If $V(F)$ is a vector space of dimension n, then coordinate vector of $\mathbf{0} \in V$ relative to any basis of V is always the zero n-tuple $(0, 0, \ldots, 0)$.

Illustrative Examples

Example 10: *Show that the set*

$$S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$$

is a basis of $\mathbf{R}^3$ $(\mathbf{R})$ *where* $\mathbf{R}$ *is the field of real numbers. Hence find the coordinates of the vector* (a, b, c) *with respect to the above basis.*

Solution: The dimension of the vector space $\mathbf{R}^3$ $(\mathbf{R})$ is 3. If the set S is linearly independent, then S will form a basis of $\mathbf{R}^3$ $(\mathbf{R})$ because S contains 3 vectors. Let x, y, z be scalars in $\mathbf{R}$ such that

$$x(1, 0, 0) + y(1, 1, 0) + z(1, 1, 1) = \mathbf{0} = (0, 0, 0)$$

$$\Rightarrow \quad (x + y + z, y + z, z) = (0, 0, 0)$$

$\Rightarrow \qquad x = 0, y = 0, z = 0$

$\Rightarrow$ the set S is linearly independent.

$\therefore$ S is a basis of $\mathbf{R}^3$ **(R).**

Now to find the coordinates of (a, b, c) with respect to the ordered basis S. Let p, q, r be scalars in $\mathbf{R}$ such that

$$(a, b, c) = p\,(1, 0, 0) + q\,(1, 1, 0) + r\,(1, 1, 1)$$

$$\Rightarrow \quad (a, b, c) = (p + q + r, q + r, r)$$

$$\Rightarrow \quad p + q + r = a, q + r = b, r = c$$

$$\Rightarrow \quad r = c, q = b - c, p = a - b.$$

Hence the coordinates of the vector (a, b, c) are (p, q, r) *i.e.,* $(a - b, b - c, c)$.

Example 11: *Find the coordinates of the vector* $(2, 1, -6)$ *of* $\mathbf{R}^3$ *relative to the basis*

$$\alpha_1 = (1, 1, 2), \alpha_2 = (3, -1, 0), \alpha_3 = (2, 0, -1).$$

Solution: To find the coordinates of the vector $(2, 1, -6)$ relative to the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$, we shall express the vector $(2, 1, -6)$ as a linear combination of the vectors $\alpha_1, \alpha_2, \alpha_3$. Let p, q, r be scalars in $\mathbf{R}$ such that

$$(2, 1, -6) = p\alpha_1 + q\alpha_2 + r\alpha_3$$

$$\Rightarrow \quad (2, 1, -6) = p\,(1, 1, 2) + q\,(3, -1, 0) + r\,(2, 0, -1)$$

$$\Rightarrow \quad (2, 1, -6) = (p + 3q + 2r, p - q, 2p - r).$$

$$\therefore \quad p + 3q + 2r = 2, \; p - q = 1, \qquad \ldots(1)$$

and $\quad 2p - r = -6.$

Solving the equations (1), we have

$$p = -7/8, q = -15/8$$

and $\quad r = 17/4.$

Hence the requried coordinates of the vector $(2, 1, -6)$ relative to the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$ are (p, q, r) *i.e.,* $(-7/8, -15/8, 17/4)$.

Comprehensive Exercise 3

1. Show that the set $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $\mathbf{C}^3$ **(C)** where $\mathbf{C}$ is the field of complex numbers. Hence find the coordinates of the vector $(3 + 4i, 6i, 3 + 7i)$ in $\mathbf{C}^3$ with respect to the above basis.
2. Find the co-ordinate vector of $(3, 5, -2)$ relative to the basis $\{e_1, e_2, e_3\}$ where $e_1 = (1, 1, 1), e_2 = (0, 2, 3), e_3 = (0, 2, -1)$.
3. Let $B = \{\alpha_1, \alpha_2, \alpha_3\}$ be an ordered basis for $\mathbf{R}^3$, where
$$\alpha_1 = (1, 0, -1), \alpha_2 = (1, 1, 1), \alpha_3 = (1, 0, 0).$$
Obtain the coordinates of the vector (a, b, c) in the ordered basis B.

equal to two from the field of real numbers **R** into itself. For a fixed $t \in \mathbf{R}$, let

$g_1(x) = 1,\ g_2(x) = x + t,\ g_3(x) = (x + t)^2$.

Prove that $\{g_1, g_2, g_3\}$ is a basis for V and obtain the coordinates of $c_0 + c_1 x + c_2 x^2$ in this ordered basis.

Answers 3

1. $(3 - 2i, -3 - i, 3 + 7i)$
2. $(3, -1, 2)$
3. $(b - c, b, a - 2b + c)$
4. $(c_0 - c_1 t + c_2 t^2, c_1 - 2c_2 t, c_2)$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. If W be a subspace of a finite dimensional vector space $V(F)$, then $\dim V/W =$

   (a) $\dim V - \dim W$ (b) $\dim V + \dim W$
   (c) $(\dim V)/(\dim W)$ (d) None of these.

   (Garhwal 2010, 12)

2. Two finite dimensional vector spaces U and V over a field F are isomorphic if

   (a) $\dim U = \dim V$ (b) $\dim U \neq \dim V$
   (c) $\dim U < \dim V$ (d) $\dim U > \dim V$ **(Garhwal 2006)**

3. The set $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $\mathbf{R}^3$ $(\mathbf{R})$, then coordinates of the vector $(a\ b, c)$ with respect to the basis are

   (a) $(a + b, b - c, c)$ (b) $(a - b, b - c, c)$
   (c) $(a - b, b + c, c)$ (d) $(a - b, b - c, -c)$

4. Let $U(F)$ and $V(F)$ be two vector spaces. For a mapping $f : U \to V$ to be isomorphic which one of the following is not required

   (a) T is one-one (b) T is onto
   (c) T is linear (d) T is into **(Garhwal 2008)**

5. The dimension of $\dfrac{V}{W}$, where $V = R^3$, $W = \{(a, 0, 0) : a \in R\}$ is

   (a) 1 (b) 3
   (c) 2 (d) 4

Fill in the blanks "......" so that the following statements are complete and correct.

1. Every n-dimensional vector space $V(F)$ is isomorphic to ....
2. Let V be the vector space of all functions from the real field $\mathbf{R}$ into $\mathbf{R}$. Let W_1 be the subspace of even functions and W_2 be the subspace of odd functions. Then $V =$ ....
3. If W is an m-dimensional subspace of an n-dimensional vector space V, then the dimension of the quotient space $\frac{V}{W}$ is .... **(Garhwal 2006)**
4. Two subspaces W_1 and W_2 of the vector space $V(F)$ are said to be disjoint if $W_1 \cap W_2 =$ ....
5. If a finite dimensional vector space $V(F)$ is a direct sum of two subspaces W_1 and W_2, then $\dim V =$ ....

True or False

Write 'T' for true and 'F' for false statement.

1. If a finite dimensional vector space $V(F)$ is a direct sum of two subspaces W_1 and W_2, then $\dim V = \dim W_1 + \dim W_2$.
2. If V is a finite dimensional vector space and f is an isomorphism of V into V. Then f must map V onto V.
3. The dimension of $\frac{V}{W}$, where $V = \mathbf{R}^3$, $W = \{(a, 0, 0) : a \in \mathbf{R}\}$ is 3.
4. If $W_1 = \{(x, 0, z) : x, z \in \mathbf{R}\}$, $W_2 = \{(0, y, z) : y, z \in \mathbf{R}\}$ be two subspaces of $\mathbf{R}^3$ then $\mathbf{R}^3 \neq W_1 + W_2$.

Answers

Multiple Choice Questions

1. (a) 2. (a) 3. (b) 4. (d) 5. (c)

Fill in the Blank(s)

1. $V_n(F)$ 2. $W_1 \oplus W_2$ 3. $n - m$ 4. $\{\mathbf{0}\}$
5. $\dim W_1 + \dim W_2$

True or False

1. T 2. T 3. F 4. F

□

Chapter

5

Operations with Linear Transformations

5.1 Linear Transformation or Vector Space Homomorphism

(Garhwal 2010)

Definition: *Let $U(F)$ and $V(F)$ be two vector spaces over the* **same field F.** *A* **linear transformation from U into V** *is a function T from U into V such that*

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) \qquad \ldots(1)$$

for all α, β in U and for all a, b in F.

The condition (1) is also called **linearity property.** It can be easily seen that the condition (1) is equivalent to the condition

$$T(a\alpha + \beta) = aT(\alpha) + T(\beta)$$

for all α, β in U and all scalars a in F.

Linear operator. Definition. *Let $V(F)$ be a vector space. A linear operator on V is a function T from V into V such that*

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta)$$

for all α, β in V and for all a, b in F.

Illustration 1. *The function*

$$T : V_3 \ (\mathbf{R}) \to V_2 \ (\mathbf{R})$$

defined by $T \ (a, b, c) = (a, b) \ \forall \ a, b, c \in \mathbf{R}$ *is a linear transformation from* $V_3 \ (\mathbf{R})$ *into* $V_2 \ (\mathbf{R})$.

Let $\alpha = (a_1, b_1, c_1), \beta = (a_2, b_2, c_2) \in V_3 \ (\mathbf{R})$.

If $a, b \in \mathbf{R}$, then

$$\begin{aligned} T \ (a\alpha + b\beta) &= T \ [a \ (a_1, b_1, c_1) + b \ (a_2, b_2, c_2)] \\ &= T \ (aa_1 + ba_2, ab_1 + bb_2, ac_1 + bc_2) \\ &= (aa_1 + ba_2, ab_1 + bb_2) && \text{[by def. of } T \text{]} \\ &= (aa_1, ab_1) + (ba_2, bb_2) \\ &= a \ (a_1, b_1) + b \ (a_2, b_2) \\ &= aT \ (a_1, b_1, c_1) + bT \ (a_2, b_2, c_2) \\ &= aT \ (\alpha) + bT \ (\beta). \end{aligned}$$

$\therefore$ T is a linear transformation from $V_3 \ (\mathbf{R})$ into $V_2 \ (\mathbf{R})$.

Illustration 2. *Let $V \ (F)$ be the vector space of all $m \times n$ matrices over the field F. Let P be a fixed $m \times m$ matrix over F, and let Q be a fixed $n \times n$ matrix over F. The correspondence T from V into V defined by*

$$T \ (A) = PAQ \ \forall \ A \in V$$

is a linear operator on V.

If A is an $m \times n$ matrix over the field F, then PAQ is also an $m \times n$ matrix over the field F. Therefore T is a function from V into V. Now let $A, B \in V$ and $a, b \in F$. Then

$$\begin{aligned} T \ (aA + bB) &= P \ (aA + bB) \ Q && \text{[by def. of } T \text{]} \\ &= (aPA + bPB) \ Q = a \ PAQ + b \ PBQ = aT \ (A) + bT \ (B). \end{aligned}$$

$\therefore$ T is a linear transformation from V into V. Thus T is a linear operator on V.

Illustration 3. *Let $V \ (F)$ be the vector space of all polynomials over the field F. Let $f \ (x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n \in V$ be a polynomial of degree n in the indeterminate x. Let us define*

$$Df \ (x) = a_1 + 2a_2 x + \ldots + na_n x^{n-1} \text{ if } n > 1$$

and $\quad Df \ (x) = 0$ *if $f \ (x)$ is a constant polynomial.*

Then the correspondence D from V into V is a linear operator on V.

If $f \ (x)$ is a polynomial over the field F, then $Df \ (x)$ as defined above is also a polynomial over the field F. Thus if $f \ (x) \in V$, then $Df \ (x) \in V$. Therefore D is a function from V into V.

Also if $\quad f \ (x), g \ (x) \in V$ and $a, b \in F$, then

$$D \ [a \ f \ (x) + bg \ (x)] = a \ Df \ (x) + b \ Dg \ (x).$$

$\therefore$ D is a linear transformation from V into V.

for polynomials the definition of differentiation can be given purely algebraically, and does not require the usual theory of limiting processes.

Illustration 4. *Let V ($\mathbf{R}$) be the vector space of all continuous functions from $\mathbf{R}$ into $\mathbf{R}$. If $f \in V$ and we define T by*

$$(Tf)(x) = \int_0^x f(t)\, dt \;\; \forall \;\; x \in \mathbf{R},$$

then T is a linear transformation from V into V.

If f is real valued continuous function, then Tf, as defined above, is also a real valued continuous function. Thus

$$f \in V \;\Rightarrow\; Tf \in V.$$

Also the operation of integration satisfies the linearity property. Therefore T is a linear transformation from V into V.

5.2 Some Particular Transformations

1. Zero Transformation. *Let U (F) and V (F) be two vector spaces. The function T, from U into V defined by*

$$T(\alpha) = \mathbf{0} \text{ (zero vector of } V) \;\forall\; \alpha \in U$$

is a linear transformation from U into V.

Let $\alpha, \beta \in U$ and $a, b \in F$. Then $a\alpha + b\beta \in U$.

We have $T(a\alpha + b\beta) = \mathbf{0}$ [by def. of T]

$$= a\mathbf{0} + b\mathbf{0} = aT(\alpha) + bT(\beta).$$

$\therefore$ T is a linear transformation from U into V. It is called **zero transformation** and we shall in future denote it by $\hat{\mathbf{0}}$.

2. Identity operator. *Let V (F) be a vector space. The function I from V into V defined by $I(\alpha) = \alpha \;\forall\; \alpha \in V$ is a linear transformation from V into V.*

If $\alpha, \beta \in V$ and $a, b \in F$, then $a\alpha + b\beta \in V$ and we have

$$I(a\alpha + b\beta) = a\alpha + b\beta \qquad \text{[by def. of } I\text{]}$$

$$= aI(\alpha) + bI(\beta).$$

$\therefore$ I is a linear transformation from V into V. The transformation I is called **identity operator** on V and we shall always denote it by I.

3. Negative of a linear transformation. *Let U (F) and V (F) be two vector spaces. Let T be a linear transformation from U into V. The correspondence $-T$ defined by*

$$(-T)(\alpha) = -[T(\alpha)] \;\; \forall \;\; \alpha \in U$$

is a linear transformation from U into V. **(Garhwal 2008)**

Since $T(\alpha) \in V \;\Rightarrow\; -T(\alpha) \in V$, therefore $-T$ is a function from U into V.

Let $\alpha, \beta \in U$ and $a, b \in F$. Then $a\alpha + b\beta \in U$ and we have

$= -[aT(\alpha) + bT(\beta)]$ [$\because$ T is a linear transformation]

$= a[-T(\alpha)] + b[-T(\beta)] = a[(-T)\alpha] + b[(-T)\beta]$.

$\therefore$ $-T$ is a linear transformation from U into V. The linear transformation $-T$ is called the **negative** of the linear transformation T.

5.3 Properties of Linear Transformations.

Theorem: *Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$. Then*

(i) $T(\mathbf{0}) = \mathbf{0}$ where $\mathbf{0}$ on the left hand side is zero vector of U and $\mathbf{0}$ on the right hand side is zero vector of V.

(ii) $T(-\alpha) = -T(\alpha)\ \forall\ \alpha \in U$.

(iii) $T(\alpha - \beta) = T(\alpha) - T(\beta)\ \forall\ \alpha, \beta \in U$.

(iv) $T(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) = a_1T(\alpha_1) + a_2T(\alpha_2) + \ldots + a_nT(\alpha_n)$

where $\alpha_1, \alpha_2, \ldots \alpha_n \in U$ and $a_1, a_2, \ldots, a_n \in F$.

Proof: (i) Let $\alpha \in U$. Then $T(\alpha) \in V$. We have

$T(\alpha) + \mathbf{0} = T(\alpha)$ [$\because$ $\mathbf{0}$ is zero vector of V and $T(\alpha) \in V$]

$= T(\alpha + \mathbf{0})$ [$\because$ $\mathbf{0}$ is zero vector of U]

$= T(\alpha) + T(\mathbf{0})$ [$\because$ T is a linear transformation]

Now in the vector space V, we have

$$T(\alpha) + \mathbf{0} = T(\alpha) + T(\mathbf{0})$$

$\Rightarrow$ $\mathbf{0} = T(\mathbf{0})$, by left cancellation law for addition in V.

Note: When we write $T(\mathbf{0}) = \mathbf{0}$, there should be no confusion about the vector $\mathbf{0}$. Here T is a function from U into V. Therefore if $\mathbf{0} \in U$, then its image under T *i.e.*, $T(\mathbf{0}) \in V$. Thus in $T(\mathbf{0}) = \mathbf{0}$, the zero on the right hand side is zero vector of V.

(ii) We have $T[\alpha + (-\alpha)] = T(\alpha) + T(-\alpha)$

[$\because$ T is a linear transformation]

But $T[\alpha + (-\alpha)] = T(\mathbf{0}) = \mathbf{0} \in V$. [by (i)]

Thus in V, we have

$$T(\alpha) + T(-\alpha) = \mathbf{0}$$

$\Rightarrow$ $T(-\alpha) = -T(\alpha)$.

(iii) $T(\alpha - \beta) = T[\alpha + (-\beta)]$

$= T(\alpha) + T(-\beta)$ [$\because$ T is linear]

$= T(\alpha) + [-T(\beta)]$ [by (ii)]

$= T(\alpha) - T(\beta)$.

(iv) We shall prove the result by induction on n, the number of vectors in the linear combination $a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$. Suppose

$$T(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_{n-1}\alpha_{n-1}) = a_1T(\alpha_1) + a_2T(\alpha_2)$$

Then $T\,(a_1\alpha_1 + a_2\,\alpha_2 + \ldots + a_n\,\alpha_n)$

$$= [T\,(a_1\alpha_1 + a_2\,\alpha_2 + \ldots + a_{n-1}\,\alpha_{n-1}) + a_n\,\alpha_n]$$
$$= T\,(a_1\alpha_1 + a_2\,\alpha_2 + \ldots + a_{n-1}\,\alpha_{n-1}) + a_n\,T\,(\alpha_n)$$
$$= [a_1\,T\,(\alpha_1) + a_2\,T\,(\alpha_2) + \ldots + a_{n-1}\,T\,(\alpha_{n-1})] + a_n\,T\,(\alpha_n) \quad [\text{by } (1)]$$
$$= a_1\,T\,(\alpha_1) + a_2\,T\,(\alpha_2) + \ldots + a_{n-1}\,T\,(\alpha_{n-1}) + a_n\,T\,(\alpha_n).$$

Now the proof is complete by induction since the result is true when the number of vectors in the linear combination is 1.

Note: On account of this property sometimes we say that a linear transformation **preserves linear combinations.**

5.4 Range and Null Space of a Linear Transformation

Range of a linear transformation.

Definition. *Let $U\,(F)$ and $V\,(F)$ be two vector spaces and let T be a linear transformation from U into V. Then the range of T written as $R\,(T)$ is the set of all vectors β in V such that $\beta = T\,(\alpha)$ for some α in U.*

Thus the range of T is the image set of U under T *i.e.,*

Range $(T) = \{T\,(\alpha) \in V : \alpha \in U\}$.

Theorem 1: *If $U\,(F)$ and $V\,(F)$ are two vector spaces and T is a linear transformation from U into V, then range of T is a subspace of V.*

Proof: Obviously $R\,(T)$ is a non-empty subset of V.

Let $\beta_1, \beta_2 \in R\,(T)$. Then there exist vectors α_1, α_2 in U such that

$$T\,(\alpha_1) = \beta_1, T\,(\alpha_2) = \beta_2.$$

Let a, b be any elements of the field F. We have

$$a\beta_1 + b\beta_2 = a\,T\,(\alpha_1) + b\,T\,(\alpha_2)$$
$$= T\,(a\alpha_1 + b\alpha_2) \qquad [\because\ T \text{ is a linear transformation}]$$

Now U is a vector space. Therefore $\alpha_1, \alpha_2 \in U$ and

$$a, b \in F \Rightarrow a\alpha_1 + b\alpha_2 \in U.$$

Consequently $T\,(a\alpha_1 + b\alpha_2) = a\beta_1 + b\beta_2 \in R\,(T)$.

Thus $a, b \in F$ and $\beta_1, \beta_2 \in R\,(T) \Rightarrow a\beta_1 + b\beta_2 \in R\,(T)$.

Therefore $R\,(T)$ is a subspace of V.

Null space of linear transformation.

Definition. *Let $U\,(F)$ and $V\,(F)$ be two vector spaces and let T be a linear transformation from U into V. Then the null space of T written as $N\,(T)$ is the set of all vectors α in U such that $T\,(\alpha) = \mathbf{0}$ (zero vector of V). Thus*

$$N\,(T) = \{\alpha \in U : T\,(\alpha) = \mathbf{0} \in V\}.$$

homomorphism of U into V, then the null space of T is also called the **kernel of** T.

Theorem 2: *If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V, then the kernel of T or the null space of T is a subspace of U.*

Proof: Let $N(T) = \{\alpha \in U : T(\alpha) = \mathbf{0} \in V\}$.

Since $T(\mathbf{0}) = \mathbf{0} \in V$, therefore at least $\mathbf{0} \in N(T)$.

Thus $N(T)$ is a non-empty subset of U.

Let $\alpha_1, \alpha_2 \in N(T)$. Then $T(\alpha_1) = \mathbf{0}$ and $T(\alpha_2) = \mathbf{0}$.

Let $a, b \in F$. Then $a\alpha_1 + b\alpha_2 \in U$ and

$$T(a\alpha_1 + b\alpha_2) = a\,T(\alpha_1) + b\,T(\alpha_2) \quad [\because T \text{ is a linear transformation}]$$
$$= a\mathbf{0} + b\mathbf{0} = \mathbf{0} + \mathbf{0} = \mathbf{0} \in V.$$

$\therefore \quad a\alpha_1 + b\alpha_2 \in N(T)$.

Thus $a, b \in F$ and $\alpha_1, \alpha_2 \in N(T) \Rightarrow a\alpha_1 + b\alpha_2 \in N(T)$. Therefore $N(T)$ is a subspace of U.

5.5 Rank and Nullity of a Linear Transformation

Theorem 1: *Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$. If U is finite dimensional, then the range of T is a finite dimensional subspace of V.*

Proof: Since U is finite dimensional, therefore there exists a finite subset of U, say $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ which spans U.

Let $\beta \in$ range of T. Then there exists α in U such that $T(\alpha) = \beta$.

Now $\alpha \in U \Rightarrow \exists\, a_1, a_2, \ldots, a_n \in F$ such that

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$$
$$\Rightarrow \quad T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n)$$
$$\Rightarrow \quad \beta = a_1 T(\alpha_1) + a_2 T(\alpha_2) + \ldots + a_n T(\alpha_n). \quad \ldots(1)$$

Now the vectors $T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n)$ are in the range of T. If β is any vector in the range of T, then from (1), we see that β can be expressed as a linear combination of $T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n)$.

Therefore range of T is spanned by the vectors

$$T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n).$$

Hence range of T is finite dimensional.

Now we are in a position to define **rank** and **nullity** of a linear transformation.

Rank and nullity of a linear transformation.

Definition. *Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$ with U as finite dimensional. The rank of T denoted by $\rho(T)$ is the dimension of the range of T i.e.,* *The nullity of T denoted by $\nu(T)$ is the dimension of the null space of T i.e.,*

$$\nu(T) = dim\ N(T).$$ **(Garhwal 2007, 10)**

Theorem 2: *Let U and V be vector spaces over the field F and let T be a linear transformation from U into V. Suppose that U is finite dimensional. Then*

$$rank\ (T) + nullity\ (T) = dim\ U.$$ **(Garhwal 2006)**

Proof: Let N be the null space of T. Then N is a subspace of U. Since U is finite dimensional, therefore N is finite dimensional. Let dim N = nullity $(T) = k$ and let $\{\alpha_1, \alpha_2, \ldots, \alpha_k\}$ be a basis for N.

Since $\{\alpha_1, \alpha_2, \ldots, \alpha_k\}$ is a linearly independent subset of U, therefore we can extend it to form a basis of U. Let dim $U = n$ and let

$$\{\alpha_1, \alpha_2, \ldots, \alpha_k, \alpha_{k+1}, \ldots, \alpha_n\}$$

be a basis for U.

The vectors $T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_k), T(\alpha_{k+1}), \ldots, T(\alpha_n)$ are in the range of T. We claim that $\{T(\alpha_{k+1}), T(\alpha_{k+2}), \ldots, T(\alpha_n)\}$ is a basis for the range of T.

(i) First we shall prove that the vectors

$$T(\alpha_{k+1}), T(\alpha_{k+2}), \ldots, T(\alpha_n) \text{ span the range of } T.$$

Let $\beta \in$ range of T. Then there exists $\alpha \in U$ such that

$$T(\alpha) = \beta.$$

Now $\alpha \in U \Rightarrow \exists\ a_1, a_2, \ldots, a_n \in F$ such that

$$\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$$

$$\Rightarrow \quad T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n)$$

$$\Rightarrow \quad \beta = a_1T(\alpha_1) + \ldots + a_k T(\alpha_k) + a_{k+1} T(\alpha_{k+1}) + \ldots + a_n T(\alpha_n)$$

$$\Rightarrow \quad \beta = a_{k+1} T(\alpha_{k+1}) + a_{k+2} T(\alpha_{k+2}) + \ldots + a_n T(\alpha_n)$$

$$[\because \alpha_1, \alpha_2, \ldots, \alpha_k \in N \Rightarrow T(\alpha_1) = \mathbf{0}, \ldots, T(\alpha_k) = \mathbf{0}]$$

$\therefore$ the vectors $T(\alpha_{k+1}), \ldots, T(\alpha_n)$ span the range of T.

(ii) Now we shall show that the vectors

$$T(\alpha_{k+1}), \ldots, T(\alpha_n)$$

are linearly independent.

Let $c_{k+1}, \ldots, c_n \in F$ be such that

$$c_{k+1} T(\alpha_{k+1}) + \ldots + c_n T(\alpha_n) = \mathbf{0}$$

$$\Rightarrow \quad T(c_{k+1}\alpha_{k+1} + \ldots + c_n\alpha_n) = \mathbf{0}$$

$$\Rightarrow \quad c_{k+1}\alpha_{k+1} + \ldots + c_n\alpha_n \in \text{null space of } T \text{ i.e., } N$$

$$\Rightarrow \quad c_{k+1}\alpha_{k+1} + \ldots + c_n\alpha_n = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_k\alpha_k$$

for some $b_1, b_2, \ldots, b_k \in F$

[$\because$ each vector in N can be expressed as a linear combination of the vectors $\alpha_1, \ldots, \alpha_k$ forming a basis of N]

$$\Rightarrow \quad b_1\alpha_1 + \ldots + b_k\alpha_k - c_{k+1}\alpha_{k+1} - \ldots - c_n\alpha_n = \mathbf{0}$$

[$\because$ $\alpha_1, \alpha_2, \ldots, \alpha_k, \alpha_{k+1}, \ldots, \alpha_n$ are linearly independent, being a basis for U]

$\Rightarrow$ the vectors $T(\alpha_{k+1}), \ldots, T(\alpha_n)$ are linearly independent.

$\therefore$ the vectors $T(\alpha_{k+1}), \ldots, T(\alpha_n)$ form a basis of range of T.

$\therefore$ rank T = dim of range of $T = n - k$.

$\therefore$ rank (T) + nullity $(T) = (n - k) + k = n = \dim U$.

Note: If in place of the vector space V, we take the vector space U *i.e.*, if T is a linear transformation on an n-dimensional vector space U, even then as a special case of the above theorem,

$$\rho(T) + \nu(T) = n.$$

Illustrative Examples

Example 1: *Show that the mapping $T : V_3(\mathbf{R}) \to V_2(\mathbf{R})$ defined as*

$$T(a_1, a_2, a_3) = (3a_1 - 2a_2 + a_3, a_1 - 3a_2 - 2a_3)$$

is a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$.

Solution: Let $\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3) \in V_3(\mathbf{R})$.

Then $\quad T(\alpha) = T(a_1, a_2, a_3) = (3a_1 - 2a_2 + a_3, a_1 - 3a_2 - 2a_3)$

and $\quad T(\beta) = (3b_1 - 2b_2 + b_3, b_1 - 3b_2 - 2b_3)$.

Also let $a, b \in \mathbf{R}$. Then $a\alpha + b\beta \in V_3(\mathbf{R})$. We have

$$\begin{aligned} T(a\alpha + b\beta) &= T[a(a_1, a_2, a_3) + b(b_1, b_2, b_3)] \\ &= T(aa_1 + bb_1, aa_2 + bb_2, aa_3 + bb_3) \\ &= (3(aa_1 + bb_1) - 2(aa_2 + bb_2) + aa_3 + bb_3, \\ &\qquad aa_1 + bb_1 - 3(aa_2 + bb_2) - 2(aa_3 + bb_3)) \\ &= (a(3a_1 - 2a_2 + a_3) + b(3b_1 - 2b_2 + b_3), \\ &\qquad a(a_1 - 3a_2 - 2a_3) + b(b_1 - 3b_2 - 2b_3)) \\ &= a(3a_1 - 2a_2 + a_3, a_1 - 3a_2 - 2a_3) \\ &\qquad + b(3b_1 - 2b_2 + b_3, b_1 - 3b_2 - 2b_3) \\ &= aT(\alpha) + bT(\beta). \end{aligned}$$

Hence T is a linear transformation from $V_3(\mathbf{R})$ into $V_2(\mathbf{R})$.

Example 2: *Show that the mapping $T : V_2(\mathbf{R}) \to V_3(\mathbf{R})$ defined as*

$$T(a, b) = (a + b, a - b, b)$$

is a linear transformation from $V_2(\mathbf{R})$ into $V_3(\mathbf{R})$. Find the range, rank, null-space and nullity of T. **(Garhwal 2006)**

Solution: Let $\quad \alpha = (a_1, b_1), \beta = (a_2, b_2) \in V_2(\mathbf{R})$.

Then $\quad T(\alpha) = T(a_1, b_1) = (a_1 + b_1, a_1 - b_1, b_1)$

and $\quad T(\beta) = (a_2 + b_2, a_2 - b_2, b_2)$.

Also let $a, b \in \mathbf{R}$.

$= T\ (aa_1 + ba_2\ , ab_1 + bb_2)$
$= (aa_1 + ba_2 + ab_1 + bb_2\ , aa_1 + ba_2 - ab_1 - bb_2\ , ab_1 + bb_2)$
$= (a\,[a_1 + b_1] + b\,[a_2 + b_2], a\,[a_1 - b_1] + b\,[a_2 - b_2], ab_1 + bb_2)$
$= a\,(a_1 + b_1, a_1 - b_1, b_1) + b\,(a_2 + b_2\ , a_2 - b_2\ , b_2)$
$= aT\ (\alpha) + bT\ (\beta).$

$\therefore$ T is a linear transformation from V_2 (**R**) into V_3 (**R**).

Now $\{(1, 0), (0, 1)\}$ is a basis for V_2 (**R**).

We have $T\ (1, 0) = (1 + 0, 1 - 0, 0) = (1, 1, 0)$

and $T\ (0, 1) = (0 + 1, 0 - 1, 1) = (1, -1, 1).$

The vectors $T\ (1, 0), T\ (0, 1)$ span the range of T.

Thus the range of T is the subspace of V_3 (**R**) spanned by the vectors $(1, 1, 0), (1, -1, 1)$.

Now the vectors $(1, 1, 0), (1, -1, 1) \in V_3$ (**R**) are linearly independent because if $x, y \in$ **R**, then

$$x\ (1, 1, 0) + y\ (1, -1, 1) = (0, 0, 0)$$

$\Rightarrow$ $(x + y, x - y, y) = (0, 0, 0)$

$\Rightarrow$ $x + y = 0, x - y = 0, y = 0 \Rightarrow x = 0, y = 0.$

$\therefore$ the vectors $(1, 1, 0), (1, -1, 1)$ form a basis for the range of T.

Hence rank T = dim of range of $T = 2$.

Nullity of T = dim of V_2 (**R**) $-$ rank $T = 2 - 2 = 0$.

$\therefore$ null space of T must be the zero subspace of V_2 (**R**).

Otherwise, $(a, b) \in$ null space of T

$\Rightarrow$ $T\ (a, b) = (0, 0, 0)$

$\Rightarrow$ $(a + b, a - b, b) = (0, 0, 0)$

$\Rightarrow$ $a + b = 0, a - b = 0, b = 0$

$\Rightarrow$ $a = 0, b = 0.$

$\therefore$ $(0, 0)$ is the only element of V_2 (**R**) which belongs to the null space of T.

$\therefore$ null space of T is the zero subspace of V_2 (**R**).

Example 3:. *Let V be the vector space of all $n \times n$ matrices over the field F and let B be a fixed $n \times n$ matrix. If*

$$T\ (A) = AB - BA\ \forall\ A \in V,$$

verify that T is a linear transformation from V into V.

Solution: If $A \in V$, then $T\ (A) = AB - BA \in V$ because $AB - BA$ is also an $n \times n$ matrix over the field F. Thus T is a function from V into V.

Let $A_1, A_2 \in V$ and $a, b \in F$.

Then $aA_1 + bA_2 \in V$

and $T\ (aA_1 + bA_2) = (aA_1 + bA_2)\ B - B\ (aA_1 + bA_2)$

$$= a\,(A_1 B - BA_1) + b\,(A_2 B - BA_2)$$
$$= aT\,(A_1) + bT\,(A_2).$$

$\therefore$ T is a linear transformation from V into V.

Example 4: *Let V be an n-dimensional vector space over the field F and let T be a linear transformation from V into V such that the range and null space of T are identical. Prove that n is even. Give an example of such a linear transformation.*

Solution: Let N be the null space of T. Then N is also the range of T.

Now $\rho\,(T) + \nu\,(T) = \dim V$

i.e., dim of range of T + dim of null space of T = $\dim V = n$

i.e., $2 \dim N = n$ $\quad$ [$\because$ range of T = null space of $T = N$]

i.e., n is even.

Example of such a transformation.

Let $T : V_2\,(\mathbf{R}) \to V_2\,(\mathbf{R})$ be defined by

$$T\,(a, b) = (b, 0) \;\; \forall\, a, b \in \mathbf{R}.$$

Let $\alpha = (a_1, b_1), \beta = (a_2, b_2) \in V_2\,(\mathbf{R})$ and let $x, y \in \mathbf{R}$.

Then $T\,(x\alpha + y\beta) = T\,(x\,(a_1, b_1) + y\,(a_2, b_2)) = T\,(xa_1 + ya_2, xb_1 + yb_2)$

$$= (xb_1 + yb_2, 0) = (xb_1, 0) + (yb_2, 0) = x\,(b_1, 0) + y\,(b_2, 0)$$
$$= xT\,(a_1, b_1) + yT\,(a_2, b_2) = xT\,(\alpha) + yT\,(\beta).$$

$\therefore$ T is a linear transformation from $V_2\,(\mathbf{R})$ into $V_2\,(\mathbf{R})$.

Now $\{(1, 0), (0, 1)\}$ is a basis of $V_2\,(\mathbf{R})$.

We have $T\,(1, 0) = (0, 0)$ and $T\,(0, 1) = (1, 0)$.

Thus the range of T is the subspace of $V_2\,(\mathbf{R})$ spanned by the vectors $(0, 0)$ and $(1, 0)$. The vector $(0, 0)$ can be omitted from this spanning set because it is zero vector. Therefore the range of T is the subspace of $V_2\,(\mathbf{R})$ spanned by the vector $(1, 0)$. Thus the range of $T = \{a\,(1, 0) : a \in \mathbf{R}\} = \{(a, 0) : a \in \mathbf{R}\}$.

Now let $(a, b) \in N$ (the null space of T).

Then $(a, b) \in N \;\Rightarrow\; T\,(a, b) = (0, 0) \;\Rightarrow\; (b, 0) = (0, 0) \;\Rightarrow\; b = 0.$

$\therefore$ null space of $T = \{(a, 0) : a \in \mathbf{R}\}$.

Thus range of T = null space of T.

Also we observe that $\dim V_2\,(\mathbf{R}) = 2$ which is even.

Example 5: *Let V be a vector space and T a linear transformation from V into V. Prove that the following two statements about T are equivalent :*

(i) *The intersection of the range of T and the null space of T is the zero subspace of V i.e., $R\,(T) \cap N\,(T) = \{\mathbf{0}\}$.*

(ii) $T\,[T\,(\alpha)] = \mathbf{0} \;\Rightarrow\; T\,(\alpha) = \mathbf{0}$.

Solution: First we shall show that (i) $\Rightarrow$ (ii).

We have $T\,[T\,(\alpha)] = \mathbf{0} \;\Rightarrow\; T\,(\alpha) \in N\,(T)$

$\Rightarrow \qquad T(\alpha) = \mathbf{0}$ because $R(T) \cap N(T) = \{\mathbf{0}\}$.

Now we shall show that (ii) $\Rightarrow$ (i).

Let $\alpha \neq \mathbf{0}$ and $\alpha \in R(T) \cap N(T)$.

Then $\alpha \in R(T)$ and $\alpha \in N(T)$.

Since $\alpha \in N(T)$, therefore $T(\alpha) = \mathbf{0}$. ...(1)

Also $\alpha \in R(T) \Rightarrow \exists\ \beta \in V$ such that $T(\beta) = \alpha$.

Now $T(\beta) = \alpha$

$\Rightarrow \qquad T[T(\beta)] = T(\alpha) = \mathbf{0}$ [From (1)]

Thus $\exists\ \beta \in V$ such that $T[T(\beta)] = \mathbf{0}$ but $T(\beta) = \alpha \neq \mathbf{0}$.

This contradicts the given hypothesis (ii).

Therefore there exists no $\alpha \in R(T) \cap N(T)$ such that $\alpha \neq \mathbf{0}$.

Hence $\qquad R(T) \cap N(T) = \{\mathbf{0}\}$.

Example 6: *Consider the basis* $S = \{\alpha_1, \alpha_2, \alpha_3\}$ *of* $\mathbf{R}^3$ *where*

$$\alpha_1 = (1, 1, 1), \alpha_2 = (1, 1, 0), \alpha_3 = (1, 0, 0).$$

Express $(2, -3, 5)$ *in terms of the basis* $\alpha_1, \alpha_2, \alpha_3$.

Let $\qquad T : \mathbf{R}^3 \to \mathbf{R}^2$ *be defined as*

$$T(\alpha_1) = (1, 0), T(\alpha_2) = (2, -1), T(\alpha_3) = (4, 3).$$

Find $T(2, -3, 5)$. **(Meerut 2003)**

Solution: Let $(2, -3, 5) = a\alpha_1 + b\alpha_2 + c\alpha_3$

$$= a(1, 1, 1) + b(1, 1, 0) + c(1, 0, 0).$$

Then $\qquad a + b + c = 2, a + b = -3, a = 5.$

Solving these equations, we get

$$a = 5, b = -8, c = 5.$$

$\therefore \qquad (2, -3, 5) = 5\alpha_1 - 8\alpha_2 + 5\alpha_3.$

Now $\qquad T(2, -3, 5) = T(5\alpha_1 - 8\alpha_2 + 5\alpha_3)$

$$= 5T(\alpha_1) - 8T(\alpha_2) + 5T(\alpha_3)$$

[$\because$ T is a linear transformation]

$$= 5(1, 0) - 8(2, -1) + 5(4, 3)$$

$$= (5, 0) - (16, -8) + (20, 15)$$

$$= (9, 23).$$

Comprehensive Exercise 1

1. Show that the mapping $T : V_3(\mathbf{R}) \to V_2(\mathbf{R})$ defined as

$$T(a_1, a_2, a_3) = (a_1 - a_2, a_1 - a_3)$$

is a linear transformation.

linear.

3. Show that the following functions are linear :

(i) $T : \mathbf{R}^2 \to \mathbf{R}^2$ defined by $T(a, b) = (b, a)$

(ii) $T : \mathbf{R}^2 \to \mathbf{R}^2$ defined by $T(a, b) = (a + b, a)$

(iii) $T : \mathbf{R}^3 \to \mathbf{R}$ defined by $T(a, b, c) = 2a - 3b + 4c$.

4. Show that the following mappings T are not linear :

(i) $T : \mathbf{R}^2 \to \mathbf{R}$ defined by $T(x, y) = xy$;

(ii) $T : \mathbf{R}^2 \to \mathbf{R}^2$ defined by $T(x, y) = (1 + x, y)$;

(iii) $T : \mathbf{R}^3 \to \mathbf{R}^2$ defined by $T(x, y, z) = (|x|, 0)$;

(iv) $T : \mathbf{R}^2 \to \mathbf{R}$ defined by $T(x, y) = |x - y|$.

5. Show that the mapping $T : \mathbf{R}^2 \to \mathbf{R}^3$ defined as

$$T(a, b) = (a - b, b - a, -a)$$

is a linear transformation from $\mathbf{R}^2$ into $\mathbf{R}^3$. Find the range, rank, null space and nullity of T.

6. Let F be the field of complex numbers and let T be the function from F^3 into F^3 defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2 + 2x_3, 2x_1 + x_2 - x_3, -x_1 - 2x_2).$$

Verify that T is a linear transformation. Describe the null space of T.

(Garhwal 2007)

7. Let F be the field of complex numbers and let T be the function from F^3 into F^3 defined by

$$T(a, b, c) = (a - b + 2c, 2a + b, -a - 2b + 2c).$$

Show that T is a linear transformation. Find also the rank and the nullity of T.

8. Let $T : \mathbf{R}^3 \to \mathbf{R}^3$ be the linear transformation defined by :

$$T(x, y, z) = (x + 2y - z, y + z, x + y - 2z).$$

Find a basis and the dimension of (i) the range of T (ii) the null space of T.

(Garhwal 2011)

9. Let $T : V_4(\mathbf{R}) \to V_3(\mathbf{R})$ be a linear transformation defined by

$$T(a, b, c, d) = (a - b + c + d, a + 2c - d, a + b + 3c - 3d).$$

Then obtain the basis and dimension of the range space of T and null space of T.

10. Let V be the vector space of polynomials in x over $\mathbf{R}$. Let $D : V \to V$ be the differential operator : $D(f) = \dfrac{df}{dx} \cdot$ Find the image (*i.e.* range) and kernel of D.

11. Let V be the vector space of $n \times n$ matrices over the field F and let E be an arbitrary matrix in V. Let $T : V \to V$ be defined as $T(A) = AE + EA$, $A \in V$. Show that T is linear.

Let $T : V \to V$ be the linear function defined by $T(A) = MA$ for $A \in V$. Find a basis and the dimension of (i) the kernel of T and (ii) the range of T.

13. Let V be the vector space of 2×2 matrices over $\mathbf{R}$ and let $M = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Let $T : V \to V$ be the linear transformation defined by $T(A) = AM - MA$. Find a basis and the dimension of the kernel of T.

14. Let V be the space of $n \times 1$ matrices over a field F and let W be the space of $m \times 1$ matrices over F. Let A be a fixed $m \times n$ matrix over F and let T be the linear transformation from V into W defined by $T(X) = AX$.
Prove that T is the zero transformation if and only if A is the zero matrix.

15. Let $U(F)$ and $V(F)$ be two vector spaces and let T_1, T_2 be two linear transformations from U to V. Let x, y be two given elements of F. Then the mapping T defined as $T(\alpha) = x\,T_1(\alpha) + y\,T_2(\alpha)\ \forall\ \alpha \in U$ is a linear transformation from U into V.

Answers 1

5. Null space of $T = \{\mathbf{0}\}$; nullity of $T = 0$, rank $T = 2$

The set $\{(1, -1, -1), (-1, 1, 0)\}$ is a basis set for $R(T)$

6. Null space of $T = \{\mathbf{0}\}$

7. Rank $T = 2$; nullity of $T = 1$

8. (i) $\{(1, 0, 1), (2, 1, 1)\}$ is a basis of $R(T)$ and dim. $R(T) = 2$

(ii) $\{(3, -1, 1)\}$ is a basis of $N(T)$ and dim $N(T) = 1$

9. $\{(1, 1, 1), (0, 1, 2)\}$ is a basis of $R(T)$ and dim. $R(T) = 2$

$\{(1, 2, 0, 1), (2, 1, -1, 0)\}$ is a basis of $N(T)$ and dim. $N(T) = 2$

10. The image of D is the whole space V

The kernel of D is the set of constant polynomials

12. $\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}\right\}$ is a basis of the kernel T and dim (kernel T) = 2

$\left\{\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}\right\}$ is a basis for $R(T)$ and dim $R(T) = 2$

13. $\left\{\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$ is a basis of the kernel of T and dim (kernel T) = 2

Let $L(U, V)$ be the set of all linear transformations from a vector space $U(F)$ into a vector space $V(F)$. Sometimes we denote this set by Hom (U, V). Now we want to impose a vector space structure on the set $L(U, V)$ over the same field F. For this purpose we shall have to suitably define addition in $L(U, V)$ and scalar multiplication in $L(U, V)$ over F.

Theorem 1: *Let U and V be vector spaces over the field F. Let T_1 and T_2 be linear transformations from U into V. The function $T_1 + T_2$ defined by*

$$(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) \ \forall \ \alpha \in U$$

is a linear transformation from U into V. If c is any element of F, the function (cT) defined by

$$(cT)(\alpha) = cT(\alpha) \ \forall \ \alpha \in U$$

is a linear transformation from U into V. The set $L(U, V)$ of all linear transformations from U into V, together with the addition and scalar multiplication defined above is a vector space over the field F.

Proof: Suppose T_1 and T_2 are linear transformations from U into V and we define $T_1 + T_2$ as follows :

$$(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) \ \forall \ \alpha \in U. \qquad \ldots(1)$$

Since $T_1(\alpha) + T_2(\alpha) \in V$, therefore $T_1 + T_2$ is a function from U into V.

Let $a, b \in F$ and $\alpha, \beta \in U$. Then

$$(T_1 + T_2)(a\alpha + b\beta) = T_1(a\alpha + b\beta) + T_2(a\alpha + b\beta) \qquad \text{[by (1)]}$$
$$= [aT_1(\alpha) + bT_1(\beta)] + [aT_2(\alpha) + bT_2(\beta)]$$
$$[\because T_1 \text{ and } T_2 \text{ are linear transformations}]$$
$$= a[T_1(\alpha) + T_2(\alpha)] + b[T_1(\beta) + T_2(\beta)] \qquad [\because V \text{ is a vector space}]$$
$$= a(T_1 + T_2)(\alpha) + b(T_1 + T_2)(\beta) \qquad \text{[by (1)]}$$

$\therefore$ $T_1 + T_2$ is a linear transformation from U into V. Thus

$$T_1, T_2 \in L(U, V) \Rightarrow T_1 + T_2 \in L(U, V).$$

Therefore $L(U, V)$ is closed with respect to addition defined in it.

Again let $T \in L(U, V)$ and $c \in F$. Let us define cT as follows :

$$(cT)(\alpha) = cT(\alpha) \ \forall \ \alpha \in U. \qquad \ldots(2)$$

Since $cT(\alpha) \in V$, therefore cT is a function from U into V.

Let $a, b \in F$ and $\alpha, \beta \in U$. Then

$$(cT)(a\alpha + b\beta) = cT(a\alpha + b\beta) \qquad \text{[by (2)]}$$
$$= c[aT(\alpha) + bT(\beta)] \qquad [\because T \text{ is a linear transformation}]$$
$$= c[aT(\alpha)] + c[bT(\beta)] = (ca)T(\alpha) + (cb)T(\beta)$$
$$= (ac)T(\alpha) + (bc)T(\beta) = a[cT(\alpha)] + b[cT(\beta)]$$
$$= a[(cT)(\alpha)] + b[(cT)(\beta)].$$

$\therefore$ cT is a linear transformation from U into V. Thus

$$T \in L(U, V) \text{ and } c \in F \Rightarrow cT \in L(U, V).$$

Associativity of addition in $L(U, V)$.

Let $T_1, T_2, T_3 \in L(U, V)$. If $\alpha \in U$, then

$$[T_1 + (T_2 + T_3)](\alpha) = T_1(\alpha) + (T_2 + T_3)(\alpha)$$

[by (1) *i.e.*, by def. of addition in $L(U, V)$]

$$= T_1(\alpha) + [T_2(\alpha) + T_3(\alpha)] \quad \text{[by (1)]}$$
$$= [T_1(\alpha) + T_2(\alpha)] + T_3(\alpha) \quad [\because \text{ addition in } V \text{ is associative}]$$
$$= (T_1 + T_2)(\alpha) + T_3(\alpha) \quad \text{[by (1)]}$$
$$= [(T_1 + T_2) + T_3](\alpha) \quad \text{[by (1)]}$$

$\therefore \quad T_1 + (T_2 + T_3) = (T_1 + T_2) + T_3$

[by def. of equality of two functions]

Commutativity of addition in $L(U, V)$. Let $T_1, T_2 \in L(U, V)$. If α is any element of U, then

$$(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha) \quad \text{[by (1)]}$$
$$= T_2(\alpha) + T_1(\alpha) \quad [\because \text{ addition in } V \text{ is commutative}]$$
$$= (T_2 + T_1)(\alpha) \quad \text{[by (1)]}$$

$\therefore \quad T_1 + T_2 = T_2 + T_1$ [by def. of equality of two functions]

Existence of additive identity in $L(U, V)$. Let $\hat{\mathbf{0}}$ be the zero function from U into V *i.e.*, $\hat{\mathbf{0}}(\alpha) = \mathbf{0} \in V \ \forall \ \alpha \in U$.

Then $\hat{\mathbf{0}} \in L(U, V)$. If $T \in L(U, V)$ and $\alpha \in U$, we have

$$(\hat{\mathbf{0}} + T)(\alpha) = \hat{\mathbf{0}}(\alpha) + T(\alpha) \quad \text{[by (1)]}$$
$$= \mathbf{0} + T(\alpha) \quad [\text{by def. of } \hat{\mathbf{0}}]$$
$$= T(\alpha) \quad [\mathbf{0} \text{ being additive identity in } V]$$

$\therefore \quad \hat{\mathbf{0}} + T = T \ \forall \ T \in L(U, V)$.

$\therefore \quad \hat{\mathbf{0}}$ is the additive identity in $L(U, V)$.

Existence of additive inverse of each element in $L(U, V)$.

Let $T \in L(U, V)$. Let us define $-T$ as follows :

$$(-T)(\alpha) = -T(\alpha) \ \forall \ \alpha \in U.$$

Then $-T \in L(U, V)$. If $\alpha \in U$, we have

$$(-T + T)(\alpha) = (-T)(\alpha) + T(\alpha)$$

[by def. of addition in $L(U, V)$]

$$= -T(\alpha) + T(\alpha) \quad [\text{by def. of } -T]$$
$$= \mathbf{0} \in V$$
$$= \hat{\mathbf{0}}(\alpha) \quad [\text{by def. of } \hat{\mathbf{0}}]$$

$\therefore \quad -T + T = \hat{\mathbf{0}}$ for every $T \in L(U, V)$.

Therefore $L(U, V)$ is an **abelian group** with respect to addition defined in it.

Further we make the following observations :

(i) Let $c \in F$ and $T_1, T_2 \in L(U, V)$. If α is any element in U, we have

$$[c(T_1 + T_2)](\alpha) = c[(T_1 + T_2)(\alpha)] \quad \text{[by (2) } i.e., \text{ by def. of scalar multiplication in } L(U, V)]$$
$$= c[T_1(\alpha) + T_2(\alpha)] \quad \text{[by (1)]}$$
$$= cT_1(\alpha) + cT_2(\alpha)$$
$$[\because c \in F \text{ and } T_1(\alpha), T_2(\alpha) \in V \text{ which is a vector space}]$$
$$= (cT_1)(\alpha) + (cT_2)(\alpha) \quad \text{[by (2)]}$$
$$= (cT_1 + cT_2)(\alpha) \quad \text{[by (1)]}$$

$\therefore$ $$c(T_1 + T_2) = cT_1 + cT_2.$$

(ii) Let $a, b \in F$ and $T \in L(U, V)$. If $\alpha \in U$, we have

$$[(a + b)T](\alpha) = (a + b)T(\alpha) \quad \text{[by (2)]}$$
$$= aT(\alpha) + bT(\alpha) \quad [\because V \text{ is a vector space}]$$
$$= (aT)(\alpha) + (bT)(\alpha) \quad \text{[by (2)]}$$
$$= (aT + bT)(\alpha) \quad \text{[by (1)]}$$

$\therefore$ $$(a + b)T = aT + bT.$$

(iii) Let $a, b \in F$ and $T \in L(U, V)$. If $\alpha \in U$, we have

$$[(ab)T](\alpha) = (ab)T(\alpha) \quad \text{[by (2)]}$$
$$= a[bT(\alpha)] \quad [\because V \text{ is a vector space}]$$
$$= a[(bT)(\alpha)] \quad \text{[by (2)]}$$
$$= [a(bT)](\alpha) \quad \text{[by (2)]}$$

$\therefore$ $$(ab)T = a(bT).$$

(iv) Let $1 \in F$ and $T \in L(U, V)$. If $\alpha \in U$, we have

$$(1T)(\alpha) = 1T(\alpha) \quad \text{[by (2)]}$$
$$= T(\alpha) \quad [\because V \text{ is a vector space}]$$

$\therefore$ $$1T = T.$$

Hence $L(U, V)$ is a vector space over the field F.

Note: If in place of the vector space V, we take U, then we observe that the set of all linear operators on U forms a vector space with respect to addition and scalar multiplication defined as above.

Dimension of $L(U, V)$. Now we shall prove that if $U(F)$ and $V(F)$ are finite dimensional, then the vector space of linear transformations from U into V is also finite dimensional. For this purpose we shall require an **important result** which we prove in the following theorem :

$B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be an ordered basis for U. Let V be a vector space over the same field F and let $\beta_1, \dots, \beta_n$ be any vectors in V. Then there exists a unique linear transformation T from U into V such that

$$T(\alpha_i) = \beta_i, \; i = 1, 2, \dots, n.$$

Proof: Existence of T

Let $\quad \alpha \in U.$

Since $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a basis for U, therefore there exist unique scalars $x_1, x_2, \dots, x_n$ such that

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \dots + x_n\alpha_n.$$

For this vector α, let us define

$$T(\alpha) = x_1\beta_1 + x_2\beta_2 + \dots + x_n\beta_n.$$

Obviously $T(\alpha)$ as defined above is a unique element of V. Therefore T is a well-defined rule for associating with each vector α in U a unique vector $T(\alpha)$ in V. Thus T is a function from U into V.

The unique representation of $\alpha_i \in U$ as a linear combination of the vectors belonging to the basis B is

$$\alpha_i = 0\alpha_1 + 0\alpha_2 + \dots + 1\alpha_i + 0\alpha_{i+1} + \dots + 0\alpha_n.$$

Therefore according to our definition of T, we have

$$T(\alpha_i) = 0\beta_1 + 0\beta_2 + \dots + 1\beta_i + 0\beta_{i+1} + \dots + 0\beta_n$$

i.e., $\quad T(\alpha_i) = \beta_i, \; i = 1, 2, \dots, n.$

Now to show that T is a linear transformation.

Let $a, b \in F$ and $\alpha, \beta \in U$. Let

$$\alpha = x_1\alpha_1 + \dots + x_n\alpha_n \quad \text{and} \quad \beta = y_1\alpha_1 + \dots + y_n\alpha_n.$$

Then
$$\begin{aligned} T(a\alpha + b\beta) &= T[a(x_1\alpha_1 + \dots + x_n\alpha_n) + b(y_1\alpha_1 + \dots + y_n\alpha_n)] \\ &= T[(ax_1 + by_1)\alpha_1 + \dots + (ax_n + by_n)\alpha_n] \\ &= (ax_1 + by_1)\beta_1 + \dots + (ax_n + by_n)\beta_n \qquad \text{[by def. of } T\text{]} \\ &= a(x_1\beta_1 + \dots + x_n\beta_n) + b(y_1\beta_1 + \dots + y_n\beta_n) \\ &= aT(\alpha) + bT(\beta) \qquad \text{[by def. of } T\text{]} \end{aligned}$$

$\therefore$ T is a linear transformation from U into V. Thus there exists a linear transformation T from U into V such that

$$T(\alpha_i) = \beta_i, \; i = 1, 2, \dots, n.$$

Uniqueness of T: Let T' be a linear transformation from U into V such that

$$T'(\alpha_i) = \beta_i, \; i = 1, 2, \dots, n.$$

For the vector $\alpha = x_1\alpha_1 + \dots + x_n\alpha_n \in U$, we have

$$\begin{aligned} T'(\alpha) &= T'(x_1\alpha_1 + \dots + x_n\alpha_n) \\ &= x_1T'(\alpha_1) + \dots + x_nT'(\alpha_n) \\ &\qquad [\because T' \text{ is a linear transformation}] \\ &= x_1\beta_1 + \dots + x_n\beta_n \qquad \text{[by def. of } T'\text{]} \end{aligned}$$

Thus $T'(\alpha) = T(\alpha)\ \forall\ \alpha \in U$.

$\therefore$ $T' = T$.

This shows the uniqueness of T.

Note: From this theorem we conclude that if T is a linear transformation from a finite dimensional vector space $U(F)$ into a vector space $V(F)$, then T is completely defined if we mention under T the images of the elements of a basis set of U. If S and T are two linear transformations from U into V such that

$S(\alpha_i) = T(\alpha_i)\ \forall\ \alpha_i$ belonging to a basis of U, then

$S(\alpha) = T(\alpha)\ \forall\ \alpha \in U$, *i.e.,* $S = T$.

Thus two linear transformations from U into V are equal if they agree on a basis of U.

Theorem 3: *Let U be an n-dimensional vector space over the field F, and let V be an m-dimensional vector space over F. Then the vector space L (U, V) of all linear transformations from U into V is finite dimensional and is of dimension mn.*

Proof: Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \ldots, \beta_m\}$ be ordered basis for U and V respectively. By theorem 2, there exists a unique linear transformation T_{11} from U into V such that

$$T_{11}(\alpha_1) = \beta_1, T_{11}(\alpha_2) = \mathbf{0}, \ldots, T_{11}(\alpha_n) = \mathbf{0}$$

where $\beta_1, \mathbf{0}, \ldots, \mathbf{0}$ are vectors in V.

In fact, for each pair of integers (p, q) with $1 \le p \le m$ and $1 \le q \le n$, there exists a unique linear transformation T_{pq} from U into V such that

$$T_{pq}(\alpha_i) = \begin{cases} \mathbf{0}, & \text{if} \quad i \ne q \\ \beta_p, & \text{if} \quad i = q \end{cases}$$

i.e., $$T_{pq}(\alpha_i) = \delta_{iq}\ \beta_p, \qquad \ldots(1)$$

where $\delta_{iq} \in F$ is Kronecker delta *i.e.,* $\delta_{iq} = 1$ if $i = q$ and $\delta_{iq} = 0$ if $i \ne q$.

Since p can be any of $1, 2, \ldots, m$ and q any of $1, 2, \ldots, n$, there are mn such T_{pq}'s. Let B_1 denote the set of these mn transformations T_{pq}'s. We shall show that B_1 is a basis for $L(U, V)$.

(i) First we shall show that $L(U, V)$ is a linear span of B_1.

Let $T \in L(U, V)$. Since $T(\alpha_1) \in V$ and any element in V is a linear combination of $\beta_1, \beta_2, \ldots, \beta_m$, therefore

$$T(\alpha_1) = a_{11}\beta_1 + a_{21}\beta_2 + \ldots + a_{m1}\beta_m,$$

for some $a_{11}, a_{21}, \ldots, a_{m1} \in F$. In fact for each i, $1 \le i \le n$,

$$T(\alpha_i) = a_{1i}\beta_1 + a_{2i}\beta_2 + \ldots + a_{mi}\beta_m = \sum_{p=1}^{m} a_{pi}\beta_p \qquad \ldots(2)$$

Now consider $S = \sum_{p=1}^{m} \sum_{q=1}^{n} a_{pq} T_{pq}$.

Obviously S is a linear combination of elements of B_1 which is a subset of $L(U, V)$. Since $L(U, V)$ is a vector space, therefore $S \in L(U, V)$ *i.e.,* S is also a linear transformation from U into V. We shall show that $S = T$.

$$S(\alpha_i) = \left[\sum_{p=1}^{m} \sum_{q=1}^{n} a_{pq} T_{pq}\right](\alpha_i) = \sum_{p=1}^{m} \sum_{q=1}^{n} a_{pq} T_{pq}(\alpha_i)$$

$$= \sum_{p=1}^{m} \sum_{q=1}^{n} a_{pq} \delta_{iq} \beta_p \qquad \text{[From (1)]}$$

$$= \sum_{p=1}^{m} a_{pi} \beta_p$$

[On summing with respect to q. Remember that $\delta_{iq} = 1$ when $q = i$ and $\delta_{iq} = 0$ when $q \neq i$]

$$= T(\alpha_i). \qquad \text{[From (2)]}$$

Thus $S(\alpha_i) = T(\alpha_i) \ \forall \ \alpha_i \in B$. Therefore S and T agree on a basis of U. So we must have $S = T$. Thus T is also a linear combination of the elements of B_1. Therefore $L(U, V)$ is a linear span of B_1.

(ii) Now we shall show that B_1 is linearly independent. For b_{pq}'s $\in F$, let

$$\sum_{p=1}^{m} \sum_{q=1}^{n} b_{pq} T_{pq} = \hat{\mathbf{0}} \text{ i.e., zero vector of } L(U, V)$$

$\Rightarrow$ $\left[\sum_{p=1}^{m} \sum_{q=1}^{n} b_{pq} T_{pq}\right](\alpha_i) = \hat{\mathbf{0}}(\alpha_i) \ \forall \ \alpha_i \in B$

$\Rightarrow$ $\sum_{p=1}^{m} \sum_{q=1}^{n} b_{pq} T_{pq}(\alpha_i) = \mathbf{0} \in V$ [$\because$ $\hat{\mathbf{0}}$ is zero transformation]

$\Rightarrow$ $\sum_{p=1}^{m} \sum_{q=1}^{n} b_{pq} \delta_{iq} \beta_p = \mathbf{0}$ $\Rightarrow$ $\sum_{p=1}^{m} b_{pi} \beta_p = \mathbf{0}$

$\Rightarrow$ $b_{1i} \beta_1 + b_{2i} \beta_2 + \ldots + b_{mi} \beta_m = \mathbf{0}, 1 \leq i \leq n$

$\Rightarrow$ $b_{1i} = 0, b_{2i} = 0, \ldots, b_{mi} = 0, 1 \leq i \leq n$

[$\because$ $\beta_1, \beta_2, \ldots, \beta_m$ are linearly independent]

$\Rightarrow$ $b_{pq} = 0$ where $1 \leq p \leq m$ and $1 \leq q \leq n$

$\Rightarrow$ B_1 is linearly independent. Therefore B_1 is a basis of $L(U, V)$.

$\therefore$ dim $L(U, V)$ = number of elements in $B_1 = mn$.

Corollary: *The vector space $L(U, U)$ of all linear operators on an n-dimensional vector space U is of dimension n^2.*

Note: Suppose $U(F)$ is an n-dimensional vector space and $V(F)$ is an m-dimensional vector space. If $U \neq \{\mathbf{0}\}$ and $V \neq \{\mathbf{0}\}$, then $n \geq 1$ and $m \geq 1$. Therefore $L(U, V)$ does not just consist of the element $\hat{\mathbf{0}}$, because dimension of $L(U, V)$ is $mn \geq 1$.

5.7 Product of Linear Transformations

Theorem 1: *Let U, V and W be vector spaces over the field F. Let T be a linear transformation from U into V and S a linear transformation from V into W. Then the composite function ST (called product of linear transformations) defined by* *is a linear transformation from U into W.*

Proof: T is a function from U into V and S is a function from V into W.

So $\quad \alpha \in U \Rightarrow T(\alpha) \in V$. Further

$T(\alpha) \in V \Rightarrow S[T(\alpha)] \in W$. Thus $(ST)(\alpha) \in W$.

Therefore ST is a function from U into W. Now to show that ST is a linear transformation from U into W.

Let $a, b \in F$ and $\alpha, \beta \in U$. Then

$$\begin{aligned} (ST)(a\alpha + b\beta) &= S[T(a\alpha + b\beta)] && \text{[by def. of product of two functions]} \\ &= S[aT(\alpha) + bT(\beta)] && [\because T \text{ is a linear transformation}] \\ &= aS[T(\alpha)] + bS[T(\beta)] && [\because S \text{ is a linear transformation}] \\ &= a(ST)(\alpha) + b(ST)(\beta). \end{aligned}$$

Hence ST is a linear transformation from U into W.

Note: If T and S are linear operators on a vector space $V(F)$, then both the products ST as well as TS exist and each is a linear operator on V. However, in general $TS \neq ST$ as is obvious from the following examples.

Illustrative Examples

Example 7: *Let T_1 and T_2 be linear operators on $\mathbf{R}^2$ defined as follows :*

$$T_1(x_1, x_2) = (x_2, x_1) \quad \textit{and} \quad T_2(x_1, x_2) = (x_1, 0).$$

Show that $\quad T_1T_2 \neq T_2T_1$.

Solution: We have

$$\begin{aligned} (T_1T_2)(x_1, x_2) &= T_1[T_2(x_1, x_2)] \\ &= T_1(x_1, 0), && \text{[by def. of } T_2] \\ &= (0, x_1), && \text{[by def. of } T_1]. \end{aligned}$$

Also
$$\begin{aligned} (T_2T_1)(x_1, x_2) &= T_2[T_1(x_1, x_2)], && \text{[by def. of } T_2T_1] \\ &= T_2(x_2, x_1), && \text{[by def. of } T_1] \\ &= (x_2, 0), && \text{[by def. of } T_2]. \end{aligned}$$

Thus we see that $(T_1T_2)(x_1, x_2) \neq (T_2T_1)(x_1, x_2) \; \forall \; (x_1, x_2) \in \mathbf{R}^2$.

Hence by the definition of equality of two mappings, we have

$$T_1T_2 \neq T_2T_1.$$

Example 8: *Let $S(\mathbf{R})$ be the vector space of all polynomial functions in x with coefficients as elements of the field $\mathbf{R}$ of real numbers. Let D and T be two linear operators on V defined by*

$$D(f(x)) = \frac{d}{dx} f(x) \qquad \ldots(1)$$

and

$$T(f(x)) = \int_0^x f(x)\, dx \qquad \ldots(2)$$

Then show that $DT = I$ (identity operator) and $TD \neq I$.

Solution: Let $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots \in V$.

We have $(DT)(f(x)) = D[T(f(x))]$

$$= D\left[\int_0^x f(x)\,dx\right] = D\left[\int_0^x (a_0 + a_1 x + a_2 x^2 + \ldots)\,dx\right]$$

$$= D\left[a_0 x + \frac{a_1}{2} x^2 + \frac{a_2}{3} x^3 + \ldots\right]_0^x = \frac{d}{dx}\left[a_0 x + \frac{a_1}{2} x^2 + \ldots\right]$$

$$= a_0 + a_1 x + a_2 x^2 + \ldots = f(x) = I[f(x)].$$

Thus we have $(DT)[f(x)] = I[f(x)]\ \forall\ f(x) \in V$. Therefore $DT = I$.

Now $(TD)\, f(x) = T[D\, f(x)]$

$$= T\left[\frac{d}{dx}(a_0 + a_1 x + a_2 x^2 + \ldots)\right] = T(a_1 + 2a_2 x + \ldots)$$

$$= \int_0^x (a_1 + 2a_2 x + \ldots)\,dx = [a_1 x + a_2 x^2 + \ldots]_0^x$$

$$= a_1 x + a_2 x^2 + \ldots$$

$$\neq f(x) \text{ unless } a_0 = 0.$$

Thus $\exists\ f(x) \in V$ such that $(TD)[f(x)] \neq I[f(x)]$.

$\therefore$ $\qquad TD \neq I$.

Hence $\qquad TD \neq DT$,

showing that product of linear operators is not in general commutative.

Example 9: *Let $V(\mathbf{R})$ be the vector space of all polynomials in x with coefficients in the field $\mathbf{R}$. Let D and T be two linear transformations on V defined as*

$$D[f(x)] = \frac{d}{dx} f(x)\ \forall\ f(x) \in V \quad \text{and} \quad T[f(x)] = x\, f(x)\ \forall\ f(x) \in V.$$

Then show that $DT \neq TD$.

Solution: We have

$$(DT)[f(x)] = D[T(f(x))] = D[x\, f(x)] = \frac{d}{dx}[x\, f(x)]$$

$$= f(x) + x \frac{d}{dx} f(x). \qquad \ldots(1)$$

Also

$$(TD)[f(x)] = T[D(f(x))] = T\left[\frac{d}{dx}(f(x))\right]$$

$$= x \frac{d}{dx} f(x). \qquad \ldots(2)$$

From (1) and (2), we see that $\exists\ f(x) \in V$ such that

$$(DT)(f(x)) \neq (TD)(f(x)) \quad \Rightarrow \quad DT \neq TD.$$

Also we see that

$$(DT - TD)(f(x)) = (DT)(f(x)) - (TD)(f(x))$$

$$= f(x) = I(f(x)).$$

Theorem 2: *Let $V(F)$ be a vector space and A, B, C be linear transformations on V. Then*

(i) $A\hat{\mathbf{0}} = \hat{\mathbf{0}} = \hat{\mathbf{0}}A$ *(ii)* $AI = A = IA$

(iii) $A(BC) = (AB)C$ *(iv)* $A(B + C) = AB + AC$

(v) $(A + B)C = AC + BC$

(vi) $c(AB) = (cA)B = A(cB)$ *where c is any element of F.* **(Garhwal 2006)**

Proof: Just for the sake of convenience we first mention here our definitions of addition, scalar multiplication and product of linear transformations :

$$(A + B)(\alpha) = A(\alpha) + B(\alpha) \qquad \ldots(1)$$

$$(cA)(\alpha) = cA(\alpha) \qquad \ldots(2)$$

$$(AB)(\alpha) = A[B(\alpha)] \qquad \ldots(3)$$

$\forall\ \alpha \in V$ and $\forall\ c \in F$.

Now we shall prove the above results.

(i) We have $\forall\ \alpha \in V$, $(A\hat{\mathbf{0}})(\alpha) = A[\hat{\mathbf{0}}(\alpha)]$ [by (3)]

$= A(\mathbf{0})$ [$\because$ $\hat{\mathbf{0}}$ is zero transformation]

$= \mathbf{0} = \hat{\mathbf{0}}(\alpha)$.

$\therefore$ $A\hat{\mathbf{0}} = \hat{\mathbf{0}}$. [by def. of equality of two functions]

Similarly we can show that $\hat{\mathbf{0}}A = \hat{\mathbf{0}}$.

(ii) We have $\forall\ \alpha \in V$,

$(AI)(\alpha) = A[I(\alpha)]$

$= A(\alpha)$ [$\because$ I is identity transformation]

$\therefore$ $AI = A$.

Similarly we can show that $IA = A$.

(iii) We have $\forall\ \alpha \in V$

$[A(BC)](\alpha) = A[(BC)(\alpha)]$ [by (3)]

$= A[B(C(\alpha))]$ [by (3)]

$= (AB)[C(\alpha)]$ [by (3)]

$= [(AB)C](\alpha)$. [by (3)]

$\therefore$ $A(BC) = (AB)C$.

(iv) We have $\forall\ \alpha \in V$,

$[A(B + C)](\alpha) = A[(B + C)(\alpha)]$ [by (3)]

$= A[B(\alpha) + C(\alpha)]$ [by (1)]

$= A[B(\alpha)] + A[C(\alpha)]$ [$\because$ A is a linear transformation and $B(\alpha), C(\alpha) \in V$]

$= (AB)(\alpha) + (AC)(\alpha)$ [by (3)]

$\therefore \qquad A(B+C) = AB + AC.$

(v) We have $\forall \; \alpha \in V$,

$$[(A+B)C](\alpha) = (A+B)[C(\alpha)] \qquad \text{[by (3)]}$$
$$= A[C(\alpha)] + B[C(\alpha)] \qquad \text{[by (1) since } C(\alpha) \in V\text{]}$$
$$= (AC)(\alpha) + (BC)(\alpha) \qquad \text{[by (3)]}$$
$$= (AC + BC)(\alpha) \qquad \text{[by (1)]}$$

$\therefore \qquad (A+B)C = AC + BC.$

(vi) We have $\forall \; \alpha \in V$,

$$[c(AB)](\alpha) = c[(AB)(\alpha)] \qquad \text{[by (2)]}$$
$$= c[A(B(\alpha))] \qquad \text{[by (3)]}$$
$$= (cA)[B(\alpha)] \qquad \text{[by (2) since } B(\alpha) \in V\text{]}$$
$$= [(cA)B](\alpha) \qquad \text{[by (3)]}$$

$\therefore \qquad c(AB) = (cA)B.$

Again $\qquad [c(AB)](\alpha) = c[(AB)(\alpha)] \qquad$ [by (2)]

$$= c[A(B(\alpha))] \qquad \text{[by (3)]}$$
$$= A[cB(\alpha)]$$

[$\because$ A is a linear transformation and $B(\alpha) \in V$]

$$= A[(cB)(\alpha)] \qquad \text{[by (2)]}$$
$$= [A(cB)](\alpha). \qquad \text{[by (3)]}$$

$\therefore \qquad c(AB) = A(cB).$

5.8 Ring of Linear Operators on a Vector Space

Ring. Definition. *A non-empty set R with two binary operations, to be denoted additively and multiplicatively, is called a ring if the following postulates are satisfied :*

R_1. *R is closed with respect to addition i.e.,*
$a + b \in R \; \forall \; a, b \in R.$

R_2. $(a+b)+c = a+(b+c) \; \forall \; a, b, c \in R.$

R_3. $a + b = b + a \; \forall \; a, b \in R.$

R_4. *$\exists$ an element 0 (called zero element) in R such that*
$0 + a = a \; \forall \; a \in R.$

R_5. $a \in R \;\Rightarrow\; \exists \, -a \in R$ *such that*
$(-a) + a = 0.$

R_6. *R is closed with respect to multiplication i.e.,*
$ab \in R, \; \forall \; a, b \in R$

R_7. $(ab)c = a(bc) \; \forall \; a, b, c \in R.$

R_8. *Multiplication is distributive with respect to addition, i.e.,*
$a(b+c) = ab + ac$ *and* $(a+b)c = ac + bc \; \forall \; a, b, c \in R.$

If in a ring R there exists an element $1 \in R$ *such that*

$$1a = a = a1 \; \forall \; a \in R,$$

then R is called a ring with unity element. The element 1 is called the unity element of the ring.

Theorem: *The set* $L(V, V)$ *of all linear transformations from a vector space* $V(F)$ *into itself is a ring with unity element with respect to addition and multiplication of linear transformations defined as below :*

$$(S + T)(\alpha) = S(\alpha) + T(\alpha)$$

and $$(ST)(\alpha) = S[T(\alpha)] \; \forall \; S, T \in L(V, V) \text{ and } \forall \alpha \in V.$$

Proof: The students should themselves write the complete proof of this theorem. We have proved all the steps here and there. They should show here that all the ring postulates are satisfied in the set $L(V, V)$. The transformation $\hat{\mathbf{0}}$ will act as the zero element and the identity transformation I will act as the unity element of this ring.

5.9 Algebra or Linear Algebra

(Garhwal 2006, 12)

Definition: *Let F be a field. A vector space V over F is called a linear algebra over F if there is defined an additional operation in V called* **multiplication of vectors** *and satisfying the following postulates :*

1. $\alpha\beta \in V \; \forall \; \alpha, \beta \in V$
2. $\alpha(\beta\gamma) = (\alpha\beta)\gamma \; \forall \; \alpha, \beta, \gamma \in V$
3. $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$ *and* $(\alpha + \beta)\gamma = \alpha\gamma + \beta\gamma \; \forall \; \alpha, \beta, \gamma \in V.$
4. $c(\alpha\beta) = (c\alpha)\beta = \alpha(c\beta) \; \forall \; \alpha, \beta \in V$ *and* $c \in F.$

If there is an element 1 in V such that

$$1\alpha = \alpha = \alpha 1 \; \forall \; \alpha \in V,$$

then we call V a **linear algebra with identity over** F. Also 1 is then called the identity of V. The algebra V is **Commutative** if

$$\alpha\beta = \beta\alpha \; \forall \; \alpha, \beta \in V.$$

Theorem: *Let* $V(F)$ *be a vector space. The vector space* $L(V, V)$ *over F of all linear transformations from V into V is a linear algebra with identity with respect to the product of linear transformations as the multiplication composition in* $L(V, V)$.

Proof: The students should write the complete proof here. All the necessary steps have been proved here and there.

5.10 Polynomials

Let T be a linear transformation on a vector space $V(F)$. Then $T\,T$ is also a linear transformation on V. We shall write $T^1 = T$ and $T^2 = T\,T$. Since the product of linear transformations is an associative operation, therefore if m is a positive integer, we shall define

$$T^m = T\,T\,T \ldots \text{upto } m \text{ times.}$$

Obviously T^m is a linear transformation on V.

Also we define $T^0 = I$ (identity transformation).

If m and n are non-negative integers, it can be easily seen that

$$T^m\,T^n = T^{m+n} \text{ and } (T^m)^n = T^{mn}.$$

The set $L(V, V)$ of all linear transformations on V is a vector space over the field F. If $a_0, a_1, \ldots, a_n \in F$, then

$$p(T) = a_0 I + a_1 T + a_2 T^2 + \ldots + a_n T^n \in L(V, V)$$

i.e., $p(T)$ is also a linear transformation on V because it is a linear combination over F of elements of $L(V, V)$. We call $p(T)$ as a polynomial in linear transformation T. The polynomials in a linear transformation behave like ordinary polynomials.

5.11 Invertible Linear Transformations

(Garhwal 2006)

Definition: *Let U and V be vector spaces over the field F. Let T be a linear transformation from U into V such that T is one-one onto. Then T is called invertible.*

If T is a function from U into V, then T is said to be 1-1 if

$$\alpha_1, \alpha_2 \in U \text{ and } \alpha_1 \neq \alpha_2 \;\Rightarrow\; T(\alpha_1) \neq T(\alpha_2).$$

In other words T is said to be 1-1 if

$$\alpha_1, \alpha_2 \in U \text{ and } T(\alpha_1) = T(\alpha_2) \;\Rightarrow\; \alpha_1 = \alpha_2.$$

Further T is said to be onto if

$$\beta \in V \;\Rightarrow\; \exists\, \alpha \in U \text{ such that } T(\alpha) = \beta.$$

If T is one-one and onto, then we define a function from V into U, called the inverse of T and denoted by T^{-1} as follows :

Let β be any vector in V. Since T is onto, therefore

$$\beta \in V \;\Rightarrow\; \exists\, \alpha \in U \text{ such that } T(\alpha) = \beta.$$

Also α determined in this way is a unique element of U because T is one-one and therefore

$$\alpha_0, \alpha \in U \text{ and } \alpha_0 \neq \alpha \;\Rightarrow\; \beta = T(\alpha) \neq T(\alpha_0).$$

We define $T^{-1}(\beta)$ to be α. Thus

$$T^{-1} : V \Rightarrow U \text{ such that}$$

$$T^{-1}(\beta) = \alpha \;\Leftrightarrow\; T(\alpha) = \beta.$$

prove that T^{-1} is a linear transformation from V into U.

Theorem 1: *Let U and V be vector spaces over the field F and let T be a linear transformation from U into V. If T is one-one and onto, then the inverse function T^{-1} is a linear transformation from V into U.*

Proof: Let $\beta_1, \beta_2 \in V$ and $a, b \in F$.

Since T is one-one and onto, therefore there exist unique vectors $\alpha_1, \alpha_2 \in U$ such that $T(\alpha_1) = \beta_1, T(\alpha_2) = \beta_2$.

By definition of T^{-1}, we have

$$T^{-1}(\beta_1) = \alpha_1, T^{-1}(\beta_2) = \alpha_2.$$

Now $a\alpha_1 + b\alpha_2 \in U$ and we have by linearity of T,

$$T(a\alpha_1 + b\alpha_2) = aT(\alpha_1) + bT(\alpha_2)$$
$$= a\beta_1 + b\beta_2 \in V.$$

$\therefore$ by def. of T^{-1}, we have

$$T^{-1}(a\beta_1 + b\beta_2) = a\alpha_1 + b\alpha_2$$
$$= aT^{-1}(\beta_1) + bT^{-1}(\beta_2).$$

$\therefore$ T^{-1} is a linear transformation from V into U.

Theorem 2: *Let T be an invertible linear transformation on a vector space $V(F)$. Then*

$$T^{-1}T = I = TT^{-1}.$$

Proof. Let α be any element of V and let $T(\alpha) = \beta$. Then

$$T^{-1}(\beta) = \alpha.$$

We have $T(\alpha) = \beta$

$\Rightarrow$ $T^{-1}[T(\alpha)] = T^{-1}(\beta)$ $\Rightarrow$ $(T^{-1}T)(\alpha) = \alpha$

$\Rightarrow$ $(T^{-1}T)(\alpha) = I(\alpha)$ $\Rightarrow$ $T^{-1}T = I$.

Let β be any element of V. Since T is onto, therefore $\beta \in V \Rightarrow \exists\, \alpha \in V$ such that $T(\alpha) = \beta$. Then $T^{-1}(\beta) = \alpha$.

Now $T^{-1}(\beta) = \alpha$

$\Rightarrow$ $T[T^{-1}(\beta)] = T(\alpha)$

$\Rightarrow$ $(TT^{-1})(\beta) = \beta$

$\Rightarrow$ $(TT^{-1})(\beta) = \beta = I(\beta)$

$\Rightarrow$ $TT^{-1} = I$.

Theorem 3: *If A, B and C are linear transformations on a vector space $V(F)$ such that*

$$AB = CA = I,$$

then A is invertible and $A^{-1} = B = C$.

and onto.

(i) A is one-one.

Let $\alpha_1, \alpha_2 \in V$. Then

$$A(\alpha_1) = A(\alpha_2)$$

$$\Rightarrow \quad C[A(\alpha_1)] = C[A(\alpha_2)]$$

$$\Rightarrow \quad (CA)(\alpha_1) = (CA)(\alpha_2)$$

$$\Rightarrow \quad I(\alpha_1) = I(\alpha_2)$$

$$\Rightarrow \quad \alpha_1 = \alpha_2.$$

$\therefore$ A is **one-one.**

(ii) A is onto.

Let β be any element of V. Since B is a linear transformation on V, therefore $B(\beta) \in V$. Let $B(\beta) = \alpha$. Then

$$B(\beta) = \alpha$$

$$\Rightarrow \quad A[B(\beta)] = A(\alpha)$$

$$\Rightarrow \quad (AB)(\beta) = A(\alpha)$$

$$\Rightarrow \quad I(\beta) = A(\alpha) \qquad [\because \; AB = I]$$

$$\Rightarrow \quad \beta = A(\alpha).$$

Thus $\beta \in V \Rightarrow \exists\, \alpha \in V$ such that $A(\alpha) = \beta$.

$\therefore$ A is onto.

Since A is one-one and onto therefore A is invertible *i.e.*, A^{-1} exists.

(iii) Now we shall show that $A^{-1} = B = C$.

We have $AB = I$

$$\Rightarrow \quad A^{-1}(AB) = A^{-1} I \quad \Rightarrow (A^{-1} A) B = A^{-1}$$

$$\Rightarrow \quad IB = A^{-1} \quad \Rightarrow \quad B = A^{-1}.$$

Again $CA = I$

$$\Rightarrow \quad (CA) A^{-1} = IA^{-1} \quad \Rightarrow \quad C(AA^{-1}) = A^{-1}$$

$$\Rightarrow \quad CI = A^{-1} \quad \Rightarrow \quad C = A^{-1}.$$

Hence the theorem.

Theorem 4: *The necessary and sufficient condition for a linear transformation A on a vector space $V(F)$ to be invertible is that there exists a linear transformation B on V such that*

$$AB = I = BA.$$

Proof: The condition is necessary. For proof see theorem 2.

The condition is sufficient. For proof see theorem 3. Take B in place of C.

Also we note that $B = A^{-1}$ and $A = B^{-1}$.

on a vector space $V(F)$. Then A possesses unique inverse.

Proof: Let B and C be two inverses of A. Then

$$AB = I = BA \quad \text{and} \quad AC = I = CA.$$

We have $\quad C(AB) = CI = C.$ ...(1)

Also $\quad (CA)B = IB = B.$ ...(2)

Since product of linear transformations is associative, therefore from (1) and (2), we get

$$C(AB) = (CA)B$$

$\Rightarrow \quad C = B.$

Hence the inverse of A is unique.

Theorem 6: *Let $V(F)$ be a vector space and let A, B be linear transformations on V. Then show that*

(i) If A and B are invertible, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

(ii) If A is invertible and $a \neq 0 \in F$, then aA is invertible and

$$(aA)^{-1} = \frac{1}{a}A^{-1}.$$

(iii) If A is invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.

Proof: (i) We have

$$(B^{-1}A^{-1})(AB) = B^{-1}[A^{-1}(AB)] = B^{-1}[(A^{-1}A)B]$$
$$= B^{-1}(IB) = B^{-1}B = I.$$

Also $\quad (AB)(B^{-1}A^{-1}) = A[B(B^{-1}A^{-1})] = A[(BB^{-1})A^{-1}]$

$$= A(IA^{-1}) = AA^{-1} = I.$$

Thus $\quad (AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB).$

$\therefore$ By theorem 3, AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

(ii) We have $(aA)\left(\frac{1}{a}A^{-1}\right) = a\left[A\left(\frac{1}{a}A^{-1}\right)\right]$

$$= a\left[\frac{1}{a}(AA^{-1})\right] = \left(a\,\frac{1}{a}\right)(AA^{-1}) = 1I = I.$$

Also $\quad \left(\frac{1}{a}A^{-1}\right)(aA) = \frac{1}{a}[A^{-1}(aA)] = \frac{1}{a}[a(A^{-1}A)]$

$$= \left(\frac{1}{a}\,a\right)(A^{-1}A) = 1I = I.$$

Thus $\quad (aA)\left(\frac{1}{a}A^{-1}\right) = I = \left(\frac{1}{a}A^{-1}\right)(aA).$

$\therefore$ by theorem 3, aA is invertible and

(iii) Since A is invertible, therefore

$$AA^{-1} = I = A^{-1}A.$$

$\therefore$ by theorem 3, A^{-1} is invertible and

$$A = (A^{-1})^{-1}.$$

Singular and Non-singular transformations.

Definition: *Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$. Then T is said to be* **non-singular** *if the null space of T (i.e., ker T) consists of the zero vector alone i.e., if*

$$\alpha \in U \text{ and } T(\alpha) = \mathbf{0} \Rightarrow \alpha = \mathbf{0}.$$

If there exists a vector $\mathbf{0} \neq \alpha \in U$ such that $T(\alpha) = \mathbf{0}$, then T is said to be singular.

(Garhwal 2006, 07)

Theorem 7: *Let T be a linear transformation from a vector space $U(F)$ into a vector space $V(F)$. Then T is non-singular if and only if T is one-one.*

Proof: Given that T is non-singular. Then to prove that T is one-one.

Let $\alpha_1, \alpha_2 \in U$. Then

$$T(\alpha_1) = T(\alpha_2)$$

$\Rightarrow$ $T(\alpha_1) - T(\alpha_2) = \mathbf{0} \Rightarrow T(\alpha_1 - \alpha_2) = \mathbf{0}$

$\Rightarrow$ $\alpha_1 - \alpha_2 = \mathbf{0}$ [$\because$ T is non-singular]

$\Rightarrow$ $\alpha_1 = \alpha_2$.

$\therefore$ T is one-one.

Conversely let T be one-one. We know that $T(\mathbf{0}) = \mathbf{0}$. Since T is one-one, therefore $\alpha \in U$ and $T(\alpha) = \mathbf{0} = T(\mathbf{0}) \Rightarrow \alpha = \mathbf{0}$. Thus the null space of T consists of zero vector alone. Therefore T is non-singular.

Theorem 8: *Let T be a linear transformation from U into V. Then T is non-singular if and only if T carries each linearly independent subset of U onto a linearly independent subset of V.*

Proof: First suppose that T is non-singular.

Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$

be a linearly independent subset of U. Then image of B under T is the subset B' of V given by

$$B' = \{T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n)\}.$$

To prove that B' is linearly independent.

Let $a_1, a_2, \ldots, a_n \in F$ and let

$$a_1 T(\alpha_1) + \ldots + a_n T(\alpha_n) = \mathbf{0}$$

$\Rightarrow$ $T(a_1\alpha_1 + \ldots + a_n\alpha_n) = \mathbf{0}$ [$\because$ T is linear]

$\Rightarrow$ $a_1\alpha_1 + \ldots + a_n\alpha_n = \mathbf{0}$ [$\because$ T is non-singular]

Thus the image of B under T is linearly independent.

Conversely suppose that T carries independent subsets onto independent subsets. Then to prove that T is non-singular.

Let $\alpha \neq \mathbf{0} \in U$. Then the set $S = \{\alpha\}$ consisting of the one non-zero vector α is linearly independent. The image of S under T is the set

$$S' = \{T(\alpha)\}.$$

It is given that S' is also linearly independent. Therefore $T(\alpha) \neq \mathbf{0}$ because the set consisting of zero vector alone is linearly dependent. Thus

$$\mathbf{0} \neq \alpha \in U \Rightarrow T(\alpha) \neq \mathbf{0}.$$

This shows that the null space of T consists of the zero vector alone. Therefore T is non-singular.

Theorem 9: *(i) A linear transformation T on a finite dimensional vector space is invertible iff T is non-singular.*

(ii) A linear transformation T on a finite dimensional vector space is invertible iff T is onto.

Proof: (i) Let $V(F)$ be a finite dimensional vector space of dimension n. Let T be a linear transformation on V.

If T is invertible, then T must be one-one. Hence T must be non-singular.

Conversely if T is non-singular, then T must be one-one. Now T will be invertible if we prove that T is onto. For proof see example 3 after 4.2 in chapter 'Linear Transformations'.

(ii) If T is invertible, then T must be onto.

Conversely let T be onto. Then T will be invertible if we prove that T is one-one. For proof see example 4 after 4.2 in chapter 'Linear Transformations'.

Important Note: In the above theorem, we have proved that if T is a linear transformation on a finite dimensional vector space V, then T is one-one implies T must be onto. Also T is onto implies T must be one-one. However this theorem fails if V is not finite dimensional. This is clear from the following Illustration.

Illustration: Let $V(F)$ be the vector space of all polynomials in x with coefficients in F. Let D be the differentiation operator on V. The range of D is all of V and so D is onto. However D is not one-one because all constant polynomials are sent into $\mathbf{0}$ by D.

Now let T be the linear transformation on V defined by

$$T[f(x)] = x f(x) \ \forall \ f(x) \in V.$$

Then T is one-one.

But T is not onto V. If $g(x)$ is a non-zero constant polynomial in V, then $g(x)$ will not be the image of any element in V under the function T.

$\dim U = \dim V$. If T is a linear transformation from U into V, the following are equivalent.

(i) *T is invertible.*

(ii) *T is non-singular.*

(iii) *The range of T is V.*

(iv) *If $\{\alpha_1, \ldots, \alpha_n\}$ is any basis for U, then $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ is a basis for V.*

(v) *There is some basis $\{\alpha_1, \ldots, \alpha_n\}$ for U such that $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ is a basis for V.*

Proof: (i) $\Rightarrow$ (ii).

If T is invertible, then T is one-one. Therefore T is non-singular.

(ii) $\Rightarrow$ (iii).

Let T be non-singular. Let $\{\alpha_1, \ldots, \alpha_n\}$ be a basis for U. Then $\{\alpha_1, \ldots, \alpha_n\}$ is a linearly independent subset of U. Since T is non-singular therefore $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ is a linearly independent subset of V and it contains n vectors. Since $\dim V$ is also n, therefore this set of vectors is a basis for V. Now let β be any vector in V. Then there exist scalars $a_1, \ldots, a_n \in F$ such that

$$\beta = a_1 T(\alpha_1) + \ldots + a_n T(\alpha_n) = T(a_1 \alpha_1 + \ldots + a_n \alpha_n)$$

which shows that β is in the range of T because

$$a_1 \alpha_1 + \ldots + a_n \alpha_n \in U.$$

Thus every vector in V is in the range of T. Hence range of T is V.

(iii) $\Rightarrow$ (iv).

Now suppose that range of T is V *i.e.*, T is onto. If $\{\alpha_1, \ldots, \alpha_n\}$ is any basis for U, then the vectors $T(\alpha_1), \ldots, T(\alpha_n)$ span the range of T which is equal to V. Thus the vectors $T(\alpha_1), \ldots, T(\alpha_n)$ which are n in number span V whose dimension is also n. Therefore $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ must be a basis set for V.

(iv) $\Rightarrow$ (v).

Since U is finite dimensional, therefore there exists a basis for U. Let $\{\alpha_1, \ldots, \alpha_n\}$ be a basis for U. Then $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ is a basis for V as it is given in (iv).

(v) $\Rightarrow$ (i).

Suppose there is some basis $\{\alpha_1, \ldots, \alpha_n\}$ for U such that $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ is a basis for V. The vectors $\{T(\alpha_1), \ldots, T(\alpha_n)\}$ span the range of T. Also they span V. Therefore the range of T must be all of V *i.e.*, T is onto.

If $\alpha = c_1 \alpha_1 + \ldots + c_n \alpha_n$ is in the null space of T, then

$$T(c_1 \alpha_1 + \ldots + c_n \alpha_n) = \mathbf{0}$$

$\Rightarrow$ $c_1 T(\alpha_1) + \ldots + c_n T(\alpha_n) = \mathbf{0}$

$\Rightarrow$ $c_i = 0, 1 \le i \le n$ because $T(\alpha_1), \ldots, T(\alpha_n)$ are linearly independent

$\Rightarrow$ $\alpha = \mathbf{0}$.

$\therefore$ T is non-singular and consequently T is one-one. Hence T is invertible.

Example 10: *Describe explicitly the linear transformation* $T : \mathbf{R}^2 \to \mathbf{R}^2$ *such that* $T\,(2, 3) = (4, 5)$ *and* $T\,(1, 0) = (0, 0)$.

Solution: First we shall show that the set $\{(2, 3), (1, 0)\}$ is a basis of $\mathbf{R}^2$. For linear independence of this set let

$$a\,(2, 3) + b\,(1, 0) = (0, 0), \text{ where } a, b \in \mathbf{R}.$$

Then $\quad (2a + b, 3a) = (0, 0)$

$\Rightarrow \quad 2a + b = 0, 3a = 0$

$\Rightarrow \quad a = 0, b = 0.$

Hence the set $\{(2, 3), (1, 0)\}$ is linearly independent.

Now we shall show that the set $\{(2, 3), (1, 0)\}$ spans $\mathbf{R}^2$. Let $(x_1, x_2) \in \mathbf{R}^2$ and let $(x_1, x_2) = a\,(2, 3) + b\,(1, 0) = (2a + b, 3a)$.

Then $2a + b = x_1, 3a = x_2$. Therefore

$$a = \frac{x_2}{3}, b = \frac{3x_1 - 2x_2}{3}.$$

$$\therefore \quad (x_1, x_2) = \frac{x_2}{3}\,(2, 3) + \frac{3x_1 - 2x_2}{3}\,(1, 0). \qquad \ldots(1)$$

From the relation (1) we see that the set $\{(2, 3), (1, 0)\}$ spans $\mathbf{R}^2$. Hence this set is a basis for $\mathbf{R}^2$.

Now let (x_1, x_2) be any member of $\mathbf{R}^2$. Then we are to find a formula for $T\,(x_1, x_2)$ under the conditions that $T\,(2, 3) = (4, 5), T\,(1, 0) = (0, 0)$.

We have $\quad T\,(x_1, x_2) = T\left[\frac{x_2}{3}\,(2, 3) + \frac{3x_1 - 2x_2}{3}\,(1, 0)\right]$, by (1)

$$= \frac{x_2}{3}\,T\,(2, 3) + \frac{3x_1 - 2x_2}{3}\,T\,(1, 0), \quad \text{by linearity of } T$$

$$= \frac{x_2}{3}\,(4, 5) + \frac{3x_1 - 2x_2}{3}\,(0, 0) = \left(\frac{4x_2}{3}, \frac{5x_2}{3}\right).$$

Example 11: *Describe explicitly a linear transformation from* $V_3\,(\mathbf{R})$ *into* $V_3\,(\mathbf{R})$ *which has its range the subspace spanned by* $(1, 0, -1)$ *and* $(1, 2, 2)$.

Solution: The set $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a basis for $V_3\,(\mathbf{R})$.

Also $\{(1, 0, -1), (1, 2, 2), (0, 0, 0)\}$ is a subset of $V_3\,(\mathbf{R})$. It should be noted that in this subset the number of vectors has been taken the same as is the number of vectors in the set B.

There exists a unique linear transformation T from $V_3\,(\mathbf{R})$ into $V_3\,(\mathbf{R})$ such that and $$\left.\begin{array}{l} T\ (0,1,0)=(1,2,2), \\ T\ (0,0,1)=(0,0,0). \end{array}\right\} \qquad \ldots(1)$$

Now the vectors $T\ (1,0,0), T\ (0,1,0), T\ (0,0,1)$ span the range of T. In other words the vectors

$$(1,0,-1),(1,2,2),(0,0,0)$$

span the range of T. Thus the range of T is the subspace of $V_3\ (\mathbf{R})$ spanned by the set $\{(1,0,-1),(1,2,2)\}$ because the zero vector $(0,0,0)$ can be omitted from the spanning set. Therefore T defined in (1) is the required transformation.

Now let us find an explicit expression for T. Let (a,b,c) be any element of $V_3\ (\mathbf{R})$. Then we can write

$$(a,b,c)=a\ (1,0,0)+b\ (0,1,0)+c\ (0,0,1).$$

$$\begin{aligned} \therefore \qquad T\ (a,b,c) &= aT\ (1,0,0)+bT\ (0,1,0)+cT\ (0,0,1) \\ &= a\ (1,0,-1)+b\ (1,2,2)+c\ (0,0,0) \qquad \text{[from (1)]} \\ &= (a+b,2b,2b-a). \end{aligned}$$

Example 12: *Let T be a linear operator on $V_3\ (\mathbf{R})$ defined by*

$$T\ (a,b,c)=(3a,a-b,2a+b+c)\ \ \forall\ \ (a,b,c)\in V_3\ (\mathbf{R}).$$

Is T invertible ? If so, find a rule for T^{-1} like the one which defines T.

Solution: Let us see that T is one-one or not.

Let $\alpha=(a_1,b_1,c_1),\beta=(a_2,b_2,c_2)\in V_3\ (\mathbf{R})$.

$$\begin{aligned} &\text{Then} && T\ (\alpha)=T\ (\beta) \\ &\Rightarrow && T\ (a_1,b_1,c_1)=T\ (a_2,b_2,c_2) \\ &\Rightarrow && (3a_1,a_1-b_1,2a_1+b_1+c_1)=(3a_2,a_2-b_2,2a_2+b_2+c_2) \\ &\Rightarrow && 3a_1=3a_2,a_1-b_1=a_2-b_2,2a_1+b_1+c_1=2a_2+b_2+c_2 \\ &\Rightarrow && a_1=a_2,b_1=b_2,c_1=c_2. \end{aligned}$$

$\therefore \quad T$ is one-one.

Now T is a linear transformation on a finite dimensional vector space $V_3\ (\mathbf{R})$ whose dimension is 3. Since T is one-one, therefore T must be onto also and thus T is invertible.

If $\qquad T\ (a,b,c)=(p,q,r)$, then $T^{-1}\ (p,q,r)=(a,b,c)$.

$$\begin{aligned} &\text{Now} && T\ (a,b,c)=(p,q,r) \\ &\Rightarrow && (3a,a-b,2a+b+c)=(p,q,r) \\ &\Rightarrow && p=3a,q=a-b,r=2a+b+c \\ &\Rightarrow && a=\frac{p}{3},b=\frac{p}{3}-q,c=r-2a-b=r-\frac{2p}{3}-\frac{p}{3}+q=r-p+q. \\ &\therefore && T^{-1}\ (p,q,r)=\left(\frac{p}{3},\frac{p}{3}-q,r-p+q\right)\ \forall\ (p,q,r)\in V_3\ (\mathbf{R}) \end{aligned}$$

is the rule which defines T^{-1}.

$$T(a, b) = (\alpha a + \beta b, \gamma a + \delta b),$$

where $\alpha, \beta, \gamma, \delta$ *are fixed elements of* **C**. *Prove that* T *is invertible if and only if* $\alpha\delta - \beta\gamma \neq 0$.

Solution: The vector space $V_2(\mathbf{C})$ is of dimension 2. Therefore T is a linear transformation on a finite-dimensional vector space. T will be invertible if and only if the null space of T consists of zero vector alone. The zero vector of the space $V_2(\mathbf{C})$ is the ordered pair $(0, 0)$. Thus T is invertible

iff $T(x, y) = (0, 0) \Rightarrow x = 0, y = 0$

i.e., iff $(\alpha x + \beta y, \gamma x + \delta y) = (0, 0) \Rightarrow x = 0, y = 0$

i.e., iff $\alpha x + \beta y = 0, \gamma x + \delta y = 0 \Rightarrow x = 0, y = 0$.

Now the necessary and sufficient condition for the equations $\alpha x + \beta y = 0$, $\gamma x + \delta y = 0$ to have the only solution $x = 0, y = 0$ is that

$$\begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} \neq 0.$$

Hence T is invertible iff $\alpha\delta - \beta\gamma \neq 0$.

Example 14: *Find two linear operators* T *and* S *on* $V_2(\mathbf{R})$ *such that*

$$TS = \hat{\mathbf{0}} \quad \textit{but} \quad ST \neq \hat{\mathbf{0}}.$$

Solution: Consider the linear transformations T and S on $V_2(\mathbf{R})$ defined by

$T(a, b) = (a, 0) \ \forall \ (a, b) \in V_2(\mathbf{R})$

and $S(a, b) = (0, a) \ \forall \ (a, b) \in V_2(\mathbf{R})$.

We have $(TS)(a, b) = T[S(a, b)] = T(0, a) = (0, 0)$

$= \hat{\mathbf{0}}(a, b) \ \forall \ (a, b) \in V_2(\mathbf{R})$.

$\therefore$ $TS = \hat{\mathbf{0}}$.

Again $(ST)(a, b) = S[T(a, b)] = S(a, 0) = (0, a)$

$\neq \hat{\mathbf{0}}(a, b) \ \forall \ (a, b) \in V_2(\mathbf{R})$.

Thus $ST \neq \hat{\mathbf{0}}$.

Example 15: *Let* V *be a vector space over the field* F *and* T *a linear operator on* V. *If* $T^2 = \hat{\mathbf{0}}$, *what can you say about the relation of the range of* T *to the null space of* T? *Give an example of a linear operator* T *on* $V_2(\mathbf{R})$ *such that* $T^2 = \hat{\mathbf{0}}$ *but* $T \neq \hat{\mathbf{0}}$.

Solution: We have $T^2 = \hat{\mathbf{0}}$

$\Rightarrow$ $T^2(\alpha) = \hat{\mathbf{0}}(\alpha) \ \forall \ \alpha \in V$

$\Rightarrow$ $T[T(\alpha)] = \mathbf{0} \ \forall \ \alpha \in V$

$\Rightarrow$ $T(\alpha) \in$ null space of $T \ \forall \ \alpha \in V$.

But $T(\alpha) \in$ range of $T \ \forall \ \alpha \in V$.

For the second part of the question, consider the linear transformation T on V_2 (**R**) defined by

$$T(a, b) = (0, a) \quad \forall (a, b) \in V_2(\mathbf{R}).$$

Then obviously $T \neq \hat{\mathbf{0}}$.

We have $\quad T^2(a, b) = T[T(a, b)] = T(0, a) = (0, 0)$

$$= \hat{\mathbf{0}}(a, b) \ \forall (a, b) \in V_2(\mathbf{R}).$$

$\therefore \quad T^2 = \hat{\mathbf{0}}.$

Example 16: *If $T : U \to V$ is a linear transformation and U is finite dimensional, show that U and range of T have the same dimension iff T is non-singular. Determine all non-singular linear transformations*

$$T : V_4(\mathbf{R}) \to V_3(\mathbf{R}).$$

Solution: We know that

$$\dim U = \text{rank}(T) + \text{nullity}(T)$$
$$= \dim \text{ of range of } T + \dim \text{ of null space of } T.$$

$\therefore \quad \dim U = \dim$ of range of T

iff dim of null space of T is zero

i.e., iff null space of T consists of zero vector alone

i.e., iff T is non-singular.

Let T be a linear transformation from V_4 (**R**) into V_3 (**R**). Then T will be non-singular iff

$$\dim \text{ of } V_4(\mathbf{R}) = \dim \text{ of range of } T.$$

Now $\dim V_4(\mathbf{R}) = 4$ and dim of range of $T \leq 3$ because range of $T \subseteq V_3(\mathbf{R})$.

$\therefore \quad \dim V_4(\mathbf{R})$ cannot be equal to dim of range of T.

Hence T cannot be non-singular. Thus there can be no non-singular linear transformation from V_4 (**R**) into V_3 (**R**).

Example 17: *If A and B are linear transformations (on the same vector space), then a necessary and sufficient condition that both A and B be invertible is that both AB and BA be invertible.*

Solution: Let A and B be two invertible linear transformations on a vector space V.

We have $(AB)(B^{-1} A^{-1}) = I = (B^{-1} A^{-1})(AB)$.

$\therefore \quad AB$ is invertible.

Also we have $(BA)(A^{-1} B^{-1}) = I = (A^{-1} B^{-1})(BA)$.

$\therefore \quad BA$ is also invertible.

Thus the condition is necessary.

and onto.

First we shall show that A is invertible.

A is one-one. Let $\alpha_1, \alpha_2 \in V$. Then

$$A(\alpha_1) = A(\alpha_2) \quad \Rightarrow \quad B[A(\alpha_1)] = B[A(\alpha_2)]$$

$\Rightarrow$ $(BA)(\alpha_1) = (BA)(\alpha_2) \quad \Rightarrow \quad \alpha_1 = \alpha_2$. [$\because$ BA is one-one]

$\therefore$ A is one-one.

A is onto.

Let $\beta \in V$. Since AB is onto, therefore there exists $\alpha \in V$ such that

$$(AB)(\alpha) = \beta \quad \Rightarrow \quad A[B(\alpha)] = \beta.$$

Thus $\beta \in V \Rightarrow \exists\, B(\alpha) \in V$ such that $A[B(\alpha)] = \beta$.

$\therefore$ A is onto.

$\therefore$ A is invertible.

Interchanging the roles played by AB and BA in the above proof, we can prove that B is invertible.

Example 18: *If A is a linear transformation on a vector space V such that*

$$A^2 - A + I = \hat{\mathbf{0}},$$

then A is invertible. **(Garhwal 2007, 12)**

Solution: If $A^2 - A + I = \hat{\mathbf{0}}$, then $A^2 - A = -I$.

First we shall prove that A is one-one.

Let $\alpha_1, \alpha_2 \in V$. Then

$$A(\alpha_1) = A(\alpha_2) \qquad \ldots(1)$$

$\Rightarrow$ $A[A(\alpha_1)] = A[A(\alpha_2)]$

$\Rightarrow$ $A^2(\alpha_1) = A^2(\alpha_2) \qquad \ldots(2)$

$\Rightarrow$ $A^2(\alpha_1) - A(\alpha_1) = A^2(\alpha_2) - A(\alpha_2)$ [From (2) and (1)]

$\Rightarrow$ $(A^2 - A)(\alpha_1) = (A^2 - A)(\alpha_2)$

$\Rightarrow$ $(-I)(\alpha_1) = (-I)(\alpha_2) \Rightarrow -[I(\alpha_1)] = -[I(\alpha_2)]$

$\Rightarrow$ $-\alpha_1 = -\alpha_2 \Rightarrow \alpha_1 = \alpha_2$.

$\therefore$ A is one-one.

Now to prove that A is onto.

Let $\alpha \in V$. Then $\alpha - A(\alpha) \in V$.

We have $A[\alpha - A(\alpha)] = A(\alpha) - A^2(\alpha) = (A - A^2)(\alpha)$

$= I(\alpha)$ [$\because A^2 - A = -I \Rightarrow A - A^2 = I$]

$= \alpha$.

Thus $\alpha \in V \Rightarrow \exists\; \alpha - A(\alpha) \in V$ such that

$$A[\alpha - A(\alpha)] = \alpha.$$

Hence A is invertible.

Example 19: *Let V be a finite dimensional vector space and T be a linear operator on V. Suppose that rank (T^2) = rank (T). Prove that the range and null space of T are disjoint i.e., have only the zero vector in common.*

Solution: We have

$$\dim V = \text{rank}\,(T) + \text{nullity}\,(T)$$

and
$$\dim V = \text{rank}\,(T^2) + \text{nullity}\,(T^2).$$

Since rank (T) = rank (T^2), therefore we get

$$\text{nullity}\,(T) = \text{nullity}\,(T^2)$$

i.e., dim of null space of T = dim of null space of T^2.

Now $T(\alpha) = \mathbf{0}$

$\Rightarrow$ $\quad T[T(\alpha)] = T(\mathbf{0}) \Rightarrow T^2(\alpha) = \mathbf{0}.$

$\therefore$ $\quad \alpha \in$ null space of $T \Rightarrow \alpha \in$ null space of T^2.

$\therefore$ $\quad$ null space of $T \subseteq$ null space of T^2.

But null space of T and null space of T^2 are both subspaces of V and have the same dimension.

$\therefore$ $\quad$ null space of T = null space of T^2.

$\therefore$ $\quad$ null space of $T^2 \subseteq$ null space of T

i.e., $\quad T^2(\alpha) = \mathbf{0} \Rightarrow T(\alpha) = \mathbf{0}.$

$\therefore$ $\quad$ range and null space of T are disjoint. [See Example 1 after 5.5]

Example 20: *Let V be a finite dimensional vector space over the field F. Let $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ and $\{\beta_1, \beta_2, \ldots, \beta_n\}$ be two ordered basis for V. Show that there exists a unique invertible linear transformation T on V such that*

$$T(\alpha_i) = \beta_i, \ 1 \le i \le n.$$

Solution: We have proved in one of the previous theorems that there exists a unique linear transformation T on V such that

$$T(\alpha_i) = \beta_i, \ 1 \le i \le n.$$

Here we are to show that T is invertible. Since V is finite dimensional therefore in order to prove that T is invertible, it is sufficient to prove that T is non-singular.

Let $\quad \alpha \in V$ and $T(\alpha) = \mathbf{0}.$

Let $\quad \alpha = a_1\alpha_1 + \ldots + a_n\alpha_n$ where $a_1, \ldots, a_n \in F$.

We have $\quad T(\alpha) = \mathbf{0}$

$\Rightarrow$ $\quad T(a_1\alpha_1 + \ldots + a_n\alpha_n) = \mathbf{0}$

$\Rightarrow$ $\quad a_1 T(\alpha_1) + \ldots + a_n T(\alpha_n) = \mathbf{0}$

$\Rightarrow$ $\quad a_1\beta_1 + \ldots + a_n\beta_n = \mathbf{0}$

$\Rightarrow$ $\quad a_i = 0$ for each $1 \le i \le n$ [$\because$ $\beta_1, \ldots, \beta_n$ are linearly independent]

$\therefore$ T is non-singular because null space of T consists of zero vector alone. Hence T is invertible.

Example 21: *Let T_1 be a linear transformation on a vector space $V(F)$. Prove that the set of all linear transformations S on V for which $T_1 S = \hat{\mathbf{0}}$ is a subspace of the vector space of all linear transformations.*

Solution: Let L denote the vector space of all linear transformations on the vector space V. Let $W = \{S : S \text{ is a linear transformation on } V \text{ and } T_1 S = \hat{\mathbf{0}}\}$. To prove that W is a subspace of L.

Let $a, b \in F$ and $S_1, S_2 \in W$. Then by def. of W, we have $T_1 S_1 = \hat{\mathbf{0}}$ and $T_1 S_2 = \hat{\mathbf{0}}$.

We shall show that $T_1 (aS_1 + bS_2) = \hat{\mathbf{0}}$.

Let $\alpha \in V$. Then

$$[T_1 (aS_1 + bS_2)](\alpha) = T_1 [(aS_1 + bS_2)(\alpha)],$$

by def. of product of linear transformations

$$= T_1 [(aS_1)(\alpha) + (bS_2)(\alpha)] = T_1 [aS_1(\alpha) + bS_2(\alpha)]$$
$$= aT_1 [S_1(\alpha)] + bT_1 [S_2(\alpha)], \text{ since } T_1 \text{ is a linear transformation}$$
$$= a(T_1 S_1)(\alpha) + b(T_1 S_2)(\alpha) = a\,\hat{\mathbf{0}}(\alpha) + b\,\hat{\mathbf{0}}(\alpha)$$
$$= a\,\mathbf{0} + b\,\mathbf{0} = \mathbf{0} = \hat{\mathbf{0}}(\alpha).$$

Thus $[T_1 (aS_1 + bS_2)](\alpha) = \hat{\mathbf{0}}(\alpha) \ \forall \ \alpha \in V$.

Therefore $T_1 (aS_1 + bS_2) = \hat{\mathbf{0}}$. Consequently by def. of W, $aS_1 + bS_2 \in W$. Thus $a, b \in F$ and $S_1, S_2 \in W \Rightarrow aS_1 + bS_2 \in W$. Hence W is a subspace of L.

Comprehensive Exercise 2

1. Describe explicitly the linear transformation T from F^2 to F^2 such that $T(e_1) = (a, b), T(e_2) = (c, d)$ where $e_1 = (1, 0), e_2 = (0, 1)$.
2. Find a linear transformation $T : \mathbf{R}^2 \to \mathbf{R}^2$ such that $T(1, 0) = (1, 1)$ and $T(0, 1) = (-1, 2)$. Prove that T maps the square with vertices $(0, 0), (1, 0), (1, 1)$ and $(0, 1)$ into a parallelogram.
3. Let $T : \mathbf{R}^2 \to \mathbf{R}$ be the linear transformation for which $T(1, 1) = 3$ and $T(0, 1) = -2$. Find $T(a, b)$.
4. Describe explicitly a linear transformation from $V_3(\mathbf{R})$ into $V_4(\mathbf{R})$ which has its range the subspace spanned by the vectors $(1, 2, 0, -4), (2, 0, -1, -3)$.
5. Find a linear mapping $T : \mathbf{R}^3 \to \mathbf{R}^4$ whose image is generated by $(1, -1, 2, 3)$ and $(2, 3, -1, 0)$.

$T(a, b) = (a + b, a)$. Show that T is invertible and find a rule for T^{-1} like the one which defines T.

7. Show that the operator T on $\mathbf{R}^3$ defined by $T(x, y, z) = (x + z, x - z, y)$ is invertible and find similar rule defining T^{-1}.
8. For the linear operator T of Solved Example 12, after 5.11, prove that
$$(T^2 - I)(T - 3I) = \hat{\mathbf{0}}.$$
9. Let T and U be the linear operators on $\mathbf{R}^2$ defined by $T(a, b) = (b, a)$ and $U(a, b) = (a, 0)$. Give rules like the one defining T and U for each of the linear transformation $(U + T), UT, TU, T^2, U^2$.
10. Let T be the (unique) linear operator on $\mathbf{C}^3$ for which
$$T(1, 0, 0) = (1, 0, i),\ T(0, 1, 0) = (0, 1, 1), T(0, 0, 1) = (i, 1, 0).$$
Show that T is not invertible.
11. Show that if two linear transformations of a finite dimensional vector space coincide on a basis of that vector space, then they are identical.
12. If T is a linear transformation on a finite dimensional vector space V such that range (T) is a proper subset of V, show that there exists a non-zero element α in V with $T(\alpha) = \mathbf{0}$.
13. Let $T : \mathbf{R}^3 \to \mathbf{R}^3$ be defined as $T(a, b, c) = (0, a, b)$. Show that
$$T \neq \hat{\mathbf{0}},\ T^2 \neq \hat{\mathbf{0}} \text{ but } T^3 = \hat{\mathbf{0}}.$$
14. Let T be a linear transformation from a vector space U into a vector space V with Ker $T \neq \mathbf{0}$. Show that there exist vectors α_1 and α_2 in U such that $\alpha_1 \neq \alpha_2$ and $T\alpha_1 = T\alpha_2$.
15. Let T be a linear transformation from V_3 **(R)** into V_2 **(R)**, and let S be a linear
transformation from V_2 **(R)** into V_3 **(R)**. Prove that the transformation ST is not invertible.
16. Let A and B be linear transformations on a finite dimensional vector space V and let $AB = I$. Then A and B are both invertible and $A^{-1} = B$. Give an example to show that this is false when V is not finite dimensional.
17. If A and B are linear transformations (on the same vector space) and if $AB = I$, then A is called a left inverse of B and B is called a right inverse of A. Prove that if A has exactly one right inverse, say B, then A is invertible.
18. Prove that the set of invertible linear operators on a vector space V with the operation of composition forms a group. Check if this group is commutative.
19. Let V and W be vector spaces over the field F and let U be an isomorphism of V onto W. Prove that $T \to UTU^{-1}$ is an isomorphism of $L(V, V)$ onto $L(W, W)$.
20. If $\{\alpha_1, \ldots, \alpha_k\}$ and $\{\beta_1, \ldots \beta_k\}$ are linearly independent sets of vectors in a finite dimensional vector space V, then there exists an invertible linear transformation T on V such that
$$T(\alpha_i) = \beta_i, i = 1, \ldots, k.$$

any linear transformation $T : V \to W$, the following statements are all equivalent :

(i) T is invertible.

(ii) $R(V) = W$,

(iii) There is a basis $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ for V such that $\{T\alpha_1, T\alpha_2, \ldots, T\alpha_n\}$ is a basis for W.

22. If $V(F)$ be the vector space of all polynomials in x and D and T be the two linear operators on V defined as

$$D[f(x)] = \frac{df(x)}{dx}, T[f(x)] = xf(x)$$

for each $f(x) \in V$. Then show that the product of these two operators is not commutative and $(TD)^2 = TD + T^2 D^2$.

23. A linear operator T on a vector space V is said to be nilpotent if there exists a positive integer r such that $T^r = \hat{\mathbf{0}}$.

If V is the space of all polynomials of degree less than or equal to n over a field F, prove that the differentiation operator on V is nilpotent.

Answers 2

1. $T(x_1, x_2) = (x_1 a + x_2 c, x_1 b + x_2 d)$ **2.** $T(x_1, x_2) = (x_1 - x_2, x_1 + 2x_2)$

3. $T(a, b) = 5a - 2b$ **4.** $T(a, b, c) = (a + 2b, 2a - b, -4a - 3b)$

5. $T(a, b, c) = (a + 2b, -a + 3b, 2a - b, 3a)$

6. $T^{-1}(p, q) = (q, p - q)$ **7.** $T^{-1}(x, y, z) = \left(\frac{1}{2}x + \frac{1}{2}y, z, \frac{1}{2}x - \frac{1}{2}y\right)$

9. $(U + T)(a, b) = (a + b, a)$; $(UT)(a, b) = (b, 0)$; $(TU)(a, b) = (0, \alpha)$

$T^2(a, b) = (a, b)$; $U^2(a, b) = (a, 0)$

18. Not commutative

Objective Type Questions

Multiple Choice Qeustions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V, then range of T is a subspace of

(a) U (b) V

(c) $U \cup V$ (d) None of these.

$V(F)$ with U as finite dimensional. The rank of T is the dimension of the

(a) range of T (b) null space of T

(c) vector space U (d) vector space V.

(Garhwal 2007, 08)

3. Let U be an n-dimensional vector space over the field F, and let V be an m-dimensional vector space over F. Then the vector space $L(U, V)$ of all linear transformations from U into V is finite dimensional and is of dimension

(a) m (b) n

(c) mn (d) none of these.

4. If $T : V_2(\mathbf{R}) \to V_3(\mathbf{R})$ defined as $T(a, b) = (a + b, a - b, b)$ is a linear transformation, then nullity of T is

(a) 0 (b) 1

(c) 2 (d) none of these.

5. If T is a linear transformation from a vector space V into a vector space W, then the condition for T^{-1} to be a linear transformation from W to V is

(a) T should be one-one (b) T should be onto

(c) T should be one-one and onto (d) none of these.

6. Let F be any field and let T be a linear operator on F^2 defined by $T(a, b) = (a + b, a)$. Then $T^{-1}(a, b) =$

(a) $(b, a - b)$ (b) $(a - b, b)$

(c) $(a, a + b)$ (d) none of these.

7. If T is a linear transformation $T(a, b) = (\alpha a + \beta b, \gamma a + \gamma b)$, the T is invertible if

(a) $\alpha\beta - \gamma\delta = 0$ (b) $\alpha\beta - \gamma\delta \neq 0$

(c) $\alpha\delta - \beta\gamma = 0$ (d) $\alpha\delta - \beta\gamma \neq 0$ **(Garhwal 2007, 12)**

8. Let $V(F)$ be a vector space and let T_1, T_2 be linear Transformations on V. If T_1 and T_2 are invertible then

(a) $(T_1 T_2)^{-1} = T_2 \, T_1^{-1}$ (b) $(T_1 T_2)^{-1} = T_1 \, T_2^{-1}$

(c) $(T_1 T_2)^{-1} = T_2^{-1} T_1^{-1}$ (d) $(T_1 T_2)^{-1} = T_1^{-1} T_2^{-1}$

(Garhwal 2008, 10)

Fill in the Blank(s)

Fill in the blanks "......" so that the following statements are complete and correct.

1. Let $V(F)$ be a vector space. A linear operator on V is a function T from V into V such that $T(a\alpha + b\beta) =$ ...... for all α, β in V and for all a, b in F.

2. If $U(F)$ and $V(F)$ are two vector spaces and T is a linear transformation from U into V, then the kernel of T is a subspace of ....

3. Let U and V be vector spaces over the field F and let T be a linear transformation from U into V. Suppose that U is finite dimensional. Then

rank (T) + nullity (T) = ... **(Garhwal 2010)** transformation from V into V such that the range and null space of T are identical. Then n is ....

5. The vector space of all linear operators on an n-dimensional vector space U is of dimension ....

6. Let $V(F)$ be a vector space and let A, B be linear transformations on V. If A and B are invertible then AB is invertible and $(AB)^{-1} = \ldots\ldots$

7. A linear operator T on $\mathbf{R}^2$ defined by $T(x, y) = (ax + by, cx + dy)$ will be invertible iff ....

True or False

Write 'T' for true and 'F' for false statement.

1. Let $U(F)$ and $V(F)$ be two vector spaces and let T be a linear transformation from U into V. Then the null space of T is the set of all vectors α in U such that $T(\alpha) = \alpha$.

2. The function $T : \mathbf{R}^2 \to \mathbf{R}^2$ defined by $T(a, b) = (1 + a, b)$ is a linear transformation.

3. Two linear transformations from U into V are equal if they agree on a basis of U.

4. For two linear operators T and S on $\mathbf{R}^2$, $TS = \hat{\mathbf{0}} \to ST = \hat{\mathbf{0}}$.

5. If S and T are linear operators on a vector space U, then
$$(S + T)^2 = S^2 + 2ST + T^2.$$

6. The identity operator on a vector space is always invertible.

Answers

Multiple Choice Questions

1. (b)	2. (a)	3. (c)	4. (a)
5. (c)	6. (a)	7. (d)	8. (c)

Fill in the Blank(s)

1. $aT(\alpha) + bT(\beta)$	2. U	3. dim U
4. even	5. n^2	6. $B^{-1}A^{-1}$
7. $ad - bc \neq 0$		

True or False

1. F	2. F	3. T	4. F	5. F
6. T				

□

Chapter

6

Matrices and Linear Transformations

6.1 Matrix

Definition: *Let F be any field. A set of mn elements of F arranged in the form of a rectangular array having m rows and n columns is called an $m \times n$ matrix over the field F.*

An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

In a compact form the above matrix is represented by $A = [a_{ij}]_{m \times n}$. The element a_{ij} is called the $(i, j)^{\text{th}}$ element of the matrix A. In this element the first suffix i will always denote the number of row in which this element occurs.

If in a matrix A the number of rows is equal to the number of columns and is equal to n, then A is called a **square matrix** of order n and the elements a_{ij} for which $i = j$ constitute its **principal diagonal.**

Unit matrix: *A square matrix each of whose diagonal elements is equal to* 1 *and each of whose non-diagonal elements is equal to zero is called a unit matrix or an identity matrix.* We shall denote it by I. Thus if I is unit matrix of order n, then $I = [\delta_{ij}]_{n \times n}$ where δ_{ij} is Kronecker delta.

Diagonal matrix: *A square matrix is said to be a diagonal matrix if all the elements lying above and below the principal diagonal are equal to* 0. For example,

$$\begin{bmatrix} 0 & 2+i & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

is a diagonal matrix of order 4 over the field of complex numbers.

Null matrix. The $m \times n$ matrix whose elements are all zero is called the null matrix or (zero matrix) of the type $m \times n$.

Equality of two matrices. Definition.

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{m \times n}$. Then

$A = B$ if $a_{ij} = b_{ij}$ for each pair of subscripts i and j.

Addition of two matrices. Definition.

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$. Then we define

$$A + B = [a_{ij} + b_{ij}]_{m \times n}.$$

Multiplication of a matrix by a scalar. Definition.

Let $A = [a_{ij}]_{m \times n}$ and $a \in F$ *i.e.*, a be a scalar. Then we define

$$aA = [aa_{ij}]_{m \times n}.$$

Multiplication of two matrices. Definition.

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{jk}]_{n \times p}$ *i.e.*, the number of columns in the matrix A is equal to the number of rows in the matrix B. Then we define

$$AB = \left[\sum_{j=1}^{n} a_{ij} b_{jk} \right]_{m \times p}$$

i.e., AB is an $m \times p$ matrix whose $(i, k)^{\text{th}}$ element is equal to $\sum_{j=1}^{n} a_{ij} b_{jk}$.

If A and B are both square matrices of order n, then both the products AB and BA exist but in general $AB \neq BA$.

Transpose of a matrix. Definition:

Let $A = [a_{ij}]_{m \times n}$. The $n \times m$ matrix A^T obtained by interchanging the rows and columns of A is called the transpose of A. Thus $A^T = [b_{ij}]_{n \times m}$, where $b_{ij} = a_{ji}$, *i.e.*, the $(i, j)^{\text{th}}$ element of A^T is the $(j, i)^{\text{th}}$ element of A. If A is an $m \times n$ matrix and B is an $n \times p$ matrix, it can be shown that $(AB)^T = B^T A^T$. The transpose of a matrix A is also denoted by A^t or by A'.

Determinant of a square matrix: Let P_n denote the group of all permutations of degree n on the set $\{1, 2, \ldots, n\}$. If $\theta \in P_n$, then $\theta(i)$ will denote the image of i under θ. The symbol $(-1)^{\theta}$ for $\theta \in P_n$ will mean $+1$ if θ is an *even* permutation and -1 if θ is an *odd* permutation.

Definition: Let $A = [a_{ij}]_{n \times n}$. Then the *determinant* of A, written as det A or $|A|$ or $|a_{ij}|_{n \times n}$ is the element

The number of terms in this summation is n !because there are n !permutations in the set P_n .

We shall often use the notation

$$\begin{vmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}$$

for the determinant of the matrix $[a_{ij}]_{n\times n}$.

The following properties of determinants are worth to be noted :

(i) The determinant of a unit matrix is always equal to 1.

(ii) The determinant of a null matrix is always equal to 0.

(iii) If $A = [a_{ij}]_{n\times n}$, $B = [b_{ij}]_{n\times n}$, then det (AB) = (det A) (det B).

Cofactors. Definition: Let $A = [a_{ij}]_{n\times n}$. We define

$$A_{ij} = \text{cofactor of } a_{ij} \text{ in } A$$
$$= (-1)^{i+j}. \text{ [determinant of the matrix of order } n-1 \text{ obtained by deleting the row and column of } A \text{ passing through } a_{ij}].$$

It should be noted that

$$\sum_{i=1}^{n} a_{ik} A_{ij} = 0 \text{ if } k \neq j$$

or $$= \det A \text{ if } k = j.$$

Adjoint of a square matrix. Definition: Let $A = [a_{ij}]_{n\times n}$.

The $n \times n$ matrix which is the transpose of the matrix of cofactors of A is called the adjoint of A and is denoted by adj A.

It should be remembered that

$$A\,(\text{adj } A) = (\text{adj } A)\, A = (\det A)\, I$$

where I is the unit matrix of order n.

Inverse of a square matrix. Definition: Let A be a square matrix of order n. If there exists a square matrix B of order n such that

$$AB = I = BA$$

then A is said to be invertible and B is called the inverse of A.

Also we write $B = A^{-1}$.

The following results should be remembered :

(i) The necessary and sufficient condition for a square matrix A to be invertible is that det $A \neq 0$.

(ii) If A is invertible, then A^{-1} is unique and

$$A^{-1} = \frac{1}{\det A} (\text{adj. } A).$$

(iii) If A and B are invertible square matrices of order n, then AB is also invertible and $(AB)^{-1} = B^{-1} A^{-1}$.

(iv) If A is invertible, so is A^{-1} and $(A^{-1})^{-1} = A$.

Elementary row operations on a matrix.

Definition: Let A be an $m \times n$ matrix over the field F. The following three operations are called elementary row operations :

(1) multiplication of any row of A by a non-zero element c of F.

(2) addition to the elements of any row of A the corresponding elements of any other row of A multiplied by any element a in F.

(3) interchange of two rows of A.

Row equivalent matrices. Definition: If A and B are $m \times n$ matrices over the field F, then B is said to be row equivalent to A if B can be obtained from A by a finite sequence of elementary row operations. It can be easily seen that the relation of being row equivalent is an equivalence relation in the set of all $m \times n$ matrices over F.

Row reduced Echelon matrix. Definition:

An $m \times n$ matrix R is called a row reduced echelon matrix if :

(1) Every row of R which has all its entries 0 occurs below every row which has a non-zero entry.

(2) The first non-zero entry in each non-zero row is equal to 1.

(3) If the first non-zero entry in row i appears in column k_i, then all other entries in column k_i are zero.

(4) If r is the number of non-zero rows, then

$$k_1 < k_2 < \ldots < k_r$$

(*i.e.*, the first non-zero entry in row i is to the left of the first non-zero entry in row $i + 1$).

Row and column rank of a matrix.

Definition. Let $A = [a_{ij}]_{m \times n}$ be an $m \times n$ matrix over the field F. The row vectors of A are the vectors $\alpha_1, \ldots, \alpha_m \in V_n(F)$ defined by

$$\alpha_i = (a_{i1}, a_{i2}, \ldots, a_{in}), 1 \leq i \leq m.$$

The row space of A is the subspace of $V_n(F)$ spanned by these vectors. The **row rank** of A is the dimension of the row space of A.

The column vectors of A are the vectors $\beta_1, \ldots, \beta_n \in V_m(F)$ defined by

$$\beta_j = (a_{1j}, a_{2j}, \ldots, a_{mj}), 1 \leq j \leq n.$$

column rank of A is the dimension of the column space of A.

The following two results are to be remembered :

(1) Row equivalent matrices have the same row space.

(2) If R is a non-zero row reduced Echelon matrix, then the non-zero row vectors of R are linearly independent and therefore they form a basis for the row space of R.

In order to find the row rank of a matrix A, we should reduce it to row reduced Echelon matrix R by elementary row operations. The number of non-zero rows in R will give us the row rank of A.

6.2 Representation of Transformations by Matrices

Martix of a linear transformation. Let U be an n-dimensional vector space over the field F and let V be an m-dimensional vector space over F. Let

$$B = \{\alpha_1, \dots, \alpha_n\} \text{ and } B' = \{\beta_1, \dots, \beta_m\}$$

be ordered basis for U and V respectively. Suppose T is a linear transformation from U into V. We know that T is completely determined by its action on the vectors α_j belonging to a basis for U. Each of the n vectors $T(\alpha_j)$ is uniquely expressible as a linear combination of $\beta_1, \dots, \beta_m$ because $T(\alpha_j) \in V$ and these m vectors form a basis for V. Let for $j = 1, 2, \dots, n$,

$$T(\alpha_j) = a_{1j}\beta_1 + a_{2j}\beta_2 + \dots + a_{mj}\beta_m = \sum_{i=1}^{m} a_{ij}\beta_i .$$

The scalars $a_{1j}, a_{2j}, \dots, a_{mj}$ are the coordinates of $T(\alpha_j)$ in the ordered basis B'. The $m \times n$ matrix whose j^{th} column ($j = 1, 2, \dots, n$) consists of these coordinates is called the **matrix of the linear transformation T relative to the pair of ordered bases B and B'**. We shall denote it by the symbol $[T ; B ; B']$ or simply by $[T]$ if the bases are understood. Thus

$$\begin{aligned} [T] &= [T ; B ; B'] \\ &= \text{matrix of } T \text{ relative to the ordered bases } B \text{ and } B' \\ &= [a_{ij}]_{m \times n} \end{aligned}$$

where $\quad T(\alpha_j) = \sum_{i=1}^{m} a_{ij}\beta_i$, for each $j = 1, 2, \dots, n$. ...(1)

The coordinates of $T(\alpha_1)$ in the ordered basis B' form the first column of this matrix, the coordinates of $T(\alpha_2)$ in the ordered basis B' form the second column of this matrix and so on.

The $m \times n$ matrix $[a_{ij}]_{m \times n}$ completely determines the linear transformation T through the formulae given in (1). Therefore the matrix $[a_{ij}]_{m \times n}$ represents the transformation T.

into itself. Then in order to represent T by a matrix, it is most convenient to use the same ordered basis in each case, *i.e.*, to take $B = B'$. The representing matrix will then be called the **matrix of T relative to the ordered basis B** and will be denoted by $[T ; B]$ or sometimes also by $[T]_B$.

Thus if $B = \{\alpha_1, \ldots, \alpha_n\}$ is an ordered basis for V, then

$$[T]_B \text{ or } [T ; B] = \text{matrix of } T \text{ relative to the ordered basis } B$$
$$= [a_{ij}]_{n \times n},$$

where $$T(\alpha_j) = \sum_{i=1}^{n} a_{ij} \alpha_i, \text{ for each } j = 1, 2, \ldots, n.$$

Illustrative Examples

Example 1: *Let T be a linear transformation on the vector space $V_2(F)$ defined by $T(a, b) = (a, 0)$.*

Write the matrix of T relative to the standard ordered basis of $V_2(F)$.

Solution: Let $B = \{\alpha_1, \alpha_2\}$ be the standard ordered basis for $V_2(F)$. Then $\alpha_1 = (1, 0), \alpha_2 = (0, 1)$.

We have $\quad T(\alpha_1) = T(1, 0) = (1, 0)$.

Now let us express $T(\alpha_1)$ as a linear combination of vectors in B. We have

$$T(\alpha_1) = (1, 0) = 1(1, 0) + 0(0, 1) = 1\alpha_1 + 0\alpha_2.$$

Thus $1, 0$ are the coordinates of $T(\alpha_1)$ with respect to the ordered basis B. These coordinates will form the first column of matrix of T relative to ordered basis B.

Again $$T(\alpha_2) = T(0, 1) = (0, 0) = 0(1, 0) + 0(0, 1).$$

Thus $0, 0$ are the coordinates of $T(\alpha_2)$ and will form second column of matrix of T relative to ordered basis B.

Thus matrix of T relative to ordered basis B

$$= [T]_B \quad \text{or} \quad [T ; B] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Example 2: *Let $V(\mathbf{R})$ be the vector space of all polynomials in x with coefficients in $\mathbf{R}$ of the form*

$$f(x) = a_0 x^0 + a_1 x + a_2 x^2 + a_3 x^3$$

i.e., the space of polynomials of degree three or less. The differentiation operator D is a linear transformation on V. The set

$$B = \{\alpha_1, \ldots, \alpha_4\} \text{ where } \alpha_1 = x^0, \alpha_2 = x^1, \alpha_3 = x^2, \alpha_4 = x^3$$

is an ordered basis for V. Write the matrix of D relative to the ordered basis B.

Solution: We have

$$= 0\alpha_1 + 0\alpha_2 + 0\alpha_3 + 0\alpha_4$$

$$D(\alpha_2) = D(x^1) = x^0 = 1x^0 + 0x^1 + 0x^2 + 0x^3$$

$$= 1\alpha_1 + 0\alpha_2 + 0\alpha_3 + 0\alpha_4$$

$$D(\alpha_3) = D(x^2) = 2x^1 = 0x^0 + 2x^1 + 0x^2 + 0x^3$$

$$= 0\alpha_1 + 2\alpha_2 + 0\alpha_3 + 0\alpha_4$$

$$D(\alpha_4) = D(x^3) = 3x^2 = 0x^0 + 0x^1 + 3x^2 + 0x^3$$

$$= 0\alpha_1 + 0\alpha_2 + 3\alpha_3 + 0\alpha_4.$$

$\therefore$ the matrix of D relative to the ordered basis B

$$= [D\,;B] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 4}.$$

Theorem 1. *Let U be an n-dimensional vector space over the field F and let V be an m-dimensional vector space over F. Let B and B' be ordered basis for U and V respectively. Then cooresponding to every matrix $[a_{ij}]_{m\times n}$ of mn scalars belonging to F there corresponds a unique linear transformation T from U into V such that*

$$[T\,;B\,;B'] = [a_{ij}]_{m\times n}.$$

Proof: Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \ldots, \beta_m\}$.

Now $\quad \sum_{i=1}^{m} a_{i1}\,\beta_i, \; \sum_{i=1}^{m} a_{i2}\,\beta_i, \ldots, \sum_{i=1}^{m} a_{in}\,\beta_i$

are vectors belonging to V because each of them is a linear combination of the vectors belonging to a basis for V. It should be noted that the vector $\sum_{i=1}^{m} a_{ij}\,\beta_i$ has been obtained with the help of the j^{th} column of the matrix $[a_{ij}]_{m\times n}$.

Since B is a basis for U, therefore by the theorem 2 of article 5.6 of chapter 5 there exists a unique linear transformation T from U into V such that

$$T(\alpha_j) = \sum_{i=1}^{m} a_{ij}\,\beta_i \text{ where } j = 1, 2, \ldots, n. \qquad \ldots(1)$$

By our definition of matrix of a linear transformation, we have from (1)

$$[T\,;B\,;B'] = [a_{ij}]_{m\times n}.$$

Note: If we take $V = U$, then in place of B', we also take B. In that case the above theorem will run as :

Let V be an n-dimensional vector space over the field F and B be an ordered basis or **co-ordinate system** *for V. Then corresponding to every matrix $[a_{ij}]_{n\times n}$ of n^2 scalars belonging to F there corresponds a unique linear transformation T from V into V such that*

$$[T\,;B] \text{ or } [T]_B = [a_{ij}]_{n\times n}.$$

our aim is to establish a formula which will give us the image of any vector under a linear transformation T in terms of its matrix.

Theorem 2: *Let T be a linear transformation from an n-dimensional vector space U into an m-dimensional vector space V and let B and B' be ordered bases for U and V respectively. If A is the matrix of T relative to B and B' then $\forall\ \alpha \in U$, we have*

$$[T(\alpha)]_{B'} = A\,[\alpha]_B$$

where $[\alpha]_B$ is the co-ordinate matrix of α with respect to ordered basis B and $[T(\alpha)]_{B'}$ is co-ordinate matrix of $T(\alpha) \in V$ with respect to B'.

Proof: Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$

and $B' = \{\beta_1, \beta_2, \ldots, \beta_m\}$.

Then $A = [T\,;\,B\,;\,B'] = [a_{ij}]_{m \times n}$,

where $$T(\alpha_j) = \sum_{i=1}^{m} a_{ij}\,\beta_i\,,\ j = 1, 2, \ldots, n. \qquad \ldots(1)$$

If $\alpha = x_1\alpha_1 + \ldots + x_n\,\alpha_n$ is a vector in U, then

$$T(\alpha) = T\left(\sum_{j=1}^{n} x_j\,\alpha_j\right)$$

$$= \sum_{j=1}^{n} x_j\,T(\alpha_j) \qquad [\because\ T \text{ is a linear transformation}]$$

$$= \sum_{j=1}^{n} x_j \sum_{i=1}^{m} a_{ij}\,\beta_i \qquad [\text{From } (1)]$$

$$= \sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}\,x_j\right)\beta_i\,. \qquad \ldots(2)$$

The co-ordinate matrix of $T(\alpha)$ with respect to ordered basis B' is an $m \times 1$ matrix. From (2), we see that the i^{th} entry of this column matrix $[T(\alpha)]_{B'}$

$$= \sum_{j=1}^{n} a_{ij}\,x_j$$

i.e., the coefficient of β_i in the linear combination (2) for $T(\alpha)$.

If X is the co-ordinate matrix $[\alpha]_B$ of α with respect to ordered basis B, then X is an $n \times 1$ matrix. The product AX will be an $m \times 1$ matrix. The i^{th} entry of this column matrix AX will be

$$= \sum_{j=1}^{n} a_{ij}\,x_j\,.$$

$\therefore$ $$[T(\alpha)]_{B'} = AX = A\,[\alpha]_B = [T\,;\,B\,;\,B']\,[\alpha]_B\,.$$

Note: If we take $U = V$, then the above result will be

$$[T(\alpha)]_B = [T]_B\,[\alpha]_B\,.$$

Theorem 3: *Let $V(F)$ be an n-dimensional vector space and B be any ordered basis for V. If I be the identity transformation and $\hat{\mathbf{0}}$ be the zero transformation on V, then*

(i) $[I\,;B]=I$ *(unit matrix of order n) and*

(ii) $[\hat{\mathbf{0}}\,;B]=$ *null matrix of the type $n\times n$.*

Proof. Let $B=\{\alpha_1,\alpha_2,\ldots,\alpha_n\}$.

(i) We have
$$\begin{aligned} I(\alpha_j) &= \alpha_j\,,\ j=1,2,\ldots,n \\ &= 0\alpha_1+\ldots+1\alpha_j+0\alpha_{j+1}+\ldots+0\alpha_n \\ &= \sum_{i=1}^{n}\delta_{ij}\,\alpha_i\,, \text{ where } \delta_{ij} \text{ is Kronecker delta.}\end{aligned}$$

$\therefore$ By def. of matrix of a linear transformation, we have
$$[I\,;B]=[\delta_{ij}]_{n\times n}=I \quad i.e., \text{ unit matrix of order } n.$$

(ii) We have $\hat{\mathbf{0}}(\alpha_j)=\mathbf{0},\ j=1,2,\ldots,n$
$$\begin{aligned} &= 0\alpha_1+0\alpha_2+\ldots+0\alpha_n \\ &= \sum_{i=1}^{n}o_{ij}\,\alpha_i\,, \text{ where each } o_{ij}=0.\end{aligned}$$

$\therefore$ By def. of matrix of a linear transformation, we have
$$[\hat{\mathbf{0}}\,;B]=[o_{ij}]_{n\times n}=\text{null matrix of the type } n\times n.$$

Theorem 4: *Let T and S be linear transformations from an n-dimensional vector space U into an m-dimensional vector space V and let B and B' be ordered bases for U and V respectively. Then*

(i) $[T+S\,;B\,;B']=[T\,;B\,;B']+[S\,;B\,;B']$

(ii) $[cT\,;B\,;B']=c\,[T\,;B\,;B']$ *where c is any scalar.*

Proof: Let $B=\{\alpha_1,\alpha_2,\ldots,\alpha_n\}$ and $B'=\{\beta_1,\beta_2,\ldots,\beta_m\}$.

Let $[a_{ij}]_{m\times n}$ be the matrix of T relative to B, B'. Then
$$T(\alpha_j)=\sum_{i=1}^{m}a_{ij}\,\beta_i\,,\ j=1,2,\ldots,n.$$

Also let $[b_{ij}]_{m\times n}$ be the matrix of S relative to B, B'. Then
$$S(\alpha_j)=\sum_{i=1}^{m}b_{ij}\,\beta_i\,,\ j=1,2,\ldots,n.$$

(i) We have
$$\begin{aligned}(T+S)(\alpha_j) &= T(\alpha_j)+S(\alpha_j),\ \ j=1,2,\ldots,n \\ &= \sum_{i=1}^{m}a_{ij}\,\beta_i+\sum_{i=1}^{m}b_{ij}\,\beta_i=\sum_{i=1}^{m}(a_{ij}+b_{ij})\,\beta_i\,.\end{aligned}$$

$\therefore$ matrix of $T+S$ relative to $B, B'=[a_{ij}+b_{ij}]_{m\times n}$
$$=[a_{ij}]_{m\times n}+[b_{ij}]_{m\times n}\,,$$

(ii) We have $(cT)(\alpha_j) = cT(\alpha_j), j = 1, 2, \dots, n$

$$= c \sum_{i=1}^{m} a_{ij} \beta_i = \sum_{i=1}^{m} (ca_{ij}) \beta_i .$$

$\therefore$ $[cT ; B ; B'] =$ matrix of cT relative to B, B'

$$= [ca_{ij}]_{m \times n} = c [a_{ij}]_{m \times n} = c [T ; B ; B'].$$

Theorem 5: *Let U, V and W be finite dimensional vector spaces over the field F ; let T be a linear transformation from U into V and S a linear transformation from V into W. Further let B, B' and B'' be ordered basis for spaces U, V and W respectively. If A is the matrix of T relative to the pair B, B' and D is the matrix of S relative to the pair B', B'' then the matrix of the composite transformation ST relative to the pair B, B'' is the product matrix $C = DA$.*

Proof: Let $\dim U = n, \dim V = m$ and $\dim W = p$. Further let

$$B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}, B' = \{\beta_1, \beta_2, \dots, \beta_m\}$$

and $B'' = \{\gamma_1, \gamma_2, \dots, \gamma_p\}$.

Let $A = [a_{ij}]_{m \times n}$, $D = [d_{ki}]_{p \times m}$ and $C = [c_{kj}]_{p \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^{m} a_{ij} \beta_i , j = 1, 2, \dots, n, \qquad \dots(1)$$

$$S(\beta_i) = \sum_{k=1}^{p} d_{ki} \gamma_k , i = 1, 2, \dots, m. \qquad \dots(2)$$

and $$(ST)(\alpha_j) = \sum_{k=1}^{p} c_{kj} \gamma_k , j = 1, 2, \dots, n. \qquad \dots(3)$$

We have $(ST)(\alpha_j) = S[T(\alpha_j)], j = 1, 2, \dots, n$

$$= S\left(\sum_{i=1}^{m} a_{ij} \beta_i\right) \qquad \text{[From (1)]}$$

$$= \sum_{i=1}^{m} a_{ij} S(\beta_i) \qquad [\because S \text{ is linear}]$$

$$= \sum_{i=1}^{m} a_{ij} \sum_{k=1}^{p} d_{ki} \gamma_k \qquad \text{[From (2)]}$$

$$= \sum_{k=1}^{p} \left(\sum_{i=1}^{m} d_{ki} a_{ij}\right) \gamma_k . \qquad \dots(4)$$

Therefore from (3) and (4), we have

$$c_{kj} = \sum_{i=1}^{m} d_{ki} a_{ij}, j = 1, 2, \dots, n ; k = 1, 2, \dots, p.$$

$\therefore$ $$[c_{kj}]_{p \times n} = \left[\sum_{i=1}^{m} d_{ki} a_{ij}\right]_{p \times n}$$

$$= [d_{ki}]_{p \times m} [a_{ij}]_{m \times n}, \quad \text{by def. of product of two matrices.}$$

Note: If $U = V = W$, then the statement and proof of the above theorem will be as follows :

Let V be an n-dimensional vector space over the field F; let T and S be linear transformations of V. Further let B be an ordered basis for V. If A is the matrix of T relative to B, and D is the matrix of S relative to B, then the matrix of the composite transformation ST relative to B is the product matrix

$$C = DA \text{ i.e., } [ST]_B = [S]_B \, [T]_B .$$

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$.

Let $A = [a_{ij}]_{n \times n}$, $D = [d_{ki}]_{n \times n}$ and $C = [c_{kj}]_{n \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^{n} a_{ij} \, \alpha_i, \; j = 1, 2, \dots, n, \qquad \dots(1)$$

$$S(\alpha_i) = \sum_{k=1}^{n} d_{ki} \, \alpha_k, \; i = 1, 2, \dots, n, \qquad \dots(2)$$

and

$$(ST)(\alpha_j) = \sum_{k=1}^{n} c_{kj} \, \alpha_k, \; j = 1, 2, \dots, n. \qquad \dots(3)$$

We have $(ST)(\alpha_j) = S[T(\alpha_j)]$

$$= S\left(\sum_{i=1}^{n} a_{ij} \, \alpha_i\right) = \sum_{i=1}^{n} a_{ij} \, S(\alpha_i) = \sum_{i=1}^{n} a_{ij} \sum_{k=1}^{n} d_{ki} \, \alpha_k$$

$$= \sum_{k=1}^{n} \left(\sum_{i=1}^{n} d_{ki} \, a_{ij}\right) \alpha_k . \qquad \dots(4)$$

$\therefore$ from (3) and (4), we have

$$c_{kj} = \sum_{i=1}^{n} d_{ki} \, a_{ij} .$$

$$\therefore \quad [c_{kj}]_{n \times n} = \left[\sum_{i=1}^{n} d_{ki} \, a_{ij}\right]_{n \times n} = [d_{ki}]_{n \times n} \, [a_{ij}]_{n \times n} .$$

$\therefore \quad C = DA.$

Theorem 6: *Let U be an n-dimensional vector space over the field F and let V be an m-dimensional vector space over F. For each pair of ordered bases B, B' for U and V respectively, the function which assigns to a linear transformation T its matrix relative to B, B' is an isomorphism between the space $L(U, V)$ and the space of all $m \times n$ matrices over the field F.*

Proof. Let $B = \{\alpha_1, \dots, \alpha_n\}$ and $B' = \{\beta_1, \dots, \beta_m\}$.

Let M be the vector space of all $m \times n$ matrices over the field F. Let

$$\psi : L(U, V) \to M \text{ such that}$$

$$\psi(T) = [T ; B ; B'] \;\; \forall \, T \in L(U, V).$$

Let $T_1, T_2 \in L(U, V)$; and let

Then $\quad T_1(\alpha_j) = \sum_{i=1}^{m} a_{ij}\,\beta_i,\ j = 1, 2, \ldots, n$

and $\quad T_2(\alpha_j) = \sum_{i=1}^{m} b_{ij}\,\beta_i,\ j = 1, 2, \ldots, n.$

To prove that ψ **is one-one.**

We have $\quad \psi(T_1) = \psi(T_2)$

$\Rightarrow \quad [T_1 ; B ; B'] = [T_2 ; B ; B']$ $\quad$ [by def. of ψ]

$\Rightarrow \quad [a_{ij}]_{m \times n} = [b_{ij}]_{m \times n}$

$\Rightarrow \quad a_{ij} = b_{ij}$ for $i = 1, \ldots, m$ and $j = 1, \ldots, n$

$\Rightarrow \quad \sum_{i=1}^{m} a_{ij}\,\beta_i = \sum_{i=1}^{m} b_{ij}\,\beta_i$ for $j = 1, \ldots, n$

$\Rightarrow \quad T_1(\alpha_j) = T_2(\alpha_j)$ for $j = 1, \ldots, n$

$\Rightarrow \quad T_1 = T_2$ $\quad$ [$\because$ T_1 and T_2 agree on a basis for U]

$\therefore \quad \psi$ is one-one.

ψ is onto.

Let $[c_{ij}]_{m \times n} \in M$. Then there exists a linear transformation T from U into V such that

$$T(\alpha_j) = \sum_{i=1}^{m} c_{ij}\,\beta_i,\ j = 1, 2, \ldots, n.$$

We have $[T ; B ; B'] = [c_{ij}]_{m \times n} \quad \Rightarrow \quad \boldsymbol{\psi(T) = [c_{ij}]_{m \times n}}$.

$\therefore \quad \psi$ is onto.

ψ is a linear transformation.

If $a, b \in F$, then

$$\begin{aligned} \psi(aT_1 + bT_2) &= [aT_1 + bT_2 ; B ; B'] && \text{[by def. of } \psi] \\ &= [aT_1 ; B ; B'] + [bT_2 ; B ; B'] && \text{[by theorem 4]} \\ &= a\,[T_1 ; B ; B'] + b\,[T_2 ; B ; B'] && \text{[by theorem 4]} \\ &= a\psi(T_1) + b\psi(T_2), \text{ by def. of } \psi. \end{aligned}$$

$\therefore \quad \psi$ is a linear transformation.

Hence ψ is an isomorphism from $L(U, V)$ onto M.

Note: It should be noted that in the above theorem if $U \to V$, then ψ also preserves products and I *i.e.*,

$$\psi(T_1 T_2) = \psi(T_1)\,\psi(T_2)$$

and $\quad \psi(I) = I \quad$ *i.e.*, unit matrix.

Theorem 7: *Let T be a linear operator on an n-dimensional vector space V and let B be an ordered basis for V. Prove that T is invertible iff $[T]_B$ is an invertible matrix. Also if T is invertible, then*

$$[T^{-1}]_B = [T]_B^{-1},$$

Proof: Let T be invertible. Then T^{-1} exists and we have

$$T^{-1} T = I = T T^{-1} \quad \Rightarrow \quad [T^{-1} T]_B = [I]_B = [T T^{-1}]_B$$

$$\Rightarrow \quad [T^{-1}]_B [T]_B = I = [T]_B [T^{-1}]_B$$

$$\Rightarrow \quad [T]_B \text{ is invertible and } ([T]_B)^{-1} = [T^{-1}]_B .$$

Conversely, let $[T]_B$ be an invertible matrix. Let $[T]_B = A$. Let $C = A^{-1}$ and let S be the linear transformation of V such that

$$[S]_B = C.$$

We have $C A = I = AC$

$$\Rightarrow \quad [S]_B [T]_B = I = [T]_B [S]_B \quad \Rightarrow \quad [ST]_B = [I]_B = [TS]_B$$

$$\Rightarrow \quad ST = I = TS \quad \Rightarrow \quad T \text{ is invertible.}$$

Change of basis. Suppose V is an n-dimensional vector space over the field F. Let B and B' be two ordered bases for V. If α is any vector in V, then we are now interested to know what is the relation between its coordinates with respect to B and its coordinates with respect to B'. **(Garhwal 2010)**

Theorem 8: *Let $V(F)$ be an n-dimensional vector space and let B and B' be two ordered bases for V. Then there is a unique necessarily invertible, $n \times n$ matrix A with entries in F such that*

(1) $[\alpha]_B = A [\alpha]_{B'}$ **(Garhwal 2008)**

(2) $[\alpha]_{B'} = A^{-1} [\alpha]_B$ *for every vector α in V.*

Proof: Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$.

Then there exists a unique linear transformation T from V into V such that

$$T(\alpha_j) = \beta_j, \quad j = 1, 2, \dots, n. \qquad \dots(1)$$

Since T maps a basis B onto a basis B', therefore T is necessarily invertible. The matrix of T relative to B *i.e.*, $[T]_B$ will be a unique $n \times n$ matrix with elements in F. Also this matrix will be invertible because T is invertible.

Let $[T]_B = A = [a_{ij}]_{n \times n}$. Then

$$T(\alpha_j) = \sum_{i=1}^{n} a_{ij} \alpha_i, \; j = 1, 2, \dots, n. \qquad \dots(2)$$

Let $x_1, x_2, \dots, x_n$ be the coordinates of α with respect to B and $y_1, y_2, \dots, y_n$ be the coordinates of α with respect to B'. Then

$$\alpha = y_1 \beta_1 + y_2 \beta_2 + \dots + y_n \beta_n = \sum_{j=1}^{n} y_j \beta_j$$

$$= \sum_{j=1}^{n} y_j \, T(\alpha_j) \qquad \text{[From (1)]}$$

$$= \sum_{j=1}^{n} y_j \sum_{i=1}^{n} a_{ij} \alpha_i \qquad \text{[From (2)]}$$

$$i=1 \; (j=1 \quad , \quad ,)$$

Also $\qquad \alpha = \sum_{i=1}^{n} x_i \, \alpha_i$.

$\therefore \qquad x_i = \sum_{j=1}^{n} a_{ij} \; y_j$

because the expression for α as a linear combination of elements of B is unique.

Now $[\alpha]_B$ is a column matrix of the type $n \times 1$. Also $[\alpha]_{B'}$ is a column matrix of the type $n \times 1$. The product matrix $A \, [\alpha]_{B'}$ will also be of the type $n \times 1$.

The i^{th} entry of $[\alpha]_B = x_i = \sum_{j=1}^{n} a_{ij} \; y_j = i^{\text{th}}$ entry of $A \, [\alpha]_{B'}$.

$$\therefore \qquad [\alpha]_B = A \, [\alpha]_{B'} \qquad \Rightarrow \qquad A^{-1} \, [\alpha]_B = A^{-1} \, A \, [\alpha]_{B'}$$

$$\Rightarrow \qquad A^{-1} \, [\alpha]_B = I \, [\alpha]_B \qquad \Rightarrow \qquad A^{-1} \, [\alpha]_B = [\alpha]_{B'} \, .$$

Note: The matrix $A = [T\,]_B$ is called the transition matrix from B to $B\,'$. It expresses the coordinates of each vector in V relative to B in terms of its coordinates relative to B'.

How to write the transition matrix from one basis to another ?

Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \ldots, \beta_n\}$ be two ordered basis for the n-dimensional vector space $V\,(F\,)$. Let A be the transition matrix from the basis B to the basis B'. Let T be the linear transformation from V into V which maps the basis B onto the basis $B\,'$. Then A is the matrix of T relative to B *i.e.*, $A = [T\,]_B$. So in order to find the matrix A, we should first express each vector in the basis $B\,'$ as a linear combination over F of the vectors in B. Thus we write the relations

$$\begin{aligned}
\beta_1 &= a_{11}\,\alpha_1 + a_{21}\,\alpha_2 + \ldots + a_{n1}\,\alpha_n \\
\beta_2 &= a_{12}\,\alpha_1 + a_{22}\,\alpha_2 + \ldots + a_{n2}\,\alpha_n \\
\ldots & \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\
\ldots & \quad \ldots \quad \ldots \quad \ldots \quad \ldots \\
\beta_n &= a_{1n}\,\alpha_1 + a_{2n}\,\alpha_2 + \ldots + a_{nn}\,\alpha_n \, .
\end{aligned}$$

Then the matrix $A = [a_{ij}]_{n \times n}$ *i.e.*, A is the transpose of the matrix of coefficients in the above relations. Thus

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}.$$

Now suppose α is any vector in V. If $[\alpha]_B$ is the coordinate matrix of α relative to the basis B and $[\alpha]_{B'}$ its coordinate matrix relative to the basis $B\,'$ then

$$[\alpha]_B = A \, [\alpha]_{B'}$$

Theorem 9: *Let* $B = \{\alpha_1, \dots, \alpha_n\}$ *and* $B' = \{\beta_1, \dots, \beta_n\}$ *be two ordered basis for an n-dimensional vector space* $V(F)$. *If* $(x_1, \dots, x_n)$ *is an ordered set of n scalars, let* $\alpha = \sum_{i=1}^{n} x_i \alpha_i$ *and* $\beta = \sum_{i=1}^{n} x_i \beta_i$.

Then show that $T(\alpha) = \beta$, *where T is the linear operator on V defined by*

$$T(\alpha_i) = \beta_i, \quad i = 1, 2, \dots, n.$$

Proof: We have $T(\alpha) = T\left(\sum_{i=1}^{n} x_i \alpha_i\right) = \sum_{i=1}^{n} x_i T(\alpha_i)$ [$\because$ T is linear]

$$= \sum_{i=1}^{n} x_i \beta_i = \beta.$$

Similarity.

Similarity of matrices. Definition: *Let A and B be square matrices of order n over the field F. Then B is said to be similar to A if there exists an* $n \times n$ *invertible square matrix C with elements in F such that*

$$B = C^{-1} AC.$$

(Garhwal 2010)

Theorem 10: *The relation of similarity is an equivalence relation in the set of all* $n \times n$ *matrices over the field F.*

Proof: If A and B are two $n \times n$ matrices over the field F, then B is said to be similar to A if there exists an $n \times n$ invertible matrix C over F such that

$$B = C^{-1} AC.$$

Reflexive. Let A be any $n \times n$ matrix over F. We can write $A = I^{-1} AI$, where I is $n \times n$ unit matrix over F.

$\therefore$ A is similar to A because I is definitely invertible.

Symmetric. Let A be similar to B. Then there exists an $n \times n$ invertible matrix P over F such that

$$A = P^{-1} BP$$

$$\Rightarrow \quad PAP^{-1} = P(P^{-1}BP)P^{-1}$$

$$\Rightarrow \quad PAP^{-1} = B \quad \Rightarrow \quad B = PAP^{-1}$$

$$\Rightarrow \quad B = (P^{-1})^{-1} AP^{-1}$$

[$\because$ P is invertible means P^{-1} is invertible and $(P^{-1})^{-1} = P$]

$\Rightarrow$ B is similar to A.

Transitive. Let A be similar to B and B be similar to C. Then

$$A = P^{-1} BP \quad \text{and} \quad B = Q^{-1} CQ,$$

where P and Q are invertible $n \times n$ matrices over F.

$$= (P^{-1} Q^{-1}) C (QP)$$
$$= (QP)^{-1} C (QP) \quad [\because P \text{ and } Q \text{ are invertible means } QP \text{ is invertible and } (QP)^{-1} = P^{-1}Q^{-1}]$$

$\therefore$ A is similar to C.

Hence similarity is an equivalence relation on the set of $n \times n$ matrices over the field F.

Theorem 11: *Similar matrices have the same determinant.*

Proof: Let B be similar to A. Then there exists an invertible matrix C such that

$$B = C^{-1} AC$$

$\Rightarrow$ $\det B = \det (C^{-1} AC) \Rightarrow \det B = (\det C^{-1}) (\det A) (\det C)$

$\Rightarrow$ $\det B = (\det C^{-1}) (\det C) (\det A) \Rightarrow \det B = (\det C^{-1} C) (\det A)$

$\Rightarrow$ $\det B = (\det I) (\det A) \Rightarrow \det B = 1 (\det A) \Rightarrow \det B = \det A.$

Similarity of linear transformations. Definition: *Let A and B be linear transformations on a vector space $V (F)$. Then B is said to be similar to A if there exists an invertible linear transformation C on V such that*

$$B = CAC^{-1}.$$

(Garhwal 2007)

Theorem 12: *The relation of similarity is an equivalence relation in the set of all linear transformations on a vector space $V (F)$.*

Proof: If A and B are two linear transformations on the vector space $V (F)$, then B is said to be similar to A if there exists an invertible linear transformation C on V such that

$$B = CAC^{-1}.$$

Reflexive. Let A be any linear transformation on V. We can write

$$A = IAI^{-1},$$

where I is identity transformation on V.

$\therefore$ A is similar to A because I is definitely invertible.

Symmetric. Let A be similar to B. Then there exists an invertible linear transformation P on V such that

$$A = PBP^{-1}$$

$\Rightarrow$ $P^{-1} AP = P^{-1} (PBP^{-1}) P$

$\Rightarrow$ $P^{-1} AP = B \Rightarrow B = P^{-1} AP$

$\Rightarrow$ $B = P^{-1} A (P^{-1})^{-1}$

$\Rightarrow$ B is similar to A.

Transitive. Let A be similar to B and B be similar to C.

and $\qquad B = QCQ^{-1}$,

where P and Q are invertible linear transformations on V.

We have $\quad A = PBP^{-1} = P\,(QCQ^{-1})\,P^{-1}$

$$= (PQ)\,C\,(Q^{-1}\,P^{-1}) = (PQ)\,C\,(PQ)^{-1}.$$

$\therefore \quad A$ is similar to C.

Hence similarity is an equivalence relation on the set of all linear transformations on $V(F)$.

Theorem 13: *Let T be a linear operator on an n-dimensional vector space $V(F)$ and let B and B' be two ordered basis for V. Then the matrix of T relative to B' is similar to the matrix of T relative to B.* **(Garhwal 2006, 12)**

Proof: Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ and $B' = \{\beta_1, \ldots, \beta_n\}$.

Let $\quad A = [a_{ij}]_{n \times n}$ be the matrix of T relative to B

and $\quad C = [c_{ij}]_{n \times n}$ be the matrix of T relative to B'. Then

$$T(\alpha_j) = \sum_{i=1}^{n} a_{ij}\,\alpha_i,\ j = 1, 2, \ldots, n \qquad \ldots(1)$$

and

$$T(\beta_j) = \sum_{i=1}^{n} c_{ij}\,\beta_i,\ j = 1, 2, \ldots, n. \qquad \ldots(2)$$

Let S be the linear operator on V defined by

$$S(\alpha_j) = \beta_j,\ j = 1, 2, \ldots, n. \qquad \ldots(3)$$

Since S maps a basis B onto a basis B', therefore S is necessarily invertible. Let P be the matrix of S relative to B. Then P is also an invertible matrix.

If $\quad P = [p_{ij}]_{n \times n}$, then

$$S(\alpha_j) = \sum_{i=1}^{n} p_{ij}\,\alpha_i,\ j = 1, 2, \ldots, n \qquad \ldots(4)$$

We have $\quad T(\beta_j) = T[S(\alpha_j)]$ [From (3)]

$$= T\left(\sum_{k=1}^{n} p_{kj}\,\alpha_k\right)$$

[From (4), on replacing i by k which is immaterial]

$$= \sum_{k=1}^{n} p_{kj}\,T(\alpha_k)$$

[$\because$ T is linear]

$$= \sum_{k=1}^{n} p_{kj} \sum_{i=1}^{n} a_{ik}\,\alpha_i$$

[From (1), on replacing j by k]

$$= \sum_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik}\,p_{kj}\right)\alpha_i\,. \qquad \ldots(5)$$

Also $\quad T(\beta_j) = \sum_{k=1}^{n} c_{kj}\,\beta_k$ [From (2), on replacing i by k]

$$= \sum_{k=1}^{n} c_{kj} \sum_{i=1}^{n} p_{ik} \alpha_i \qquad \text{[From (4), on replacing } j \text{ by } k\text{]}$$

$$= \sum_{i=1}^{n} \left(\sum_{k=1}^{n} p_{ik} c_{kj} \right) \alpha_i \,. \qquad \ldots(6)$$

From (5) and (6), we have

$$\sum_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik} p_{kj} \right) \alpha_i = \sum_{i=1}^{n} \left(\sum_{k=1}^{n} p_{ik} c_{kj} \right) \alpha_i$$

$$\Rightarrow \quad \sum_{k=1}^{n} a_{ik} p_{kj} = \sum_{k=1}^{n} p_{ik} c_{kj}$$

$$\Rightarrow \quad [a_{ik}]_{n \times n} [p_{kj}]_{n \times n} = [p_{ik}]_{n \times n} [c_{kj}]_{n \times n}$$

[by def. of matrix multiplication]

$$\Rightarrow \quad AP = PC$$

$$\Rightarrow \quad P^{-1} AP = P^{-1} PC \qquad [\because \; P^{-1} \text{ exists}]$$

$$\Rightarrow \quad P^{-1} AP = IC \Rightarrow P^{-1} AP = C$$

$\Rightarrow$ $\quad C$ is similar to A.

Note: Suppose B and B' are two ordered bases for an n-dimensional vector space $V(F)$. Let T be a linear operator on V. Suppose A is the matrix of T relative to B and C is the matrix of T relative to B'. If P is the transition matrix from the basis B to the basis B', then $C = P^{-1} AP$.

This result will enable us to find the matrix of T relative to the basis B' when we already knew the matrix of T relative to the basis B.

Theorem 14: *Let V be an n-dimensional vector space over the field F and T_1 and T_2 be two linear operators on V. If there exist two ordered bases B and B' for V such that $[T_1]_B = [T_2]_{B'}$, then show that T_2 is similar to T_1.*

Proof: Let $B = \{\alpha_1, \ldots, \alpha_n\}$ and $B' = \{\beta_1, \ldots, \beta_n\}$.

Let $\quad [T_1]_B = [T_2]_{B'} = A = [a_{ij}]_{n \times n}$. Then

$$T_1(\alpha_j) = \sum_{i=1}^{n} a_{ij} \alpha_i, \; j = 1, 2, \ldots, n, \qquad \ldots(1)$$

and

$$T_2(\beta_j) = \sum_{i=1}^{n} a_{ij} \beta_i, \; j = 1, 2, \ldots, n. \qquad \ldots(2)$$

Let S be the linear operator on V defined by

$$S(\alpha_j) = \beta_j, j = 1, 2, \ldots, n. \qquad \ldots(3)$$

Since S maps a basis of V onto a basis of V, therefore S is invertible.

We have $\quad T_2(\beta_j) = T_2[S(\alpha_j)] \qquad$ [From (3)]

$$= (T_2 S)(\alpha_j). \qquad \ldots(4)$$

$$= \sum_{i=1}^{n} a_{ij} \, S(\alpha_i) \qquad \text{[From (3)]}$$

$$= S\left(\sum_{i=1}^{n} a_{ij} \, \alpha_i\right) \qquad [\because \; S \text{ is linear}]$$

$$= S\,[T_1 \, (\alpha_j)] \qquad \text{[From (1)]}$$

$$= (ST_1)\,(\alpha_j). \qquad \ldots(5)$$

From (4) and (5), we have

$$(T_2 S)\,(\alpha_j) = (ST_1)\,(\alpha_j),\; j = 1, 2, \ldots, n.$$

Since $T_2 \; S$ and ST_1 agree on a basis for V, therefore we have

$$T_2 \; S = ST_1$$

$$\Rightarrow \quad T_2 \; SS^{-1} = ST_1 \; S^{-1} \;\Rightarrow\; T_2 \; I = ST_1 \; S^{-1}$$

$$\Rightarrow \quad T_2 = ST_1 \; S^{-1} \;\Rightarrow\; T_2 \text{ is similar to } T_1.$$

Determinant of a linear transformation on a finite dimensional vector space. Let T be a linear operator on an n-dimensional vector space $V(F)$. If B and B' are two ordered basis for V, then

$$[T]_B \text{ and } [T]_{B'}$$

are similar matrices. Also similar matrices have the same determinant. This enables us to make the following definition :

Definition: *Let T be a linear operator on an n-dimensional vector space $V(F)$. Then the determinant of T is the determinant of the matrix of T relative to any ordered basis for V.*

By the above discussion the determinant of T as defined by us will be a unique element of F and thus our definition is sensible.

Scalar Transformation. Definition: *Let $V(F)$ be a vector space. A linear transformation T on V is said to be a scalar transformation of V if*

$$T(\alpha) = c\alpha \;\forall\; \alpha \in V,$$

where c is a fixed scalar in F.

Also then we write $T = c$ and we say that the linear transformation T is equal to the scalar c.

Also obviously if the linear transformation T is equal to the scalar c, then we have $T = cI$, where I is the identity transformation on V.

Trace of a Matrix. Definition: *Let A be a square matrix of order n over a field F. The sum of the elements of A lying along the principal diagonal is called the trace of A. We shall write the trace of A as* **trace A.** Thus if $A = [a_{ij}]_{n \times n}$, then

$$\text{tr } A = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \ldots + a_{nn}.$$

trace function.

Theorem 15: *Let A and B be two square matrices of order n over a field F and* $\lambda \in F$. *Then*

(i) $tr\,(\lambda\; A) = \lambda\; tr\; A$;

(ii) $tr\,(A + B) = tr\; A + tr\; B$;

(iii) $tr\,(AB) = tr\,(BA)$.

Proof: Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$.

(i) We have $\lambda A = [\lambda a_{ij}]_{n \times n}$, by def. of multiplication of a matrix by a scalar.

$$\therefore \qquad \text{tr}\,(\lambda\; A) = \sum_{i=1}^{n} \lambda a_{ii} = \lambda \sum_{i=1}^{n} a_{ii} = \lambda\; \text{tr}\; A.$$

(ii) We have $A + B = [a_{ij} + b_{ij}]_{n \times n}$.

$$\therefore \qquad \text{tr}\,(A + B) = \sum_{i=1}^{n} (a_{ii} + b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = \text{tr}\; A + \text{tr}\; B.$$

(iii) We have $AB = [c_{ij}]_{n \times n}$ where $c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$.

Also $BA = [d_{ij}]_{n \times n}$ where $d_{ij} = \sum_{k=1}^{n} b_{ik}\, a_{kj}$.

$$\text{Now tr}\,(AB) = \sum_{i=1}^{n} c_{ii} = \sum_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik}\, b_{ki} \right)$$

$$= \sum_{k=1}^{n} \sum_{i=1}^{n} a_{ik}\, b_{ki},$$

interchanging the order of summation in the last sum

$$= \sum_{k=1}^{n} \left(\sum_{i=1}^{n} b_{ki}\, a_{ik} \right) = \sum_{k=1}^{n} d_{kk}$$

$$= d_{11} + d_{22} + \ldots + d_{nn} = \text{tr}\,(BA).$$

Theorem 16: *Similar matrices have the same trace.*

Proof: Suppose A and B are two similar matrices. Then there exists an invertible matrix C such that $B = C^{-1}\; AC$.

Let $\qquad C^{-1}\; A = D.$

Then $\qquad \text{tr}\; B = \text{tr}\,(DC)$

$= \text{tr}\,(CD)$ [by theorem 15]

$= \text{tr}\,(CC^{-1}\; A) = \text{tr}\,(IA) = \text{tr}\; A.$

Trace of a linear transformation on a finite dimensional vector space. Let T be a linear operator on an n-dimensional vector space $V\,(F)$. If B and B' are two ordered basis for V, then

$[T]_B$ and $[T]_{B'}$

make the following definition.

Definition of trace of a linear transformation. *Let T be a linear operator on an n-dimensional vector space $V(F)$. Then the trace of T is the trace of the matrix of T relative to any ordered basis for V.*

By the above discussion the trace of T as defined by us will be a unique element of F and thus our definition is sensible.

Illustrative Examples

Example 3: *Find the matrix of the linear transformation T on $V_3(\mathbf{R})$ defined as $T(a, b, c) = (2b + c, a - 4b, 3a)$, with respect to the ordered basis B and also with respect to the ordered basis B' where*

(i) $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

(ii) $B' = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$.

Solution: (i) We have

$$T(1, 0, 0) = (0, 1, 3) = 0\,(1, 0, 0) + 1\,(0, 1, 0) + 3\,(0, 0, 1)$$

$$T(0, 1, 0) = (2, -4, 0) = 2\,(1, 0, 0) - 4\,(0, 1, 0) + 0\,(0, 0, 1)$$

and $$T(0, 0, 1) = (1, 0, 0) = 1\,(1, 0, 0) + 0\,(0, 1, 0) + 0\,(0, 0, 1).$$

$\therefore$ by def. of matrix of T with respect to B, we have

$$[T]_B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -4 & 0 \\ 3 & 0 & 0 \end{bmatrix}.$$

Note: In order to find the matrix of T relative to the standard ordered basis B, it is sufficient to compute $T(1, 0, 0), T(0, 1, 0)$ and $T(0, 0, 1)$. There is no need of further expressing these vectors as linear combinations of $(1, 0, 0), (0, 1, 0)$ and $(0, 0, 1)$. Obviously the co-ordinates of the vectors $T(1, 0, 0), T(0, 1, 0)$ and $T(0, 0, 1)$ respectively constitute the first, second and third columns of the matrix $[T]_B$.

(ii) We have $T(1, 1, 1) = (3, -3, 3)$.

Now our aim is to express $(3, -3, 3)$ as a linear combination of vectors in B'.

Let $$(a, b, c) = x\,(1, 1, 1) + y\,(1, 1, 0) + z\,(1, 0, 0)$$
$$= (x + y + z, x + y, x).$$

Then $x + y + z = a, x + y = b, x = c$

i.e., $$x = c, y = b - c, z = a - b. \qquad \text{...(1)}$$

Putting $a = 3, b = -3$, and $c = 3$ in (1), we get

$$x = 3, \ y = -6 \text{ and } z = 6.$$

$\therefore$ $$T(1, 1, 1) = (3, -3, 3) = 3\,(1, 1, 1) - 6\,(1, 1, 0) + 6\,(1, 0, 0).$$

Putting $a = 2, b = -3$ and $c = 3$ in (1), we get

$$T(1,1,0) = (2, -3, 3) = 3(1,1,1) - 6(1,1,0) + 5(1,0,0).$$

Finally, $T(1,0,0) = (0,1,3)$.

Putting $a = 0, b = 1$ and $c = 3$ in (1), we get

$$T(1,0,0) = (0,1,3) = 3(1,1,1) - 2(1,0,0) - 1(1,0,0).$$

$$\therefore \qquad [T]_{B'} = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 6 & 5 & -1 \end{bmatrix}.$$

Example 4: *Let T be the linear operator on $\mathbf{R}^3$ defined by*

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

What is the matrix of T in the ordered basis $\{\alpha_1, \alpha_2, \alpha_3\}$ where

$$\alpha_1 = (1,0,1), \alpha_2 = (-1,2,1) \quad \textit{and} \quad \alpha_3 = (2,1,1)\,?$$

Solution: By def. of T, we have

$$T(\alpha_1) = T(1,0,1) = (4, -2, 3).$$

Now our aim is to express $(4, -2, 3)$ as a linear combination of the vectors in the basis $B = \{\alpha_1, \alpha_2, \alpha_3\}$. Let

$$\begin{aligned}(a, b, c) &= x\,\alpha_1 + y\alpha_2 + z\alpha_3 \\ &= x(1,0,1) + y(-1,2,1) + z(2,1,1) \\ &= (x - y + 2z, 2y + z, x + y + z).\end{aligned}$$

Then $x - y + 2z = a,\ 2y + z = b, x + y + z = c$.

Solving these equations, we have

$$x = \frac{-a - 3b + 5c}{4}, \quad y = \frac{b + c - a}{4}, \quad z = \frac{b - c + a}{2}. \qquad \ldots(1)$$

Putting $a = 4, b = -2, c = 3$ in (1), we get

$$x = \frac{17}{4}, y = -\frac{3}{4}, z = -\frac{1}{2}.$$

$$\therefore \qquad T(\alpha_1) = \frac{17}{4}\alpha_1 - \frac{3}{4}\alpha_2 - \frac{1}{2}\alpha_3.$$

Also $\qquad T(\alpha_2) = T(-1,2,1) = (-2, 4, 9)$.

Putting $a = -2, b = 4, c = 9$ in (1), we get $x = \dfrac{35}{4}, y = \dfrac{15}{4}, z = \dfrac{-7}{2}$.

$$\therefore \qquad T(\alpha_2) = \frac{35}{4}\alpha_1 + \frac{15}{4}\alpha_2 - \frac{7}{2}\alpha_3.$$

Finally $T(\alpha_3) = T(2,1,1) = (7, -3, 4)$.

Putting $a = 7, b = -3, c = 4$ in (1), we get $x = \dfrac{11}{2}, y = -\dfrac{3}{2}, z = 0$.

$$\therefore \qquad T(\alpha_3) = \frac{11}{2}\alpha_1 - \frac{3}{2}\alpha_2 + 0\alpha_3.$$

$$\therefore \qquad [T]_B = \begin{bmatrix} \cdots & \cdots & \cdots \\ -\frac{3}{4} & \frac{15}{4} & -\frac{3}{2} \\ -\frac{1}{2} & -\frac{7}{2} & 0 \end{bmatrix}.$$

Example 5: *Let T be a linear operator on $\mathbf{R}^3$ defined by*

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

Prove that T is invertible and find a formula for T^{-1}.

Solution: Suppose B is the standard ordered basis for $\mathbf{R}^3$. Then

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

Let $A = [T]_B$ *i.e.*, let A be the matrix of T with respect to B. First we shall compute A. We have

$$T(1, 0, 0) = (3, -2, -1), T(0, 1, 0) = (0, 1, 2) \text{ and}$$

$T(0, 0, 1) = (1, 0, 4)$.

$$\therefore \qquad A = [T]_B = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}.$$

Now T will be invertible if the matrix $[T]_B$ is invertible. [See theorem 7 of article 6.2].

We have $\det A = |A| = \begin{vmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{vmatrix} = 3(4 - 0) + 1(-4 + 1) = 9.$

Since $\det A \neq 0$, therefore the matrix A is invertible and consequently T is invertible.

Now we shall compute the matrix A^{-1}. For this let us first find adj. A.

The cofactors of the elements of the first row of A are

$$\begin{vmatrix} 1 & 0 \\ 2 & 4 \end{vmatrix}, -\begin{vmatrix} -2 & 0 \\ -1 & 4 \end{vmatrix}, \begin{vmatrix} -2 & 1 \\ -1 & 2 \end{vmatrix} \quad i.e., \; 4, 8, -3.$$

The cofactors of the elements of the second row of A are

$$-\begin{vmatrix} 0 & 1 \\ 2 & 4 \end{vmatrix}, \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix}, -\begin{vmatrix} 3 & 0 \\ -1 & 2 \end{vmatrix} \quad i.e., \; 2, 13, -6.$$

The cofactors of the elements of the third row of A are

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}, -\begin{vmatrix} 3 & 1 \\ -2 & 0 \end{vmatrix}, \begin{vmatrix} 3 & 0 \\ -2 & 1 \end{vmatrix} \quad i.e., \; -1, -2, 3.$$

$$\therefore \qquad \text{Adj. } A = \text{transpose of the matrix} \begin{bmatrix} 4 & 8 & -3 \\ 2 & 13 & -6 \\ -1 & -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \cdot & \cdot & \cdot \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}.$$

$$\therefore \qquad A^{-1} = \frac{1}{\det A} \text{Adj. } A = \frac{1}{9}\begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}.$$

Now $\qquad [T^{-1}]_B = ([T]_B)^{-1} = A^{-1}.$ [See theorem 7 of article 6.2]

We shall now find a formula for T^{-1}. Let $\alpha = (a, b, c)$ be any vector belonging to $\mathbf{R}^3$. Then

$$[T^{-1}(\alpha)]_B = [T^{-1}]_B\, [\alpha]_B \text{ [See Note of Theorem 2, article 6.2]}$$

$$= \frac{1}{9}\begin{bmatrix} 4 & 2 & -1 \\ 8 & 13 & -2 \\ -3 & -6 & 3 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 4a + 2b - c \\ 8a + 13b - 2c \\ -3a - 6b + 3c \end{bmatrix}$$

Since B is the standard ordered basis for $\mathbf{R}^3$,

$$\therefore \qquad T^{-1}(\alpha) = T^{-1}(a, b, c) = \frac{1}{9}(4a + 2b - c, 8a + 13b - 2c, -3a - 6b + 3c).$$

Example 6: *Let T be the linear operator on* $\mathbf{R}^3$ *defined by*

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

(i) What is the matrix of T in the standard ordered basis B for $\mathbf{R}^3$ *?*

(ii) Find the transition matrix P from the ordered basis B to the ordered basis $B' = \{\alpha_1, \alpha_2, \alpha_3\}$ *where* $\alpha_1 = (1, 0, 1), \alpha_2 = (-1, 2, 1),$ *and* $\alpha_3 = (2, 1, 1)$. *Hence find the matrix of T relative to the ordered basis B′.*

Solution: (i) Let $A = [T]_B$. Then

$$A = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}. \qquad \text{[For calculation work see Example 5]}$$

(ii) Since B is the standard ordered basis, therefore the transition matrix P from B to B' can be immediately written as

$$P = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Now $\qquad [T]_{B'} = P^{-1}[T]_B\, P.$ [See note of theorem 13, article 6.2]

In order to compute the matrix P^{-1}, we find that $\det P = -4$.

$$\text{Therefore} \qquad P^{-1} = \frac{1}{\det P}\text{Adj. } P = -\frac{1}{4}\begin{bmatrix} 1 & 3 & -5 \\ 1 & -1 & -1 \\ -2 & -2 & 2 \end{bmatrix}.$$

$$\therefore \quad [T]_{B'} = -\frac{1}{4}\begin{bmatrix} 1 & -1 & -1 \\ -2 & -2 & 2 \end{bmatrix}\begin{bmatrix} -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}\begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= -\frac{1}{4}\begin{bmatrix} 2 & -7 & -19 \\ 6 & -3 & -3 \\ -4 & 2 & 6 \end{bmatrix}\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= -\frac{1}{4}\begin{bmatrix} -17 & -35 & -22 \\ 3 & -15 & 6 \\ 2 & 14 & 0 \end{bmatrix} = \begin{bmatrix} \frac{17}{4} & \frac{35}{4} & \frac{11}{2} \\ -\frac{3}{4} & \frac{15}{4} & -\frac{3}{2} \\ -\frac{1}{2} & -\frac{7}{2} & 0 \end{bmatrix}.$$

[Note that this result tallies with that of Example 4].

Example 7: *Let T be a linear operator on $\mathbf{R}^2$ defined by :*

$$T\,(x, y) = (2y, 3x - y).$$

Find the matrix representation of T relative to the basis $\{(1, 3), (2, 5)\}$. **(Garhwal 2010)**

Solution: Let $\alpha_1 = (1, 3)$ and $\alpha_2 = (2, 5)$. By def. of T, we have

$$T\,(\alpha_1) = T\,(1, 3) = (2\,.\,3, 3\,.\,1 - 3) = (6, 0)$$

and $$T\,(\alpha_2) = T\,(2, 5) = (2\,.\,5, 3\,.\,2 - 5) = (10, 1).$$

Now our aim is to express the vectors $T\,(\alpha_1)$ and $T\,(\alpha_2)$ as linear combinations of the vectors in the basis $\{\alpha_1, \alpha_2\}$.

Let $\quad (a, b) = p\alpha_1 + q\alpha_2 = p\,(1, 3) + q\,(2, 5) = (p + 2q, 3p + 5q).$

Then $\quad p + 2q = a, 3p + 5q = b.$

Solving these equations, we get

$$p = -5a + 2b, q = 3a - b. \qquad \ldots(1)$$

Putting $a = 6, b = 0$ in (1), we get $p = -30, q = 18$.

$$\therefore \quad T\,(\alpha_1) = (6, 0) = -30\alpha_1 + 18\alpha_2\,. \qquad \ldots(2)$$

Again putting $a = 10, b = 1$ in (1), we get

$$p = -48, q = 29.$$

$$\therefore \quad T\,(\alpha_2) = (10, 1) = -48\alpha_1 + 29\alpha_2\,. \qquad \ldots(3)$$

From the relations (2) and (3), we see that the matrix of T relative to the basis

$$\{\alpha_1, \alpha_2\} \text{ is} = \begin{bmatrix} -30 & -48 \\ 18 & 29 \end{bmatrix}.$$

Example 8: *Consider the vector space $V\,(\mathbf{R})$ of all 2×2 matrices over the field $\mathbf{R}$ of real numbers. Let T be the linear transformation on V that sends each matrix X onto AX, where $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Find the matrix of T with respect to the ordered basis $B = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ for V where*

Solution: We have

$$T(\alpha_1) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

$$T(\alpha_2) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T(\alpha_3) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

and
$$T(\alpha_4) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

∴
$$[T]_B = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Example 9: *If the matrix of a linear transformation T on V_2 $(\mathbf{C})$, with respect to the ordered basis $B = \{(1, 0), (0, 1)\}$ is $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, what is the matrix of T with respect to the ordered basis $B' = \{(1, 1), (1, -1)\}$?*

Solution: Let us first define T explicitly. It is given that

$$[T]_B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

∴ $\quad T(1, 0) = 1(1, 0) + 1(0, 1) = (1, 1),$

and $\quad T(0, 1) = 1(1, 0) + 1(0, 1) = (1, 1).$

If $(a, b) \in V_2$ $(\mathbf{C})$, then we can write

$$(a, b) = a(1, 0) + b(0, 1).$$

∴
$$T(a, b) = aT(1, 0) + bT(0, 1)$$
$$= a(1, 1) + b(1, 1) = (a + b, a + b).$$

This is the explicit expression for T.

We have $T(1, 1) = (2, 2)$.

Let $\quad (2, 2) = x(1, 1) + y(1, -1) = (x + y, x - y)$.

Then $\quad x + y = 2, x - y = 2$

$\Rightarrow \quad x = 2, y = 0$.

$\therefore \quad (2, 2) = 2(1, 1) + 0(1, -1)$.

Also $\quad T(1, -1) = (0, 0) = 0(1, 1) + 0(1, -1)$.

$$\therefore \quad [T]_{B'} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

Note: If P is the transition matrix from the basis B to the basis B', then

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

We can compute $[T]_{B'}$ by using the formula

$$[T]_{B'} = P^{-1} [T]_B P.$$

Example 10: *Show that the vectors* $\alpha_1 = (1, 0, -1), \alpha_2 = (1, 2, 1), \alpha_3 = (0, -3, 2)$ *form a basis for* $\mathbf{R}^3$. *Express each of the standard basis vectors as a linear combination of* $\alpha_1, \alpha_2, \alpha_3$.

Solution: Let a, b, c be scalars *i.e.*, real numbers such that

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \mathbf{0}$$

i.e., $\quad a(1, 0, -1) + b(1, 2, 1) + c(0, -3, 2) = (0, 0, 0)$

i.e., $\quad (a + b + 0c, 0a + 2b - 3c, -a + b + 2c) = (0, 0, 0)$

$$\text{i.e.,} \quad \left.\begin{aligned} a + b + 0c &= 0, \\ 0a + 2b - 3c &= 0, \\ -a + b + 2c &= 0. \end{aligned}\right\} \qquad \ldots(1)$$

The coefficient matrix A of these equations is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 2 \end{bmatrix}.$$

We have $\det A = |A| = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 2 \end{vmatrix}$.

$$= 1(4 + 3) - 1(0 - 3) = 7 + 3 = 10.$$

Since $\det A \neq 0$, therefore the matrix A is non-singular and rank $A = 3$ *i.e.*, equal to the number of unknowns a, b, c. Hence $a = 0, b = 0, c = 0$ is the only solution of the equations (1). Therefore the vectors $\alpha_1, \alpha_2, \alpha_3$ are linearly independent over $\mathbf{R}$. Since $\dim \mathbf{R}^3 = 3$, therefore the set $\{\alpha_1, \alpha_2, \alpha_3\}$ containing three linearly independent vectors forms a basis for $\mathbf{R}^3$.

Now let $B = \{e_1, e_2, e_3\}$ be the standard ordered basis for $\mathbf{R}^3$.

Let $\qquad B' = \{\alpha_1, \alpha_2, \alpha_3\}$.

We have $\qquad \alpha_1 = (1, 0, -1) = 1e_1 + 0e_2 - 1e_3$

$$\alpha_2 = (1, 2, 1) = 1e_1 + 2e_2 + 1e_3$$

$$\alpha_3 = (0, -3, 2) = 0e_1 - 3e_2 + 2e_3.$$

If P is the transition matrix from the basis B to the basis B', then

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & -3 \\ -1 & 1 & 2 \end{bmatrix}.$$

Let us find the matrix P^{-1}. For this let us first find Adj. P.

The cofactors of the elements of the first row of P are

$$\begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}, -\begin{bmatrix} 0 & -3 \\ -1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}, \text{ i.e., } 7, 3, 2.$$

The cofactors of the elements of the second row of P are

$$-\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \begin{vmatrix} 1 & 0 \\ -1 & 2 \end{vmatrix}, -\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} \text{ i.e., } -2, 2, -2.$$

The cofactors of the elements of the third row of P are

$$\begin{vmatrix} 1 & 0 \\ 2 & -3 \end{vmatrix}, -\begin{vmatrix} 1 & 0 \\ 0 & -3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix}, \text{ i.e, } -3, 3, 2$$

$\therefore \qquad$ Adj P = transpose of the matrix $\begin{bmatrix} 7 & 3 & 2 \\ -2 & 2 & -2 \\ -3 & 3 & 2 \end{bmatrix}$

$$= \begin{bmatrix} 7 & -2 & -3 \\ 3 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix}.$$

$$\therefore \qquad P^{-1} = \frac{1}{\det P} \text{ Adj } P = \frac{1}{10} \begin{bmatrix} 7 & -2 & -3 \\ 3 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix}.$$

Now $\qquad e_1 = 1e_1 + 0e_2 + 0e_3$.

$\therefore \qquad$ Coordinate matrix of e_1 relative to the basis B

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

$$= [e_1]_{B'} = P^{-1}\begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$= \frac{1}{10}\begin{bmatrix}7 & -2 & -3\\3 & 2 & 3\\2 & -2 & 2\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$= \frac{1}{10}\begin{bmatrix}7\\3\\2\end{bmatrix} = \begin{bmatrix}7/10\\3/10\\2/10\end{bmatrix}.$$

$\therefore$ $$e_1 = \frac{7}{10}\alpha_1 + \frac{3}{10}\alpha_2 + \frac{2}{10}\alpha_3.$$

Also $$[e_2]_B = \begin{bmatrix}0\\1\\0\end{bmatrix} \text{ and } [e_3]_B = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

$\therefore$ $$[e_2]_{B'} = P^{-1}\begin{bmatrix}0\\1\\0\end{bmatrix}$$

and $$[e_3]_{B'} = P^{-1}\begin{bmatrix}0\\0\\1\end{bmatrix}.$$

Thus $$[e_2]_{B'} = \frac{1}{10}\begin{bmatrix}-2\\2\\-2\end{bmatrix}, \; [e_3]_{B'} = \frac{1}{10}\begin{bmatrix}-3\\3\\2\end{bmatrix}.$$

$\therefore$ $$e_2 = -\frac{2}{10}\alpha_1 + \frac{2}{10}\alpha_2 - \frac{2}{10}\alpha_3$$

and $$e_3 = -\frac{3}{10}\alpha_1 + \frac{3}{10}\alpha_2 + \frac{2}{10}\alpha_3.$$

Example 11: *Let A be an $m \times n$ matrix with real entries. Prove that $A = 0$ (null matrix) if and only if trace $(A^t A) = 0$.*

Solution: Let $A = [a_{ij}]_{m \times n}$.

Then $A^t = [b_{ij}]_{n \times m}$, where $b_{ij} = a_{ji}$.

Now $A^t A$ is a matrix of the type $n \times n$.

Let $A^t A = [c_{ij}]_{n \times n}$. Then the i^{th} row of A^t and the i^{th} column of A

$$= b_{i1}\ a_{1i} + b_{i2}\ a_{2i} + \ldots + b_{im}\ a_{mi}$$

$$= a_{1i}\ a_{1i} + a_{2i}\ a_{2i} + \ldots + a_{mi}\ a_{mi} \qquad [\because \ b_{ij} = a_{ji}]$$

$$= a_{1i}^2 + a_{2i}^2 + \ldots + a_{mi}^2 .$$

Now trace $(A^t\ A) = \sum_{i=1}^{n} c_{ii}$

$$= \sum_{i=1}^{n} \{a_{1i}^2 + a_{2i}^2 + \ldots + a_{mi}^2\}$$

= the sum of the squares of all the elements of A.

Now the elements of A are all real numbers.

Therefore trace $(A^t\ A) = 0$

$\Rightarrow$ the sum of the squares of all the elements of A is zero

$\Rightarrow$ each element of A is zero

$\Rightarrow$ A is a null matrix.

Conversely if A is a null matrix, then $A^t\ A$ is also a null matrix and so trace $(A^t\ A) = 0$.

Hence trace $(A^t\ A) = 0$ iff $A = 0$.

Example 12: *Show that the only matrix similar to the identity matrix I is I itself.*

Solution: The identity matrix I is invertible and we can write $I = I^{-1}\ I\ I$. Therefore I is similar to I. Further let B be a matrix similar to I. Then there exists an invertible matrix P such that

$$B = P^{-1}\ IP$$

$$\Rightarrow \quad B = P^{-1}\ P \qquad [\because \ P^{-1}\ I = P^{-1}]$$

$$\Rightarrow \quad B = I.$$

Hence the only matrix similar to I is I itself.

Example 13: *Let T and S be linear operators on the finite dimensional vector space $V\ (F)$, prove that*

(i) *det $(TS) =$ (det T) (det S);*

(ii) *T is invertible iff det $T \neq 0$.*

Solution: (i) Let B be any ordered basis for V.

We have $[TS]_B = [T]_B\ [S]_B$

$\therefore$ $\det\ [TS]_B = \det\ ([T]_B\ [S]_B)$

$= (\det\ [T]_B)\ (\det\ [S]_B)$

[$\because$ determinant of the product of two matrices is equal to the product of their determinants].

matrix with respect to any ordered basis.

$\therefore \qquad \det (TS) = (\det T)(\det S).$

(ii) Suppose T is invertible. Then there exists a linear transformation T^{-1} on V such that $T^{-1} T = I = T T^{-1}$.

$\therefore \qquad \det (T T^{-1}) = \det I$

$\Rightarrow \qquad (\det T)(\det T^{-1}) = \det [I]_B,$

where B is any ordered basis for V

$\Rightarrow \qquad (\det T)(\det T^{-1}) = 1.$

[$\because$ $[I]_B$ is unit matrix and the determinant of a unit matrix is equal to 1].

Now $\det T$ and $\det T^{-1}$ are elements of F. In a field the product of two elements can be 0 iff at least one of them is 0.

$\therefore \qquad (\det T)(\det T^{-1}) = 1$

$\Rightarrow \qquad \det T \neq 0.$

Conversely suppose that $\det T \neq 0$.

Then $\det [T]_B \neq 0$, where B is any ordered basis for V.

Now $\det [T]_B \neq 0$ implies that the matrix $[T]_B$ is invertible.

$\therefore \quad T$ is also invertible.

Example 14: *If T and S are similar linear transformations on a finite dimensional vector space $V(F)$, then $\det T = \det S$.*

Solution: Since T and S are similar, therefore there exists an invertible linear transformation P on V such that $T = PSP^{-1}$.

Therefore $\det T = \det (PSP^{-1}) = (\det P)(\det S)(\det P^{-1})$

$$= (\det P)(\det P^{-1})(\det S) = [\det (PP^{-1})](\det S)$$
$$= (\det I)(\det S) = 1 (\det S) = \det S.$$

Comprehensive Exercise 3

1. Let $V = \mathbf{R}^3$ and $T : V \to V$ be a linear mapping defined by
$$T(x, y, z) = (x + z, -2x + y, -x + 2y + z).$$
What is matrix of T relative to the basis $B = \{(1, 0, 1), (-1, 1, 1), (0, 1, 1)\}$?

2. Find matrix representation of linear mapping $T : \mathbf{R}^3 \to \mathbf{R}^3$ given by
$T(x, y, z) = (z, y + z, x + y + z)$ relative to the basis
$B = \{(1, 0, 1), (-1, 2, 1), (2, 1, 1)\}$.

vectors

$$\alpha_1 = (1, 1, 0, 0), \alpha_2 = (1, 0, 1, 1), \alpha_3 = (2, 0, 0, 2), \alpha_4 = (0, 0, 2, 2).$$

4. Let T be the linear operator on $\mathbf{R}^2$ defined by

$$T(x, y) = (4x - 2y, 2x + y).$$

Compute the matrix of T relative to the basis $\{\alpha_1, \alpha_2\}$ where $\alpha_1 = (1, 1)$, $\alpha_2 = (-1, 0)$.

5. Let T be the linear operator on $\mathbf{R}^2$ defined by

$$T(x, y) = (4x - 2y, 2x + y).$$

(i) What is the matrix of T in the standard ordered basis B for $\mathbf{R}^2$?

(ii) Find the transition matrix P from the ordered basis B to the ordered basis $B' = \{\alpha_1, \alpha_2\}$ where $\alpha_1 = (1, 1), \alpha_2 = (-1, 0)$. Hence find the matrix of T relative to the ordered basis B'.

6. Let T be the linear operator on $\mathbf{R}^2$ defined by $T(a, b) = (a, 0)$. Write the matrix of T in the standard ordered basis $B = \{(1, 0), (0, 1)\}$.If $B' = \{(1, 1), (2, 1)\}$ is another ordered basis for $\mathbf{R}^3$, find the transition matrix P from the basis B to the basis B'. Hence find the matrix of T relative to the basis B'.

7. The matrix of a linear transformation T on $V_3(\mathbf{C})$ relative to the basis

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ is } \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}.$$

What is the matrix of T relative to the basis

$$B' = \{(0, 1, -1), (1, -1, 1), (-1, 0, 1)\}?$$

8. Find the matrix relative to the basis

$$\alpha_1 = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right), \ \alpha_2 = \left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right), \alpha_3 = \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$$

of $\mathbf{R}^3$, of the linear transformation $T : \mathbf{R}^3 \to \mathbf{R}^3$ whose matrix relative to the standard ordered basis is

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

9. Explain what is meant by the matrix of a linear transformation of V relative to a basis of V. Let F be a field and V, the set of all polynomials in x over F of degree ≤ 5. If $D : V \to V$ is defined by $D[f(x)] = f''(x)$ where $f'(x)$ is the matrix of D in the basis $\{1, x, x^2, x^3, x^4\}$.

10. Let V be the vector space of those polynomial functions from the reals into itself which have degree ≤ 3. Let $B = \{f_1, f_2, f_3, f_4\}$ where $f_i(x) = x^{i-1}$ $(1 \leq i \leq 4)$. Show that B forms a basis for V. For any real number t let $g_i(x) = (x+t)^{i-1}$. Show that $B' = \{g_1, g_2, g_3, g_4\}$ is also a basis for V. If D is the differentiation operator on V, write the matrices of D in the ordered basis B and B'.

11. Find the matrix representation of the linear mappings relative to the usual basis for $\mathbf{R}^n$.

(i) $T : \mathbf{R}^3 \to \mathbf{R}^2$ defined by $T(x, y, z) = (2x - 4y + 9z, 5x + 3y - 2z)$.

(ii) $T : \mathbf{R} \to \mathbf{R}^2$ defined by $T(x) = (3x, 5x)$.

(iii) $T : \mathbf{R}^3 \to \mathbf{R}^3$ defined by $T(x, y, z) = (x, y, 0)$.

(iv) $T : \mathbf{R}^3 \to \mathbf{R}^3$ defined by $T(x, y, z) = (z, y + z, x + y + z)$.

12. Let $B = \{(1, 0), (0, 1)\}$ and $B' = \{(1, 2), (2, 3)\}$ be any two bases of $\mathbf{R}^2$ and $T(x, y) = (2x - 3y, x + y)$.

(i) Find the transition matrices P and Q from B to B' and from B' to B respectively.

(ii) Verify that $[\alpha]_B = P[\alpha]_{B'}$ $\forall\, \alpha \in \mathbf{R}^2$

(iii) Verify that $P^{-1}[T]_B P = [T]_{B'}$. **(Garhwal 2012)**

13. Let V be the space of all 2×2 matrices over the field F and let P be a fixed 2×2 matrix over F. Let T be the linear operator on V defined by

$$T(A) = PA, \forall\, A \in V.$$

Prove that trace $(T) = 2$ trace (P).

14. Let V be the space of 2×2 matrices over $\mathbf{R}$ and let $M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Let T be the linear operator on V defined by $T(A) = MA$. Find the trace of T.

15. Find the trace of the operator T on $\mathbf{R}^3$ defined by

$$T(x, y, z) = (a_1 x + a_2 y + a_3 z, b_1 x + b_2 y + b_3 z, c_1 x + c_2 y + c_3 z).$$

16. Show that the only matrix similar to the zero matrix is the zero matrix itself.

17. If two linear transormations A and B on $V(F)$ are similar, then show that A^2 and B^2 are also similar and if A, B are invertible, then A^{-1}, B^{-1} are also similar.

18. If A and B are linear transformations on the same vector space and if at least one of them is invertible, then AB and BA are similar.

19. If T and S are linear transformations on a finite dimensional vector space V such that

20. If $\{\alpha_1, \ldots, \alpha_n\}$ and $\{\beta_1, \ldots, \beta_n\}$ are bases in the same finite dimensional vector space $V(F)$, and if T is a linear transformation such that

$$T(\alpha_i) = \beta_i, \; i = 1, \ldots, n, \text{ then } \det T \neq 0.$$

21. If A and B are $n \times n$ complex matrices, show that $AB - BA = I$ is impossible.

Answers 3

1. $\begin{bmatrix} 2 & 1 & 2 \\ 0 & 1 & 1 \\ -2 & 2 & 0 \end{bmatrix}$

2. $\begin{bmatrix} \frac{3}{2} & 0 & \frac{13}{4} \\ \frac{1}{2} & 1 & \frac{5}{4} \\ 0 & 1 & -\frac{1}{2} \end{bmatrix}$

3. $\left(1, 0, \frac{1}{2}, \frac{3}{2}\right)$

4. $\begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix}$

5. (i) $\begin{bmatrix} 4 & -2 \\ 2 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}; \begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix}$

6. $[T]_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}; [T]_{B'} = \begin{bmatrix} -1 & -2 \\ 1 & 2 \end{bmatrix}$

7. $[T]_{B'} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 0 & -2 \\ 0 & 0 & -3 \end{bmatrix}$

8. $\begin{bmatrix} 3 & -2/3 & -2/3 \\ -2/3 & 10/3 & 0 \\ -2/3 & 0 & 8/3 \end{bmatrix}$

9. $\begin{bmatrix} 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 12 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

10. $[D\,;B]=\begin{bmatrix}0&0&2&0\\0&0&0&3\\0&0&0&0\end{bmatrix}$; $[D\,;B']=\begin{bmatrix}0&0&2&0\\0&0&0&3\\0&0&0&0\end{bmatrix}$

11. (i) $\begin{bmatrix}2&-4&9\\5&3&-2\end{bmatrix}$ (ii) $\begin{bmatrix}3\\5\end{bmatrix}$

(iii) $\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$ (iv) $\begin{bmatrix}0&0&1\\0&1&1\\1&1&1\end{bmatrix}$

12. (i) $P=\begin{bmatrix}1&2\\2&3\end{bmatrix}$; $Q=\begin{bmatrix}-3&2\\2&-1\end{bmatrix}$

14. trace $(T)=10$

15. trace $(T)=a_1+b_2+c_3$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. Let T be a linear transformation on the vector space $V_2\,(F)$ defined by

$$T\,(a,b)=(a,0).$$

The matrix of T relative to the ordered basis $\{(1,0),(0,1)\}$ of $V_2\,(F)$ is

(a) $\begin{bmatrix}1&0\\0&0\end{bmatrix}$ (b) $\begin{bmatrix}0&1\\0&0\end{bmatrix}$

(c) $\begin{bmatrix}0&0\\1&0\end{bmatrix}$ (d) $\begin{bmatrix}0&0\\0&1\end{bmatrix}$.

(Garhwal 2007)

2. The transition matrix P from the standard ordered basis to the ordered basis $\{(1,1),(-1,0)\}$ is

(a) $\begin{bmatrix}1&1\\-1&0\end{bmatrix}$ (b) $\begin{bmatrix}1&-1\\1&0\end{bmatrix}$

(c) $\begin{bmatrix}0&1\\1&0\end{bmatrix}$ (d) $\begin{bmatrix}1&0\\0&1\end{bmatrix}$.

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Let T be the linear operator on V defined by $T(A) = MA$. Then the trace of T is

(a) 5 (b) 10

(c) 0 (d) None of these.

Fill in the Blank(s)

Fill in the blanks "……" so that the following statements are complete and correct.

1. If T is a linear operator on $\mathbf{R}^2$ defined by

$$T(x, y) = (x - y, y), \text{ then } T^2(x, y) = \text{……} .$$

2. Let A be a square matrix of order n over a field F. The sum of the elements of A lying along the principal diagonal is called the …… of A.
3. Let T and S be similar linear operators on the finite dimensional vector space $V(F)$, then det (T) …… det (S).
4. Let $V(F)$ be an n-dimensional vector space and B be any ordered basis for V. If I be the identity transformation on V, then $[I\ ;\ B] =$ …… .
5. Let T be a linear operator on an n-dimensional vector space $V(F)$ and let B and B' be two ordered bases for V. Then the matrix of T relative to B' is …… to the matrix of T relative to B.

True or False

Write 'T' for true and 'F' for false statement.

1. The relation of similarity is an equivalence relation in the set of all linear transformations on a vector space $V(F)$.
2. Similar matrices have the same trace.

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (b)

Fill in the Blank(s)

1. $(x - 2y, y)$ 2. trace

3. $=$ 4. unit matrix of order n 5. similar

True or False

1. T 2. T

□

Chapter

7

Linear Functionals

7.1 Linear Functionals

Let $V(F)$ be a vector space. We know that the field F can be regarded as a vector space over F. This is the vector space $F(F)$ or F^1. We shall simply denote it by F. A linear transformation from V into F is called a **linear functional** on V. We shall now give independent definition of a linear functional.

Linear Functionals. Definition. *Let $V(F)$ be a vector space. A function f from V into F is said to be a linear functional on V if*

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) \ \forall\ a, b \in F \text{ and } \forall\ \alpha, \beta \in V.$$

If f is a linear functional on $V(F)$, then $f(\alpha)$ is in F for each α belonging to V. Since $f(\alpha)$ is a scalar, therefore a linear functional on V is a **scalar valued function.**

Illustration 1. *Let $V_n(F)$ be the vector space of ordered n-tuples of the elements of the field F.*

Let $x_1, x_2, \ldots, x_n$ be n field elements of F. If let f be a function from $V_n(F)$ into F defined by

$$f(\alpha) = x_1 a_1 + x_2 a_2 + \ldots + x_n a_n .$$

Let $\beta = (b_1, b_2, \ldots, b_n) \in V_n(F)$. If $a, b \in F$, we have

$$\begin{aligned} f(a\alpha + b\beta) &= f[a(a_1, \ldots, a_n) + b(b_1, \ldots, b_n)] \\ &= f(aa_1 + bb_1, \ldots, aa_n + bb_n) \\ &= x_1(aa_1 + bb_1) + \ldots + x_n(aa_n + bb_n) \\ &= a(x_1 a_1 + \ldots + x_n a_n) + b(x_1 b_1 + \ldots + x_n b_n) \\ &= af(a_1, \ldots, a_n) + bf(b_1, \ldots, b_n) \\ &= af(\alpha) + bf(\beta). \end{aligned}$$

$\therefore$ f is a linear functional on $V_n(F)$.

Illustration 2. Now we shall give a very important example of a linear functional.

We shall prove that *the trace function is a linear functional on the space of all $n \times n$ matrices over a field F.*

Let n be a positive integer and F a field. Let $V(F)$ be the vector space of all $n \times n$ matrices over F. If $A = [a_{ij}]_{n \times n} \in V$, then the trace of A is the scalar

$$\text{tr } A = a_{11} + a_{22} + \ldots + a_{nn} = \sum_{i=1}^{n} a_{ii} .$$

Thus the trace of A is the scalar obtained by adding the elements of A lying along the principal diagonal.

The trace function is a linear functional on V because if

$$a, b \in F \text{ and } A = [a_{ij}]_{n \times n}, B = [b_{ij}]_{n \times n} \in V, \text{then}$$

$$\text{tr}(aA + bB) = \text{tr}(a[a_{ij}]_{n \times n} + b[b_{ij}]_{n \times n}) = \text{tr}([aa_{ij} + bb_{ij}]_{n \times n})$$

$$= \sum_{i=1}^{n}(aa_{ii} + bb_{ii}) = a\sum_{i=1}^{n} a_{ii} + b\sum_{i=1}^{n} b_{ii} = a(\text{tr } A) + b(\text{tr } B).$$

Illustration 3. *Now we shall give another important example of a linear functional.*

Let V be a finite-dimensional vector space over the field F and let B be an ordered basis for V. The function f_i which assigns to each vector α in V the i^{th} coordinate of α relative to the ordered basis B is a linear functional on V.

Let $B = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$.

If $\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n \in V$, then by definition of f_i, we have

$$f_i(\alpha) = a_i .$$

Similarly if $\beta = b_1\alpha_1 + \ldots + b_n\alpha_n \in V$, then

$$f_i(\beta) = b_i .$$

If $a, b \in F$, we have

$$\begin{aligned} f_i(a\alpha + b\beta) &= f_i[a(a_1\alpha_1 + \ldots + a_n\alpha_n) + b(b_1\alpha_1 + \ldots + b_n\alpha_n)] \\ &= f_i[(aa_1 + bb_1)\alpha_1 + \ldots + (aa_n + bb_n)\alpha_n] \\ &= aa_i + bb_i = af_i(\alpha) + bf_i(\beta). \end{aligned}$$

Illustration 4. *Let V be the vector space of polynomials in x over* $\mathbf{R}$*. Let $T : V \to$* $\mathbf{R}$ *be the integral operator defined by $T(f(x)) = \int_0^1 f(x)\,dx$. Then T is linear and hence it is a linear functional on V.*

7.2 Some Particular Linear Functionals

1. Zero functional. *Let V be a vector space over the field F. The function f from V into F defined by*

$$f(\alpha) = 0 \text{ (zero of } F) \ \forall\, \alpha \in V$$

is a linear functional on V.

Proof: Let $\alpha, \beta \in V$ and $a, b \in F$. We have

$$f(a\alpha + b\beta) = 0 \qquad \text{(by def. of } f)$$
$$= a0 + b0 = af(\alpha) + bf(\beta).$$

$\therefore$ f is a linear functional on V. It is called the zero functional and we shall in future denote it by $\hat{\mathbf{0}}$

2. Negative of a linear functional. *Let V be a vector space over the field F. Let f be a linear functional on V. The correspondence $-f$ defined by*

$$(-f)(\alpha) = -[f(\alpha)] \ \forall\, \alpha \in V$$

is a linear functional on V.

Proof: Since $f(\alpha) \in F \Rightarrow -f(\alpha) \in F$, therefore $-f$ is a function from V into F.

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$(-f)(a\alpha + b\beta) = -[f(a\alpha + b\beta)] \qquad [\text{by def. of } -f]$$
$$= -[af(\alpha) + bf(\beta)] \qquad [\because f \text{ is a linear functional}]$$
$$= a[-f(\alpha)] + b[-f(\beta)]$$
$$= a[(-f)(\alpha)] + b[(-f)(\beta)].$$

$\therefore$ $-f$ is a linear functional on V.

Properties of a linear functional.

Theorem 1: *Let f be a linear functional on a vector space $V(F)$. Then*

(i) $f(\mathbf{0}) = 0$ where $\mathbf{0}$ on the left hand side is zero vector of V, and 0 on the right hand side is zero element of F.

(ii) $f(-\alpha) = -f(\alpha) \ \forall\, \alpha \in V$.

Proof. (i) Let $\alpha \in V$. Then $f(\alpha) \in F$.

We have $f(\alpha) + 0 = f(\alpha) \qquad [\because 0 \text{ is zero element of } F]$

$$= f(\alpha + \mathbf{0}) \qquad [\because \mathbf{0} \text{ is zero element of } V]$$

Now F is a field. Therefore

$$f(\alpha) + 0 = f(\alpha) + f(\mathbf{0})$$

$\Rightarrow$ $f(\mathbf{0}) = 0$, by left cancellation law for addition in F.

(ii) We have $f[\alpha + (-\alpha)] = f(\alpha) + f(-\alpha)$ [$\because$ f is a linear functional]

But $f[\alpha + (-\alpha)] = f(\mathbf{0}) = 0$ [by (i)]

Thus in F, we have

$$f(\alpha) + f(-\alpha) = 0$$

$$\Rightarrow \quad f(-\alpha) = -f(\alpha).$$

7.3 Dual Spaces

(Garhwal 2006, 08)

Let V' be the set of all linear functionals on a vector space $V(F)$. Sometimes we denote this set by V^*. Now our aim is to impose a vector space structure on the set V' over the same field F. For this purpose we shall have to suitably define addition in V' and scalar multiplication in V' over F.

Theorem: *Let V be a vector space over the field F. Let f_1 and f_2 be linear functionals on V. The function $f_1 + f_2$ defined by*

$$(f_1 + f_2)(\alpha) = f_1(\alpha) + f_2(\alpha) \ \forall \alpha \in V$$

is a linear functional on V. If c is any element of F, the function cf defined by

$$(cf)(\alpha) = cf(\alpha) \ \forall \alpha \in V$$

is a linear functional on V. The set V' of all linear functionals on V, together with the addition and scalar multiplication defined as above is a vector space over the field F.

Proof: Suppose f_1 and f_2 are linear functionals on V and we define $f_1 + f_2$ as follows :

$$(f_1 + f_2)(\alpha) = f_1(\alpha) + f_2(\alpha) \ \forall \alpha \in V. \quad \text{...(1)}$$

Since $f_1(\alpha) + f_2(\alpha) \in F$, therefore $f_1 + f_2$ is a function from V into F.

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$(f_1 + f_2)(a\alpha + b\beta) = f_1(a\alpha + b\beta) + f_2(a\alpha + b\beta) \quad \text{[by (1)]}$$

$$= [af_1(\alpha) + bf_1(\beta)] + [af_2(\alpha) + bf_2(\beta)]$$

[$\because f_1$ and f_2 are linear functionals]

$$= a[f_1(\alpha) + f_2(\alpha)] + b[f_1(\beta) + f_2(\beta)]$$

$$= a[(f_1 + f_2)(\alpha)] + b[(f_1 + f_2)(\beta)] \quad \text{[by (1)]}$$

$\therefore$ $(f_1 + f_2)$ is a linear functional on V. Thus

$$f_1, f_2 \in V' \Rightarrow f_1 + f_2 \in V'.$$

Therefore V' is closed with respect to addition defined in it.

Again let $f \in V'$ and $c \in F$. Let us define cf as follows :

$$(cf)(\alpha) = cf(\alpha) \ \forall \ \alpha \in V. \quad \text{...(2)}$$

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$\begin{aligned}
(cf)(a\alpha + b\beta) &= cf(a\alpha + b\beta) && \text{[by (2)]} \\
&= c[af(\alpha) + bf(\beta)] && [\because f \text{ is linear functional}] \\
&= c[af(\alpha)] + c[bf(\beta)] && [\because F \text{ is a field}] \\
&= (ca) f(\alpha) + (cb) f(\beta) \\
&= (ac) f(\alpha) + (bc) f(\beta) \\
&= a[cf(\alpha)] + b[cf(\beta)] \\
&= a[(cf)(\alpha)] + b[(cf)(\beta)].
\end{aligned}$$

$\therefore$ cf is a linear functional on V. Thus

$$f \in V' \text{ and } c \in F \Rightarrow cf \in V'.$$

Therefore V' is closed with respect to scalar multiplication defined in it.

Associativity of addition in V'.

Let $f_1, f_2, f_3 \in V'$. If $\alpha \in V$, then

$$\begin{aligned}
&[f_1 + (f_2 + f_3)](\alpha) = f_1(\alpha) + (f_2 + f_3)(\alpha) && \text{[by (1)]} \\
&= f_1(\alpha) + [f_2(\alpha) + f_3(\alpha)] && \text{[by (1)]} \\
&= [f_1(\alpha) + f_2(\alpha)] + f_3(\alpha) && [\because \text{ addition in } F \text{ is associative}] \\
&= (f_1 + f_2)(\alpha) + f_3(\alpha) && \text{[by (1)]} \\
&= [(f_1 + f_2) + f_3](\alpha) && \text{[by (1)]}
\end{aligned}$$

$\therefore$ $f_1 + (f_2 + f_3) = (f_1 + f_2) + f_3$

[by def. of equality of two functions]

Commutativity of addition in V'. Let $f_1, f_2 \in V'$. If α is any element of V, then

$$\begin{aligned}
(f_1 + f_2)(\alpha) &= f_1(\alpha) + f_2(\alpha) && \text{[by (1)]} \\
&= f_2(\alpha) + f_1(\alpha) && [\because \text{ addition in } F \text{ is commutative}] \\
&= (f_2 + f_1)(\alpha) && \text{[by (1)]}
\end{aligned}$$

$\therefore$ $f_1 + f_2 = f_2 + f_1$.

Existence of additive identity in V'. Let $\hat{\mathbf{0}}$ be the zero linear functional in V *i.e.*,

$$\hat{\mathbf{0}}(\alpha) = 0 \ \forall \ \alpha \in V.$$

Then $\hat{\mathbf{0}} \in V'$. If $f \in V'$ and $\alpha \in V$, we have

$$\begin{aligned}
(\hat{\mathbf{0}} + f)(\alpha) &= \hat{\mathbf{0}}(\alpha) + f(\alpha) && \text{[by (1)]} \\
&= 0 + f(\alpha) && [\text{by def. of } \hat{\mathbf{0}}] \\
&= f(\alpha) && [0 \text{ being additive identity in } F]
\end{aligned}$$

$\therefore \qquad \hat{\mathbf{0}}$ is the additive identity in V'.

Existence of additive inverse of each element in V'.

Let $f \in V'$. Let us define $-f$ as follows :

$$(-f)(\alpha) = -f(\alpha) \ \forall \ \alpha \in V.$$

Then $-f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} (-f + f)(\alpha) &= (-f)(\alpha) + f(\alpha) && \text{[by (1)]} \\ &= -f(\alpha) + f(\alpha) && \text{[by def. of } -f\text{]} \\ &= 0 \\ &= \hat{\mathbf{0}}(\alpha) && \text{[by def. of } \hat{\mathbf{0}}\text{]} \end{aligned}$$

$\therefore \qquad -f + f = \hat{\mathbf{0}}$ for every $f \in V'$.

Thus each element in V' possesses additive inverse. Therefore V' is an **abelian group** with respect to addition defined in it.

Further we make the following observations :

(i) Let $c \in F$ and $f_1, f_2 \in V'$. If α is any element in V, we have

$$\begin{aligned} [c(f_1 + f_2)](\alpha) &= c[(f_1 + f_2)(\alpha)] && \text{[by (2)]} \\ &= c[f_1(\alpha) + f_2(\alpha)] && \text{[by (1)]} \\ &= cf_1(\alpha) + cf_2(\alpha) \\ &= (cf_1)(\alpha) + (cf_2)(\alpha) && \text{[by (2)]} \\ &= (cf_1 + cf_2)(\alpha) && \text{[by (1)]} \end{aligned}$$

$\therefore \qquad c(f_1 + f_2) = cf_1 + cf_2$.

(ii) Let $a, b \in F$ and $f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} [(a + b)f](\alpha) &= (a + b)f(\alpha) && \text{[by (2)]} \\ &= af(\alpha) + bf(\alpha) && [\because F \text{ is a field}] \\ &= (af)(\alpha) + (bf)(\alpha) && \text{[by (2)]} \\ &= (af + bf)(\alpha) && \text{[by (1)]} \end{aligned}$$

$\therefore \qquad (a + b)f = af + bf$.

(iii) Let $a, b \in F$ and $f \in V'$. If $\alpha \in V$, we have

$$\begin{aligned} [(ab)f](\alpha) &= (ab)f(\alpha) && \text{[by (2)]} \\ &= a[bf(\alpha)] && [\because \text{ multiplication in } F \text{ is associative}] \\ &= a[(bf)(\alpha)] && \text{[by (2)]} \\ &= [a(bf)](\alpha) && \text{[by (2)]} \end{aligned}$$

$\therefore \qquad (ab)f = a(bf)$.

(iv) Let 1 be the multiplicative identity of F and $f \in V'$. If $\alpha \in V$, we have

$$(1f)(\alpha) = 1f(\alpha) \qquad \text{[by (2)]}$$

$\therefore \qquad 1f = f.$

Hence V' is a vector space over the field F.

Dual Space.

Definition: *Let V be a vector space over the field F. Then the set V' of all linear functionals on V is also a vector space over the field F. The vector space V' is called the dual space of V.* **(Garhwal 2006, 08, 12)**

Sometimes V^* and $\hat{V}$ are also used to denote the dual space of V. The dual space of V is also called the **conjugate space** of V.

7.4 Dual Bases

Theorem 1: *Let V be an n-dimensional vector space over the field F and let $B = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis for V. If $\{x_1, \dots, x_n\}$ is any ordered set of n scalars, then there exists a unique linear functional f on V such that*

$$f(\alpha_i) = x_i, \; i = 1, 2, \dots, n.$$

Proof: Existence of f. Let $\alpha \in V$.

Since $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a basis for V, therefore there exist unique scalars $a_1, a_2, \dots, a_n$ such that

$$\alpha = a_1\alpha_1 + \dots + a_n \alpha_n.$$

For this vector α, let us define

$$f(\alpha) = a_1 x_1 + \dots + a_n x_n.$$

Obviously $f(\alpha)$ as defined above is a unique element of F. Therefore f is a well-defined rule for associating with each vector α in V a unique scalar $f(\alpha)$ in F. Thus f is a function from V into F.

The unique representation of $\alpha_i \in V$ as a linear combination of the vectors belonging to the basis B is

$$\alpha_i = 0\alpha_1 + 0\alpha_2 + \dots + 1\alpha_i + 0\alpha_{i+1} + \dots + 0\alpha_n.$$

Therefore according to our definition of f, we have

$$f(\alpha_i) = 0x_1 + 0x_2 + \dots + 1x_i + 0x_{i+1} + \dots + 0x_n$$

i.e., $\qquad f(\alpha_i) = x_i, \; i = 1, 2, \dots, n.$

Now to show that f is a linear functional.

Let $a, b \in F$ and $\alpha, \beta \in V$. Let

$$\alpha = a_1\alpha_1 + \dots + a_n \alpha_n,$$

and $\qquad \beta = b_1\alpha_1 + \dots + b_n \alpha_n.$

Then $f(a\alpha + b\beta) = f[a(a_1\alpha_1 + \dots + a_n \alpha_n) + b(b_1\alpha_1 + \dots + b_n \alpha_n)]$

$$= f[(aa_1 + bb_1)\alpha_1 + \dots + (aa_n + bb_n)\alpha_n]$$

$$= (aa_1 + bb_1)x_1 + \dots + (aa_n + bb_n)x_n \qquad \text{[by def. of } f\text{]}$$

$\therefore$ f is a linear functional on V. Thus there exists a linear functional f on V such that $f(\alpha_i) = x_i$, $i = 1, 2, \ldots, n$.

Uniqueness of f. Let g be a linear functional on V such that

$$g(\alpha_i) = x_i, \quad i = 1, 2, \ldots, n.$$

For any vector $\alpha = a_1\alpha_1 + \ldots + a_n \alpha_n \in V$, we have

$$\begin{aligned} g(\alpha) &= g(a_1\alpha_1 + \ldots + a_n \alpha_n) \\ &= a_1 g(\alpha_1) + \ldots + a_n g(\alpha_n) && [\because g \text{ is linear}] \\ &= a_1 x_1 + \ldots + a_n x_n && [\text{by def. of } g] \\ &= f(\alpha). && [\text{by def. of } f] \end{aligned}$$

Thus $g(\alpha) = f(\alpha) \ \forall \ \alpha \in V$.

$\therefore$ $g = f$.

This shows the uniqueness of f.

Remark: From this theorem we conclude that if f is a linear functional on a finite dimensional vector space V, then f is completely determined if we mention under f the images of the elements of a basis set of V. If f and g are two linear functionals on V such that $f(\alpha_i) = g(\alpha_i)$ for all α_i belonging to a basis of V, then

$$f(\alpha) = g(\alpha) \ \forall \ \alpha \in V \ \ i.e., \ f = g.$$

Thus two linear functionals of V are equal if they agree on a basis of V.

Theorem 2: *Let V be an n-dimensional vector space over the field F and let $B = \{\alpha_1, \ldots, \alpha_n\}$ be a basis for V. Then there is a uniquely determined basis $B' = \{f_1, \ldots, f_n\}$ for V' such that $f_i(\alpha_j) = \delta_{ij}$. Consequently the dual space of an n-dimensional space is n-dimensional.*

The basis B' is called the **dual basis** of B.

Proof: $B = \{\alpha_1, \ldots, \alpha_n\}$ is an ordered basis for V. Therefore by theorem 1, there exists a unique linear functional f_1 on V such that

$$f_1(\alpha_1) = 1, f_1(\alpha_2) = 0, \ldots, f_1(\alpha_n) = 0$$

where $\{1, 0, \ldots, 0\}$ is an ordered set of n scalars.

In fact, for each $i = 1, 2, \ldots, n$ there exists a unique linear functional f_i on V such that

$$f_i(\alpha_j) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

i.e., $$f_i(\alpha_j) = \delta_{ij}, \qquad \ldots(1)$$

where $\delta_{ij} \in F$ is Kronecker delta *i.e.,* $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

Let $B' = \{f_1, \ldots, f_n\}$. Then B' is a subset of V' containing n distinct elements of V'. We shall show that B' is a basis for V'.

First we shall show that B' is linearly independent.

$\Rightarrow$ $(c_1 f_1 + \ldots + c_n f_n)(\alpha) = \hat{\mathbf{0}}(\alpha) \ \forall \alpha \in V$

$\Rightarrow$ $c_1 f_1(\alpha) + \ldots + c_n f_n(\alpha) = 0 \ \forall \alpha \in V$ $[\because \hat{\mathbf{0}}(\alpha) = 0]$

$\Rightarrow$ $\sum_{i=1}^{n} c_i f_i(\alpha) = 0 \ \forall \alpha \in V$

$\Rightarrow$ $\sum_{i=1}^{n} c_i f_i(\alpha_j) = 0, j = 1, 2, \ldots, n$

[Putting $\alpha = \alpha_j$ where $j = 1, 2, \ldots, n$]

$\Rightarrow$ $\sum_{i=1}^{n} c_i \delta_{ij} = 0, j = 1, 2, \ldots, n$

$\Rightarrow$ $c_j = 0, j = 1, 2, \ldots, n$

$\Rightarrow$ $f_1, f_2, \ldots, f_n$ are linearly independent.

In the second place, we shall show that the linear span of B' is equal to V'.

Let f be any element of V'. The linear functional f will be completely determined if we define it on a basis for V. So let

$$f(\alpha_i) = a_i, i = 1, 2, \ldots, n. \qquad \ldots(2)$$

We shall show that

$$f = a_1 f_1 + \ldots + a_n f_n = \sum_{i=1}^{n} a_i f_i.$$

We know that two linear functionals on V are equal if they agree on a basis of V. So let $\alpha_j \in B$ where $j = 1, \ldots, n$. Then

$$\left[\sum_{i=1}^{n} a_i f_i\right](\alpha_j) = \sum_{i=1}^{n} a_i f_i(\alpha_j)$$

$$= \sum_{i=1}^{n} a_i \delta_{ij} \qquad \text{[from (1)]}$$

$= a_j$, on summing with respect to i and remembering that $\delta_{ij} = 1$ when $i = j$ and $\delta_{ij} = 0$ when $i \neq j$

$$= f(\alpha_j) \qquad \text{[from (2)]}$$

Thus $\left[\sum_{i=1}^{n} a_i f_i\right](\alpha_j) = f(\alpha_j) \ \forall \alpha_j \in B.$

Therefore $f = \sum_{i=1}^{n} a_i f_i$. Thus every element f in V' can be expressed as a linear combination of $f_1, \ldots, f_n$.

$\therefore$ V' = linear span of B'. Hence B' is a basis for V'.

Now dim V' = number of distinct elements in $B' = n$.

Corollary: *If V is an n-dimensional vector space over the field F, then V is isomorphic to its dual space V'.*

$\therefore \qquad V$ is isomorphic to V'.

Theorem 3: *Let V be an n-dimensional vector space over the field F and let $B = \{\alpha_1, \ldots, \alpha_n\}$ be a basis for V. Let $B' = \{f_1, \ldots, f_n\}$ be the dual basis of B. Then for each linear functional f on V, we have*

$$f = \sum_{i=1}^{n} f(\alpha_i) f_i$$

and for each vector α in V we have

$$\alpha = \sum_{i=1}^{n} f_i(\alpha)\, \alpha_i.$$

Proof: Since B' is dual basis of B, therefore

$$f_i(\alpha_j) = \delta_{ij}. \qquad \ldots(1)$$

If f is a linear functional on V, then $f \in V'$ for which B' is basis. Therefore f can be expressed as a linear combination of $f_1, \ldots, f_n$. Let $f = \sum_{i=1}^{n} c_i f_i$.

Then $f(\alpha_j) = \left(\sum_{i=1}^{n} c_i f_i\right)(\alpha_j) = \sum_{i=1}^{n} c_i f_i(\alpha_j)$

$$= \sum_{i=1}^{n} c_i \delta_{ij} \qquad \text{[From (1)]}$$

$$= c_j,\ j = 1, 2, \ldots, n.$$

$$\therefore \qquad f = \sum_{i=1}^{n} f(\alpha_i) f_i.$$

Now let α be any vector in V. Let

$$\alpha = x_1\alpha_1 + \ldots + x_n \alpha_n. \qquad \ldots(2)$$

Then $f_i(\alpha) = f_i\left(\sum_{j=1}^{n} x_j \alpha_j\right)$ $\qquad \left[\text{From (2)}, \alpha = \sum_{j=1}^{n} x_j \alpha_j\right]$

$$= \sum_{j=1}^{n} x_j f_i(\alpha_j) \qquad [\because f_i \text{ is linear functional}]$$

$$= \sum_{j=1}^{n} x_j \delta_{ij} \qquad \text{[From (1)]}$$

$$= x_i.$$

$$\therefore \qquad \alpha = f_1(\alpha)\alpha_1 + \ldots + f_n(\alpha)\alpha_n = \sum_{i=1}^{n} f_i(\alpha)\alpha_i.$$

Important: It should be noted that if $B = \{\alpha_1, \ldots, \alpha_n\}$ is an ordered basis for V and $B' = \{f_1, \ldots, f_n\}$ is the dual basis, then f_i is precisely the function which assigns to each vector α in V the i^{th} coordinate of α relative to the ordered basis B.

vector in V, there exists a linear functional f on V such that $f(\alpha) \neq 0$.

Proof: Since $\alpha \neq \mathbf{0}$, therefore $\{\alpha\}$ is a linearly independent subset of V. So it can be extended to form a basis for V. Thus there exists a basis $B = \{\alpha_1, \ldots, \alpha_n\}$ for V such that $\alpha_1 = \alpha$.

If $B' = \{f_1, \ldots, f_n\}$ is the dual basis, then

$$f_1(\alpha) = f_1(\alpha_1) = 1 \neq 0.$$

Thus there exists linear functional f_1 such that

$$f_1(\alpha) \neq 0.$$

Corollary: *Let V be an n-dimensional vector space over the field F. If*

$$f(\alpha) = 0 \ \forall\ f \in V', \textit{then } \alpha = \mathbf{0}.$$

Proof: Suppose $\alpha \neq \mathbf{0}$. Then there is a linear functional f on V such that $f(\alpha) \neq 0$. This contradicts the hypothesis that

$$f(\alpha) = 0 \ \ \forall\ f \in V'.$$

Hence we must have $\alpha = \mathbf{0}$.

Theorem 5: *Let V be an n-dimensional vector space over the field F. If α, β are any two different vectors in V, then there exists a linear functional f on V such that $f(\alpha) \neq f(\beta)$.*

Proof: We have $\alpha \neq \beta \ \Rightarrow \ \alpha - \beta \neq \mathbf{0}$.

Now $\alpha - \beta$ is a non-zero vector in V. Therefore by theorem 4, there exists a linear functional f on V such that

$$f(\alpha - \beta) \neq 0 \quad \Rightarrow \quad f(\alpha) - f(\beta) \neq 0 \quad \Rightarrow \quad f(\alpha) \neq f(\beta)$$

Hence the result.

7.5 Reflexivity

Second dual space (or Bi-dual space). We know that every vector space V possesses a dual space V' consisting of all linear functionals on V. Now V' is also a vector space. Therefore it will also possess a dual space $(V')'$ consisting of all linear functionals on V'. This dual space of V' is called the **Second dual space** or **Bi-dual space** of V and for the sake of simplicity we shall denote it by V''.

If V is finite-dimensional, then

$$\dim V = \dim V' = \dim V''$$

showing that they are isomorphic to each other.

Theorem 1: *Let V be a finite dimensional vector space over the field F. If α is any vector in V, the function L_α on V' defined by*

$$L_\alpha(f) = f(\alpha) \ \ \forall\ f \in V'$$

is a linear functional on V' i.e., $L_\alpha \in V''$.

Also the mapping $\alpha \Rightarrow L_\alpha$ is an isomorphism of V onto V''.

correspondence L_α defined by

$$L_\alpha\ (f) = f\ (\alpha)\ \ \forall\ f \in V' \qquad \ldots(1)$$

is a function from V' into F.

Let $a, b \in F$ and $f, g \in V'$, then

$$L_\alpha\ (af + bg) = (af + bg)\ (\alpha) \qquad \text{[From (1)]}$$
$$= (af)\ (\alpha) + (bg)\ (\alpha)$$
$$= a f\ (\alpha) + b\, g\ (\alpha)$$

[by scalar multiplication of linear functionals]

$$= a\ [L_\alpha\ (f)] + b\ [L_\alpha\ (g)]. \qquad \text{[From (1)]}$$

Therefore L_α is a linear functional on V' and thus $L_\alpha \in V''$.

Now let ψ be the function from V into V'' defined by

$$\psi\ (\alpha) = L_\alpha\ \forall\ \ \alpha \in V.$$

ψ is one-one. If $\alpha, \beta \in V$, then

$$\psi\ (\alpha) = \psi\ (\beta)$$

$\Rightarrow$ $L_\alpha = L_\beta \ \Rightarrow\ L_\alpha\ (f) = L_\beta\ (f)\ \ \forall\ f \in V'$

$\Rightarrow$ $f\ (\alpha) = f\ (\beta)\ \ \forall\ f \in V'$ [From (1)]

$\Rightarrow$ $f\ (\alpha) - f\ (\beta) = 0\ \ \forall\ \ f \in V'$

$\Rightarrow$ $f\ (\alpha - \beta) = 0\ \ \forall\ f \in V'$

$\Rightarrow$ $\alpha - \beta = \mathbf{0}$

[$\because$ by theorem 4 of 2.16, if $\alpha - \beta \neq \mathbf{0}$, then $\exists$ a linear functional f on V such that $f\ (\alpha - \beta) \neq 0$. Here we have $f\ (\alpha - \beta) = 0\ \ \forall\ \ f \in V'$ and so $\alpha - \beta$ must be $\mathbf{0}$]

$\Rightarrow$ $\alpha = \beta$.

$\therefore$ ψ is one-one.

ψ is a linear transformation.

Let $a, b \in F$ and $\alpha, \beta \in V$. Then

$$\psi\ (a\alpha + b\beta) = L_{a\alpha + b\beta} \qquad \text{[by def. of } \psi]$$

For every $f \in V'$, we have

$$L_{a\alpha + b\beta}\ (f) = f\ (a\alpha + b\beta)$$
$$= af\ (\alpha) + bf\ (\beta) \qquad \text{[From (1)]}$$
$$= aL_\alpha\ (f) + bL_\beta\ (f) \qquad \text{[From (1)]}$$
$$= (aL_\alpha)\ (f) + (bL_\beta)\ (f) = (aL_\alpha + bL_\beta)\ (f).$$

$\therefore$ $L_{a\alpha + b\beta} = aL_\alpha + bL_\beta = a\,\psi\ (\alpha) + b\ \psi\ (\beta).$

Thus $\psi\ (a\alpha + b\beta) = a\,\psi\ (\alpha) + b\,\psi\ (\beta)$.

$\therefore$ ψ is a linear transformation from V into V''. We have $\dim V = \dim V''$. Therefore ψ is one-one implies that ψ must also be onto.

Hence ψ is an isomorphism of V onto V''.

called the **natural correspondence** between V and V''. It is important to note that the above theorem shows not only that V and V'' are isomorphic—this much is obvious from the fact that they have the same dimension — but that the natural correspondence is an isomorphism. This property of vector spaces is called **reflexivity.** Thus in the above theorem we have proved that *every finite-dimensional vector space is reflexive.*

In future we shall identify V'' with V through the natural isomorphism $\alpha \leftrightarrow L_\alpha$. We shall say that the element L of V'' is the same as the element α of V iff $L = L_\alpha$ *i.e.*, iff

$$L(f) = f(\alpha) \ \forall \ f \in V'.$$

It will be in this sense that we shall regard $V'' = V$.

Theorem 2: *Let V be a finite dimensional vector space over the field F. If L is a linear functional on the dual space V' of V, then there is a unique vector α in V such that*

$$L(f) = f(\alpha) \ \forall \ f \in V'.$$

Proof: This theorem is an immediate corollary of theorem 1. We should first prove theorem 1. Then we should conclude like this :

The correspondence $\alpha \to L_\alpha$ is a one-to-one correspondence between V and V''. Therefore if $L \in V''$, there exists a unique vector α in V such that $L = L_\alpha$ *i.e.*, such that

$$L(f) = f(\alpha) \ \forall \ f \in V'.$$

Theorem 3: *Let V be a finite dimensional vector space over the field F. Each basis for V' is the dual of some basis for V.*

Proof: Let $B' = \{f_1, f_2, \ldots, f_n\}$ be a basis for V. Then there exists a dual basis $(B')' = \{L_1, L_2, \ldots, L_n\}$ for V'' such that

$$L_i(f_j) = \delta_{ij}. \qquad \ldots(1)$$

By previous theorem, for each i there is a vector α_i in V such that

$$L_i = L_{\alpha_i} \text{ where } L_{\alpha_i}(f) = f(\alpha_i) \ \forall \ f \in V'. \qquad \ldots(2)$$

The correspondence $\alpha \leftrightarrow L_\alpha$ is an isomorphism of V onto V''. Under an isomorphism a basis is mapped onto a basis. Therefore $B = \{\alpha_1, \ldots, \alpha_n\}$ is a basis for V because it is the image set of a basis for V'' under the above isomorphism.

Putting $f = f_j$ in (2), we get

$$f_j(\alpha_i) = L_{\alpha_i}(f_j) = L_i(f_j)$$
$$= \delta_{ij}. \qquad \text{[From (1)]}$$

$\therefore \qquad B' = \{f_1, \ldots, f_n\}$ is the dual of the basis B.

Hence the result.

Theorem 4: *Let V be a finite dimensional vector space over the field F. Let B be a basis for V and B' be the dual basis of B. Then show that*

$$B'' = (B')' = B.$$

$B' = \{f_1, \ldots, f_n\}$ be the dual basis of B in V' and

$B'' = (B')' = \{L_1, \ldots, L_n\}$ be the dual basis of B in V''. Then

$$f_i(\alpha_j) = \delta_{ij},$$

and $\quad L_i(f_j) = \delta_{ij}, \; i = 1, \ldots, n; \; j = 1, \ldots, n.$

If $\quad \alpha \in V$, then there exists $L_\alpha \in V''$ such that

$$L_\alpha(f) = f(\alpha) \;\; \forall \; f \in V'.$$

Taking α_i in place of α, we see that for each $j = 1, \ldots, n$,

$$L_{\alpha_i}(f_j) = f_j(\alpha_i) = \delta_{ij} = L_i(f_j).$$

Thus L_{α_i} and L_i agree on a basis for V'. Therefore

$$L_{\alpha_i} = L_i.$$

If we identify V'' with V through the natural isomorphism $\alpha \leftrightarrow L_\alpha$, then we consider L_α as the same element as α.

So $\quad L_i = L_{\alpha_i} = \alpha_i$ where $i = 1, 2, \ldots, n.$

Thus $\quad B'' = B.$

Illustrative Examples

Example 1: *Find the dual basis of the basis set*

$$B = \{(1, -1, 3), (0, 1, -1), (0, 3, -2)\}$$

for $V_3(\mathbf{R})$. **(Garhwal 2008, 12)**

Solution: Let $\alpha_1 = (1, -1, 3), \alpha_2 = (0, 1, -1), \alpha_3 = (0, 3, -2)$.

Then $\quad B = \{\alpha_1, \alpha_2, \alpha_3\}$.

If $\quad B' = \{f_1, f_2, f_3\}$ is dual basis of B, then

$$f_1(\alpha_1) = 1, f_1(\alpha_2) = 0, f_1(\alpha_3) = 0,$$

$$f_2(\alpha_1) = 0, f_2(\alpha_2) = 1, f_2(\alpha_3) = 0,$$

and $\quad f_3(\alpha_1) = 0, f_3(\alpha_2) = 0, f_3(\alpha_3) = 1.$

Now to find explicit expressions for f_1, f_2, f_3.

Let $(a, b, c) \in V_3(\mathbf{R})$.

Let $\quad (a, b, c) = x(1, -1, 3) + y(0, 1, -1) + z(0, 3, -2)$ ...(1)

$$= x\alpha_1 + y\alpha_2 + z\alpha_3.$$

Then $f_1(a, b, c) = x, f_2(a, b, c) = y,$ and $f_3(a, b, c) = z.$

Now to find the values of x, y, z.

From (1), we have

$$x = a, -x + y + 3z = b, 3x - y - 2z = c.$$

Solving these equations, we have

$$x = a, y = 7a - 2b - 3c, z = b + c - 2a.$$

and $\quad f_3 (a, b, c) = -2a + b + c.$

Therefore $B' = \{f_1, f_2, f_3\}$ is a dual basis of B where f_1, f_2, f_3 are as defined above.

Example 2: *The vectors* $\alpha_1 = (1, 1, 1), \alpha_2 = (1, 1, -1)$, *and* $\alpha_3 = (1, -1, -1)$ *form a basis of* $V_3(\mathbf{C})$. *If* $\{f_1, f_2, f_3\}$ *is the dual basis and if* $\alpha = (0, 1, 0)$, *find* $f_1(\alpha), f_2(\alpha)$ *and* $f_3(\alpha)$.

Solution: Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$. Then

$$f_1(\alpha) = a_1, f_2(\alpha) = a_2, f_3(\alpha) = a_3.$$

Now $\quad \alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$

$\Rightarrow \quad (0, 1, 0) = a_1(1, 1, 1) + a_2(1, 1, -1) + a_3(1, -1, -1)$

$\Rightarrow \quad (0, 1, 0) = (a_1 + a_2 + a_3, a_1 + a_2 - a_3, a_1 - a_2 - a_3)$

$\Rightarrow \quad a_1 + a_2 + a_3 = 0, a_1 + a_2 - a_3 = 1, a_1 - a_2 - a_3 = 0$

$\Rightarrow \quad a_1 = 0, a_2 = \frac{1}{2}, a_3 = -\frac{1}{2}.$

$\therefore \quad f_1(\alpha) = 0, f_2(\alpha) = \frac{1}{2}, f_3(\alpha) = -\frac{1}{2}.$

Example 3: *If f is a non-zero linear functional on a vector space V and if x is an arbitrary scalar, does there necessarily exist a vector α in V such that $f(\alpha) = x$?*

Solution: f is a non-zero linear functional on V. Therefore there must be some non-zero vector β in V such that

$$f(\beta) = y \text{ where } y \text{ is a non-zero element of } F.$$

If x is any element of F, then

$$x = (xy^{-1})\, y = (xy^{-1})\, f(\beta)$$

$$= f[(xy^{-1})\beta] \qquad [\because f \text{ is linear functional}]$$

Thus there exists $\alpha = (xy^{-1})\beta \in V$ such that $f(\alpha) = x$.

Note: If f is a non-zero linear functional on $V(F)$, then f is necessarily a function from V onto F.

Important Note: In some books $f(\alpha)$ is written as $[\alpha, f]$.

Example 4. *Prove that if f is a linear functional on an n-dimensional vector space $V(F)$, then the set of all those vectors α for which $f(\alpha) = 0$ is a subspace of V, what is the dimension of that subspace?*

Solution: Let $N = \{\alpha \in V : f(\alpha) = 0\}$.

N is not empty because at least $\mathbf{0} \in N$. Remember that

$$f(\mathbf{0}) = 0.$$

Let $\alpha, \beta \in N$. Then $f(\alpha) = 0, f(\beta) = 0$.

If $\quad a, b \in F$, we have

$$f(a\alpha + b\beta) = af(\alpha) + bf(\beta) = a0 + b0 = 0.$$

Thus $\quad a, b \in F$ and $\alpha, \beta \in N$

$\Rightarrow \quad a\alpha + b\beta \in N.$

$\therefore \quad N$ is a subspace of V. This subspace N is the null space of f.

We know that dim V = dim N + dim (range of f).

(i) If f is zero linear functional, then range of f consists of zero element of F alone. Therefore dim (range of f) = 0 in this case.

$\therefore$ In this case, we have

$$\dim V = \dim N + 0$$

$$\Rightarrow \quad n = \dim N.$$

(ii) If f is a non-zero linear functional on V, then f is onto F. So range of f consists of all F in this case. The dimension of the vector space F' is 1.

$\therefore$ In this case we have

$$\dim V = \dim N + 1$$

$$\Rightarrow \quad \dim N = n - 1.$$

Example 5: *Let V be a vector space over the field F. Let f be a non-zero linear functional on V and let N be the null space of f. Fix a vector α_0 in V which is not in N. Prove that for each α in V there is a scalar c and a vector β in N such that $\alpha = c\alpha_0 + \beta$. Prove that c and β are unique.*

Solution: Since f is a non-zero linear functional on V, therefore there exists a non-zero vector α_0 in V such that $f(\alpha_0) \neq 0$. Consequently $\alpha_0 \notin N$. Let $f(\alpha_0) = y \neq 0$.

Let α be any element of V and let $f(\alpha) = x$.

We have $\quad f(\alpha) = x$

$\Rightarrow \quad f(\alpha) = (x y^{-1}) y \qquad [\because \; 0 \neq y \in F \Rightarrow y^{-1}$ exists]

$\Rightarrow \quad f(\alpha) = cy$ where $c = x y^{-1} \in F$

$\Rightarrow \quad f(\alpha) = c f(\alpha_0)$

$\Rightarrow \quad f(\alpha) = f(c\,\alpha_0) \qquad [\because f$ is a linear functional]

$\Rightarrow \quad f(\alpha) - f(c\,\alpha_0) = 0$

$\Rightarrow \quad f(\alpha - c\,\alpha_0) = 0$

$\Rightarrow \quad \alpha - c\,\alpha_0 \in N$

$\Rightarrow \quad \alpha - c\,\alpha_0 = \beta$ for some $\beta \in N$.

$\Rightarrow \quad \alpha = c\,\alpha_0 + \beta.$

If possible, let

$$\alpha = c'\alpha_0 + \beta' \text{ where } c' \in F \text{ and } \beta' \in N.$$

Then $c\,\alpha_0 + \beta = c'\alpha_0 + \beta' \qquad \ldots(1)$

$\Rightarrow \quad (c - c')\,\alpha_0 + (\beta - \beta') = \mathbf{0}$

$\Rightarrow \quad f[(c - c')\,\alpha_0 + (\beta - \beta')] = f(\mathbf{0})$

$\Rightarrow \qquad (c - c') f(\alpha_0) = 0$

$$[\because \beta, \beta' \in N \Rightarrow \beta - \beta' \in N \text{ and thus } f(\beta - \beta') = 0]$$

$\Rightarrow \qquad (c - c') = 0 \qquad [\because f(\alpha_0) \text{ is a non-zero element of } F]$

$c = c'.$

Putting $c = c'$ in (1), we get

$$c\alpha_0 + \beta = c\alpha_0 + \beta'$$

$\Rightarrow \qquad \beta = \beta'.$

Hence c and β are unique.

Example 6: *If f and g are in V' such that $f(\alpha) = 0 \Rightarrow g(\alpha) = 0$, prove that $g = kf$ for some $k \in F$.*

Solution: It is given that $f(\alpha) = 0 \Rightarrow g(\alpha) = 0$. Therefore if α belongs to the null space of f, then α also belongs to the null space of g. Thus null space of f is a subset of the null space of g.

(i) If f is zero linear functional, then null space of f is equal to V. Therefore in this case V is a subset of null space of g. Hence null space of g is equal to V. So g is also zero linear functional. Hence we have

$$g = kf \ \forall \ k \in F.$$

(ii) Let f be non-zero linear functional on V. Then there exists a non-zero vector $\alpha_0 \in V$ such that $f(\alpha_0) = y$ where y is a non-zero element of F.

Let $\qquad k = \dfrac{g(\alpha_0)}{f(\alpha_0)} \cdot$

If $\quad \alpha \in V$, then we can write

$$\alpha = c\,\alpha_0 + \beta \text{ where } c \in F \text{ and } \beta \in \text{null space of } f.$$

We have $\quad g(\alpha) = g(c\,\alpha_0 + \beta) = cg(\alpha_0) + g(\beta)$

$$= cg(\alpha_0)$$

$$[\because \beta \in \text{null space of } f \Rightarrow f(\beta) = 0 \text{ and so } g(\beta) = 0]$$

Also $\quad (kf)(\alpha) = kf(\alpha) = kf(c\alpha_0 + \beta)$

$$= k[cf(\alpha_0) + f(\beta)]$$

$$= kc\,f(\alpha_0) \qquad [\because f(\beta) = 0]$$

$$= \frac{g(\alpha_0)}{f(\alpha_0)} cf(\alpha_0) = cg(\alpha_0).$$

Thus $\qquad g(\alpha) = (kf)(\alpha) \ \forall \ \alpha \in V.$

$\therefore \qquad g = kf.$

Comprehensive Exercise 1

1. Let $f : \mathbf{R}^2 \to \mathbf{R}$ and $g : \mathbf{R}^2 \to \mathbf{R}$ be the linear functionals defined by $f(x, y) = x + 2y$ and $g(x, y) = 3x - y$. Find (i) $f + g$ (ii) $4f$ (iii) $2f - 5g$.
2. Let $f : \mathbf{R}^2 \to \mathbf{R}$ and $g : \mathbf{R}^2 \to \mathbf{R}$ be the linear functionals defined by $f(x, y, z) = 2x - 3y + z$ and $g(x, y, z) = 4x - 2y + 3z$. Find (i) $f + g$ (ii) $3f$ (iii) $2f - 5g$.
3. Find the dual basis of the basis set $\{(2, 1), (3, 1)\}$ for $\mathbf{R}^2$.
4. Find the dual basis of the basis set
$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \text{ for } V_3 \text{ } (\mathbf{R}).$$
5. Find the dual basis of the basis set
$$B = \{(1, -2, 3), (1, -1, 1), (2, -4, 7)\} \text{ of } V_3 (\mathbf{R}).$$
6. If $B = \{(-1, 1, 1), (1, -1, 1), (1, 1, -1)\}$ is a basis of V_3 $(\mathbf{R})$, then find the dual basis of B.
7. Let V be the vector space of polynomials over $\mathbf{R}$ of degree ≤ 1 *i.e.,*
$$V = \{a + bx : a, b \in \mathbf{R}\}.$$
Let ϕ_1 and ϕ_2 be linear functionals on V defined by
$$\phi_1 [f(x)] = \int_0^1 f(x) \, dx \text{ and } \phi_2 [f(x)] = \int_0^2 f(x) \, dx.$$
Find the basis $\{f_1, f_2\}$ of V which is dual to $\{\phi_1, \phi_2\}$.
8. Let V be the vector space of polynomials over $\mathbf{R}$ of degree ≤ 2. Let ϕ_1, ϕ_2 and ϕ_3 be the linear functionals on V defined by
$$\phi_1 [f(x)] = \int_0^1 f(x) \, dx, \quad \phi_2 [f(x)] = f'(1), \phi_3 [f(x)] = f(0).$$
Here $f(x) = a + bx + cx^2 \in V$ and $f'(x)$ denotes the derivative of $f(x)$. Find the basis $\{f_1(x), f_2(x), f_3(x)\}$ of V which is dual to $\{\phi_1, \phi_2, \phi_3\}$.
9. Prove that every finite dimensional vector space V is isomorphic to its second conjugate space V^{**} under an isomorphism which is independent of the choice of a basis in V.
10. Define a non-zero linear functional f on $\mathbf{C}^3$ such that $f(\alpha) = 0 = f(\beta)$ where $\alpha = (1, 1, 1)$ and β $(1, 1, -1)$.

Answers 1

1.	(i) $4x + y$	(ii) $4x + 8y$	(iii) $-13x + 9y$
2.	(i) $6x - 5y + 4z$	(ii) $6x - 9y + 3z$	(iii) $-16x + 4y - 13z$

4. $B' = \{f_1, f_2, f_3\}$ where $f_1\ (a, b, c) = a,\ f_2\ (a, b, c) = b,\ f_3\ (a, b, c) = c$

5. $B' = \{f_1, f_2, f_3\}$ where $f_1\ (a, b, c) = -3a - 5b - 2c,\ f_2\ (a, b, c) = 2a + b,$
$f_3\ (a, b, c) = a + 2b + c$

6. $B' = \{f_1, f_2, f_3\}$, where $f_1\ (a, b, c) = \frac{1}{2}(b + c),\ f_2\ (a, b, c) = \frac{1}{2}(a + c),$
$f_3\ (a, b, c) = \frac{1}{2}(a + b)$

7. $f_1\ (x) = 2 - 2x,\ f_2\ (x) = -\frac{1}{2} + x$

8. $f_1\ (x) = 3x - \frac{3}{2}x^2,\ f_2\ (x) = -\frac{1}{2}x + \frac{3}{4}x^2,\ f_3\ (x) = 1 - 3x + \frac{3}{2}x^2$

10. $f\ (a, b, c) = x\ (a - b)$, where x is any non-zero scalar.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. Let V be an n-dimensional vector space over the field F. The dimension of the dual space of V is
   (a) n (b) n^2
   (c) $\frac{1}{2}n$ (d) none of these

2. If the dual basis of the basis set $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for V_3 $(\mathbf{R})$ is $B' = \{f_1, f_2, f_3\}$, then
   (a) $f_1\ (a, b, c) = a,\ f_2\ (a, b, c) = b,\ f_3\ (a, b, c) = c$
   (b) $f_1\ (a, b, c) = b,\ f_2\ (a, b, c) = c,\ f_3\ (a, b, c) = a$
   (c) $f_1\ (a, b, c) = c,\ f_2\ (a, b, c) = a,\ f_3\ (a, b, c) = b$
   (d) None of these.

3. If $V(F)$ is a vector space and f be a linear functional from $V \to F$, then
   (a) $f\ (\mathbf{0}) = 0$ (b) $f\ (\mathbf{0}) \neq 0$
   (c) $f\ (0) = \mathbf{0}$ (d) $f\ (0) \neq \mathbf{0}$

4. If $V\ (F)$ is a vector space and f is a linear functional from $V \to F$, then
   (a) $f\ (-x) = f\ (x)$ (b) $f\ (-x) = -f\ (x)$
   (c) $f\ (-x) \neq -f\ (x)$ (d) $f\ (-x) \neq f\ (x)$

5. Let V be a vector space over a field F. A linear functional on V is a linear mapping from
   (a) F into V (b) V into F
   (c) V into itself (d) none of these

(Garhwal 2006, 12)

Fill in the blanks "......" so that the following statements are complete and correct.

1. Let V be a vector space over the field F. The dual space of V is also called the ...... space of V.
2. If V is an n-dimensional vector space over the field F, then V is ...... to its dual space V'.

True or False

Write 'T' for true and 'F' for false statement.

1. Let $V(F)$ be a vector space. A linear functional on V is a vector valued function.
2. Every finite dimensional vector space is not reflexive.
3. If $V(F)$ is a vector space, then mapping $f: V \to F: f(\alpha) = 0$, then f is a linear functional.

Answers

Multiple Choice Questions

1. (a) 2. (a) 3. (a) 4. (b) 5. (b)

Fill in the Blank(s)

1. conjugate 2. isomorphic

True or False

1. F 2. F 3. T

□

SECTION

B

LINEAR PROGRAMMING

Chapters

Chapter

1

Mathematical Preliminaries

(Some Basic Concepts on Matrices and Linear Algebra)

1.1 Some Basic Concepts

Consider the system of equations

$$\begin{aligned} 2x + 9y + 7z &= 4, \\ 3x + 4y - 3z &= 5, \\ 6x + 8y - 3z &= 1, \\ 4x - 2y + z &= 2 \end{aligned}$$

Here x, y and z are unknowns and their coefficients are all numbers. Arranging the coefficients in the order in which they occur in the equations and enclosing them in square brackets, we obtain a rectangular array of the form

$$\begin{bmatrix} 2 & 9 & 7 \\ 3 & 4 & -3 \\ 6 & 8 & -3 \\ 4 & -2 & 1 \end{bmatrix}$$

This rectangular array is an example of a **matrix.** The horizontal lines ($\rightarrow$) are called **rows** or *row vectors,* and the vertical lines ($\downarrow$) are called **columns** or *column* a matrix of the type 4×3. The numbers $3, 4, -3, 2$ etc., constituting this matrix are called its **elements.** The difference between a matrix and a number should be clearly understood. A matrix is not a number. It has got no numerical value. It is a new thing formed with the help of numbers. It is just an ordered collection of numbers arranged in the form of a rectangular array. Simply 7 is a number. But in our notation of matrices $[7]$ is a matrix of the type 1×1 and we cannot have $7 = [7]$. We cannot have a relation of equality between a matrix and a number.

We shall use capital letters (in bold type or in italic type) to denote matrices.

Thus $\quad \mathbf{A} = \begin{bmatrix} 5 & 0 & 1 \\ 6 & 1 & 7 \end{bmatrix}_{2\times3}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times3}$

are both matrices. They are of the type 2×3 and 3×3 respectively.

Sometimes we also use the brackets ( ) or the double bars, || || in place of the square brackets [ ] to denote matrices.

Thus $\quad \mathbf{A} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 3+5i & 9 \\ -4 & 3-5i \end{pmatrix}, \mathbf{C} = \begin{Vmatrix} 7 & 7 \\ 7 & 7 \end{Vmatrix},$

are all matrices each of the type 2×2.

1.2 Matrix

A set of mn numbers (real or complex) arranged in the form of a rectangular array having m rows and n columns is called an $m \times n$ matrix [to be read as 'm by n' matrix'].

An $m \times n$ matrix is usually written as

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \ldots a_{1n} \\ a_{21} & a_{22} & \ldots a_{2n} \\ a_{31} & a_{32} & \ldots a_{3n} \\ \ldots & \ldots & \ldots\ldots \\ \ldots & \ldots & \ldots\ldots \\ a_{m1} & a_{m2} & a_{mn} \end{bmatrix}$$

In a compact form the above matrix is represented by $\mathbf{A} = [a_{ij}], i = 1, 2, \ldots m, j = 1, 2, \ldots n$ or simply by $[a_{ij}]_{m\times n}$. We write the general element of the matrix and enclose it in brackets of the type [ ] or of the type ( ).

The numbers a_{11}, a_{12} etc. of this rectangular array are called the **elements** of the matrix. The element a_{ij} belongs to the i^{th} row and the j^{th} column and is sometimes called the $(i, j)^{th}$ element of the matrix. Thus in the element a_{ij} the first suffix i will always denote the number of the row and the second suffix j, the rows and the columns need not be equal.

1.3 Special Types of Matrices

1.3.1 Square Matrix

A $m \times n$ matrix for which $m = n$ (*i.e.*, the number of rows is equal to the number of columns) is called a square matrix of order n. It is also called an n-rowed square matrix. The elements $a_{11}, a_{22}, a_{33}, \ldots, a_{nn}$ are called the **diagonal elements** and the line along which they lie is called the **principal diagonal** of the matrix. For example the matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 3 & 1 & 0 \\ 5 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}_{4\times4}$$

is a square matrix of order 4. The elements 0, 3, 1, 2 constitute the principal diagonal of this matrix.

1.3.2 Unit Matrix or Identity Matrix

A square matrix each of whose diagonal elements is 1 and each of whose non-diagonal elements is equal to zero is called a **unit matrix** or an **identity matrix** and is denoted by I. I_n will denote a unit matrix of order n. Thus a square matrix $\mathbf{A} = [a_{ij}]$ is a unit matrix if $a_{ij} = 1$ when $i = j$ and $a_{ij} = 0$ when $i \neq j$.

1.3.3 Null Matrix or Zero Matrix

A $m \times n$ matrix whose elements are all 0 is called the **null matrix** (or **zero matrix**) of the type $m \times n$. It is usually denoted by $\mathbf{O}$ or more clearly by $\mathbf{O}_{m,\, n}$. Often a null matrix is simply denoted by the symbol 0 read as 'zero'.

1.3.4 Row Matrices and Column Matrices

Any $1 \times n$ matrix which has only one row and n columns is called a **row matrix** or a **row vector**. Similarly any $m \times 1$ matrix which has m rows and only one column is a **column matrix** or a **column vector.**

For example, $\mathbf{X} = [2 \quad 7 \quad -8 \quad 5 \quad 11]_{1\times5}$ is a row matrix of the type 1×5 while

$$\mathbf{Y} = \begin{bmatrix} 2 \\ -9 \\ 11 \end{bmatrix}_{3\times1}$$ is a column matrix of the type 3×1.

1.4 Submatrices of a Matrix

Any matrix obtained by omitting some rows and columns from a given $(m \times n)$ matrix **A** is a called a **submatrix** of **A**.

The matrix **A** itself is a sub-matrix of **A** as it can be obtained from **A** by omitting no rows or columns.

A square submatrix of a square matrix **A** is called a **principal submatrix**, if its diagonal elements are also the diagonal elements of the matrix **A**. Principal submatrices are obtained only by omitting corresponding rows and columns.

For Example: The matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}$ is a submatrix of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 7 & 11 & 6 & 5 \\ 0 & 2 & 1 & 8 \end{bmatrix}$ as it can be obtained from **A** by omitting the second row and the fourth column.

1.5 Equality of Two Matrices

Two matrices $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ are said to be equal if

1. they are of the same size and
2. the elements in the corresponding places of the two matrices are the same *i.e.*, $a_{ij} = b_{ij}$ for each pair of subscripts i and j.

If two matrices **A** and **B** are equal, we write $\mathbf{A} = \mathbf{B}$. If two matrices **A** and **B** are not equal, we write $\mathbf{A} \neq \mathbf{B}$. If two matrices are not of the same size, they cannot be equal.

1.6 Addition of Matrices

Let **A** and **B** be two matrices of the same type $m \times n$. Then their sum (to be denoted by $\mathbf{A} + \mathbf{B}$) is defined to be the matrix of the type $m \times n$ obtained by adding the corresponding elements of **A** and **B**. Thus if $\mathbf{A} = [a_{ij}]_{m\times n}$ and $\mathbf{B} = [b_{ij}]_{m\times n}$, then $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]_{m\times n}$.

Note that $\mathbf{A} + \mathbf{B}$ is also a matrix of the type $m \times n$.

For example, if

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & -1 \\ 4 & -3 & 1 \end{bmatrix}_{2\times 3} \text{ and } \mathbf{B} = \begin{bmatrix} 1 & -2 & 7 \\ 3 & 2 & -1 \end{bmatrix}_{2\times 3}$$

then $$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 3+1 & 2-2 & -1+7 \\ 4+3 & -3+2 & 1-1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 6 \\ 7 & -1 & 0 \end{bmatrix}_{2\times 3}$$

which are of the same size. If two matrices **A** and **B** are of the same size, they are said to be **conformable for addition.** If the matrices **A** and **B** are not of the same size, we cannot find their sum.

1.7 Properties of Matrix Addition

1. **Matrix Addition is Commutative :** If **A** and **B** be two $m \times n$ matrices, then $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
2. **Matrix Addition is Associative :** If **A, B, C** be three matrices each of the type $m \times n$, then
$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$
3. **Existence of Additive Identity :** If **O** be the $m \times n$ matrix each of whose elements is zero, then $\mathbf{A} + \mathbf{O} = \mathbf{A} = \mathbf{O} + \mathbf{A}$ for every $m \times n$ matrix **A**.
4. **Existence of the Additive Inverse**
Negative of a Matrix : Let $\mathbf{A} = [a_{ij}]_{m \times n}$. Then the negative of the matrix **A** is defined as the matrix $[-a_{ij}]_{m \times n}$ and is denoted by $-\mathbf{A}$.
The matrix $-\mathbf{A}$ is the additive inverse of the matrix **A**. Obviously,
$$-\mathbf{A} + \mathbf{A} = \mathbf{O} = \mathbf{A} + (-\mathbf{A}).$$
Here **O** is the null matrix of the type $m \times n$. It is identity element for matrix addition.
Subtraction of Two Matrices : If **A** and **B** are two $m \times n$ matrices, then we define
$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}).$$
Thus the difference $\mathbf{A} - \mathbf{B}$ is obtained by subtracting from each element of **A** the corresponding element of **B**.
5. **Cancellation laws** hold good in the case of addition of matrices *i.e.*, if A, B, C are three $m \times n$ matrices, then
$$\mathbf{A} + \mathbf{B} = \mathbf{A} + \mathbf{C} \Rightarrow \mathbf{B} = \mathbf{C} \quad \textbf{(left cancellation law)}$$
and $$\mathbf{B} + \mathbf{A} = \mathbf{C} + \mathbf{A} \Rightarrow \mathbf{B} = \mathbf{C} \quad \textbf{(right cancellation law)}$$
6. The equation $\mathbf{A} + \mathbf{X} = \mathbf{O}$ has a unique solution $\mathbf{X} = -\mathbf{A}$ in the set of all $m \times n$ matrices.

1.8 Multiplication of a Matrix by a Scalar

Let **A** be any $m \times n$ matrix and k any number (real or complex) called scalar. The $m \times n$ matrix obtained by multiplying every element of the matrix **A** by k is called the scalar multiple of **A** by k and is denoted by $k\,\mathbf{A}$ or $\mathbf{A}\,k$. Symbolically, if $\mathbf{A} = [a_{ij}]_{m \times n}$, then $k\,\mathbf{A} = \mathbf{A}\,k = [ka_{ij}]_{m \times n}$.

For example, if $k = 2$ and $A = \begin{bmatrix} 3 & 2 & -1 \\ 4 & -3 & 1 \end{bmatrix}_{2\times 3}$,

then

$$2\mathbf{A} = \begin{bmatrix} 2\times 3 & 2\times 2 & 2\times -1 \\ 2\times 4 & 2\times -3 & 2\times 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 & -2 \\ 8 & -6 & 2 \end{bmatrix}_{2\times 3}.$$

1.8.1 Properties of Multiplication of a Matrix by a Scalar

Theorem 1: If **A** and **B** are two matrices each of the type $m \times n$, then $k(\mathbf{A}+\mathbf{B}) = k\mathbf{A} + k\mathbf{B}$ *i.e.*, scalar multiplication of matrices distributes over the addition of matrices.

Theorem 2: If p and q are two scalars and **A** is any $m \times n$ matrix, then $(p+q)\mathbf{A} = p\mathbf{A} + q\mathbf{A}$.

Theorem 3: If p and q are two scalars and **A** is any $m \times n$ matrix, then $p(q\mathbf{A}) = (pq)\mathbf{A}$.

Theorem 4: If **A** be any $m \times n$ matrix and k be any scalar, then $(-k)\mathbf{A} = -(k\mathbf{A}) = k(-\mathbf{A})$.

Theorem 5: If **A** be any $m \times n$ matrix, then

(i) $1\mathbf{A} = \mathbf{A}$

(ii) $(-1)\mathbf{A} = -\mathbf{A}$,

Theorem 6: If **A** and **B** are two $m \times n$ matrices, then $-(\mathbf{A}+\mathbf{B}) = -\mathbf{A} - \mathbf{B}$.

1.9 Multiplication of Two Matrices

Let $\mathbf{A} = [a_{ij}]_{m\times n}$ and $\mathbf{B} = [b_{jk}]_{n\times p}$ be two matrices such that the number of columns in **A** is equal to the number of rows in **B**. Then the $m \times p$ matrix $\mathbf{C} = [c_{ik}]_{m\times p}$ such that

$$c_{ik} = \sum_{j=1}^{n} a_{ij}\, b_{jk}$$

[Note that the summation is with respect to the repeated suffix j]
is called the product of the matrices **A** and **B** in that order and we write **C = AB.**

The product **AB** of two matrices **A** and **B** exists if and only if the number of columns in **A** is equal to the number of rows in **B**. Two such matrices are said to be **conformable for multiplication**. The rule of multiplication is **row-by-column multiplication**.

For example, if $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}_{3\times 2}$, $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}_{2\times 2}$

then $\quad AB = \begin{bmatrix} a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} \end{bmatrix}_{3\times 2}$

Important Note : If the product **AB** exists, then it is not necessary that the product **BA** will also exist. For example if **A** is a 4×5 matrix and **B** is a 5×3 matrix, then the product **AB** exists while the product **BA** does not exist.

1.10 Properties of Matrix Multiplication

1. Matrix multiplication is associative, if conformability is assured; *i.e.*, $\mathbf{A(BC)} = \mathbf{(AB)C}$ if **A, B, C** are $m \times n,\ n \times p,\ p \times q$ matrices respectively.
2. Multiplication of matrices is distributive with respect to addition of matrices *i.e.*,

$$\mathbf{A(B+C)} = \mathbf{AB} + \mathbf{AC},$$

where **A, B, C** are any three, $m \times n,\ n \times p,\ n \times p$ matrices respectively.

3. The multiplication of matrices is not always commutative.

   Whenever $\mathbf{AB} = \mathbf{BA}$, the matrices **A** and **B** are said to ***commute.***

   If $\mathbf{AB} = -\mathbf{BA}$ the matrices **A** and **B** are said to ***anti-commute.***
4. If **A** be any $m \times n$ matrix and $\mathbf{O}_{n,\,p}$ be an $n \times p$ null matrix, then $\mathbf{AO}_{n,\,p} = \mathbf{O}_{m,\,p}$ where $\mathbf{O}_{m,\,p}$ is an $m \times p$ null matrix.
   Similarly if $\mathbf{O}_{m,\,n}$ be an $m \times n$ null matrix and **A** be any $n \times p$ matrix, then $\mathbf{O}_{m,\,n}\mathbf{A} = \mathbf{O}_{m,\,p}$.
   If **A** be any n-rowed square matrix and **O** be an n- rowed null matrix, then $\mathbf{AO} = \mathbf{OA} = \mathbf{O}$.
5. The equation $\mathbf{AB} = \mathbf{O}$ does not necessarily imply that at least one of the matrices **A** and **B** must be a zero matrix. Or
   The product of two matrices can be a zero matrix while neither of them is a zero matrix.
6. In the case of matrix multiplication if $\mathbf{AB} = \mathbf{O}$, then it does not necessarily imply that $\mathbf{BA} = \mathbf{O}$.
7. If **A** be an $m \times n$ matrix, $\mathbf{I}_n$ denotes the n-rowed unit matrix, it can be easily seen that

$$\mathbf{AI}_n = \mathbf{A} = \mathbf{I}_m\mathbf{A}.$$

1.11 Determinant of a Square Matrix

Let **A** be any square matrix. The determinant formed by the elements of **A** is said to be the determinant of matrix **A**. This is denoted by $|\mathbf{A}|$ or det **A**. Since in a

only square matrices can have determinants.

Hence, if $\mathbf{A} = \begin{bmatrix} 12 & 0 & 13 \\ 15 & 12 & 11 \\ 13 & 11 & 14 \end{bmatrix}$, then det $\mathbf{A} = |\mathbf{A}| = \begin{vmatrix} 12 & 0 & 13 \\ 15 & 12 & 11 \\ 13 & 11 & 14 \end{vmatrix}$

1.11.1 Difference between a Matrix and a Determinant

1. A matrix "**A**" cannot be reduced to a number whereas the determinant can be reduced to a number.
2. The number of rows may or may not be equal to number of columns in a matrix while in a determinant the number of rows is equal to the number of columns.
3. Interchanging the rows and columns, a different matrix is formed while in a determinant, an interchange of rows and columns does not change the value of the determinant.

1.12 Non-Singular and Singular Matrices

A square matrix **A** is said to be non-singular or singular according as $|\mathbf{A}| \neq 0$ or $|\mathbf{A}| = 0$.

1.13 Transpose of a Matrix

Let $\mathbf{A} = [a_{ij}]_{m \times n}$. Then the $n \times m$ matrix obtained from **A** by changing its rows into columns and its columns into rows is called the transpose of **A** and is denoted by the symbol $\mathbf{A}'$ or $\mathbf{A}^T$.

The operation of interchanging rows with columns is called transposition. Symbolically if

$$\mathbf{A} = [a_{ij}]_{m \times n},$$

then $\quad \mathbf{A}' = [b_{ji}]_{n \times m}$, where $b_{ji} = a_{ij}$,

i.e., the $(j, i)^{th}$ element of $\mathbf{A}'$ is the $(i, j)^{th}$ element of **A**.

For example, the transpose of the 3×4 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 2 & 1 \end{bmatrix}_{3 \times 4} \quad \text{is the } 4 \times 3 \text{ matrix } \mathbf{A}' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 2 \\ 4 & 1 & 1 \end{bmatrix}_{4 \times 3}$$

The first row of **A** is the first column of $\mathbf{A}'$. The second row of **A** is the second column of $\mathbf{A}'$. The third row of **A** is the third column of $\mathbf{A}'$.

1. $(A')' = A$;
2. $(A+B)' = A'+B'$, A and B being of the same size.
3. $(kA)' = kA'$, k being any complex number.
4. $(AB)' = B'A'$, A and B being conformable to multiplication.

The above law (4) is called the reversal law for transposes *i.e.*, the transpose of the product is the product of the transposes taken in the reverse order.

1.14 Adjoint or Adjugate of a Square Matrix

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be any $n \times n$ matrix. The transpose $\mathbf{B}'$ of the matrix $\mathbf{B} = [C_{ij}]_{n \times n}$, where C_{ij} denotes the cofactor of the element a_{ij} in the determinant $|\mathbf{A}|$, is called the adjoint of the matrix **A** and is denoted by the symbol adj **A**.

Thus the adjoint of a matrix **A** is the transpose of the matrix formed by the cofactors of **A** *i.e.*, if

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix},$$

then $\quad$ Adj **A** = the transpose of the matrix $\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$

= the matrix $\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$

Note : Sometimes the adjoint of a matrix is also called the **adjugate** of that matrix.

Theorem : If **A** is any square matrix of order n, then $\mathbf{A}(adj\,\mathbf{A}) = |\mathbf{A}|\,\mathbf{I}_n = (adj\,\mathbf{A})\,\mathbf{A}$.

1.15 Invertible Matrices : Inverse or Reciprocal of a Matrix

Let **A** be any n-rowed square matrix. Then a matrix **B**, if it exists, such that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$$

is called the inverse of matrix **A**.

are both square matrices of the same order. Thus **non-square matrices cannot possess inverse.**

Existence of the Inverse, Theorem : The necessary and sufficient condition for a square matrix **A** to possess the inverse is that $|\mathbf{A}| \neq 0$.

Important :

1. If **A** be an invertible matrix, then the inverse of **A** is $\frac{1}{|\mathbf{A}|} \, Adj\ \mathbf{A}$. It is usual to denote the inverse of **A** by $\mathbf{A}^{-1}$.

   Thus, $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \, Adj\ \mathbf{A}$, provided $|\mathbf{A}| \neq 0$.

2. **Computation of the Inverse by Partitioning :** A non-singular square matrix in partitioned form is expressed as follows :

$$M = \begin{bmatrix} I & Q \\ O & R \end{bmatrix}$$

Where, I (unit matrix) and R are square submatrices and Q, O (null matrix) are two matrices.

If R^{-1} exists and is known, then the inverse of the matrix M is given by

$$M^{-1} = \begin{bmatrix} I & -QR^{-1} \\ O & R^{-1} \end{bmatrix}$$

Theorem 1: If **A**,**B** be two n-rowed non-singular matrices, then **AB** is also non-singular and

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1},$$

i.e., the inverse of a product is the product of the inverses taken in the reverse order.

Theorem 2: If **A** be an $n \times n$ non-singular matrix, then $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$ *i.e.*, the operations of transposing and inverting are commutative.

1.16 Vectors

Any ordered n-tuple of numbers is called an n-vector. By an ordered n-tuple we mean a set consisting of n numbers in which the place of each number is fixed. If $x_1, x_2, \ldots, x_n$ be any n numbers, then the ordered n-tuple $\mathbf{X} = (x_1, x_2, \ldots., x_n)$ is called an n-vector. The ordered triad (x_1, x_2, x_3) is called a 3-vector. Similarly (1, 0, 1, –1) and (1, 8, –5, 7) are 4-vectors. The n numbers $x_1, x_2, \ldots., x_n$ are called components of the n-vector $\mathbf{X} = (x_1, x_2, \ldots., x_n)$. A vector may be written either as a *row vector* or as a *column vector*. If **A** be a matrix of the type $m \times n$, then each row of **A** will be an n-vector and each column of **A** will be an m-vector. A vector whose components are all zero is called a *zero vector* and will be denoted by **O**.

If k be any number and $\mathbf{X}$ be any vector, then relative to the vector $\mathbf{X}$, k is called a scalar.

Algebra of Vectors : Since an n-vector is nothing but a row matrix or a column matrix, therefore we can develop an algebra of vectors in the same manner as the algebra of matrices.

Equality of Two Vectors : Two n-vectors $\mathbf{X}$ and $\mathbf{Y}$ where $\mathbf{X} = (x_1, x_2, ..., x_n)$ and $\mathbf{Y} = (y_1, y_2, ..., y_n)$ are said to be **equal** if and only if their corresponding components are equal *i.e.,* if $x_i = y_i$, for all $i = 1, 2, ..., n$

For example if

$$\mathbf{X} = (1, 4, 7) \text{ and } \mathbf{Y} = (1, 4, 7).$$

then $\mathbf{X} = \mathbf{Y}$

Multiplication of a Vector by a Scalar (Number) :

If k be any number and $\mathbf{X} = (x_1, x_2, ..., x_n)$,

then by definition $k\,\mathbf{X} = (kx_1, kx_2, ..., kx_n)$

The vector $k\,\mathbf{X}$ is called the scalar multiple of the vector $\mathbf{X}$ by the scalar k.

Properties of Addition and Scalar Multiplication of Vectors : If $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ be any three n-vectors and p, q be any two numbers, then obviously

1. $\mathbf{X} + \mathbf{Y} = \mathbf{Y} + \mathbf{X}$
2. $\mathbf{X} + (\mathbf{Y} + \mathbf{Z}) = (\mathbf{X} + \mathbf{Y}) + \mathbf{Z}.$
3. $p(\mathbf{X} + \mathbf{Y}) = p\mathbf{X} + p\mathbf{Y}$
4. $(p + q)\mathbf{X} = p\mathbf{X} + q\mathbf{X}$
5. $p(q\mathbf{X}) = (pq)\mathbf{X}$

1.17 Linear Dependence and Linear Independence of Vectors

Linearly Dependent Set of Vectors : A set of r n-vector $\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_r$ is said to be linearly dependent if there exist r scalars (numbers) $k_1, k_2, ..., k_r$ not all zero, such that

$$k_1\mathbf{X}_1 + k_2\mathbf{X}_2 + ... + k_r\mathbf{X}_r = \mathbf{O},$$

where, $\mathbf{O}$, denotes the n-vector whose components are all zero.

Linearly Independent Set of Vectors : A set of r, n-vectors $\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_r$ is said to be linearly independent if every relation of the type

$$k_1\mathbf{X}_1 + k_2\mathbf{X}_2 + ... k_r\mathbf{X}_\mathbf{r} = \mathbf{O}$$

implies $k_1 = k_2 = k_3 = ... = k_r = 0$

1.18 A Vector as a Linear Combination of Vectors

A vector **X** *which can be expressed in the form*

$$\mathbf{X} = k_1 \mathbf{X}_1 + k_2 \mathbf{X}_2 + \ldots. + k_r \mathbf{X}_r ,$$

is said to be a linear combination of the set of vectors $\mathbf{X}_1, \mathbf{X}_2, \ldots.\mathbf{X}_r$.

Here $k_1, k_2, \ldots, k_r$ are numbers.

The following two results are quite obvious :

1. If a set of vectors is linearly dependent, then at least one member of the set can be expressed as a linear combination of the remaining members.
2. If a set of vectors is linearly independent then no member of the set can be expressed as a linear combination of the remaining members.

1.19 Systems of Linear Non-Homogeneous Equations

Sometimes we think that we can solve every two simultaneous equations of the type

$$\left.\begin{aligned} a_1 x + b_1 y &= c_1 \\ a_2 x + b_2 y &= c_2 \end{aligned}\right\}.$$

but it is not so. For example, consider the simultaneous equations

$$\left.\begin{aligned} 3x + 4y &= 5 \\ 6x + 8y &= 13 \end{aligned}\right\}.$$

There is no set of values of x and y which satisfies both these equations. Such equations are said to be **inconsistent.**

Let us take another example. Consider the simultaneous equations

$$\left.\begin{aligned} 3x + 4y &= 5 \\ 6x + 8y &= 10 \end{aligned}\right\}.$$

These equations are consistent since there exist values of x and y which satisfy both of these equations. We see that $x = -\frac{4}{3}c + \frac{5}{3},\ y = c$ constitute a solution of these equations, where c is arbitrary. Thus these equations possess an infinite number of solutions.

Now we shall discuss the nature of solutions of a system of non-homogeneous linear equations. Let

$$\left.\begin{array}{l} a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m \end{array}\right\} \quad \ldots(1)$$

be a system of m non-homogeneous equations in n unknowns $x_1, x_2, \ldots, x_n$.

If we write

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}_{m\times n}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ x_n \end{bmatrix}_{n\times 1}, \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \ldots \\ b_m \end{bmatrix}_{m\times 1}$$

where $\mathbf{A}, \mathbf{X}, \mathbf{B}$ are $m \times n$, $n \times 1$ and $m \times 1$ matrices respectively, the above equations can be written in the form of a single matrix equations $\mathbf{AX} = \mathbf{B}$.

Any set of values of $x_1, x_2, \ldots, x_n$ which simultaneously satisfy all these equations is called a solution of the system (1). When the system of equations has one or more solutions, the equations are said to be **consistent**, otherwise they are said to be **inconsistent.**

The matrix $[\mathbf{A} \quad \mathbf{B}] = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{bmatrix}$

is called the **augmented matrix** of the given system of equations.

1.20 Condition for a System of n Equations in n Unknowns to have a Unique Solution

Theorem 1: ***If* A *be an n-rowed non-singular matrix,* X *be an* $n \times 1$ *matrix,* B *be an* $n \times 1$ *matrix, the system of equations* AX = B *has a unique solution.***

Proof: If $\mathbf{A}$ be an n-rowed non-singular matrix, the ranks of the matrices $\mathbf{A}$ and $[\mathbf{A} \quad \mathbf{B}]$ are both n. Therefore the system of equations $\mathbf{AX} = \mathbf{B}$ is consistent *i.e.,* possesses a solution.

Pre-multiplying both sides of $\mathbf{AX} = \mathbf{B}$ by $\mathbf{A}^{-1}$, we have

$$\mathbf{A}^{-1}\mathbf{AX} = \mathbf{A}^{-1}\mathbf{B} \quad \Rightarrow \quad \mathbf{IX} = \mathbf{A}^{-1}\mathbf{B} \Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

is a solution of the equation $\mathbf{AX} = \mathbf{B}$.

To show that the solution is unique, let us suppose that $\mathbf{X}_1$ and $\mathbf{X}_2$ be two solutions of $\mathbf{AX} = \mathbf{B}$.

Then $\quad \mathbf{AX}_1 = \mathbf{B},\ \mathbf{AX}_2 = \mathbf{B} \Rightarrow \mathbf{AX}_1 = \mathbf{AX}_2 \Rightarrow \mathbf{A}^{-1}\mathbf{AX}_1 = \mathbf{A}^{-1}\mathbf{AX}_2$

$\Rightarrow \quad \mathbf{IX}_1 = \mathbf{IX}_2 \Rightarrow \mathbf{X}_1 = \mathbf{X}_2$.

Hence the solution is unique.

1.21 Working Rule for Finding the Solution of the Equation $AX = B$

Suppose the coefficient matrix **A** is of the type $m \times n$, *i.e.,* we have m equations in n unknowns. Write the augmented matrix $[\mathbf{A}\ \ \mathbf{B}]$ and reduce it to a Echelon form by applying only E-row transformations on it. This Echelon form will enable us to know the ranks of the augmented matrix $[\mathbf{A}\ \ \mathbf{B}]$ and the coefficient matrix **A**. Then the following different cases arise :

Case I : Rank $\mathbf{A} <$ Rank $[\mathbf{A}\ \ \mathbf{B}]$.

In this case the equations $\mathbf{AX} = \mathbf{B}$ are inconsistent *i.e.,* they have no solution.

Case II : Rank $\mathbf{A}$ = Rank $[\mathbf{A}\ \ \mathbf{B}] = r$ (say).

In this case the equations $\mathbf{AX}=\mathbf{B}$ are consistent *i.e.,* they possess a solution. If $r < m$, then in the process of reducing the matrix $[\mathbf{A}\ \ \mathbf{B}]$ to Echelon form, $(m - r)$ equations will be eliminated. The given system of m equations will then be replaced by an equivalent system of r equations. From these r equations we shall be able to express the values of some r unknowns in terms of the remaining $n - r$ unknowns which can be given any arbitrarily chosen values.

If $r = n$, then $n - r = 0$, so that no variable is to be assigned arbitrary values and therefore in this case there will be a unique solution.

If $r < n$, the $n - r$ variables can be assigned arbitrary values. So in this case there will be an infinite number of solutions. Only $n - r + 1$ solutions will be linearly independent and the rest of the solutions will be linear combinations of them.

If $m < n$, then $r \le m < n$. Thus in this case $n - r > 0$. Therefore when the number of equations is less than the number of unknowns, the equations will always have an infinite number of solutions, provided they are consistent.

1.22 Some Useful Results of Linear Algebra

1. **Basis :** A subset S of E^n is said to be the basis of E^n if
   (i) S is linearly independent
   (ii) S generates E^n *i.e.,* $E^n = L(S)$.
2. **Generating Set :** A set of vectors $\alpha_1, \alpha_2, ..., \alpha_k \in E^n$ is said to generate E^n if every vector belonging to E^n can be expressed as a linear combination of $\alpha_1, \alpha_2, ..., \alpha_k$.
3. Any two basis of E^n have same number of elements.
4. Each set of $n + 1$ or more vectors of E^n is L.D.
5. If a set of n vectors of E^n spans E^n then S is a basis of E^n.
6. If S is a basis of E^n then every vector $\alpha \in E^n$ can be uniquely expressed as a linear combination of vectors of S.

□

Chapter

2

Linear Programming Problem
(Formulation & Graphical Soluton)

2.1 Introduction

Linear Programming (L.P.) is a branch of Operations Research. In order to understand the importance of its study, we must know something of its history and evolution. Its main origin was during the second world war (1939-1945). At that time, the military management in England called upon a team of scientists to study the strategic and tactical problems related to air and land defence of the country. Since they were having very limited military resources, it was necessary to decide upon the most effective utilization of them *e.g.,* the efficient transportation, effective bombing, adequate food supply, etc.

Following the end of war, the success of military teams attracted the attention of **Industrial** managers who were seeking solutions to their complex executive-type problems. The most common problem was to find: what methods should be adopted so that the total cost is minimum or the total profit is maximum? The

Programming, was developed and applications have been developed through the efforts and co-operation of interested individuals in academic institutions and industry both.

While making use of the techniques of Linear Programming, a mathematical model of the problem is formulated. This model is actually a simplified representation of the problems in which only the most important features are considered for reasons of simplicity. Then an **optimal** or most **favourable solution** is found.

Thus this technique is concerned with the optimization problems. ***A decision, which taking into account all the circumstances can be considered the best one, is called an optimal decision.***

Thus, Linear Programming is the mathematical tool used in solving problems in business, industry, commerce, military operations, etc. Here the problem before us is usually to maximize the profit or minimize the cost keeping in mind the restrictions imposed upon us by the limitations of various resources.

2.2 Linear Programming & Linear Programming Problem

Linear Programming : *Linear programming is an important optimization (maximization or minimization) technique used in decision making in business and every-day life for obtaining the maximum or minimum values as required of a linear expression subject to satisfying certain number of given linear restrictions.*

Linear Programming Problem (L.P.P.) : *The linear programming problem in general calls for optimizing (Maximizing/minimizing) a linear function of variables called the objective function subject to a set of linear equations and/or linear inequations called the constraints or restrictions.*

The term *linear* means that all the relations governing the problem are linear and the term *programming* refers to the process of determining a particular programme or plan of action.

Objective Function : The function which is to be optimized (maximized or minimized) is called an *objective function.*

Constraints : The system of linear inequations (or equations) under which the objective function is to be optimized are called the *constraints.*

2.5 Basic Requirements of a Linear Programming Problem (L.P.P.)

1. **Well-defined Objective Function :** The objective function of the L.P.P. should be clear and well-defined. The objective may be either to maximize the profit by utilizing the available resources or it may be to minimize the cost by using a limited amount of various resources required for production.

2. **Quantitative Measurement of Elements :** It is necessary that each element may be capable of being measured quantitatively. Numerical data must be given so that the relationships among the elements may be considered.

3. **Presence of Constraints and Restrictions :** The available resources which are to be allocated among various competing activities should be limited.

4. **Alternate Lines of Action :** There must be different lines of actions available to perform the job.

5. **Non-negative Restrictions :** All the variables considered for making decisions assume non-negative values.

6. **Linearity :** A primary requirement of a L.P.P. is that boh the objective function and all the constraints must be expressed in terms of linear equations and inequations.

   Linearity implies that the products of variables such as xy, $x^2 y$ etc., powers of variables such as x^2, x^3 etc., and combinations of variables of the type $ax + b \log y$ etc. are not allowed.

7. **Additivity :** Additivity means that if it takes x minutes on machine A to make product P and y minutes to make product Q, then the total time on machine A devoted to this production is $x + y$, provided the time required to change the machine from product P to product Q is negligible.

8. **Multiplicativity :** Multiplicativity means that if the cost of production of 1 unit of an item is ₹ 10, then the cost of production of x units of the same item is ₹ $10x$.

   In terms of profit we can say that the profit from selling a given number of units is the profit per unit multiplied by the number of units sold.

9. **Divisibility :** Divisibility means that the fractional levels of variables must be permissible besides integral values.

10. **Finite Number of Constraints :** The number of activities involved in a problem should be finite, thus leading to a finite number of constraints in the problem.

2.4 Mathematical Description of a General Linear Programming Problem

A general linear programming problem can be stated as follows :

Find $x_1, x_2, \ldots, x_n$ which optimize the linear function

$$Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n \qquad \ldots(1)$$

subject to the constraints

$$\left.\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & +\ldots+ & a_{1n}x_n & (\le = \ge) & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & +\ldots+ & a_{2n}x_n & (\le = \ge) & b_2 \\ \cdots & \cdots & \cdots & \cdots\cdots\cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots\cdots\cdots & \cdots & \cdots & \cdots \\ a_{m1}x_1 & + & a_{m2}x_2 & +\ldots+ & a_{mn}x_n & (\le = \ge) & b_m \end{array}\right\} \qquad \ldots(2)$$

and the non-negative restrictions

$$x_1, x_2, \ldots, x_n \ge 0, \qquad \ldots(3)$$

where all $a_{11}, a_{12}, \ldots, a_{mn}; b_1, b_2, \ldots, b_m; c_1, c_2, \ldots, c_n$ are constants and $x_1, x_2, \ldots, x_n$ are variables.

The function Z given in (1) is called the **objective function**, the conditions given in (2) are called the **linear constraints** and the conditions given in (3) are called the **non-negative restrictions** of the linear programming problem.

This linear programming problem may be stated in matrix form as follows :

Find $x_1, x_2, \ldots, x_n$, so as to optimize

$$Z = \mathbf{cx}$$

subject to $A\mathbf{x} (\le = \ge) \mathbf{b}$

and $\mathbf{x} \ge \mathbf{0}$,

where $A = [a_{ij}]_{m \times n}$ is called the **coefficient matrix,**

$\mathbf{c} = [c_1, c_2, \ldots, c_n]$ is a row matrix known as **price vector**,

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ \cdots \\ b_m \end{bmatrix} = [b_1, b_2, \ldots, b_m]^T$$

is a column matrix called the **requirement vector.**

$$\mathbf{x} = \begin{bmatrix} x_2 \\ \cdots \\ \cdots \\ x_n \end{bmatrix} = [x_1, x_2, \ldots, x_n]^T$$

is a column **matrix of variables**

and **0** is a null matrix of the type $n \times 1$.

Note : For the matrix form all constraints in L.P.P. should have the same inequality (*i.e.*, $\geq$ or $\leq$) or equality sign.

2.5 Mathematical Formulation of a Linear Programming Problem

To make use of the linear programming technique it is important to recognize a problem which can be handled by linear programming and then to formulate its mathematical model.

Working Rule to form a Linear Programming Problem : The following steps are helpful in the formulation of a linear programming problem :

1. Identify the variables in L.P.P. and denote them by x_1, x_2, x_3 etc.
2. Identify the objective function and express it as a linear function of the variables x_1, x_2, x_3 etc.
3. Find the type of the objective function. It may be in the form of maximizing profits or minimizing costs.
4. Identify all the constraints and express them as linear inequations or equations.

Illustrative Examples

Example 1: *A goldsmith manufactures necklaces and bracelets. The total number of necklaces and bracelets that he can handle per day is at most 24. It takes one hour to make a bracelet and half an hour to make a necklace. It is assumed that he can work for a maximum of 16 hours a day. Further the profit on a bracelet is ₹ 300 and the profit on a necklace is ₹ 100. Formulate this problem as a linear programming problem so as to maximize the profit.*

Solution: Suppose the goldsmith manufactures x_1 necklaces and x_2 bracelets per day.

Since the profit on a necklace is ₹ 100 and profit on a bracelet is ₹ 300, therefore the total profit Z in ₹ is given by

$$Z = 100x_1 + 300x_2 \qquad \ldots(1)$$

necklaces = $(1/2)x_1$ hours.

Again it takes one hour to make one bracelet, so the time required to make x_2 bracelets

$$= 1.x_2 \text{ hours} = x_2 \text{ hours.}$$

Therefore, total time required to make x_1 necklaces and x_2 bracelets

$$= (x_1/2 + x_2) \text{ hours.} \qquad ...(2)$$

Since total time available per day is 16 hours, therefore

$$x_1/2 + x_2 \leq 16 \quad \text{or} \quad x_1 + 2x_2 \leq 32. \qquad ...(3)$$

The total number of necklaces and bracelets that the goldsmith can manufacture in a day is atmost 24, so we have

$$x_1 + x_2 \leq 24. \qquad ...(4)$$

Also the number of necklaces and bracelets manufactured can never be negative, therefore

$$x_1 \geq 0, x_2 \geq 0. \qquad ...(5)$$

Hence, the linear programming problem formulated from the given problem is as follows:

$$\text{Maximize } Z = 100x_1 + 300x_2$$

subject to the constraints

$$x_1 + 2x_2 \leq 32$$
$$x_1 + x_2 \leq 24$$

and the non-negative restrictions $x_1 \geq 0, x_2 \geq 0$.

Example 2: *A tyre factory produces three types of tyres T_1, T_2, T_3. Three different types of chemicals say C_1, C_2, C_3 are required for production. One T_1 tyre needs 2 units of C_1, 3 units of C_3; one T_2 tyre needs 3 units of C_1, 2 units of C_2 and 2 units of C_3; and one T_3 tyre needs 5 units of C_2 and 4 units of C_3. The factory has only a stock of 20 units of C_1, 25 units of C_2 and 30 units of C_3. Further the profit from the sale of one tyre T_1 is ₹ 6, one tyre T_2 is ₹ 10 and of one tyre T_3 is ₹ 8. Assuming that the factory can sell all that it produces, formulate a linear programming problem to maximize its profit.*

Solution: Let the factory produce x_1 tyres of type T_1, x_2 tyres of type T_2 and x_3 tyres of type T_3.

The give informations can be put in a tabular form as given below :

Chemicals	**Tyre T_1** x_1	**Tyre T_2** x_2	**Tyre T_3** x_3	**Total Chemical Available**
Chemical C_1	2	3	0	20
Chemical C_2	0	2	5	25
Chemical C_3	3	2	4	30
Profit in ₹ (per tyre)	6	10	8	

The total profit Z in ₹ is given by

$$Z = 6x_1 + 10x_2 + 8x_3. \quad \text{...(1)}$$

The total quantity of chemical C_1 required = $(2x_1 + 3x_2)$ units.

Since the factory has a stock of 20 units of chemical C_1, therefore we have

$$2x_1 + 3x_2 \le 20. \quad \text{...(2)}$$

Similarly, considering the total quantity of the chemicals C_2 and C_3 required, we have

$$2x_2 + 5x_3 \le 25 \quad \text{...(3)}$$

and $\quad 3x_1 + 2x_2 + 4x_3 \le 30 \quad$ ...(4)

Also, we have

$$x_1 \ge 0,\ x_2 \ge 0,\ x_3 \ge 0, \quad \text{...(5)}$$

since the factory cannot produce negative quantities.

Hence, the linear programming problem formulated from the given problem is as follows:

$$\text{Maximize } Z = 6x_1 + 10x_2 + 8x_3$$

subject to the constraints

$$2x_1 + 3x_2 \le 20$$

$$2x_2 + 5x_3 \le 25$$

$$3x_1 + 2x_2 + 4x_3 \le 30$$

and the non-negative restrictions

$$x_1 \ge 0,\ x_2 \ge 0,\ x_3 \ge 0.$$

Example 3: *The objective of a diet problem is to ascertain the quantities of certain foods that should be eaten to meet certain nutritional requirement at a minimum cost. The consideration is limited to milk, green vegetables and eggs, and to vitamins A, B, C. The number of milligrams of each of these vitamins contained within a unit of each food and their daily minimum requirements along with the cost of each food is given in the table below :*

Vitamin	**Litre of Milk**	**Kg. of Vegetables**	**Dozen of Eggs**	**Minimum Daily Requirement**
A	1	1	10	1 mg.
B	100	10	10	50 mg.
C	10	100	10	10 mg.
Cost in ₹	20	10	8	

Solution: Let the daily diet consist of x_1 litres of milk, x_2 kgs. of vegetables and x_3 dozens of eggs.

$\therefore$ the total cost Z per day in ₹ is given by

$$Z = 20x_1 + 10x_2 + 8x_3 \quad \text{...(1)}$$

Total amount of vitamin A in the daily diet is

$$(x_1 + x_2 + 10x_3) \text{ mg.,}$$

which should be at least equal to 1 mg., therefore

$$x_1 + x_2 + 10x_3 \geq 1. \quad \text{...(2)}$$

Similarly, considering the total amounts of vitamins B and C in the daily diet, we have

$$100x_1 + 10x_2 + 10x_3 \geq 50 \quad \text{...(3)}$$

and

$$10x_1 + 100x_2 + 10x_3 \geq 10 \quad \text{...(4)}$$

Since the quantities of different food items to be consumed cannot be negative, therefore we have

$$x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0.$$

Hence, the linear programming problem formulated from the given diet problem is :

$$\text{Minimize } Z = 20x_1 + 10x_2 + 8x_3,$$

subject to the constraints

$$x_1 + x_2 + 10x_3 \geq 1$$
$$100x_1 + 10x_2 + 10x_3 \geq 50$$
$$10x_1 + 100x_2 + 10x_3 \geq 10$$

and the non-negative restrictions

$$x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0.$$

Example 4: *According to the medical experts it is necessary for an adult to consume at least 75 gms of proteins, 85 gms of fats and 300 gms of carbohydrates daily. The following table gives the analysis of the food items readily available in the market with their respective costs.*

Food Type	Food value (in gm.) per 100 gms			Cost in ₹ per kg.
	Proteins	Fats	Carbohydrates	
A	18.0	15.0	—	3.0
B	16.0	4.0	7.0	4.0
C	4.0	20.0	2.5	2.0
D	5.0	8.0	40.0	1.5
Minimum daily requirement	75.0	85.0	300.0	

Solution: Let the daily diet consist of x_1 kg. of food A, x_2 kg. of food B, x_3 kg. of food C and x_4 kg. of food D.

Then the total cost per day in ₹ is

$$Z = 3x_1 + 4x_2 + 2x_3 + 1.5x_4. \qquad ...(1)$$

Total amount of proteins in the daily diet is

$$(180x_1 + 160x_2 + 40x_3 + 50x_4)$$

Since the minimum daily requirement of proteins is 75 gms, therefore we have

$$180x_1 + 160x_2 + 40x_3 + 50x_4 \geq 75 \qquad ...(2)$$

Similarly, considering the total amounts of fats and carbohydrates in the diet, we have

$$150x_1 + 40x_2 + 200x_3 + 80x_4 \geq 85 \qquad ...(3)$$

and

$$70x_2 + 25x_3 + 400x_4 \geq 300. \qquad ...(4)$$

Since, the daily diet cannot contain quantities with negative values of any food item, therefore

$$x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0,\ x_4 \geq 0. \qquad ...(5)$$

Hence, the linear programming problem formulated for the given diet problem is :

$$\text{Minimize } Z = 3x_1 + 4x_2 + 2x_3 + 1.5x_4$$

subject to the constraints

$$180x_1 + 160x_2 + 40x_3 + 50x_4 \geq 75$$

$$150x_1 + 40x_2 + 200x_3 + 80x_4 \geq 85$$

$$70x_2 + 25x_3 + 400x_4 \geq 300$$

and the non-negative restrictions

$$x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \text{ and } x_4 \geq 0.$$

Example 5: *A manufacturer of patent medicines is proposed to prepare a production plan for medicines A and B. There are sufficient ingredients available to make 20,000 bottles of medicine A and 40,000 bottles of medicine B, but there are only 45,000 bottles into which either of the medicines can be filled. Further, it takes three hours to prepare enough material to fill 1,000 bottles of medicine A and one hour to prepare enough material to fill 1,000 bottles of medicine B and there are 66 hours available for this operation. The profit is ₹ 8 per bottle for medicine A and ₹ 7 per bottle for medicine B. Formulate this problem as a L.P.P..*

Solution: Suppose, the manufacturer produces x_1 thousand bottles of medicine A and x_2 thousand bottles of medicine B.

Since there are only 45,000 bottles available for filling the medicines A and B, therefore we have $x_1 + x_2 \leq 45$.

Therefore, the time required to fill x_1 thousand bottles of medicine A is $3x_1$ hours.

Similarly, the time required to fill x_2 thousand bottles of medicine B is x_2 hours.

Thus, the total time required to fill x_1 thousand bottles of medicine A and x_2 thousand bottles of medicine B is $(3x_1 + x_2)$ hours.

The total time available for this operation is 66 hours. So, we have $3x_1 + x_2 \le 66$.

There are sufficient ingredients available to make 20 thousand bottles of medicine A and 40 thousand bottles of medicine B, therefore we have $x_1 \le 20,\ x_2 \le 40$.

Since the number of bottles cannot be negative, therefore we have $x_1 \ge 0,\ x_2 \ge 0$.

The profit per bottle for medicine A is ₹ 8 and for medicine B is ₹ 7. So, the total profit in ₹ on x_1 thousand bottles of medicine A and x_2 thousand bottles of medicine B is given by

$$Z = 8 \times 1{,}000x_1 + 7 \times 1{,}000x_2 = 8{,}000x_1 + 7{,}000x_2.$$

Hence, the linear programming problem formulated for the given problem is

$$\text{Maximize } Z = 8{,}000x_1 + 7{,}000x_2,$$

subject to the constraints :

$$x_1 + x_2 \le 45$$
$$3x_1 + x_2 \le 66$$
$$x_1 \le 20$$
$$x_2 \le 40$$

and the non-negative restrictions $x_1 \ge 0,\ x_2 \ge 0$.

Example 6: *A firm manufactures two types of products A and B and sells them at a profit of ₹ 2 on type A and ₹ 3 on type B. Each product is processed on two machines E and F. Type A requires one minute of processing time on E and two minutes on F, type B required one minute on E and one minute on F. The machine E is available for not more than 6 hours 40 minutes while machine F is available for 10 hours during any working day. Formulate the problem as a linear programming problem.* **(Garhwal 2012)**

Solution: Let the manufacturer produce x_1 units of the product of type A and x_2 units of the product of type B.

The given information can be systematically arranged in the form of the following table :

	Type A (x_1 units)	Type B (x_2 units)	(minutes)
E	1	1	400
F	2	1	600
Profit per unit	₹ 2	₹ 3	

Since the machine E takes 1 minute time for processing a unit of type A and 1 minute time for processing a unit of type B, therefore the total time required on machine E is $(x_1 + x_2)$ minutes.

But the machine E is available for not more than 6 hours and 40 minutes = 400 minutes, therefore we have

$$x_1 + x_2 \leq 400.$$

Similarly, the total time (in minutes) required on machine F is $2x_1 + x_2$.

Since the machine F is available for not more than 10 hours = 600 minutes, therefore we have

$$2x_1 + x_2 \leq 600.$$

Since it is not possible to produce negative quantities, so we have $x_1 \geq 0,\ x_2 \geq 0.$

The profit on type A is ₹ 2 per unit so the profit on selling x_1 units of type A will be ₹ $2x_1$. Similarly, the profit on selling x_2 units of type B will be ₹ $3x_2$.

Therefore, the total profit (in ₹) on selling x_1 units of type A and x_2 units of type B is

$$Z = 2x_1 + 3x_2.$$

Hence, the required linear programming problem formulated for the given problem is

$$\text{Maximize } Z = 2x_1 + 3x_2$$

subject to the conditions

$$x_1 + x_2 \leq 400$$

$$2x_1 + x_2 \leq 600$$

$$x_1 \geq 0,\ x_2 \geq 0.$$

Example 7: *A manufacturer produces three models (I, II, and III) of a certain product. He uses two types of raw materials (A and B) of which 4,000 and 6,000 units respectively are available. The raw material requirements per unit of the three models are given below :*

Raw Materials			
	I	II	III
A	2	3	5
B	4	2	7

The labour time for each unit of model I is twice that of model II and three times that of model III. The entire labour force of the factory can produce the equivalent of 2,500 units of model I. A market survey indicates that the minimum demand of the three models are 500, 500 and 375 units respectively. However, the ratio of the number of units produced must be equal to 3:2:5. Assume that the profits per unit of models I, II and III are ₹ 60, 40 and 100 respectively. Formulate the problem as a L.P.P. in order to determine the number of units of each product which will maximize profit.

Solution: Let the manufacturer produce x_1, x_2, x_3 units of models I, II and III respectively. Since the profit per unit on model I, II and III is ₹ 60, ₹ 40 and ₹ 100 respectively, the objective function is to maximize the profit :

$$Z = 60x_1 + 40x_2 + 100x_3. \qquad ...(1)$$

The raw materials used and available give rise to the following constraints respectively.

$$2x_1 + 3x_2 + 5x_3 \leq 4{,}000 \qquad ...(2)$$

$$4x_1 + 2x_2 + 7x_3 \leq 6{,}000 \qquad ...(3)$$

Now if t be the labour time required for one unit of model I, then the time required for one unit of model II will be $\frac{1}{2}t$ and that for the model III will be $\frac{1}{3}t$. As the factory can produce 2,500 units of model I, so the restriction on the production time will be

$$t\,x_1 + \frac{1}{2}t\,x_2 + \frac{1}{3}t\,x_3 \leq 2{,}500\,t \text{ i.e., } 6x_1 + 3x_2 + 2x_3 \leq 15{,}000. \qquad ...(4)$$

Now according to the market demand, we must have

$$x_1 \geq 500,\ x_2 \geq 500,\ x_3 \geq 375. \qquad ...(5)$$

Further the ratio of the number of units of different types of models is 3:2:5 i.e.,

$$x_1 = 3k,\ x_2 = 2k,\ x_3 = 5k.$$

These give rise to the constraints

3 2

and $$\frac{1}{2}x_2 = \frac{1}{5}x_3 \qquad \text{...(6)}$$

Thus, the linear programming problem is as follows :

$$\text{Maxmize } Z = 60\,x_1 + 40x_2 + 100x_3$$

subject to the conditions :

$$2x_1 + 3x_2 + 5x_3 \le 4{,}000$$

$$4x_1 + 2x_2 + 7x_3 \le 6{,}000$$

$$6x_1 + 3x_2 + 2x_3 \le 15{,}000$$

$$2x_1 = 3\,x_2\,,\; 5x_2 = 2x_3$$

and $$x_1 \ge 500,\; x_2 \ge 500,\; x_3 \ge 375.$$

Note : x_1, x_2, x_3 will automatically be non-negative due to (5). So we need not to mention that $x_1, x_2, x_3 \ge 0$. Even if we write then these will be redundant constraints.

Example 8: *A complete unit of a certain product consists of four units of component A and three units of component B. The two components (A and B) are manufactured from two different raw materials of which 100 units and 200 units respectively are available. Three departments are engaged in the production process with each department using a different method for manufacturing the components. The following table gives the raw material requirements per production run and the resulting units of each component. The objective is to determine the number of production runs for each department which will maximize the total number of component units of the final product:*

Department	**Input per run (units)**		**Output per run (units)**	
	Raw Material		**Components**	
	I	**II**	***A***	***B***
1	7	5	6	4
2	4	8	5	8
3	2	7	7	3

Formulate a linear programming model to the above problem.

Solution: Let x_1, x_2, x_3 be the number of production runs for the departments 1, 2, 3 respectively.

The raw materials available give the restrictions

$$7x_1 + 4x_2 + 2x_2 \leq 100 \qquad \text{...(1)}$$

$$5x_1 + 8x_2 + 7x_3 \leq 200 \qquad \text{...(2)}$$

The total number of units produced by three departments :

$$6x_1 + 5x_2 + 7x_3 \text{ (component } A\text{)}$$

and $$4x_1 + 8x_2 + 3x_3 \text{ (component } B\text{)}.$$

Now final product requires 4 units of components A and 3 units of component B. So the maximum number of units of the final product cannot exceed the smaller value of

$$\frac{1}{4}(6x_1 + 5x_2 + 7x_3) \text{ and } \frac{1}{3}(4x_1 + 8x_2 + 3x_3).$$

Thus the objective function becomes

$$\text{Max. } Z = \min\left[\frac{1}{4}(6x_1 + 5x_2 + 7x_3), \frac{1}{3}(4x_1 + 8x_2 + 3x_3)\right]$$

Since the objective function is not linear, a suitable transformation can be used to convert this into a L.P.P.

Let, $$\min.\left[\frac{1}{4}(6x_1 + 5x_2 + 7x_3), \frac{1}{3}(4x_1 + 8x_2 + 3x_3)\right] = v.$$

Then $$\frac{1}{4}(6x_1 + 5x_2 + 7x_3) \geq v$$

and $$\frac{1}{3}(4x_1 + 8x_2 + 3x_3) \geq v.$$

Thus we get the L.P.P. as follows :

Maximize $Z = v$,

subject to the constraints :

$$\begin{aligned} 6x_1 + 5x_2 + 7x_3 - 4v &\geq 0 \\ 4x_1 + 8x_2 + 3x_3 - 3v &\geq 0 \\ 7x_1 + 4x_2 + 2x_3 &\leq 100 \\ 5x_1 + 8x_2 + 7x_3 &\leq 200 \end{aligned}$$

and the non-negative restrictions $x_1, x_2, x_3, v \geq 0$.

Example 9: *The owner of the Metro Sports wishes to determine how many advertisements to place in the selected three monthly magazines A, B and C. His objective is to advertise in such a way that total exposure to principal buyers of expensive sports good is maximized. Percentages of readers for each magazine are known. Exposure in any particular magazine is the number of advertisements placed multiplied by the number of principal buyers. The following data may be used :*

Exposure Category	**Magazines**		
	A	**B**	**C**
Readers (in Lakhs)	1	0.6	0.4
Principal Buyers	10%	15%	7%
Cost per advertisement in (₹)	5,000	4,500	4,250

The budgeted amount is at most ₹ 1 lakh for the advertisement. The owner has already decided that magazine A should have no more than 6 advertisement and that B and C each have at least two advertisements. Formulate a LP model for the problem.

Solution: Let x_1, x_2, x_3 be the number of advertisement in magazines *A, B* and *C* respectively.

Then the *LP* formulation will be as follows :

Maximize

$$Z = (10\% \text{ of } 1{,}00{,}000)\, x_1 + (15\% \text{ of } 60{,}000)\, x_2 + (7\% \text{ of } 40{,}000)\, x_3$$

or Maximize $Z = 10{,}000\, x_1 + 9{,}000\, x_2 + 2{,}800\, x_3$

subject the constraints : $5{,}000 x_1 + 4{,}500 x_2 + 4{,}250 x_3 \le 1{,}00{,}000$

$$x_1 \le 6,\ x_2 \ge 2,\ x_3 \ge 2$$

and the non-negativity restrictions $x_1, x_2, x_3 \ge 0$

Example 10: *A manufacturer of biscuits is considering four types of gift packs containing three types of biscuits : orange cream (OC), chocolate cream (CC), and wafers (W). Market research conducted recently to assess the preferences of the consumers shows the following types of assortments to be in good demand :*

Assortments	**Contents**	**Selling price per kg (Rs.)**
A	Not less than 40% of *OC* Not more than 20% of *CC* Any quantity of *W*	20
B	Not less than 20% of *OC* Not more than 40% of *CC* Any quantity of *W*	25

C	Not less than 50% of *OC* Not more than 10% of *CC* Any quantity of *W*	22
D	No restrictions	12

For the biscuits, the manufacturing capacity and costs are given below :

Biscuits variety	***OC***	***CC***	***W***
Plant capacity (kg/day)	200	200	150
Manufacturing cost (Rs./kg.)	8	9	7

Formulate a linear programming model to find the production schedule which maximizes the profit assuming that there are no market restrictions.

Solution: Let x_{ij} kg. be the quantity used in i-th assortment of the j-th type of biscuits, where $i = A, B, C, D$ and J = *OC*, *CC*, *W* (numbers 1, 2, 3 respectively).

Then in assortment *A*, x_{A1}, x_{A2}, x_{A3} denote the quantity in kg. of *OC*, *CC* and *W* type of biscuits respectively. Similarly we can interpret for other assortments.

Now the given data can be put in the form of L.P.P. as follows :

$$\begin{aligned}\text{Maximize } Z = {} & 20(x_{A1}+x_{A2}+x_{A3})+25(x_{B1}+x_{B2}+x_{B3}) \\ & +22(x_{C1}+x_{C2}+x_{C3})+12(x_{D1}+x_{D2}+x_{D3}) \\ & -8(x_{A1}+x_{B1}+x_{C1}+x_{D1})-9(x_{A2}+x_{B2}+x_{C2}+x_{D2}) \\ & -7(x_{A3}+x_{B3}+x_{C3}+x_{D3})\end{aligned}$$

or
$$\begin{aligned}\text{Maximize } Z = {} & 12x_{A1}+11x_{A2}+13x_{A3}+17x_{B1}+16x_{B2}+18x_{B3} \\ & +14x_{C1}+13x_{C2}+15x_{C3}+4x_{D1}+3x_{D2}+5x_{D3}\end{aligned}$$

Subject to the constraints :

$$\left.\begin{aligned} x_{A1} &\geq 0.40(x_{A1}+x_{A2}+x_{A3}) \\ x_{A2} &\leq 0.20(x_{A1}+x_{A2}+x_{A3}) \end{aligned}\right\} \text{gift pack } A$$

$$\left.\begin{aligned} x_{B1} &\geq 0.20(x_{B1}+x_{B2}+x_{B3}) \\ x_{B2} &\leq 0.40(x_{B1}+x_{B2}+x_{B3}) \end{aligned}\right\} \text{gift pack } B$$

$$\left.\begin{aligned} x_{C1} &\geq 0.50(x_{C1}+x_{C2}+x_{C3}) \\ x_{C2} &\leq 0.10(x_{C1}+x_{C2}+x_{C3}) \end{aligned}\right\} \text{gift pack } C$$

Plant capacity constraints are : $x_{A1}+x_{B1}+x_{C1}+x_{D1} \leq 200$

$$x_{A2} + x_{B2} + x_{C2} + x_{D2} \le 200$$

$$x_{A3} + x_{B3} + x_{C3} + x_{D3} \le 150$$

and non-negativity restrictions are $x_{ij} \ge 0\,(i = A, B, C, D; j = 1, 2, 3)$.

Example 11: *A company has two grades of inspectors, I and II, who are to be assigned for a quality control inspection. It is required that at least 2,000 pieces be inspected per 8 hour day. Grade I inspectors can check pieces at the rate of 50/hour with the accuracy of 97% grade II inspectors can check prices at the rate of 40/hour with the accuracy 95%. The wage rate of Grade I inspector is ₹ 4.50/hour and that of Grade II is ₹ 2.50/hour. Each time an error is made by an inspector, the cost to the company is one rupee. The company has available for the inspection job, 10 grade I and 5 grade II inspectors. Formulate the problem to minimize the total cost of inspection.*

Solution: Let the no. of inspectors of grade I and II be x_1 and x_2 respectively. They will inspect $(8 \times 50)x_1 + (8 \times 40)x_2$ pieces daily. As the company requires at least 2,000 items to be inspected daily, so the constraint is

$$(8 \times 50)x_1 + (8 \times 40)x_2 \ge 2000 \quad \textit{i.e.,} \quad 5x_1 + 4x_2 \ge 25. \qquad ...(1)$$

The limitations on the available numbers of inspectors give the constraints

$$x_1 \le 10,\ x_2 \le 5 \qquad ...(2)$$

Further the company is bearing two types of costs, the wages of inspectors and the costs of inspection errors.

The cost of each inspector per hour is as follows :

Grade I : $$₹\left[4.50 + 1 \times \left(\frac{3}{100} \times 50\right)\right] = ₹\ 6.00 \text{ / hour}$$

Grade II : $$₹\left[2.50 + 1 \times \left(\frac{5}{100} \times 40\right)\right] = ₹\ 4.50 \text{ / hour}$$

Now the objective of the company is to minimize the total cost of inspection at the above rates for each day.

Thus the objective function is

$$\text{Minimize } Z = 8 \times 6.00 \times x_1 + 8 \times 4.50 \times x_2 = 48x_1 + 36x_2 \qquad ...(3)$$

Hence the required L.P.P. is

$$\text{Minimize } Z = 48x_1 + 36x_2$$

subject to the constraints : $5x_1 + 4x_2 \ge 25$

$$x_1 \le 10,\ x_2 \le 5$$

and the non-negativity restrictions $x_1, x_2 \ge 0$.

Comprehensive Exercise 1

1. A furniture dealer deals in two items, *viz.,* tables and chairs. He has ₹ 10,000 to invest and a space to store at most 60 pieces (including both tables and chairs). A table costs him ₹ 500 and a chair ₹ 100. He can sell all the items that he buys, earning a profit of ₹ 50 for each table and ₹ 15 for each chair. Formulate this problem as a LPP so that he maximizes the profit.

2. A factory produces two products A and B. Each of the product A requires 2 hrs of moulding, 3 hrs of grinding and 4 hrs for polishing and each of the product B requires 4 hrs for moulding, 2 hrs for grinding and 2 hrs for polishing. The moulding machine can work for 20 hrs, grinding machine for 24 hrs and polishing machine available for 13 hrs. The profit is ₹ 5 per unit of A and ₹ 3 per unit of B. Assuming that the factory can sell all that it produces, formulate the problem as a LPP to maximize the profit.

3. A dietician decides a certain minimum intake of vitamins A, B and C for a family. The minimum daily needs of the vitamins A, B, C are 30, 20, 16 units respectively. For the supply of these, the dietician depends on two types of foods X and Y. The first one gives 7, 5, 2 units per gram of vitamins A, B, C respectively. The second one gives, 2, 4, 8 units per gram of these vitamins respectively. The first food costs ₹ 2 per gram and the second ₹ 1 per gram.

   How many grams of each food stuff should the family buy every day to keep the food expense at a minimum? Formulate a linear programming problem for this problem.

4. A firm can produce two products A and B during a given period of time. Each of these products requires four different operations, *viz.,* Grinding, Turning, Assembling and Testing. The requirement in hours per unit of manufacturing of these products is as given :

	Grinding	**Turning**	**Assembling**	**Testing**
A	1	3	4	5
B	2	1	3	4

   The available capacities of these operations in hours for the given time period are : 30 for grinding, 60 for turning, 200 for assembling and 200 for testing.

   Profit on each unit of A is ₹ 3 and that for each unit of B is ₹ 2. Formulate the problem as a linear programming model to maximize the profit assuming that the firm can sell all the items that it produces at the prevailing market price.

5. A toy company manufactures two types to dolls; an ordinary doll A and a deluxe doll B. Each type of doll B takes twice as long to produce as one of

type A. It is given that the company would have time to make a maximum of 2000 dolls per day if it produces only the ordinary version. The supply of plastic is sufficient to produce 1500 dolls per day (both A and B combined). The deluxe version requires a fancy dress of which there are only 600 pieces per day available. If the company makes profit of ₹ 3 and ₹ 5 per doll respectively on doll A and doll B, formulate the problem as a linear programming problem to maximize the profit.

6. A furniture firm manufactures chairs and tables, each requiring the use of three machines A, B and C. Production of one chair requires 2 hours on machine A, 1 hour on machine B and 1 hour on machine C. Each table requires 1 hour each on machines A and B and 3 hours on machine C. The profit realized by selling one chair is ₹ 30 while for a table the figure is ₹ 60. The total time available per week on machine A is 70 hours, on machine B is 40 hours, and on machine C is 90 hours. Formulate the linear programming problem to maximize the profit.

7. A diet is to contain at least 4000 units of carbohydrates, 500 units of fat and 300 unit of protein. Two foods A and B are available. Food A costs ₹ 2 per unit and food B costs ₹ 4 per unit. A unit of food A contains 10 units of carbohydrates, 20 units of fat and 15 units of protein. A unit of food B contains 25 units of carbohydrates, 10 units of fat and 20 units of protein. Formulate the problem as a LPP so as to find the minimized cost for a diet that consists of a mixture of these two foods and also meets the minimum nutrition requirements.

8. A resourceful home decorator manufactures two types of lamps say A and B. Both lamps go through two technicians, first a cutter, second a finisher. Lamp A requires 2 hours of the cutter's time and 1 hour of the finisher's time. Lamp B requires 1 hour of cutter's and 2 hours of finisher's time. The cutter has 104 hours and finisher has 76 hours of time available each month. Profit on the lamp A is ₹ 6 and on the lamp B is ₹ 11. Assuming that he can sell all that he produces, how many of each type of lamps should he manufacture per month to obtain the best returns?
Formulate a LPP for this problem.

9. The manager of an oil refinery must decide on the optimal mix of 2 possible blending processes of which the inputs and outputs per production run are as follows :

Process	Input		Output	
	Crude A	Crude B	Gasoline X	Gasoline Y
1	6	4	6	9
2	5	6	5	5

The maximum amounts available of crudes A and B are 500 units and 400 units respectively. Market demand shows that at least 300 units of gasoline

X and 260 units of gasoline Y must be produced. The profits per production run from process 1 and 2 are ₹. 40 and ₹ 50 respectively. Formulate the LPP for maximizing the profit.

10. A factory produces two products A and B. To manufacture one unit of product A, a machine has to work for $1\frac{1}{2}$ hours and a craftsman has to work for 2 hours. To manufacture one unit of product B, the machine has to work for 3 hours and the craftsman for one hour. In a week the factory can avail of 80 hours of machine time and 70 hours of craftsman's time. The profit on the sale of each unit of A and B is of ₹ 10 and ₹ 8 respectively. If the manufacturer can sell all the items produced, how many of each should be produced to get the maximum profit per week ?

Formulate the problem as a LPP.

11. A company manufactures two kinds of leather purses, A and B. A is a high quality purse and B is lower quality. The sales of each of these purses A and B earn profit of ₹ 4 and ₹ 3 respectively. Each purse of type A requires twice as much time as a purse of type B, and if all purses are to type B, the company could make 1000 purses per day. The supply of leather is sufficient for only 800 purses per day (both A and B combined). Purse A requires a fancy buckle, and only 400 buckles per day are available. There are only 700 buckles available for purse B. What should be the daily production of each type of purse to get the maximum profit ? Formulate the problem as a LPP.

12. A firm can produce three types of cloth, say : A, B and C. Three kinds of wool are required for it, say red wool, green wool and blue wool. One unit length of type A cloth needs 2 yards of red wool and 3 yards of blue wool; one unit length of type B cloth needs 3 yards of red wool, 2 yards of green wool and 2 yards of blue wool and one unit length of type C cloth needs 5 yards of green wool and 4 yards of blue wool. The firm has only a stock of 8 yards of red wool, 10 yards of green wool and 15 yards of blue wool. It is assumed that the income obtained from one unit length of type A cloth is ₹ 3.00, of type B cloth is ₹ 5.00 and of type C cloth is ₹ 4.00. Formulate this problem as a linear programming model to maximize the income from the finished cloth.

13. A firm manufactures 3 products A, B and C. The profits are ₹ 3, ₹ 2 and ₹ 4 respectively . The firm has 2 machines and below is the required processing time in minutes for each machine on each product :

Machine	**Product**		
	A	***B***	***C***
G	4	3	5
H	2	2	4

Machines G and H have 2,000 and 2,500 machines minutes, respectively. The firm must manufacture 100 A's, 200 B's and 50 C's but not more than 150 A's. Set up a linear programming problem to maximize profit.

(Garhwal 2006)

14. A farmer has 100 acre farm. He can sell all tomatoes, lettuce or radishes he can raise. The price he can obtain is ₹ 1.00 per kg for tomatoes, ₹ 0.75 a head for lettuce and ₹ 2.00 per kg for radishes. The average yield per-acre is 2,000 kg of tomatoes, 3,000 heads of lettuce and 1,000 kg of radishes. Fertilizer is available at ₹ 0.50 per kg and the amount required per acre is 100 kg each for tomatoes and lettuce and 50 kg for radishes. Labour required for sowing, cultivating and harvesting per acre is 5 man-days for tomatoes and radishes, and 6 man-days for lettuce. A total of 400 man-days of labour are available at ₹ 20.00 per man-day. Formulate this problem as a linear programming model to maximize the farmer's total profit.

15. A city hospital has the following minimal daily requirements for nurses :

Period	**Clock time (24 hr. day)**	**Minimum number of nurse required**
1	6 A.M. — 10 A.M.	2
2	10 A.M. — 2 P.M.	7
3	2 P.M. — 6 P.M.	15
4	6 P.M. — 10 P.M.	8
5	10 P.M. — 2 A.M.	20
6	2 A.M. — 6 A.M.	6

Nurses report to the hospital at the beginning of each period and work for 8 consecutive hours. The hospital wants to determine the minimum number of nurses available for each period. Formulate this as a L.P.P. by setting up appropriate constraints and objective function.

Answers 1

1. Maximize $Z = 50x + 15y$,
subject to the constraints
$$5x + y \le 100$$
$$x + y \le 60$$
and the non-negative restrictions
$x \ge 0, y \ge 0$

2. Maximize $Z = 5x + 3y$,
subject to the constraints
$$2x + 4y \le 20$$
$$3x + 2y \le 24$$
$$4x + 2y \le 13$$
and $x \ge 0, y \ge 0$

3. Minimize $Z = 2x + y$,
subject to the constraints
$$7x + 2y \ge 30$$
$$5x + 4y \ge 20$$
$$2x + 8y \ge 16$$
and $x \ge 0, y \ge 0$

4. Maximize $Z = 3x + 2y$,
subject to the constraints
$$x + 2y \ge 30$$
$$3x + y \le 60$$
$$4x + 3y \le 200$$
$$5x + 4y \le 200$$
and $x \ge 0, y \ge 0$

5. Maximize $Z = 3x + 5y$,
subject to the constraints
$$x + 2y \le 2000$$
$$x + y \le 1500$$
$$y \le 600$$
and $x \ge 0, y \ge 0$

6. Maximize $Z = 30x + 60y$,
subject to the constraints
$$2x + y \le 70$$
$$x + y \le 40$$
$$x + 3y \le 90$$
and $x \ge 0, y \ge 0$

7. Minimize $Z = 2x + 4y$,
subject to the constraints
$$10x + 25y \ge 4000$$
$$20x + 10y \ge 500$$
$$15x + 20y \ge 300$$
and $x, y \ge 0$

8. Maximize $Z = 6x + 11y$,
subject to the constraints
$$2x + y \le 104$$
$$x + 2y \ge 76$$
and $x \ge 0, y \ge 0$

9. Maximize $Z = 40x + 50y$,
subject to the constraints
$$6x + 5y \le 500$$
$$4x + 6y \le 400$$
$$6x + 5y \ge 300$$
$$9x + 5y \ge 260$$
and $x, y \ge 0$

10. Maximize $Z = 10x + 8y$,
subject to the constraints
$$1.5x + 3y \le 80$$
$$2x + y \le 70$$
and $x \ge 0, y \ge 0$

11. Maximize $Z = 4x + 3y$,
subject to the constraints
$$2x + y \le 1000$$
$$x + y \le 800$$
$$x \le 400$$
$$y \le 700$$
and $x \ge 0, y \ge 0$

12. Maximize $Z = 3x_1 + 5x_2 + 4x_3$,
subject to the constraints
$$2x_1 + 3x_2 \le 8$$
$$2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$
and $x_1, x_2, x_3 \ge 0$

13. Maximize $Z = 3x_1 + 2x_2 + 4x_3$,
subject to the constraints
$$4x_1 + 3x_2 + 5x_3 \le 2000$$
$$2x_1 + 2x_2 + 4x_3 \le 2500$$
$$100 \le x_1 \le 150, x_2 \ge 200, x_3 \ge 50.$$

14. Maximize
$Z = 1850x_1 + 2080x_2 + 1875x_3$,
subject to the constraints
$$x_1 + x_2 + x_3 \le 100$$
$$5x_1 + 6x_2 + 5x_3 \le 400$$
and $x_1, x_2, x_3 \ge 0.$

15. Maximize $Z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$,
subject to the constraints
$$x_1 + x_2 \ge 7,\ x_2 + x_3 \ge 15,$$
$$x_3 + x_4 \ge 8,\ x_4 + x_5 \ge 20,$$
$$x_5 + x_6 \ge 6,\ x_6 + x_1 \ge 2$$
and $x_1, x_2, x_3, x_4, x_5, x_6 \ge 0.$

2.6 Some Important Definitions

1. **Solution of a L.P.P. :** A set of values of the variables $x_1, x_2, \ldots, x_n$ satisfying the constraints of a L.P.P. is called **a solution** of the L.P.P.
2. **Feasible Solution of a L.P.P. :** A set of values of the variables $x_1, x_2, \ldots, x_n$ satisfying the constraints and the non-negative restrictions of a L.P.P. is called **a feasible solution** of the L.P.P.
3. **Optimal (or optimum) Solution of a L.P.P. :** A feasible solution of a L.P.P. is said to be **optimal** (or **optimum**) if it also optimizes (*i.e.*, maximizes or minimizes as the case may be) the objective function of the problem.
4. **Unbounded Solution :** If the value of the objective function can be increased or decreased indefinitely, such solutions are called **unbounded solutions.** In this case we say that the problem has unbounded solution.
5. **Fundamental Extreme Point Theorem :** An optimum solution of a L.P.P., if it exists, occurs at one of the extreme points (*i.e.,* corner points) of the convex polygon of the set of all feasible solutions.

2.7 Solution of Simultaneous Linear Inequations

The **graph** or the **solution set of a system of simultaneous linear inequations** is the region containing the points (x, y) which satisfy all the inequations of the given system simultaneously.

To draw the graph of the simultaneous linear inequations *i.e.*, to find the solution set of the simultaneous linear inequations, we find the region of the *xy*-plane, common to all the portions comprising the solution sets of the given inequations. If there is no region common to all the solutions of the given inequations, we say that the solution set of the system of ineqations is empty. It should be noted that the solution set of simultaneous linear inequations may be an empty set or it may be the region bounded by the straight lines corresponding to given linear inequations or it may be an unbounded region with straight line boundaries.

Illustrative Examples

Example 12: *Draw the graph of the solution set of the ineqautions*

$$x + y \leq 5,\ 4x + y \geq 4,\ x + 5y \geq 5,\ x \leq 4,\ y \leq 3.$$

Solution: Consider the equations

$$x + y = 5,\ 4x + y = 4,\ x + 5y = 5,\ x = 4,\ y = 3.$$

First we shall consider each inequation separately.

Region Represented by $x + y \leq 5$: The straight line $x + y = 5$ meets the x-axis at $(5, 0)$ and y-axis at $(0, 5)$. Since the given inequality is not strict, we join these points by a thick line. Clearly, the point (0,0) not lying on the line $x + y = 5$ satisfies the inequation $x + y \leq 5$. Therefore, out of the two portions of the xy-plane divided by the line $x + y = 5$, the one containing the origin (0, 0) along with the line represents the solution set of the inequation $x + y \leq 5$.

Region Represented by $4x + y \geq 4$: The straight line $4x + y = 4$ meets x-axis at $(1, 0)$ and y-axis at $(0, 4)$. Since the given inequality is not strict, we join these point by a thick line. Clearly, the point (0, 0) not lying on the line $4x + y = 4$ does not satisfy the ineqations $4x + y \geq 4$. Therefore, out of the two portions of the xy-plane divided by the line $4x + y = 4$, the portion not containing the origin along with the line represents the solution set of the inequations $4x + y \geq 4$.

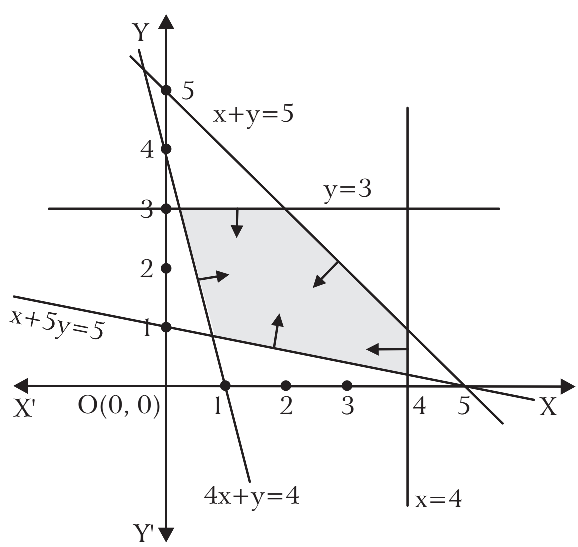

Fig. 2.1

Region Represented by $x + 5y \geq 5$: The straight line $x + 5y = 5$ meets the x-axis at $(5, 0)$ and the y-axis at $(0, 1)$. Since the given inequality is not strict, we join these points by a thick line. Clearly, the point $(0, 0)$ not lying on the line $x + 5y = 5$ does not satisfy the inequation $x + 5y \geq 5$. Therefore, out of the two portions of the xy-plane divided by the line $x + 5y = 5$, the portion not containing the origin along with the line represents the solution set of the inequation $x + 5y \geq 5$.

Region Represented by $x \leq 4$: Clearly, the line $x = 4$ is parallel to y-axis at a distance of 4 unit from the origin lying to the right hand side of the y-axis. Since the given inequality is not strict, we draw this line as a thick line. The point $(0, 0)$ not lying on the line $x = 4$ satisfies the inequation $x \leq 4$, therefore out of the two portions of the xy-plane divided by this line, the portion containing the origin along with the line represents the solution set of the inequation $x \le 4$.

Region Represented by $y \le 3$: Clearly, the line $y = 3$ is parallel to x-axis at a distance of 3 units from the origin lying above x-axis. Since the inequality is not strict, we draw this line as a thick line. The point (0, 0) not lying on the line $y = 3$ satisfies the inequation $y \le 3$, therefore out of the two portions of the xy-plane divided by this line, the portion containing the origin along with the line represents the solution set of the inequation $y \le 3$.

Hence, the shaded region *i.e.,* the region common to the given five inequations, represents the solution set of the given system of inequations.

Example 13: ***Exhibit graphically the solution set of the linear inequations*** $\boldsymbol{3x + 2y \ge 6, x \ge 1, y \ge 1}$.

Solution: Consider the equations $3x + 2y = 6$, $x = 1$, $y = 1$.

To find the solution set of the given system of inequations, we first find the solution sets of the given distinct inequations separately.

Region Represented by $3x + 2y \ge 6$: The straight line $3x + 2y = 6$ meets x-axis at (2, 0)and y-axis at (0, 3). Since the given inequality is not strict, we join these points by a thick line. Clearly, the point (0, 0) not lying on the line $3x + 2y = 6$ does not satisfy the inequation $3x + 2y \ge 6$.

Therefore out of the portions of the xy-plane divided by the line $3x + 2y = 6$, the portion not containing the origin along with the line represents the solution set of the inequation $3x + 2y \ge 6$.

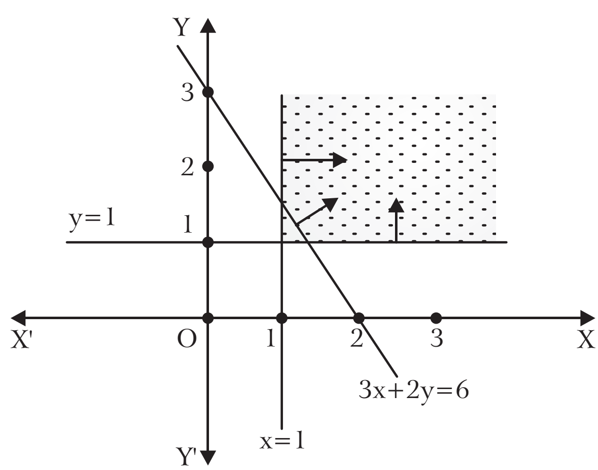

Fig. 2.2

Region Represented by $x \ge 1$: Clearly, the line $x = 1$ is parallel to y-axis at a distance of 1 unit from the origin lying to the right hand side of y-axis. Since the given inequality is not strict, we draw this line as a thick line. The point (0, 0) not lying on the line $x = 1$ does not satisfy the inequation $x \ge 1$, therefore out of the two portions of the xy-plane divided by this line, the portion not containing the origin along with the line represents the solution set of the inequation $x \ge 1$.

Region Represented by $y \geq 1$: Clearly, the line $y = 1$ is parallel to x-axis at a distance of 1 unit from the origin lying above x-axis. Since the given inequality is not strict, we draw this line as a thick line. The point (0, 0) not lying on the line $y = 1$ does not satisfy the inequation $y \geq 1$, therefore out of the two portions of the xy-plane divided by this line, the portions not containing the origin along with the line represents the solution set of the inequation $y \geq 1$.

Hence, the shaded region common to the given three inequations represents the solution set of the given system of inequations. We observe that the solution set of the given system of linear inequations is an **unbounded region.**

Example 14: *Draw the diagram of the solution set of the system of linear inequations :*

$$\mathbf{2x + 3y \leq 6, x + 4y \leq 4, x \geq 0, y \geq 0}.$$

Solution: Changing the given inequations into equations, we get

$$2x + 3y = 6, x + 4y = 4, x = 0, y = 0.$$

Region Represented by $2x + 3y \leq 6$: The line $2x + 3y = 6$ meets x-axis at (3, 0) and y-axis at (0, 2). Join the points (3, 0) and (0, 2) by a thick line to obtain the graph of the line $2x + 3y = 6$. Since the point (0, 0) satisfies the inequation $2x + 3y \leq 6$, therefore the portion of the xy-plane divided by the line $2x + 3y = 6$ containing the origin along with the line represents the solution set of the inequation $2x + 3y \leq 6$.

Region represented by $x + 4y \leq 4$: The line $x + 4y = 4$ meets x-axis at (4, 0) and y-axis at (0, 1). Join the points (4, 0) and (0, 1) by a thick line to obtain the graph of the line $x + 4y = 4$. The point (0, 0) does not lie on the line $x + 4y = 4$. Since (0, 0) satisfies the inequation $x + 4y \leq 4$, therefore the portion of the xy-plane divided by the line $x + 4y = 4$ containing the origin along with the line represents the solution set of the inequation $x + 4y \leq 4$.

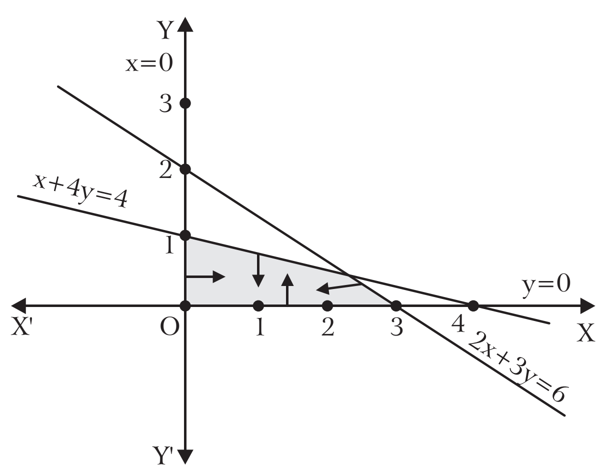

Fig. 2.3

Region Represented by $x \geq 0$: The inequation $x \geq 0$ represents the region lying on the right hand side of the y-axis including y-axis.

Region Represented by $y \geq 0$: The inequation $y \geq 0$ represents the region lying above the x-axis.

The common region (shaded region in the figure) of the above four regions represents the solution set of the given system of inequations.

Example 15: *Exhibit graphically the solution set of the following system of linear inequations :*

$$x+y\ge 1,\ -3x-4y\ge -12,\ -x+2y\ge -2,\ x\ge 0,\ y\ge 0.$$

Solution: **Region Represented by $x+y\ge 1$:** The straight line $x+y=1$ meets x-axis at $(1, 0)$ and y-axis at $(0, 1)$. We join the points $(1, 0)$ and $(0, 1)$ by a thick line. Since the origin $(0, 0)$ does not satisfy the inequation $x+y\ge 1$, therefore the portion of the xy-plane divided by the line $x+y=1$ which does not contain the origin along with the line represents the solution set of the inequation $x+y\ge 1$.

Region Represented by $-3x-4y\ge -12$ i.e., by $3x+4y\le 12$: The straight line $3x+4y=12$ meets x-axis at $(4, 0)$ and y-axis at $(0, 3)$. We join the points $(4, 0)$ and $(0, 3)$ by a thick line. Since the point $(0, 0)$ satisfies the inequation $3x+4y\le 12$, therefore the portion of the xy-plane divided by the line $3x+4y=12$ which contains the origin along with the line represents the inequation $-3x-4y\ge -12$.

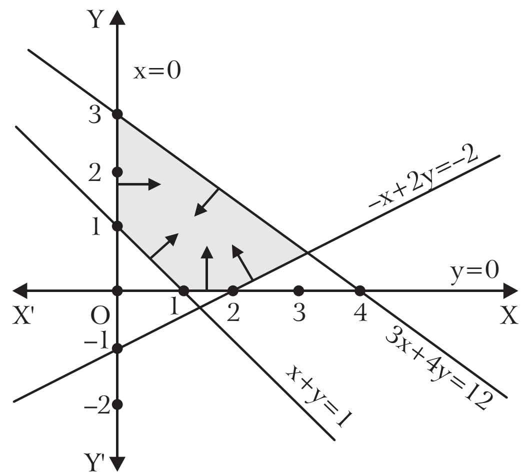

Fig. 2.4

Region Represented by $-x+2y\ge -2$: The straight line $-x+2y=-2$ meets x-axis at $(2, 0)$ and y-axis at $(0, -1)$. We join the points $(2, 0)$ and $(0, -1)$ by a thick line. Since the origin $(0, 0)$ satisfies the inequation $-x+2y\ge -2$, therefore the portion of the xy-plane divided by the line $-x+2y=-2$ which contains the origin along with the line represents the inequation $-x+2y\ge -2$.

Region Represented by $x\ge 0$: The inequation $x\ge 0$ represents the region lying on the right hand side of the y-axis including y-axis.

Region Represented by $y\ge 0$: The inequation $y\ge 0$ representes the region lying above the x-axis including x-axis.

Hence, the shaded region *i.e.*, the region common to the given five inequations, represented the solution set of the given system of inequations.

2.8 Methods for the Solution of a L.P.P.

In general we use the following three methods for the solution of a L.P.P.

1. **Graphical (or geometrical) Method :** If the objective function Z is a function of two variables only then the problem can be solved by graphical method. A problem of three variables can also be solved by this method but it is complicated enough.

2. **Analytic Method : (Trial and Error Method) :** The L.P.P. having more than two variables cannot be solved by graphical method as even the problem of three variables becomes complicated enough. In such cases, the analytic method (trial and error method) can be useful.
3. **Simplex Method :** This is the most powerful method to solve a L.P.P. as any problem can be solved by this method. This method is an algebraic procedure which progressively approaches the optimal solution. This method is discussed in chapter four.

2.9 Graphical Method to Solve a L.P.P.

There are two techniques of solving a L.P.P. by graphical method :

(i) Corner-Point method and

(ii) Iso-Profit or Iso-Cost Method.

The graphical method of solving a L.P.P. is based on the principle of extreme point theorem.

While solving a L.P.P. graphically by this method, we first obtain the region in the *xy*-plane containing all points that simultaneously satisfy all constraints including non-negative restrictions. This polygonal region so obtained is called the **convex polygon** of the set of all feasible solutions of the L.P.P. It is also called the **permissible region** for the values of the variables.

Now we determine the vertices (or corner points) of this convex polygon. These vertices are called the **extreme points** of the set of all feasible solutions of the L.P.P.

After obtaining the extreme points we find the values of the objective function Z at all these points. The point where the objective function attains its optimum value (maximum of minimum value of Z as the case may be) gives the optimal (or optimum) value of the given L.P.P.

If the two vertices of the convex polygon give the same optimal value of the objective function, then all point on the line segment joining these two vertices give the optimal value of the objective function and L.P.P. is said to have **infinite number** of optimal solutions.

Procedure to the Solve a L.P.P by Graphical Method : In this method we proceed as follows :

Step 1 : Consider each constraint as an equation.

Step 2 : Draw lines in the plane corresponding to each equation (obtained in step1) and non-negative restrictions.

Method to draw lines : Putting $x_2 = 0$ in the equation of the line find x_1 and then putting $x_1 = 0$ find x_2. Thus we get the points of intersection *i.e.*, $(x_1, 0)$, $(0, x_2)$, of the line with the axes. The line is drawn by joining these two points on the axes.

Step 3 : Now we find the permissible region for the values of the variables which is the region bounded by these lines such that every point of this region satisfies all the constraints and the non-negative restrictions.

Working Rule for Finding Permissible Region (Feasible Region) : Consider the constraint $ax + by \leq$ or $\geq c$, where $c > 0$. The line $ax + by = c$ (drawn in step 2) divide the *xy*-plane in two regions, one containing and the other not containing the origin. Since (0, 0) satisfy the inequality $ax + by \leq c$, so for the inequality $ax + by \leq c$, the feasible region is the region which contains the origin. Also (0, 0) does not satisfy the inequality $ax + by \geq c$, so for the inequality $ax + by \geq c$, the feasible region is the region which does not contain the origin. See fig. 2.5(a).

Again consider the constraint $y - mx \leq$ or ≥ 0 where $m > 0$.

The line $y - mx = 0$ (drawn in step 2) divide the *xy*-plane in two regions, one containing the +ve *x*-axis and the other containing the +ve *y*-axis. For the inequality $y - mx < 0$, the feasible region is the region which contains the positive *x*-axis for the region $y - mx > 0$, the feasible region is the region which contains the positive *y*-axis. See fig. 2.5(b).

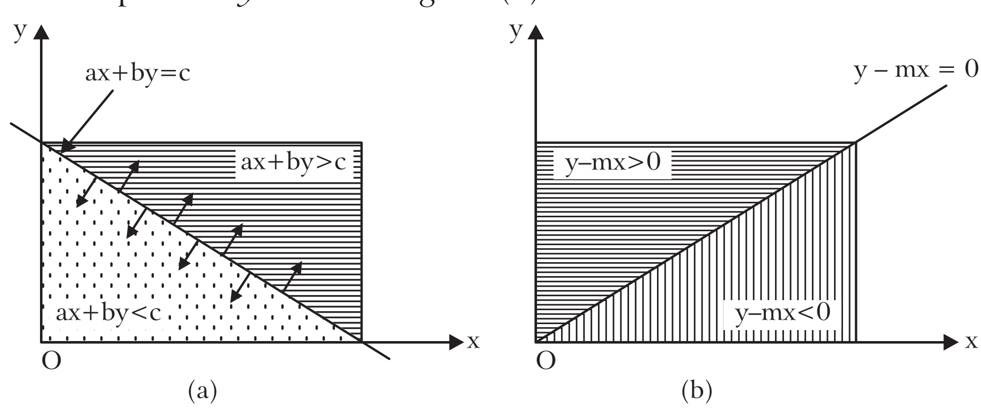

Fig. 2.5

Thus, we find feasible corresponding to all inequalities. Then the region which is common to all these regions is the **permissible region** (*i.e.*, **feasible region**) for the values of the variables. This permissible region is shaded.

Step 4 : Here we find the point in the permissible region (obtained in step 3) which gives the *optimum value* of the objective function Z. The point will be one of the extreme points (vertices) of the convex polygon enclosing the permissible region.

This point, can be attained by one of the following two methods.

Method 1 : Corner Point Method : Determine the vertices of the convex polygon, which are the points of intersection of the straight lines passing through them. These vertices are called the **extreme points** of the set of all feasible solution of the **L.P.P.** Then find the values of the objective function Z at all these points. The point where the objective function Z attains its optimum value (maximum or minimum value as the case may be) gives the optimum (or optimal) value of the **L.P.P.**.

If two vertices of the convex polygon given the same optimum value of the objective function Z, then all points on the line segment joining these two vertices will give the optimum value of the objective function Z. In this case the L.P.P. is said to have **infinite number** of optimum solutions.

Method 2 : Iso-Profit or Iso-Cost Method : Here, to find the vertex of the convex polygon, which gives the *optimum value* of the objective function Z, draw a straight line in the feasible region corresponding to the equation obtained by giving some convenient value k to the objective function. This line is called an **iso-profit** or **iso-cost line**, since every point on the line within the permissible region will yield the same value of Z. We can also take $k = 0$. Which give a line passing through the origin and parallel to iso-profit line. Thus, we draw the line (dotted line) through the origin corresponding to $Z = 0$. Then for the maximization problem the extreme point of the permissible region which is farthest away (*i.e.*, at greatest distance) from this line and for the minimization problem the extreme point of the permissible region which is nearest to this line gives the optimum values of Z. To obtain this extreme point of the permissible region, giving the optimum value of the objective function Z, we go on drawing lines parallel to the line $Z = 0$. The farthest extreme point is the vertex of the permissible region through which one of the parallel lines passes and after which it leaves this region and the nearest extreme point is the vertex of the permissible region through which the parallel line enters this region.

Note : If there is no permissible region then we say that the problem has no solution.

2.10 Limitations of the Graphical Method

This method can be applied to problems involving only two variables while most of the practical situations do involve more than two variables. Therefore it is not a powerful tool for the solution of L.P.P.

Illustrative Examples

Example 16: *Solve by graphical method, the linear programming problem :*

$$\text{Minimize } Z = 20x_1 + 10x_2$$

subject to the constraints, $x_1 + 2x_2 \le 40$

$$3x_1 + x_2 \ge 30$$

$$4x_1 + 3x_2 \ge 60$$

and the non-negative restrictions $x_1, x_2 \ge 0.$

Solution: **Step 1:** Considering the constraints as equations, we get the following equations

$$x_1 + 2x_2 = 40,\ 3x_1 + x_2 = 30,\ 4x_1 + 3x_2 = 60.$$

Step 2 : Draw lines corresponding to each equation.

Step 3 : The shaded region *PQRSP* is the permissible region of the problem.

Step 4 : By corner-point method : Solving simultaneously the equations of the corresponding intersecting lines, the coordinates of the vertices of the convex polygon are

$$P(40,0),\ Q(4,18), R(6,12) \text{ and } S(15,0)$$

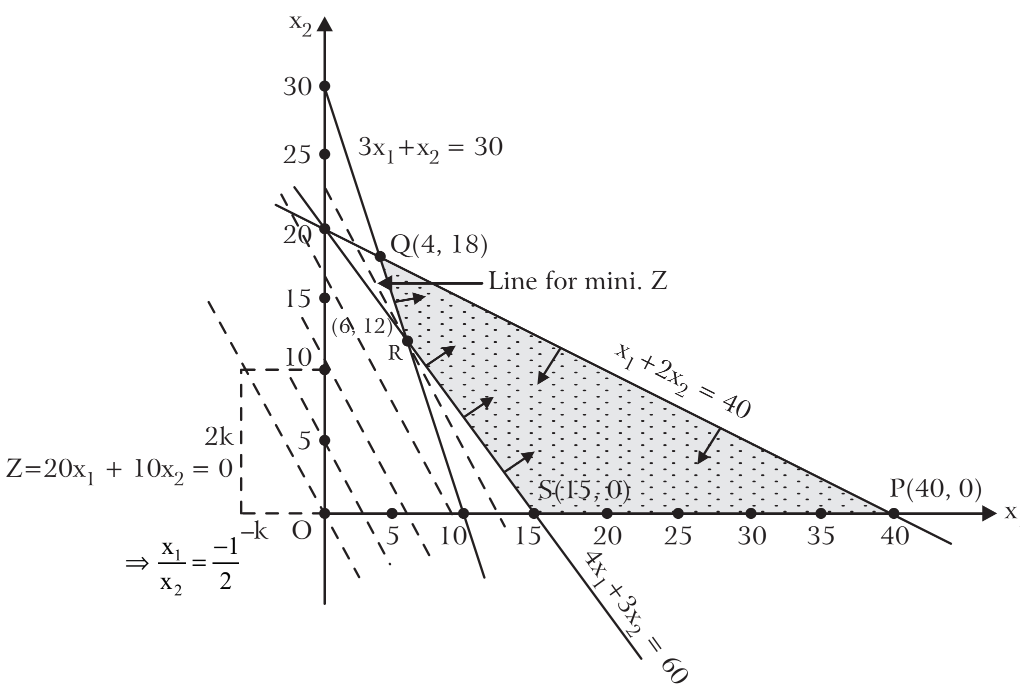

Fig. 2.6

Now the values of the objectives function Z at these vertices (corner points) are as given in the table below :

Point (x, y)	**Value of the objective function $Z = 20x_1 + 10x_2$**
$P(40,0)$	$Z = 20 \times 40 + 10 \times 0 = 800$
$Q(4,18)$	$Z = 20 \times 4 + 10 \times 18 = 260$
$R(6,12)$	$Z = 20 \times 6 + 10 \times 12 = 240$ (Mini)
$Q(15,0)$	$Z = 20 \times 15 + 0 = 300$

Obviously, Z is minimum at $R(6,12)$.

Hence, the optimal solution of the given L.P.P. is

$$x_1 = 6, x_2 = 12 \text{ and minimum } Z = 240.$$

By Iso-profit Method : Here, we draw the line through the origin corresponding to $Z = 0$, which is parallel to iso-profit line.

$$Z = 0 \Rightarrow 20x_1 + 10x_2 = 0 \Rightarrow x_1/x_2 = (-1)/2$$

The dotted line through the origin is shown in the figure. Drawing parallel lines away from the origin (note) O, we see that the nearest line (since it is minimization problem) in the permissible region passes through the vertex $R(6,12)$.

Hence, the optimal solution is

$$x_1 = 6, x_2 = 12 \text{ and Minimum } Z = 20 \times 6 + 10 \times 12 = 240.$$

Example 17: *Using Iso-profit line method, compute the maximum value of the expression* $2x_1 + 3x_2$ *subject to the conditions :*

$$2x_1 + x_2 \leq 5$$

$$x_1 - x_2 \leq 1$$

$$x_2 \leq 2$$

and the non-negative restrictions $x_1 \geq 0, x_2 \geq 0$.

Verify the maximum value by computing the values at the boundary points of the polygon.

Solution: Converting the given conditions into equation, and drawing these lines, the permissible region is $OPQRSO$ (shaded region) which is the set of all feasible solution of the problem.

Here, we are to maximize the objective function

$$Z = 2x_1 + 3x_2.$$

Taking $Z = 0$ we get $x_1 / x_2 = (-3)/2$.

The line corresponding to $Z = 0$ is shown by dotted line through the origin O, which is parallel to the Iso-profit line.

Now drawing parallel lies away from the origin (since it is maximization problem), shown by dotted lines, we see that the farthest line from the origin O passes through the vertex $R(3/2, 2)$.

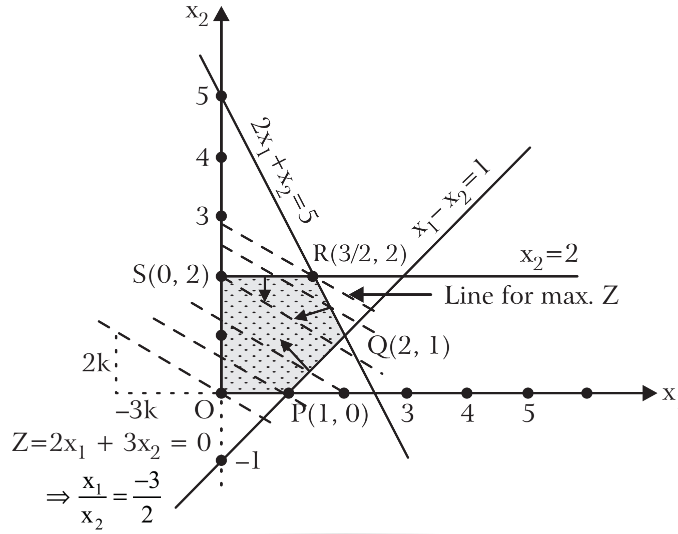

Fig. 2.7

Hence, the optimal solution is

$$x_1 = 3/2, \; x_2 = 2$$

and $\qquad$ Max. $Z = 2 \times 3/2 + 3 \times 2 = 9.$

Verification : Solving simultaneously the equations of the corresponding intersecting lines of the permissible region, the coordinates of the vertices (corners) of the convex polygon are $O(0,0)$, $P,(1,0)$, $Q(2,1)$, $R\left(\frac{3}{2}, 2\right)$ and $S(0,2)$.

The values of the objective function Z at these corner points are as follows :

Point (x_1, x_2)	**Value of the objective function $Z = 2x_1 + 3x_2$**
$O(0,0)$	$Z = 0 + 0 = 0$
$P,(1,0)$	$Z = 2 \times 1 + 0 = 2$
$Q(2,1)$	$Z = 2 \times 2 + 3 \times 1 = 7$
$R\left(\frac{3}{2}, 2\right)$	$Z = 2 \times 3/2 + 3 \times 2 = 9$ (Max.)
$S(0,2)$	$Z = 0 + 3 \times 2 = 6.$

Obviously, the maximum value of Z is 9 at $R\left(\frac{3}{2}, 2\right)$ *i.e.*, when $x_1 = \frac{3}{2}$, $x_2 = 2.$

Example 18: *A goldsmith manufactures necklaces and bracelets. The total number of necklaces and braclets that he can handle per day is at most 24. It takes one hour to make a bracelet and half an hour to make a necklace. It is assumed that he can work for a maximum of 16 hours a day. Further the profit on a bracelet is ₹ 300 and the profit on a necklace is ₹ 100. Find how many of each should be produced daily to maximize the profit.*

Solution: **Formation of the problem as linear programming problem :** Proceeding as in Example 1 page 21, the mathematical form of the given problem as a L.P.P. is as follows:

$$\text{Maximize } Z = 100x_1 + 300x_2$$

subject to the constraints $x_1 + 2x_2 \le 32$

$$x_1 + x_2 \le 24$$

and the non-negative restrictions

$$x_1 \ge 0,\ x_2 \ge 0.$$

where x_1 and x_2 are the numbers of necklaces and bracelets respectively to be manufactured per day.

Solution of the Problem : Proceeding stepwise, the permissible region is the shaded region *OSTPO* which is the set of all points which simultaneously satisfy all the constraints and the non-negative restrictions. Solving simultaneously the equations of the corresponding intersecting lines, we get the co-ordinates of the vertices of the shaded region as

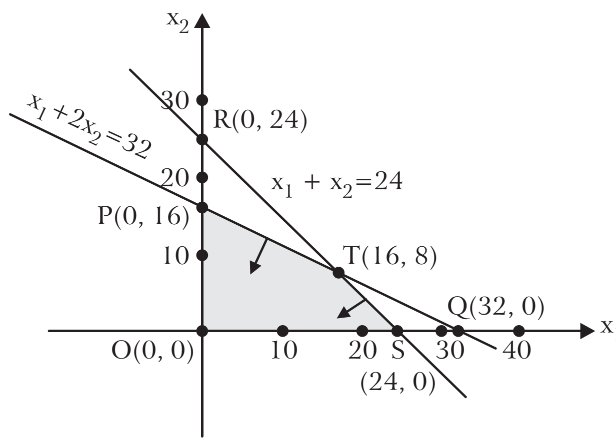

Fig. 2.8

$$P(0,16),\ O(0,0),\ S(24,0),\ T(16,8).$$

Now the values of the objective function at these corner points are as given in the table below :

Point (x_1, x_2)	**Value of the objective function $Z = 100x_1 + 300x_2$**
$P(0,16)$	$100 \times 0 + 300 \times 16 = 4800$ (Max.)
$O(0,0)$	$100 \times 0 + 300 \times 0 = 0$
$S(24,0)$	$100 \times 24 + 300 \times 0 = 2400$
$T(16,8)$	$100 \times 16 + 300 \times 8 = 4000$

Clearly, Z is maximum at $P(0,16)$. Hence, $x_1 = 0, x_2 = 16$ is the optimal solution of the given problem and the optimal value of Z is 4800, *i.e.*, the goldsmith earns maximum profit when he makes 16 bracelets and 0 necklace a day and that his maximum profit in this case is ₹ 4800.

Example 19: *A toy company manufactures two types of dolls; a basic version-doll A and a deluxe version-doll B. Each doll of type B takes twice as long to produce as one of type A, and the company would have time to make a maximum 2000 per day if it produces only the basic version. The supply of plastic is sufficient to produce 1500 dolls per day (both A and B combined). The deluxe version requires a fancy dress of which there are only 600 pieces per day available. If the company makes a profit of ₹ 3 and ₹ 5. per doll respectively, on dolls A and B, how many of each should be produced per day in order to maximize profit?*

Solution: **Formulation of the problem as linear programming problem :** Proceeding as in Ques. 5, page 34, the mathematical form of the given problem as a L.P.P. is as follows :

$$\text{Maximize } Z = 3x_1 + 5x_2$$

subject to the constraints $x_1 + 2x_2 \le 2000$

$$x_1 + x_2 \le 1500$$

$$x_2 \le 600$$

and the non-negative restrictions

$$x_1 \ge 0,\ x_2 \ge 0.$$

Solution of the Problem : Proceeding stepwise, the permissible region is the shaded region *OPQRSO* which is the set of all feasible solutions of the problem.

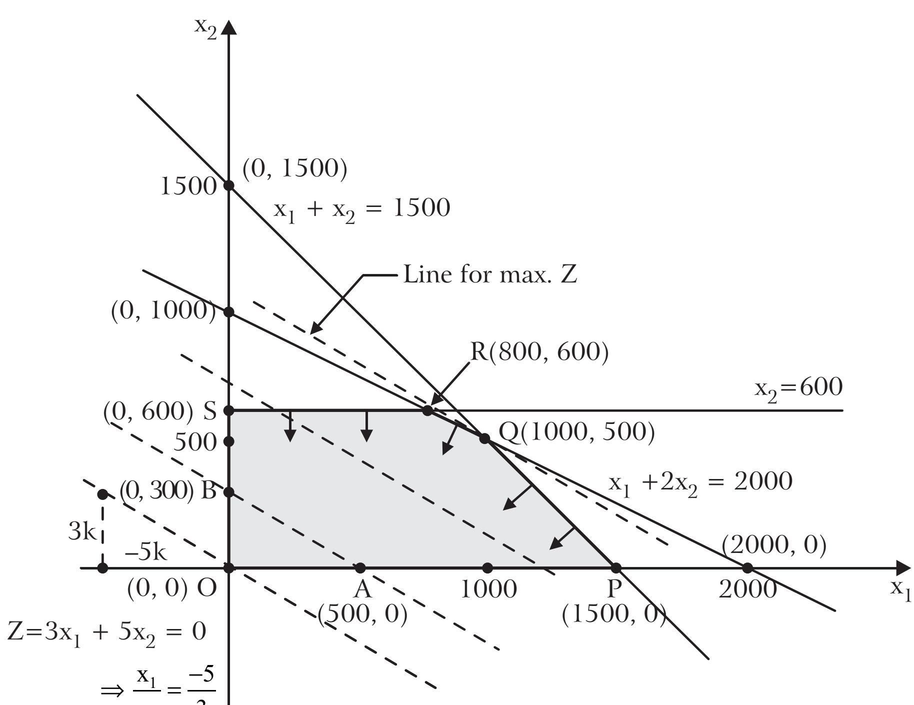

Fig. 2.9

To find the maximum value of the objective function Z.

By Corner Point Method : Solving simultaneously the equations of the corresponding intersection lines of the feasible region, we get coordinates of the corner points of the feasible region as $O(0,0)$, $P(1500,0)$, $Q(1000,500)$, $R(800,600)$ and $S(0,600)$.

The values of the objective function Z at these corner points are as given below :

Point (x_1, x_2)	Value of the objective function $Z = 3x_1 + 5x_2$
$O(0,0)$	$Z = 3\times0+5\times0 = 0$
$P(1500,0)$	$Z = 3\times1500+5\times0 = 4500$
$Q(1000,500)$	$Z = 3\times1000+5\times500 = 5500$ (Max.)
$R(800,600)$	$Z = 3\times800+5\times600 = 5400$
$S(0,600)$	$Z = 3\times0+5\times600 = 3000$

Thus, Z is maximum at the corner point Q *i.e.,* for $x_1 = 1000$ and $x_2 = 500$ and the maximum value of Z is 5500. The optimal solution of the problem is $x_1 = 1000$, $x_2 = 500$.

Hence, 1000 dolls of type A and 500 dolls of type B should be produced per day to maximize the profit and the maximum profit per day is ₹ 5500.

The optimal solution may also be found by the following alternative method.

By Iso-profit Line Method : To find the maximum value of the objective function Z, we draw the dotted line $Z = 3x_1 + 5x_2 = 0$ passing through the origin, which is parallel to iso-profit line. We now move this line parallel to itself away from the origin so that its distance from the origin becomes maximum yet it has at least one point in the feasible region. We see that in this position this line passes through only one point $Q(1000,500)$ of the feasible region. This line is an iso-profit line having only one point *viz.,* Q in the feasible region which will yield the maximum value of Z.

Hence, Z is maximum for $x_1 = 1{,}000$ and $x_2 = 500$ and the maximum value of Z is

$$3\times1{,}000+5\times500 = ₹\ 5{,}500.$$

Example 20: *A dietician mixes two types of food in such a way that the vitamin contents of the mixture contain at least 8 units of vitamin A and 10 units of vitamin C. Food X contains 2 units/kg of vitamin A and 1 unit/kg of vitamin C while food Y contains 1 unit/kg of vitamin A and 2 units/kg of vitamin C. One kg of food X costs ₹ 5 whereas one kg of food Y costs ₹ 7. Determine the minimum cost of such a mixture.*

Solution: **Formulation of the problem as L.P.P. :** Let the dietician mix x_1 kg of food X and x_2 kg of food Y. Then

$$x_1 \geq 0,\ x_2 \geq 0$$

Since the minimum requirement of vitamin A is 8 units, therefore

$$2x_1 + x_2 \geq 8$$

Similarly, since the minimum requirement of vitamin C is 10 units, therefore

$$x_1 + 2x_2 \geq 10.$$

The total cost Z in ₹ of purchasing x_1 kg of food X and x_2 kg of food Y is

$$Z = 5x_1 + 7x_2.$$

Hence, the mathematical form of the given problem as a L.P.P. is as follows :

$$\text{Minimize } Z = 5x_1 + 7x_2$$

subject to the constraints $2x_1 + x_2 \geq 8$

$$x_1 + 2x_2 \geq 10$$

and the non-negative restrictions

$$x_1 \geq 0,\ x_2 \geq 0.$$

Solution of the Problem : Proceeding stepwise, the permissible region is the shaded region, which is unbounded and is the set of all feasible solutions of the problem.

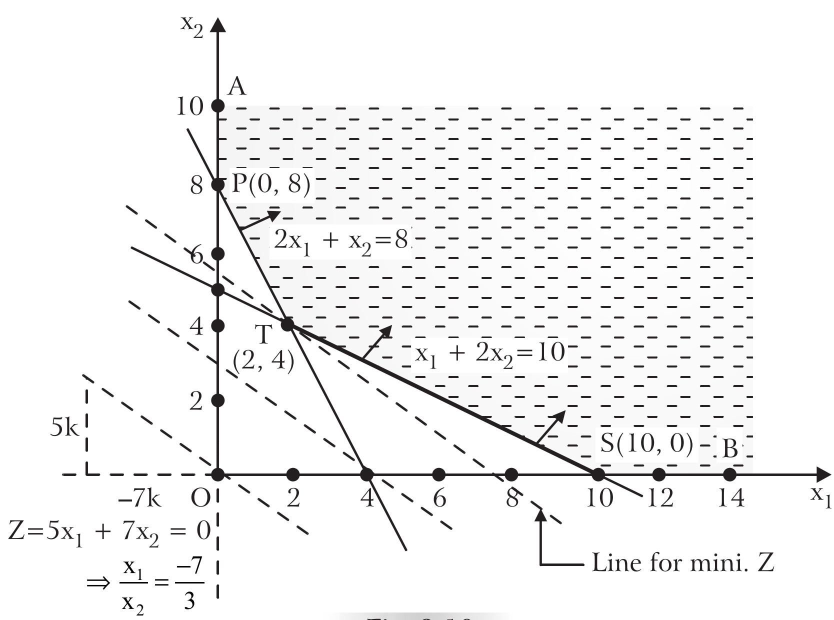

Fig. 2.10

To find the minimum value of the objective function Z.

By Corner Point Method : Solving simultaneously the equation of the corresponding intersecting lines of the permissible region, the co-ordinates of the corner points of this region are $P(0,8)$, $T(2,4)$ and $S(10,0)$.

Now the values of the objective function at these corner points are as given in the table below :

Point (x_1, x_2)	Value of the objective function $Z = 5x_1 + 7x_2$
$P(0,8)$	$Z = 5\times0+7\times8=56$
$T(2,4)$	$Z = 5\times2+7\times4=38$ (Min.)
$S(10,0)$	$Z = 5\times10+7\times0=50$

Clearly, Z is minimum at $T(2,4)$. Hence $x_1 = 2$ and $x_2 = 4$ is the optimal solution of the given problem and the optimal value is $Z = 38$.

Hence, the dietician should mix 2 kg of food X and 4 kg of food Y to make the required mixture at minimum cost. The minimum cost in this case is ₹ 38.

By Iso-cost Method : To find the minimum value of $Z,$ we draw the dotted line corresponding to $Z = 5x_1 + 7x_2 = 0$ passing through the origin which is parallel to iso-cost line. We now move this line parallel to itself so that it passes through only one point [here the corner point $T(2,4)$] of the feasible region. This line is an iso-cost line having only one point viz. T in the feasible region which gives the minimum value of Z.

Hence, Z is minimum for $x_1 = 2$ and $x_2 = 4$ and the minimum value of Z is

$$5\times2+7\times4 = ₹\ 38.$$

Example 21: *A farm is engaged in breeding hens. In view of the need to ensure certain nutrients (say x_1, x_2, x_3), it is necessary to buy two types of food, say A and B. One unit of food A contains 36 units of x_1, 3 units of x_2 and 20 units of x_3. One unit of food B contains 6 units of x_1, 12 units of x_2 and 10 units of x_3. The minimum daily requirement of x_1, x_2 and x_3 is 108, 36 and 100 units respectively. The cost of food A is ₹ 20 per unit whereas food B costs ₹ 40 per unit. Find the minimum food cost so as to meet the minimum daily requirement of nutrients.*

Solution: **Formulation of the Problem as L.P.P. :** Let x units of food A and y units of food B be bought to fulfill the minimum requirement of the nutrients x_1, x_2, x_3 and to minimize the cost.

If Z be the cost in ₹ of the two types of food bough, then $Z = 20x + 40y$.

According to the minimum daily requirements of x_1, x_2, x_3 we have

$$36x + 6y \geq 108,\ 3x + 12y \geq 36,\ 20x + 10y \geq 100$$

Also, since the quantity of each type of food bought cannot be negative, therefore $x \geq 0, y \geq 0$.

Hence, the mathematical form of the given problem as a L.P.P. is as follows :

$$\text{Minimize } Z = 20x + 40y$$

subject to the constraints $36x + 6y \geq 108$, $3x + 12y \geq 36$, $20x + 10y \geq 100$

and the non-negative restrictions $x \geq 0, y \geq 0$.

Solution of the Problem : Proceeding stepwise, the permissible region (the set of all points satisfying all the constraints and the non-negative restrictions) consists of the shaded region *YLSTPX* which is unbounded.

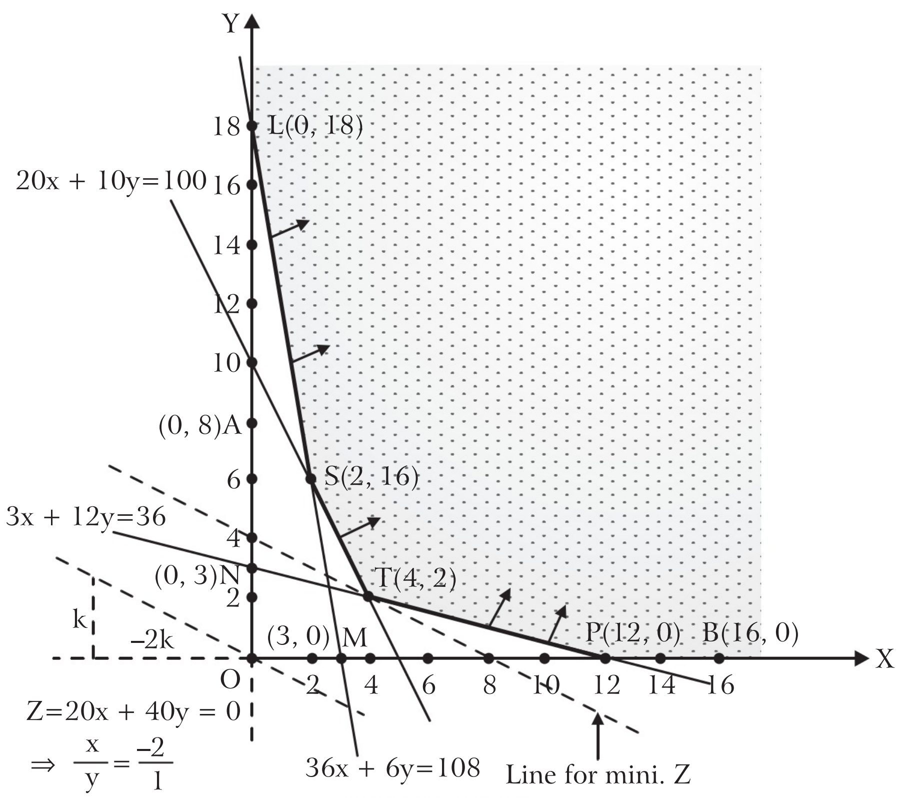

Fig. 2.11

To find the minimum value of Z, we draw the dotted line through the origin corresponding to $Z = 20x + 40y = 0$, which is parallel to iso-cost line. We now move this line parallel to itself so that it enters the permissible region through a point [here the corner point T (4,2)]. This line is an iso-cost line having only one point *viz.* T in the feasible region which will give the minimum value of Z.

Therefore, Z is minimum for $x = 4$ and $y = 2$ and the minimum value of Z is $20 \times 4 + 40 \times 2 =$ ₹ 160.

Hence, 4 units of food A and 2 units of food B should be bought to fulfill the minimum requirements of x_1, x_2, x_3 at a minimum cost of ₹ 160.

Example 22: *Old hens can be bough at ₹ 2 each and young ones at ₹ 5 each. The old hens lay 3 eggs per week and the young ones lay 5 eggs per week, each egg being worth 30 paise. A hen (young or old) costs ₹ 1 per week to feed. I have only ₹ 80 to spend for hens. How many of each kind should I buy to give me a maximum profit, assuming that I cannot house more than 20 hens ?*

Solution: **Formulation of the Problem as L.P.P. :** Suppose I buy x_1 old hens and x_2 young hens.

Since old hens lay 3 eggs per week and the young ones lay 5 eggs per weak, the total number of eggs I have per week is $3x_1 + 5x_2$. Consequently, each egg being worth 30 paise, my total income per week is ₹ 0.3 $(3x_1 + 5x_2)$.

Also, the expenses for feeding $(x_1 + x_2)$ hens, at the rate of ₹ 1 per hen per week

$$= ₹\ (x_1 + x_2)$$

Thus, the total profit Z in ₹ . earned per week is given as

$$Z = 0.3(3x_1 + 5x_2) - (x_1 + x_2) = -0.1x_1 + 0x_2$$

Since the cost of one old hen is ₹ 2 and that of one young hen is ₹ 5 and I have only ₹ 80 to spend for hens, therefore $2x_1 + 5x_2 \le 80$

Since I cannot house more than 20 hens, therefore $x_1 + x_2 \le 20$.

Again, the number of hens purchased, whether old ones or young ones, cannot be negative.

Therefore, $x_1 \ge 0$ and $x_2 \ge 0$.

Hence, the given problem formulated as L.P.P. is as follows :

Maximize $Z = -0.1x_1 + 0.5x_2$

subject to the constraints $2x_1 + 5x_2 \le 80, x_1 + x_2 \le 20$

and the non-negative restrictions $x_1 \ge 0$ and $x_2 \ge 0$.

Solution of the Problem

Proceeding stepwise, the permissible region (the set of all points satisfying all the constraints and the non-negative restrictions) consists of the shaded region *OBECO*.

To find the maximum value of the objective function Z, we draw the dotted line through the origin corresponding to $Z = -0.1x_1 + 0.5x_2 = 0$, which is the iso-cost line.

We now move this line parallel to itself away from the origin so that is leaves the region through only one point [here the corner point $B(0,16)$] of the feasible region. This line is an iso-profit line having only one point *viz.* B in the feasible region which will yield the maximum value of Z.

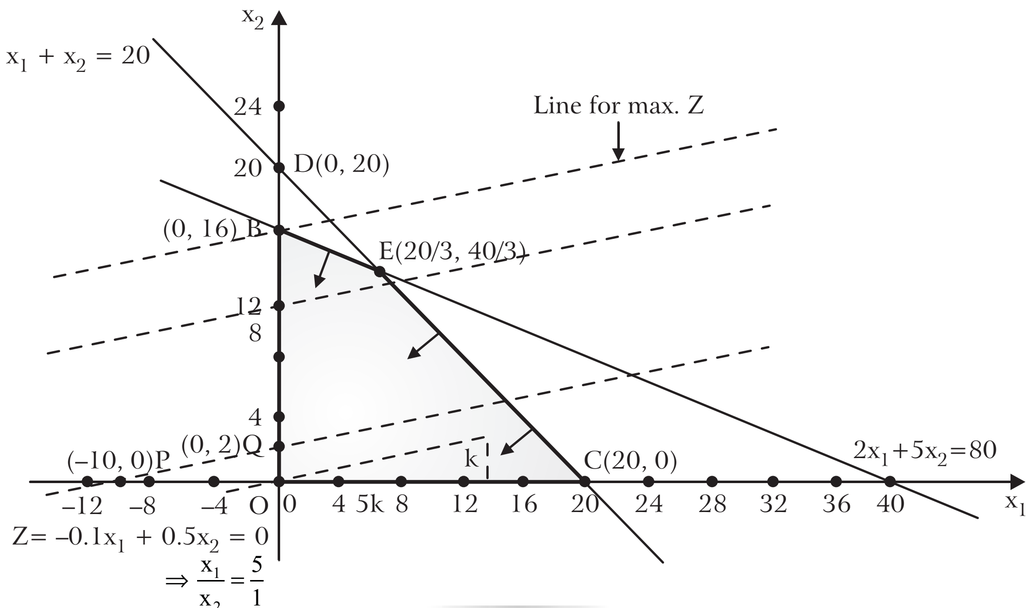

Fig. 2.12

Thus, Z is maximum for $x_1 = 0$ and $x_2 = 16$ and the maximum value of Z

$$= 0.5 \times 16 - 0.1 \times 0 = ₹\ 8.$$

Hence, he should buy only 16 young hens and no old hen in order to get the maximum profit of ₹ 8 per week.

Note : If the dotted line through the origin is moved in the opposite direction then it will pass through the vertex $C(20,0)$ for which

$$Z = -0.1 \times 20 + 0.5 \times 0 = -2 < 8$$

$\therefore$ Z is not max. at $C(20,0)$.

By Corner Point Method : The point giving the maximum value of the objective function may also be located by finding the values of the objective function at all the different corner points of the feasible region.

Solving simultaneously the equations of the corresponding intersecting lines of the feasible region, we get the co-ordinates of the vertices of the feasible region as $O(0,0)$, $B(0,16)$, $E(20/3, 40/3)$ and $C(20,0)$. The values of the objective function at these corner points are as given below :

Point (x_1, x_2)	**Value of the objective function $Z = -0.1x_1 + 0.5x_2$**
$O(0,0)$	$Z = -0.1 \times 0 + 0.5 \times 0 = 0$
$B(0,16)$	$Z = -0.1 \times 0 + 0.5 \times 16 = 8$ (Max.)
$E\left(\frac{20}{3}, \frac{40}{3}\right)$	$Z = -0.1 \times \frac{20}{3} + 0.5 \times \frac{40}{3} = 6$
$C(20,0)$	$Z = -0.1 \times 20 + 0.5 \times 0 = -2$

Thus, the maximum value of Z is 8 and is attained when $x_1 = 0$ and $x_2 = 16$.

Example 23: *Suresh has two factories of toys, one located in city X and the other in city Y. Both these factories manufacture the same type of toys. From these locations a certain number of toys are delivered to each of the three depots situated at places A, B and C. The weekly requirements of the depots are respectively 5, 5 and 4 units, while the production capacity of the factories at X and Y are respectively 8 and 6 units. The transportation cost in ₹ per unit from a factory to a depot is as given in the table.*

From Factory \ To Depot	A	B	C
X	16	10	15
Y	10	12	10

How many units should be transported from each factory to each depot in order that the transportation cost is minimum ?

Solution: **Formulation of the problem as L.P.P. :** Let x_1 and x_2 units of toys be transported from the factory at X to the depots at A and B respectively. Since the production capacity of the factory at X is 8 units, so $8 - x_1 - x_2$ units will be transported from the factory at X to the depot at C.

Now the number of units supplied to the depots cannot be negative, therefore

$$x_1 \geq 0,\ x_2 \geq 0 \text{ and } 8 - x_1 - x_2 \geq 0 \Rightarrow x_1 + x_2 \leq 8$$

The weekly requirement of the depot at A is 5 units. Since x_1 units are supplied from the factory at X, the remaining $5 - x_1$ units are to be supplied from the factory at Y.

$\therefore \qquad 5 - x_1 \geq 0 \Rightarrow x_1 \leq 5.$

Similarly, $5 - x_2$ units are to be supplied from the factory at Y to the depot at B.

$\therefore \qquad 5 - x_2 \geq 0 \Rightarrow x_2 \leq 5$

Since the production capacity of the factory at Y is 6 units, therefore

$$6 - (5 - x_1 + 5 - x_2), \quad i.e., \ x_1 + x_2 - 4$$

Units of toys are to be supplied from this factory to the depot at C.

On account of the non-negative restrictions, we must have

$$x_1 + x_2 - 4 \geq 0 \Rightarrow x_1 + x_2 \geq 4.$$

The total transportation cost Z. in ₹ is given by

$$Z = 16x_1 + 10x_2 + 15(8 - x_1 - x_2) + 10(5 - x_1) + 12(5 - x_2) + 10(x_1 + x_2 - 4)$$

or $\qquad Z = x_1 - 7x_2 + 190.$

Hence the given problem formulated as L.P.P. is as follows ;

Minimize $Z = x_1 - 7x_2 + 190$

subject to the constraints $x_1 + x_2 \le 8$

$$x_1 \le 5$$

$$x_2 \le 5$$

$$x_1 + x_2 \ge 4$$

and the non-negative restrictions $x_1 \ge 0, x_2 \ge 0.$

Solution of the Problem : Proceeding stepwise, the permissible region (the set of all points satisfying all the constraints and the non-negative restrictions) consists of the shaded region *EFCGHDE.*

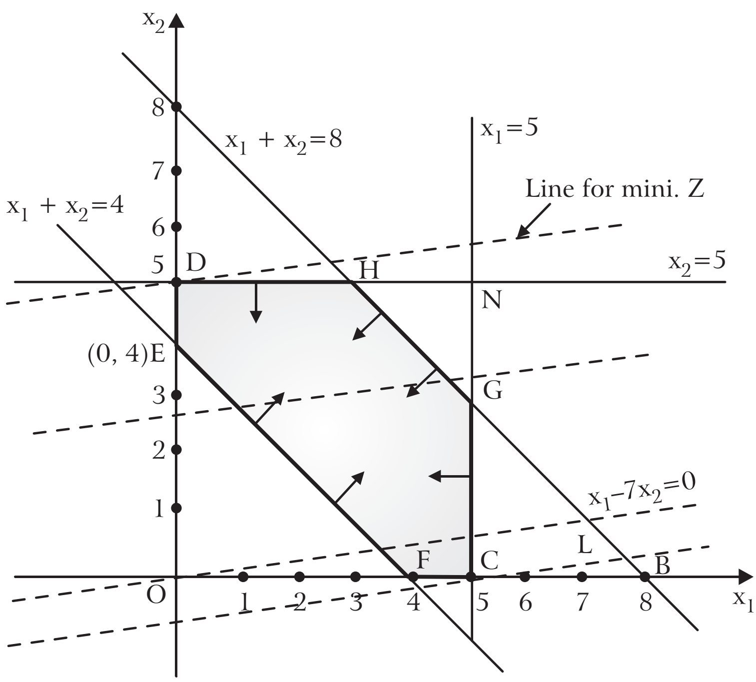

Fig. 2.13

To find the minimum value of the objective function Z, we draw a dotted line through the origin corresponding to $Z = 0$ (Note) which is an iso-cost line. When we move this line parallel to itself in the direction of x_2 decreasing, so that it passes through only one point [here the corner point $C(5,0)$] of the feasible region. At $C(5,0)$, $Z = 195 > 190$. So we move the line OL parallel to itself away from the origin towards the positive direction of x_2-axis so that it passes through only one point [here the corner point $D(0,5)$] of the feasible region. This line is an iso-cost line having only one point *viz.* D in the feasible region which will give the minimum value of Z.

Thus, Z is minimum for $x_1 = 0$ and $x_2 = 5$ and the minimum value of Z in this case is $Z = 0 - 7 \times 5 + 190 = 155$.

Hence, the optimal transportation strategy is to supply 0, 5 and 3 units from the factory at X and 5, 0 and 1 units from the factory at Y to the depots at A, B and C respectively. In this case the transportation cost is minimum and is ₹ 155.

Example 24: *The post master of a local post office wishes to hire extra helpers during the deepawali season, because of a large increase in the volume of mail handling and delivery. Keeping in view the limited office space and the budgetary condition, the number of temporary helpers must not exceed 10. According to past experience, a man can handle 300 letters and 80 packages per day, and a woman can handle 400 letters and 50 packages per day. It is believed that the daily volume of extra mail and packages will be no less than 3400 and 680 respectively. A man receives ₹ 25 a day and a woman receives ₹ 22 a day. How many men and woman helpers should be hired to keep the pay-roll at a minimum ?*

Solution: **Formulation of the Problem as L.P.P. :** Let x_1 men and x_2 women be hired by the post master to keep the pay-roll at a minimum.

If Z be the daily pay-roll in ₹ of the extra helpers, then

$$Z = 25x_1 + 22x_2$$

Since the maximum number of helpers can be only 10, therefore $x_1 + x_2 \le 10$.

Given that a man can handle 300 letters daily and a woman can handle 400 letters daily and that the number of extra letters expected daily is not less than 3400, we must have

$$300x_1 + 400x_2 \ge 3400.$$

Similarly, $80x_1 + 50x_2 \le 680$.

Since the number of men and women hired cannot be negative, we have

$$x_1 \ge 0,\ x_2 \ge 0.$$

Hence, the L.P.P. formulated for the given problem is as follows :

$$\text{Minimize } Z = 25x_1 + 22x_2$$

subject to the constraints $x_1 + x_2 \le 10$

$$300\,x_1 + 400x_2 \ge 3400$$

$$80x_1 + 50x_2 \ge 680$$

and the non-negative restrictions $x_1 \ge 0, x_2 \ge 0$.

Solution of the Problem : Proceeding stepwise, the permissible region (the set of all points satisfying all the constraints and the non-negative restrictions) consists of the point $G(6,4)$ only.

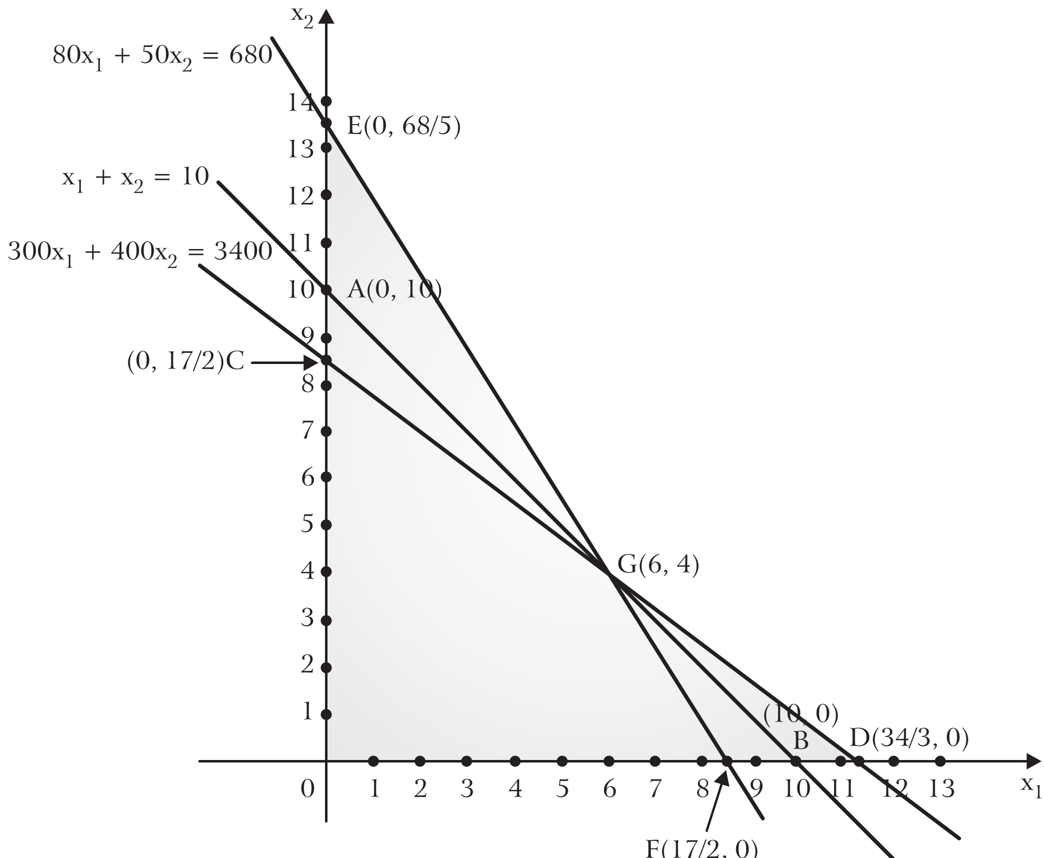

Fig. 2.14

Therefore, the optimal solution is $x_1 = 6, x_2 = 4$ and the optimal value of

$$Z = 25 \times 6 + 22 \times 4 = 238.$$

Hence, 6 men and 4 women helpers should be hired by the courier company to meet its seasonal requirements and keep the pay-roll at a minimum of ₹ 238.

Example 25: *Solve by graphical method the L.P.P.*

$$\textit{Minimize } Z = 5x_1 + 6x_2$$

subject to the constraints $\quad x_1 + x_2 \geq 50,\ x_1 + 2x_2 \leq 40,\ 3x_1 + 4x_2 \leq 100$

and the non-negative restrictions $x_1 \geq 0$ *and* $x_2 \geq 0.$

Solution: Solving simultaneously the inequations of the constraints and the non-negative restrictions by graphical method, we see that there exist no values of x_1 and x_2 that simultaneously satisfy all the constraints and the non-negative restrictions.

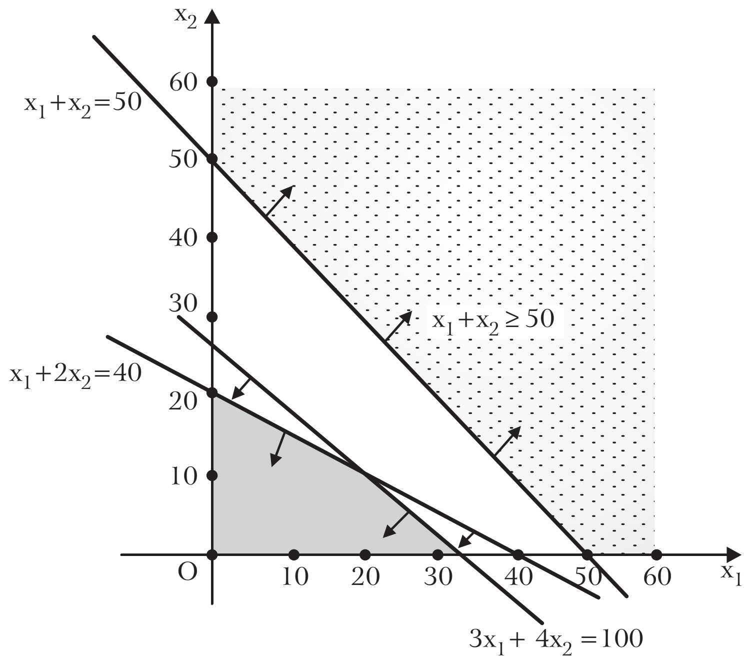

Fig. 2.15

Hence, the given problem does not have any feasible solution.

Problems Having Unbounded Solution

Example 26: *Solve graphically the following L.P.P.*

$$Max.\ Z = 3x_1 + 2x_2$$

subject to $x_1 - x_2 \le 1,$

$$x_1 + x_2 \ge 3$$

and $x_1, x_2 \ge 0.$

Solution: Proceeding stepwise, the permissible region is shaded in the figure which is unbounded. From the figure it is clear that the dotted line through the origin representing $Z = 0$ can be moved parallel to itself in the direction of Z increasing and still have some points in the permissible region.

Thus Z can be made arbitrarily large and so the problem has no finite maximum value of Z. Hence the problem has unbounded solution.

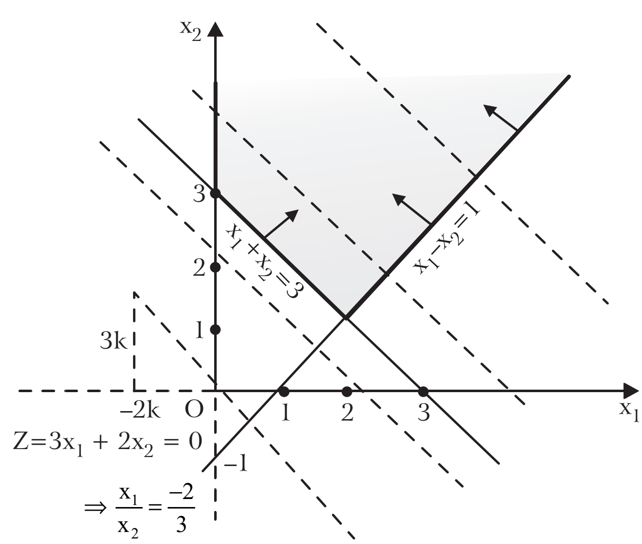

Fig. 2.16

Example 27: *Solve by graphical method, the L.P.P.*

$$\text{Maximize } Z = -3x_1 + 2x_2$$

subject to $x_1 \le 3$

$x_1 - x_2 \le 0$

and $x_1, x_2 \ge 0.$

Solution: Proceeding stepwise, the permissible region is shaded in the figure which is unbounded. From the figure it is obvious that the dotted line through the origin representing $Z = 0$ can be moved parallel to itself in the direction of Z increasing and still have some points in the permissible region.

Thus, Z can be made arbitrarily large and so the problem has no finite maximum value of Z.

Hence, the problem has unbounded solution.

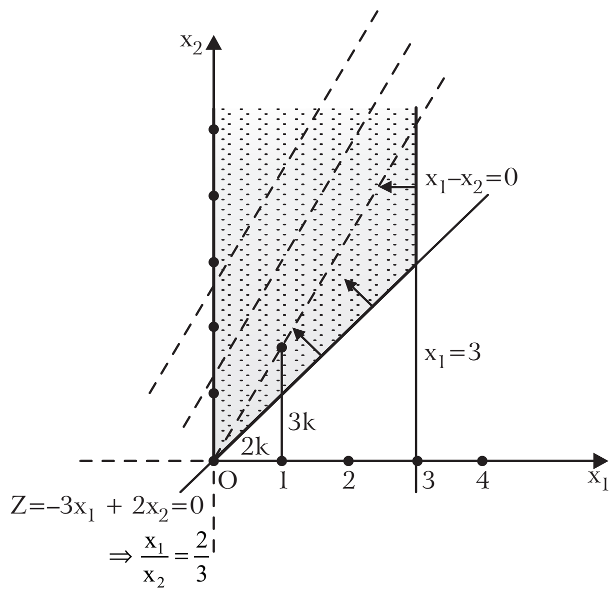

Fig. 2.17

Comprehensive Exercise 2

Solve graphically the following linear programming problems :

1. Max. $Z = x_1 + x_2$
subject to $x_1 + x_2 \le 2000$
$x_1 + x_2 \le 1500,$
$x_2 \le 600$
and $x_1, x_2 \ge 0.$ **(Garhwal 2008, 12)**

2. Max. $Z = 8x_1 + 7x_2$
subject to $3x_1 + x_2 \le 66{,}000$
$x_1 + x_2 \le 45{,}000$
$x_1 \le 20{,}000,\ x_2 \le 40{,}000$
and $x_1, x_2 \ge 0.$

3. Min. $Z = 1.5x_1 + 2.5x_2$
subject to $x_1 + 3x_2 \ge 3,$
$x_1 + x_2 \ge 2$
and $x_1, x_2 \ge 0.$

4. Max. $Z = 6x_1 + 11x_2$
subject to $2x_1 + x_2 \le 104,$
$x_1 + 2x_2 \le 76$
and $x_1, x_2 \ge 0.$

5. Max. $Z = 5x_1 + 7x_2$
subject to $x_1 + x_2 \le 4,$
$3x_1 + 8x_2 \le 24,$
$10x_1 + 7x_2 \le 35$
and $x_1, x_2 \ge 0.$

6. Min. $Z = 3x_1 + 5x_2$
subject to $-3x_1 + 4x_2 \le 12$
$2x_1 - x_2 \ge -2,$
$2x_1 + 3x_2 \ge 12,$
$x_1 \le 4,\ x_2 \ge 2$
and $x_1, x_2 \ge 0.$

7. Max. $Z = 3x_1 + 2x_2$
subject to $x_1 + x_2 \le 4$
$x_1 - x_2 \le 2$
and $x_1, x_2 \ge 0.$

8. Max. $Z = 3x_1 + 4x_2$
subject to $x_1 - x_2 \le -1$
$-x_1 + x_2 \le 0$
and $x_1, x_2 \ge 0.$

9. Min. $Z = x_1 + x_2,$
subject to $5x_1 + 10x_2 \le 50$
$x_1 + x_2 \ge 1$
$x_2 \le 4,$
and $x_1, x_2 \ge 0.$

10. Max. $Z = 0.75x_1 + x_2$
subject to $x_1 - x_2 \ge 0$
$-0.5x_1 + x_2 \le 1$
and $x_1, x_2 \ge 0$

11. Max. $Z = 6x_1 - 2x_2$
subject to $2x_1 - x_2 \le 2$
$x_1 \le 3$
and $x_1, x_2 \ge 0.$

12. Max. $Z = 3x_1 + 4x_2$
subject to $5x_1 + 4x_2 \le 200$
$3x_1 + 5x_2 \le 150$
$5x_1 + 4x_2 \ge 100$
$8x_1 + 4x_2 \ge 80$
$x_1, x_2 \ge 0.$

13. Max. $Z = x_1 + \frac{1}{2}x_2$
subject to $3x_1 + 2x_2 \le 12$
$5x_1 \le 10$
$x_1 + x_2 \ge 8,$
$-x_1 + x_2 \ge 4$
and $x_1, x_2 \ge 0.$

14. Min. $Z = 2x_1 - 10x_2$
subject to $x_1 - x_2 \ge 0$
$x_1 - 5x_2 \ge -5$
and $x_1, x_2 \ge 0.$

15. Max. $Z = 3x_1 - 2x_2$
subject to $x_1 + x_2 \le 1$
$2x_1 + 2x_2 \ge 4$
and $x_1, x_2 \ge 0.$

16. Max. $Z = x_1 + x_2$
subject to $x_1 - x_2 \ge 0$
$-3x_1 + x_2 \ge 3$
and $x_1, x_2 \ge 0.$

17. Max. $Z = 7x_1 + 3x_2$
subject to $x_1 + 2x_2 \le 3,\ x_1 + x_2 \le 4,$
$0 \le x_1 \le \frac{5}{2}, 0 \le x_2 \le \frac{3}{2}.$

18. Min. $Z = -x_1 + 2x_2$
subject to $-x_1 + 3x_2 \le 10$
$x_1 + x_2 \le 6,\ x_1 - x_2 \le 2$
and $x_1, x_2 \ge 0.$

19. Min. $Z = 5x_1 - 2x_2$
subject to $2x_1 + 3x_2 \geq 1$
$x_1, x_2 \geq 0.$

20. Max. $Z = 5x_1 + 3x_2$
subject to $3x_1 + 5x_2 \leq 15$
$5x_1 + 2x_2 \leq 10$
$x_1, x_2 \geq 0.$

21. Max. $Z = 2x_1 + 3x_2$
subject to $x_1 + x_2 \leq 1$
$3x_1 + x_2 \leq 4$
and $x_1, x_2 \geq 0.$

22. Max. $Z = 2x_1 + x_2$
subject to $x_1 + 2x_2 \leq 10,$
$x_1 + x_2 \leq 6$
$x_1 - x_2 \leq 2,$
$x_1 - 2x_2 \leq 1$
and $x_1, x_2 \geq 0.$

23. Obtain graphically the maximum value of
$Z = [\min(3x_1 - 10), \min(-5x_1 + 5)], 0 \leq x_1 \leq 5.$

24. A soft drink plant two bottling machines A and B. It produces and sells 8 ounce and 16 ounce bottles. The following data is available :

Machine	8 Ounce	16 Ounce
A	100/minute	40/minute
B	60/minute	75/minute

The machines can be run 8 hrs. per day, 5 days per week. Weekly production of the drinks cannot exceed 3,00,000 ounces and the market can absorb 25,000 eight ounce bottles and 7,000 sixteen ounce bottles per week. Profit on these bottles is 15 paise and 25 paise per bottle respectively. The planner wishes to maximize his profit subject to all the production and marketing restrictions. Formulate it as a linear programming problem and solve graphically.

25. The *ABC* Electric Appliance Company produces two products; refrigerators and coolers. Production takes place in two separate departments. Refrigerators are produced in Department I and coolers are produced in Department II. The company's two products are produced and sold on a weekly basis. The weekly production cannot exceed 25 refrigerators in Department I and 35 coolers in Department II, because of the limited available facilities in these two departments. The company regularly employs a total of 60 workers in the two departments. A refrigerator requires 2 man-week of labour, while a cooler requires 1 man-week of labour. A refrigerator contributes a profit of ₹ 60 and a cooler contributes a profit of ₹ 40. How many units of refrigerators and coolers should the company produce to realize maximum profit ?

26. A firm manufactures two types of products A and B and sells each of them at a profit of ₹ 2 per product. Each product is processed on two machines P and Q. Type A product requires one minute of processing time on machine P and two minutes on machine Q. Type B product requires one minute on machine P and one minute on machine Q. The machine P is available for not more than 6 hours and 40 minutes while machine Q is available for 10 hours during any working day.

Formulate the given problem as a LPP and find how many products of each type should the firm produce each day in order to get maximum profit.

27. An automobile manufacturer makes automobiles and trucks in a factory that is divided into two shops. Shop A, which performs the basic assembly operation, must work 5 man-days on each truck but only 2 man-days on each automobile. Shop B, which performs finishing operation, must work 3 man-days for each automobile or truck that it produces. Because of men and machine limitations, shop A has 180 man-days per week available while shop B has 135 man-days per week. If the manufacturer makes a profit of ₹ 300 on each truck and ₹ 200 on each automobile, how many of each should he produce to maximize his profit ?

28. A pineapple firm produces two products-canned pineapple and canned juice. The specific amounts of material, labour and equipment required to produce each product and the availability of each of these resources are shown in the table given below :

	Canned juice	**Canned Pineapple**	**Available Resources**
Labour (man-hour)	3	2.0	12.0
Equipment (machine hours)	1	2.3	6.9
Material (unit)	1	1.4	4.9

Assuming one unit each of canned juice and canned pineapple has profit margins of ₹ 2 and ₹ 1 respectively. Determine the product mix that will maximize the profit.

29. Diet Problem : Consider two different types of foodstuffs, say F_1 and F_2. Assume that these foodstuffs contain vitamins V_1, V_2 and V_3 respectively. Minimum daily requirement of three vitamins are 1 mg. of V_1, 50 mg. of V_2 and 10 mg. of V_3. Suppose that the food stuff F_1 contains 1 mg. of V_1, 100 mg of V_2 and 10 mg. of V_3. Whereas the food stuff F_2 contains 1 mg. of V_1, 10 mg. of V_2 100 mg. of V_3. Cost of one unit of foodstuff F_1 is ₹ 1 and that of F_2 is ₹ 1.5.

Find the minimum cost diet that would supply the body at least the minimum requirements of each vitamin by graphical method.

30. A company sells two different products A and B. The company makes a profit of ₹ 40 and ₹ 30 per unit on products A and B respectively. The products are produced in a common production process and are sold in two different markets. The production process has a capability of 30,000 **man-hours.** It takes 3 hours to produce one unit of A and one hour to produce one unit of B. The market has been surveyed, and company officials feel that the maximum number of units of A that can be sold is 8,000 and the maximum of B is 12,000 units. Subject to these limitations, the products can be sold in any convex combination. Formulate the above problem as a L.P.P. and solve it by graphical method.

31. A farm is engaged in breeding pigs. The pigs are fed on various products grown on the farm. In view of the need to ensure certain nutrient constituents, it is necessary to buy products (say A and B) in addition. The contents of the various products, per unit, in nutrients are vitamins, proteins etc. given in the following table :

Nutrients	Nutrient content in		Min. amount of nutrient
	A	B	
M_1	36	6	108
M_2	3	12	36
M_3	20	10	100

The last column of the above table gives the minimum amount of nutrient constituents M_1, M_2, M_3 which must be given to the pigs. If the products A and B cost ₹ 20 and ₹ 40 per unit respectively, how much each of these two products should be bought so that the total cost is minimized ?

32. A company produces two types of leather belts, say type A and B. Belt A is of superior quality and belt B is of a lower quality. Profits on the two types of belts are 40 and 30 paise per belt, respectively. Each belt of type A requires twice as much time as required by a belt of type B. If all belts were of type B, the company would produce 1000 belts per day. But the supply of leather is sufficient only for 800 per day. Belt A requires a fancy buckle and 400 fancy buckles are available for this, per day. For belt of type B, only 700 buckles are available per day. How should the company manufacture the two types of belts in order to have maximum overall profit?

33. A person requires 10, 12 and 12 units of chemicals A, B and C respectively for his garden. A liquid product contains 5, 2 and 1 units of A, B and C respectively per jar. A dry product contains 1, 2 and 4 units of A, B and C per carton. If the liquid product sells for ₹ 3 per jar and the dry product sells for ₹ 2 per carton, how many of each should be purchased to minimize the cost and meet the requirements?

34. A manufacturer has employed 5 skilled men and 10 semi-skilled men and makes an article in two qualities : deluxe model and an ordinary model. The making of a deluxe model requires 2 hrs work by a skilled man and 2 hrs work by a semi-skilled man. The ordinary model requires 1 hr by a skilled man and 3 hrs by a semi-skilled man. By union rules no man may work more than 8 hrs per day. The manufacturer's clear profit on deluxe model is ₹ 15 and on an ordinary model is ₹ 10. How many of each type should be made in order to maximize his total daily profit ?

Answers 2

1. Infinite number of solution. One is $x_1 = 1000,\ x_2 = 500;Z = 1500.$
2. $x_1 = 10{,}500,\ x_2 = 34{,}500;$ $Z = 3{,}25{,}500.$
3. $x_1 = \frac{3}{2},\ x_2 = \frac{1}{2};Z = 3.5.$
4. $x_1 = 44,\ x_2 = 16;Z = 440.$
5. $x_1 = 1.6,\ x_2 = 2.4;Z = 24.8.$
6. $x_1 = 3,\ x_2 = 2;Z = 19.$
7. $x_1 = 3,\ x_2 = 1;Z = 11.$
8. No feasible solution.
9. Infinite number of solutions.
10. Unbounded solution.
11. $x_1 = 3,\ x_2 = 4;Z = 10.$
12. $x_1 = \frac{400}{13},\ x_2 = 150/13;$ $Z = 1800/13.$
13. No feasible solution.
14. $x_1 = \frac{5}{4},\ x_2 = \frac{5}{4};Z = -10.$
15. No solution.
16. No feasible solution.
17. $x_1 = 5/2,\ x_2 = 1/4;Z = 73/4.$
18. $x_1 = 2,\ x_2 = 0;Z = -2.$
19. $x_1 = 0,\ x_2 = 1/3;Z = -2/3.$
20. $x_1 = \frac{20}{19},\ x_2 = \frac{45}{19};Z = \frac{235}{19}.$
21. $x_1 = 0,\ x_2 = 1;Z = 3.$
22. $x_1 = 4,\ x_2 = 2;Z = 10.$

23. Max. $Z = -10$ for $x_1 = 0$.

24. Let x_1 and x_2 be the number of bottles of 8 and 16 ounces respectively.
L.P.P. is Max. $Z = 0.15x_1 + 0.25x_2$
s.t. $8x_1 + 16x_2 \le 3{,}00{,}000$,
$\frac{x_1}{100} + \frac{x_2}{40} \le 2{,}400, \frac{x_1}{60} + \frac{x_2}{75} \le 2{,}400$,
$x_1 \le 25{,}000,\ x_2 \le 7{,}000$ and $x_1, x_2 \ge 0$.
Solution is $x_1 = 25{,}000,\ x_2 = 6{,}250, Z = 5{,}312.50$.

25. Refrigerators = 12.5, Coolers = 35 ; Max. profit = ₹ 2,150.

26. Let x_1 and x_2 be the numbers of products A and B to be manufactured.
L.P.P. is Max. $Z = 2x_1 + 2x_2$
s.t. $x_1 + x_2 \le 400,\ 2x_1 + x_2 \le 600$
and $x_1, x_2 \ge 0$.
Solutions is $x_1 = 200,\ x_2 = 200$
or $x_1 = 0,\ x_2 = 400$
or any point on line segment joining these two points. $Z =$ ₹ 800.

27. Trucks = 30, Automobiles = 15 per week : Max. Profit = ₹ 12,000

28. Canned juice = 4, Pineapple juice = 0 ; profit = ₹ 8.

29. $F_1 = 1$ unit, $F_2 = 0$ unit ; cost = ₹ 1.

30. If two products A and B be x_1 and x_2 units respectively then *L.P.P.* is
Max. $Z = 40x_1 + 30x_2$
s.t. $3x_1 + x_2 \le 30{,}000,\ x_1 \le 8{,}000,\ x_2 \le 12{,}000,\ x_1, x_2 \ge 0$.
Solution is $x_1 = 6{,}000,\ x_2 = 12{,}000,\ Z =$ ₹ 60,000

31. 4 units of product A, 2 units or product B ; cost = ₹ 160.

32. 200 belts of type A, 600 belts of type B ; profit = ₹ 260.

33. 1 unit of liquid product, 5 cartons of dry product, cost = ₹ 13.

34. Maximum profit is ₹ 350 when 20 ordinary and 10 deluxe models are made.

2.11 Slack and Surplus Variables

2.11.1 Slack Variables

If a constraint has a sing ≤, then in order to make it an equality we have to add some thing positive to the left hand side.

The non-negative variables which are added to L.H. sides of the constraints to convert them into equalities are called the slack variables.

For example, consider the linear programming problem

$$\text{Max. } Z = 2x_1 + 3x_2 + 4x_3$$

$$s.t. \quad \left.\begin{aligned} x_1 + x_2 + x_3 &\le b_1 \\ 2x_1 + 4x_2 - x_3 &\le b_2 \end{aligned}\right\} \qquad \ldots(1)$$

$$x_1, x_2, x_3 \ge 0.$$

In order to convert constraints (1) into equalities (equations) we add two non-negative variables x_4 and x_5 on L.H.S. of (1), then we have

$$x_1 + x_2 + x_3 + x_4 = b_1$$

and $\quad 2x_1 + 4x_2 - x_3 + x_5 = b_2.$

Hence x_4 and x_5 are called the **slack variables.**

2.11.2 Surplus Variables

If a constraint has a sign ≥, then in order to make it an equality we have to subtract something positive from its L.H.S. **The non-negative variables which are subtracted from the L.H. sides of the constraints to convert them into equalities are called the surplus variables.**

For example, consider the linear programming problem

$$\text{Max. } Z = 2x_1 + 4x_2 + 4x_3$$

$$s.t. \quad \left.\begin{aligned} x_1 + x_2 + 2x_3 &\le b_1 \\ 2x_1 + 4x_2 + 6x_3 &\ge b_2 \\ x_1 - 2x_2 + 4x_3 &\ge b_3 \end{aligned}\right\} \qquad \ldots(2)$$

The constraints (2) are inequalities. In order to convert them into equalities we have to add something in the L.H.S. of first and subtract something from the L.H. sides of the second and the third inequalities of (2), *i.e.*, we have following equalities

$$x_1 + x_2 + 2x_3 + x_4 = b_1$$

$$2x_1 + 4x_2 + 6x_3 - x_5 = b_2$$
$$x_1 - 2x_2 + 4x_3 - x_6 = b_3$$

Here, x_4 is the slack variable while x_5 and x_6 are the surplus variables.

2.12 Standard form of Linear Programming Problem

The standard form of a L.P.P. has the following characteristics :

1. The objective function is of maximization form.
2. All the constraints are equations.
3. The right hand side elements of all the constraints are non-negative.
4. All the variables are non-negative *i.e.*, ≥ 0

Working Rule to find the Standard form of a L.P.P. : To convert a L.P.P. to the standard form proceed stepwise as follows :

Step 1 : ***To convert the objective function to the maximization form :***

If the objective function is

$$\text{Min. } Z = f(x)$$

Then multiply both sides of the objective function by -1 and put $-Z = Z'$ to get the corresponding maximization form of the objective function.

i.e., the corresponding objective function in maximization form is

$$\text{Max. } Z' = -Z = -f(x)$$

Step 2 : ***To make right hand side element of each constraint non-negative, if not.***

If the right hand side of a constraint is negative, then it is converted to positive sign by multiplying both sides of this constraint by -1.

Remember that, if a constraint contain inequality sign (*i.e.*, $\geq$ or $\leq$ sign), then multiplication of both sides of it by -1 will reverse the sign of the inequality.

e.g., the multiplication of both sides by -1, the inequality $2x_1 - 3x_2 \geq -3$ reduces to $-2x_1 + 3x_2 \leq 3$.

Step 3 : ***To convert all the constraints to equations if not.***

All the constraints containing inequality sign one converted to equations by the introduction of non-negative variables called *slack* or *surplus variables.*

1. The constraint with $\leq$ sign, is reduced to an equation by adding a non-negative variable on the left land side.

e.g., the constraint $2x_1 + 5x_2 \le 7$ is reduced to equation $2x_1 + 5x_2 + x_3 = 7$, by adding the non-negative variable (called *slack variable*) on the left hand side.

2. The constraint with $\ge$ sign, is reduced to an equation by subtracting a non-negative variable from the left hand side.

   e.g., the constraint $3x_1 - 5x_2 \ge 9$ is reduced to equation $3x_1 - 5x_2 - x_4 = 9$, by subtracting the non-negative variable x_4 (called *surplus variable*) from the left hand side.

Step 4 : *To reduce the variable to non-negative values*

If a variable say x is unrestriced in sign *i.e.*, it can have positive, negative or zero value, then it is replaced by the difference of two non-negative variables.

i.e., write $x = x' - x''$ where $x' \ge 0$ and $x'' \ge 0$ in the objective function and in all equations obtained in step 3.

2.13 Matrix form of a Linear Programming Problem

A L.P. Problem in standard form can be expressed in matrix form as follows :

$$\text{Max. } Z = \mathbf{c}\,\mathbf{x}$$

subject to $\quad \mathbf{A}\,\mathbf{x} = \mathbf{b}, \qquad \boldsymbol{b} \ge \mathbf{0}$

and $\quad \boldsymbol{x} \ge \mathbf{0}$

where, $\mathbf{c}$ = row matrix (row vector) of coefficients of variables (given, slack and surplus) in objective function Z.

$\mathbf{x}$ = Column matrix (or column vector) of the variables (given, slack and surplus)

$\mathbf{b}$ = Column matrix (or column vector) of R.H.S elements in the constraint equations

and $\mathbf{A}$ = Matrix of the coefficients of variables in the constraint equations.

For clear understanding of the method, see the following examples.

Illustrative Examples

Example 28: *Convert the following L.P.P. into standard form.*

$$\text{Max. } Z = 2x_1 + 3x_2 + 5x_3$$

subject to,

$$5x_1 - 4x_2 + 3x_3 \le 7$$
$$2x_1 + 5x_2 - 4x_3 \ge 2$$
$$4x_1 + 3x_2 + 7x_3 \le 8$$

and $\quad x_1, x_2, x_3 \ge 0$

Also express the L.P.P. in the matrix form.

Solution: For clear understanding we shall convert the given L.P.P. into standard form stepwise as discussed in article 2.12.

Step 1 : Here the given objection function $Z = 2x_1 + 3x_2 + 5x_3$ is of maximization form.

Step 2 : In the given problem, right hand side element of each constraint is positive (non-negative)

Step 3 : Adding the slack variables x_4 and x_6 (both non-negative) in the first and third constraint respectively and subtracting the surplus variable x_5 (non-negative) from the second constraint, the constraint of the given L.P.P. as equations are

$$\begin{aligned} 5x_1 - 4x_2 + 3x_3 + x_4 &= 7 \\ 2x_1 + 5x_2 - 4x_3 - x_5 &= 2 \\ 4x_1 + 3x_2 + 7x_3 + x_6 &= 8 \end{aligned}$$

Step 4 : All the variables, $x_1, x_2, x_3, x_4, x_5, x_6$ are non-negative *i.e.*, ≥ 0.

Taking coefficients 0 for slack and surplus variables in the objective function Z, the *standard form* of the given L.P.P. is

$$\text{Max. } Z = 2x_1 + 3x_2 + 5x_3 + 0.x_4 + 0.x_5 + 0.x_6$$

subject to,

$$\begin{aligned} 5x_1 - 4x_2 + 3x_3 + x_4 &= 7 \\ 2x_1 + 5x_2 - 4x_3 - x_5 &= 2 \\ 4x_1 + 3x_2 + 7x_3 + x_6 &= 8 \end{aligned}$$

and $\quad x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$

$\therefore$ The **matrix form** of the L.P.P. is

$$\text{Max. } Z = \mathbf{c\,x}$$

subject to, $\quad \mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$

where, $\quad \mathbf{c} = [2\ \ 3\ \ 5\ \ 0\ \ 0\ \ 0]$

$$\mathbf{x} = [x_1\ \ x_2\ \ x_3\ \ x_4\ \ x_5\ \ x_6]^T,\ \mathbf{b} = [7\ \ 2\ \ 8]^T$$

and $$\mathbf{A} = \begin{bmatrix} 5 & -4 & 3 & 1 & 0 & 0 \\ 2 & 5 & -4 & 0 & -1 & 0 \\ 4 & 3 & 7 & 0 & 0 & 1 \end{bmatrix}$$

Example 29: *Convert the following L.P.P. into standard form.*

$$Min.\ Z = 2x_1 + x_2 + 4x_3$$

subject to, $-2x_1 + 4x_2 \leq 4$

$$x_1 + 2x_2 + x_3 \geq 5$$

$$2x_1 + 3x_3 \leq 2$$

and $x_1, x_2 \geq 0,\ x_3$ *unrestricted in sign.*

Also express the L.P.P. in the matrix form.

Solution: Stepwise solution of the given L.P.P. is as follows :

Step 1 : Taking Z' as the objective function in maximization form, we have

$$\text{Max. } Z' = -Z = -2x_1 - x_2 - 4x_3 \quad \text{...(1)}$$

Step 2 : In the given L.P.P. right hand side of each constraint is positive.

Step 3 : Adding the slack variables x_4 and x_6 (both non-negative) in first and third constraint respectively and subtracting the surplus variable (non-negative) from the second constraint, the constraints of the L.P.P. as equations are

$$\left.\begin{array}{r} -2x_1 + 4x_2 + x_4 = 4 \\ x_1 + 2x_2 + x_3 - x_5 = 5 \\ 2x_1 + 3x_3 + x_6 = 2 \end{array}\right\} \quad \text{...(2)}$$

Step 4 : Here x_3 is unrestricted in sign, so putting $x_3 = x_3' - x_3''$, where $x_3', x_3'' \geq 0$, in (1) and (2), the standard form of the given L.P.P. is

$$\text{Max. } Z' = -2x_1 - x_2 - 4x_3' + 4x_3'' + 0.x_4 + 0.x_5 + 0.x_6$$

subject to,
$$\begin{array}{rrrrrrrl} -2x_1 & + 4x_2 & & & +x_4 & & & = 4 \\ x_1 & + 2x_2 & + x_3' & - x'' & & -x_5 & & = 5 \\ 2x_1 & & + 3x_3' & - 3x_3'' & & & +x_6 & = 2 \end{array}$$

and $x_1, x_2, x_3', x_3'', x_4, x_5, x_6 \geq 0$

$\therefore$ The matrix form of the given L.P.P. is

$$\text{Max. } Z = \mathbf{c}\ \mathbf{x}$$

subject to $\mathbf{A}\,\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$

where, $\mathbf{c} = [-2 \quad -1 \quad -4 \quad 4 \quad 0 \quad 0 \quad 0]$

$$\mathbf{x} = [x_1 \quad x_2 \quad x_3' \quad x_3'' \quad x_4 \quad x_5 \quad x_6]^T, \quad \mathbf{b} = [4 \quad 5 \quad 2]^T$$

and $$\boldsymbol{A} = \begin{bmatrix} -2 & 4 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & -1 & 0 & -1 & 0 \\ 2 & 0 & 3 & -3 & 0 & 0 & 1 \end{bmatrix}.$$

Comprehensive Exercise 3

Reduce the following L.P.P. into standard form. Also find their matrix form.

1. Max. $Z = x_1 + 2x_2$
subject to, $2x_1 + 3x_2 \le 6$
$x_1 + 7x_2 \ge 4$
and $x_1, x_2 \ge 0$

2. Min. $Z = 12x_1 + 5x_2$
subject to $5x_1 + 3x_2 \ge 15$
$7x_1 - 2x_2 \le 14$
and $x_1, x_2 \ge 0$

3. Max. $Z = 2x_1 + 6x_2 + 5x_3$
subject to $5x_1 - 2x_2 + 3x_3 \le 9$
$x_1 + x_2 + x_3 \ge 5$
$x_1 + 2x_2 = 7$
and $x_1, x_2, x_3 \ge 0$

4. Max. $Z = 3x_1 + 5x_2 + 7x_3$
subject to $6x_1 - 4x_2 \le 5$
$3x_1 + 2x_2 + 5x_3 \ge 11$
$4x_1 + 3x_3 \le 2$
$x_1, x_2 \ge 0$ and x_3 unrestricted in sign

5. Min. $Z = x_1 - 2x_2 + x_3$
subject to $2x_1 + 3x_2 + 4x_3 \ge -4$
$3x_1 + 5x_2 + 2x_3 \ge 7$
$x_1 \ge 0,\ x_2 \ge 0,\ x_3$ in unrestricted in sign

Answers 3

1. Max. $Z = x_1 + 2x_2 + 0.x_3 + 0x_4$

subject to $2x_1 + 3x_2 + x_3 = 6$

$x_1 + 7x_2 - x_4 = 4$

and $x_1, x_2, x_3, x_4 \geq 0$

Matrix form Max. $Z = \mathbf{c}\ \mathbf{x}$

subject to $A\,\mathbf{x} = \mathbf{b}$ where

$\mathbf{c} = [1,\ 2,\ 0,\ 0]$,

$\mathbf{x} = [x_1, x_2, x_3, x_4]^T$

$\mathbf{b} = [6, 4]^T$,

$$A = \begin{bmatrix} 2 & 3 & 1 & 0 \\ 1 & 7 & 0 & -1 \end{bmatrix} \text{ and } \mathbf{x} \geq \mathbf{0}$$

2. Max.

$Z' = -12x_1 - 5x_2 + 0.x_3 + 0.x_4$

subject to $5x_1 + 3x_2 - x_3 = 15$

$7x_1 - 2x_2 + x_4 = 14$

and $x_1, x_2, x_3, x_4 \geq 0$

Matrix form Max. $Z' = \mathbf{c}\ \mathbf{x}$

subject to $A\ \mathbf{x} = \mathbf{b}$, where

$\mathbf{c} = [-12,\ -5,\ 0,\ 0]$,

$\mathbf{x} = [x_1, x_2, x_3, x_4]^T$

$\mathbf{b} = [15, 14]^T$,

$$A = \begin{bmatrix} 5 & 3 & -1 & 0 \\ 7 & -2 & 0 & 1 \end{bmatrix}, \mathbf{x} \geq \mathbf{0}$$

3. Max. $Z = 2x_1 + 6x_2 + 5x_3 + 0.x_4 + 0.x_5$

subject to

$5x_1 - 2x_2 + 3x_3 + x_4 = 9$

$x_1 + x_2 + x_3 - x_5 = 5$

$x_1 + 2x_2 = 7$

and $x_1, x_2, x_3, x_4, x_5 \geq 0$

Matrix form Max. $Z = \mathbf{c}\,\mathbf{x}$

subject to $A\ \mathbf{x} = \mathbf{b}$, where

$\mathbf{c} = [2,\ 6,\ 5,\ 0,\ 0]$,

$\mathbf{x} = [x_1, x_2, x_3, x_4, x_5]^T$

$\mathbf{b} = [9, 5, 7]^T$,

$$A = \begin{bmatrix} 5 & -2 & 3 & 1 & 0 \\ 1 & 1 & 1 & 0 & -1 \\ 1 & 2 & 0 & 0 & 0 \end{bmatrix}, \mathbf{x} \geq \mathbf{0}$$

4 Max.

$Z = 3x_1 + 5x_2 + 7x_3' - 7x_3'' + 0.x_4 + 0.x_5 + 0.x_6$

subject to $6x_1 - 4x_2 + x_4 = 5$

$3x_1 + 2x_2 + 5x_3' - 5x_3'' - x_5 = 11$

$4x_1 + 3x_3' - 3x_3'' + x_6 = 2$

and $x_1, x_2, x_3', x_3'', x_4, x_5, x_6 \geq 0$

Matrix form : Max. $Z = \mathbf{c}\ \mathbf{x}$

subject to $A\ \mathbf{x} = \mathbf{b}$, where

$\mathbf{c} = [3,\ 5,\ 7,\ -7,\ 0,\ 0,\ 0]$

$\mathbf{x} = [x_1, x_2, x_3', x_3'', x_4, x_5, x_6]^T$,

$\mathbf{b} = [5,\ 11,\ 2]^T$

$$A = \begin{bmatrix} 6 & -4 & 0 & 0 & 1 & 0 & 0 \\ 3 & 2 & 5 & -5 & 0 & -1 & 0 \\ 4 & 0 & 3 & -3 & 0 & 0 & 1 \end{bmatrix},$$

$\mathbf{x} \geq \mathbf{0}$

5. Max. $Z' = -x_1 + 2x_2 - x_3'$
$+x_3'' + 0.x_4 + 0.x_5$

s.t.
$-2x_1 - 3x_2 - 4x_3' + 4x_3'' + x_4 = 4$

$3x_1 + 5x_2 + 2x_3' - 2x_3'' - x_5 = 7$
and $x_1, x_2, x_3', x_3'', x_4, x_5 \geq 0$
Matrix form : Max. $Z' = \mathbf{c}\ \mathbf{x}$
subject to $A\,\mathbf{x} = \mathbf{b}$ where
$\mathbf{c} = [-1,\ 2,\ -1,\ 1,\ 0,\ 0]$
$\mathbf{x} = [x_1, x_2, x_3', x_3'', x_4, x_5]^T$,
$\mathbf{b} = [4,\ 7]^T$
$$A = \begin{bmatrix} -2 & -3 & -4 & 4 & 1 & 0 \\ 3 & 5 & 2 & -2 & 0 & -1 \end{bmatrix},$$
$\mathbf{x} \geq \mathbf{0}$

2.14 Basic Solution (B.S.)

Consider a system $A\mathbf{x} = \mathbf{b}$ of m equations in n unknowns $(n > m)$ and $r(A) = r(A\mathbf{b}) = m$ *i.e.*, none of the equations is redundant.

A solution obtained by setting any $(n - m)$ variables to zero is called a basic solution, provided the determinant of the coefficients of the remaining m variables is not zero.

Such m variables (any of them may be zero) are called **basic variables** and remaining $(n - m)$ zero valued variables are called **non-basic variables.**

Thus for a solution to be basic, at least $n - m$ variables must be zero.

Here we note that the matrix formed by the coefficients of the m basic variables, or say formed by the vectors associated to the basic variables is non-singular as its determinant does not vanish. *Hence the vectors associated to the basic variables are L.I.*

Thus *a solution in which the vectors associated associated to m variables are L.I. and remaining $n - m$ variables are zero is called a basic solution.*

Hence a basic solution can be constructed by selecting the m L.I. vectors out of n and setting the variables associated to the remaining $(n - m)$ columns to zero. If B is the matrix of m L.I. vectors of A and $\mathbf{x}_B$ is the column vector of the corresponding variables (basic variables), then the basic solution is given by

$$B\mathbf{x}_B = \mathbf{b} \quad \text{or} \quad \mathbf{x}_B = B^{-1}\mathbf{b}.$$

The number of basic solutions thus obtained will be at the most ${}^nC_m = \dfrac{n!}{m!(n-m)!}$ since m vectors out of n can be selected in ${}^nC_m$ ways. Note that a B.S. corresponds to some basis. *Basic solutions are of two types : Non-degenerate basic solutions and degenerate basic solutions.*

Degenerate Basic Solution : A basic solution is said to be *degenerate basic solution* if at least one of the basic variables is zero.

Non-degenerate Basic Solution : A basic solution is said to be *non-degenerate basic solution* if none of the basic variables is zero.

Thus a non-degenerate basic solution contains exactly m non-zero and $(n-m)$ zero variables.

2.15 Theorem

A necessary and sufficient condition for the existence and non-degeneracy of all the basic solutions of $A\mathbf{x} = \mathbf{b}$ is that every set of m columns of the augmented matrix $A\,\mathbf{b} = [A, \mathbf{b}]$ is L.I.

Proof : The condition is necessary : Suppose all the basic solutions of the system $A\mathbf{x} = \mathbf{b}$ exist and are non-degenerate. Therefore, every set of m-column vectors of A are L.I.

Let $\alpha_1, \alpha_2, \ldots, \alpha_m$ be one set of m-column vectors of A, then the given system gives

$$\alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_m x_m = \mathbf{b}.$$

Now since each solution is non-degenerate, $x_1 \neq 0$ and hence by replacement theorem vector $\mathbf{b}$ can replace α_1 in the basis $\alpha_1, \alpha_2, \ldots, \alpha_m$, Thus vectors $\mathbf{b}$, $\alpha_2, \alpha_3, \ldots \alpha_m$ also form a basis and hence they are L.I. Similarly $\mathbf{b}$ can replace α_2 as x_2 is also non-zero. Hence $a_1, \mathbf{b}, \alpha_3, \ldots, \alpha_m$ are also L.I. In a similar way we can show that $\mathbf{b}$ with any $(m-1)$ vectors of A forms a L.I. set. Hence every set of m columns of the augmented matrix $A\,\mathbf{b} = [A, \mathbf{b}]$ is L.I.

The Condition is Sufficient : Suppose every set of m column vectors of $A\,\mathbf{b} = [A, \mathbf{b}]$ is L.I. Then obviously every set of m-column vectors of A is also L.I. and hence all the basic solutions exist.

Now consider any m-column vectors $\alpha_1, \alpha_2, \ldots, \alpha_m$ of A which are L.I. and let $x_1, x_2, \ldots, x_m$ be the corresponding basic solution. Then

$$x_1\,\alpha_1 + x_2\,\alpha_2 + \ldots + x_m\,\alpha_m = \mathbf{b}$$

or $$-1.\mathbf{b} + x_1\,\alpha_1 + x_2\,\alpha_2 + \ldots + x_m\,\alpha_m = \mathbf{0}$$

Now if $x_1 = 0$, then we have

$$-1.\mathbf{b} + x_2\,\alpha_2 + \ldots + x_m\,\alpha_m = \mathbf{0},$$

which shows that the vectors $\mathbf{b}, \alpha_2, \ldots, \alpha_m$ are L.D.

But this is a contradiction as we have already assumed that m vectors of $[A, \mathbf{b}]$ are L.I.

Hence, $\quad x_1 \neq 0.$

Similarly, we can show that $x_2, \ldots, x_m$ are also different from zero *i.e.,* the solution is non-degenerate.

In this way, it can be shown that all the basic solutions are non-degenerate.

Corollary 1: The necessary and sufficient condition for any given basic solution $\mathbf{x}_B = B^{-1}\mathbf{b}$ to be non-degenerate is the linear independence of $\mathbf{b}$ and every $(m-1)$ columns from B.

Proof: The condition is necessary : We know that if a solution is non-degenerate $(x_i \neq 0, i = 1, 2, \ldots, m)$, $\mathbf{b}$ can replace any column of B and still a basis is maintained.

Hence $\mathbf{b}$ and any $(m-1)$ columns of B are L.I.

The condition is sufficient : Let $\mathbf{b}$ and any $(m-1)$ columns of B be L.I. Any column of B can be replaced by $\mathbf{b}$ and still a basis is maintained. Hence $x_1, \ldots, x_m$ are all non-zero.

2.16 Some Definitions

1. **A Basic Feasible Solution (B.F.S.) :** In a L.P.P. a feasible solution which is also basic is called a **basic feasible solution.** In other words, a feasible solution to a L.P.P. in which the vectors associated to non-zero variables (*basic variables*) are L.I. is called a **basic feasible solution.**

   In a L.P.P. having m constraints, at most m vectors may be L.I. Hence a B.F.S. cannot have more than m non-zero (*i.e.,* positive) variables. Thus for a F.S. to be a B.F.S. at least $(n-m)$ variables must vanish.

   B.F.S. are also finite in number and their maximum number is ${}^nC_m$, where n is the number of variables and m the number of constraints $(m \leq n)$.

2. **Basic Variables :** *The variables associated to B.F.S. are called* **basic variables.**

3. **Non-degenerate B.F.S. :** *A B.F.S. of a L.P.P. is said to be* **non-degenerate B.F.S.** *if none of the basic variables is zero.*

4. **Degenerate B.F.S. :** *A B.F.S. of a L.P.P. is said to be* **degenerate B.F.S.** *if at least one of the basic variables is zero.*

Illustrative Examples

Example 30: *Show that the feasible solution* $x_1 = 1, x_2 = 0, x_3 = 1$ *and* $Z = 6$ *to the system of equations*

$$x_1 + x_2 + x_3 = 2$$
$$x_1 - x_2 + x_3 = 2$$
$$x_i \geq 0$$

which minimizes $Z = 2x_1 + 3x_2 + 4x_3$ *is not basic.*

Solution: In the given F.S. there are only two non-zero variables namely x_1 and x_3. If vectors associated to these variables be α_1 and α_3 then the determinant of column vectors corresponding to variables x_1, x_3 is $\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1 - 1 = 0$

$\therefore$ the vectors α_1, α_3 are L.D.

Hence the solution is not a basic solution.

Example 31: *Find all the basic solutions of the following system :*

$$x_1 + 2x_2 + x_3 = 4$$
$$2x_1 + x_2 + 5x_3 = 5$$

and prove that they are non-degenerate.

Solution: The given system of equations can be expressed in matrix form as A $\mathbf{x} = \mathbf{b}$ where $A = (\alpha_1, \alpha_2, \alpha_3)$

$$\alpha_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Here n = number of variables = 3 and m = number of equations = 2. Hence there can be at most ${}^3C_2 = 3$ basic solutions.

Three sets of two vectors out of $\alpha_1, \alpha_2, \alpha_3$ are as follows :

$$B_1 = [\alpha_1, \alpha_2] = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, B_2 = [\alpha_1, \alpha_3] = \begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}, B_3 = [\alpha_2, \alpha_3] = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}$$

We have $\quad |B_1| = -3 \neq 0, |B_2| = 3 \neq 0, |B_3| = 9 \neq 0.$

If follows that every set of two vectors of A is L.I. Hence all the three basic solutions exist. If $\mathbf{x}_{B_1}, \mathbf{x}_{B_2}, \mathbf{x}_{B_3}$ are the vectors of corresponding basic variables, then

$$\mathbf{x}_{B_1} = B_1^{-1}\,\mathbf{b} = -\frac{1}{3}\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\mathbf{x}_{B_2} = B_2^{-1}\,\mathbf{b} = \frac{1}{3}\begin{bmatrix} 5 & -1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

$$\mathbf{x}_{B_3} = B_3^{-1}\,\mathbf{b} = \frac{1}{9}\begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 2/3 \end{bmatrix}$$

To find all the basic Solutions : In basis B_1, basic vectors are α_1, α_2. So

$$\mathbf{x}_{B_1} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \Rightarrow x_1 = 2,\ x_2 = 1.$$

These two variables are the basic variables and the remaining variable x_3 is non-basic. It will have the value zero. Thus the basic solution associated to the basis B_1 is given by (2, 1, 0). Similarly other basic solutions are (5, 0, – 1) and (0, 5/3, 2/3).

In all the three basic solutions none of the basic variables is zero, hence they are all non-degenerate.

Example 32: *Which of the following vectors* $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6, x_7]$ *is a B.F.S. of the system*

$$\begin{aligned} x_1 + 2x_2 + x_3 + 3x_4 + x_5 &= 9 \\ 2x_1 + x_2 + 3x_4 + x_6 &= 9 \\ -x_1 + x_2 + x_3 + x_7 &= 0, \\ x_1, x_2, \ldots, x_7 &\geq 0 \end{aligned}$$

(i) $\mathbf{x}_1 = [2,\ 2,\ 0,\ 1,\ 0,\ 0,\ 0]$ *(ii)* $\mathbf{x}_2 = [0,\ 0,\ 9,\ 0,\ 0,\ 9,\ -9]$

(iii) $\mathbf{x}_3 = [3,\ 3,\ 0,\ 0,\ 0,\ 0,\ 0]$ *(iv)* $\mathbf{x}_1 = [0,\ 0,\ 0,\ 0,\ 9,\ 9,\ 0]$

(v) $\mathbf{x}_5 = [1,\ 0,\ 0,\ 0,\ 8,\ 7,\ 1]$ *(vi)* $\mathbf{x}_6 = [0,\ 0,\ 0,\ 3,\ 0,\ 0,\ 0]$

Solution: We have denoted the column vector by symbol [ ].

(i) The vectors associated to non-zero variables are

$$\alpha_1 = [1, \; 2, \; -1], \alpha_2 = [2, \; 1, \; 1], \alpha_4 = [3, \; 3, \; 0].$$

Now $$|\alpha_1 \quad \alpha_2 \quad \alpha_4| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{vmatrix} = 0$$

$\therefore$ vectors $\alpha_1, \; \alpha_2, \; \alpha_4$ associated to non-zero variables are L.D.

Hence $\mathbf{x}_1$ is not a B.F.S.

(ii) The vectors associated to non-zero variables are

$$\alpha_3 = [1, \; 0, \; 1], \alpha_6 = [0 \quad 1, \; 0], \alpha_7 = [0, \; 0, \; 1].$$

Now $$|\alpha_3 \quad \alpha_6 \quad \alpha_7| = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 1 \neq 0$$

$\therefore$ vectors $\alpha_3, \; \alpha_6, \; \alpha_7$ are L.I.

Hence $\mathbf{x}_2$ is a basic solution but not a basic feasible solution as x_7 is negative.

(iii) This solution contains 5 zero variables. So it may be a B.F.S. Now vectors associated to non-zero variables are

$$\alpha_1 = [1, \; 2, \; -1], \alpha_2 = [2, \; 1, \; 1].$$

Now these vectors together with the vector

$\alpha_5 = [1, \; 0, \; 0]$ are L.I. as

$$|\alpha_1 \quad \alpha_2 \quad \alpha_5| = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ -1 & 1 & 0 \end{vmatrix} = 3 \neq 0$$

Hence $\mathbf{x}_3$ is a B.F.S. with x_1, x_2, x_5 as basic variables.

Since one of the basic variables namely x_5 vanishes, so this is a degenerate B.F.S.

(iv) This solution contains 5 zero variables. So it may be a B.F.S. Now vectors associated to non-zero variables are $\alpha_5 = [1, \; 0, \; 0], \alpha_6 = [0, \; 1, \; 0]$. Obviously these vectors together with $a_7 = [0, \; 0, \; 1]$ are L.I. Hence $\mathbf{x}_4$ is a

B.F.S. with x_5, x_6, x_7 as basic variables. Since one of the basic variables namely x_7 vanishes, so this is degenerate B.F.S.

(v) This solution does not contain alteast $7 - 3 = 4$ variables zero or it contains more than 3 (no. of equations) non-zero variables, so it is not a B.F.S.

(vi) This solution contains one non-zero variable. The vector associated to this variable is $a_4 = [3, \; 3, \; 0]$.

We have $$|\alpha_4 \quad \alpha_6 \quad \alpha_7| = \begin{vmatrix} 3 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 3 \neq 0$$

$\therefore$ vectors $\alpha_4, \alpha_6, \alpha_7$ are L.I.

Hence $\mathbf{x}_6$ is a B.F.S. with x_4, x_6, x_7 as basic variables. Since basic variables x_6, x_7 also vanish, so it is a degenerate B.F.S.

2.17 Analytic Method (Trial and Error Method)

As already stated, the Graphical method can be used to solve the L.P.P. in two variables only. L.P.P. in three variables can also be solved by Graphical method but it is complicated enough. Simplex method (discussed in chapter 3) is the powerful method (or technique) to solve a L.P.P. in any number of variables.

Here we give *analytic method* (*trial and error method*) to solve a L.P.P. We know that for a system of m equations in n unknowns ($n > m$), a solution in which at least $n - m$ variables have their values equal to zero is called a *basic solution.* These $(n - m)$ variables are called *non-basic variables.* The remaining m variables whose values may or may not be equal to zero are called basic variables, the vectors in the coefficient matrix corresponding to these basic variables should be L.I.

In analytical method giving zero values to the non-basic variables in the given equations, we get equations in basic variables. Solving these equations we can get the basic solutions of the problem. Then the basic solutions **(B.S.)** which also satisfy the condition $x_i \geq 0, \forall \, i$ called the *basic feasible solutions* **(B.F.S.)** are obtained. For all these B.F. solutions the values of objective function Z are computed and the *B.F.S.* giving the optimal value (maximum or minimum value as required) of Z is taken as the optimal solution of the given L.P.P.

Drawbacks of Analytical Method

1. For large values of m and n, it is extremely difficult and time consuming to solve various sets of simultaneous equations.
2. Some of the sets given infeasible solutions. There is no technique to detect all such sets and avoid to solve them.

For clear understanding of the method, see following illustrative examples.

Illustrative Examples

Example 33: *Find an optimal solution of the following L.P.P. without using the simlex algorithm.*

$$\text{Max. } Z = 2x_1 + 3x_2 + 4x_3 + 7x_4$$

$$\text{subject to} \quad 2x_1 + 3x_2 - x_3 + 4x_4 = 8$$

$$x_1 - 2x_2 + 6x_3 - 7x_4 = -3$$

$$\text{and} \quad x_i \geq 0,\ i = 1, 2, 3, 4.$$

Solution: In matrix form the above system of equations can be written as

$$A\,\mathbf{x} = \mathbf{b}$$

where
$$A = \begin{bmatrix} 2 & 3 & -1 & 4 \\ 1 & -2 & 6 & -7 \end{bmatrix} = (\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4),$$

where
$$\alpha_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \alpha_3 = \begin{bmatrix} -1 \\ 6 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 4 \\ -7 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 8 \\ -3 \end{bmatrix}.$$

Here order of A is 2×4 *i.e.*, $m = 2$, $n = 4$.

Since $n - m = 4 - 2 = 2$, so the basic solutions are obtained by taking 2 variables equal to zero at a time. In this problem out of 4 variables 2 variables can be taken equal to zero in ${}^4C_2 = 6$ number of ways. Hence the problem can have at the most 6 basic solutions, which can be computed as follows :

S.N.	Basic variables	Non-basic variables	Equation to obtain B.S.	Basic solution (If exists)	Is it B.F.S.	For B.F.S. value of Z
1.	x_1, x_2	$x_3 = x_4 = 0$	$2x_1 + 3x_2 = 8$ $x_1 - 2x_2 = -3$	$x_1 = 1$ $x_2 = 2$ $x_3 = x_4 = 0$	Yes.	5
2.	x_1, x_3	$x_2 = x_4 = 0$	$2x_1 - x_3 = 8$ $x_1 + 6x_3 = -3$	$x_1 = 45/13$ $x_3 = -14/13$ $x_2 = x_4 = 0$	No.	—
3.	x_1, x_4	$x_2 = x_3 = 0$	$2x_1 + 4x_4 = 8$ $x_1 - 7x_4 = -3$	$x_1 = 22/9$ $x_4 = 7/9$ $x_2 = x_3 = 0$	Yes.	31/3
4.	x_2, x_3	$x_1 = x_4 = 0$	$3x_2 - x_3 = 8$ $-2x_2 + 6x_3 = -3$	$x_2 = 45/16$ $x_3 = 7/16$ $x_1 = x_4 = 0$	Yes.	163/16
5.	x_2, x_4	$x_1 = x_3 = 0$	$3x_2 + 4x_4 = 8$ $-2x_2 - 7x_4 = -3$	$x_2 = 44/13$ $x_4 = -7/13$ $x_1 = x_3 = 0$	No.	—
6.	x_3, x_4	$x_1 = x_2 = 0$	$-x_3 + 4x_4 = 8$ $6x_3 - 7x_4 = -3$	$x_3 = 44/17$ $x_4 = 45/17$ $x_1 = x_2 = 0$	Yes.	491/17 (Max.)

Hence the optimal solution of the given L.P.P. is

$x_1 = x_2 = 0$, $x_3 = 44/17$, $x_4 = 45/17$ and maximum $Z = 491/17$.

Example 34: *Find all the optimal B.F. solutions of the following L.P.P. without using the simplex method.*

$$\text{Max. } Z = 6x_1 + 2x_2 + 9$$

subject to $3x_1 + x_2 \le 6$

$$x_1 + 3x_2 \le 9$$

and $x_1, x_2 \ge 0.$

Solution: Introducing the slack variables x_3 and x_4, the problem reduces to

$$\text{Max. } Z = 6x_1 + 2x_2 + 9$$

subject to $3x_1 + x_2 + x_3 = 6$

$$x_1 + 3x_2 + x_4 = 6$$

and $\quad x_i \geq 0, \forall\ i = 1, 2, 3, 4.$

In matrix form the above system of equations can be written as

$$A\,\mathbf{x} = \mathbf{b}$$

where $\quad A = \begin{bmatrix} 3 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{bmatrix} = (\alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4).$

Here $\quad \alpha_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}.$

Here $n = 4$ and $m = 2$. So the basic solutions are obtained by taking $n - m = 4 - 2 = 2$, variables equal to zero at a time. Here out of 4 variables 2 can be taken to zero in ${}^4C_2 = 6$ number of ways.

Hence the problem can have at the most 6 basic solutions which may be B.F.S. also. These solutions are computed as follows :

S.N.	Basic variables	Non-basic variables	Equations to obtain B.S.	Basic solution (If exists)	Is it B.F.S.	For B.F.S. value of Z
1.	x_1, x_2	$x_3 = x_4 = 0$	$3x_1 + x_2 = 6$ $x_1 + 3x_2 = 6$	$x_1 = 3/2, x_2 = 3/2$ $x_3 = 0, x_4 = 0$	Yes.	21 (Max.)
2.	x_1, x_3	$x_2 = x_4 = 0$	$3x_1 + x_3 = 6$ $x_1 = 6$	$x_1 = 6, x_3 = -12$ $x_2 = 0, x_4 = 0$	No.	—
3.	x_1, x_4	$x_2 = x_3 = 0$	$3x_1 = 6$ $x_1 + x_4 = 6$	$x_1 = 2, x_4 = 4$ $x_2 = 0, x_3 = 0$	Yes.	21 (Max.)
4.	x_2, x_3	$x_1 = x_4 = 0$	$x_2 + x_3 = 6$ $3x_2 = 6$	$x_2 = 2, x_3 = 4$ $x_1 = 0, x_4 = 0$	Yes.	13
5.	x_2, x_4	$x_1 = x_3 = 0$	$x_2 = 6$ $3x_2 + x_4 = 6$	$x_2 = 6, x_4 = -12$	No.	—
6.	x_3, x_4	$x_1 = x_2 = 0$	$x_3 = 6$ $x_4 = 6$	$x_3 = 6, x_4 = 6$ $x_1 = 0, x_2 = 0$	Yes.	9

Here we obtain two optimal basic feasible solutions of the problem

(i) $x_1 = 3/2, x_2 = 3/2$ and (ii) $x_1 = 2, x_2 = 0$

For both B.F.S.'s. Optimal value of $Z = 21$

Note : If we solve this L.P.P. by Graphical method then we see that every point on the line segment joining points (2, 0) and (3/2, 3/2) gives B.F.S. and optimal $Z = 21$.

Comprehensive Exercise 4

1. Find all basic solutions for the system of simultaneous equations :

$$2x_1 + 3x_2 + 4x_3 = 5$$
$$3x_1 + 4x_2 + 5x_3 = 6.$$

2. Determine all basic feasible solutions of the system of equations :

$$2x_1 + x_2 + 4x_3 = 11$$
$$3x_1 + x_2 + 5x_3 = 14.$$

3. Is $x_1 = 1, x_2 = \frac{1}{2}, x_3 = x_4 = x_5 = 0$ a basic solution to the following equations :

$$x_1 + 2x_2 + x_3 + x_4 = 2,\ x_1 + 2x_2 + \frac{1}{2}x_3 + x_5 = 2.$$

4. If $a_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, a_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, a_3 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

then determine whether all possible basic solutions exist for the following set of equations :

$$[a_1 \quad a_2 \quad a_3]\,\mathbf{x} = \mathbf{b}.$$

5. Find a B.F.S. of the system

$$x_1 + 2x_2 + 3x_3 + 4x_4 = 7$$
$$2x_1 + x_2 + x_3 + 2x_4 = 3,$$
$$x_1, x_2, x_3, x_3 \ge 0$$

6. Do the system $x_1 + x_2 = 1,\ x_1 - x_3 = 2,\ x_1, x_2, x_3 \ge 0$ have a F.S.

7. Find all the B.F.S. of the following system :

$$8x_1 + 6x_2 + 13x_3 + x_4 + x_5 = 6$$
$$9x_1 + x_2 + 2x_3 + 6x_4 + 10x_5 = 10.$$

8. Determine two different B.F.S. of the L.P.P.

$$\text{Max. } Z = 2x_1 + 3x_2 + x_3 + 0.x_4 + 0.x_5 - M.x_6$$

subject to $2x_1 + 3x_2 + 4x_3 + x_4 = 6$

$$x_1 + 2x_2 + 2x_3 - x_5 + x_6 = 3$$

and $x_1, x_2, x_3, x_4, x_5, x_6 \ge 0.$

Answers 4

1. $(0, -1, 2), \left(-\frac{1}{2}, 0, \frac{3}{2}\right), (-2, 3, 0).$
2. (i) $x_1 = 3, x_2 = 5, x_3 = 0$ (ii) $x_1 = \frac{1}{2}, x_2 = 0, x_3 = \frac{5}{2}.$
3. No
4. Yes
5. (−1/3, 11/3, 0, 0), (2/5, 0, 11/5, 0), (−1/3, 0, 0, 11/6), (0, 2, 1, 0), (0, 0, 1, 1).
6. No
7. $\left(\frac{2}{3}, 0, 0, \frac{2}{3}, 0\right), \left(\frac{50}{71}, 0, 0, 0, \frac{26}{71}\right), \left(0, \frac{26}{35}, 0, \frac{54}{35}, 0\right), \left(0, \frac{50}{59}, 0, 0, \frac{54}{59}\right),$ $\left(0, 0, \frac{13}{38}, \frac{59}{38}, 0\right), \left(0, 0, \frac{25}{64}, 0, \frac{59}{64}\right).$
8. (0, 2, 0, 0, 1, 0) and (3, 0, 0, 0, 0, 0). Max. $Z = 26$.

2.18 Applications of Linear programming Technique

In this section we discuss important applications of linear programming in our life.

1. **Personnel Assignment Problem :** Linear programming technique can be used to allocate the available man power to various jobs in the best way (which minimizes the total cost or maximizes the total profit).
2. **Transportation Problem :** This type of problem occurs very frequently in practical life or say matches with many real situations. In general transportation problems are concerned with the distributions of certain products from several sources to number of destinations at minimum cost.
3. **Production Management :** Linear programming can be applied in production management for determining product mix, product smoothing and assembly time balancing.
4. **Agricultural Applications :** Linear programming can be applied in agricultural planning for allocating the available limited resources such as acreage, labour, water supply and working capital to the various crops so that the total net revenue may be maximum.
5. **Military Applications :** Linear programming technique may be applied in selecting an air weapon system against gurillas so as to keep them pinned down utilising the minimum amount of aviation gasoline ; in maximizing

the total tonnage of bombs dropped on the enemy's targets ; in determining the transportation schedule of the essential items such as weapons, medicines etc. in least time ; in determining the number of defence units used in the attack in order to provide the required level of protection at the lowest possible cost.

6. **Blending Problems :** Such problems arise when a new product is to be prepared by mixing the certain other things, maintaining some composition formula of the new product. By using the L.P. technique, we can determine the most economical blend of the new product subject to the availability of the raw materials and the minimum or maximum limitations of the constituents. Such problems occur frequently in the petroleum, paint and steel industries etc.

7. **Marketing Management :** Linear programming helps in analysing the effectiveness of advertising compaign and time based on the available advertising media. It also helps traveling salesman in finding the shortest route for his tour.

8. **Feed Mix :** The problem of finding the best diet *i.e.,* the combination of food that can be supplied at minimum cost, satisfying the daily requirement of the nutrients is also a L.P.P.

9. **Trim Loss Problems :** If an item is made in a particular size by cutting it from a sheet of standard size, the optimum combination of requirements can be determined so that trim loss may be minimum.

10. **Efficiencing on Operation of System of Dams :** In this problem, we determine variations in water storage of dams which generate power so as to maximize the energy obtained from the entire system. The physical limitations of storage appear as inequalities.

2.19 Limitations of Linear Programming

In spite of wide area of applications, in practical situations, the linear programming technique has certain limitations as follows :

1. The L.P. technique is applicable to that class of problems in which the objective function and the constraints both are linear. But in many practical problems, it is not possible to express both the objective function and the constraints in linear form.
2. There is no guarantee of having integral valued solutions. For example, in finding out how many men and machines would be required to perform a particular job, rounding off the solution to the nearest integer will not give an optimal solution.
3. The effect of time and uncertainty is not considered in linear programming model. Thus the model should be defined in such a way that any change due to internal as well as external factors can be incorporated.

4. Parameters appearing in the model are assumed to be constant while in real life situations they are neither constant nor deterministic.
5. Linear programming deals with only single objective whereas in real life situations problems occur with multi objectives.
6. All practical L.P.P. involve extensive calculations and hence computer facility is required. In many situations the computations become tedious even on large digital computers.

2.20 Advantages of Linear Programming

It indicates how the available resources can be used in the best way. It improves the quality of decisions. Thus it helps in attaining the optimum use of productive resources and man-power. It also reflects the drawbacks of the production process. The modification in mathematical solution is also possible by using linear programming method.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The extreme point of the convex set of feasible solutions of the L.P.P. Max $Z = 10x_1 + 15x_2$ s.t. $x_1 + x_2 = 2$, $3x_1 + 2x_2 \le 6$, $x_1, x_2 \ge 0$ are

(a) (2, 0), (0, 2) (b) (2, 0), (0, 3)
(c) (0, 2), (0, 3) (d) (0, 0), (0, 3)

2. If there is no feasible region in a L.P.P. then we say that the problem has

(a) infinite solutions (b) no solution
(c) unbounded solution (d) none of these

3. The solution of the L.P.P. Max. $Z = 5x_1 + 7x_2$ s.t. $3x_1 + 2x_2 \le 12$, $2x_1 + 3x_2 \le 13$, $x_1, x_2 \ge 0$ is

(a) $x_1 = 2, x_2 = 3, Z = 31$ (b) $x_1 = 4, x_2 = 0, Z = 20$
(c) $x_1 = 0, x_2 = \frac{13}{3}, Z = \frac{91}{3}$ (d) none of these

4. Given the following set of equations :
$x_1 + 4x_2 - x_3 = 3$, $5x_1 + 2x_2 + 3x_3 = 4$
The B.F.S. involving x_1 and x_2 is

(a) $\left(\frac{5}{9}, \frac{11}{18}, 0\right)$ (b) $\left(\frac{5}{9}, 0, 0\right)$
(c) $\left(0, \frac{11}{18}, 0\right)$ (d) none of these

5. The maximum number of basic solutions to a set of m simultaneous equations in n unknowns $(n \geq m)$ is

(a) m (b) $n - m$

(c) ${}^{n}C_{m}$ (d) none of these

Fill in the Blank

Fill in the blanks "……" so that the following statements are complete and correct.

1. In L.P.P. the function which is to be optimized is called an .... function.
2. The linear inequations (or equations) under which the objective function is to be optimized are called ....
3. A set of values of the variables $x_1, x_2, \ldots, x_n$ satisfying the constraints and non-negative restrictions of a L.P.P. is called a .... solution of the L.P.P.
4. If the value of the objective function Z can be increased or decreased indefinitely, such solutions are called .... solutions.
5. A necessary and sufficient condition for the existence and non-degeneracy of all the basic solutions of $A\mathbf{x} = \mathbf{b}$ (m equations in an unknowns), is that every set of m columns of the augmented matrix $[A, \mathbf{b}]$ is ....
6. The non-negative variables which are added to L.H.S. to the constraints to convert them into equalities are called the .... variables.
7. The non-negative variables which are subtracted from the L.H.S. of the constraints to convert them into equalities are called the .... variables.
8. The non-basic variables in a basic solution are .... valued variables.
9. A feasible solution to a L.P.P. in which the vectors associated to non-zero variables are .... is called a B.F.S.
10. A L.P.P. of .... variables can be solved graphically.

True or False

Write 'T' for true and 'F' for false statement.

1. A basic feasible solution is said to be optimum, if it optimizes the objective function.
2. A basic feasible solution is called degenerate if one or more basic variables are zero-valued.
3. The L.P.P. optimize $Z = x_1 + x_2$, s.t. $x_1 + x_2 \leq 1$, $3x_1 + x_2 \geq 3$, $x_1, x_2 \geq 0$ has single point as its feasible region.
4. The L.P.P. Max. $Z = x_1 + x_2$, s.t. $x_1 - x_2 \geq 0$, $3x_1 - x_2 \leq -3$, $x_1, x_2 \geq 0$ has a feasible solution.
5. By graphical method any L.P.P. can be solved easily.

Answers

Multiple Choice Questions

1.	(a)	**2.**	(b)	**3.**	(a)	**4.**	(a)	**5.**	(c)

Fill in the Blank

1.	Objective	**2.**	Constraints
3.	Feasible	**4.**	Unbounded
5.	Linearly independent	**6.**	Slack
7.	Surplus	**8.**	Zero
9.	L.I.	**10.**	Two

True or False

1.	*T*	**2.**	*T*	**3.**	*T*	**4.**	*F*	**5.**	*F*

□

Chapter

3

Simplex Method
(Theory and Application)

3.1 Introduction

In chapter 2 we have discussed the formulation of linear programming problems and the graphic method of solving them. It was observed in the graphical approach to the solution of L.P.P. that in a given situation the set of constraints determines the feasible region and objective function gives the optimal point, the one that minimizes or maximizes, as the case may be. But the graphical method suffers from a great limitation that it can handle problems involving only two decision variables. Whereas in real world situations, we frequently encounter such problems where more than two variables are involved.

It is sometimes impossible or requires great labour to search for optimal solution from amongst all the feasible solutions which may be infinite in number. Fundamental theorem of linear programming makes it easy to find an optimal solution because it deals with basic feasible solutions only. But it is also not an easy job to enumerate all the B.F.S. even for small values of m (number of constraints) and n (number of variables). To overcome this difficulty **Simplex**

Method or **Simplex Algorithm** was developed by **George Dantzig** in 1947. This method provides an efficient technique which can be applied for solving linear programming problems of any magnitude (involving two or more decision variables).

The simplex method is an iterative procedure, for finding, in a systematic manner the optimal solution to a linear programming problem. The simplex algorithm according to its iterative search selects the optimal solution from among the set of feasible solutions to the problem.

The simplex method consists of the following main steps (iterations) :

1. Finding a trial B.F.S. to given problem.
2. Testing whether this B.F.S. is optimal or not.
3. Improving the first trial solution (if it is not optimal) by a set of rules and repeating the above procedure till an optimal solution is attained.

In simplex method, we concentrate on maximization problem only because we can easily convert the minimization problem to maximization.

In this chapter, we shall deal with only those problems for which initial basic feasible solution is **non-degenerate**.

3.2 Some Definitions and Notations

Here we shall introduce some notations and definitions which are extremely useful in the discussion of simplex algorithm.

Consider a linear programming problem which after introducing slack and surplus variables is as follows :

$$\text{Max. } Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n + 0x_{n+1} + 0x_{n+2} + \ldots + 0x_{n+m}$$

subject to

$$\begin{aligned}
&a_{11} x_1 + a_{12} x_2 + \ldots + a_{1j} x_j + \ldots + a_{1n} x_n + x_{n+1} = b_1\\
&a_{21} x_1 + a_{22} x_2 + \ldots + a_{2j} x_j + \ldots + a_{2n} x_n + x_{n+2} = b_2\\
&\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots\\
&\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots\\
&a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mj} x_j + \ldots + a_{mn} x_n + x_{n+m} = b_m,\\
&x_i \geq 0 \text{ for all } i = 1, 2, \ldots, N, \text{ where } N = n + m
\end{aligned}$$

$b_1, b_2, \ldots, b_m$ are all positive.

The above linear programming problem can be easily converted to matrix form

$$\text{Max. } Z = \mathbf{c}\,\mathbf{x}$$

subject to $\quad A\mathbf{x} = \mathbf{b},\ \mathbf{x} \geq \mathbf{0},$

where $\quad A = [a_{ij}]_{m \times N}$ is the coefficient matrix of order $m \times N$,

$$\mathbf{x} = [x_1, x_2, x_3, \ldots, x_n, \ldots, x_N]_{N \times 1}$$

$$\mathbf{c} = (c_1, c_2, \ldots, c_n, 0, 0, \ldots, 0)_{1 \times N}$$

and $\quad \mathbf{b} = [b_1, b_2, \ldots, b_m]_{m \times 1}$

where $\mathbf{c}$ is a row vector of order $1 \times N$, $\mathbf{x}$ and $\mathbf{b}$ are column vectors of order $N \times 1$ and $m \times 1$ respectively.

If we denote the j-th column of the matrix A by α_j, $j = 1, 2, \ldots, N$, we can write

$$A = (\alpha_1, \alpha_2, \ldots, \alpha_N)$$

Let B be a non-singular sub-matrix of A of order $m \times m$ whose column vectors are m linearly independent columns selected from A. If we denote these columns by $\beta_1, \beta_2, \ldots, \beta_m$, then

$$B = (\beta_1, \beta_2, \ldots, \beta_m)$$

and is called the **basis matrix.**

Since the vectors $\beta_1, \beta_2, \ldots, \beta_m$ are linearly independent they form a basis for E^m.

Therefore each $\alpha_j \in A \subset E^m$ can be expressed as a linear combination of vectors of B. Thus, we can write

$$\alpha_j = \beta_1 y_{1j} + \beta_2 y_{2j} + \ldots + \beta_m y_{mj}$$

$$= (\beta_1, \beta_2, \ldots, \beta_m) \begin{bmatrix} y_{1j} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix}$$

or $$\alpha_j = BY_j, \quad \text{where} \quad Y_j = \begin{bmatrix} y_{1j} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix}_{m \times 1},$$

$y_{1j}, y_{2j}, \ldots, y_{mj}$ are the scalars required to express α_j in such a form.

Also $\quad \alpha_j = BY_j \Rightarrow Y_j = B^{-1}\alpha_j.$

The vector Y_j will change if the columns of A forming B change.

Any basis matrix B will provide a **basic solution** to $A\mathbf{x} = \mathbf{b}$.

The variables corresponding to $\beta_1, \beta_2, \ldots, \beta_m$ are denoted by $x_{B1}, x_{B2}, \ldots, x_{Bm}$ respectively and are called **basic variables**.

We denote the column vector of these m basic variables by $\mathbf{x}_B$

$\therefore$ $\quad\quad \mathbf{x}_B = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$

and $\quad\quad \mathbf{x}_B = B^{-1}\,\mathbf{b}.$

This is called the **Basic Feasible Solution (B.F.S.)** of the L.P.P.

Since $x_{B1}, x_{B2}, \ldots, x_{Bm}$ are the basic variables, the remaining $N - m$ variables belonging to $\mathbf{x}$ are called **non-basic variables.**

We shall denote the coefficients of basic variables $x_{B1}, x_{B2}, \ldots, x_{Bm}$ in the objective function Z by $c_{B1}, c_{B2}, \ldots, c_{Bm}$.

Corresponding to any $\mathbf{x}_B$, $\mathbf{c}_B$ will represent the row vector containing the constants $c_{B1}, c_{B2}, \ldots, c_{Bm}$.

$\therefore$ $\quad\quad \mathbf{c}_B = (c_{B1}, c_{B2}, \ldots, c_{Bm}).$

Since for any basic feasible solution all non-basic variables are zero therefore the objective function Z becomes

$$Z = c_{B1}x_{B1} + c_{B2}x_{B2} + \ldots + c_{Bm}x_{Bm} + \mathbf{0}$$

$$= (c_{B1}, c_{B2}, \ldots, c_{Bm})[x_{B1}, x_{B2}, \ldots, x_{Bm}]$$

$$= \mathbf{c}_B\,\mathbf{x}_B.$$

Finally, we introduce a new variable Z_j, given by

$$\begin{aligned} Z_j &= y_{1j}\,c_{B1} + y_{2j}\,c_{B2} + \ldots + y_{mj}\,c_{Bm} \\ &= (c_{B1}, c_{B2}, \ldots, c_{Bm})\;[y_{1j}, y_{2j}, \ldots, y_{mj}] \\ &= \mathbf{c}_B Y_j. \end{aligned}$$

There exists Z_j for each α_j and it changes as the columns of A forming B change.

Note : For convenience, we shall represent column vectors by [ ] without using transpose symbol and row vector by ( ). There should be no confusion in understanding scalar multiplication of two vectors $\mathbf{c}$ and $\mathbf{x}$ which is written as $\mathbf{c}\,\mathbf{x}$ in place of $\mathbf{c}.\mathbf{x}$.

3.3 Fundamental Theorem of Linear Programming

Theorem : *If a linear programming problem*

$$\textit{Max. } Z = \mathbf{c}\,\mathbf{x} \textit{ subject to } A\mathbf{x} = \mathbf{b},\; \mathbf{x} \geq 0,$$

where A is an $m \times N, (N = m + n)$ matrix of coefficients given by $A = (\alpha_1, \alpha_2, \ldots, \alpha_N)$ has an optimal feasible solution, then at least one basic feasible solution must be optimal.

Proof: Let $\mathbf{x}^* = [x_1, x_2, \ldots\ldots, x_N]$

be an optimal feasible solution of the given linear programming problem and

$$Z^* = \sum_{i=1}^{N} c_i x_i$$

be the corresponding optimum value of the objective function.

Also suppose that k $(k \le N)$ variables in $\mathbf{x}^*$ are non-zero and the remaining $N - k$ variables are zero. For the sake of convenience, we can assume that the first k variables of $\mathbf{x}^*$ are non-zero.

Thus $$\mathbf{x}^* = [x_1, x_2, \ldots, x_k, 0, \overbrace{0, \ldots, 0}^{N-k}]$$

$\therefore$ $$\sum_{i=1}^{k} x_i \alpha_i = \mathbf{b} \qquad \ldots(1)$$

and $$Z^* = \sum_{i=1}^{k} c_i x_i. \qquad \ldots(2)$$

Now two possibilities may arise :

The vectors $\alpha_1, \alpha_2, \ldots, \alpha_k$ may be either linearly independent or dependent.

Case 1: If $\alpha_1, \alpha_2, \ldots, \alpha_k$ are linearly independent.

Any feasible solution for which the vectors α_i, $i = 1, 2, \ldots, k$ associated with non-zero variables x_i, $i = 1, 2, \ldots, k$ are L.I. is called a basic feasible solution. Thus $\mathbf{x}^*$ is a B.F.S. which is also optimal.

Hence the result of the theorem is true.

The solution $\mathbf{x}^*$ is **degenerate** if $k < m$ and **non-degenerate** if $k = m$.

Case 2: If $\alpha_1, \alpha_2, \ldots, \alpha_k$ are linearly dependent and $k > m$.

In this case, we shall reduce the number of non-zero variables step by step until we obtain a B.F.S. from feasible solution.

Since $\alpha_1, \alpha_2, \ldots, \alpha_k$ are linearly dependent therefore there exist scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$ such that

$$\lambda_1 \alpha_1 + \lambda_2 \alpha_2 + \ldots + \lambda_k \alpha_k = \mathbf{0}$$

or $$\sum_{i=1}^{k} \lambda_i \alpha_i = \mathbf{0} \qquad \ldots(3)$$

and at least one $\lambda_i \ne 0$.

We can assume that at least one λ_i is positive because if none is positive, then we can multiply the equation (3) by -1 and get positive λ_i.

Now, suppose

$$v = \max_{1 \le i \le k} \left(\frac{\lambda_i}{x_i} \right). \qquad \text{...(4)}$$

Obviously v is positive since $x_i > 0$, $i = 1, 2, \ldots, k$ and at least one λ_i is positive.

Multiplying both sides of (3) by $\frac{1}{v}$ and subtracting from (1), we get

$$\sum_{i=1}^{k} \left(x_i - \frac{\lambda_i}{v} \right) \alpha_i = \mathbf{b}. \qquad \text{...(5)}$$

(5) gives

$$\mathbf{x}' = \left[x_1 - \frac{\lambda_1}{v}, \, x_2 - \frac{\lambda_2}{v}, \ldots, x_k - \frac{\lambda_k}{v}, 0, 0, \ldots, 0 \right]$$

A new solution of the matrix equation $A\,\mathbf{x} = \mathbf{b}$.

Also from (4) for $i = 1, 2, \ldots, k$, we have

$$v \ge \frac{\lambda_i}{x_i} \quad \text{or} \quad x_i \ge \frac{\lambda_i}{v} \qquad [\because x_i > 0, \; v > 0]$$

or
$$x_i - \frac{\lambda_i}{v} \ge 0.$$

Since all the components of $\mathbf{x}'$ are non-negative therefore $\mathbf{x}'$ is a feasible solution to the given L.P.P.

Also $v = \frac{\lambda_i}{x_i}$, for at least one i, $i = 1, 2, \ldots, k$.

i.e., for this value of i, $1 \le i \le k$, $v - \frac{\lambda_i}{x_i} = 0$, therefore the new feasible solution $\mathbf{x}'$ cannot have more than $k - 1$ non-zero variables.

In this way, we have derived a new feasible solution from the given optimal feasible solution which contains less number of non-zero variables. This solution is B.F.S., if the column vectors associated to non-zero variables in this new solution are L.I. If these associated vectors are not L.I., we shall repeat the whole reduction procedure as explained above. Continuing in this way for finite number of times, we will derive a solution in which columns corresponding to positive variables are L.I. *i.e.,* we will obtain a B.F.S. of the system.

Now it remains to prove that $\mathbf{x}'$ is also an optimal solution.

Let Z' be the new value of the objective function corresponding to the new solution $\mathbf{x}'$. Then

$$Z' = \sum_{i=1}^{k} c_i \left(x_i - \frac{\lambda_i}{v} \right) = \sum_{i=1}^{k} c_i x_i - \frac{1}{v} \sum_{i=1}^{k} c_i \lambda_i$$

or $$Z' = Z^* - \frac{1}{v} \sum_{i=1}^{k} c_i \lambda_i, \text{ from (2)} \qquad \ldots(6)$$

Now for optimality, we must have

$$Z' = Z^*$$

i.e., $\mathbf{x}'$ will be optimal solution only if

$$\sum_{i=1}^{k} c_i \lambda_i = 0. \qquad \ldots(7)$$

We shall prove this result by contradiction.

If possible, let us assume that

$$\sum_{i=1}^{k} c_i \lambda_i \neq 0.$$

Then either $$\sum_{i=1}^{k} c_i \lambda_i < 0 \qquad \ldots(8)$$

or $$\sum_{i=1}^{k} c_i \lambda_i > 0. \qquad \ldots(9)$$

In either of these two equations (8) and (9), we can find a real number r such that

$$r \sum_{i=1}^{k} c_i \lambda_i > 0$$

[in equation (8) r will be negative and in equation (9) r will be positive]

or $$\sum_{i=1}^{k} c_i (r \lambda_i) > 0$$

or $$\sum_{i=1}^{k} c_i (r\lambda_i) + \sum_{i=1}^{k} c_i x_i > \sum_{i=1}^{k} c_i x_i$$

or $$\sum_{i=1}^{k} c_i (x_i + r \lambda_i) > Z^*, \quad \text{from (2)} \qquad \ldots(10)$$

Multiplying equation (3) by r and adding to (1), we get

$$\sum_{i=1}^{k} x_i \alpha_i + \sum_{i=1}^{k} r\lambda_i \alpha_i = \mathbf{b}$$

or $$\sum_{i=1}^{k} (x_i + r\lambda_i)\alpha_i = \mathbf{b}.$$

Therefore, $$[x_1 + r\lambda_1, x_2 + r\lambda_2, \ldots, x_k + r\lambda_k, \overbrace{0, 0, \ldots, 0}^{N-k}]$$

is also a solution of the matrix equation $A\mathbf{x} = \mathbf{b}$ for all values of r.

Furthermore, we can choose r in infinitely many ways for which the above solution also satisfies the non-negative restrictions.

Let us choose r such that

$$x_i + r\lambda_i \geq 0, \; i = 1, 2, \ldots, k$$

or $$r\lambda_i \geq -x_i$$

$\therefore$ $$r \geq -\frac{x_i}{\lambda_i} \quad \text{if} \quad \lambda_i > 0,$$

$$r \leq -\frac{x_i}{\lambda_i} \quad \text{if} \quad \lambda_i < 0$$

and r is unrestricted if $\lambda_i = 0$.

Thus $x_i + r\lambda_i \geq 0$ for $i = 1, 2, \ldots k$ if we select r such that

$$\max_{\substack{i \\ (\lambda_i > 0)}} \left(-\frac{x_i}{\lambda_i}\right) \leq r \leq \max_{\substack{i \\ (\lambda_i < 0)}} \left(-\frac{x_i}{\lambda_i}\right) \qquad \ldots(11)$$

Furthermore, we have

$$\max_{\substack{i \\ (\lambda_i > 0)}} \left(-\frac{x_i}{\lambda_i}\right) < 0 \text{ and } \min_{\substack{i \\ (\lambda_i < 0)}} \left(-\frac{x_i}{\lambda_i}\right) > 0$$

which means that interval given by (11) is non-empty.

Thus r lies in the non-empty interval given in (11).

Consequently, an infinite number of solutions given by

$$[x_1 + r\lambda_1, x_2 + r\lambda_2, \ldots, x_k + r\lambda_k, 0, \overbrace{0, \ldots, 0}^{N-k}]$$

satisfy the non-negative restrictions as well.

Now, returning to result (10), we find that $\sum_{i=1}^{k} c_i (x_i + r\lambda_i)$ yields the value of the objective function which is strictly greater than the greatest value (or optimal value) Z^* of objective function, which is impossible.

$\therefore$ We must have $\sum_{i=1}^{k} c_i \lambda_i = 0$

or $Z' = Z^*$.

Hence
$$\mathbf{x}' = \left[x_1 - \frac{\lambda_1}{v}, x_2 - \frac{\lambda_2}{v}, \ldots, x_k - \frac{\lambda_k}{v}, \overbrace{0, 0, \ldots, 0}^{N-k}\right]$$

is also an optimal solution.

This proves the theorem.

3.4 Reduction of Feasible Solution to Basic Feasible Solution

Theorem: *If a linear programming problem*

$$\textit{Max. } Z = \mathbf{c}\,\mathbf{x} \textit{ subject to } A\mathbf{x} = \mathbf{b},\ \mathbf{x} \geq \mathbf{0},$$

where $A = (\alpha_1, \alpha_2, \ldots, \alpha_N)$ *is the coefficient matrix of order* $m \times N$, $(N = m + n)$, *has at least one feasible solution, then it has at least one basic feasible solution also.*

Proof: Consider an arbitrary feasible solution of the given linear programming problem

$$\mathbf{x}^* = (x_1, x_2, \ldots, x_N),\ x_i \geq 0. \qquad \ldots(1)$$

Let us assume that k, $(k \leq N)$ variables in $\mathbf{x}^*$ have positive values and the remaining $N - k$ variables are zero. We can also assume that the variables have been numbered in such a way that the first k variables are non-zero.

Thus
$$\mathbf{x}^* = [x_1, x_2, \ldots, x_k, 0, \overbrace{0, \ldots, 0}^{N-k}]$$

Also, we have $\sum_{i=1}^{k} x_i \alpha_i = \mathbf{b}$. ...(2)

Now two possibilities may arise :

The vectors $\alpha_1, \alpha_2, \ldots, \alpha_k$ may be either linearly independent or dependent.

Case 1: If $\alpha_1, \alpha_2, \ldots, \alpha_k$ are linearly independent.

Any feasible solution for which the vectors α_i, $i = 1, 2, \ldots, k$ associated with non-zero variables x_i, $i = 1, 2, \ldots, k$ are L.I. is called a *basic feasible solution*. Thus $\mathbf{x}^*$ is a B.F.S. which is also optimal.

Hence, the result of the theorem is true.

The solution $\mathbf{x}^*$ is degenerate if $k < m$ and non-degenerate if $k = m$.

Case 2: If $\alpha_1, \alpha_2, \ldots, \alpha_k$ are linearly dependent and $k > m$.

In this case we shall reduce the number of non-zero variables step by step until we obtain a B.F.S. from feasible solution.

Since $\alpha_1, \alpha_2, \ldots., \alpha_k$ are linearly dependent therefore there exist scalars $\lambda_1, \lambda_2, \ldots., \lambda_k$ such that

$$\lambda_1 \alpha_1 + \lambda_2 \alpha_2 + \ldots. + \lambda_k \lambda_k = \mathbf{0}$$

or
$$\sum_{i=1}^{k} \lambda_i \alpha_i = \mathbf{0} \qquad \ldots(3)$$

and at least one $\lambda_i \neq 0$.

We can assume that at least one λ_i is positive because if none is positive, then we can multiply the equation (3) by -1 and get positive λ_i.

Now, suppose

$$v = \max_{1 \leq i \leq k} \left(\frac{\lambda_i}{x_i} \right) \qquad \ldots(4)$$

Obviously, v is positive since $x_i > 0$, $i = 1, 2, \ldots., k$ and at least one λ_i is positive.

Multiplying both sides of (3) by $\frac{1}{v}$ and subtracting from (2), we get

$$\sum_{i=1}^{k} \left(x_i - \frac{\lambda_i}{v} \right) \alpha_i = \mathbf{b}. \qquad \ldots(5)$$

(5) gives
$$\mathbf{x}' = \left[x_1 - \frac{\lambda_1}{v}, x_2 - \frac{\lambda_2}{v}, \ldots, x_k - \frac{\lambda_k}{v}, 0, 0, \ldots, 0 \right]$$

A new solution of the matrix equation $A\mathbf{x} = \mathbf{b}$.

Also from (4) for $i = 1, 2, \ldots, k$, we have

$$v \geq \frac{\lambda_i}{x_i} \quad \text{or} \quad x_i \geq \frac{\lambda_i}{v} \qquad [\because x_i, v > 0]$$

or $$x_i - \frac{\lambda_i}{v} \geq 0.$$

Since all the components of $\mathbf{x}'$ are non-negative therefore $\mathbf{x}'$ is a feasible solution to the given L.P.P.

Also $v = \frac{\lambda_i}{x_i}$, for at least one i, $1 \leq i \leq k$.

i.e., for this value of i, $1 \leq i \leq k$,

$v - \frac{\lambda_i}{x_i} = 0$, therefore the new feasible solution $\mathbf{x}'$ cannot have more thank $k-1$ non-zero variables.

In this way, we have derived a new feasible solution from the given optimal feasible solution which contains less number of non-zero variables. This solution is B.F.S., if the column vectors associated to non-zero variables in this new solution are L.I. If these associated vectors are not L.I., we shall repeat the whole reduction procedure as explained above. Continuing in this way for finite number of times we will derive a solution in which columns corresponding to positive variables are L.I. *i.e.,* we will obtain a B.F.S. of the system.

Hence, the theorem is proved.

Note : 1. By applying the above mentioned procedure we can obtain a basic feasible solution from any feasible solution.

2. If the given feasible solution is optimal, then basic feasible solution is also optimal.

Illustrative Examples

Example 1: *If* $x_1 = 1, x_2 = 0, x_3 = 1$ *be a feasible solution of the L.P.P.*

$$\textit{Min. } Z = 2x_1 + 3x_2 + 4x_3$$

subject to $$x_1 + x_2 + x_3 = 2$$
$$x_1 - x_2 + x_3 = 0,\ x_1, x_2, x_3 \geq 0,$$

then show that the given feasible solution is not basic.

Solution: The given system of constraint equations can be written in matrix form $A\mathbf{x} = \mathbf{b}$.

or $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

$$\therefore \qquad \alpha_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

The given feasible solution $x_1 = 1, x_2 = 0, x_3 = 1$ will not be basic if the vectors associated to non-zero variables are not linearly independent.

i.e., if α_1 and α_3 are linearly dependent.

If we choose $\lambda_1 = -1$ and $\lambda_2 = 1$, we have

$$\lambda_1 \alpha_1 + \lambda_2 \alpha_3 = -1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1+1 \\ -1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

$\therefore$ The vectors α_1 and α_3 are linearly dependent *i.e.,* are not L.I.

Hence the given feasible solution is not basic.

Aliter : $|(\alpha_1, \alpha_3)| = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \Rightarrow$ the vectors α_1 and α_3 are not L.I.

Example 2: *If $x_1 = 2, x_2 = 3, x_3 = 1$ be a feasible solution of the linear programming problem*

$$\textit{Max. } Z = x_1 + 2x_2 + 4x_3$$

$$\textit{subject to} \quad 2x_1 + x_2 + 4x_3 = 11$$

$$3x_1 + x_2 + 5x_3 = 14$$

$$x_1, x_2, x_3 \geq 0,$$

then find a basic feasible solution from the given feasible solution.

Solution: The given L.P.P. can be expressed a

$$\text{Max. } Z = x_1 + 2x_2 + 4x_3,$$

subject to $\quad A\mathbf{x} = \mathbf{b}$

or
$$\begin{bmatrix} 2 & 1 & 4 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ 14 \end{bmatrix}$$

or
$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 \alpha_3 = \mathbf{b}, \qquad \ldots(1)$$

where
$$\alpha_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 11 \\ 14 \end{bmatrix}.$$

Since $x_1 = 2, x_2 = 3, x_3 = 1$ is a feasible solution to the given L.P.P. Therefore, from (1)

$$2\alpha_1 + 3\alpha_2 + \alpha_3 = \mathbf{b}. \quad \text{...(2)}$$

Now the vectors $\alpha_1, \alpha_2, \alpha_3$ associated with non-zero variables x_1, x_2, x_3 will be linearly dependent if one of these vectors can be expressed as the linear combination of the remaining two vectors.

Let $$\alpha_1 = a\alpha_2 + b\alpha_3 \quad \text{...(3)}$$

or $$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = a\begin{bmatrix} 1 \\ 1 \end{bmatrix} + b\begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} a+4b \\ a+5b \end{bmatrix}$$

$\Rightarrow$ $$a + 4b = 2$$

and $$a + 5b = 3 \Rightarrow a = -2, b = 1$$

Substituting the values of a and b in (3), we get

$$\alpha_1 = -2\alpha_2 + \alpha_3$$

or $$\alpha_1 + 2\alpha_2 - \alpha_3 = 0 \quad \text{...(4)}$$

or $$\sum_{i=1}^{3} \lambda_i \alpha_i = \mathbf{0} \text{ which gives } \lambda_1 = 1, \lambda_2 = 2, \lambda_3 = -1.$$

Now we shall determine which of the three variables x_1, x_2, x_3 should be zero.

Let $$v = \max_{1 \le i \le 3} (\lambda_i / x_i) = \max\ (\lambda_1/x_1, \lambda_2/x_2, \lambda_3/x_3)$$

$$= \max (1/2, 2/3, -1/1) = 2/3 = \lambda_2/x_2.$$

$\therefore$ x_2 should be zero, for which we should eliminate α_2 between (2) and (4).

Eliminating α_2 between (2) and (4), we get

$$2\alpha_1 - 3(\alpha_1 - \alpha_3)/2 + \alpha_3 = \mathbf{b}$$

or $$(1/2)\alpha_1 + (5/2)\alpha_3 = \mathbf{b}.$$

$\therefore$ The new F.S. is $x_1 = 1/2, x_2 = 0, x_3 = 5/2$.

Now the column vectors α_1 and α_3 corresponding to basic (non-zero) variables x_1 and x_3 are L.I. as

$$|(\alpha_1, \alpha_3)| = \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = -2 \neq 0,$$

so this new F.S. is B.F.S.

Hence, the B.F.S. obtained from the given F.S. is

$$x_1 = 1/2, x_2 = 0, x_3 = 5/2.$$

Note:

1. We can also find the new B.F.S as follows.

   We have $v = 2/3$.

   Since $x_i - \lambda_i / v \geq 0$ for $\forall\, i = 1, 2, 3$.

   $\therefore$ The new F.S. is

   $$(x_1 - \lambda_1/v,\ x_2 - \lambda_2/v,\ x_3 - \lambda_3/v).$$

   We have $x_1 - \lambda_1/v = 2 - 1/(2/3) = 1/2$, $x_2 - \lambda_2/v = 0$, $x_3 - \lambda_3/v = 5/2$.

   Hence, the new F.S. is (1/2, 0, 5/2), which is also B.F.S.

2. Another new B.F.S. can be obtained as follows :

   Equation (4), can be written as

   $$-\alpha_1 - 2\alpha_2 + \alpha_3 = \mathbf{0} \qquad \text{...(5)}$$

   $$\therefore \qquad \lambda_1 = -1,\ \lambda_2 = -2,\ \lambda_3 = 1$$

   $$\therefore \qquad v = \max_{1 \leq i \leq 3} (\lambda_i / x_i) = 1/1 = \lambda_3 / x_3 .$$

   $\therefore$ x_3 should be zero and α_3 should be eliminated between (2) and (5).

   Eliminating α_3 between (2) and (5), we have

   $$2\alpha_1 + 5\alpha_2 = \mathbf{0}.$$

   $\therefore$ New F.S. which is also B.F.S. is

   $$x_1 = 2,\ x_2 = 5,\ x_3 = 0.$$

Example 3: *Consider the set of equations*

$$5x_1 - 4x_2 + 3x_3 + x_4 = 3$$
$$2x_1 + x_2 + 5x_3 - 3x_4 = 0$$
$$x_1 + 6x_2 - 4x_3 + 2x_4 = 15,$$
$$x_1, x_2, x_3, x_4 \geq 0.$$

If $x_1 = 1,\ x_2 = 2,\ x_3 = 1,\ x_4 = 3$ *is a feasible solution, then find a basic feasible solution.*

Solution: The given set of equations can be expressed in the matrix form

$$A\,\mathbf{x} = \mathbf{b}$$

or
$$\begin{bmatrix} 5 & -4 & 3 & 1 \\ 2 & 1 & 5 & -3 \\ 1 & 6 & -4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 15 \end{bmatrix}$$

$$\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 = \mathbf{b} \qquad \text{...(1)}$$

where $\alpha_1 = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 3 \\ 5 \\ -4 \end{bmatrix}, \alpha_4 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ 15 \end{bmatrix}.$

Since $x_1 = 1, x_2 = 2, x_3 = 1, x_4 = 3$ is a feasible solution to the given set of equations, therefore from (1)

$$\alpha_1 + 2\alpha_2 + \alpha_3 + 3\alpha_4 = \mathbf{b}. \quad \text{...(2)}$$

Now the vectors $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ associated with non-zero variables x_1, x_2, x_3, x_4 will be linearly dependent if one of these vectors can be expressed as the linear combination of the remaining three vectors.

Let $$\alpha_1 = a\alpha_2 + b\alpha_3 + c\alpha_4 \quad \text{...(3)}$$

or $$\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} = a\begin{bmatrix} -4 \\ 1 \\ 6 \end{bmatrix} + b\begin{bmatrix} 3 \\ 5 \\ -4 \end{bmatrix} + c\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

or $$\begin{bmatrix} -4a + 3b + c \\ a + 5b - 3c \\ 6a - 4b + 2c \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}$$

or $$\begin{aligned} -4a + 3b + c &= 5 \\ a + 5b - 3c &= 2 \\ 6a - 4b + 2c &= 1 \end{aligned}$$

Solving these, we get $a = 22/43$, $b = 139/86$, $c = 189/86$.

Substituting these values in (3), we get

$$\frac{22}{43}\alpha_2 + \frac{139}{86}\alpha_3 + \frac{189}{86}\alpha_4 = \alpha_1$$

or $$86\alpha_1 - 44\alpha_2 - 139\alpha_3 - 189\alpha_4 = \mathbf{0} \quad \text{...(4)}$$

or $\sum_{i=1}^{4} \lambda_i \alpha_i = \mathbf{0}$, which gives $\lambda_1 = 86, \lambda_2 = -44, \lambda_3 = -139, \lambda_4 = -189.$

Using $v = \max_{1 \le i \le 4}\left(\frac{\lambda_i}{x_i}\right)$, we have

$$v = \max\left(\frac{\lambda_1}{x_1}, \frac{\lambda_2}{x_2}, \frac{\lambda_3}{x_3}, \frac{\lambda_4}{x_4}\right) = \max\left(\frac{86}{1}, -\frac{44}{2}, -\frac{139}{1}, \frac{-189}{3}\right) = \frac{86}{1} = \frac{\lambda_1}{x_1}.$$

$\therefore$ The variable x_1 should be zero.

i.e., The vector α_1 should be eliminated between (2) and (4).

Eliminating α_1, between (2) and (4), we get

$$(44\alpha_2 + 139\alpha_3 + 189\alpha_4)/86 + 2\alpha_2 + \alpha_3 + 3\alpha_4 = \mathbf{b}$$

or $$0.\alpha_1 + \frac{216}{86}\alpha_2 + \frac{225}{86}\alpha_3 + \frac{447}{86}\alpha_4 = \mathbf{b}$$

$\therefore$ $x_1 = 0, x_2 = \frac{216}{86}, x_3 = \frac{225}{86}, x_4 = \frac{447}{86}$ is the new feasible solution.

Since the vectors $\alpha_2, \alpha_3, \alpha_4$ corresponding to non-zero (basic) variables are L.I. as

$$|(\alpha_2 \ \ \alpha_3 \ \ \alpha_4)| = \begin{vmatrix} -4 & 3 & 1 \\ 1 & 5 & -3 \\ 6 & -4 & 2 \end{vmatrix} \neq 0.$$

$\therefore$ This new F.S. is B.F.S.

Note : Writing (4) as

$$-86\alpha_1 + 44\alpha_2 + 139\alpha_3 + 189\alpha_4 = 0$$

Proceeding as above, another B.F.S. obtained is

$$x_1 = 225/139, x_2 = 234/139, x_3 = 0, x_4 = 228/139.$$

Comprehensive Exercise 1

1. State and prove the fundamental theorem of linear programming.
2. Show that if a linear programming problem has a feasible solution, it also has basic feasible solution.
3. Prove that if the system $A\mathbf{x} = \mathbf{b}$ of m linear equations in n unknowns $(m \leq n)$ rank $A = m$ has a feasible solution, then it has a basic feasible solution also.
4. If $x_1 = 2, x_2 = 4$ and $x_3 = 1$ be a F.S. to the system of equations
$$2x_1 - x_2 + 2x_3 = 2, \ x_1 + 4x_2 = 18,$$
then find two basic feasible solutions.
5. (2, 1, 3) is a feasible solution of the set of equations
$$4x_1 + 2x_2 - 3x_3 = 1, \ 6x_1 + 4x_2 - 5x_3 = 1.$$
Reduce the feasible solution to a B.F.S. of the set.
6. $x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 0$ is a feasible solution to the system of equations
$$x_1 + 2x_2 + 4x_3 + x_4 = 7, \ 2x_1 - x_2 + 3x_3 - 2x_4 = 4.$$
Reduce the feasible solution to two different basic feasible solutions.
7. $x_1 = 2, x_2 = 4, x_3 = 5$ is a F.S. of th system of equations
$$2x_1 - x_2 + 2x_3 = 10$$

$$x_1 + 4x_2 = 18$$

$$x_1, x_2, x_3 \geq 0$$

Reduce this F.S. to B.F.S.

8. (1, 1, 1) is a feasible solution of the system of equations

$$x_1 + x_2 + 2x_3 = 4$$

$$2x_1 - x_2 + x_3 = 2$$

Reduce the F.S. to B.F.S.

9. Consider the system of equations

$$2x_1 - 3x_2 + 4x_3 + 6x_4 = 25$$

$$x_1 + 2x_2 + 3x_3 - 3x_4 + 5x_5 = 12$$

If $x_1 = 2$, $x_2 = 1$, $x_3 = 3$, $x_4 = 2$, $x_5 = 1$ is a F.S., then reduce it to a B.F.S. of the system.

10. Consider the set of equations

$$2x_1 - 3x_2 + 4x_3 + 6x_4 = 21$$

$$x_1 + 2x_2 + 3x_3 - 3x_4 + 5x_5 = 9$$

If $x_1 = 2$, $x_2 = 1$, $x_3 = 2$, $x_4 = 2$, $x_5 = 1$ is a F.S., then reduce it to two different basic feasible solutions.

Answers 1

4. $x_1 = 26/9$, $x_2 = 34/9$, $x_3 = 0$; $x_1 = 0$, $x_2 = 9/2$, $x_3 = 13/4$

5. $x_1 = 1$, $x_2 = 0$, $x_3 = 1$

6. $x_1 = 0$, $x_2 = 1/2$, $x_3 = 3/2$, $x_4 = 0$; $x_1 = 3$, $x_2 = 2$, $x_3 = 0$, $x_4 = 0$

7. $x_1 = 58/9$, $x_2 = 26/9$, $x_3 = 0$

8. $x_1 = 0$, $x_2 = 0$, $x_3 = 2$; $x_1 = 2$, $x_2 = 2$, $x_3 = 0$

9. $x_1 = 147/2$, $x_2 = 0$, $x_3 = 0$, $x_4 = 1/12$, $x_5 = 0$

10. $x_1 = 0$, $x_2 = 0$, $x_3 = 39/10$, $x_4 = 9/10$, $x_5 = 0$

and $x_1 = 39/4$, $x_2 = 0$, $x_3 = 0$, $x_4 = 1/4$, $x_5 = 0$

3.5 To Determine Improved B.F.S. from a Given B.F.S.

The following theorem helps us to develop a procedure for determining another basic feasible solution from a given B.F.S. which gives a better value of objective function.

Theorem: *Let* $\mathbf{x}_B = B^{-1}\mathbf{b}$ *be a B.F.S. of a linear programming problem with* $Z = \mathbf{c}_B \mathbf{x}_B$ *as the value of the objective function. If for any column* α_j *in A, but not in B, the condition* $c_j - Z_j > 0$ *or* $Z_j - c_j < 0$ *holds and if at least one* $y_{ij} > 0$, $i = 1, 2, \ldots, m$, *then it is possible to obtain a new B.F.S. by replacing one of the columns in B by* α_j *and if* Z' *is the new value of objective function, then* $Z' \geq Z$.

If the initial B.F.S. $\mathbf{x}_B = B^{-1}\mathbf{b}$ *is non-degenerate, then* $Z' > Z$.

Proof: Consider the L.P.P.

$$\text{Max. } Z = \mathbf{c}\,\mathbf{x}, \text{ subject to } A\,\mathbf{x} = \mathbf{b},\ \mathbf{x} \geq \mathbf{0},$$

where $A = (\alpha_1, \alpha_2, \ldots, \alpha_N)$, $N = m + n$,

basis matrix $B = (\beta_1, \beta_2, \ldots, \beta_m)$.

Let $\mathbf{x}_B = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$ be a B.F.S. of the given L.P.P.

Since the vectors $\beta_1, \beta_2, \ldots, \beta_m$ are in the basis of A, therefore we can express α_j as a linear combination of β's.

$$\therefore \quad \alpha_j = \sum_{i=1}^{m} y_{ij}\,\beta_i = y_{1j}\,\beta_1 + y_{2j}\,\beta_2 + \ldots + y_{mj}\,\beta_m. \qquad \ldots(1)$$

If $y_{rj} \neq 0$, then α_j can replace β_r in B and B is still a basis matrix.

Let $y_{rj} \neq 0$, then from (1), we have

$$\beta_r = \frac{1}{y_{rj}}\,\alpha_j - \frac{y_{1j}}{y_{rj}}\beta_1 - \ldots - \frac{y_{(r-1)j}}{y_{rj}}\beta_{r-1} - \frac{y_{(r+1)j}}{y_{rj}}\beta_{r+1} - \ldots - \frac{y_{mj}}{y_{rj}}\beta_m$$

or
$$\beta_r = \frac{1}{y_{rj}}\,\alpha_j - \sum_{\substack{i=1 \\ i \neq r}}^{m} \frac{y_{ij}}{y_{rj}}\beta_i. \qquad \ldots(2)$$

Now, we have

$$B\,\mathbf{x}_B = \mathbf{b}$$

or
$$(\beta_1, \beta_2, \ldots, \beta_r, \ldots, \beta_m)\,[x_{B1}, x_{B2}, \ldots, x_{Br}, \ldots, x_{Bm}] = \mathbf{b}$$

or
$$\beta_1\,x_{B1} + \beta_2\,x_{B2} + \ldots + \beta_r\,x_{Br} + \ldots + \beta_m\,x_{Bm} = \mathbf{b}$$

or $$\sum_{\substack{i=1 \\ i \neq r}}^{m} \beta_i \, x_{Bi} + \beta_r \, x_{Br} = \mathbf{b} \qquad ...(3)$$

Putting for β_r from (2) in (3), we get

$$\sum_{\substack{i=1 \\ i \neq r}}^{m} \beta_i \left[x_{Bi} - \frac{y_{ij}}{y_{rj}} x_{Br} \right] + \frac{x_{Br}}{y_{rj}} \alpha_j = \mathbf{b} \qquad ...(4)$$

or $$\sum_{\substack{i=1 \\ i \neq r}}^{m} x'_{Bi} \, \beta_i + x'_{Br} \, \alpha_j = \mathbf{b}, \qquad ...(5)$$

where $$x'_{Bi} = x_{Bi} - x_{Br} \frac{y_{ij}}{y_{rj}}, \quad i = 1, 2, \ldots, m, \; i \neq r$$

and $$x'_{Br} = \frac{x_{Br}}{y_{rj}}, \; i = r \qquad ...(6)$$

Comparing (3) and (5), we observe that the new basic solution of $A\mathbf{x} = \mathbf{b}$ is given by

$$\mathbf{x}'_B = [x'_{Bi}, x'_{Br}], \; i = 1, 2, \ldots., m, \; i \neq r$$

where the values of x'_{Bi}, x'_{Br} are given by (6).

This basic solution will be feasible if we have

$$x_{Bi} - \frac{y_{ij}}{y_{rj}} x_{Br} \geq 0, \quad i = 1, 2, \ldots, m, \; i \neq r$$

and $$\frac{x_{Br}}{y_{rj}} \geq 0, \; i = r. \qquad ...(7)$$

Since $\mathbf{x}_B$ is the initial B.F.S., we have

$$x_{Br} \geq 0, \; r = 1, 2, \ldots., m.$$

Thus we observe that (7) will hold only if

$$y_{rj} > 0 \text{ and } y_{ij} \leq 0, \; i = 1, 2, \ldots, m, \; i \neq r.$$

If $y_{rj} > 0$ and $y_{ij} > 0$, then (7) is satisfied only if

$$\frac{x_{Bi}}{y_{ij}} - \frac{x_{Br}}{y_{rj}} \geq 0$$

or $$\frac{x_{Br}}{y_{rj}} \leq \frac{x_{Bi}}{y_{ij}}$$

or $$\frac{x_{Br}}{y_{rj}} = \min_i \left[\frac{x_{Bi}}{y_{ij}},\ y_{ij} > 0\right] = v. \quad ...(8)$$

Thus a new B.F.S. can be obtained from the initial B.F.S. by removing the column vector β_r of the basis matrix B by α_j if r is to be selected such that

$$v = \frac{x_{Br}}{y_{rj}} = \min_i \left\{\frac{x_{Bi}}{y_{ij}},\ y_{ij} > 0\right\}.$$

If we have $v = 0$, which is possible only when $x_{Br} = 0$ then it means that the initial B.F.S. is degenerate.

Now we shall prove $Z' \geq Z$.

The value of the objective function for the initial B.F.S. $\mathbf{x}_B$ is

$$\begin{aligned} Z &= \mathbf{c}_B\, \mathbf{x}_B \\ &= (c_{B1}, c_{B2}, \ldots., c_{Bm})\ [x_{B1}, x_{B2}, \ldots., x_{Bm}] \\ &= \sum_{i=1}^{m} c_{Bi}\, x_{Bi}. \end{aligned}$$

Corresponding to the new B.F.S. $\mathbf{x}'_B$, the value of the objective function is Z', so we have

$$Z' = \sum_{i=1}^{m} c'_{Bi}\, x'_{Bi}, \quad ...(9)$$

where c'_{Bi} are the coefficients of the basic variables x'_{Bi} $(i = 1, 2, \ldots., m)$ in the objective function.

Obviously, $\quad c'_{Bi} = c_{Bi}, i = 1, 2, \ldots., m,\ i \neq r$

and $\quad c'_{Br} = c_j.$

$\therefore$ We can write

$$Z' = \sum_{\substack{i=1 \\ i \neq r}}^{m} c_{Bi}\, x'_{Bi} + c_j\, x'_{Br}. \quad ...(10)$$

Substituting for x'_{Bi}, x'_{Br} from (6) in (10), we get

$$\begin{aligned} Z' &= \sum_{\substack{i=1 \\ i \neq r}}^{m} c_{Bi} \left(x_{Bi} - x_{Br}\frac{y_{ij}}{y_{rj}}\right) + c_j \frac{x_{Br}}{y_{rj}} \\ &= \sum_{i=1}^{m} c_{Bi} \left(x_{Bi} - x_{Br}\frac{y_{ij}}{y_{rj}}\right) + c_j \frac{x_{Br}}{y_{rj}} \end{aligned}$$

Since the term $c_{Bi}\left(x_{Bi} - x_{Br}\dfrac{y_{ij}}{y_{rj}}\right) = 0$ when $i = r$ therefore it can be included in the summation without changing the value of Z'.

$$\therefore \qquad Z' = \sum_{i=1}^{m} c_{Bi}\, x_{Bi} - \frac{x_{Br}}{y_{rj}} \sum_{i=1}^{m} c_{Bi}\, y_{ij} + \frac{x_{Br}}{y_{rj}} c_j$$

$$= Z + \frac{x_{Br}}{y_{rj}}(c_j - Z_j), \qquad \text{where} \qquad Z_j = \sum_{i=1}^{m} c_{Bi}\, y_{ij}$$

$$= Z + v(c_j - Z_j), \text{from (8)} \qquad \ldots(11)$$

From (11), we observe that the new value of the objective function is equal to the original value of the objective function plus the quantity $v(c_j - Z_j)$.

We have $\quad Z' \geq Z$ if $v(c_j - Z_j) \geq 0$.

Since $\quad v \geq 0$ therefore $Z' \geq Z$ only if $c_j - Z_j \geq 0$.

Hence by choosing the vector α_j *for which* $c_j - Z_j > 0$ *and at least one* $y_{ij} > 0$, *we obtain a new improved value of the objective function.*

If the initial B.F.S. is non-degenerates then $v > 0$ and in that case $Z' > Z$.

Note :

1. $B' = (\beta_1', \beta_2', \ldots, \beta_m')$ is the new non-singular matrix obtained from B by replacing β_r with α_j, then we have $\beta_i' = \beta_i,\, i \neq r$ and $\beta_r' = \alpha_j$.
2. If $v = 0$, then the initial B.F.S. is degenerate. From (6), we have

$$x_{Bi}' = x_{Bi},\, i = 1, 2, \ldots, m,\, i \neq r$$

$$x_{Br}' = x_{Br} = 0,\, i = r.$$

Since the values of the variables common to the initial and new solutions do not change therefore the new B.F.S. is also degenerate.

3.6 Conditions for the Existence of Unbounded Solutions

In article 3.5, we have proved that if for any column α_j in A but not in B the condition $c_j - Z_j > 0$ holds and if at least one $y_{ij} > 0$, $i = 1, 2, \ldots, m$, then it is possible to find an improved B.F.S.

Now the question is, that, what will be the result if for at least one α_j all $y_{ij} \leq 0, i = 1, 2, \ldots, m$.

To get the answer of this question we shall prove the following theorem :

Theorem: *(Unbounded Solutions) : If for any basic feasible solution* $\mathbf{x}_B = B^{-1}\mathbf{b}$ *to* $A\mathbf{x} = \mathbf{b}$ *there is some column* α_j *in A but not in B for which* $c_j - Z_j > 0$ *and* $y_{ij} \leq 0$,

$i = 1, 2, \ldots, m$, then if the objective function is to be maximized, the problem has an unbounded solution.

Proof: Consider the linear programming problem

$$\text{Max. } Z = \mathbf{c}\,\mathbf{x} \quad \text{subject to} \quad A\mathbf{x} = \mathbf{b},\ \mathbf{x} \geq \mathbf{0},$$

where $A = (\alpha_1, \alpha_2, \ldots, \alpha_N),\ N = m + n,$

basis matrix $B = (\beta_1, \beta_2, \ldots, \beta_m)$.

Let $\mathbf{x}_B = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$ be a B.F.S. of the given L.P.P. Then, we have

$$B\mathbf{x}_B = \mathbf{b}$$

or
$$\sum_{i=1}^{m} x_{Bi}\, \beta_i = \mathbf{b} \qquad \ldots(1)$$

and the corresponding value of the objective function is

$$Z = \mathbf{c}_B\, \mathbf{x}_B = \sum_{i=1}^{m} c_{Bi}\, x_{Bi} \qquad \ldots(2)$$

Adding and subtracting $\lambda\alpha_j$ in (1), we get

$$\sum_{i=1}^{m} x_{Bi}\, \beta_i - \lambda\alpha_j + \lambda\alpha_j = \mathbf{b} \qquad \ldots(3)$$

where λ is some scalar.

Let $\alpha_j \in A$ s.t. $\alpha_j \notin B$.

Since the vectors $\beta_1, \beta_2, \ldots, \beta_m$ are in the basis of A therefore we can express α_j as the linear combination of β's

$$\therefore \qquad \alpha_j = \sum_{i=1}^{m} y_{ij}\, \beta_i$$

or
$$-\lambda\alpha_j = -\lambda \sum_{i=1}^{m} y_{ij}\, \beta_i.$$

Substituting the above value of $-\lambda\alpha_j$ in (3), we get

$$\sum_{i=1}^{m} x_{Bi}\, \beta_i - \lambda \sum_{i=1}^{m} y_{ij}\, \beta_i + \lambda\alpha_j = \mathbf{b}$$

or
$$\sum_{i=1}^{m} (x_{Bi} - \lambda\, y_{ij})\beta_i + \lambda\alpha_j = \mathbf{b}$$

Thus, we obtain a new solution

$$\mathbf{x}'_B = [x'_{B1}, x'_{B2}, \ldots, x'_{Bm}, \lambda] \qquad \ldots(4)$$

where $\quad x'_{Bi} = x_{Bi} - \lambda y_{ij}, i = 1, 2, \ldots, m.$

When $\lambda > 0$, $x_{Bi} - \lambda\, y_{ij} \geq 0$ since $y_{ij} \leq 0$, $i = 1, 2, \ldots, m$ therefore (4) gives the feasible solution in which the number of positive variables is less than or equal to $(m+1)$. It may be less than $(m+1)$ because some $x_{Bi} - \lambda\, y_{ij}$ may be zero for some i. In case, the number of positive variables in this solution is equal to $(m+1)$, then this solution will be non-basic feasible solution.

If Z' is the new value of the objective function corresponding to this new solution, then we have

$$Z' = \sum_{i=1}^{m} c_{Bi} (x_{Bi} - \lambda y_{ij}) + c_j \lambda$$

or
$$Z' = \sum_{i=1}^{m} c_{Bi}\, x_{Bi} + \lambda (c_j - c_{Bi} y_{ij}) = Z + \lambda (c_j - Z_j).$$

Since $c_j - Z_j > 0$ (given) therefore Z' can be made as large as we please by giving sufficiently large values to λ. We know that a L.P.P. has an unbounded solution if the value of its objective function can be increased or decreased arbitrarily.

Hence, the given L.P.P. has an unbounded solution.

Note : If for some α_j, $Z_j - c_j > 0$ and $y_{ij} \leq 0$, $i = 1, 2, \ldots, m$, then the L.P.P. has an unbounded solution if the objective function is to be minimised.

3.7 Condition for Improved Basic Feasible Solution to Become Optimal

Theorem: *Let $\mathbf{x}_B = B^{-1}\,\mathbf{b}$ be the basic (degenerate or non-degenerate) feasible solution of the L.P.P. $Z = \mathbf{c}\,\mathbf{x}$ s.t $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$. Let $Z^* = \mathbf{c}_B\, \mathbf{x}_B$ be the value of the objective function at any iteration of simplex method. If $c_j - Z_j \leq 0$ for every column α_j in A but not in B, then Z^* is the optimum value of the objective function Z and $\mathbf{x}_B$ is an optimal basic feasible solution.*

Proof: Suppose $\mathbf{x}_B = [x_{B1}, x_{B2}, \ldots, x_{Bm}]$ is the B.F.S. of the given L.P.P.

We have

$$A = (\alpha_1, \alpha_2, \ldots, \alpha_N),\ N = m + n$$

and the basis matrix $B = (\beta_1, \beta_2, \ldots, \beta_m)$.

$\therefore \quad B\mathbf{x}_B = \mathbf{b}$

or $\quad \mathbf{x}_B = B^{-1}\,\mathbf{b}. \qquad \ldots(1)$

The value of the objective function for the B.F.S. $\mathbf{x}_B$ is

$$Z^* = \mathbf{c}_B \, \mathbf{x}_B = \sum_{i=1}^{m} c_{Bi} \, x_{Bi}. \qquad \text{...(2)}$$

Also, we are given that $c_j - Z_j \le 0$ for every column α_j in A but not in B.

Let $\mathbf{x}' = [x_1', x_2', ..., x_N']$ be any feasible solution and Z' be the value of the objective function for this solution.

Then, we have

$$A\mathbf{x}' = \mathbf{b} \qquad \text{...(3)}$$

and

$$Z' = \mathbf{c}\,\mathbf{x}' = \sum_{i=1}^{N} c_j \, x_j'. \qquad \text{...(4)}$$

Now Z^* will be the optimum value of the objective function if we have $Z^* \ge Z'$.

To prove $Z^* \ge Z'$ we proceed in the following manner :

We have $\mathbf{x}_B = B^{-1} \mathbf{b} = B^{-1} (A\mathbf{x}')$, from (3)

$$= (B^{-1} A)\mathbf{x}'$$

$$= Y \, \mathbf{x}', \text{ where } Y = (Y_1, Y_2, ..., Y_N)$$

$$= (Y_1, Y_2, ..., Y_N) \, [x_1', x_2', ..., x_N']$$

$$= \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1j} & \cdots & y_{1N} \\ y_{21} & y_{22} & \cdots & y_{2j} & \cdots & y_{2N} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ y_{m1} & y_{m2} & \cdots & y_{mj} & \cdots & y_{mN} \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_j' \\ \vdots \\ x_N' \end{bmatrix}$$

or $[x_{B1}, x_{B2}, ..., x_{Bi}, ..., x_{Bm}]$

$$= \left[\sum_{j=1}^{N} y_{1j} \, x_j', \; \sum_{j=1}^{N} y_{2j} \, x_j', ..., \sum_{j=1}^{N} y_{ij} \, x_j', ..., \sum_{j=1}^{N} y_{mj} \, x_j' \right]$$

Equating i-th component on both sides, we get

$$x_{Bi} = \sum_{j=1}^{N} y_{ij} \; x'_j. \qquad \text{...(5)}$$

We have $c_j - Z_j \le 0$ for all j for which α_j is not in B.

If this inequality also holds for all those j for which α_j is in B, then $c_j - Z_j \le 0$ for all j for which α_j also belongs to B.

Now if $\alpha_j = \beta_i$, then

$$\alpha_j = \beta_i = 0 . \beta_1 + 0 . \beta_2 + \dots . + 0 . \beta_{i-1} + 1 . \beta_i + 0 . \beta_{i+1} + \dots + 0 . \beta_m$$

which shows that $Y_j = e_i$, a vector whose i-th component is unity and all other components are zero.

Again $\qquad \alpha_j = \beta_i$ gives $c_j = c_{Bi}$

$\therefore \qquad c_j - Z_j = c_j - \mathbf{c}_B Y_j = c_j - \mathbf{c}_B \, e_i = c_j - c_{Bi} = 0$

Thus $\qquad c_j - Z_j = 0$ for all those j for which $\alpha_j \in B$.

$\therefore \qquad c_j - Z_j \le 0$ for all α_j in A

or $\qquad c_j \le Z_j$

or $\qquad c_j \, x'_j \le Z_j \, x'_j, \qquad [\because x'_j \ge 0]$

or $$\sum_{i=1}^{N} c_j \, x'_j \le \sum_{j=1}^{N} Z_j \, x'_j$$

or $$Z' \le \sum_{j=1}^{N} x'_j \, (\mathbf{c}_B Y_j), \text{ from (4)}$$

$$= \sum_{j=1}^{N} x'_j \left(\sum_{i=1}^{m} c_{Bi} \, y_{ij} \right)$$

$$= \sum_{i=1}^{m} c_{Bi} \left(\sum_{j=1}^{N} x'_j \, y_{ij} \right)$$

$$= \sum_{i=1}^{m} c_{Bi} \, x_{Bi}, \text{ from (5)}$$

$\therefore \qquad Z' \le Z^*.$

Hence, Z^* is the optimum value of the objective function.

3.8 Alternative Optimal Solutions

The given linear programming problem is said to possess an alternative optimal solution if the set of variables giving the optimal value of the objective function is not unique.

Theorem: *(Conditions for Alternative Optimum Solutions)*

Suppose there exists an optimal B.F.S. to a L.P.P. and

(i) If for some α_j *in A but not in B,* $c_j - Z_j = 0$ *and* $y_{ij} \le 0$ *for all* $i = 1, 2, \ldots, m$*, then a non-basic alternative optimal solution will exist.*

(ii) If for some α_j *in A but not in B,* $c_j - Z_j = 0$ *and* $y_{ij} > 0$ *for at least one i, then an alternative basic optimum solution will exist.*

Proof: (i) We have discussed in article 3.6 that if we introduce the column vector α_j in B where α_j is in A but not in B and $y_{ij} \le 0$ for all $i = 1, 2, \ldots, m$, then a non-basic feasible solution $\mathbf{x}'_B$ with $(m+1)$ number of positive variables is given by

$$\mathbf{x}'_B = [x'_{B1}, x'_{B2}, \ldots, x'_{Bm}, \lambda]$$

where $$x'_{Bi} = x_{Bi} - \lambda\, y_{ij},\ i = 1, 2, \ldots, m,\ \lambda > 0$$

and the value of the objective function for this new F.S. is given by

$$Z' = Z + \lambda\,(c_j - Z_j).$$

If $$c_j - Z_j = 0, \text{then } Z' = Z$$

i.e., the value of the objective function for this new non-basic F.S. is also equal to Z' (optimal value). Hence, this new non-basic F.S. is an alternative optimal solution of the given L.P.P.

(ii) We have shown in theorem of article 3.5 that if $y_{ij} > 0$ for at least one $i = 1, 2, \ldots, m$, then by replacing one column β_r in B by the column α_j which is in A but not in B, we obtain a new B.F.S. $\mathbf{x}'_B$ is given by

$$\mathbf{x}'_B = [x'_{B1}, x'_{B2}, \ldots, x'_{Bm}]$$

where $$x'_{Bi} = x_{Bi} - \frac{y_{ij}}{y_{rj}}\, x_{Br},\ i = 1, 2, \ldots, m,\ i \ne r$$

$$x'_{Br} = \frac{x_{Br}}{y_{rj}},\ i = r$$

and $$\frac{x_{Br}}{y_{rj}} = \operatorname*{Min}_{i}\left(\frac{x_{Bi}}{y_{ij}},\ y_{ij} > 0\right).$$

The value of objective function for this new B.F.S. is given by

$$Z' = Z + \frac{x_{Br}}{y_{rj}}(c_j - Z_j)$$

i.e., the value of the objective function for this new B.F.S. is also equal to Z' (optimal value). Hence, this new B.F.S. is an alternative optimal B.F.S.

3.9 Inconsistency & Redundancy In Constraint Equations

3.9.1 Redundancy in Constraint Equations

By redundancy in constraint equations we mean that the system has more than enough number of constraint equations, in other words, it has more constraint equations than the number of variables.

This is the situation when

$$r(A) = r(A\mathbf{b}) = k \le n < m.$$

In this case, there will be $(m - k)$ redundant equations.

3.9.2 Inconsistency

As already defined, the set of constraints (linear equations) is said to be inconsistent if $r(A) \ne r(A\mathbf{b})$.

Before solving a L.P. problem by simplex method, we should have $r(A) = r(A\mathbf{b})$ *i.e.,* the constraint equations (after introducing the slack and artificial variables) should be consistent. Since in simplex method we always have $r(A) = r(A\mathbf{b}) = m$.

If the system $A\mathbf{x} = \mathbf{b}$ involves artificial variables, then we cannot say whether this system is consistent or there is any redundancy. Below we give the cases (without proof) to decide about the consistency and redundancy in such systems.

Case 1 : If the basis B contains no artificial vector and the optimality condition is satisfied (at any iteration), then the current solution is a **B.F.S.** of the problem.

Case 2 : If one or more artificial vector appears in the basis B at a zero level *i.e.*, the value of the artificial variables corresponding to artificial vectors in B are zero and the optimality condition is satisfied (at any iteration), then the system is consistent. Furthermore if $y_{ij} = 0 \ \forall j$ and $x_{Br} = 0$ and r corresponds to the row containing an artificial vector, then the r-th constraint equation is **redundant.**

Case 3 : If at least one artificial vector appears in the basis B at a positive level *i.e.*, the value of at least one artificial variable corresponding to artificial vector in B is non-zero and the optimality condition is satisfied (at any iteration), then there exists **no feasible** solution of the problem.

3.10 To Determine Starting B.F.S.

In this article, we are going to discuss a method for finding a most convenient initial B.F.S. to a linear programming problem.

In the constraints of a general linear programming problem, there may be any of the three signs $\leq, =, \geq$. Let us assume that the requirement vector $\mathbf{b} \geq \mathbf{0}$ (if any of the b_i's is negative, multiply the corresponding constraint by -1).

Case 1 : To find initial B.F.S. when all the original constraints have $\leq$ sign.

To convert all the constraints into equations we insert slack variables only. The equations obtained are as follows :

$$\begin{array}{lllll} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n + 1.x_{n+1} & & & & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n + & 1.x_{n+2} & & & = b_2 \\ \ldots \quad \ldots \quad \ldots & \ldots & & & \\ \ldots \quad \ldots \quad \ldots & & \ldots & & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n + & & + & 1.x_{n+m} & = b_m \end{array}$$

Here $x_{n+1}, x_{n+2}, \ldots, x_{n+m}$ are slack variables.

In matrix form these equations can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} & 1 & 0\ldots0 \\ a_{21} & a_{22} & \ldots & a_{2n} & 0 & 1\ldots0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & 0 & 0\ldots1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Taking the initial basis matrix

$$B = I_m, \text{(unit matrix of order } m \times m).$$

$\therefore$ The initial basic solution is given by

$$\mathbf{x}_B = B^{-1}\,\mathbf{b} = I_m\,\mathbf{b} = \mathbf{b} \geq 0$$

Thus, the initial B.F.S. is

$$x_{n+1} = x_{B1} = b_1,\ x_{n+2} = x_{B2} = b_2, \ldots, x_{n+m} = x_{Bm} = b_m,$$

which can be obtained by writing all the non-basic variables (*i.e.*, given variables) $x_1, x_2, \ldots, x_n$ equal to zero and solving the equations for the remaining variables (*i.e.*, slack variables) $x_{n+1}, x_{n+2}, \ldots, x_{n+m}$.

Case 2 : To find initial B.F.S. when all the original constraints have ≥ sign.

First we convert all the constraints into equations by inserting surplus variables. The equations obtained are as follows :

$$
\begin{array}{llllll}
a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n - x_{n+1} & & & & = b_1 \\
a_{21}x_1 + a_{22}x_2 + \ldots a_{2n}x_n & -x_{n+2} & & & = b_2 \\
\ldots & \ldots & \ldots & \ldots & \ldots \\
\ldots & \ldots & \ldots & \ldots & \ldots \\
a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n & & & -x_{n+m} & = b_m.
\end{array}
$$

Here $x_{n+1}, x_{n+2}, \ldots, x_{n+m}$ are surplus variables.

In matrix form, these equations can be written as

$$
\begin{bmatrix}
a_{11} & a_{12} & \ldots & a_{1n} & -1 & 0 & \ldots & 0 \\
a_{21} & a_{22} & \ldots & a_{2n} & 0 & -1 & \ldots & 0 \\
\ldots & \ldots & & & \ldots & & \ldots & \ldots \\
\ldots & \ldots & & & \ldots & & \ldots & \\
a_{m1} & a_{m2} & \ldots & a_{mn} & 0 & 0 & \ldots & -1
\end{bmatrix}
\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{bmatrix}
=
\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}
$$

If we take the initial basis matrix $B = -I_m$,

we have $\quad \mathbf{x}_B = B^{-1}\,\mathbf{b} = -I_m\,\mathbf{b} = -\mathbf{b} \leq \mathbf{0}$, which is not a B.F.S.

To avoid this difficulty we add one more variable to each constraint. These variables are called "**artificial variables**".

Adding surplus and artificial variables, the constraints of the given L.P.P. change to the following equations :

$$
\begin{array}{lllll}
a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n & -x_{n+1} & & +x_{n+m+1} & \\
= b_1 & & & & \\
a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n & & -x_{n+2} & +x_{n+m+2} & \\
= b_2 & & & & \\
\ldots & \ldots & \ldots & \ldots & \ldots \\
\ldots & \ldots & \ldots & \ldots & \ldots \\
a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n & & -x_{n+m} & +x_{n+m+m} & = b_m
\end{array}
$$

Here $x_{n+m+1}, x_{n+m+2}, \ldots, x_{n+m+m}$ are the artificial variables.

In matrix form, these equations can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & -1 & 0 & \dots & 0 & 1 & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 & -1 & \dots & 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & 0 & 0 & \dots & -1 & 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \\ x_{n+m+1} \\ \vdots \\ \vdots \\ x_{n+m+m} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Now taking the basis matrix $B = I_m$.

$$\therefore \qquad \mathbf{x}_B = B^{-1}\ \mathbf{b} = I_m\ \mathbf{b} = \mathbf{b} \geq \mathbf{0},$$

which is a basic feasible solution.

$\therefore$ The B.F.S. is $x_{n+m+1} = x_{B1} = b_1,\ x_{n+m+2} = x_{B2} = b_2, \dots, x_{n+m+m} = x_{Bm} = b_m$, which can be obtained by writing all the non-basic variables (*i.e.*, given variables and surplus variables) $x_1, x_2, \dots, x_n;\ x_{n+1}, x_{n+2}, \dots, x_{n+m}$ equal to zero and solving the equations for the remaining basic variables (*i.e.*, artificial variables) $x_{n+m+1}, x_{n+m+2}, \dots, x_{n+m+m}$.

Case 3 : To find initial B.F.S. when the constraints have '≤', '≥' and '=' signs.

In this case, we convert the constraints into equations by inserting slack, surplus and artificial variables. Here the basis matrix $B = I_m$ is obtained by introducing unit column vectors corresponding to the slack and artificial variables.

To obtain the initial B.F.S. we put all the non-basic variables equal to zero and solve for remaining basic variables. Here also the initial B.F.S. $\mathbf{x}_B = \mathbf{b} \geq \mathbf{0}$.

Note :

1. It is important to note that in all the three cases the initial B.F.S. consists of the constants $b_1, b_2, \dots, b_m$, (note that $b_1, b_2, \dots, b_m$ are all non-negative).
2. Sometimes an identity is present in A without introducing the artificial variables. So there is no need to introduce the artificial variables in such cases.

3.11 Computational Procedure of Simplex Method (The Maximization Problem)

(Garhwal 2010)

It consists of the following steps systematically :

Step 1: If the given problem is of minimization, convert it into the maximization problem.

For this, multiply both sides of the objective function by -1 and put $-Z = Z'$.

If v is the maximum value of Z', then $-v$ will be the minimum value of Z.

Step 2 : Make all the b_i's non-negative.

The R.H.S. of each of the constraints should be non-negative. If the L.P.P. has a constraint for which a negative b_i is given, it should be converted into positive value by multiplying both sides of the constraints by -1.

For example, if the given constraint is $7x_1 - 8x_2 \geq -3$, it will change to $-7x_1 + 8x_2 \leq 3$.

Step 3 : Convert inequalities of constraints into equations.

For this introduce slack or surplus variables. The coefficients of slack or surplus variables in the objective function are zero. Introduce artificial variables, if necessary.

Step 4 : Find the initial B.F.S.

Follow the method discussed in article 3.10 to find the initial B.F.S. If artificial variables are introduced in the L.P.P., then follow the two-phase method or Big M-method for solving such problems.

Step 5 : Construct the initial simplex table (starting simplex table).

(See the initial simplex table on next page)

It should be remembered that the values of non-basic variables are always zero at each iteration.

Step 6 : Test the initial B.F.S. for optimality.

Compute the net evaluation Δ_j for each variable x_j by using the formula $\Delta_j = c_j - \mathbf{c}_B Y_j$.

Note that in the starting simplex table Δ_j's are same as c_j's. Also Δ_j's corresponding to basic variables are zero.

Optimality Test :

(i) If $\Delta_j \leq 0$ for all j, the solution under consideration is **optimal.**

(a) Alternative optimal solution will exist if any Δ_j (for non-basic variable) is also zero.

Initial Simplex Table

		$c_j \rightarrow$	c_1	c_2	…	c_k	…	c_n	c_{n+1}	c_{n+2}	…	c_{n+m}	Min.
B	$\mathbf{c}_B$	$\mathbf{x}_B \downarrow$	$Y_1(=\alpha_1)$	$Y_2(=\alpha_2)$		$Y_k(=\alpha_k)$		$Y_n(=\alpha_n)$	$Y_{n+1}(\beta_1)$	$Y_{n+2}(\beta_2)$		$Y_{n+m}(\beta_m)$	ratio
Y_{n+1}	$c_{B1}=0$	$x_{B1}=b_1$	$y_{11}=a_{11}$	$y_{12}=a_{12}$	…	$y_{1k}=a_{1k}$	…	$y_{1n}=a_{1n}$	1	0	…	0	
Y_{n+2}	$c_{B2}=0$	$x_{B2}=b_2$	$y_{21}=a_{21}$	$y_{22}=a_{22}$	…	$y_{2k}=a_{2k}$	…	$y_{2n}=a_{2n}$	0	1	…	0	
⋮	⋮	⋮	⋮	⋮	…	⋮	…	⋮	⋮	⋮	…	⋮	
Y_{n+r}	$c_{Br}=0$	$x_{Br}=b_r$	$y_{r1}=a_{r1}$	$y_{r2}=a_{r2}$	…	$\boxed{y_{rk}=a_{rk}}$	…	$y_{rn}=a_{rn}$	⋮	⋮	…	⋮	$\rightarrow$
⋮	⋮	⋮	⋮	⋮	…	⋮	…	⋮	⋮	⋮	…	⋮	
Y_{n+m}	$c_{Bm}=0$	$x_{Bm}=b_m$	$y_{m1}=a_{m1}$	$y_{m2}=a_{m2}$	…	$y_{mk}=a_{mk}$	…	$y_{mn}=a_{mn}$	0	0	…	1	
$Z=\mathbf{c}_B\mathbf{x}_B=0$		Δ_j	Δ_1	Δ_2		$\Delta_k \uparrow$		Δ_n	Δ_{n+1}	Δ_{n+2}		Δ_{n+m}	

(b) If $\Delta_j < 0$ for all j, corresponding to non-basic variables, then the solution is **unique optimal solution.**

(ii) If $\Delta_j > 0$ for any j *i.e.*, if at least one Δ_j is positive the solution under test is **not optimal.** In this case, we must proceed to *improve the solution* (step 7).

(iii) If corresponding to maximum positive Δ_j, all elements of the column Y_j are negative or zero, the solution under test will be **unbounded.**

(iv) If optimality condition is satisfied but the value of at least one artificial variable present in the basis is non-zero, the problem will have no **feasible solution.**

Step 7 : Select the entering (incoming) vector and departing (outgoing) vector.

To improve the initial B.F.S. we select the vector entering the basis matrix and the vector to be removed from the basis matrix by the following rules :

(i) To find Incoming Vector : The incoming vector α_k is always selected corresponding to the largest positive value of Δ_j.
If maximum value of Δ_j occurs for more than one α_j, then we can select any of these vectors as incoming vector.

(ii) To find Outgoing Vector : The departing vector β_r is selected corresponding to that value of r for which

$$\frac{x_{Br}}{y_{rk}} = \min_i \left\{ \frac{x_{Bi}}{y_{ik}},\ y_{ik} > 0 \right\},$$

if α_k is the incoming vector.

Note : If the above mentioned minimum value is not unique, then more than one variable will vanish in the next solution. As a result the next solution will be a degenerate B.F.S. for which the outgoing vector is selected in a different way.

Step 8 : If α_k is the entering vector and β_r is the outgoing vector, then the element y_{rk} which lies at the intersection of minimum ratio arrow ($\rightarrow$) and incoming vector arrow $\uparrow$ is called the pivot element (key element).

We put this element in ÿ.

In order to bring β_r in place of incoming vector Y_k, unity must occupy the place ÿ. In other words, key element y_{rk} should be 1. If it is not so, then divide all the elements of this row by the key element y_{rk}. Then subtract suitable multiples of the row containing the key element from all other rows and obtain zero at all other positions of this column Y_k. Now bring β_r in place of α_r and construct the revised simplex table.

Thus, we get improved basic feasible solution.

Step 9 : Test the improved B.F.S. for optimality.

If it is not optimum, repeat steps 7 and 8 until we obtain an optimum solution.

Illustrative Examples

Example 4: *Maximize* $Z = 40x_1 + 35x_2$

subject to $2x_1 + 3x_2 \le 60$

$4x_1 + 3x_2 \le 96,$

$x_1, x_2 \ge 0.$

For the clear understanding of the simplex method the solution to this problem is illustrated below in a step-wise manner.

Solution: Step 1: The given problem is of maximization and all the b_i's are already positive.

Step 2 : Converting the inequalities of constraints into equations by introducing slack variables x_3 and x_4, we get

$$2x_1 + 3x_2 + x_3 = 60 \quad \text{and } 4x_1 + 3x_2 + x_4 = 96$$

The coefficients of slack variables are zero in the objective function. Therefore, the given problem becomes

Maximize $Z = 40\,x_1 + 35x_2 + 0x_3 + 0x_4$

subject to $2x_1 + 3x_2 + x_3 + 0x_4 = 60$

$4x_1 + 3x_2 + 0x_3 + x_4 = 96$

$x_1, x_2, x_3, x_4 \ge 0.$

This is the standardised form of the given problem.

Step 3 : The simplex method begins with an initial basic feasible solution in which the values of basic variables are zero.

$\therefore \quad x_1 = 0,\ x_2 = 0$ which gives $x_3 = 60,\ x_4 = 96.$

Step 4 : Constructing the initial simplex table.

First Simplex Table

B	c_B	c_j	40	35	0	0	Min. ratio
		x_B	Y_1	Y_2	$Y_3(\beta_1)$	$Y_4(\beta_2)$	$x_B/Y_1,\ y_{i1} > 0$
Y_3	0	60	2	3	1	0	$60/2 = 30$
Y_4	0	96	$\boxed{4}$	3	0	1	$96/4 = 24(\min) \rightarrow$
$Z = c_B x_B = 0$		Δ_j	40 $\uparrow$	35	0	0 $\downarrow$	

Step 5 : Computing Δ_j for all non-basic variables x_j, $j = 1, 2$, using the formula

$$\Delta_j = c_j - \mathbf{c}_B Y_j$$

$$\Delta_1 = c_1 - \mathbf{c}_B\, Y_1 = 40 - (0, 0)\ (2, 4) = 40$$

$$\Delta_2 = c_2 - \mathbf{c}_B\, Y_2 = 35 - (0, 0)\ (3, 3) = 35.$$

Since all Δ_j's are not less than or equal to zero therefore the solution is not optimal. So we proceed to next step to improve the solution.

Step 6 : Incoming and outgoing vectors.

Since $\Delta_1 = 40$ is the maximum value of Δ_j, $j = 1, 2$, therefore $\alpha_1 = Y_1$ is the **incoming vector.**

To find outgoing vector

Now $$\frac{\mathbf{x}_B}{Y_1} = \left(\frac{x_{B1}}{y_{11}}, \frac{x_{B2}}{y_{21}}\right) = \left(\frac{60}{2}, \frac{96}{4}\right) = (30, 24).$$

$$\frac{x_{Br}}{y_{r1}} = \min_i\left[\frac{x_{Bi}}{y_{i1}},\ y_{i1} > 0\right] = \min_i\left(\frac{x_{B1}}{y_{11}}, \frac{x_{B2}}{y_{21}}\right) = 24 = \frac{x_{B2}}{y_{21}}$$

$\therefore$ $r = 2$ which gives $\beta_2 = Y_4$ as the **outgoing vector**.

$\therefore$ $y_{21} = a_{21}$ is the key element which is equal to 4.

To bring Y_1 in place of Y_4 we proceed in the following manner :

Dividing the second row containing the key element $a_{21} = 4$ by 4 to make it unity.

Now we shall subtract appropriate multiples of this new row from the other remaining row so as to obtain zeros in the remaining positions of Y_1.

Subtracting 2 times of the second row obtained after dividing by 4 from the first row and subtract 40 times of the this second row from the row containing Δ_j's to obtain new values of Δ_j's.

Constructing the second simplex table in which β_2 (Y_4) is replaced by $\alpha_1 (= Y_1)$.

Second Simplex Table

B	$\mathbf{c}_B$	c_j	**10**	**35**	**0**	**0**	**Min. ratio**
		$\mathbf{x}_B$	$Y_1\,(\beta_2)$	Y_2	$Y_3\,(\beta_1)$	Y_4	$\mathbf{x}_B/Y_2$, $y_{i2} > 0$
Y_3	0	12	0	$\boxed{3/2}$	1	$-1/2$	8 (min) $\rightarrow$
Y_1	40	24	1	3/4	0	1/4	32
$Z = \mathbf{c}_B \mathbf{x}_B = 960$		Δ_j	0	5 $\uparrow$	0 $\downarrow$	-10	

Also computing Δ_j by using the formula $\Delta_j = c_j - \mathbf{c}_B Y_j$ for x_2, x_4, we get

$$\Delta_2 = c_2 - \mathbf{c}_B Y_2 = 35 - (0,40)\left(\frac{3}{2}, \frac{3}{4}\right) = 5$$

$$\Delta_4 = c_4 - \mathbf{c}_B Y_4 = 0 - (0,40)\left(-\frac{1}{2}, \frac{1}{4}\right) = -10$$

Δ_1 and Δ_3 are zero as they correspond to unit column vectors.

Since all Δ_j's are not less than or equal to zero therefore the solution obtained is not optimal.

$\Delta_2 = 5$ is the maximum value of Δ_j.

$\therefore$ $\alpha_2 = Y_2$ is the **incoming vector.**

Now
$$\frac{x_{Br}}{y_{r2}} = \min_i \left[\frac{x_{Bi}}{y_{i2}}, y_{i2} > 0\right]$$

$$= \min\left[\frac{x_{B1}}{y_{12}}, \frac{x_{B2}}{y_{22}}\right] = \min[8,32] = 8 = \frac{x_{B1}}{y_{12}}$$

$\therefore$ $r = 1$ which gives $\beta_1 = Y_3$ as the outgoing vector.

$\therefore$ $y_{12} = a_{12} = \frac{3}{2}$ is the key element.

To bring Y_2 in place of Y_3 we proceed in the following manner :

Dividing the first row by 3/2 to make the key element unity.

Subtracting 3/4 times of the first row thus obtained from the second row and subtracting 5 times of it from the row containing Δ_j's to obtain new values of Δ_j's.

Constructing the third simplex table in which Y_3 is replaced by Y_2.

Third Simplex Table

B	$\mathbf{c}_B$	c_j	40	35	0	0	Min. ratio
		$\mathbf{x}_B$	$Y_1(\beta_2)$	$Y_2(\beta_1)$	Y_3	Y_4	
Y_2	35	8	0	1	2/3	−1/3	
Y_1	40	18	1	0	−1/2	1/2	
$Z = \mathbf{c}_B \mathbf{x}_B = 1000$		Δ_j	0	0	−10/3	−25/3	

Since all $\Delta_j \leq 0$, therefore the above solution is optimal.

Hence the optimal solution is

$$x_1 = 18, x_2 = 8 \text{ and Max. } Z = 1000.$$

This solution is unique optimal solution.

$\because$ Both $\Delta_3, \Delta_4 < 0$ corresponding to non-basic variables x_3, x_4.

The above solution with its different computational steps can be more conveniently represented by a single table as shown below :

B	$\mathbf{c}_B$	c_j	40	35	0	0	Min. ratio
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	$\mathbf{x}_B/Y_1, Y_{i1} > 0$
Y_3	0	60	2	3	1	0	$60/2 = 30$
Y_4	0	96	$\boxed{4}$	3	0	1	$96/4 = \underset{\rightarrow}{24}$ (min)
$Z = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	40 $\uparrow$	35	0	0 $\downarrow$	$\mathbf{x}_B/Y_2, Y_{i2} > 0$
Y_3	0	12	0	$\boxed{3/2}$	1	$-1/2$	$\underset{\rightarrow}{8}$ (min)
Y_1	40	24	1	3/4	0	1/4	32
$Z = \mathbf{c}_B\mathbf{x}_B = 960$		Δ_j	0	5 $\uparrow$	0 $\downarrow$	-10	
Y_2	35	8	0	1	2/3	$-1/3$	
Y_1	40	18	1	0	$-1/2$	1/2	
$Z = \mathbf{c}_B\mathbf{x}_B = 1000$		Δ_j	0	0	$-10/3$	$-25/3$	

Example 5: *Solve by simplex method the following L.P.P. :*

Maximize $Z = 3x_1 + 5x_2 + 4x_3$

subject to $2x_1 + 3x_2 \leq 8,\ 2x_2 + 5x_3 \leq 10$

$3x_1 + 2x_2 + 4x_3 \leq 15,\ x_1, x_2, x_3 \geq 0.$

Solution: The given problem is of maximization and all the b_i's are positive.

Converting the inequalities of constraints into equations by introducing slack variables x_4, x_5, x_6. Also the coefficients of slack variables are zero in the objective function. Thus the given problem becomes

$$\text{Max } Z = 3x_1 + 5x_2 + 4x_3 + 0x_4 + 0x_5 + 0x_6$$

subject to
$$\begin{aligned} 2x_1 + 3x_2 + 0x_3 + x_4 \quad &= 8 \\ 0x_1 + 2x_2 + 5x_3 \quad + x_5 &= 10 \\ 3x_1 + 2x_2 + 4x_3 \quad + x_6 &= 15 \end{aligned}$$

$$x_1, x_2, \ldots, x_6 \geq 0.$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_4 = 8, x_5 = 10, x_6 = 15$, which is the initial B.F.S.

The solution to the problem using simplex method is given in the following table :

B	c_B	c_j / x_B	3 / Y_1	5 / Y_2	4 / Y_3	0 / Y_4	0 / Y_5	0 / Y_6	Min. ratio x_B/Y_2, $y_{i2} > 0$
Y_4	0	8	2	$\boxed{3}$	0	1	0	0	8/3 $\underset{\rightarrow}{(}$ min)
Y_5	0	10	0	2	5	0	1	0	10/2 = 5
Y_6	0	15	3	2	4	0	0	1	15/2
$Z = c_B x_B = 0$		Δ_j	3	5 ↑	4	0 ↓	0	0	x_B/Y_3, $y_{i3} > 0$
Y_2	5	8/3	2/3	1	0	1/3	0	0	—
Y_5	0	14/3	−4/3	0	$\boxed{5}$	−2/3	1	0	14/15 $\underset{\rightarrow}{(}$ min)
Y_6	0	29/3	5/3	0	4	−2/3	0	1	29/12
$Z = c_B x_B = 40/3$		Δ_j	−1/3	0	4 ↑	−5/3	0 ↓	0	x_B / Y_1, $y_{i1} > 0$
Y_2	5	8/3	2/3	1	0	1/3	0	0	4
Y_3	4	14/15	−4/15	0	1	−2/15	1/5	0	—
Y_6	0	89/15	$\boxed{41/15}$	0	0	−2/15	−4/5	1	89/41$\underset{\rightarrow}{(}$min)
$Z = c_B x_B = 256/15$		Δ_j	11/15 ↑	0	0	−17/15	−4/5	0 ↓	
Y_2	5	50/41	0	1	0	15/41	8/41	−10/41	
Y_3	4	62/41	0	0	1	−6/41	5/41	4/41	
Y_1	3	89/41	1	0	0	−2/41	−12/41	15/41	
$Z = c_B x_B = 765/41$		Δ_j	0	0	0	−45/41	−24/41	−11/41	

In the last table all Δ_j's ≤ 0, therefore it gives optimal solution.

$\therefore$ Optimal solution is

$$x_1 = 89/41,\ x_2 = 50/41,\ x_3 = 62/41 \quad \text{and} \quad \text{Max. } Z = 765/41$$

Example 6: *Solved by simplex method the following L.P.P. :*

Maximize $\quad Z = 3x_1 + 2x_2 + 5x_3$

$$\text{subject to} \quad \begin{aligned} x_1 + 2x_2 + x_3 &\le 430 \\ 3x_1 + 2x_3 &\le 460 \\ x_1 + 4x_2 &\le 420, \\ x_1, x_2, x_3 &\ge 0. \end{aligned}$$

(Garhwal 2006)

Solution: The given problem is of maximization and all the b_i's are positive.

Converting the inequalities of the constraints into equations by introducing slack variables x_4, x_5 and x_6 the given problem becomes

$$\text{Max.} \quad Z = 3x_1 + 2x_2 + 5x_3 + 0x_4 + 0x_5 + 0x_6$$

$$\text{subject to} \quad \begin{aligned} x_1 + 2x_2 + x_3 + x_4 &= 430 \\ 3x_1 + 0x_2 + 2x_3 + x_5 &= 460 \\ x_1 + 4x_2 + 0x_3 + x_6 &= 420 \\ x_1, x_2, \dots, x_6 &\ge 0. \end{aligned}$$

Taking $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, we get $x_4 = 430$, $x_5 = 460$, $x_6 = 420$ which is the initial B.F.S.

The solution to the problem using simplex algorithm is given in the following table :

B	$\mathbf{c}_B$	c_j / $\mathbf{x}_B$	3 / Y_1	2 / Y_2	5 / Y_3	0 / Y_4	0 / Y_5	0 / Y_6	Min. ratio $\mathbf{x}_B/Y_3$, $y_{i3} > 0$
Y_4	0	430	1	2	1	1	0	0	430
Y_5	0	460	3	0	2	0	1	0	230 (min) $\rightarrow$
Y_6	0	420	1	4	0	0	0	1	—
$Z = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	3	2	5 $\uparrow$	0	0 $\downarrow$	0	$\mathbf{x}_B/Y_2$, $y_{i2} > 0$
Y_4	0	200	−1/2	2	0	1	−1/2	0	100 (min) $\rightarrow$
Y_3	5	230	3/2	0	1	0	1/2	0	—
Y_6	0	420	1	4	0	0	0	1	420/4
$Z = 1150$		Δ_j	−9/2	2 $\uparrow$	0	0 $\downarrow$	−5/2	0	
Y_2	2	100	−1/4	1	0	1/2	−1/4	0	
Y_3	5	230	3/2	0	1	0	1/2	0	
Y_6	0	20	2	0	0	−2	1	1	
$Z = 1350$		Δ_j	−4	0	0	−1	−2	0	

In the last table all $\Delta_j \le 0$, therefore the solution is optimal.

Hence, the optimal solution is

$$x_1 = 0,\ x_2 = 100,\ x_3 = 230 \text{ and Max. } Z = 1350.$$

Example 7: *Solve by simplex method the following L.P.P. :*

$$\begin{aligned} \textit{Minimize} \quad & Z = x_2 - 3x_3 + 2x_5 \\ \textit{subject to} \quad & 3x_2 - x_3 + 2x_5 \le 7 \\ & -2x_2 + 4x_3 \le 12 \\ & -4x_2 + 3x_3 + 8x_5 \le 10, \\ & x_2, x_3, x_5 \ge 0. \end{aligned}$$

(Garhwal 2008)

Solution: The given problem is of minimization. Convert it to maximization problem by taking the objective function as $Z' = -Z$.

The objective function becomes

$$\text{Max.} \quad Z' = -Z = -x_2 + 3x_3 - 2x_5.$$

Converting the inequalities of constraints into equations introducing slack variables x_1, x_4 and x_6, the given problem becomes

$$\text{Max.} \quad Z' = -Z = 0\,.\,x_1 - x_2 + 3x_3 + 0\,.\,x_4 - 2x_5 + 0\,.\,x_6$$

$$\begin{aligned} \text{subject to} \quad & x_1 + 3x_2 - x_3 + 0x_4 + 2x_5 + 0x_6 = 7 \\ & 0x_1 - 2x_2 + 4x_3 + x_4 + 0x_5 + 0x_6 = 12 \\ & 0x_1 - 4x_2 + 3x_3 + 0x_4 + 8x_5 + x_6 = 10, \\ & x_1, x_2, \ldots, x_6 \ge 0. \end{aligned}$$

Taking $x_2 = 0, x_3 = 0, x_5 = 0$, we get $x_1 = 7, x_4 = 12, x_6 = 10$, which is the initial B.F.S.

The solution to the problem using simplex algorithm is given below :

B	$\mathbf{c}_B$	c_j	**0**	**−1**	**3**	**0**	**−2**	**0**	**Min. ratio**
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	$\mathbf{x}_B/Y_3,\ y_{i3} > 0$
Y_1	0	7	1	3	−1	0	2	0	—
Y_4	0	12	0	−2	4	$\boxed{1}$	0	0	3 (min) →
Y_6	0	10	0	−4	3	0	8	1	10/3
$Z' = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	0	−1	3 ↑	0 ↓	−2	0	$\mathbf{x}_B/Y_2,\ y_{i2} > 0$

Y_1	0	10	1	$\boxed{5/2}$	0	1/4	2	0	4 (min) →
Y_3	3	3	0	−1/2	1	1/4	0	0	—
Y_6	0	1	0	−5/2	0	−3/4	8	1	—
$Z' = 9$		Δ_j	0 ↓	1/2 ↑	0	−3/4	−2	0	
Y_2	−1	4	2/5	1	0	1/10	4/5	0	
Y_3	3	5	1/5	0	1	3/10	2/5	0	
Y_6	0	11	1	0	0	−1/2	10	1	
$Z' = 11$		Δ_j	−1/5	0	0	−4/5	−12/5	0	

Since all Δ_j's ≤ 0, therefore the solution given by last table is optimal.

Hence, the optimal solution is

$$x_2 = 4,\ x_3 = 5,\ x_5 = 0$$

and Min. $Z = -Z' = -11$

Note : Another form of Ex. 7 is

Min. $Z = x_1 - 3x_2 + 2x_3$

subject to $3x_1 - x_2 + 2x_3 \leq 7$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10,\ x_1, x_2, x_3 \geq 0$$

Example 8: *Solve by simplex method the following L.P.P. :*

Maximize $Z = 2x_1 + 5x_2 + 7x_3$

subject to $3x_1 + 2x_2 + 4x_3 \leq 100$

$$x_1 + 4x_2 + 2x_3 \leq 100$$

$$x_1 + x_2 + 3x_3 \leq 100,$$

$$x_1, x_2, x_3 \geq 0.$$

Solution: The given problem is of maximization and all b_i's are positive.

Introducing slack variables x_4, x_5, x_6 to convert constraint inequalities into equations, the given problem becomes

$$Z = 2x_1 + 5x_2 + 7x_3 + 0x_4 + 0x_5 + 0x_6$$

subject to $3x_1 + 2x_2 + 4x_3 + x_4 = 100$

$$x_1 + 4x_2 + 2x_3 + x_5 = 100$$

$$x_1 + x_2 + 3x_3 + x_6 = 100.$$

The starting B.F.S. is

$$x_1 = 0,\ x_2 = 0,\ x_3 = 0,\ x_4 = 100,\ x_5 = 100,\ x_6 = 100.$$

The solution to the problem using simplex method is given below :

B	$\mathbf{c}_B$	c_j	2	5	7	0	0	0	Min. ratio
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	$\mathbf{x}_B/Y_3$, $y_{i3} > 0$
Y_4	0	100	3	2	$\boxed{4}$	1	0	0	25 (min) →
Y_5	0	100	1	4	2	0	1	0	50
Y_6	0	100	1	1	3	0	0	1	100/3
$Z = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	2	5	7 ↑	0 ↓	0	0	$\mathbf{x}_B/Y_2$, $y_{i2} > 0$
Y_3	7	25	3/4	1/2	1	1/4	0	0	50
Y_5	0	50	−1/2	$\boxed{3}$	0	−1/2	1	0	50/3 (min) →
Y_6	0	25	−5/4	−1/2	0	−3/4	0	1	Neg.
$Z = \mathbf{c}_B\mathbf{x}_B = 175$		Δ_j	−13/4	3/2 ↑	0	−7/4	0 ↓	0	
Y_3	7	50/3	5/6	0	1	1/3	−1/6	0	
Y_2	5	50/3	−1/6	1	0	−1/6	1/3	0	
Y_6	0	100/3	−4/3	0	0	−5/6	1/6	1	
$Z = \mathbf{c}_B\mathbf{x}_B = 200$		Δ_j	−3	0	0	−3/2	−1/2	0	

In the last table all $\Delta_j \le 0$, therefore the solution is optimal.

Hence, the optimal solution is

$$x_1 = 0,\ x_2 = 50/3,\ x_3 = 50/3 \text{ and Max. } Z = 200.$$

3.12 Artificial Variables Technique

Sometimes in linear programming problems constraints may also have ≥ and = signs after ensuring that all $b_i \ge 0$. In such problems, we first introduce **surplus variables** to convert inequalities into equations. In such cases, basis matrix is not obtained as an identity matrix in the starting simplex table. To overcome this difficulty we introduce new variables, called, the **artificial variables** to each of such constraints. These variables are fictitious and cannot have any physical meaning.

The artificial variables are introduced for the limited purpose of obtaining an **initial solution.** It is not relevant whether the objective function is of the maximization or minimization type. Since artificial variables do not represent any quantity relating to the decision problem they must be driven out of the system and must not be present in the final solution (if at all they do, it represents a situation of infeasibility which is discussed later in this chapter).

The L.P.P. involving artificial variables can be solved by two methods :

1. **Method 1 :** Big M-Method (Carners M-Method)
2. **Method 2 :** Two Phase Method

3.12.1 Method 1: Big M-Method (Carners M-Method)

For the solution of the L.P.P. involving artificial variables a method known as **Big-M Method or Carners M-Method** was developed by A. Carners. In this method, a very large negative price say $-M$ is assigned to each artificial variable in the objective function.

∴ The objective function is written as

$$Z = \mathbf{c}\,\mathbf{x} + 0\,.\,\mathbf{x}_{\text{slack}\,.\,V} + 0\,.\,\mathbf{x}_{\text{surplus}\,.\,V} - M\,.\,\mathbf{x}_{\text{art}\,.\,V}$$

Due to this large negative price $-M$ in the objective function, it cannot be improved in the presence of the artificial variables. So first we shall have to remove these artificial vector from the basis matrix. Once artificial column vectors corresponding to the artificial variables leave the basis, we forget about it forever and never consider it as a vector to enter into the basis at any iteration.

For clear understanding of the procedure see following examples :

Note : A solution to the problem which does not contain an artificial variable in the basis, represents a feasible solution to the problem.

Illustrative Examples

Example 9: Apply Big-M Method to solve the following L.P.P. :

Max. $Z = 2x_1 + 4x_2$

subject to $2x_1 + x_2 \le 18$

$3x_1 + 2x_2 \ge 30$

$x_1 + 2x_2 = 26$

$x_1, x_2 \ge 0.$

Solution: The problem is of maximization and all b_i's are positive.

Introducing the necessary slack variable x_3, surplus variable x_4 and artificial variables, x_{a_1}, x_{a_2} and assigning large negative costs $-M$ to artificial variables, the problem reduces to the form

$$\text{Max.} \quad Z = 2x_1 + 4x_2 + 0.x_3 + 0.x_4 - Mx_{a_1} - Mx_{a_2}$$

$$\begin{aligned} \text{subject to} \quad 2x_1 + x_2 + x_3 &= 18 \\ 3x_1 + 2x_2 - x_4 + x_{a_1} &= 30 \\ x_1 + 2x_2 + x_{a_2} &= 26. \end{aligned}$$

Taking $x_1 = 0, x_2 = 0, x_4 = 0$, we get $x_3 = 18, x_{a_1} = 30, x_{a_2} = 26$, which is the initial (starting) B.F.S.

The solution to the problem using simplex method is given in the following table :

B	$\mathbf{c}_B$	c_j	2	4	0	0	$-M$	$-M$	Min. ratio
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	A_1	A_2	$\mathbf{x}_B/Y_2$, $y_{i2} > 0$
Y_3	0	18	2	1	1	0	0	0	18
A_1	$-M$	30	3	2	0	-1	1	0	15
A_2	$-M$	26	1	2	0	0	0	1	13 (min) →
$Z = \mathbf{c}_B\mathbf{x}_B = -56M$		Δ_j	$2+4M$	$4+4M$ ↑	0	$-M$	0	0 ↓	$\mathbf{x}_B/Y_1$, $y_{i1} > 0$
Y_3	0	5	3/2	0	1	0	0		10/3
A_1	$-M$	4	2	0	0	-1	1		2 (min) →
Y_2	4	13	1/2	1	0	0	0		26
$Z = 52 - 4M$		Δ_j	$2M$ ↑	0	0	$-M$	0 ↓		
Y_3	0	2	0	0	1	3/4			
Y_1	2	2	1	0	0	$-1/2$			
Y_2	4	12	0	1	0	1/4			
$Z = 52$		Δ_j	0	0	0	0			

Computation of Δ_j

For the first table

$$\Delta_1 = c_1 - \mathbf{c}_B Y_1 = 2 - (0, -M, -M)(2, 3, 1) = 2 + 4M$$

$$\Delta_2 = c_2 - \mathbf{c}_B Y_2 = 4 - (0, -M, -M)(1, 2, 2) = 4 + 4M$$

$$\Delta_4 = c_4 - \mathbf{c}_B Y_4 = 0 - (0, -M, -M)(0, -1, 0) = -M$$

For the second table,

$$\Delta_1 = c_1 - \mathbf{c}_B Y_1 = 2 - (0, -M, 4)(3/2, 2, 1/2) = 2M$$

$$\Delta_2 = 0 = \Delta_3 = \Delta_5, \Delta_4 = c_4 - \mathbf{c}_B Y_4 = 0 - (0, -M, 4)(0, -1, 0) = -M$$

For the third table

$$\Delta_1 = 0 = \Delta_2 = \Delta_3, \Delta_4 = c_4 - \mathbf{c}_B Y_4 = 0 - (0, 2, 4)(3/4, -1/2, 1/4) = 0.$$

In the last table all $\Delta_j \le 0$, and no artificial variable appears in the basis, therefore this solution is optimal.

Hence, the optimal solution is $x_1 = 2$, $x_2 = 12$ and Max. $Z = 52$.

Example 10: *Apply Big-M Method to solve the following L.P.P. :*

$$\begin{aligned} \textit{Max.} \quad & Z = x_1 + 2x_2 + 3x_3 - x_4 \\ \textit{subject to} \quad & x_1 + 2x_2 + 3x_3 = 15 \\ & 2x_1 + x_2 + 5x_3 = 20 \\ & x_1 + 2x_2 + x_3 + x_4 = 10, \\ & x_1, x_2, x_3, x_4 \ge 0. \end{aligned}$$

Solution: The given problem is of maximization and all b_i's are positive. Also the constraints are equations. Examining the constraints we observe that in order to obtain a unit matrix of order 3 we need two more unit vectors as one unit vector is formed by the coefficients of x_4. Therefore introducing two artificial variables x_{a_1} and x_{a_2} in the first two constraints and assigning large negative cost $-M$ to the artificial variables, the given problem becomes

$$\begin{aligned} \text{Max.} \quad & Z = x_1 + 2x_2 + 3x_3 - x_4 - Mx_{a_1} - Mx_{a_2} \\ \text{subject to} \quad & x_1 + 2x_2 + 3x_3 + x_{a_1} = 15 \\ & 2x_1 + x_2 + 5x_3 + x_{a_2} = 20 \\ & x_1 + 2x_2 + x_3 + x_4 = 10. \end{aligned}$$

Taking $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, we get $x_{a_1} = 15$, $x_{a_2} = 20$, $x_4 = 10$, which is the initial B.F.S.

The solution to the problem using simplex method is given in the following table :

B	$\mathbf{c}_B$	c_j	1	2	3	-1	$-M$	$-M$	**Min. ratio**
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	A_1	A_2	$\mathbf{x}_B/Y_3,$ $y_{i3}>0$
A_1	$-M$	15	1	2	3	0	1	0	5
A_2	$-M$	20	2	1	$\boxed{5}$	0	0	1	4 (min) $\rightarrow$
Y_4	-1	10	1	2	1	1	0	0	10
$Z=\mathbf{c}_B\mathbf{x}_B$ $=-35M-10$		Δ_j	$3M+2$	$3M+4$	$8M+4$ $\uparrow$	0	0	0 $\downarrow$	$\mathbf{x}_B/Y_2,$ $y_{i2}>0$
A_1	$-M$	3	$-1/5$	$\boxed{7/5}$	0	0	1		15/7 (min) $\rightarrow$
Y_3	3	4	2/5	1/5	1	0	0		20
Y_4	-1	6	3/5	9/5	0	1	0		10/3
$Z=\mathbf{c}_B\mathbf{x}_B$ $=-3M+6$		Δ_j	$\frac{(-M+2)}{5}$	$\frac{(7M+16)}{5}$ $\uparrow$	0	0	0 $\downarrow$		$\mathbf{x}_B/Y_1,$ $y_{i1}>0$
Y_2	2	15/7	$-1/7$	1	0	0			Neg.
Y_3	3	25/7	3/7	0	1	0			25/3
Y_4	-1	15/7	$\boxed{6/7}$	0	0	1			5/2 (min) $\rightarrow$
$Z=\mathbf{c}_B\mathbf{x}_B$ $=90/7$		Δ_j	6/7 $\uparrow$	0	0	0 $\downarrow$			
Y_2	2	5/2	0	1	0	1/6			
Y_3	3	5/2	0	0	1	$-1/2$			
Y_1	1	5/2	1	0	0	7/6			
$Z=\mathbf{c}_B\mathbf{x}_B$ $=15$		Δ_j	0	0	0	-1			

Computation of Δ_j. By $\Delta_j = c_j - \mathbf{c}_B Y_j$

For the first table,

$$\Delta_1 = c_1 - \mathbf{c}_B Y_1 = 1-(-M,-M,-1)(1, 2, 1) = 3M+2,$$

$$\Delta_2 = c_2 - \mathbf{c}_B Y_2 = 2-(-M,-M,-1)(2, 1, 2) = 3M+4$$

$$\Delta_3 = c_3 - \mathbf{c}_B Y_3 = 3-(-M,-M,-1)(3, 5, 1) = 8M+4$$

$$\Delta_4 = 0 = \Delta_5 = \Delta_6$$

For the second table,

$$\Delta_1 = 1-(-M, 3,-1)(-1/5, 2/5, 3/5) = \frac{1}{5}(-M+2),$$

$$\Delta_2 = 2-(-M, 3,-1)(7/5, 1/5, 9/5) = \frac{1}{5}(7M+16)$$

$$\Delta_3 = 0 = \Delta_4 = \Delta_5$$

For the third table,

$$\Delta_1 = 1-(2, 3, -1)(-1/7, 3/7, 6/7) = 6/7, \Delta_2 = 0 = \Delta_3 = \Delta_4$$

In the last table all $\Delta_j \le 0$, therefore this solution is optimal.

Hence, the optimal solution is

$$x_1 = 5/2 = x_2 = x_3,\ x_4 = 0 \text{ and Max. } Z = 15$$

Infeasible Solution

Example 11: *Apply Big-M method to solve the L.P.P.*

Max. $Z = -x_1 - x_2$

subject to $3x_1 + 2x_2 \ge 30$

$-2x_1 + 3x_2 \le -30$

$x_1 + x_2 \le 5,$

$x_1, x_2 \ge 0.$

Solution: The given problem is of maximization.

Since b_i in second constraint is negative therefore multiplying both sides by -1, we get

$$2x_1 - 3x_2 \ge 30.$$

To convert inequalities of constraints into equations, introducing slack, surplus and artificial variables wherever necessary, the given problem becomes

Max. $Z = -x_1 - x_2 + 0x_3 + 0x_4 + 0x_5 - Mx_{a_1} - Mx_{a_2}$

subject to $3x_1 + 2x_2 - x_3 + x_{a_1} = 30$

$2x_1 - 3x_2 - x_4 + x_{a_2} = 30$

$x_1 + x_2 + x_5 = 5$

where $x_1, x_2, x_3, x_4, x_5, x_{a_1}, x_{a_2} \ge 0.$

Taking $x_1 = 0,\ x_2 = 0,\ x_3 = 0,\ x_4 = 0$, we get $x_5 = 5,\ A_1 = 30,\ A_2 = 30$ which is the initial B.F.S.

The solution to the problem using simplex method is given below :

B	c_B	c_j	-1	-1	0	0	0	$-M$	$-M$	**Min. ratio**
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	A_1	A_2	x_B/Y_1, $y_{i1}>0$
A_1	$-M$	30	3	2	-1	0	0	1	0	10
A_2	$-M$	30	2	-3	0	-1	0	0	1	15
Y_5	0	5	$\boxed{1}$	1	0	0	1	0	0	5 (min) →
$Z=c_Bx_B$ $=-60M$		Δ_j	$5M-1$ ↑	$-M$	$-M$	$-M$	0 ↓	0	0	
A_1	$-M$	15	0	-1	-1	0	-3	1	0	
A_2	$-M$	20	0	-5	0	-1	-2	0	1	
Y_1	-1	5	1	1	0	0	1	0	0	
$Z=-35M-5$		Δ_j	0	$-6M$	$-M$	$-M$	$-5M$	0	0	

Since all $\Delta_j \le 0$, therefore the solution obtained is optimal. But the artificial column vectors A_1, A_2 (corresponding to artificial variables x_{a_1}, x_{a_2}) appear in the basis at positive levels which implies that *the given L.P.P. has no feasible solution.*

Note : Sometimes the constraints may be inconsistent so that there is no feasible solution to the problem. Such a situation is called **infeasibility**. In the case of infeasibility, one or more artificial variables appear in the basis at positive level in the find simplex table.

3.12.2 Method 2: Two Phase Method

As an alternative to the Big-M method, there is another method for dealing with linear programming problems involving artificial variables. This is called the **two phase method** and as its name suggests it separates the solution procedure into two phases. **In phase I,** all the artificial variables are eliminated from the basis.

In phase II, we use the solution from phase I as the initial basic feasible solution and use the simplex method to determine the optimal solution.

3.12.2.1 Computational Procedure of Two Phase Method (Garhwal 2010)

Phase I : Step 1 : After making all b_i's positive we convert each of the constraints into equations, by introducing slack, surplus and artificial variables.

Step 2 : Assign zero coefficient to each of the primary, slack and surplus variables and the coefficient (–1) to each of the artificial variables in the objective function. As a result the new objective function is

$$Z' = -\text{ (sum of the artificial variables)}$$

Step 3 : Solve the problem formed in step 2 by applying the simplex method. If the original problem has a feasible solution, then this problem shall have an optimal solution with optimal value of the objective function Z' equal to zero as each of the artificial variables will be equal to zero.

If max $Z' < 0$ and at least one artificial variable appears in the optimum basis at a positive level, then the given problem does not possess any feasible solution.

If max $Z' = 0$ and at least one artificial variable appears in the optimum basis at zero level, we proceed to phase II.

If max $Z' = 0$ and no artificial variable appears in the optimum basis then also we proceed to phase II.

Phase II : In phase II start with the optimal solution contained in the final simplex table of the phase I. Assign the actual costs to the variables in the objective function and a zero cost to every slack and surplus variable. Eliminate the artificial variables which are non-basic at the end of the phase-I. Remove c_j row values of the optimum table and replace them by c_j values of the original problem. Now apply simplex algorithm to the problem contained in the new table to obtain the optimal solution.

3.12.3 Disadvantages of Big-M Method Over Two Phase Method

1. We can always use Big-M method to check the existence of a feasible solution but its computational procedure may be inconvenient because of the manipulation of the constant M. Two phase method eliminates the artificial variables in the beginning.
2. When we solve a problem on a digital computer we have to assign some numerical value to M which must be larger than the values $c_1, c_2, \ldots$ present in the objective function. But a computer has only a fixed number of digits.

Illustrative Examples

Example 12: Solve the following problem by using the two phase method :

$$\begin{aligned} \text{Min.} \quad & Z = 40x_1 + 24x_2 \\ \text{subject to} \quad & 20x_1 + 50x_2 \geq 4800 \\ & 80x_1 + 50x_2 \geq 7200,\ x_1, x_2 \geq 0. \end{aligned}$$

Solution: The given problem is of minimization. Converting it to maximization by taking the objective function as

$$Z' = -Z = -40x_1 - 24x_2$$

Introducing the surplus variables x_3, x_4 and artificial variables x_{a_1}, x_{a_2}, the constraint inequalities, reduce to the following equations :

$$20x_1 + 50x_2 - x_3 \qquad + x_{a_1} \qquad = 4800$$

$$80x_1 + 50x_2 \qquad - x_4 \qquad + x_{a_2} = 7200,$$

$$x_1, x_2, x_3, x_4, x_{a_1}, x_{a_2} \geq 0.$$

Phase I : Assigning cost -1 to artificial variables, cost 0 to all other variables, the new objective function of auxiliary problem becomes

$$\text{Max. } Z'' = 0x_1 + 0x_2 + 0x_3 + 0x_4 - x_{a_1} - x_{a_2}$$

subject to the constraints given above.

Taking $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$, we get $x_{a_1} = 4800$, $x_{a_2} = 7200$, which is the initial B.F.S.

Now applying the simplex method in the usual manner, we have the following table :

B	$\mathbf{c}_B$	c_j	**0**	**0**	**0**	**0**	**−1**	**−1**	**Min. ratio**
		X_B	Y_1	Y_2	Y_3	Y_4	A_1	A_2	$\mathbf{x}_B/Y_1, y_{i1} > 0$
A_1	−1	4800	20	50	−1	0	1	0	4800/20 = 240
A_2	−1	7200	$\boxed{80}$	50	0	−1	0	1	7200/80 = 90 (min) →
$Z'' = \mathbf{c}_B \mathbf{x}_B = -12000$		Δ_j	100 ↑	100	−1	−1	0	0 ↓	$\mathbf{x}_B/y_2, y_{i2} > 0$
A_1	−1	3000	0	$\boxed{75/2}$	−1	1/4	1		80 (min) →
Y_1	0	90	1	5/8	0	−1/80	0		144
$Z'' = -2910$		Δ_j		0 ↑	75/2	0	1/4 ↓	0	
Y_2	0	80	0	1	−2/75	1/50			
Y_1	0	40	1	0	1/60	−1/60			
$Z'' = 0$		Δ_j	0	0	0	0			

Since all $\Delta_j \leq 0$ and no artificial variable appears in the basis therefore an optimum solution to the auxiliary problem has been attained.

Phase II : Now assigning the actual costs to the original variables and cost zero to the surplus variables, the objective function becomes

$$\text{Max. } Z' = -40x_1 - 24x_2 + 0x_3 + 0x_4.$$

Replace the c_j row values in the final simplex table of phase I by the c_j values of the original objective function.

Also delete the artificial variables from the final simplex table of phase I, we write the first simplex table to phase II. Now solution of the problem applying simplex method in the usual manner is given in the following table :

B	$\mathbf{c}_B$	c_j	–40	–24	0	0	Min. ratio
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	$\mathbf{x}_B/Y_3,\ y_{i3} > 0$
Y_2	–24	80	0	1	–2/75	1/50	Neg.
Y_1	–40	40	1	0	$\boxed{1/60}$	–1/60	2400 (min) →
$Z' = \mathbf{c}_B\mathbf{x}_B = -3520$		Δ_j	0 ↓	0	2/75 ↑	–38/75	
Y_2	–24	144	8/5	1	0	–1/50	
Y_3	0	2400	60	0	1	–1	
$Z' = -3456$		Δ_j	–8/5	0	0	–12/25	

Since all $\Delta_j \le 0$, therefore the solution obtained is optimal.

Hence the optimal solution is $x_1 = 0$, $x_2 = 144$ and Min. $Z = -Z' = 3456$.

Infeasible Solution

Example 13: *Solve the following L.P.P. by using the two phase method :*

$$\begin{aligned} \textit{Min.} \quad & Z = x_1 - 2x_2 - 3x_3 \\ \textit{subject to} \quad & -2x_1 + x_2 + 3x_3 = 2 \\ & 2x_1 + 3x_2 + 4x_3 = 1 \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

Solution: Converting the objective function to maximization form by substituting $Z = -Z'$, we get

$$\text{Max. } Z' = -x_1 + 2x_2 + 3x_3$$

Introducing the artificial variables x_{a_1} and x_{a_2}, the constraint equations become

$$-2x_1 + x_2 + 3x_3 \quad + x_{a_1} \quad = 2$$
$$2x_1 + 3x_2 + 4x_3 \quad + x_{a_2} \quad = 1$$

where $\quad x_1, x_2, x_3, x_{a_1}, x_{a_2} \geq 0.$

Phase I : Assigning costs -1 to artificial variables and costs 0 to all other variables, the new objective function of auxiliary problem is

$$\text{Max } Z' = 0x_1 + 0x_2 + 0x_3 - x_{a_1} - x_{a_2},$$

subject to the constraints mentioned above.

Taking $x_1 = 0,\ x_2 = 0,\ x_3 = 0$, we get $x_{a_1} = 2,\ x_{a_2} = 1$, which is the starting B.F.S.

Now apply simplex method in the usual manner to remove artificial variables.

B	c_B	c_j	0	0	0	−1	−1	**Min. ratio**
		x_B	Y_1	Y_2	Y_3	A_1	A_2	$x_B/Y_3,\ y_{i3} > 0$
A_1	−1	2	−2	1	3	1	0	2/3
A_2	−1	1	2	3	$\boxed{4}$	0	1	1/4 (min) →
$Z' = -3$		Δ_j	0	4	7 ↑	0	0 ↓	
A_1	−1	5/4	−7/2	−5/4	0	1	−3/4	
Y_3	0	1/4	1/2	3/4	1	0	1/4	
$Z' = -5/4$		Δ_j	−7/4	−5/4	0	0	−3/4	

Since all Δ_j's are negative or zero, an optimum basic feasible solution to the auxiliary problem has been attained. But *the artificial variable x_{a_1} (column vector A_1) appears in the basic solution at a positive level. Hence the original L.P.P. does not possess any feasible solution.*

3.13 Miscellaneous Problems

L.P.P. Having Unbounded Solution

A L.P.P. has unbounded solution if the feasible region is unbounded such that value of the objective function can be increased or decreased (as required) indefinitely. It is, however, not necessary that an unbounded feasible region should yield an unbounded value for the objective function.

For clear understanding see the following examples :

Illustrative Examples

L.P.P. Having Unbounded Feasible Region But Bounded Optimal Solution

Example 14: *Max.* $Z = 6x_1 - 2x_2$

subject to $2x_1 - x_2 \le 2$

$x_1 \le 4$

$x_1, x_2 \ge 0.$

Solution: The given problem is of maximization and all b_i's are positive.

To convert inequalities of constraints into equations introducing slack variables x_3 and x_4, given problem becomes

$$\text{Max.} \quad Z = 6x_1 - 2x_2 + 0x_3 + 0x_4$$

$$\text{subject to} \quad 2x_1 - x_2 + x_3 = 2$$

$$x_1 + x_4 = 4.$$

Taking $x_1 = 0$, $x_2 = 0$, we get $x_3 = 2$, $x_4 = 4$ which is the initial B.F.S.

The solution to the problem using simplex method is given below :

B	$\mathbf{c}_B$	c_j / $\mathbf{x}_B$	**6** Y_1	**–2** Y_2	**0** Y_3	**0** Y_4	**Min. ratio** $\mathbf{x}_B/Y_1$, $y_{i1} > 0$
Y_3	0	2	$\boxed{2}$	–1	1	0	1(min) →
Y_4	0	4	1	0	0	1	4
$Z = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	6 ↑	–2	0 ↓	0	$\mathbf{x}_B/Y_2$, $y_{i2} > 0$
Y_1	6	1	1	–1/2	1/2	0	—
Y_4	0	3	0	$\boxed{1/2}$	–1/2	1	6 (min) →
$Z = 6$		Δ_j	0	1 ↑	–3	0 ↓	
Y_1	6	4	1	0	0	1	
Y_2	–2	6	0	1	–1	2	
$Z = 12$		Δ_j	0	0	–2	–2	

Since all $\Delta_j \le 0$, therefore the solution obtained is optimal.

Hence the optimal solution is $x_1 = 4$, $x_2 = 6$ and Max. $Z = 12$.

From the first simplex table, we observe that the elements of column of Y_2 are either negative or zero which indicates that the feasible region is not bounded.

Hence, a linear programming problem having unbounded feasible region may have a bounded optimal solution.

L.P.P. Having Unbounded Solutions

Example 15: *Max.* $Z = 10x_1 + 20x_2$

subject to $2x_1 + 4x_2 \geq 16$

$x_1 + 5x_2 \geq 15,\ x_1,\ x_2 \geq 0.$

Solution: The given problem is of maximization and all b_i's are positive.

Introducing surplus variables x_3, x_4 and artificial variables x_{a_1}, x_{a_2}, the given problem becomes

$$\text{Max. } Z = 10x_1 + 20x_2 + 0x_3 + 0x_4 - Mx_{a_1} - Mx_{a_2}$$

subject to $2x_1 + 4x_2 - x_3 + x_{a_1} = 16$

$x_1 + 5x_2 - x_4 + x_{a_2} = 15, \quad x_1, x_2, x_3, x_4, x_{a_1}, x_{a_2} \geq 0.$

Taking $x_1 = 0,\ x_2 = 0,\ x_3 = 0,\ x_4 = 0$, we get $x_{a_1} = 16,\ x_{a_2} = 15$, which is the starting B.F.S.

The solution to the problem using simplex method is given in the following table :

B	c_B	c_j / x_B	10 / Y_1	20 / Y_2	0 / Y_3	0 / Y_4	$-M$ / A_1	$-M$ / A_2	Min. ratio x_B/Y_2, $y_{i2} > 0$
A_1	$-M$	16	2	4	-1	0	1	0	16/4
A_2	$-M$	15	1	$\boxed{5}$	0	-1	0	1	15/5 (min) →
$Z = -31M$		Δ_j	$3M+10$	$9M+20$ ↑	$-M$	$-M$	0	0 ↓	x_B/Y_1, $y_{i1} > 0$
A_1	$-M$	4	$\boxed{6/5}$	0	-1	4/5	1		10/3 (min) →
Y_2	20	3	1/5	1	0	$-1/5$	0		15
$Z = -4M+60$		Δ_j	$\frac{6}{5}(M+5)$ ↑	0	$-M$	$\frac{4}{5}(M+5)$	0 ↓		x_B/Y_3
Y_1	10	10/3	1	0	$-5/6$	2/3			—
Y_2	20	7/3	0	1	$\boxed{1/6}$	$-1/3$			14 (min) →
$Z = 80$		Δ_j	0	0 ↓	5 ↑	0			

B	c_B	c_j	10	20	0	0	$-M$	$-M$	Min. ratio x_B/Y_1, $y_{i1} > 0$
		x_B	Y_1	Y_2	Y_3	Y_4	A_1	A_2	
Y_1	10	15	1	5	0	-1			
Y_3	0	14	0	6	1	-2			
$Z = 150$		Δ_j	0	-30	0	10 ↑			

In the last table we observe that Y_4 is the incoming vector but we cannot find outgoing vector because all the elements in this column are negative. Therefore, the solution is unbounded.

Note : If in a situation there is at least one Δ_j greater than zero but there is no non-negative ratios or min. ratios $\to \infty$, then also we have unbounded solution.

L.P.P. Having More than One Optimum Solution.

Example 16: *Max.* $Z = 6x_1 + 4x_2$

subject to
$$2x_1 + 3x_2 \le 30$$
$$3x_1 + 2x_2 \le 24,$$
$$x_1 + x_2 \ge 3,\ x_1,\ x_2 \ge 0.$$

Solution: To convert inequalities into equations, introducing slack, surplus and artificial variables, the problem becomes

$$\text{Max.} \quad Z = 6x_1 + 4x_2 + 0x_3 + 0x_4 + 0x_5 - Mx_a$$

subject to
$$2x_1 + 3x_2 + x_3 = 30$$
$$3x_1 + 2x_2 + x_4 = 24$$
$$x_1 + x_2 - x_5 + x_a = 3$$
$$x_1, x_2, x_3, x_4, x_5, x_a \ge 0.$$

The starting B.F.S. is

$$x_1 = 0,\ x_2 = 0,\ x_3 = 30,\ x_4 = 24,\ x_5 = 0,\ x_a = 3.$$

The solution to the problem using simplex method is given below :

B	$\mathbf{c}_B$	c_j / $\mathbf{x}_B$	6 / Y_1	4 / Y_2	0 / Y_3	0 / Y_4	0 / Y_5	$-M$ / A_1	Min. ratio $\mathbf{x}_B/Y_1$, $y_{i0}>0$
Y_3	0	30	2	3	1	0	0	0	15
Y_4	0	24	3	2	0	1	0	0	8
A_1	$-M$	3	$\boxed{1}$	1	0	0	-1	1	3 (min) →
$Z=\mathbf{c}_B\mathbf{x}_B=-3M$		Δ_j	$6+M$ ↑	$4+M$	0	0	$-M$	0 ↓	$\mathbf{x}_B/Y_5$, $y_{i5}>0$
Y_3	0	24	0	1	1	0	2		12
Y_4	0	15	0	-1	0	1	$\boxed{3}$		5 (min) →
Y_1	6	3	1	1	0	0	-1		—
$Z=18$		Δ_j	0	-2	0	0 ↓	0 ↑		$\mathbf{x}_B/Y_2$, $y_{i2}>0$
Y_3	0	14	0	$\boxed{5/3}$	1	$-2/3$	0		42/5 →
Y_5	0	5	0	$-1/3$	0	1/3	1		—
Y_1	6	8	1	2/3	0	1/3	0		12
$Z=48$		Δ_j	0	0 ↑	0 ↓	-2	0		

Since all $\Delta_j \le 0$, therefore the solution is optimal.

The optimal solution is $x_1 = 8$, $x_2 = 0$, Max. $Z = 48$.

Here corresponding to non-basic variable x_2 in the last table $\Delta_2 = 0$ and Y_2 is not in the basis B, therefore an alternative solution also exists. Thus, the *problem does not have unique solution.*

Taking Y_2 as incoming and Y_3 as outgoing vector we obtain the following simplex table :

B	$\mathbf{c}_B$	c_j / $\mathbf{x}_B$	6 / Y_1	4 / Y_2	0 / Y_3	0 / Y_4	0 / Y_5	Min. ratio
Y_2	4	42/5	0	1	3/5	$-2/5$	0	
Y_5	0	39/5	0	0	1/5	1/5	1	
Y_1	6	12/5	1	0	$-2/5$	3/5	0	
$Z=\mathbf{c}_B\mathbf{x}_B=48$		Δ_j	0	0	0	-2	0	

Since all $\Delta_j \leq 0$, therefore this solution is also optimal having the same maximum value of Z.

Second optimal solution is

$$x_1 = 12/5,\ x_2 = 42/5 \text{ and Max. } Z = 48$$

Hence, two optimal solutions of the problem are

1. $x_1 = 8,\ x_2 = 0$
2. $x_1 = 12/5,\ x_2 = 42/5$ and Max. $Z = 200$.

We know that the convex combination of B.F.S. is also an optimal solution.

Thus if we obtain two alternative optimum solutions, then we can obtain any number of optimum solutions.

For any arbitrary value of λ such that $0 \leq \lambda \leq 1$ the following table gives different optimum solutions which are infinite in number :

Variables	I Sol.	II Sol.	Gen. Sol.
x_1	8	12/5	$8\lambda + (12/5)(1-\lambda)$
x_2	0	42/5	$0\lambda + (42/5)(1-\lambda)$

Taking a particular value of $\lambda, \lambda = 1/2$ (say),

the third optimal solution is

$$x_1 = 26/5,\ x_2 = 21/5 \text{ and Max. } Z = 48.$$

L.P.P. with Unrestricted Variables

Example 17: Max. $Z = 2x_1 + 3x_2$

subject to $-x_1 + 2x_2 \leq 4$

$x_1 + x_2 \leq 6$

$x_1 + 3x_2 \leq 9$, x_1, x_2 are unrestricted.

Solution: The given problem is of maximization and all b_i's are positive. We know that the simplex method is applicable if the variables are non-negative. In the given problem, x_1 and x_2 are unrestricted (may be +, – or zero). In such cases, we replace all these variables by the difference of two non-negative variables and solve the problem in usual manner.

$\therefore$ Making the transformation

$$x_1 = x_1' - x_1'' \text{ and } x_2 = x_2' - x_2''$$

such that $x_1', x_1'', x_2', x_2'' \geq 0.$

The given problem becomes

$$\text{Max. } Z = 2(x_1' - x_1'') + 3(x_2' - x_2'')$$

$$\text{subject to } -(x_1' - x_1'') + 2(x_2' - x_2'') \leq 4$$

$$(x_1' - x_1'') + (x_2' - x_2'') \leq 6$$

$$(x_1' - x_1'') + 3(x_2' - x_2'') \leq 9,\ x_1', x_2', x_1'', x_2'' \geq 0.$$

To change constraint inequalities into equations introducing slack variables x_3, x_4 and x_5, the given L.P.P. becomes

$$\text{Max. } Z = 2x_1' - 2x_1'' + 3x_2' - 3x_2'' + 0.x_3 + 0.x_4 + 0.x_5$$

$$-x_1' + x_1'' + 2x_2' - 2x_2'' + x_3 = 4$$

$$x_1' - x_1'' + x_2' - x_2'' + x_4 = 6$$

$$x_1' - x_1'' + 3x_2' - 3x_2'' + x_5 = 9.$$

Taking $x_1' = 0, x_1'' = 0, x_2' = 0, x_2'' = 0$ we get $x_3 = 4, x_4 = 6, x_5 = 9$, which is the starting B.F.S.

The solution to the problem using simplex algorithm is given below :

B	c_B	c_j	2	–2	3	–3	0	0	0	Min. ratio
		x_B	Y_1'	Y_1''	Y_2'	Y_2''	Y_3	Y_4	Y_5	x_B/Y_2', $y_{i2}' > 0$
Y_3	0	4	–1	1	$\boxed{2}$	–2	1	0	0	2 (min) →
Y_4	0	6	1	–1	1	–1	0	1	0	6
Y_5	0	9	1	–1	3	–3	0	0	1	3
$Z = c_B x_B = 0$		Δ_j	2	–2	3 ↑	–3	0 ↓	0	0	x_B/Y_1', $y_{i1}' > 0$
Y_2'	3	2	–1/2	1/2	1	–1	1/2	0	0	—
Y_4	0	4	3/2	–3/2	0	0	–1/2	1	0	8/3
Y_5	0	3	$\boxed{5/2}$	–5/2	0	0	–3/2	0	1	6/5 (min) →
$Z = c_B x_B = 6$		Δ_j	7/2 ↑	–7/2	0	0	–3/2	0	0 ↓	x_B/Y_3

Y_2' 3	13/5	0	0	1	−1	1/5	0	1/5	13
Y_4 0	11/5	0	0	0	0	2/5	1	−3/5	11/2 (min) →
Y_1' 2	6/5	1	−1	0	0	−3/5	0	2/5	
$Z = \mathbf{c}_B\mathbf{x}_B = 51/5$	Δ_j	0	0	0	0	3/5 ↑	0 ↓	−7/5	
Y_2' 3	3/2	0	0	1	−1	0	−1/2	1/2	
Y_3 0	11/2	0	0	0	0	1	5/2	−3/2	
Y_1' 2	9/2	1	−1	0	0	0	3/2	−1/2	
$Z = \mathbf{c}_B\mathbf{x}_B = 27/2$	Δ_j	0	0	0	0	0	−3/2	−1/2	

Since all $\Delta_j \leq 0$, therefore the solution is optimal.

$\therefore$ The optimal solution is $x_1' = 9/2$, $x_1'' = 0$, $x_2' = 3/2$, $x_2'' = 0$,

Max. $Z = 27/2$

$\Rightarrow$ $x_1 = x_1' - x_1'' = 9/2$, $x_2 = x_2' - x_2'' = 3/2$, Max. $Z = 27/2$.

Example 18: *Minimize* $Z = x_1 + x_2 + x_3$

subject to

$$x_1 - 3x_2 + 4x_3 = 5$$

$$x_1 - 2x_2 \leq 3$$

$$2x_2 - x_3 \geq 4,$$

$x_1, x_2 \geq 0$ *and* x_3 *is unrestricted.* **(Garhwal 2009)**

Solution: First we shall convert the minimization problem into maximization by substituting $Z' = -Z$.

The objective function changes to

$$\text{Max. } Z' = -x_1 - x_2 - x_3.$$

Since x_3 is unrestricted therefore replacing it by

$x_3' - x_3''$ where $x_3', x_3'' \geq 0$.

Also introducing slack variable x_4, surplus variable x_5 and artificial variables x_{a_1} and x_{a_2}, the given problem becomes

$$\text{Max. } Z' = -x_1 - x_2 - x_3' + x_3'' + +0x_4 + 0x_5 - Mx_{a_1} - Mx_{a_2}$$

subject to $x_1 - 3x_2 + 4x_3' - 4x_3'' + x_{a_1} = 5$

$x_1 - 2x_2 \quad + x_4 = 3$

$$0x_1 + 2x_2 - x_3' + x_3'' - x_5 + x_{a_2} = 4,$$
$$x_1, x_2, x_3', x_3'', x_4, x_5, x_{a_1}, x_{a_2} \geq 0.$$

Now solve the problem using simplex algorithm.

(Proceed as in Ex. 17 above).

The optimal solution is

$$x_1 = 0,\ x_2 = 21/5,\ x_3 = 22/5 \text{ and Min. } Z = 43/5.$$

L.P.P. with Some Constant Term in the Objective Function

In a L.P.P., when the objective function contains a constant term, then the simplex method is applied by leaving this constant term in the beginning and optimal solution is obtained. In the end, the constant term (which was left initially) is added to the optimal value of the objective function.

For clear understanding of the method see the following examples :

Example 19: *Solve the L.P.P.*

$$\textit{Max.}\quad Z = 2x_1 - x_2 + x_3 + 50$$

$$\textit{subject to}\quad 2x_1 + 2x_2 - 6x_3 \leq 16$$
$$12x_1 - 3x_2 + 3x_3 \geq 6$$
$$-2x_1 - 3x_2 + x_3 \leq 4$$
$$\textit{and}\quad x_1, x_2, x_3 \geq 0.$$

Solution: Leaving the constant term 50 from the objective function in the beginning, introducing slack, surplus and artificial variables, the given problem reduces to

$$\text{Max. } Z' = 2x_1 - x_2 + x_3 + 0x_4 + 0x_5 + 0x_6 - M x_{a_1}$$

$$\text{subject to}\quad 2x_1 + 2x_2 - 6x_3 + x_4 = 16$$
$$12x_1 - 3x_2 + 3x_3 - x_5 + x_{a_1} = 6$$
$$-2x_1 - 3x_2 + x_3 + x_6 = 4$$
$$\text{and}\quad x_1, x_2, x_3, x_4, x_5, x_6, x_{a_1} \geq 0.$$

Taking $x_1 = 0 = x_2 = x_3 = x_5$, we get $x_4 = 16, x_{a_1} = 6, x_6 = 4$ which is the starting B.F.S.

The solution of the problem by simplex method is given in the following table :

B	c_B	c_j	2	-1	1	0	0	0	$-M$	**Min. ratio**
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	A	x_B/Y_1, $y_{i1} > 0$
Y_4	0	16	2	2	-6	1	1	0	0	16/2
A	$-M$	6	$\boxed{12}$	-3	3	0	-1	0	1	6/12 (min.) $\rightarrow$
Y_6	0	4	-2	-3	1	0	0	1	0	—
$Z' = c_B x_B = -6M$		Δ_j	$2+12M$ $\uparrow$	$-1-3M$	$1+3M$	0	$-M$	0	0 $\downarrow$	x_B/Y_3, $y_{i3} > 0$
Y_4	0	15	0	5/2	$-13/2$	1	1/6	0		—
Y_1	2	1/2	1	$-1/4$	$\boxed{1/4}$	0	$-1/2$	0		2 (min.) $\rightarrow$
Y_6	0	5	0	$-7/2$	3/2	0	$-1/6$	1		10/3
$Z' = x_B x_B = 1$		Δ_j	0 $\downarrow$	$-1/2$	1/2 $\uparrow$	0	1/6	0		x_B/Y_5, $y_{i5} > 0$
Y_4	0	28	26	-4	0	1	-2	0		—
Y_3	1	2	4	-1	1	0	$-1/3$	0		—
Y_6	0	2	-6	-2	0	0	$\boxed{1/3}$	1		6 (min.) $\rightarrow$
$Z' = x_B x_B = 2$		Δ_j	-2	0	0	0	1/3 $\uparrow$	0 $\downarrow$		x_B/Y_1, $y_{i1} > 0$
Y_4	0	40	-10	-16	0	1	0	6		—
Y_3	1	4	-2	-3	1	0	0	1		—
Y_5	0	6	-18	-6	0	0	1	3		—
$Z' = C_B x_B = 4$		Δ_j	4 $\uparrow$	2	0	0	0	-1		

Here Y_1 is the entering vector but all elements in column one are –ve, so we cannot select the outgoing vector. Hence, the solution of the problem is unbounded.

L.P.P. with Lower Bounds of Some or All Variables, Other than Zero

In some L.P.P., the lower bounds may be specified for examples, in a L.P.P. it may be stipulated that $x_1 \geq c_1$, $x_2 \geq c_2$ etc. In such problems, we substitute $x_1 = c_1 + y_1$, $x_2 = c_2 + y_2$ etc. and then solve the problem in terms of y_1 and y_2, where $y_1 \geq 0, y_2 \geq 0$ etc.

For clear understanding of the method see the following example :

Example 20: *Solve the L.P.P.*

$$\text{Max. } Z = 3x_1 + 5x_2 + 4x_3$$

subject to $2x_1 - 3x_2 \le 8$

$2x_2 + 5x_3 \le 10$

$3x_1 + 2x_2 + 4x_3 \le 15$

$x_1 \ge 2,\ x_2 \ge 4,\ x_3 \ge 0.$

Solution: Taking $x_1 = y_1 + 2,\ x_2 = y_2 + 4,\ x_3 = y_3$; the given problem reduces to

$$\text{Max. } Z = 3y_1 + 5y_2 + 4y_3 + 26$$

subject to $2y_1 - 3y_2 \le 16,\ 2y_2 + 5y_3 \le 2,$

$3y_1 + 2y_2 + 4y_3 \le 1$

and $y_1, y_2, y_3 \ge 0.$

Introducing the slack variables y_4, y_5, y_6 and leaving constant term 26 from Z in the beginning, the above L.P.P. reduces to

$$\text{Max. } Z' = 3y_1 + 5y_2 + 4y_3 \text{ where } Z = Z' + 26$$

subject to $2y_1 - 3y_2 + y_4 = 16$

$2y_2 + 5y_3 + y_5 = 2$

$3y_1 + 2y_2 + 4y_3 + y_6 = 1$

and $y_1, y_2, y_3 \ge 0.$

Taking $y_1 = 0 = y_2 = y_3$, we get $y_4 = 16, y_5 = 2, y_6 = 1$ which is the starting B.F.S.

The solution by simplex method is shown in the following table :

B	$\mathbf{c}_B$	c_j	3	5	4	0	0	0	Min. ratio $\mathbf{x}_B/Y_2$, $y_{i2} > 0$
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_4	0	16	2	−3	0	1	0	0	—
Y_5	0	2	0	2	5	0	1	0	2/2
Y_6	0	1	3	$\boxed{2}$	4	0	0	1	1/2 (min.) →
$Z' = \mathbf{c}_B\,\mathbf{x}_B = 0$		Δ_j	3	5 ↑	4	0	0	0 ↓	

Y_4	0	35/2	13/2	0	6	1	0	3/2	
Y_5	0	1	−3	0	1	0	1	−1	
Y_2	5	1/2	3/2	1	2	0	0	1/2	
$Z' = \mathbf{c}_B \mathbf{x}_B = 5/2$		Δ_j	−9/2	0	−6	0	0	−5/2	

Since no $\Delta_j > 0$, so this solution is optimal.

∴ Optimal solution is

$$y_1 = 0, y_2 = 1/2, y_3 = 0, \text{ Max. } Z' = 5/2$$

∴ $$x_1 = y_1 + 2 = 2,\ x_2 = y_2 + 4 = 9/2,\ \text{Max. } Z = Z' + 26 = 57/2$$

i.e., $$x_1 = 2,\ x_2 = 9/2, \quad \text{Max. } Z = 57/2.$$

3.14 Solution of Simultaneous Linear Equations by Simplex Method

To solve a system of n simultaneous linear equations by simplex method, if non-negative restriction of variables is not given, then we replace each variable by the difference of two non-negative variables. Also we add an artificial variable (non-negative) in each equation (if required) to obtain a basis matrix (identity matrix). A dummy objective function is also introduced in which every given variable is assigned 0 (zero) cost and every artificial variable is assigned cost −1.

The reformulated L.P.P. is, then solved by using simplex method.

For clear understanding of the method, see the following examples :

Illustrative Examples

Example 21: *Solve the following system of linear equations by using simplex method :*

$$x_1 - x_3 + 4x_4 = 3, \qquad 2x_1 - x_2 = 3$$

$$3x_1 - 2x_2 - x_4 = 1, \qquad x_1, x_2, x_3, x_4 \geq 0.$$

Solution: Here the variables are non-negative. Adding the artificial variables (non-negative) $x_{a_1}, x_{a_2}, x_{a_3}$ in the given equations and introducing the dummy objective function Z with costs zero to each given variable and cost–1 to each artificial variable the given system can be written as a L.P.P. in the following form :

$$\text{Max. } Z = 0x_1 + 0x_2 + 0x_3 + 0x_4 - 1x_{a_1} - 1x_{a_2} - 1x_{a_3}$$

subject to $$x_1 - x_3 + 4x_4 + x_{a_1} = 3$$

$$2x_1 - x_2 + x_{a_2} = 3$$

$$3x_1 - 2x_2 - x_4 + x_{a_3} = 1$$

and $$x_1, x_2, x_3, x_4, x_{a_1}, x_{a_2} \geq 0.$$

Taking $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$, we get $x_{a_1} = 3$, $x_{a_2} = 3$, $x_{a_3} = 1$, which is the initial B.F.S.

The solution to the problem using simplex algorithm is given in the following table :

B	c_B	c_j	0	0	0	0	–1	–1	–1	**Min. ratio**
		x_B	Y_1	Y_2	Y_3	Y_4	A_1	A_2	A_3	x_B/Y_1, $y_{i1} > 0$
A_1	–1	3	1	0	–1	4	1	0	0	3
A_2	–1	3	2	–1	0	0	0	1	0	3/2
A_3	–1	1	$\boxed{3}$	–2	0	–1	0	0	1	1/3 (min) →
$Z = c_B x_B = -7$		Δ_j	6 ↑	–3	–1	3	0	0	0 ↓	x_B/Y_4, $y_{i4} > 0$
A_1	–1	8/3	0	2/3	–1	$\boxed{13/3}$	1	0		8/13 (min) →
A_2	–1	7/3	0	1/3	0	2/3	0	1		7/2
Y_1	0	1/3	1	–2/3	0	–1/3	0	0		—
$Z = c_B x_B = -5$		Δ_j	0	1	–1	5 ↑	0 ↓	0		x_B/Y_2, $y_{i3} > 0$
Y_4	0	8/13	0	$\boxed{2/13}$	–3/13	1		0		4 →
A_2	–1	25/13	0	3/13	2/13	0		1		25/3
Y_1	0	7/13	1	–8/13	–1/13	0		0		—
$Z = -25/13$		Δ_j	1	3/13 ↑	2/13	0 ↓		0		x_B/Y_3, $y_{i3} > 0$
Y_2	0	4	0	1	–3/2	13/2		0		—
A_2	–1	1	0	0	$\boxed{1/2}$	–3/2		1		2 →
Y_1	0	3	1	0	–1	4		0		—
$Z = -1$		Δ_j	0	0	1/2 ↑	–3/2		0 ↓		
Y_2	0	7	0	1	0	2				
Y_3	0	2	0	0	1	–3				
Y_1	0	5	1	0	0	1				
$Z = 0$		Δ_j	0	0	0	0				

Since all $\Delta_j \le 0$, therefore this solution is optimal.

Hence, the solution of the given system is

$$x_1 = 5,\ x_2 = 7,\ x_3 = 2,\ x_4 = 0.$$

Example 22: *Solve the following system of linear equations by simplex method :*

$$2x_1 + x_2 = 1$$
$$3x_1 + 4x_2 = 12.$$

Solution: Here the two variables x_1 and x_2 are not required to be non-negative, so we write $x_1 = x_1' - x_1''$ and $x_2 = x_2' - x_2''$, such that $x_1', x_1'', x_2', x_2'' \ge 0$.

Adding the artificial variables (non-negative) x_{a_1} and x_{a_1} in the two equations and introducing the dummy objective function Z with costs zero to each variable x_1', x_1'', x_2', x_2'' and costs–1 to each artificial variable x_{a_1}, x_{a_2}, the given problem in L.P.P. is as follows :

$$\text{Max. } Z = 0x_1' + 0x_1'' + 0x_2' + 0x_2'' - x_{a_1} - x_{a_2},$$

subject to

$$2x_1' - 2x_1'' + x_2' - x_2'' + x_{a_1} = 1$$
$$3x_1' - 3x_1'' + 4x_2' - 4x_2'' + x_{a_2} = 12$$

and $\quad x_1', x_1'', x_2', x_2'', x_{a_1}, x_{a_2} \ge 0.$

Taking $x_1' = 0, x_1'' = 0, x_2' = 0, x_2'' = 0,$ we get $x_{a_1} = 1, x_{a_2} = 12$ which is the starting B.F.S.

The solution of the problem using simplex method is given in the following table :

B	c_B	c_j	0	0	0	0	–1	–1	Min. ratio $x_B/Y_2', y_{i2}' > 0$
		x_B	Y_1'	Y_1''	Y_2'	Y_2''	A_1	A_2	
A_1	–1	1	2	–2	$\boxed{1}$	–1	1	0	1/1 → (min)
A_2	–1	12	3	–3	4	–4	0	1	12/4
$Z = -13$		Δ_j	5	–5	5 ↑	–5	0	0	$x_B/Y_1'', y_{i1}'' > 0$
Y_2'	0	1	2	–2	1	–1	1	0	—
A_2	–1	8	–5	$\boxed{5}$	0	0	–4	1	8/5 →
$Z = -8$		Δ_j	–5	5 ↑	0	0	–5	0	
Y_2'	0	21/5	0	0	1	–1	–3/5	2/5	
Y_1''	0	8/5	–1	1	0	0	–4/5	1/5	
$Z = 0$		Δ_j	0	0	0	0	0	0	

Since all $\Delta_j \le 0$, therefore the solution is optimal.

The optimal solution is $x_1' = 0,\ x_1'' = 8/5,\ x_2' = 21/5,\ x_2'' = 0$

i.e., $$x_1 = x_1' - x_1'' = -8/5,\ x_2 = x_2' - x_2'' = 21/5$$

3.15 Inverse of a Matrix by Simplex Method

Let A be a $n \times n$ non-singular real matrix. Then to find the **inverse matrix** of matrix A by simplex method proceed as follows :

Introducing a dummy $n \times 1$ real matrix **b**, consider the following system of equations :

$$A\mathbf{x} = \mathbf{b},\ \mathbf{x} \ge 0$$

Now introduce the artificial variables $\mathbf{x}_a > 0$ and a dummy objective function Z, with costs zero to variables in **x** and cost -1 to each artificial variable. Then find the solution of the following L.P.P. formed, by using simplex method :

$$\text{Max. } Z = 0\,\mathbf{x} - 1\mathbf{x}_a$$

$$\text{subject to } A\mathbf{x} = \mathbf{b},\ \mathbf{x},\ \mathbf{x}_a \ge 0.$$

Then in the final simplex table which gives the optimal solution (*i.e.,* when all $\Delta_j \le 0$) and the columns of A becomes the columns of unit matrix I (*i.e.,* when A is converted to an unit matrix or when all variables of vector **x** are in the basis), the inverse of matrix A is the matrix formed by the column vectors which were the column vectors of the initial basis in proper order.

If in the final simplex table which gives the optimal solution (*i.e.,* when all $\Delta_j \le 0$), the matrix A is not converted to an unit matrix (*i.e.,* all variables of vector **x** are not in the basis and artificial variable / variables appear in the basis), then we continue the simplex method by excluding the artificial variable and introducing the remaining variable (not in the basis) into the basis. By this operation (iteration), the solution obtained must remain optimal while it can be **feasible** or **infeasible**. The process is continued till A is converted to an unit matrix. Finally, the inverse of A is obtained as above.

3.15.1 Method to Find Dummy $n \times 1$ Real Matrix b

Although there is no particular method to find dummy $n \times 1$ real matrix **b**. Even then in most of the cases it can be obtained as follows :

$$\mathbf{b} = \begin{bmatrix} b_1 = \sum_{j=1}^{n} a_{1j} \\ b_2 = \sum_{j=1}^{n} a_{2j} \\ \cdots\cdots\cdots \\ \cdots\cdots\cdots \\ b_n = \sum_{j=1}^{n} a_{nj} \end{bmatrix}$$

For clear understanding of the method see the following example :

Example 23: *Apply simplex method to find the inverse of the matrix*

$$\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}.$$

Solution: Let $A = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 = \sum a_{1j} = 4+3 \\ b_2 = \sum a_{2j} = 3+2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ be a dummy real column matrix. Consider the system of equations

$$A\mathbf{x} = \mathbf{b}$$

i.e.,

$$\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix} \text{ i.e., } \begin{array}{l} 4x_1 + 3x_2 = 7 \\ 3x_1 + 2x_2 = 5 \end{array}$$

Introducing the artificial variables x_{a_1}, x_{a_2} and the dummy objective function Z with cost 0 to each variable x_1, x_2 cost -1 to each artificial variable x_{a_1}, x_{a_2}, the resulting L.P.P. is

$$\text{Max. } Z = 0x_1 + 0x_2 - 1x_{a_1} - 1x_{a_2}$$

subject to

$$4x_1 + 3x_2 + x_{a_1} = 7$$

$$3x_1 + 2x_2 + x_{a_2} = 5$$

and

$$x_1, x_2, x_{a_1}, x_{a_2} \geq 0.$$

Taking $x_1 = 0$, $x_2 = 0$, we get $x_{a_1} = 7$, $x_{a_2} = 5$, which is the starting B.F.S.

The solution of the above L.P.P. by simplex method is given in the following table :

B	c_B	c_j / $\mathbf{x}_B$	0 / Y_1	0 / Y_2	-1 / A_1	-1 / A_2	Min. ratio $\mathbf{x}_B/Y_1, y_{i1} > 0$
A_1	-1	7	4	3	1	0	7/4
A_2	-1	5	$\boxed{3}$	2	0	1	5/3 (min.) $\rightarrow$
$Z = \mathbf{c}_B\mathbf{x}_B = -12$		Δ_j	7 $\uparrow$	5	0	0 $\downarrow$	$\mathbf{x}_B/Y_2, y_{i2} > 0$
A_1	-1	1/3	0	$\boxed{1/3}$	1	$-4/3$	1 (min.) $\rightarrow$
Y_1	0	5/3	1	2/3	0	1/3	5/2
$Z = \mathbf{c}_B\mathbf{x}_B = -\frac{1}{3}$		Δ_j	0	1/3 $\uparrow$	0 $\downarrow$	$-7/3$	

Y_2 0	1	0	1	3	-4	
Y_1 0	1	1	0	-2	3	
$Z = \mathbf{c}_B \mathbf{x}_B = 0$	Δ_j	0	0	-1	-1	

Since in the last table no $\Delta_j > 0$, therefore the solution is optimal.

The last table in proper order (*i.e.*, form) of unit matrix for Y_1, Y_2 (*i.e.*, A) can be written as

B	c_B	x_B	Y_1	Y_2	A_1	A_2
Y_1	0	1	1	0	-2	3
Y_2	0	1	0	1	3	-4

Since the given matrix A (given by columns Y_1, Y_2) has been converted to a unit matrix, therefore A^{-1} is given by the columns A_1, A_2 of the initial basis.

Hence $$A^{-1} = \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}.$$

3.16 Conditions For The Occurrence of Degeneracy in a L.P.P.

In a L.P.P. the degeneracy may appear in the following ways :

1. The degeneracy may appear in a L.P.P. at the very first stage when some b_i is zero or at any iteration when the value of some variable in the basis becomes zero.
2. If none of the component of $\mathbf{b}$ is zero at any iteration and the choice of the outgoing vector at the same iteration is not unique then the next iteration is bound to be degenerate.

For the resolution of this type of degeneracy we give the procedure in the next article.

3.17 Computational Procedure for Resolution of Degeneracy.

If a_k is the vector entering the basis and

$$\underset{i}{\text{Mini}} \left\{ \frac{y_{Bi}}{y_{ik}}, y_{ik} > 0 \right\}$$

is not unique then the following procedure is adopted.

1. Renumber the columns of the simplex tableau starting with the columns in the basis. Let $\overline{Y}_1, \overline{Y}_2, \ldots$ etc., be the new numbers of columns. Let $\overline{Y}_t$ be the new number of entering vector Y_k *i.e.*, $Y_k = \overline{Y}_t$.

2. If $\underset{i}{\text{Mini}}\left\{\dfrac{y_{Bi}}{y_{ik}}, y_{ik} > 0\right\}$ occur at $i = i_1, i_2, \ldots, i_s$ then let $I_1 = \{i_1, i_2, \ldots, i_s\}$.

   Compute $\left\{\dfrac{\overline{y}_{i1}}{\overline{y}_{it}}, \overline{y}_{it} > 0\right\}, \forall\, i \in I_1$.

   If this minimum is unique then delete the corresponding vector from the basis.
   If the minimum is also not unique then proceed to the next step 3.

3. Compute $\underset{i}{\text{Mini}}\left\{\dfrac{\overline{y}_{i2}}{\overline{y}_{u}}, \overline{y}_{u} > 0\right\}, \forall\, i \in I_2$

   where I_2 is the set of all those values of $i \leftrightarrow I_1$, for which there is a tie in (2). Clearly $I_2 \quad I_1$.
   If the minimum is unique then delete the corresponding vector from the basis.
   If this minimum is also not unique then proceed to the next step 4.

4. Compute $\underset{i}{\text{Mini}}\left\{\dfrac{\overline{y}_{i3}}{\overline{y}_{it}}, \overline{y}_{u} > 0\right\}, \forall\, i \in I_3$

   where I_3 is the set of those values of $i \in I_2$ for which there is a tie in (3). Clearly
   $I_3 \subset I_2 \subset I_1$.
   Continuing in this way we shall get a unique minimum value *i.e.*, the unique vector to be deleted from this basis. For clear understanding of the method see the following examples.

Illustrative Examples

Example 24: *Solve the following L.P.P. by simplex method*

$$\text{Max. } Z = 2x_1 + x_2,$$

subject to
$$4x_1 + 3x_2 \leq 12$$
$$4x_1 + x_2 \leq 8$$
$$4x_1 - x_2 \leq 8$$

and
$$x_1, x_2 \geq 0.$$

Solution: Introducing the slack variables x_3 , x_4 and x_5 , the given L.P.P. can be written as

$$
\begin{aligned}
\text{Max.}\quad & Z = 2x_1 + x_2 \\
\text{s.t.}\quad & 4x_1 + 3x_2 + x_3 = 12 \\
& 4x_1 + x_2 + x_4 = 8 \\
& 4x_1 - x_2 + x_5 = 8
\end{aligned}
$$

and $\quad x_1, x_2, x_3, x_4, x_5 \geq 0.$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_3 = 12, x_4 = 8, x_5 = 8$, which is the starting B.F.S.

$\therefore$ The starting simplex table is :

B	$\mathbf{c}_B$	c_j	2	1	0	0	0	Min. Ratio
		$\mathbf{x}_B$	$\overline{Y}_4$	$\overline{Y}_5$	$\overline{Y}_1$	$\overline{Y}_2$	$\overline{Y}_3$	$\mathbf{x}_B / Y_1$
			Y_1	Y_2	Y_3	Y_4	Y_5	
Y_3	0	12	4	3	1	0	0	3
Y_4	0	8	4 $(\overline{y}_{24})$	1	0 $(\overline{y}_{21})$	1 $(\overline{y}_{22})$	0	2
Y_5	0	8	$\boxed{4}$ $(\overline{y}_{34})$	-1	0 $(\overline{y}_{31})$	0 $(\overline{y}_{32})$	1	2 $\rightarrow$
$Z = \mathbf{c}_B \cdot \mathbf{x}_B = 0$		Δ_j	2 $\uparrow$	1	0	0	0 $\downarrow$	

Here $\Delta_1 = c_1 - \mathbf{c}_B Y_1 = 2$, $\Delta_2 = c_2 - \mathbf{c}_B Y_2 = 1, \Delta_3 = 0 = \Delta_4 = \Delta_5$, which are positive, therefore the solution is not optimal.

Since Max. $\Delta_j = 2 = \Delta_1$.

$\therefore$ Entering (Incoming) vector is α_1 $(= Y_1)$.

By minimum ratio rule we find that minimum is not unique but occurs for $i = 2$ and $i = 3$ both.

Thus the problem is the problem of degeneracy.

$\therefore$ To select the vector to the deleted from the basis, we proceed as explained in article 3.17.

1. First of all we renumber the columns of the table, such that the columns associated to the identity matrix come first in the proper order.

   $\therefore$ Let $\overline{Y}_1 = Y_3$, $\overline{Y}_2 = Y_4$, $\overline{Y}_3 = Y_5$, $\overline{Y}_4 = Y_1$ and $\overline{Y}_5 = Y_2$.

2. Since minimum ratio occurs for $i = 2$ and 3

$\therefore$ $I_1 = \{2, 3\}$ and incoming vector is $Y_1 = \overline{Y}_4$.

$\therefore$ $t = 4$. For $i = 2, 3$

$$\underset{i \in I_1}{\text{Mini}} \left\{ \frac{\bar{y}_{i1}}{\bar{y}_{it}}, \bar{y}_{it} > 0 \right\} = \underset{i \in I_1}{\text{Mini}} \left\{ \frac{\bar{y}_{i1}}{\bar{y}_{i4}}, \bar{y}_{i4} > 0 \right\} = \text{Mini} \left\{ \frac{\bar{y}_{21}}{\bar{y}_{24}}, \frac{\bar{y}_{31}}{\bar{y}_{34}} \right\}$$

$$= \text{Mini} \left\{ \frac{0}{4}, \frac{0}{4} \right\} = \text{Mini } \{0, 0\}.$$

Since this minimum is not unique and there is a tie at $i = 2, 3$.

$\therefore$ Let $I_2 = \{2, 3\} \subseteq I_1$ and proceed to the next step.

3. $$\underset{i \in I_2}{\text{Mini}} \left\{ \frac{\bar{y}_{i2}}{\bar{y}_{it}}, \bar{y}_{it} > 0 \right\} = \text{Mini} \left\{ \frac{\bar{y}_{22}}{\bar{y}_{24}}, \frac{\bar{y}_{32}}{\bar{y}_{34}} \right\} = \text{Mini} \left\{ \frac{1}{4}, \frac{0}{4} \right\}$$

$$= \text{Mini } \{1/4, 0\} = \bar{y}_{32} / \bar{y}_{34}$$

i.e., minimum occurs at $i = 3$.

$\therefore$ $\overline{Y}_3 = Y_5$ is the outgoing vector and key element $= y_{31} = 4$.

Thus in order to bring Y_1 in place of Y_5, we divide the third row containing the key element by 4 and then subtract its 4, 4 and 2 times from first row, second row and from the row of Δ_j's. Thus we get the following table.

B	$\mathbf{c}_B$	c_j / $\mathbf{x}_B$	2 / Y_1	1 / Y_2	0 / Y_3	0 / Y_4	0 / Y_5	Min. Ratio $\mathbf{x}_B / Y_2$
Y_3	0	4	0	4	1	0	−1	1
Y_4	0	0	0	$\boxed{2}$	0	1	−1	0 (Mini) →
Y_1	2	2	1	−1/4	0	0	1/4	Neg.
$Z = \mathbf{c}_B \cdot \mathbf{x}_B = 4$		Δ_j	0	3/2 ↑	0	0 ↓	−1/2	$\mathbf{x}_B / Y_5$
Y_3	0	4	0	0	1	−2	$\boxed{1}$	4 (Mini) →
Y_2	1	0	0	1	0	1/2	−1/2	Neg.
Y_1	2	2	1	0	0	1/8	1/8	16
$Z = \mathbf{c}_B \cdot \mathbf{x}_B = 4$		Δ_j	0	0	0 →	−3/4	1/4 ↑	
Y_5	0	4	0	0	1	−2	1	
Y_2	1	2	0	1	1/2	1/2	0	
Y_1	2	3/2	1	0	−1/8	3/8	0	
$Z = \mathbf{c}_B \cdot \mathbf{x}_B = 5$		Δ_j	0	0	−1/4	−1/2	0	

Since all $\Delta_j \le 0$, therefore the solution is optimal.

$\therefore$ Optimal solution is

$$x_1 = 3/2,\ x_2 = 2$$

and Max. $Z = 5$.

Example 25: *Solve the following L.P.P., using simplex method :*

$$\text{Max. } 6x_1 + 3x_2 + 5$$

$$\text{subject to } x_1 + 3x_2 \le 9$$

$$x_1 + x_2 \ge 5$$

$$2x_1 + x_2 \le 8$$

$$x_1, x_2 \ge 0.$$

Solution: Let $Z = 6x_1 + 3x_2 + 5 = Z' + 5$,

where $$Z' = 6x_1 + 3x_2 .$$

Obviously Max. $Z =$ Max. $Z' + 5$.

The problem is of maximization and all b_i's are positive.

Introducing the necessary slack variables x_3, x_5 surplus variables x_4 and artificial variable x_a and assigning large negative cost$-M$ to artificial variable, the problem reduces to the following form

$$\text{Max. } Z' = 6x_1 + 3x_2 + 0.x_3 + 0.x_4 + 0.x_5 - M.x_a$$

$$\begin{aligned} \text{s.t.} \quad x_1 + 3x_2 + x_3 \qquad\qquad &= 9 \\ x_1 + x_2 \qquad + x_4 \qquad + x_a &= 5 \\ 2x_1 + x_2 \qquad\qquad + x_5 \qquad &= 8 \end{aligned}$$

and $$x_1, x_2, x_3, x_4, x_5, x_a \ge 0.$$

Taking $x_1 = 0, x_2 = 0, x_4 = 0$, we get $x_3 = 9, x_a = 5, x_5 = 8$ which is the starting B.F.S.

The solution of this problem using simplex method is given in the following table:

The solution of this problem using simplex method is given in the following table:

B	$\mathbf{c}_B$	c_j	6	3	0	0	0	$-M$	Min. Ratio
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	Y_5	A	
Y_3	0	9	1	3	1	0	0	0	9/1
A	$-M$	5	1	1	0	-1	0	1	5/1
Y_5	0	8	$\boxed{2}$	1	0	0	1	0	8/2 (min) $\rightarrow$
$Z' = \mathbf{c}_B \cdot \mathbf{x}_B = -5M$		Δ_j	$M+6$ $\uparrow$	$M+3$	0	$-M$	0 $\downarrow$	0	
			$\overline{Y}_3$	$\overline{Y}_4$	$\overline{Y}_1$	$\overline{Y}_5$	$\overline{Y}_6$	$\overline{Y}_2$	$\mathbf{x}_B / Y_2$
Y_3	0	5	0	5/2 $(\bar{y}_{14})$	1 $(\bar{y}_{11})$	0	$-1/2$	0	2
A	$-M$	1	0	$\boxed{1/2}$ $(\bar{y}_{24})$	0 $(\bar{y}_{21})$	-1	$-1/2$	1	2 (min) $\rightarrow$
Y_1	6	4	1	1/2	0	0	1/2	0	8
$Z' = \mathbf{c}_B \cdot \mathbf{x}_B = -M + 24$		Δ_j	0	$\frac{1}{2}M$ $\uparrow$	0	$-M$	$-\frac{1}{2}M - 3$	0	
Y_3	0	0	0	0	1	5	2	-5	
Y_2	3	2	0	1	0	-2	-1	2	
Y_1	6	3	1	0	0	1	1	-1	
$Z' = \mathbf{c}_B \cdot \mathbf{x}_B = 24$		Δ_j	0	0	0	0	-3	$-M$	

In the second table by minimum ratio rule minimum is not unique but occur for $i = 1$ and $i = 2$ both.

$\therefore \qquad I_1 = \{1, 2\}.$

Renumbering the columns, the entering vector $Y_2 = \overline{Y}_4$

$\therefore \qquad t = 4.$

For $\quad i = 1, 2$

$$\underset{i \in I}{\text{Mini}} \left\{ \frac{\bar{y}_{i1}}{y_{it}}, \bar{y}_{it} > 0 \right\} = \underset{i \in I_1}{\text{Mini}} \left\{ \frac{\bar{y}_{i1}}{y_{i4}}, \bar{y}_{i4} > 0 \right\} = \text{Mini} \left\{ \frac{\bar{y}_{11}}{\bar{y}_{14}}, \frac{\bar{y}_{21}}{\bar{y}_{24}} \right\}$$

$$= \text{Mini} \left\{ \frac{1}{5/2}, \frac{0}{1/2} \right\} = 0$$

for $i = 2$

$\therefore$ $\bar{y}_2 = A$ is the outgoing vector and key elements $\bar{y}_{24} = y_{22}$.

In the last table all $\Delta_j \leq 0$, therefore this solution is optimal.

Hence optimal solution is

$$x_1 = 3, x_2 = 2 \text{ and Max. } Z = \text{Max. } Z' + 5 = 24 + 5 = 29.$$

Comprehensive Exercise 2

Solve the following problems using simplex method :

1. (i) Max. $Z = 3x_1 + 2x_2$

subject to $x_1 + x_2 \leq 4$

$x_1 - x_2 \leq 2,$

$x_1, x_2 \geq 0$

(Garhwal 2010)

(ii) Max. $Z = 5x_1 + 3x_2$

subject to $3x_1 + 5x_2 \leq 15$

$5x_1 + 2x_2 \leq 10$

$x_1, x_2 \geq 0$

(Garhwal 2007, 12)

(iii) Min. $Z = x_1 - 3x_2 + 2x_3$

subject to $3x_1 - x_2 + 2x_3 \leq 7$

$-2x_1 + 4x_2 \leq 12$

$-4x_1 + 3x_2 + 8x_3 \leq 10$

$x_1, x_2, x_3 \geq 0$

(iv) Max. $Z = 5x_1 + 10x_2 + 8x_3$

subject to $3x_1 + 5x_2 + 2x_3 \leq 60$

$4x_1 + 4x_2 + 4x_3 \leq 72$

$2x_1 + 4x_2 + 5x_3 \leq 100$

$x_1, x_2, x_3 \geq 0$

(v) Max. $Z = 2x_1 + 4x_2 + x_3 + x_4$

subject to

$x_1 + 3x_2 + x_4 \leq 4$

$2x_1 + x_2 \leq 3$

$x_2 + 4x_3 + x_4 \leq 3$

$x_1, x_2, x_3, x_4 \geq 0$

(vi) Max. $Z = 2x_1 + x_2$

subject to $x_1 - x_2 \leq 10$

$2x_1 - x_2 \leq 40$

$x_1, x_2 \geq 0.$

2. (i) Max. $Z = 7x_1 + 5x_2$

subject to

$$-x_1 - 2x_2 \geq -6$$
$$4x_1 + 3x_2 \leq 12$$
$$x_1, x_2 \geq 0$$

(ii) Max. $Z = 3x_1 + 5x_2$

subject to

$$3x_1 + 2x_2 \leq 18$$
$$x_1 \leq 4$$
$$x_2 \leq 6,$$
$$x_1, x_2 \geq 0$$

(iii) Max. $Z = 3x_1 + 2x_2$

subject to

$$2x_1 + x_2 \leq 40$$
$$x_1 + x_2 \leq 24$$
$$2x_1 + 3x_2 \leq 60$$
$$x_1, x_2 \geq 0$$

(iv) Max. $Z = 2x_1 + 3x_2 + 3x_3$

subject to

$$3x_1 + 2x_2 + x_3 \leq 3$$
$$2x_1 + x_2 + 2x_3 \leq 4$$
$$x_1, x_2, x_3 \geq 0.$$

3. Max. $Z = 5x_1 + 7x_2$

subject to the constraints

$$x_1 + x_2 \leq 4$$
$$3x_1 - 8x_2 \leq 24$$
$$10x_1 + 7x_2 \leq 35$$
$$x_1, x_2 \geq 0.$$

4. Max. $Z = x_1 - x_2 + 3x_3$

subject to

$$x_1 + x_2 + x_3 \leq 10$$
$$2x_1 - x_3 \leq 2$$
$$2x_1 - 2x_2 + 3x_3 \leq 0$$
$$x_1, x_2, x_3 \geq 0.$$

5. Max. $Z = 4x_1 + 10x_2$

subject to

$$2x_1 + x_2 \leq 50$$
$$2x_1 + 5x_2 \leq 100$$
$$2x_1 + 3x_2 \leq 90$$
$$x_1, x_2 \geq 0.$$

6. Max. $Z = 2x_1 + 4x_2$

subject to

$$2x_1 + 3x_2 \leq 48$$
$$x_1 + 3x_2 \leq 42$$
$$x_1 + x_2 \leq 21$$
$$x_1, x_2 \geq 0.$$

7. Max. $Z = 2x_1 + x_2$

subject to

$$x_1 + 2x_2 \leq 10$$
$$x_1 + x_2 \leq 6$$
$$x_1 - x_2 \leq 2$$
$$x_1 - 2x_2 \leq 1$$
$$x_1, x_2 \geq 0.$$

8. Max. $Z = 3x_1 + 4x_2$

subject to

$$x_1 - x_2 \leq 1$$
$$-x_1 + x_2 \leq 2$$
$$x_1, x_2 \geq 0.$$

9. Max. $Z = 4x_1 + 5x_2 + 9x_3 + 11x_4$

subject to the constraints

$$x_1 + x_2 + x_3 + x_4 \le 15$$
$$7x_1 + 5x_2 + 3x_3 + 2x_4 \le 120$$
$$3x_1 + 5x_2 + 11x_3 + 15x_4 \le 100$$
$$x_1, x_2, x_3, x_4 \ge 0.$$

10. Max. $Z = 8x_1 + 11x_2$

subject to

$$3x_1 + x_2 \le 7$$
$$x_1 + 3x_2 \le 8$$
$$x_1, x_2 \ge 0.$$

11. Min. $Z = x_1 + x_2 + 3x_3$

subject to

$$3x_1 + 2x_2 + x_3 \le 3$$
$$2x_1 + x_2 + 2x_3 \le 2$$
$$x_1, x_2, x_3 \ge 0.$$

12. Max. $Z = 3x_1 + 2x_2 - 2x_3$

subject to

$$x_1 + 2x_2 + 2x_3 \le 10$$
$$2x_1 + 4x_2 + 3x_3 \le 15$$
$$x_1, x_2, x_3 \ge 0.$$

13. Max. $Z = 7x_1 + x_2 + 2x_3$

subject to

$$x_1 + x_2 - 2x_3 \le 10$$
$$4x_1 + x_2 + x_3 \le 20$$
$$x_1, x_2, x_3 \ge 0.$$

14. Max. $Z = 2x_1 + 4x_2 + 3x_3$

subject to

$$3x_1 + 4x_2 + 2x_3 \le 60$$
$$2x_1 + x_2 + 2x_3 \le 40$$
$$x_1 + 3x_2 + 2x_3 \le 80$$
$$x_1, x_2, x_3 \ge 0.$$

15. Max.
$Z = 5x_1 + 2x_2 + 3x_3 - x_4 + x_5$

subject to the constraints

$$x_1 + 2x_2 + 2x_3 + x_4 = 8,$$
$$3x_1 + 4x_2 + x_3 + x_5 = 7,$$
$$x_1, x_2, x_3, x_4, x_5 \ge 0.$$

16. Min. $Z = 4x_1 + 8x_2 + 3x_3$

subject to

$$x_1 + x_2 \ge 2$$
$$2x_1 + x_3 \ge 5$$
$$x_1, x_2, x_3 \ge 0.$$

17. Max. $Z = 3x_1 + 4x_2$

subject to

$$x_1 + 3x_2 \le 9$$
$$2x_1 - x_2 \le 8$$
$$x_1 + x_2 \le 5$$
$$x_1, x_2 \ge 0$$

18. Max. $Z = 3x_1 + x_2 + 2x_3 + 7x_4$

subject to

$$2x_1 + 3x_2 - x_3 + 4x_4 \le 40$$
$$-2x_1 + 2x_2 + 5x_3 - x_4 \le 35$$
$$x_1 + x_2 - 2x_3 + 3x_4 \le 100$$
$$x_1 \ge 2, x_2 \ge 1, x_3 \ge 3, x_4 \ge 4$$

Solve the following L.P.P. using two phase method :

19. Min. $Z = x_1 + x_2$

subject to

$$2x_1 + x_2 \geq 4$$
$$x_1 + 7x_2 \geq 7$$
$$x_1,\ x_2 \geq 0.$$

(Garhwal 2006)

20. Max. $Z = 3x_1 - x_2$

subject to

$$2x_1 + x_2 \geq 2$$
$$x_1 + 3x_2 \leq 2$$
$$x_2 \leq 4$$
$$x_1,\ x_2 \geq 0.$$

21. Max. $Z = 5x_1 + 8x_2$

subject to

$$3x_1 + 2x_2 \geq 3$$
$$x_1 + 4x_2 \geq 4$$
$$x_1 + x_2 \leq 5$$
$$x_1,\ x_2 \geq 0.$$

22. Max. $Z = 107x_1 + x_2 + 2x_3$

subject to

$$14x_1 + x_2 - 6x_3 + 3x_4 = 7$$
$$16x_1 + \frac{1}{2}x_2 - 6x_3 \leq 5$$
$$3x_1 - x_2 - x_3 \leq 0$$
$$x_1,\ x_2,\ x_3,\ x_4 \geq 0.$$

23. Max. $Z = 3x_1 + 2x_2 + x_3 + 4x_4$

subject to

$$4x_1 + 5x_2 + x_3 + 5x_4 = 5$$
$$2x_1 - 3x_2 - 4x_3 + 5x_4 = 7$$
$$x_1 + 4x_2 + 5x_3 - 4x_4 = 6$$
$$x_1,\ x_2,\ x_3,\ x_4 \geq 0.$$

Apply Big-M method to solve the following problems :

24. Max. $Z = -2x_1 - x_2$

subject to

$$3x_1 + x_2 = 3$$
$$4x_1 + 3x_2 \geq 6$$
$$x_1 + 2x_2 \leq 4$$
$$x_1,\ x_2 \geq 0.$$

25. Max. $Z = 3x_1 - x_2$

subject to

$$2x_1 + x_2 \geq 2$$
$$x_1 + 3x_2 \leq 3$$
$$x_2 \leq 4$$
$$x_1,\ x_2 \geq 0.$$

26. Max. $Z = 4x_1 + 5x_2 - 3x_3$

subject to

$$x_1 + x_2 + x_3 = 10$$
$$x_1 - x_2 \geq 1$$
$$2x_1 + 3x_2 + x_3 \leq 40$$
$$x_1, x_2, x_3 \geq 0$$

27. Min. $Z = 5x_1 + 6x_2$

subject to

$$2x_1 + 5x_2 \geq 1500$$
$$3x_1 + x_2 \geq 1200$$
$$x_1, x_2 \geq 0.$$

28. Max. $Z = 4x_1 + 2x_2$

subject to

$$3x_1 + x_2 \leq 27$$
$$x_1 + x_2 \geq 21$$
$$x_1, x_2, \geq 0.$$

29. Max. $Z = 2x_1 + x_2 + 3x_3$

subject to

$$x_1 + x_2 + x_3 \leq 5$$
$$2x_1 + 3x_2 + 4x_2 \leq 12$$
$$x_1, x_2, x_3 \geq 0.$$

30. Min. $Z = 4x_1 + 6x_2$

subject to

$$x_1 + 2x_2 \geq 80$$
$$3x_1 + x_2 \geq 75$$
$$x_1, x_2 \geq 0.$$

31. Min. $Z = 2y_1 + 3y_2$

subject to $y_1 + y_2 \geq 5$

$$y_1 + 2y_2 \geq 6$$
$$y_1, y_2 \geq 0.$$

32. Min. $Z = x_1 + x_2 + 3x_3$

subject to $3x_1 + 2x_2 + x_3 \leq 3$

$$2x_1 + x_2 + 2x_3 \geq 3$$
$$x_1, x_2, x_3 \geq 0.$$

Solve the following system of linear equations by simplex method :

33. $3x_1 + 2x_2 = 4,\ 4x_1 - x_2 = 6.$

34. $3x_1 + 2x_2 = 5,\ 5x_1 + x_2 = 9.$

35. $x_1 + x_2 = 1,\ 2x_1 + x_2 = 3.$

36. Find the inverse of the matrix

(i) $\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}$

(iii) $\begin{bmatrix} 4 & 1 & 2 \\ 0 & 1 & 0 \\ 8 & 4 & 5 \end{bmatrix}$ (iv) $\begin{bmatrix} 4 & 1 \\ 2 & 9 \end{bmatrix}$.

37. Food A contains 20 units of vitamin X and 40 units of vitamin Y per gram. Food B contains 30 units each of vitamin X and Y. The daily minimum human requirements of vitamins X and Y are 900 units and 1200 units respectively. How many grams of each type of food should be consumed so as to minimize the cost if food A costs 60 paise per gram and food B costs 80 paise per gram?

38. A finished product must weigh exactly 150 gms. The two raw materials used in manufacturing the product are : A with cost of ₹ 2 per unit and B with a cost of ₹ 8 per unit. At least 14 units of B and not more than 20 units of A must be used. Each unit of A and B weighs 5 and 10 grams respectively.

How much of each type of raw material should be used for each unit of the final product in order to minimize the cost ?

39. A company produces two types of products say type A and B. Product B is of superior quality and product A is of a lower quality. Profits on the two types of products are ₹ 30 and ₹ 40 respectively. The data on resources required and capacity available is given below:

Types of Products	Requirements		Capacity available per month
	Product A	Product B	
Raw materials (kg)	60	120	12,000
Machining (hours per piece)	8	5	600
Assembly (man hour)	3	4	500

How should the company manufacture the two types of products in order to have a maximum overall profit ?

Answers 2

1. (i) $x_1 = 3, x_2 = 1$, Max. $Z = 11$

(ii) $x_1 = 20/19, x_2 = 45/19$, Max. $Z = 235/19$

(iii) $x_1 = 4, x_2 = 5, x_3 = 0$, Min. $Z = -11$

(iv) $x_1 = 0, x_2 = 8, x_3 = 10$, Max. $Z = 160$

(v) $x_1 = 1, x_2 = 1, x_3 = 1/2, x_4 = 0$, Max. $Z = 13/2$

(vi) $x_1 = 30, x_2 = 20$, Max. $Z = 80$

2. (i) $x_1 = 3, x_2 = 0$, Max. $Z = 21$ (ii) $x_1 = 2, x_2 = 6$, Max. $Z = 36$

(iii) $x_1 = 16, x_2 = 8$, Max. $Z = 64$

(iv) $x_1 = 0, x_2 = 2/3, x_3 = 5/3$, Max. $Z = 7$

3. $x_1 = 0, x_2 = 4$, Max. $Z = 28$

4. $x_1 = 0, x_2 = 29/5, x_3 = 21/5$, Max. $Z = 34/5$

5. $x_1 = 0, x_2 = 20$, Max. $Z = 200$ or $x_1 = 75/4, x_2 = 25/2$, Max. $Z = 200$

6. $x_1 = 6, x_2 = 12, Z = 60$

7. $x_1 = 4, x_2 = 2$, Max. $Z = 10$

8. unbounded

9. $x_1 = 50/7, x_2 = 0, x_3 = 55/7, x_4 = 0$ Max. $Z = 695/7$

10. $x_1 = 13/8, x_2 = 17/8$, Max. $Z = 291/8$

11. $x_1 = x_2 = x_3 = 0$, Max. $Z = 0$

12. $x_1 = 15/2$, $x_2 = 0$, $x_3 = 0$, Max. $Z = 45/2$

13. $x_1 = 0$, $x_2 = 0$, $x_3 = 20$, Max. $Z = 40$

14. $x_1 = 0$, $x_2 = 20/3$, $x_3 = 50/3$, Max. $Z = 230/3$

15. $x_1 = 6/5$, $x_2 = 0, x_3 = 17/5$, $x_4 = 0$, $x_5 = 0$, Max. $Z = 18/5$

16. $x_1 = 5/2$, $x_2 = 0$, $x_3 = 0$, Min. $Z = 10$

17. $x_1 = 3$, $x_2 = 2$, Max. $Z = 17$.

18. $x_1 = 71/4, x_2 = 1, x_3 = 29/2, x_4 = 4$, Max. $Z = 445/4$

19. $x_1 = 21/13$, $x_2 = 10/13$, Min. $Z = 31/13$

20. $x_1 = 2$, $x_2 = 0$, Max. $Z = 6$

21. $x_1 = 0$, $x_2 = 5$, Max. $Z = 40$

22. unbounded

23. no feasible solution

24. $x_1 = 3/5$, $x_2 = 6/5$, Max. $Z = -12/5$

25. $x_1 = 3$, $x_2 = 0$, Max. $Z = 9$

26. $x_1 = 11/2$, $x_2 = 9/2$, Max. $Z = 89/2$

27. $x_1 = 4500/13$, $x_2 = 2100/13$, Max. $Z = 2700$

28. $x_1 = 0$, $x_2 = 27$, Max. $Z = 54$

29. $x_1 = 4$, $x_2 = 0$, $x_3 = 1$, Max. $Z = 11$

30. $x_1 = 24$, $x_2 = 33$, Min. $Z = 254$

31. $x_1 = 4$, $x_2 = 1$, Min. $Z = 11$

32. $x_1 = 3/4$, $x_2 = 0$, $x_3 = 3/4$, Min. $Z = 3$

33. $x_1 = 16/11$, $x_2 = -2/11$

34. $x_1 = 13/7$, $x_2 = -2/7$

35. $x_1 = 2$, $x_2 = -1$

36. (i) $\frac{1}{4}\begin{bmatrix} -2 & 2 \\ 3 & -1 \end{bmatrix}$ (ii) $\frac{1}{11}\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$

(iii) $\frac{1}{4}\begin{bmatrix} 5 & 3 & -2 \\ 0 & 4 & 0 \\ -8 & -8 & 4 \end{bmatrix}$ (iv) $\frac{1}{34}\begin{bmatrix} 9 & -1 \\ -2 & 4 \end{bmatrix}$

37. Food $A = 15$gm, food $B = 20$ gm, total cost = ₹ 25.

38. 2, 14, min. cost = ₹ 116.

39. 200/11 units of type A, 1000/11 units of type B, max. profit = ₹ 46,000/11

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. A slack variable is introduced if the given constraint has a sign :
   (a) $\geq$ (b) $\leq$
   (c) $=$ (d) None of these

2. Simplex method to solve linear programming problems was developed by :
   (a) Newton (b) Lagrange
   (c) George Dantzig (d) None of these

3. If a constraint has $\leq$ sign, we introduce :
   (a) Surplus variable (b) Artificial variable
   (c) Slack variable (d) None of these

4. Solution of the L.P.P. :
   Max. $Z = 10x_1 + 6x_2$
   subject to
   $x_1 + x_2 \leq 2$, $2x_1 + x_2 \leq 4$, $3x_1 + 8x_2 \leq 12$,
   $x_1, x_2 \geq 0$ is
   (a) $x_1 = 2, x_2 = 0,$ Max. $Z = 20$ (b) $x_1 = 0, x_2 = 0,$ Max. $Z = 0$
   (c) $x_1 = 1, x_2 = 7,$ Max. $Z = -2$ (d) $x_1 = 0, x_2 = 3,$ Max. $Z = 1$

Fill in the Blank

Fill in the blanks "......" so that the following statements are complete and correct.

1. Simplex method was developed by .... in 1947.
2. According to fundamental theorem of linear programming, we can search the ....solution among the basic feasible solutions.
3. If a constraint has a sign $\leq$, in order to convert it into an equation we use .... variables.
4. If a constraint has a $\geq$ sign, then to change it into an equation we introduce .... variables.
5. A linear programming problem is said to have an .... solution if the objective function can be increased or decreased indefinitely.
6. To convert the problem of minimization into the maximization problem we multiply both sides by ....

7. To solve a L.P.P. by simplex method all b_i's should be ....
8. If in the simplex table all $\Delta_j \le 0$, the solution under test is ....
9. The linear programming problem has no .... solution if the solution contains one or more artificial variables as basic variables.
10. If in the final simplex table all $\Delta_j < 0$, the optimal solution is ....
11. If corresponding to maximum positive Δ_j all minimum ratios are negative or $\to \infty$, the solution under test is ....

True or False

Write 'T' for true and 'F' for false statement.

1. Fundamental theorem of L.P.P. states that if the given L.P.P. has an optimal solution, then at least one basic solution must be optimal.
2. The problem has no feasible solution if the value of at least one artificial variable present in the basis is non-zero and the optimality condition is satisfied.
3. In the phase I of two phase method, we remove artificial variables from the basis matrix.
4. If we have to solve a linear programming problem by simplex method the variable x should be unrestricted in sign.

Answers

Multiple Choice Questions

1. (b) 2. (c) 3. (c) 4. (a)

Fill in the Blank

1. George Dantzig 2. optimal 3. slack 4. surplus
5. unbounded 6. (−1) 7. non-negative
8. optimal 9. feasible 10. unique 11. unbounded

True/False

1. *T* 2. *T* 3. *T* 4. *F*

□

Chapter 4

Duality in Linear Programming

4.1 Introduction

The concept of duality was one of the most important discoveries in the early development of linear programming. It explains that corresponding to every linear programming problem there is another linear programming problem. The original problem is known as ***'primal'*** and the other, the related problem is known as the ***'dual'***. The two problems are replicas of each other. If the primal problem is of maximization type, the dual will be of the minimization type. If the optimal solution to one problem is known to us, we can easily find the optimal solution of the other. This fact is important because at times it is easier to solve the dual than the primal.

To understand the concept of duality more clearly, consider the following diet problem :

The amount of two vitamins A and B per unit present in two different foods A_1 and A_2 respectively are given below :

Vitamin	Food		Minimum daily requirement
	A_1	A_2	
A	6	9	60
B	4	13	108
Cost (per unit)	₹ 12	₹ 18	

The objective of the diet problem is to ascertain the quantities of foods A_1 and A_2 that should be eaten to meet the minimum daily requirements of vitamins A and B at a minimum cost.

Let x_1 and x_2 be the number of units of foods A_1 and A_2 to be purchased respectively. Then the problem is to find the values of x_1 and x_2 which minimize

$$Z_x = 12x_1 + 18x_2$$

subject to the constraints

$$6x_1 + 9x_2 \geq 60, \; 4x_1 + 13x_2 \geq 108, \; x_1, x_2 \geq 0$$

We shall consider this linear programming problem as the primal problem. Now we shall formulate the dual corresponding to this primal.

Suppose a wholesale dealer sells two vitamins A and B. Customers purchase the two vitamins from him in the form of two foods A_1 and A_2 (as given in the above table). The foods A_1 and A_2 have their market value only because of their vitamin contents. Now the dealer has to fix-up the maximum per unit selling prices for the two vitamins A and B in such a way that the resulting prices of the foods A_1 and A_2 do not exceed their existing market prices.

Suppose the dealer decides to fix-up the two prices y_1 and y_2 respectively.

Then the problem is to determine the values of y_1 y_2 which maximize

$$Z_y = 60y_1 + 108y_2$$

subject to

$$6y_1 + 4y_2 \leq 12$$

$$9y_1 + 13y_2 \leq 18$$

$$y_1, y_2 \geq 0$$

Observing the above primal and the dual a little closely, we find that

1. Primal is a minimization problem while the dual is a maximization problem.
2. The constraint values 60, 108 of the primal have become the coefficients of the dual variables y_1 and y_2 in the objective function of the dual in that order. The coefficients for the variables x_1 and x_2 in the objective function of the primal have become the constraint values in the dual.
3. The constraint coefficient matrix of the dual is the transpose of the constraint coefficient matrix of the primal.
4. The direction of inequalities in the dual is the reverse of that in the primal.

We can represent the primal-dual relationship in the following form :

Primal	**Dual**
Minimize	Maximize
$Z_x = \overset{\mathbf{c}}{(12 \quad 18)} \overset{\mathbf{x}}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}$	$Z_y = \overset{\mathbf{b}'}{(60 \quad 108)} \overset{\mathbf{y}}{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}}$
subject to	subject to
$\overset{A}{\begin{bmatrix} 6 & 9 \\ 4 & 13 \end{bmatrix}} \overset{\mathbf{x}}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} \ge \overset{\mathbf{b}}{\begin{bmatrix} 60 \\ 108 \end{bmatrix}}$	$\overset{A'}{\begin{bmatrix} 6 & 4 \\ 9 & 13 \end{bmatrix}} \overset{\mathbf{y}}{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}} \le \overset{\mathbf{c}'}{\begin{bmatrix} 12 \\ 18 \end{bmatrix}}$
$x_1, x_2 \ge 0$	$y_1, y_2 \ge 0$

4.2 Standard or Canonical Form of a Primal

While dealing with duality theory, the given L.P.P. (known as primal problem) should be changed to the standard (or canonical) form. A linear programming problem is said to be in standard primal form (or canonical primal form) if

1. For a maximization L.P.P. all the constraints have $\le$ sign.
2. For a minimization L.P.P. all the constraints have $\ge$ sign.

4.3 Methods to Convert a L.P.P. into its Dual

4.3.1 Dual of a Symmetrical Primal Form

Consider the following linear programming problem, we may call it the *symmetrical (standard canonical)* primal form :

Primal L.P.P. : Find the variables $x_1, x_2, \ldots, x_n$ which maximize

$$Z_p = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$$

subject to,

$$a_{11}, x_1 + a_{12} x_2 + \ldots + a_{1n} x_n \le b_1$$

$$a_{21} x_1 + a_{22} x_2 + \ldots + a_{2n} x_n \le b_2$$

$$\ldots \qquad \ldots \qquad \ldots$$

$$\ldots \qquad \ldots \qquad \ldots$$

$$a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mn} x_n \le b_m$$

and $\qquad x_1, x_2, \ldots, x_n \geq 0,$

the signs of the parameters a, b, c are arbitrary.

To obtain the dual of the above primal, following steps are required :

(i) Minimize the objective function instead of maximizing it.

(ii) Interchange the role of constant terms and the coefficients of the objective function.

(iii) Find A', where A' denotes the transpose of the coefficient matrix A.

(iv) Reverse the direction of inequalities.

Thus the dual problem is :

Find the variables $w_1, w_2, \ldots, w_m$ which minimize

$$Z_w = b_1 w_1 + b_2 w_2 + \ldots + b_m w_m$$

subject to
$$\begin{aligned} a_{11} w_1 + a_{21} w_2 + \ldots + a_{m1} w_m &\geq c_1 \\ a_{12} w_1 + a_{22} w_2 + \ldots + a_{m2} w_m &\geq c_2 \\ \ldots \qquad \ldots \qquad \ldots & \\ \ldots \qquad \ldots \qquad \ldots & \\ a_{1n} w_1 + a_{2n} w_2 + \ldots + a_{mn} w_m &\geq c_n \end{aligned}$$

and $\qquad w_1, w_2, \ldots, w_n \geq 0$

4.3.2 Matrix Form of Symmetric Primal-Dual Problem

Primal Problem : Find a column vector $\mathbf{x} \in R^n$ which maximizes

$$Z_x = \mathbf{c}\,\mathbf{x}, \mathbf{c} \in R^n$$

subject to $\qquad A\mathbf{x} \leq \mathbf{b}, \mathbf{b} \in R^m,$

$\mathbf{x} \geq \mathbf{0}$ and A is an $m \times n$ real matrix.

Dual Problem : Find a column vector $\mathbf{w} \in R^m$ which minimizes

$$Z_w = \mathbf{b}'\mathbf{w}$$

subject to $A'\mathbf{w} \geq \mathbf{c}'$

$\mathbf{w} \geq \mathbf{0}, A', \mathbf{b}', \mathbf{c}'$ are the transposes of A, $\mathbf{b}$ and $\mathbf{c}$ respectively.

For example : Consider the symmetric primal problem

$$\text{Max. } Z_x = 5x_1 + 9x_2$$

subject to $\qquad x_1 \leq 6$

$$x_1 + x_2 \le 13$$

$$x_2 \le 8,$$

$$x_1, x_2 \ge 0.$$

The corresponding dual problem is

$$\text{Min. } Z_w = 6w_1 + 13w_2 + 8w_3$$

subject to $\quad w_1 + w_2 \ge 5$

$$w_2 + w_3 \ge 9,$$

$$w_1, w_2, w_3 \ge 0.$$

Note : The following table gives a simple and convenient method to remember the primal-dual relationship :

$$(x_1, x_2, \ldots x_n) \quad \text{Min.}$$

$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \le \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\text{Max. } (c_1, c_2, \ldots, c_n)$$

For primal constraints, read across the table while for dual constraints read down the columns.

4.3.3 Dual of an Unsymmetric Primal Problems

Consider the following L.P.P. we may call it the unsymmetric primal form :

Primal Problem : Find a column vector $\mathbf{x} \in R^n$ which maximizes

$$Z_x = \mathbf{c}\,\mathbf{x}, \mathbf{c} \in R^n$$

subject to $\quad A\mathbf{x} = \mathbf{b}, \mathbf{b} \in R^m$

$\mathbf{x} \ge \mathbf{0}$ and A is an $m \times n$ real matrix.

Dual Problem : Find a column vector $\mathbf{w} \in R^m$ which minimizes

$$Z_w = \mathbf{b}'\,\mathbf{w}$$

subject to $\quad A'\,\mathbf{w} \ge \mathbf{c}'.$

In this case the *dual variables* are *unrestricted* in *sign*.

4.3.4 Dual of an L.P.P. with Mixed Restrictions

Sometimes a given L.P.P. contains a mixture of inequalities (≥, ≤), equations; non-negative variables and unrestricted variables, then to obtain its dual we proceed in the following manner :

1. If a constraint is an equation (has = sign), replace it by two constraints involving the inequalities going in opposite directions.

   For example

   The equation $2x_1 + 5x_2 = 9$ is replaced by

   In maximization problem

   $$2x_1 + 5x_2 \le 9 \quad \text{...(1)}$$

   and $$2x_1 + 5x_2 \ge 9 \quad \text{...(2)}$$

   In minimization problem

   $$2x_1 + 5x_2 \ge 9 \quad \text{...(3)}$$

   and $$2x_1 + 5x_2 \le 9 \quad \text{...(4)}$$

   In the problem of maximization, all constraints should have ≤ sign. So we multiply both sides of constraint (2) by –1.

   So in maximization problem, the equation $2x_1 + 5x_2 = 9$ is replaced by $2x_1 + 5x_2 \le 9$ and $-2x_1 - 5x_2 \le -9$.

   Similarly, if the problem is of minimization, all constraints should have ≥ sign.

   So in this case the equation say $2x_1 + 5x_2 = 9$ is replaced by $2x_1 + 5x_2 \ge 9$ and $-2x_1 - 5x_2 \ge -9$.

2. If there is some unrestricted variable, replace it by the difference of two non-negative variables.

3. Now to find the dual problem proceed as in article 8.3.1.

Note : The dual variables corresponding to primal equality constraints must be unrestricted in sign and those associated with primal inequalities must be non-negative.

Illustrative Examples

Example 1: *Find the dual of the following L.P.P. :*

$$\text{Min. } Z = 10x_1 + 20x_2$$

subject to
$$3x_1 + 2x_2 \ge 18$$
$$x_1 + 3x_2 \ge 8$$
$$2x_1 - x_2 \le 6,\ x_1, x_2 \ge 0.$$

Solution: The given L.P.P. in the standard primal form is

$$\text{Min. } Z = 10x_1 + 20x_2$$

subject to $3x_1 + 2x_2 \geq 18$

$$x_1 + 3x_2 \geq 8$$

$$-2x_1 + x_2 \geq -6$$

and $x_1, x_2 \geq 0$

The matrix form of this problem is

$$\text{Min. } Z = 10x_1 + 20x_2 = (10,20)[x_1, x_2] = \mathbf{c\,x}$$

$$\text{subject to } \begin{bmatrix} 3 & 2 \\ 1 & 3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 18 \\ 8 \\ -6 \end{bmatrix}$$

or $A\mathbf{x} \geq \mathbf{b}$, $x_1, x_2 \geq 0$.

$\therefore$ The dual of this problem is

$$\text{Max. } Z_D = \mathbf{b}'\mathbf{y} = (18,8,-6)[y_1, y_2, y_3\} = 18y_1 + 8y_2 - 6y_3$$

$$\text{subject to } A'\mathbf{y} \leq \mathbf{c}' \text{ or } \begin{bmatrix} 3 & 1 & -2 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \leq \begin{bmatrix} 10 \\ 20 \end{bmatrix}$$

or
$$\begin{bmatrix} 3y_1 + y_2 - 2y_3 \\ 2y_1 + 3y_2 + y_3 \end{bmatrix} \leq \begin{bmatrix} 10 \\ 20 \end{bmatrix}$$

or
$$3y_1 + y_2 - 2y_3 \leq 10,\ 2y_1 + 3y_2 + y_3 \leq 20,\ y_1, y_2, y_3 \geq 0.$$

Hence dual of the given problem is

$$\text{Max. } Z_D = 18y_1 + 8y_2 - 6y_3$$

subject to $3y_1 + y_2 - 2y_3 \leq 10,\ 2y_1 + 3y_2 + y_3 \leq 20,\ y_1, y_2, y_3 \geq 0.$

Example 2: *Write the dual of the following problem :*

$$\textit{Min. } Z = 3x_1 - 2x_2 + 4x_3$$

subject to $3x_1 + 5x_2 + 4x_3 \geq 7$

$$6x_1 + x_2 + 3x_3 \geq 4$$

$$7x_1 - 2x_2 - x_3 \leq 10$$

$$x_1 - 2x_2 + 5x_3 \geq 3$$

$$4x_1 + 7x_2 - 2x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0.$$

Solution: The standard primal form of the given L.P.P. is

$$\text{Min. } Z = 3x_1 - 2x_2 + 4x_3$$

subject to $3x_1 + 5x_2 + 4x_3 \geq 7$

$6x_1 + x_2 + 3x_3 \geq 4$

$-7x_1 + 2x_2 + x_3 \geq -10$

$x_1 - 2x_2 + 5x_3 \geq 3$

$4x_1 + 7x_2 - 2x_3 \geq 2,\ x_1, x_2, x_3 \geq 0.$

The matrix form of the above problem is

$$\text{Min. } Z = (3, -2, 4)[x_1, x_2, x_3] = \mathbf{c}\,.\,\mathbf{x}$$

$$\text{subject to} \quad \begin{bmatrix} 3 & 5 & 4 \\ 6 & 1 & 3 \\ -7 & 2 & 1 \\ 1 & -2 & 5 \\ 4 & 7 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 7 \\ 4 \\ -10 \\ 3 \\ 2 \end{bmatrix} \text{ or } A\mathbf{x} \geq \mathbf{b},\ x_1, x_2, x_3 \geq 0.$$

$\therefore$ The dual of the given primal is

$$\text{Max. } Z_D = \mathbf{b}'\mathbf{y} = (7, 4, -10, 3, 2)[y_1, y_2, y_3, y_4, y_5]$$

$$= 7y_1 + 4y_2 - 10y_3 + 3y_4 + 2y_5$$

subject to $A'\mathbf{y} \leq \mathbf{c}'$

or
$$\begin{bmatrix} 3 & 6 & -7 & 1 & 4 \\ 5 & 1 & 2 & -2 & 7 \\ 4 & 3 & 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} \leq \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$$

or
$$\begin{bmatrix} 3y_1 + 6y_2 - 7y_3 + y_4 + 4y_5 \\ 5y_1 + y_2 + 2y_3 - 2y_4 + 7y_5 \\ 4y_1 + 3y_2 + y_3 + 5y_4 - 2y_5 \end{bmatrix} \leq \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$$

Hence the dual problem of the given L.P.P. is

or $\quad \text{Max. } Z_D = 7y_1 + 4y_2 - 10y_3 + 3y_4 + 2y_5$

subject to $3y_1 + 6y_2 - 7y_3 + y_4 + 4y_5 \leq 3$

$5y_1 + y_2 + 2y_3 - 2y_4 + 7y_5 \leq -2$

$4y_1 + 3y_2 + y_3 + 5y_4 - 2y_5 \leq 4,$

$y_1, y_2, y_3, y_4, y_5 \geq 0.$

Example 3: *Write the dual of the following problem :*

$$Min.\ Z = 2x_2 + 5x_3$$

subject to $x_1 + x_2 \geq 2$

$$2x_1 + x_2 + 6x_3 \leq 6$$

$$x_1 - x_2 + 3x_3 = 4$$

$$x_1, x_2, x_3 \geq 0.$$

Solution: First we shall convert the given problem to standard primal form.

1. Since the problem is of minimization therefore all the constraints should have the sign $\geq$.
2. Multiplying the second constraint by -1, if becomes

$$-2x_1 - x_2 - 6x_3 \geq -6$$

3. Since the third constraint is an equality, so replacing it by the following two constraints :

$$x_1 - x_2 + 3x_3 \geq 4$$

and $x_1 - x_2 + 3x_3 \leq 4$

or $x_1 - x_2 + 3x_3 \geq 4$

and $-x_1 + x_2 - 3x_3 \geq -4$

$\therefore$ The given problem in standard primal form is

$$\text{Min. } Z = 0\,x_1 + 2x_2 + 5x_3$$

subject to $x_1 + x_2 \geq 2$

$$-2x_1 - x_2 - 6x_3 \geq -6$$

$$x_1 - x_2 + 3x_3 \geq 4$$

$$-x_1 + x_2 - 3x_3 \geq -4$$

$$x_1, x_2, x_3 \geq 0.$$

The matrix form of the above problem is

$$\text{Min. } Z = (0, 2, 5)[x_1, x_2, x_3] = \mathbf{c} \,.\, \mathbf{x}$$

subject to $\begin{bmatrix} 1 & 1 & 0 \\ -2 & -1 & -6 \\ 1 & -1 & 3 \\ -1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 2 \\ -6 \\ 4 \\ -4 \end{bmatrix}$ or $A\mathbf{x} \geq \mathbf{b}$, $x_1, x_2, x_3 \geq 0$

$\therefore$ The dual of the given primal is

$$\text{Max. } Z_D = \mathbf{b}'.\mathbf{y} = (2,-6,4,-4)[y_1, y_2, y_3', y_3'']$$

$$= 2y_1 - 6y_2 + 4(y_3' - y_3'')$$

subject to $A'\mathbf{y} \le \mathbf{c}'$

$$\begin{bmatrix} 1 & -2 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 0 & -6 & 3 & -3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3' \\ y_3'' \end{bmatrix} \le \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix}$$

or
$$\begin{bmatrix} y_1 - 2y_2 + y_3' - y_3'' \\ y_1 - y_2 - y_3' + y_3'' \\ 0.y_1 - 6y_2 + 3y_3' - 3y_3'' \end{bmatrix} \le \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix},$$

$$y_1, y_2, y_3', y_3'' \ge 0$$

or $\quad \text{Max. } Z_D = 2y_1 - 6y_2 + 4(y_3' - y_3'')$

subject to

$$y_1 - 2y_2 + (y_3' - y_3'') \le 0$$

$$y_1 - y_2 - (y_3' - y_3'') \le 2$$

$$-6y_2 + 3(y_3' - y_3'') \le 5,$$

$$y_1, y_2, y_3', y_3'' \ge 0$$

Substituting $y_3 = y_3' - y_3''$, the required dual is

$$\text{Max. } Z_D = 2y_1 - 6y_2 + 4y_3$$

subject to

$$y_1 - 2y_2 + y_3 \le 0, y_1 - y_2 - y_3 \le 2, -6y_2 + 3y_3 \le 5,$$

$y_1, y_2 \ge 0$ and y_3 is unrestricted in sign.

Note : The variable corresponding to the equality equation (here third equation) in the constraint will be unrestricted in sign in dual problem.

Example 4: *Write the dual of the following L.P.P. :*

$$\text{Max. } Z = 2x_1 + 3x_2 + x_3$$

subject to $\quad 4x_1 + 3x_2 + x_3 = 6$

$$x_1 + 2x_2 + 5x_3 = 4,$$

$$x_1, x_2, x_3 \ge 0.$$ **(Garhwal 2011)**

Solution: First we shall convert the given L.P.P. into standard primal form.

Here both constraints are equalities, so replacing each by two inequalities, we get the constraints,

$$4x_1 + 3x_2 + x_3 \le 6 \quad \text{and} \quad 4x_1 + 3x_2 + x_3 \ge 6$$

$$x_1 + 2x_2 + 5x_3 \le 4 \quad \text{and} \quad x_1 + 2x_2 + 5x_3 \ge 4.$$

Since the given problem is of maximization, so all the constraints should have the sign $\le$.

The standard primal form of the given L.P.P. is

$$\text{Max. } Z = 2x_1 + 3x_2 + x_3$$

subject to

$$4x_1 + 3x_2 + x_3 \le 6$$

$$-4x_1 - 3x_2 - x_3 \le -6$$

$$x_1 + 2x_2 + 5x_3 \le 4$$

$$-x_1 - 2x_2 - 5x_3 \le -4,$$

$$x_1, x_2, x_3 \ge 0.$$

The matrix form of the above problem is

$$\text{Max. } Z = 2x_1 + 3x_2 + x_3 = (2, 3, 1)[x_1, x_2, x_3] = \mathbf{c} . \mathbf{x}$$

subject to

$$\begin{bmatrix} 4 & 3 & 1 \\ -4 & -3 & -1 \\ 1 & 2 & 5 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \le \begin{bmatrix} 6 \\ -6 \\ 4 \\ -4 \end{bmatrix}$$

or $\quad A\mathbf{x} \le \mathbf{b},$

$$x_1, x_2, x_3 \ge 0.$$

$\therefore$ The dual to the primal is

$$\text{Min. } Z_D = \mathbf{b}' . \mathbf{y} = (6, -6, 4, -4)[y_1', y_1'', y_2', y_2'']$$

$$= 6(y_1' - y_1'') + 4(y_2' - y_2'')$$

subject to $\quad A'\mathbf{y} \ge \mathbf{c}' \quad$ or $\quad \begin{bmatrix} 4 & -4 & 1 & -1 \\ 3 & -3 & 2 & -2 \\ 1 & -1 & 5 & -5 \end{bmatrix} \begin{bmatrix} y_1' \\ y_1'' \\ y_2' \\ y_2'' \end{bmatrix} \ge \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$

or
$$\begin{bmatrix} 4y_1' - 4y_1'' + y_2' - y_2'' \\ 3y_1' - 3y_1'' + 2y_2' - 2y_2'' \\ y_1' - y_1'' + 5y_2' - 5y_2'' \end{bmatrix} \geq \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

or
$$4(y_1' - y_1'') + y_2' - y_2'' \geq 2$$
$$3(y_1' - y_1'') + 2(y_2' - y_2'') \geq 3$$
$$(y_1' - y_1'') + 5(y_2' - y_2'') \geq 1,$$
$$y_1', y_1'', y_2', y_2'' \geq 0.$$

Substituting $y_1 = y_1' - y_1''$, $y_2 = y_2' - y_2''$, the required dual is

$$\text{Min. } Z_D = 6y_1 + 4y_2$$

subject to $4y_1 + y_2 \geq 2,\ 3y_1 + 2y_2 \geq 3,\ y_1 + 5y_2 \geq 1,$

where y_1, y_2 are unrestricted in sign.

Example 5: *Write the dual of the following problem :*

$$\text{Min. } Z = x_1 + x_2 + x_3$$

subject to $x_1 - 3x_2 + 4x_3 = 5,\ x_1 - 2x_2 \leq 3,\ 2x_2 - x_3 \geq 4,$

$x_1, x_2 \geq 0, x_3$ *is unrestricted.* **(Garhwal 2009, 11)**

Solution: First we shall write the given L.P.P. in the standard primal form, substituting $x_3 = x_3' - x_3''$, $x_3' \geq 0$, $x_3'' \geq 0$. The given problem can be written in the standard primal form as

$$\text{Min. } Z = x_1 + x_2 + x_3' - x_3''$$

subject to
$$x_1 - 3x_2 + 4(x_3' - x_3'') \geq 5$$
$$-x_1 + 3x_2 - 4(x_3' - x_3'') \geq -5$$
$$-x_1 + 2x_2 \geq -3$$
$$2x_2 - (x_3' - x_3'') \geq 4,$$
$$x_1, x_2, x_3', x_3'' \geq 0.$$

The matrix form of the above problem is

$$\text{Min. } Z = (1, 1, 1, -1)[x_1, x_2, x_3', x_3''] = \mathbf{c} \,.\, \mathbf{x}$$

subject to

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ -1 & 3 & -4 & 4 \\ -1 & 2 & 0 & 0 \\ 0 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3' \\ x_3'' \end{bmatrix} \geq \begin{bmatrix} 5 \\ -5 \\ -3 \\ 4 \end{bmatrix}$$

or $$A\mathbf{x} \geq \mathbf{b},\ x_1,\ x_2,\ x_3',\ x_3'' \geq 0$$

Now the dual of the given primal is

$$\text{Max. } Z_D = \mathbf{b}'.\mathbf{y} = (5, -5, -3, 4)[y_1', y_1'', y_2, y_3]$$

$$= 5(y_1' - y_1'') - 3y_2 + 4y_3$$

subject to $$A'\mathbf{y} \leq \mathbf{c}'$$

or $$\begin{bmatrix} 1 & -1 & -1 & 0 \\ -3 & 3 & 2 & 2 \\ 4 & -4 & 0 & -1 \\ -4 & 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1' \\ y_1'' \\ y_2 \\ y_3 \end{bmatrix} \leq \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

or $$(y_1' - y_1'') - y_2 \leq 1$$

$$-3(y_1' - y_1'') + 2y_2 + 2y_3 \leq 1$$

$$4(y_1' - y_1'') - y_3 \leq 1$$

$$-4(y_1' - y_1'') + y_3 \leq -1,$$

$$y_1', y_1'', y_2, y_3 \geq 0.$$

Substituting $y_1 = y_1' - y_1''$, the required dual is

$$\text{Max. } Z_D = 5y_1 - 3y_2 + 4y_3$$

subject to $y_1 - y_2 \leq 1, -3y_1 + 2y_2 + 2y_3 \leq 1,\ 4y_1 - y_3 \leq 1, \quad 4y_1 - y_3 \geq 1$

$y_2, y_3 \geq 0$ and y_1 is unrestricted.

Hence the required dual is

$$\text{Max. } Z_D = 5y_1 - 3y_2 + 4y_3$$

subject to $y_1 - y_2 \leq 1,\ -3y_1 + 2y_2 + 2y_3 \leq 1,\ 4y_1 - y_3 = 1,$

$y_2, y_3 \geq 0, y_1$ is unrestricted in sign.

Example 6: *Write the dual of the following problem :*

$$\text{Max. } Z = 3x_1 + 5x_2 + 7x_3$$

subject to $$x_1 + x_2 + 3x_3 \leq 10$$

$$4x_1 - x_2 + 2x_3 \geq 15,$$

$x_1, x_2 \geq 0,\ x_3$ *is unrestricted.*

Solution: First we shall write the given L.P.P. in the standard primal form. Since the given problem is of maximization therefore all the constraints should have the sing $\leq$.

Substituting $x_3 = x_3' - x_3''$, $x_3' \geq 0$, $x_3'' \geq 0$, the standard primal form of the given problem is

$$\text{Max. } Z = 3x_1 + 5x_2 + 7(x_3' - x_3'')$$

subject to

$$x_1 + x_2 + 3x_3' - 3x_3'' \leq 10$$

$$-4x_1 + x_2 - 2x_3' + 2x_3'' \leq -15$$

$$x_1, x_2, x_3', x_3'' \geq 0.$$

The matrix form of the above problem is

$$\text{Max. } Z = (3, 5, 7, -7)[x_1, x_2, x_3', x_3''] = \mathbf{c}.\mathbf{x}$$

subject to

$$\begin{bmatrix} 1 & 1 & 3 & -3 \\ -4 & 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3' \\ x_3'' \end{bmatrix} \leq \begin{bmatrix} 10 \\ -15 \end{bmatrix}$$

or $\quad A\mathbf{x} \leq \mathbf{b}$, $x_1, x_2, x_3', x_3'' \geq 0$.

$\therefore$ The dual of the given problem is

$$\text{Min. } Z_D = \mathbf{b}'\mathbf{y} = (10, -15)[y_1, y_2] = 10y_1 - 15y_2$$

subject to $\quad A'\mathbf{y} \geq \mathbf{c}'$

or $$\begin{bmatrix} 1 & -4 \\ 1 & 1 \\ 3 & -2 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \geq \begin{bmatrix} 3 \\ 5 \\ 7 \\ -7 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} y_1 - 4y_2 \\ y_1 + y_2 \\ 3y_1 - 2y_2 \\ -3y_1 + 2y_2 \end{bmatrix} \geq \begin{bmatrix} 3 \\ 5 \\ 7 \\ -7 \end{bmatrix}$$

or $\quad \text{Min. } Z_D = 10y_1 - 15y_2$

subject to $\quad y_1 - 4y_2 \geq 3, y_1 + y_2 \geq 5, 3y_1 - 2y_2 \geq 7, -3y_1 + 2y_2 \geq -7$

or $\quad y_1 - 4y_2 \geq 3, y_1 + y_2 \geq 5, 3y_1 - 2y_2 \geq 7, 3y_1 - 2y_2 \leq 7$

$\quad y_1, y_2 \geq 0.$

Hence the required dual problem is

$$\text{Min. } Z_D = 10y_1 - 15y_2$$

subject to $\quad y_1 - 4y_2 \geq 3, y_1 + y_2 \geq 5, 3y_1 - 2y_2 = 7, y_1, y_2 \geq 0.$

Comprehensive Exercise 1

1. Describe a method to convert a L.P.P. into its dual.

Find the dual of the following linear programming problems :

2. Max. $Z = x_1 - x_2 + 3x_3$

subject to

$$x_1 + x_2 + x_3 \le 10$$
$$2x_1 - x_3 \le 2$$
$$2x_1 - 2x_2 + 3x_3 \le 6$$
$$x_1, x_2, x_3 \ge 0.$$

3. Max. $Z = 3x_1 + 5x_2 + 4x_3$

subject to

$$2x_1 + 3x_2 \le 8$$
$$2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$
$$x_1, x_2, x_3 \ge 0.$$

4. Max. $Z = 5x_1 + 3x_2$

subject to

$$3x_1 + 5x_2 \le 15$$
$$5x_1 + 2x_2 \le 15$$
$$x_1, x_2 \ge 0.$$

5. Min. $Z = 4x_1 + 6x_2 + 18x_3$

subject to

$$x_1 + 3x_3 \ge 3$$
$$x_2 + 2x_3 \ge 5$$
$$x_1, x_2, x_3 \ge 0.$$

6. Max. $Z = 3x_1 + 4x_2$

subject to

$$2x_1 + 6x_2 \le 16$$
$$5x_1 + 2x_2 \ge 20$$
$$x_1, x_2 \ge 0.$$

7. Min. $Z = 3x_1 + x_2$

subject to

$$2x_1 + 3x_2 \ge 2$$
$$x_1 + x_2 \ge 1$$
$$x_1, x_2 \ge 0.$$

8. Min. $Z = 2x_1 + 2x_2 + 4x_3$

subject to

$$2x_1 + 3x_2 + 5x_3 \ge 2$$
$$3x_1 + x_2 + 7x_3 \le 3$$
$$x_1 + 4x_2 + 6x_3 \le 5$$
$$x_1, x_2, x_3 \ge 0.$$

9. Min. $Z = 7x_1 + 3x_2 + 8x_3$

subject to

$$8x_1 + 2x_2 + x_2 \ge 3$$
$$3x_1 + 6x_2 + 4x_3 \ge 4$$
$$4x_1 + x_2 + 5x_3 \ge 1$$
$$x_1 + 5x_2 + 2x_3 \ge 7$$
$$x_1, x_2, x_3 \ge 0.$$

10. Max. $Z = 2x_1 + 5x_2 + 6x_3$

subject to

$$5x_1 + 6x_2 - x_3 \le 3$$
$$-2x_1 + x_2 + 4x_3 \le 4$$
$$x_1 - 5x_2 + 3x_3 \le 1$$
$$-3x_1 - 3x_2 + 7x_3 \le 6$$
$$x_1, x_2, x_3 \ge 0.$$

11. Min. $Z = 3x_1 - 2x_2 + 4x_3$

subject to

$$3x_1 + 5x_2 + 4x_3 \le 7$$
$$6x_1 + 3x_2 + 2x_3 \le 9$$
$$2x_1 + 7x_2 + 9x_3 \ge 5$$
$$4x_1 - 2x_2 + 7x_3 \ge 1$$
$$x_1, x_2, x_3 \ge 0.$$

12. Max.
$Z = 3x_1 + x_2 + 4x_3 + x_4 + 9x_5$
subject to

$4x_1 - 5x_2 - 9x_3 + x_4 - 2x_5 \leq 6$

$2x_1 + 3x_2 + 4x_3 - 5x_4 + x_5 \leq 9$

$x_1 + x_2 - 5x_3 - 7x_4 + 11x_5 \leq 10,$

$x_1, x_2, x_3, x_4, x_5 \geq 0.$

13. Max. $Z = 4x_1 - 2x_2 + 7x_3$
subject to

$3x_1 + 5x_2 + 4x_3 \geq 9$
$4x_1 + x_2 + 3x_3 \geq 7$
$7x_1 - 3x_2 + x_3 \leq 10$
$x_1 + 5x_2 - 2x_3 \geq 2$
$4x_1 - 2x_2 + 7x_3 \geq 8$
$x_1, x_2, x_3 \geq 0.$

14. Max. $Z = x_1 + 3x_2$
subject to

$3x_1 + 2x_2 \leq 6$
$3x_1 + x_2 = 4$
$x_1, x_2 \geq 0.$

15. Max. $Z = 3x_1 + x_2 + x_3 - x_4$
subject to

$x_1 + 5x_2 + 3x_2 + 4x_4 \leq 5$
$x_1 + x_2 = -1$
$x_3 - x_4 \geq -5$
$x_1, x_2, x_3, x_4 \geq 0$

16. Max. $Z = 3x_1 - 2x_2$
subject to

$x_1 + x_2 \leq 5$
$x_1 \leq 4$
$1 \leq x_2 \leq 6.$

17. Min. $Z = x_1 + x_2 + x_3$
subject to

$x_1 - 3x_2 + 4x_3 = 5$
$x_1 - 2x_2 \leq 3$
$2x_2 - x_3 \geq 4$
$x_1, x_3 \geq 0,$
x_2 is unrestricted.

18. Max. $Z = 3x_1 + x_2 + 2x_3 - x_4$
subject to

$2x_1 - x_2 + 3x_3 + x_4 = 1$
$x_1 + x_2 - x_3 + x_4 = 3$
$x_1, x_2, x_3 \geq 0,$
x_4 is unrestricted.

19. Max. $Z = 4x_1 + 2x_2$
subject to

$x_1 - 2x_2 \geq 2$
$x_1 + 2x_2 = 8$
$x_1 - x_2 \leq 10$
$x_1 \geq 0$, x_2 unrestricted in sign.

Answers 1

2. Min. $Z_D = 10y_1 + 2y_2 + 6y_3$

subject to

$$y_1 + 2y_2 + 2y_3 \geq 1$$
$$-y_1 + 2y_3 \leq 1$$
$$y_1 - y_2 + 3y_3 \geq 3$$
$$y_1, y_2, y_3 \geq 0$$

3. Min. $Z_D = 8y_1 + 10y_2 + 15y_3$

subject to

$$2y_1 + 3y_3 \geq 3$$
$$3y_1 + 2y_2 + 2y_3 \geq 5$$
$$5y_2 + 4y_3 \geq 4$$
$$y_1, y_2, y_3 \geq 0$$

4. Min. $Z_D = 15y_1 + 15y_2$

subject to

$$3y_1 + 5y_2 \geq 5$$
$$5y_1 + 2y_2 \geq 3$$
$$y_1, y_2 \geq 0$$

5. Max. $Z_D = 3y_1 + 5y_2$

subject to

$$y_1 \leq 4,\ y_2 \leq 6$$
$$3y_1 + 2y_2 \leq 18$$
$$y_1, y_2 \geq 0$$

6. Min. $Z_D = 16y_1 - 20y_2$

subject to

$$2y_1 - 5y_2 \geq 3$$
$$6y_1 - 2y_2 \geq 4$$
$$y_1, y_2 \geq 0$$

7. Max. $Z_D = 2y_1 + y_2$

subject to

$$2y_1 + y_2 \leq 3$$
$$3y_1 + y_2 \leq 1$$
$$y_1, y_2 \geq 0.$$

8. Max. $Z_D = 2y_1 - 3y_2 - 5y_3$

subject to

$$2y_1 - 3y_2 - y_3 \leq 2$$
$$3y_1 - y_2 - 4y_3 \leq 2$$
$$5y_1 - 7y_2 - 6y_3 \leq 4$$
$$y_1, y_2, y_3 \geq 0.$$

9. Max. $Z_D = 3y_1 + 4y_2 + y_3 + 7y_4$

subject to

$$8y_1 + 3y_2 + 4y_3 + y_4 \leq 7$$
$$2y_1 + 6y_2 + y_3 + 5y_4 \leq 3$$
$$y_1 + 4y_2 + 5y_3 + 2y_4 \leq 8$$
$$y_1, y_2, y_3, y_4 \geq 0.$$

10. Min.
$Z_D = 3y_1 + 4y_2 + y_3 + 6y_4$

subject to

$$5y_1 - 2y_2 + y_3 - 3y_4 \geq 2$$
$$6y_1 + y_2 - 5y_3 - 3y_4 \geq 5$$
$$-y_1 + 4y_2 + 3y_3 + 7y_4 \geq 6$$
$$y_1, y_2, y_3 y_4 \geq 0$$

11. Max. $Z_D = -7y_1 - 9y_2 + 5y_3 + y_4$

subject to

$$-3y_1 - 6y_2 + 2y_3 + 4y_4 \leq 3$$
$$5y_1 + 3y_2 - 7y_3 + 2y_4 \geq 2$$
$$-4y_1 - 2y_2 + 9y_3 + 7y_4 \leq 4$$
$$y_1, y_2, y_3, y_4 \geq 0$$

12. Min. $Z_D = 6y_1 + 9y_2 + 10y_3$

subject to

$$4y_1 + 2y_2 + y_3 \geq 3$$
$$-5y_1 + 3y_2 + y_3 \geq 1$$
$$-9y_1 + 4y_2 - 5y_3 \geq 4$$
$$y_1 - 5y_2 - 7y_3 \geq 1$$
$$-2y_1 + y_2 + 11y_3 \geq 9$$
$$y_1, y_2, y_3 \geq 0$$

13. Min. $Z_D = -9y_1 - 7y_2 + 10y_3 - 2y_4 - 8y_5$

subject to

$$-3y_1 - 4y_2 + 7y_3 - y_4 - 4y_5 \geq 4$$
$$+5y_1 + y_2 + 3y_3 + 5y_4 - 2y_5 \leq 2$$
$$-4y_1 - 3y_2 + y_3 + 2y_4 - 7y_5 \geq 7$$
$$y_1, y_2, y_3, y_4, y_5 \geq 0$$

14. Min. $Z_D = 6y_1 + 4y_2$

subject to

$$3y_1 + 3y_2 \geq 1$$
$$2y_1 + y_2 \geq 3$$
$y_1 \geq 0,\ y_2$ is unrestricted

15. Min. $Z_D = 5y_1 - y_2 + 5y_3$

$$y_1 + y_2 \geq 3$$
$$5y_1 + y_2 \geq 1$$
$$3y_1 - y_3 \geq 1$$
$$4y_1 + y_3 \geq -1$$
$y_1, y_3 \geq 0,$

y_2 is unrestricted

16. Min.
$Z_D = 5y_1 + 4y_2 - y_3 + 6y_4$

subject to

$$y_1 + y_2 = 3$$
$$-y_1 + y_3 - y_4 \leq 2$$
$$y_1, y_2, y_3, y_4 \geq 0$$

17. Max. $Z_D = 5y_1 - 3y_2 + 4y_3$

subject to

$$y_1 - y_2 \leq 1$$
$$-3y_1 + 2y_2 + 2y_3 = 1$$
$$4y_1 - y_3 \leq 1$$
$y_2, y_3 \geq 0, y_1$ is unrestricted

18. Min. $Z_D = y_1 + 3y_2$

subject to

$$2y_1 + y_2 \geq 3$$
$$-y_1 + y_2 \geq 1$$
$$3y_1 - y_2 \geq 2$$
$$y_1 + y_2 = -1$$
y_1, y_2 are unrestricted

19. Min. $Z_D = -2y_1 + 8y_2 + 10y_3$

subject to

$$-y_1 + y_2 + y_3 \geq 4$$
$$2y_1 + 2y_2 - y_3 = 2$$
$y_1, y_3 \geq 0\ \ y_2$ is unrestricted

4.4 Duality Theorems

We observe that dual of a primal problem is also a linear programming problem therefore we can also construct the dual of a dual problem.

Here we shall prove a number of fundamental theorems to describe the relation between the primal and its dual. The relationship between the primal and dual is extremely useful in the development of mathematical programming.

Theorem 1: *The dual of a dual of the given primal is the primal itself.*

(Garhwal 2009)

Proof: Suppose the given linear programming problem is

Primal Problem

$$\text{Max. } Z_P = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$$

subject to

$$\left.\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \le b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \le b_2 \\ \ldots \qquad \ldots \qquad \ldots \\ \ldots \qquad \ldots \qquad \ldots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \le b_m \\ x_1, x_2, \ldots, x_n \ge 0. \end{array}\right\} \quad \ldots(1)$$

Dual Problem

The dual of the above primal can be written as

$$\text{Min. } Z_D = b_1 w_1 + b_2 w_2 + \ldots + b_m w_m$$

subject to

$$\left.\begin{array}{l} a_{11}w_1 + a_{21}w_2 + \ldots + a_{m1}w_m \ge c_1 \\ a_{12}w_1 + a_{22}w_2 + \ldots + a_{m2}w_m \ge c_2 \\ \ldots \qquad \ldots \qquad \ldots \\ \ldots \qquad \ldots \qquad \ldots \\ a_{1n}w_1 + a_{2n}w_2 + \ldots + a_{mn}w_m \ge c_n \\ w_1, w_2, \ldots, w_m \ge 0. \end{array}\right\} \quad \ldots(2)$$

Now we have to construct the dual of the above dual. First we shall change the above dual to standard maximization form.

$$\text{Max. } (-Z_D) = -b_1 w_1 - b_2 w_2 - \ldots - b_m w_m$$

subject to

$$\left.\begin{array}{l} -a_{11}w_1 - a_{21}w_2 - \ldots - a_{m1}w_m \le -c_1 \\ -a_{12}w_1 - a_{22}w_2 - \ldots - a_{m2}w_m \le -c_2 \\ \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \\ -a_{1n}w_1 - a_{2n}w_2 - \ldots - a_{mn}w_m \le -c_n \\ w_1, w_2, \ldots, w_m \ge 0. \end{array}\right\} \quad \ldots(3)$$

Dual of the dual : Considering the above dual a primal, its dual can be written as

$$\left.\begin{array}{ll} & \text{Min.} Z_y = -c_1 y_1 - c_2 y_2 - \ldots - c_n y_n \\ \text{subject to} & \\ & -a_{11}y_1 - a_{12}y_2 - \ldots - a_{1n}y_n \ge -b_1 \\ & -a_{21}y_1 - a_{22}y_2 - \ldots - a_{2n}y_n \ge -b_2 \\ & \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \\ & -a_{m1}y_1 - a_{m2}y_2 - \ldots - a_{mn}y_n \ge -b_m. \\ & y_1, y_2, \ldots, y_n \ge 0 \end{array}\right\} \quad \ldots(4)$$

Changing the above problem to maximization and multiplying each constraint by -1, we get

$$\text{Max. } Z'_y = c_1 y_1 + c_2 y_2 + \ldots + c_n y_n, (-Z_y = Z'_y \text{ say})$$

$$\left.\begin{array}{ll} \text{subject to} & a_{11}y_1 + a_{12}y_2 + \ldots + a_{1n}y_n \le b_1 \\ & a_{21}y_1 + a_{22}y_2 + \ldots + a_{2n}y_n \le b_2 \\ & \ldots \quad \ldots \quad \ldots \\ & \ldots \quad \ldots \quad \ldots \\ & a_{m1}y_1 + a_{m2}y_2 + \ldots + a_{mn}y_n \le b_m, \\ & y_1, y_2, \ldots, y_n \ge 0 \end{array}\right\} \quad \ldots(5)$$

which is identical to the given linear programming problem (primal problem).

Hence the dual of the dual is primal.

Theorem 2: *If x is any feasible solution to the primal problem Max.* $Z_P = \mathbf{c}\,\mathbf{x}$*, subject to* $A\mathbf{x} \le \mathbf{b}, \mathbf{x} \ge \mathbf{0}$ *and w is any feasible solution to the dual problem.*

Min. $Z_D = \mathbf{b}'\mathbf{w}$ *subject to* $A'\mathbf{w} \ge \mathbf{c}'$, $\mathbf{w} \ge \mathbf{0}$, *then* $\mathbf{c}\,\mathbf{x} \le \mathbf{b}'\,\mathbf{w}$ *i.e.,* $Z_P \le Z_D$.

Proof: Consider the primal problem

$$\left.\begin{array}{ll} & \text{Max. } Z_P = \mathbf{c}\,\mathbf{x} \\ \text{subject to} & A\mathbf{x} \le \mathbf{b} \\ & \mathbf{x} \ge \mathbf{0} \end{array}\right\} \quad \ldots(1)$$

Let $\mathbf{x} = [x_1, x_2, \ldots, x_n]$ be any feasible solution to (1).

The dual of the above primal is

$$\left.\begin{array}{r}\text{Min. } Z_D = \mathbf{b}'\,\mathbf{w} \\ \text{subject to } A'\mathbf{w} \geq \mathbf{c}',\ \mathbf{w} \geq \mathbf{0}.\end{array}\right\} \quad \ldots(2)$$

Let $\mathbf{w} = [w_1, w_2, \ldots, w_m]$ be any feasible solution to the dual (2).

Now $\mathbf{w}$ (m-component column vector) is the feasible solution of the dual and $A\mathbf{x} \leq \mathbf{b}$ are the constraints of the primal (1). If we multiply both sides of $A\mathbf{x} \leq \mathbf{b}$ with $\mathbf{w}'$, the sign of the inequality remains unchanged.

$$\therefore \quad \mathbf{w}'(A\mathbf{x}) \leq \mathbf{w}'\,\mathbf{b}$$

$$\Rightarrow \quad (A'\mathbf{w})'\,\mathbf{x} \leq (\mathbf{b}'\mathbf{w})' \quad \ldots(3)$$

Similarly $\mathbf{x}$ (n-component column vector) is the feasible solution of the primal (1) and $A'\mathbf{w} \geq \mathbf{c}'$ denotes the constraints of the dual (2) so we can write

$$\mathbf{x}'\,(A'\mathbf{w}) \geq \mathbf{x}'\,\mathbf{c}'$$

$$\Rightarrow \quad \mathbf{x}'\,(\mathbf{w}'A)' \geq (\mathbf{c}\,\mathbf{x})'$$

$$\Rightarrow \quad [(\mathbf{w}'A)\,\mathbf{x}]' \geq (\mathbf{c}\,\mathbf{x})'$$

$$\Rightarrow \quad (\mathbf{w}'A)\,\mathbf{x} \geq \mathbf{c}\,\mathbf{x}$$

$$\Rightarrow \quad (A'\mathbf{w})'\,\mathbf{x} \geq \mathbf{c}\,\mathbf{x}. \quad \ldots(4)$$

From relations (3) and (4), we have

$$\mathbf{c}\,\mathbf{x} \leq (A'\mathbf{w})'\,\mathbf{x} \leq (\mathbf{b}'\mathbf{w})'$$

$$\Rightarrow \quad \mathbf{c}\,\mathbf{x} \leq (\mathbf{b}'\,\mathbf{w})' \Rightarrow \mathbf{c}\,\mathbf{x} \leq \mathbf{b}'\mathbf{w}$$

$$\Rightarrow \quad Z_P \leq Z_D.$$

Theorem 3: *If $\hat{\mathbf{x}}$ is a feasible solution to the primal problem Max. $Z_P = \mathbf{c}\,\mathbf{x}$ subject to*

$A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$ *and $\hat{\mathbf{w}}$ is a feasible solution to its dual*

Min. $Z_D = \mathbf{b}'\,\mathbf{w}$ subject to $A'\mathbf{w} \geq \mathbf{c}', \mathbf{w} \geq \mathbf{0}$

such that $\mathbf{c}\,\hat{\mathbf{x}} = b'\,\hat{\mathbf{w}}$, then $\hat{\mathbf{x}}$ is the optimal solution of the primal and $\hat{\mathbf{w}}$ is the optimal solution of the dual problem.

Or

The necessary and sufficient condition for any linear programming problem and its dual to have optimal solution is that both have feasible solutions.

Proof: Consider the primal problem

$$\left.\begin{array}{r}\text{Max.} Z_P = \mathbf{c\,x} \\ \text{subject to } A\mathbf{x} \le \mathbf{b}, \mathbf{x} \ge \mathbf{0}.\end{array}\right\} \quad ...(1)$$

Let $\mathbf{x}$ be any feasible solution to (1).

The dual of the problem (1) is

$$\left.\begin{array}{r}\text{Min.} Z_D = \mathbf{b}'\mathbf{w} \\ \text{subject to } A'\mathbf{w} \ge \mathbf{c}', \mathbf{w} \ge \mathbf{0}\end{array}\right\} \quad ...(2)$$

Let $\hat{\mathbf{w}}$ be any feasible solution to the dual (2).

Proceeding as in theorem (2) above, we have

$$\mathbf{c\,x} \le \mathbf{b}'\hat{\mathbf{w}}$$

$\Rightarrow$ $\mathbf{c\,x} \le \mathbf{c}\,\hat{\mathbf{x}}$, since $\mathbf{c}\,\hat{\mathbf{x}} = \mathbf{b}'\hat{\mathbf{w}}$

$\Rightarrow$ Value of the objective function of the primal problem at the feasible solution $\hat{\mathbf{x}}$ is greater than its value at any other feasible solution $\mathbf{x}$.

Hence $\hat{\mathbf{x}}$ is the optimal solution of the primal problem (for maximization problem).

Again suppose $\mathbf{w}$ is any feasible solution of the dual problem (2). $\hat{\mathbf{x}}$ is the given feasible solution of the primal problem (1),

Proceeding as in theorem (2), we have

$$\mathbf{c}\,\hat{\mathbf{x}} \le \mathbf{b}'\mathbf{w}$$

$\Rightarrow$ $\mathbf{b}'\hat{\mathbf{w}} \le \mathbf{b}'\,\mathbf{w}$ since $\mathbf{c}\,\hat{\mathbf{x}} = \mathbf{b}'\,\hat{\mathbf{w}}$

$\Rightarrow$

The value of the objective function of the dual problem at the given feasible solution $\hat{\mathbf{w}}$ is less than its value at any other feasible solution $\mathbf{w}$.

Hence $\hat{\mathbf{w}}$ is the optimal solution of the dual problem (for minimization problem).

Theorem 4: *(Basic Duality Theorem) : If $\mathbf{x}_0$ is an optimum solution to the primal, then there exists a feasible solution $\mathbf{w}_0$ to the dual such that*

$$\mathbf{c\,x}_0 = \mathbf{b}'\mathbf{w}_0,$$

where $\mathbf{b}'$ is the transpose of b.

Proof: Consider the following primal and dual problems :

Primal Problem

$$\left.\begin{aligned} \text{Max. } Z_P &= \mathbf{c}\,\mathbf{x} \\ \text{subject to } A\mathbf{x} &\le \mathbf{b}, \mathbf{x} \ge \mathbf{0} \end{aligned}\right\} \quad \ldots(1)$$

Dual Problem

$$\left.\begin{aligned} \text{Min. } Z_D &= \mathbf{b}'\,\mathbf{w} \\ \text{subject to } A'\mathbf{w} &\ge \mathbf{c}', \mathbf{w} \ge \mathbf{0} \end{aligned}\right\} \quad \ldots(2)$$

Suppose $\mathbf{x_0}$ is an optimum solution to the primal. Then to solve the primal problem by simplex method, introduce slack variables to each of the constraints.

Now (1) can be written as

$$\begin{aligned} \text{Max. } Z_P &= \mathbf{c}\,\mathbf{x} \\ \text{subject to } A\,\mathbf{x} + I\mathbf{x_s} &= \mathbf{b} \end{aligned}$$

where $\mathbf{x_s} \in R^m$ represents the vector of slack variables and I is the associated

$m \times m$

identity matrix.

Suppose $\mathbf{x_0} = (\mathbf{x}_B, \mathbf{0})$ is an optimum solution to the primal problem where $\mathbf{x}_B$ denotes the optimum basic feasible solution.

We know $\mathbf{x}_B = B^{-1}\,\mathbf{b}$, B is the optimal basis of A.

$\therefore$ The optimal primal objective function is

$$Z = \mathbf{c}\,\mathbf{x_0} = \mathbf{c}_B \mathbf{x}_B,$$

$\mathbf{c}_B$ is the row vector containing the prices of the basic variables.

We know

$$\begin{aligned} Z_j - c_j &= \mathbf{c}_B Y_j - c_j \\ &= \begin{cases} \mathbf{c}_B B^{-1}\alpha_j - c_j, & \forall \alpha_j \in A \\ \mathbf{c}_B B^{-1} e_j - 0, & \forall e_j \in I. \end{cases} \end{aligned}$$

Since $\mathbf{x}_0$ is an optimal solution, so we have

$$Z_j - c_j \ge 0, \forall j$$

$\Rightarrow$ $\quad \mathbf{c}_B B^{-1}\alpha_j \ge c_j, \mathbf{c}_B B^{-1} e_j > 0, \forall j$

$\Rightarrow$ $\quad \mathbf{c}_B B^{-1} A \ge \mathbf{c}$, $\mathbf{c}_B B^{-1} \ge \mathbf{0}$ (in matrix form)

$\Rightarrow$ $\quad A' B^{-1} \mathbf{c}_B{}' \ge \mathbf{c}', B^{-1}\mathbf{c}_B \ge \mathbf{0}$

$\Rightarrow$ $\quad A'\mathbf{w}_0 \ge \mathbf{c}'$; $\mathbf{w}_0 \ge \mathbf{0}$, taking $B^{-1}\mathbf{c}_B = \mathbf{w}_0 \in R^m$

$\Rightarrow$ $\mathbf{w}_0$ is a feasible solution of the dual problem. Also corresponding dual objective function is

$$\mathbf{b}'\mathbf{w}_0 = \mathbf{w}_0'\mathbf{b} = \mathbf{c}_B B^{-1}\mathbf{b} = \mathbf{c}_B \mathbf{x}_B = \mathbf{c}\mathbf{x}_0$$

Hence corresponding to a given optimal solution $\mathbf{x}_0$ of the primal there exists a feasible solution $\mathbf{w}_0$ of the dual such that

$$\mathbf{c}\,\mathbf{x}_0 = \mathbf{b}'\,\mathbf{w}_0$$

Note : Similarly we can prove the following theorem :

If $\mathbf{w}_0$ is an optimum solution to the dual, then there exists a feasible solution $\mathbf{x}_0$ to the primal such that $\mathbf{c}\,\mathbf{x}_0 = \mathbf{b}'\mathbf{w}_0$

Theorem 5: *(Fundamental Duality Theorem)*

(i) If either the primal or the dual problem has a finite optimal solution then the other problem also has a finite optimal solution and the optimal values of the objective function in both the problems are the same.

(ii) If primal (dual) problem has an unbounded optimum solution, the other problem has either no solution at all or an unbounded solution.

(Garhwal 2009)

Proof: (i) Consider the primal and the dual problems

Primal Problem

$$\left.\begin{aligned} \text{Max. } Z_P &= \mathbf{c}\,\mathbf{x} \\ \text{subject to } \; A\mathbf{x} &\le \mathbf{b}, \mathbf{x} \ge \mathbf{0} \end{aligned}\right\} \qquad ...(1)$$

Dual Problem

$$\left.\begin{aligned} \text{Min. } Z_D &= \mathbf{b}'\,\mathbf{w} \\ \text{subject to } \; A'\mathbf{w} &\ge \mathbf{c}' \\ \mathbf{w} &\ge \mathbf{0} \end{aligned}\right\} \qquad ...(2)$$

To prove the theorem we shall construct an optimal solution to the dual from a given optimal solution of the primal.

Let us assume that the primal has a finite optimal feasible solution $\mathbf{x}_B$.

To solve the primal (1) by simplex method, introduce slack variables to each of the constraints.

Now (1) can be written as

$$\left.\begin{aligned} \text{Max. } Z_P &= \mathbf{c}\,\mathbf{x} \\ \text{subject to } \; A\mathbf{x} + I\,\mathbf{x}_s &= \mathbf{b} \\ \mathbf{x} \ge \mathbf{0},\ \mathbf{x}_s &\ge \mathbf{0}, \end{aligned}\right\} \qquad ...(3)$$

where $\mathrm{x}_s \in R^m$ represents the vector of slack variables and I is the associated $m \times m$ identity matrix. B is the basis matrix and suppose $\mathbf{c}_B$ is the m-component row vector containing the prices of the basic variables.

Now $\mathbf{x}_B$ is the optimal solution to the primal so we have

$$c_j - Z_j \le 0, \forall j$$

We know $$Z_j = \mathbf{c}_B Y_j = \mathbf{c}_B B^{-1} \alpha_j$$

$\therefore$ $$c_j - \mathbf{c}_B B^{-1} \alpha_j \le 0, \forall j$$

or $$\mathbf{c}_B B^{-1} \alpha_j \ge c_j \qquad \text{...(4)}$$

$\Rightarrow$ $$\mathbf{c}_B B^{-1} (\alpha_1, \alpha_2, \ldots, \alpha_n) \ge (c_1, c_2, \ldots c_n)$$

$\Rightarrow$ $$\mathbf{c}_B B^{-1} A \ge \mathbf{c} \qquad \text{...(5)}$$

Taking $$(\hat{\mathbf{w}})' = \mathbf{c}_B B^{-1}, \quad \text{where } \hat{\mathbf{w}} = [w_1, w_2, \ldots, w_m].$$

Then from (5), we have

$$(\hat{\mathbf{w}})' A \ge \mathbf{c} \quad \Rightarrow \quad [(\hat{\mathbf{w}})' A]' \ge \mathbf{c}'$$

$\Rightarrow$ $$A'(\hat{\mathbf{w}}) \ge \mathbf{c}'$$

$\Rightarrow$ $\hat{\mathbf{w}}$ is the solution of the dual (2).

Again considering the relation (4) with α_j corresponding to slack variables, we have

$$\mathbf{c}_B \, B^{-1} (e_j) \ge \mathbf{0}, j = 1, 2, \ldots, m; \; c_j = 0,$$

$\Rightarrow$ $$(\hat{\mathbf{w}})' \, e_j \ge \mathbf{0} \Rightarrow (\hat{\mathbf{w}})' \ge \mathbf{0}$$

$\Rightarrow$ $$\hat{\mathbf{w}} \ge \mathbf{0}$$

$\Rightarrow$ $\hat{\mathbf{w}}$ is the feasible solution of the dual (2).

Now it remains to show that $\hat{\mathbf{w}}$ is an optimal solution to the dual (2).

We have

$$\begin{aligned} Z_D &= \mathbf{b}' \hat{\mathbf{w}} = [(\hat{\mathbf{w}})' \mathbf{b}]' = (\hat{\mathbf{w}})' \mathbf{b} \\ &= (\mathbf{c}_B B^{-1}) \mathbf{b} = \mathbf{c}_B \, (B^{-1} \mathbf{b}) \\ &= \mathbf{c}_B \mathbf{x}_B = Z_P. \end{aligned}$$

Since $\hat{\mathbf{w}}$ and $\mathbf{x}_B$ are the feasible solutions of the dual (2) and the primal (1) respectively and

$$Z_D = Z_P$$

Therefore $\hat{\mathbf{w}}$ is the optimal solution of the dual (2). (See the result of theorem 3)

(ii) We shall prove it by contradiction. Suppose, when the primal problem has an unbounded solution, the dual problem has a finite optimal solution.

We have proved in theorem 1 that dual of the dual is the original primal. Also in part (i) of this theorem we have proved that if a primal has a finite optimal solution, the dual also has a finite optimal solution.

Thus if we consider the dual as the primal, then its dual (which is the original primal) must have a finite optimal solution, which contradicts the hypothesis. Hence the dual has no finite optimal solution. If the primal problem has an unbounded solution, either the dual has no solution or an unbounded solution.

Similarly, we can prove that if the dual has an unbounded solution, the primal has no solution or an unbounded solution.

Theorem 6: *If any of the constraints in the primal is a perfect equality, the corresponding dual variable is unrestricted in sign.*

Proof: Suppose in the given primal the k-th constraint is an equality. Writing the primal in the standard primal form, we have

$$\text{Max. } Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$$

subject to

$$\begin{aligned}
&a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \le b_1\\
&a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \le b_2\\
&\ldots \qquad \ldots \qquad \ldots\\
&\ldots \qquad \ldots \qquad \ldots\\
&a_{k1}x_1 + a_{k2}x_2 + \ldots + a_{kn}x_n \le b_k\\
&-a_{k1}x_1 - a_{k2}x_2 - \ldots - a_{kn}x_n \le -b_k\\
&\ldots \qquad \ldots \qquad \ldots\\
&\ldots \qquad \ldots \qquad \ldots\\
&a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \le b_m,\\
&x_1, x_2, \ldots, x_n \ge 0
\end{aligned}$$

The dual of the above primal is

$$\text{Min. } Z_D = b_1w_1 + b_2w_2 + \ldots + b_k(w_k' - w_k'') + \ldots + b_mw_m$$

subject to

$$\begin{aligned}
&a_{11}w_1 + a_{21}w_2 + \ldots + a_{k1}(w_k' - w_k'') + \ldots + a_{m1}w_m \ge c_1\\
&a_{12}w_1 + a_{22}w_2 + \ldots + a_{k2}(w_k' - w_k'') + \ldots + a_{m2}w_m \ge c_2\\
&\ldots \qquad \ldots \qquad \ldots\\
&\ldots \qquad \ldots \qquad \ldots\\
&a_{1n}w_1 + a_{2n}w_2 + \ldots + a_{kn}(w_k' - w_k'') + \ldots + a_{mn}w_m \ge c_n\\
&w_1, w_2, \ldots, w_k', w_k'', \ldots, w_m \ge 0.
\end{aligned}$$

Substituting $w_k = w_k' - w_k''$ in the above dual, we get

$$\text{Min. } Z_D = b_1 w_1 + b_2 w_2 + \ldots + b_k w_k + \ldots + b_m w_m$$

subject to

$$\begin{aligned}
&a_{11} w_1 + a_{21} w_2 + \ldots + a_{k1} w_k + \ldots + a_{m1} w_m \geq c_1 \\
&a_{12} w_1 + a_{22} w_2 + \ldots + a_{k2} w_k + \ldots + a_{m2} w_m \geq c_2 \\
&\ldots \qquad \ldots \qquad \ldots \\
&\ldots \qquad \ldots \qquad \ldots \\
&a_{1n} w_1 + a_{2n} w_2 + \ldots + a_{kn} w_k + \ldots + a_{mn} w_m \geq c_n \\
&w_1, w_2, \ldots, w_{k-1}, w_{k+1}, \ldots, w_m \geq 0,
\end{aligned}$$

w_k is unrestricted in sign because

$$w_k > 0 \text{ if } w_k' > w_k'' \text{ and } w_k < 0 \text{ if } w_k' < w_k''.$$

Theorem 7: *If any variable of the primal is unrestricted in sign, the corresponding constraint in the dual will be a strict equality.*

Proof: Consider the primal in which the k-th variable is unrestricted in sign.

$$\text{Max. } Z = c_1 x_1 + c_2 x_2 + \ldots + c_k x_k + \ldots + c_n x_n$$

subject to

$$\begin{aligned}
&a_{11} x_1 + a_{12} x_2 + \ldots + a_{1k} x_k + \ldots + a_{1n} x_n \leq b_1 \\
&a_{21} x_1 + a_{22} x_2 + \ldots + a_{2k} x_k + \ldots + a_{2n} x_n \leq b_2 \\
&\ldots \qquad \ldots \qquad \ldots \\
&\ldots \qquad \ldots \qquad \ldots \\
&a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mk} x_k + \ldots + a_{mn} x_n \leq b_m \\
&x_1, x_2, \ldots, x_{k-1}, x_{k+1}, \ldots, x_n \geq 0,
\end{aligned}$$

x_k is unrestricted in sign.

Substituting $x_k = x_k' - x_k''$, $x_k' \geq 0$, $x_k'' \geq 0$ in the given primal, it changes to

$$\text{Max. } Z = c_1 x_1 + c_2 x_2 + \ldots + c_k (x_k' - x_k'') + \ldots + c_n x_n$$

subject to

$$\begin{aligned}
&a_{11} x_1 + a_{12} x_2 + \ldots + a_{1k} (x_k' - x_k'') + \ldots + a_{1n} x_n \leq b_1 \\
&a_{21} x_1 + a_{22} x_2 + \ldots + a_{2k} (x_k' - x_k'') + \ldots + a_{2n} x_n \leq b_2 \\
&\ldots \qquad \ldots \qquad \ldots \qquad \ldots \\
&\ldots \qquad \ldots \qquad \ldots \qquad \ldots \\
&a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mk} (x_k' - x_k'') + \ldots + a_{mn} x_n \leq b_m, \\
&x_1, x_2, \ldots, x_{k-1}, x_k', x_k'', x_{k+1}, \ldots, x_n \geq 0
\end{aligned}$$

Writing the dual of the above primal, we have

$$\text{Min. } Z_D = b_1 w_1 + b_2 w_2 + \ldots + b_m w_m,$$

subject to

$$\begin{aligned}
a_{11}w_1 + a_{21}w_2 + \ldots + a_{m1}w_m &\geq c_1\\
a_{12}w_1 + a_{22}w_2 + \ldots + a_{m2}w_m &\geq c_2\\
\ldots \quad \ldots \quad &\ldots\\
\ldots \quad \ldots \quad &\ldots\\
a_{1k}w_1 + a_{2k}w_2 + \ldots + a_{mk}w_m &\geq c_k\\
-a_{1k}w_1 - a_{2k}w_2 - \ldots - a_{mk}w_m &\geq -c_k\\
\ldots \quad \ldots \quad &\ldots\\
\ldots \quad \ldots \quad &\ldots\\
a_{1n}w_1 + a_{2n}w_2 + \ldots + a_{mn}w_m &\geq c_n,\\
w_1, w_2, \ldots, w_m &\geq 0
\end{aligned}$$

The two constraints

$$a_{1k}w_1 + a_{2k}w_2 + \ldots + a_{mk}w_m \geq c_k$$

and $$-a_{1k}w_1 - a_{2k}w_2 - \ldots - a_{mk}w_m \geq -c_k$$

are equivalent to the single equation

$$a_{1k}w_1 + a_{2k}w_2 + \ldots + a_{mk}w_m = c_k.$$

Hence if the k-th variable of the primal is unrestricted, the k-th constraint in the dual is an equality.

4.5 Existence Theorems

1. There exists a bounded (finite) optimum solution to a linear programming problem if an only if there exists a feasible solution to both primal and its dual.
2. If there does not exist any feasible solution to the dual (primal) but there exists at least one to the primal (dual), then there does not exist any finite optimum solution to the primal (dual).
3. If there does not exist any finite optimum solution to the primal (dual) then there does not exist any feasible solution to the dual (primal).

Proof: 1. Suppose Max. $Z = \mathbf{c}\,\mathbf{x}$

subject to $A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$ — Primal problem

and Min. $Z_D = \mathbf{b}'\mathbf{w}$

subject to $A'\mathbf{w} \geq \mathbf{c}', \mathbf{w} \geq \mathbf{0}$ — Dual problem

are the primal and dual problems respectively.

Again suppose there exists an optimum feasible solution to the primal problem. Then by fundamental theorem of duality the dual problem has at least one feasible solution.

Conversely, let us assume that both the primal and the dual possess feasible solution $\hat{\mathbf{x}}$ and $\hat{\mathbf{w}}$ respectively. Then $\mathbf{c}\,\hat{\mathbf{x}}$ and $\mathbf{b}'\,\hat{\mathbf{w}}$ are both finite and $\mathbf{c}\,\hat{\mathbf{x}} \le \mathbf{b}'\,\hat{\mathbf{w}}$ *i.e.,* $\mathbf{b}'\,\hat{\mathbf{w}}$ acts as an upper bound on $\mathbf{c}\,\hat{\mathbf{x}}$ although not necessarily the least upper bound. Hence the primal must have finite optimum solution.

2. Consider the primal problem

$$\text{Max. } Z = \mathbf{c}\,\mathbf{x},$$

subject to $\quad A\mathbf{x} \le \mathbf{b}, \mathbf{x} \ge \mathbf{0}$

The corresponding dual problem is

$$\text{Min, } Z_D = \mathbf{b}'\,\mathbf{w}$$

subject to $\quad A'\mathbf{w} \ge \mathbf{c}', \mathbf{w} \ge \mathbf{0}$

Suppose there does not exist any feasible solution to the dual but there does exist one to the primal, say $\hat{\mathbf{x}}$. Then $\mathbf{c}\,\hat{\mathbf{x}}$ is the value of the primal objective function.

If we suppose that $\hat{\mathbf{x}}$ is an optimum solution to the primal then by fundamental theorem of duality there must exist a feasible solution to the dual which contradicts the hypothesis. Therefore no feasible solution to the primal can be optimal.

Similarly, we can start with the primal.

3. Suppose the primal problem does not possess any finite optimum solution. Without loss of generality, if can be assumed that there exists a feasible solution to the primal. Now if we assume that the dual problem possesses a feasible solution then by existence theorem 1 (above) the primal problem must possess a finite optimum solution. This contradicts the hypothesis, hence the dual problem does not possess a feasible solution.

4.6 Complementary Slackness Theorems

For the optimal feasible solutions of the primal and the dual systems

1. *If the inequality occurs in the i-th relation of either system (the corresponding slack or surplus variable is positive), then the i-th variable of its dual is zero.*
2. *If the j-th variable is positive in either system, the j-th relation of its dual holds as a strict equality (i.e., the corresponding slack or surplus variable* $w_{m+j} = 0$*).*

Proof: The objective function of the primal and dual problems in explicit form can be written as

$$\text{Max. } Z_P = c_1x_1 + c_2x_2 + \ldots + c_nx_n \text{ (primal)} \qquad \ldots(1)$$

$$\text{Min. } Z_D = b_1w_1 + b_2w_2 + \ldots + b_mw_m \text{ (dual)} \qquad \ldots(2)$$

After introducing the non-negative slack variables $x_{n+1}, x_{n+2}, \ldots, x_{n+m}$ the primal constraint equations can be written as

$$\left.\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n + x_{n+1} = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n + x_{n+2} = b_2 \\ \ldots \quad \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \quad \ldots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n + x_{n+m} = b_m \\ x_1, x_2, \ldots, x_{n+m} \geq 0. \end{array}\right\} \quad \ldots(3)$$

Again introducing the non-negative surplus variables $w_{m+1}, w_{m+2}, \ldots, w_{m+n}$ the dual constraint equations can be written as

$$\left.\begin{array}{l} a_{11}w_1 + a_{21}w_2 + \ldots + a_{m1}w_m - w_{m+1} = c_1 \\ a_{12}w_1 + a_{22}w_2 + \ldots + a_{m2}w_m - w_{m+2} = c_2 \\ \ldots \quad \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \quad \ldots \\ a_{1n}w_1 + a_{2n}w_2 + \ldots + a_{mn}w_m - w_{m+n} = c_n \\ w_1, w_2, \ldots, w_m, w_{m+1}, \ldots, w_{m+n} \geq 0. \end{array}\right\} \quad \ldots(4)$$

Now, multiplying the equations (3) by $w_1, w_2, \ldots, w_m$ respectively and then adding the resulting equations, we have

$$x_1 \sum_{i=1}^{m} a_{i1} w_i + x_2 \sum_{i=1}^{m} a_{i2} w_i + \ldots + x_n \sum_{i=1}^{m} a_{in} w_i$$

$$+ x_{n+1} w_1 + x_{n+2} w_2 + \ldots + x_{n+m} w_m = \sum_{i=1}^{m} b_i w_i \quad \ldots(5)$$

Subtracting (5) from (1), we have

$$\left(c_1 - \sum_{i=1}^{m} a_{i1} w_i\right)x_1 + \left(c_2 - \sum_{i=1}^{m} a_{i2} w_i\right)x_2 + \ldots + \left(c_n - \sum_{i=1}^{m} a_{in} w_i\right)x_n$$

$$-w_1 x_{n+1} - w_2 x_{n+2} - \ldots - w_m x_{n+m} = \left(Z_P - \sum_{i=1}^{m} b_i w_i\right)$$

$$= Z_P - Z_D \quad \ldots(6)$$

From (4), we have

$$-w_{m+j} = c_j - (a_{1j}w_1 + a_{2j}w_2 + \ldots + a_{mj}w_m), j = 1, 2, \ldots, n$$

or $$-w_{m+j} = c_j - \sum_{i=1}^{m} a_{ij} w_i \text{ for all } j = 1, 2, \ldots, n. \quad \ldots(7)$$

Using (7) in (6), we have

$$-(w_{m+1}\,x_1 + w_{m+2}\,x_2 + \ldots + w_{m+n}\,x_n)$$

$$-(w_1\,x_{n+1} + w_2\,x_{n+2} + \ldots + w_m\,x_{n+m}) = Z_P - Z_D \qquad \ldots(8)$$

Now if $\mathbf{x}^* = [x_1{}^*, x_2{}^*, \ldots, x_n{}^*]$ and $\mathbf{w}^* = [w_1{}^*, w_2{}^*, \ldots, w_m{}^*]$ be the optimal solutions to the primal and the dual problems respectively, then by Duality theorem

$$Z_P{}^* = Z_D{}^*$$

Thus for the optimal solutions $\mathbf{x}^*$ and $\mathbf{w}^*$ of the primal and dual problems the corresponding slack and surplus variables are

$$x^*_{n+i} \geq 0,\ i = 1, 2, \ldots, m \text{ and } w^*_{m+j} \geq 0,\ j = 1, 2, \ldots, n.$$

From equation (8), we have

$$(w^*_{m+1}\,x_1{}^* + w^*_{m+2}\,x_2{}^* + \ldots + w^*_{m+n}\,x_n{}^*)$$

$$+(w^*_1\,x^*_{n+1} + w^*_2\,x^*_{n+2} + \ldots + w^*_m\,x^*_{n+m}) = 0 \qquad \ldots(9)$$

Since all the variables in (9) are non-negative so their product terms in (9) are also non-negative. For the validity of relation (9) each term must be individually equal to zero

i.e., $\quad w^*_{m+j}\,x^*_j = 0 \qquad (\forall\, j = 1, 2, \ldots, n) \qquad \ldots(10)$

and $\quad w^*_i\,x^*_{n+i} = 0 \qquad (\forall\, i = 1, 2, \ldots, m) \qquad \ldots(11)$

1. From (11), if $x^*_{n+i} > 0$ then we must have $w^*_i = 0$, *i.e.,* if the slack variable in the *i*-th relation of the primal is positive then *i*-th variable of the dual is zero.
   Again from (10), if $w^*_{m+j} > 0$ then we must have $x^*_j = 0$, *i.e.,* if the surplus variable in the *i*-th relation of the dual is positive then *j*-th variable of the primal (dual of the dual) is zero.
   This proves the part (1) of the theorem.

2. Also from (10), if $x^*_j > 0$, then we must have $w^*_{m+j} = 0$ *i.e.,* if the *j*-th variable in the primal is positive then the *j*-th relation in the dual is strictly an equality (as $w^*_{m+j} = 0$).

   Again from (11), if $w^*_i > 0$, then $x^*_{n+i} = 0$

   i.e., if the *i*-th variable in the dual is positive then the *j*-th relation in the primal (dual of the dual) is strictly an equality (as $x^*_{n+i} = 0$).

   This proves the part (2) of the theorem.

Alternative Statement : A necessary and sufficient condition for any pair of feasible solutions to the primal and dual to be optimal is that

$w_i x_{n+i} = 0,\ i = 1, 2, \ldots, m$ where x_{n+i} is the slack variable in the primal

and $x_j\ w_{m+j} = 0,\ j = 1, 2, \ldots, n$ where w_{m+j} is the surplus variable for the dual.

Primal and Dual Correspondence : With the help of the duality theorems developed in articles 4.5 and 4.6 we note the following correspondence between the primal and the dual problems :

S. No.	Primal	Dual
1.	Objective function Max. Z_P.	Objective function Min. Z_D.
2.	Requirement vector.	Price vector.
3.	Coefficient matrix A.	Transpose of the coefficient matrix, A' or A^T.
4.	Constraints with $\leq$ sign.	Constraints with $\geq$ sign.
5.	Relation.	Variable.
6.	i-th inequality.	i-th variable $w_i \geq 0$
7.	i-th constraint an equality.	i-th variable w_i unrestricted in sign.
8.	Variable.	Relation.
9.	i-th variable $x_i > 0$.	i-th relation a strict inequality.
10.	i-th variable x_i unrestricted in sign.	i-th constraint a strict equality.
11.	i-th slack variable positive.	i-th variable zero.
12.	i-th variable zero.	i-th surplus variable positive.
13.	Finite optimal solution.	Finite optimal solution with equal optimal value of objective function.
14.	Unbounded solution.	No solution or an unbounded solution.
15.	No finite optimal solution.	No feasible solution.

4.7 Rules for Obtaining the Solution to the Dual From the Final Simplex Table of the Primal and Vice-versa

From the final simplex table of the primal problem we can also read the optimal solution of the dual and vice-versa. For this we observe the following rules :

1. The optimal value of the primal objective function is equal to the optimal value of the dual objective function.

i.e., Max. Z_P = Min. Z_D.

This has been proved in Duality Theorem.

2. The Δ_j's ($\Delta_j = c_j - Z_j$) with sign changed for the slack (or surplus) variables in the final simplex table for the primal are the values of the corresponding optimal dual variables in the final simplex table for the dual.

   Proof: In the final simplex table (for primal) corresponding to slack and surplus variables

$$-\Delta_j = -(c_j - Z_j) = -(0 - Z_j)$$
$$= \mathbf{c}_B Y_j = \mathbf{c}_B B^{-1} \alpha_j$$

   where B is the optimal basis and $\mathbf{c}_B$, the corresponding price vector.

   But, for i-th slack variable, α_j becomes a unit vector e_i with 1 at the i-th place.

$$\therefore \qquad -\Delta_j = \mathbf{c}_B B^{-1} e_i = (w_1, w_2, \ldots, w_m) e_i = w_i$$

   where w_i is i-th dual variable.

3. If either problems has unbounded solution, then the other will have no feasible solutions.

   Note : It is always advantageous to apply the simplex method to the problem having lesser number of constraints. Therefore, first we shall solve the problem (primal or dual) with lesser number of constraints by simplex method and then read the solution of the other from the final simplex table according to the rules described above.

4.8 Some Useful Aspects of Duality

In some cases it is easier to solve linear programming problems through their duals. If the number of original variables in the primal problem is considerably less than the number of slack and surplus variables, we must opt to solve the problem through its dual.

Duality plays an important role not only in linear programming but also in physics, economics, engineering etc.

In physics, we use it in parallel circuit and series circuit theory.

In economics it is used in the formulation of input and output systems.

In game theory, we use it to find the optimal strategies of the player B when he minimizes his losses. Then, using duality, we can change the player A's problem into player B's problem and vice-versa.

Illustrative Examples

Example 7: *Write the dual of the following problem and solve it :*

$$\text{Max. } Z = 4x_1 + 2x_2$$

subject to $-x_1 - x_2 \le -3$

$-x_1 + x_2 \le -2,\ x_1,\ x_2 \ge 0.$

Hence or otherwise write down the solution of the primal.

Solution: The given problem is in standard primal form.

$\therefore$ The dual to the given primal is

$$\text{Min. } Z_D = -3w_1 - 2w_2$$

subject to $-w_1 - w_2 \ge 4$

$-w_1 + w_2 \ge 2$

$w_1,\ w_2 \ge 0.$

Changing the objective function to maximization and introducing surplus variables $w_3 \ge 0$, $w_4 \ge 0$ and artificial variables $w_{a_1}, w_{a_2} \ge 0$ the above dual problem reduces to

$$\text{Max. } Z_D' = 3w_1 + 2w_2 + 0w_3 + 0w_4 - Mw_{a_1} - Mw_{a_2}$$

subject to $-w_1 - w_2 - w_3 + w_{a_1} = 4$

$-w_1 + w_2 - w_4 + w_{a_2} = 2$

Taking $w_1 = 0 = w_2 = w_3 = w_4$, we get $w_5 = 4$, $w_6 = 2$ which is the initial B.F.S. of dual problem.

Now applying the simplex method to obtain the optimal solution, we have

B	c_B	c_j	3	2	0	0	$-M$	$-M$	**Min. ratio**
		w_B	W_1	W_2	W_3	W_4	A_1	A_2	w_B/W_2, $w_{i2} > 0$
A_1	$-M$	4	-1	-1	-1	0	1	0	—
A_2	$-M$	2	-1	$\boxed{1}$	0	-1	0	1	2 (min) $\rightarrow$
$Z_D' = -6M$		Δ_j	$3-2M$	2 $\uparrow$	$-M$	$-M$	0	0 $\downarrow$	
A_1	$-M$	6	-2	0	-1	-1	1	1	
W_2	2	2	-1	1	0	-1	0	1	
$Z_D' = 4-6M$		Δ_j	$5-2M$	0	$-M$	$2-M$	0	-2	

In the last simplex table no $\Delta_j > 0$ and a non-zero artificial variable appears in the basis therefore the dual problem does not possess any optimum basic feasible solution. Consequently, the given problem does not possess any feasible solution.

Example 8: *Write the dual of the following linear programming problem and hence solve it :*

$$Max.\ Z = 3x_1 - 2x_2$$

subject to

$$x_1 \le 4$$
$$x_2 \le 6$$
$$x_1 + x_2 \le 5$$
$$-x_2 \le -1$$
$$x_1,\ x_2 \ge 0.$$

Solution: The given problem is in standard primal form. The dual of the given primal is

$$\text{Min. } Z_D = 4w_1 + 6w_2 + 5w_3 - w_4$$

subject to

$$w_1 + w_3 \ge 3$$
$$w_2 + w_3 - w_4 \ge -2,$$
$$w_1,\ w_2,\ w_3,\ w_4 \ge 0.$$

Changing the dual problem to maximization multiplying both sides of second constraint by –1 to make R.H.S. positive, introducing surplus variable w_5 and slack variable w_6 to change the inequalities into equations, the dual problem becomes

$$\text{Max. } Z_D' = -4w_1 - 6w_2 - 5w_3 + w_4 + 0w_5 + 0w_6$$

subject to

$$w_1 + w_3 - w_5 = 3$$
$$-w_2 - w_3 + w_4 + w_6 = 2,$$
$$w_1, w_2, \ldots, w_6 \ge 0.$$

Here we have not introduced the artificial variable in the first constraint equation as w_1 will serve the purpose.

Taking $w_2 = 0, w_3 = 0, w_4 = 0, w_5 = 0$, we get $w_1 = 3, w_6 = 2$ which is the starting B.F.S.

The solution by simplex method is given in the following table :

B	c_B	c_j / w_B	−4 W_1	−6 W_2	−5 W_3	1 W_4	0 W_5	0 W_6	Min. ratio w_B/W_4, $w_{i4} > 0$
W_1	−4	3	1	0	1	0	−1	0	—
W_6	0	2	0	−1	−1	$\boxed{1}$	0	1	2 (min) →
$Z_D' = -12$		Δ_j	0	−6	−1	1 ↑	−4	0 ↓	
W_1	−4	3	1	0	1	0	−1	0	
W_4	1	2	0	−1	−1	1	0	1	
$Z_D' = -10$		Δ_j	0	−5	0	0	−4	−1	

Since all $\Delta_j \le 0$, therefore the last table gives the optimal solution of the dual.

The optimal solution of the dual is

$$w_1 = 3,\ w_2 = 0,\ w_3 = 0,\ w_4 = 2,$$

$$\text{Min. } Z_D = -\text{Max.} Z_D' = 10.$$

To read the solution of the primal from the final simplex table of the dual.

The optimal solution of the primal problem is

$$x_1 = -\Delta_5 = 4,\ x_2 = -\Delta_6 = 1 \text{ and Max. } Z = \text{Min.} Z_D = 10.$$

Example 9: *Use duality to solve*

$$\textit{Min. } Z = 3x_1 + x_2$$

subject to $\quad x_1 + x_2 \ge 1$

$$2x_1 + 3x_2 \ge 2,\ x_1,\ x_2 \ge 0.$$

Solution: The given L.P.P. is in the standard primal form. The dual problem is given by

$$\text{Max. } Z_D = w_1 + 2w_2$$

subject to $\quad w_1 + 2w_2 \le 3$

$$w_1 + 3w_2 \le 1,\ w_1,\ w_2 \ge 0.$$

Introducing slack variables w_3 and w_4 to change the constraint inequalities into equations, the dual problem becomes

$$\text{Max. } Z_D = w_1 + 2w_2 + 0w_2 + 0w_4$$

subject to $\quad w_1 + 2w_2 + w_3 = 3$

$$w_1 + 3w_2 \quad + w_4 = 1,$$

$$w_1, w_2, w_3, w_4 \geq 0.$$

Taking $w_1 = 0$, $w_2 = 0$, we get $w_3 = 3$, $w_4 = 1$, which is initial B.F.S.

The solution by simplex method is given in the following table :

B	$\mathbf{c}_B$	c_j	**1**	**2**	**0**	**0**	**Min ratio** $\mathbf{w}_B/W_2$
		$\mathbf{w}_B$	W_1	W_2	W_3	W_4	$w_{i2} > 0$
W_3	0	3	1	2	1	0	3/2
W_4	0	1	1	$\boxed{3}$	0	1	1/3 → (min)
$Z_D = 0$		Δ_j	1	2 ↑	0	0 ↓	$\mathbf{w}_B/W_1$
W_3	0	7/3	1/3	0	1	−2/3	7
W_2	2	1/3	$\boxed{1/3}$	1	0	1/3	1 → (min)
$Z_D = 2/3$		Δ_j	1/3 ↑	0 ↓	0	−2/3	
W_3	0	2	0	−1	1	−1	
W_1	1	1	1	3	0	1	
$Z_D = 1$		Δ_j	0	−1	0	−1	

In last table, since all $\Delta_j \leq 0$, therefore the dual problem has optimal solution

$$w_1 = 1, w_2 = 0, \text{ Max. } Z_D = 1$$

Now the solution of the original primal problem from the last simplex table of the dual is

$$x_1 = -\Delta_3 = 0, x_2 = -\Delta_4 = 1, \min Z = \max. Z_D = 1.$$

Example 10: *Apply simplex method to solve the following problem :*

$$\textit{Max. } Z = 30x_1 + 23x_2 + 29x_3$$

subject to

$$6x_1 + 5x_2 + 3x_3 \leq 26$$

$$4x_1 + 2x_2 + 5x_3 \leq 7$$

$$x_1, x_2, x_3 \geq 0.$$

Hence or otherwise find the solution to the dual of the above problem.

Solution: The given problem is of maximization. Introducing slack variables x_4, x_5 to change the constraint inequalities into equations, we get

$$\text{Max. } Z = 30x_1 + 23x_2 + 29x_3 + 0x_4 + 0x_5,$$

subject to $\quad 6x_1 + 5x_2 + 3x_3 + x_4 \quad = 26$

$4x_1 + 2x_2 + 5x_3 \quad + x_5 = 7,$

$x_1, x_2, \ldots, x_5 \geq 0.$

Taking $x_1 = 0,\ x_2 = 0,\ x_3 = 0$, we get $x_4 = 26,\ x_5 = 7$, which is the starting B.F.S.

The solution of the problem by simplex method is given in the following table :

B	$\mathbf{c}_B$	c_j	**30**	**23**	**29**	**0**	**0**	**Min. ratio** $\mathbf{x}_B/Y_1$
		$\mathbf{x}_B$	Y_1	Y_2	Y_3	Y_4	Y_5	$y_{i1} > 0$
Y_4	0	26	6	5	3	1	0	26/6
Y_5	0	7	$\boxed{4}$	2	5	0	1	$7/4 \rightarrow$ (min)
$Z = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	30 $\uparrow$	23	29	0	0 $\downarrow$	$\mathbf{x}_B/Y_2$ $y_{i2} > 0$
Y_4	0	31/2	0	2	$-9/2$	1	$-3/2$	31/4
Y_1	30	7/4	1	$\boxed{1/2}$	5/4	0	1/4	$7/2 \rightarrow$ (min)
$Z = 105/2$		Δ_j	0 $\downarrow$	8 $\uparrow$	$-17/2$	0	$-15/2$	
Y_4	0	17/2	-4	0	$-19/2$	1	$-5/2$	
Y_2	23	7/2	2	1	5/2	0	1/2	
$Z = 161/2$		Δ_j	-16	0	$-57/2$	0	$-23/2$	

In the last simplex table all $\Delta_j \leq 0$ therefore the optimal solution is

$$x_1 = 0,\ x_2 = 7/2,\ x_3 = 0 \text{ and Max. } Z = 161/2.$$

Dual Problem

The given problem is in standard primal form. The dual of the given problem is

$$\text{Min. } Z_D = 26w_1 + 7w_2$$

subject to $\quad 6w_1 + 4w_2 \geq 30$

$5w_1 + 2w_2 \geq 23$

$3w_1 + 5w_2 \geq 29,$

$w_1, w_2 \geq 0.$

To real the solution of the dual from the final simplex table of the primal problem.

$$w_1 = -\Delta_4 = 0,\ w_2 = -\Delta_5 = 23/2$$

and $\qquad$ Min. Z_D = Max.$Z = 161/2$

Example 11: *Write the dual of the following problem :*

$$Max.\ Z = 5x_1 - 2x_2 + 3x_3$$

subject to $\quad 2x_1 + 2x_2 - x_3 \geq 2$

$$3x_1 - 4x_2 \leq 3$$

$$x_2 + 3x_3 \leq 5, \quad and \quad x_1, x_2, x_3 \geq 0$$

Hence or otherwise write the solution of the primal.

Solution: Writing the given L.P.P. into standard primal form,

we get

$$\text{Max. } Z = 5x_1 - 2x_2 + 3x_3$$

subject to

$$-2x_1 - 2x_2 + x_3 \leq -2$$

$$3x_1 - 4x_2 \leq 3$$

$$x_2 + 3x_3 \leq 5, \quad \text{and} \quad x_1, x_2, x_3 \geq 0$$

The dual of the given primal is

$$\text{Min. } Z_D = -2w_1 + 3w_2 + 5w_3$$

subject to

$$-2w_1 + 3w_2 \geq 5$$

$$-2w_1 - 4w_2 + w_3 \geq -2$$

$$w_1 + 3w_3 \geq 3, \quad \text{and} \quad w_1, w_2, w_3 \geq 0.$$

Now to solve the dual problem by simplex method, converting it to maximization problem and multiplying both sides of the second constraint by -1, we get

$$\text{Max. } Z'_D = 2w_1 - 3w_2 - 5w_3$$

subject to $\quad -2w_1 + 3w_2 + 0w_3 \geq 5$

$$2w_1 + 4w_2 - w_3 \leq 2$$

$$w_1 + 0w_2 + 3w_3 \geq 3,$$

$$w_1, w_2, w_3 \geq 0.$$

Introducing surplus variables w_4, w_6, slack variable w_5 and artificial variables w_{a_1}, w_{a_2} the above dual problem changes to

$$\text{Max. } Z'_D = 2w_1 - 3w_2 - 5w_3 + 0w_4 + 0w_5 + 0w_6 - Mw_{a_1} - Mw_{a_2}$$

subject to

$$-2w_1 + 3w_2 + 0w_3 - w_4 + w_{a_1} = 5$$

$$2w_1 + 4w_2 - w_3 + 0w_4 + w_5 = 2$$

$$w_1 + 0w_2 + 3w_3 - w_6 + w_{a_2} = 3$$

$$w_1, w_2, \ldots, w_6, w_{a_1}, w_{a_2} \geq 0.$$

Taking $w_1 = 0 = w_2 = w_3 = w_4 = w_6$, we get $w_5 = 2$, $w_{a_1} = 5$, $w_{a_2} = 3$ which is the starting B.F.S.

The solution of this dual problem by simplex method is given in the following table :

B	c_B	c_j / w_B	2 / W_1	−3 / W_2	−5 / W_3	0 / W_4	0 / W_5	0 / W_6	−M / A_1	−M / A_2	Min. ratio
											w_B/W_2, $w_{i2} > 0$
A_1	$-M$	5	−2	3	0	−1	0	0	1	0	5/3
W_5	0	2	2	$\boxed{4}$	−1	0	1	0	0	0	2/4 (min) →
A_2	$-M$	3	1	0	3	0	0	−1	0	1	—
$Z'_D = -8M$		Δ_j	$2-M$	$-3+3M$ ↑	$-5+3M$	$-M$	0 ↓	$-M$	0	0	w_B/W_3, $w_{i3} > 0$
A_1	$-M$	7/2	−7/2	0	3/4	−1	−3/4	0	1	0	14/3
W_2	−3	1/2	1/2	1	−1/4	0	1/4	0	0	0	Neg.
A_2	$-M$	3	1	0	$\boxed{3}$	0	0	−1	0	1	1 (min) →
$Z'_D = -\frac{13}{2}M - \frac{3}{2}$		Δ_j	$\frac{7-5M}{2}$	0	$\frac{15M-23}{4}$ ↑	$-M$	$\frac{3-3M}{4}$	$-M$	0	0 ↓	w_B/W_6, $w_{i6} > 0$
A_1	$-M$	11/4	−15/4	0	0	−1	−3/4	$\boxed{1/4}$	1		11 (min) →
W_2	−3	3/4	7/12	1	0	0	1/4	−1/12	0		—
W_3	−5	1	1/3	0	1	0	0	−1/3	0		—
$Z'_D = -\frac{11}{4}M - \frac{29}{4}$		Δ_j	$\frac{65-45M}{12}$	0	0	$-M$	$\frac{3-3M}{4}$	$\frac{3M-23}{12}$ ↑	0 ↓		
W_6	0	11	−15	0	0	−4	−3	1			
W_2	−3	5/3	−2/3	1	0	−1/3	0	0			
W_3	−5	14/3	−14/3	0	1	−4/3	−1	0			
$Z'_D = -\frac{85}{3}$		Δ_j	−70/3	0	0	−23/3	−5	0			

In the last simplex table all $\Delta_j \leq 0$ therefore it gives optimal solution of the dual problem.

The optimal solution of the dual is

$$w_1 = 0,\ w_2 = 5/3,\ w_3 = 14/3$$

and $\quad$ Min. $Z_D = -\text{Max. } Z'_D = 85/3$

Now the optimal solution of the primal is

$$x_1 = -\Delta_4 = 23/3,\ x_2 = -\Delta_5 = 5,\ x_3 = -\Delta_6 = 0$$

and $\quad$ Max. $Z = \text{Min.} Z_D = 85/3.$

Example 12: *One unit of product A contributes ₹ 7 and requires 3 units of raw material and 2 hours of labour. One unit of product B contributes ₹ 5 and requires one unit of raw material and one hour of labour. Availability of the raw material at present is 48 units and there are 40 hours of labour.*

(i) Formulate the linear programming problem.

(ii) Write the dual and solve it by simplex method. Also find the optimal product mix.

Solution: The linear programming problem corresponding to the given information is

$$\text{Max. } Z = 7x_1 + 5x_2$$

subject to
$$3x_1 + x_2 \le 48$$
$$2x_1 + x_2 \le 40$$
$$x_1, x_2 \ge 0$$

The given L.P.P. is in standard primal form.

The **dual problem** can be written as

$$\text{Min. } Z_D = 48w_1 + 40w_2$$

subject to
$$3w_1 + 2w_2 \ge 7$$
$$w_1 + w_2 \ge 5,$$
$$w_1, w_2 \ge 0$$

To solve the dual problem by simplex method, changing the dual objective function to maximization and introducing surplus variables w_3, w_4 and artificial variables w_{a_1}, w_{a_2} respectively to change the constraint inequalities into equations. The dual problem can be written as

$$\text{Max. } Z'_D = -48w_1 - 40w_2 + 0w_3 + 0w_4 - Mw_{a_1} - Mw_{a_2}$$

subject to
$$3w_1 + 2w_2 - w_3 + w_{a_1} = 7$$
$$w_1 + w_2 - w_4 + w_{a_2} = 5$$
$$w_1, w_2, w_3, w_4, w_{a_1}, w_{a_2} \ge 0$$

Taking $w_1 = 0 = w_2 = w_3 = w_4$, we get $w_{a_1} = 7$, $w_{a_2} = 5$ which is the starting B.F.S.

The solution by simplex method is given in the following table :

B	c_B	c_j / w_B	-48 / W_1	-40 / W_2	0 / W_3	0 / W_4	$-M$ / A_1	$-M$ / A_2	Min. ratio w_B/W_1, $w_{i1} > 0$
A_1	$-M$	7	3	2	-1	0	1	0	$7/3 \rightarrow$ (min)
A_2	$-M$	5	$\boxed{1}$	1	0	-1	0	1	5
$Z'_D = -12M$		Δ_j	$4M-48$ ↑	$3M-40$	$-M$	$-M$	0 ↓	0	w_B/W_2, $w_{i2} > 0$
W_1	-48	$7/3$	1	$\boxed{2/3}$	$-1/3$	0	$1/3$	0	$7/2 \rightarrow$ (min)
A_2	$-M$	$8/3$	0	$1/3$	$1/3$	-1	$-1/3$	1	8
$Z'_D = -(112 + \frac{8}{3}M)$		Δ_j	0 ↓	$\frac{M}{3} - 8$ ↑	-16	$-M$	$16 - \frac{4M}{3}$	0	w_B/W_3, $w_{i3} > 0$
W_2	-40	$7/2$	$3/2$	1	$-1/2$	0	$-1/2$	0	—
A_2	$-M$	$3/2$	$-1/2$	0	$\boxed{1/2}$	-1	$-1/2$	1	3 (min) $\rightarrow$
$Z'_D = -(140 + \frac{3}{2}M)$		Δ_j	$12 - \frac{1}{2}M$	0	$\frac{1}{2}M - 20$ ↑	$-M$	$20 - \frac{3}{2}M$	0 ↓	
W_2	-40	5	1	1	0	-1	0	1	
W_3	0	3	-1	0	1	-2	-1	2	
$Z'_D = -200$		Δ_j	-8	0	0	-40	$-M$	$40 - M$	

In the last table all $\Delta_j \leq 0$, therefore this solution is optimal.

$\therefore$ Solution of the dual problem is

$$w_1 = 0, w_2 = 5, \text{ Min. } Z_D = -\text{Max. } Z'_D = 200.$$

Hence the solution of the primal problem is

$$x_1 = -\Delta_3 = 0, x_2 = -\Delta_4 = 40 \text{ and Max. } Z = 200.$$

Thus the optimal product mix is :

None of the product A and 40 units of product B for a total contribution of ₹ 200.

Comprehensive Exercise 2

1. Find the solution of the following problem by simplex method :

$$\text{Max. } Z_x = 40x_1 + 50x_2$$

subject to $2x_1 + 3x_2 \le 3,\ 8x_1 + 4x_2 \le 5,\ x_1, x_2 \ge 0.$
Write the dual of the given L.P.P. Also find the solution of the dual problem.

Use principle of duality to solve the following problems :

2. $$\text{Min. } Z = x_1 - x_2$$

subject to $2x_1 + x_2 \ge 2,\ -x_1 - x_2 \ge 1$ and $x_1, x_2 \ge 0.$

3. $$\text{Max. } Z = 3x_1 + 2x_2$$

subject to $2x_1 + x_2 \le 5,\ x_1 + x_2 \le 3$ and $x_1, x_2 \ge 0.$

4. $$\text{Max. } Z = 2x_1 + x_2$$

subject to $x_1 + 2x_2 \le 10,\ x_1 + x_2 \le 6$

$x_1 - x_2 \le 2,\ x_1 - 2x_2 \le 1$ and $x_1, x_2 \ge 0.$

5. $$\text{Min. } Z = 2x_1 + 2x_2$$

subject to $2x_1 + 4x_2 \ge 1,\ x_1 + 2x_2 \ge 1.$

$2x_1 + x_2 \ge 1,$ and $x_1, x_2 \ge 0.$

6. $$\text{Max. } Z = 3x_1 + 2x_2$$

subject to $x_1 + x_2 \ge 1,\ x_1 + x_2 \le 7,$

$x_1 + 2x_2 \le 10,\ x_2 \le 3$ and $x_1, x_2 \ge 0.$

7. Formulate the dual of the given L.P.P. and hence solve it.

$$\text{Min. } Z = 3x_1 - 2x_2 + 4x_3$$

subject to $3x_1 + 5x_2 + 4x_3 \ge 7,\ 6x_1 + x_2 + 3x_3 \ge 4,$

$7x_1 - 2x_2 - x_3 \le 10,\ x_1 - 2x_2 + 5x_3 \ge 3$

$4x_1 + 7x_2 - 2x_3 \ge 2$

and $x_1, x_2, x_3 \ge 0.$

8. Use duality theory to solve the given L.P.P.

$$\text{Min. } Z = 4x_1 + 3x_2 + 6x_3$$

subject to $x_1 + x_3 \ge 2,\ x_2 + x_3 \ge 5$ and $x_1, x_2, x_3 \ge 0.$

9. Solve the following L.P.P. using duality theory :

$$\text{Min. } Z = 50x_1 - 80x_2 - 140x_3,$$

subject to $x_1 - x_2 - 3x_3 \geq 4,\ x_1 - 2x_2 - 2x_3 \geq 3$

and $x_1, x_2, x_3 \geq 0.$

10. Apply the principle of duality to solve the following L.P.P. :

Max. $Z = 3x_1 - 2x_2$

subject to $x_1 + x_2 \leq 5,\ x_1 \geq 4,\ 1 \leq x_2 \leq 6$

and $x_1, x_2 \geq 0.$

11. Solve the following primal problem and its dual by simplex method :

Max. $Z = 5x_1 + 12x_2 + 4x_3$

subject to $x_1 + 2x_2 + x_3 \leq 5,\ 2x_1 - x_2 + 3x_3 = 2,$

and $x_1, x_2, x_3 \geq 0.$

Verify that the solution of the primal can be read from the optimal table of the dual and vice-versa.

12. Write the dual of the following primal :

Max. $Z = 40x_1 + 35x_2$

subject to $2x_1 + 3x_2 \leq 60,\ 4x_1 + 3x_2 \leq 96,$

and $x_1, x_2 \geq 0.$

Solve the primal and the dual by simplex method. Compare the optimal solutions of the two problems.

13. Use duality to solve the following L.P.P. :

Max. $Z = 4x_1 + 3x_2$

subject to
$$x_1 \leq 6,\ x_2 \leq 8,$$
$$x_1 + x_2 \leq 7,$$
$$3x_1 + x_2 \leq 15,$$
$$-x_2 \leq 1$$

and $x_1, x_2 \geq 0.$

14. Find the dual of the following L.P.P. :

Max. $Z = 2x_1 - x_2$

subject to $x_1 + x_2 \leq 10,\ -2x_1 + x_2 \leq 2$

$4x_1 + 3x_2 \geq 12,$

and $x_1, x_2 \geq 0.$

Solve the primal problem by simplex method and deduce from it the solution to the dual problem.

15. Find the dual of the following problem :

$$\text{Min. } Z = 6x + 5y + 2z$$

subject to $x + 3y + 2z \geq 5,\ 2x + 2y + z \geq 2,$

$4x - 2y + 3z \geq -1,$

and $x, y, z \geq 0.$

Hence or otherwise solve the primal problem.

16. Use duality to solve the problem

$$\text{Min. } Z = 10x_1 + 6x_2 + 2x_3$$

subject to $-x_1 + x_2 + x_3 \geq 1,$

$3x_1 + x_2 - x_3 \geq 2,$

and $x_1, x_2, x_3 \geq 0.$

17. Write the dual of the following problem :

$$\text{Max. } Z = 2x_1 + 3x_2,$$

subject to $2x_1 + 2x_2 \leq 10,$

$2x_1 + x_2 \leq 6,$

$x_1 + 2x_2 \leq 6,$

and $x_1, x_2 \geq 0.$

Solve the primal and the find then solution to the dual.

18. A company makes three products X, Y, Z out of three raw materials A, B, C. The number of units of raw materials required to produce one unit of the product is as given in the following table :

Raw material	***X***	***Y***	***Z***
A	1	2	1
B	2	1	4
C	2	5	1

The unit profit contribution of the products X, Y and Z and ₹ 40, 25 and 50 respectively. The number of units of raw materials available are 36, 60 and 45 respectively.

(i) Determine the product mix that will maximize the total profit.

(ii) Through the final simplex table write the solution to the dual problem.

Answers 2

1. $x_1 = 3/16$, $x_2 = 7/8$, max. $Z_x = 51.25$

   Dual. $w_1 = 15$, $w_2 = 5/4$, min. $Z_D = 51.25$

2. no feasible solution.

3. $x_1 = 2$, $x_2 = 1$, max. $Z = 8$

4. $x_1 = 4$, $x_2 = 2$, max. $Z = 10$

5. $x_1 = 1/3$, $x_2 = 1/3$, min. $Z = 4/3$.

6. $x_1 = 7$, $x_2 = 0$, Max. $Z = 21$.

7. no feasible solution.

8. $x_1 = 0$, $x_2 = 3$, $x_3 = 2$, min. $Z = 21$

9. no feasible solution.

10. $x_1 = 4, x_2 = 1$, max. $Z = 10$

11. $x_1 = 9/5$, $x_2 = 8/5$, $x_3 = 0$, max. $Z = 28\frac{1}{5}$

    Dual $w_1 = 29/5$, $w_2 = -2/5$, Min. $Z_D = 28\frac{1}{5}$

12. $x_1 = 18$, $x_2 = 8$, max, $Z = 1000$.

    Dual $w_1 = 10/3$, $w_2 = 25/3$, min, $Z_D = 1000$.

13. $x_1 = 4$, $x_2 = 3$, max. $Z = 25$

14. $x_1 = 10$, $x_2 = 0$, max. $Z = 20$

    Dual $w_1 = 2$, $w_2 = 0$, $w_3 = 0$, min. $Z_D = 20$

15. $x = 0$, $y = 0$, $z = 5/2$, Min. $Z = 5$

16. $x_1 = 1/4$, $x_2 = 5/4$, $x_3 = 0$, Min. $Z = 10$

17. $x_1 = 2$, $x_2 = 2$, max. $Z = 10$

    Dual $w_1 = 0$, $w_2 = 1/3$, $w_3 = 4/3$, min. $Z_D = 10$

18. $x = 20$, $y = 0$, $z = 5$, Max. $P = 1050$,

    Dual $r = 0$, $s = 10$, $t = 10$, min. $R = 1050$.

4.9 Dual Simplex Method

For a L.P.P. (maximization problem) the optimality criterion of the simplex method, $c_j - Z_j = c_j - C_B B^{-1}.\alpha_j \le 0$, for all j, where B is the basis, depends only on α_j and c_j and is independent of the requirement vector **b**. Thus, every basic solution with all $c_j - Z_j \le 0$ will not be feasible, but any basic feasible solution with all $c_j - Z_j \le 0$ will certainly be an optimal solution. The **dual simplex method** uses the above remarks. Thus, whereas the regular simplex method starts with a basic feasible but non-optimal solution and proceeds towards optimality, the dual simplex method starts with a basic infeasible but optimal solution and proceeds towards feasibility.

The dual simplex algorithm was discovered by **C.E. Lemke**, a student of Charnes while applying the simplex method to the dual of a L.P.P.

4.10 Advantage of Dual Simplex Algorithm

The advantage of the dual simplex algorithm over the other method is that here we do not require any artificial variables. Hence, a great deal of work or a lot of labour is saved whenever this method is applicable.

4.11 Computational Procedure of the Dual Simplex Algorithm

Step I : To form the given L.P.P. in standard primal form

1. If the problem is of minimization. Convert it into the maximization problem.
2. Write all the constraints in the form of inequalities involving $\le$ sign.

Note that by doing so some b_i's may change to negative values.

Step II : To find initial basic solution

Introduce the slack variables in the constraints to reduce them to equalities. Then to find the initial basic solution take all the given variables equal to zero and calculate the values of the slack variables. This solution will be the **starting (initial) basic solution,** which may not be feasible. Let $X_B = (x_{B1}, x_{B2}, \ldots, x_{Bm})$ be the initial basic solution corresponding to the basis matrix $B = (\beta_1, \beta_2, \ldots, \beta_m)$.

Step III : Construction of the starting simplex table

Construct the simplex table as usual in simplex method.

Step IV : To test the initial solution for optimality

Compute $\Delta_j = c_j - Z_j = c_j - C_B Y_j$ for each column.

1. If all $\Delta_j \le 0$ and all x_{Bi} are non-negative, the solution found above is optimum basic feasible solution.
2. If all $\Delta_j \le 0$ and at least one x_{Bi} is negative, then proceed to step V.
3. If any $\Delta_j > 0$ then the method fails.

Step V : To find the vector incoming (entering) and leaving (outgoing) the basis

Here the outgoing vector is determined first.

To determine the outgoing vector : Here β_r *i.e.,* the r-th columns in the basis *i.e.,* the corresponding vector x_r in the basis is the outgoing vector, if

$$x_{Br} = \text{Min.}\{x_{Bi}, x_{Bi} < 0\}$$

To determine the incoming vector (α_k) : If β_r is the outgoing vector, then α_k is taken as the entering (incoming) vector for the value of k, for which

$$\frac{\Delta_k}{y_{rk}} = \underset{j}{\text{Min}}\left[\frac{\Delta_j}{y_{rj}}, y_{rj} < 0\right]$$

If all $y_{rj} \ge 0$, then the problem has no F.S.

Step VI : Test of optimality

If entering the vector α_k in place of β_r in the basis all basic variables reduces to non-negative values, then this solution is optimal F.S. But if at least one basic variable is negative, then this solution is not optimal F.S. In that case repeat step V and VI, iteratively till an optimal F.S. is obtained.

Illustrative Examples

Example 13: *Solve the following L.P.P. by the simplex dual algorithm :*

$$\textit{Min. } Z = 3x_1 + x_2$$

subject to $\quad x_1 + x_2 \ge 1$

$$2x_1 + 3x_2 \ge 2$$

and $\quad x_1, x_2 \ge 0.$

Solution: For the clear understanding of the procedure we shall solve this example step by step.

Step I : First we write the given L.P.P. in the standard primal form.

Max. $\quad Z_P = -Z = -3x_1 - x_2$

subject to $\quad -x_1 - x_2 \le -1$

$$-2x_1 - 3x_2 \leq -2$$

and $\qquad x_1, x_2 \geq 0$

Since objective function is that of maximization and all $c_j < 0$.

∴ It is possible to find an infeasible but optimal basic solution and hence we can solve this problem by dual simplex algorithm.

Step II : Introducing the slack variables x_3 and x_4 the constraints reduce to the following equalities :

$$-x_1 - x_2 + x_3 \qquad = -1$$

$$-2x_1 - 3x_2 \qquad + x_4 \qquad = -2$$

Taking $x_1 = 0 = x_2$, we get $x_3 = -1$, $x_4 = -2$, which is the starting basic infeasible solution to the primal.

Step III : The **starting simplex** table is as follows :

B	$\mathbf{c}_B$	c_j	-3	-1	0	0	
		$X_B(=\mathbf{x}_B)$	Y_1	Y_2	$Y_3(\beta_1)$	$Y_4(\beta_2)$	
Y_3	0	-1	-1	-1	1	0	
Y_4	0	-2	-2	$\boxed{-3}$	0	1	→
$Z_P = \mathbf{c}_B\mathbf{x}_B = 0$		Δ_j	-3	-1 ↑	0	0 ↓	

Step IV : $\qquad \Delta_1 = c_1 - Z_1 = c_1 - \mathbf{c}_B Y_1 = -3$

$$\Delta_2 = c_2 - Z_2 = c_2 - \mathbf{c}_B Y_2 = -1$$

Thus, the starting solution $x_1 = 0$, $x_2 = 0$, $x_3 = -1$, $x_4 = -1$ is a basic solution which is infeasible but optimal.

Step V : To find the improved solution we shall determine the leaving vector and entering vector to the basis.

To determine the leaving vector (β_r)

Since $x_{Br} = \text{Min}(x_{Bi}, x_{Bi} < 0)$.

$$= \text{Min.}(x_{B1}, x_{B2}) = \text{Min.}(-1, -2) = -2 = x_{B2}$$

∴ $\qquad r = 2$ *i.e.*, $\beta_2 (= Y_4)$ is the leaving vector.

To determine the entering vector (α_k)

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{2k}} = \underset{j}{\text{Min.}}\left\{\frac{\Delta_j}{y_{2j}}, y_{2j} < 0\right\}$$

$$= \text{Min}\left\{\frac{\Delta_1}{y_{21}}, \frac{\Delta_2}{y_{22}}\right\} = \text{Min}\left\{\frac{-3}{-2}, \frac{-1}{-3}\right\} = \frac{1}{3} = \frac{\Delta_2}{y_{22}}$$

$\therefore$ $k = 2$. *i.e.*, α_2 $(= Y_2)$ is the entering vector.

$\therefore$ Key element is $y_{22} = -3$

Proceeding as usual the **second simplex table** is as follows :

B	c_B	c_j	-3	-1	0	0
		$X_B (= x_B)$	Y_1	$Y_2\ (\beta_2)$	$Y_3\ (\beta_1)$	Y_4
Y_3	0	$-1/3$	$-1/3$	0	1	$\boxed{-1/3}$ $\rightarrow$
Y_2	-1	$2/3$	$2/3$	1	0	$-1/3$
$Z_P = c_B x_B = -2/3$		Δ_j	$-7/3$	0	0 $\downarrow$	$-1/3$ $\uparrow$

The solution given in this table is $x_1 = 0$, $x_2 = 2/3$, $x_3 = -1/3$, $x_4 = 0$ which is infeasible but optimal hence it can be improved further for which we again find the leaving and entering vectors.

To determine the leaving vector (β_r)

Since $\quad x_{Br} = \text{Min.}\{x_{Bi}, x_{Bi} < 0\} = \text{Min.}\{x_{B1}\} = -1/3 = x_{B1}$

$\therefore$ $r = 1$, *i.e.*, β_1 $(= Y_3)$ is the leaving vector.

To determine the entering vector (α_k)

$$\frac{\Delta_k}{y_{rk}} = \frac{\Delta_k}{y_{1k}} = \underset{j}{\text{Min.}}\left\{\frac{\Delta_j}{y_{1j}}, y_{1j} < 0\right\} = \text{Min}\left\{\frac{\Delta_1}{y_{11}}, \frac{\Delta_4}{y_{14}}\right\}$$

$$= \text{Min}\left\{\frac{-7/3}{-1/3}, \frac{-1/3}{-1/3}\right\} = \text{Min}\{7, 1\} = 1 = \frac{\Delta_4}{y_{14}}$$

$\therefore$ $k = 4$ *i.e.*, α_4 $(= Y_4)$ is the entering vector.

$\therefore$ Key element is $y_{14} = -1/3$

Proceeding as usual the **third simplex table** is as follows

B	c_B	c_j	-3	-1	0	0
		$X_B (= x_B)$	Y_1	$Y_2\ (\beta_2)$	Y_3	$Y_4\ (\beta_1)$
Y_4	0	1	1	0	-3	1
Y_2	-1	1	1	1	-1	0
$Z_P = -1$		Δ_j	-2	0	-1	0

$\because$ All $\Delta_j \le 0$ and all $\mathbf{x}_{B1} > 0$, so the solution is optimal and feasible, which is

$$x_1 = 0,\ x_2 = 1,\ \text{max. } Z_p = -1\ i.e.,\ \min Z = 1$$

The above solution in different steps can be more conveniently represented by a single table as shown below :

B	$\mathbf{c}_B$	$\mathbf{c}_j$ / $\mathbf{x}_B$	-3 / Y_1	-1 / Y_2	0 / Y_3	0 / Y_4	Mini x_{Br} / $x_{Br} < 0$
$Y_3\ (\beta_1)$	0	-1	-1	-1	1	0	—
$Y_4\ (\beta_2)$	0	-2	-2	$\boxed{-3}$	0	1	$-2\ (x_{B2}) \rightarrow$
$Z_P = \mathbf{c}_B\,\mathbf{x}_B = 0$		Δ_j	-32	-1	0	0	
Mini ratio $\Delta_j / y_{2j}\ y_{2j} < 0$			$-3/-2 = 3/2$	$-1/-3 = 1/3 \uparrow$	—	—	
$Y_3\ (\beta_1)$	0	$-1/3$	$-1/3$	0	1	$\boxed{-1/3}$	$-1/3\ (x_{B1}) \rightarrow$
$Y_2\ (\beta_2)$	-1	$2/3$	$2/3$	1	0	$-1/3$	
$Z_P = -2/3$		Δ_j	$-7/3$	0	0	$-1/3$	
Mini ratio $\Delta_j / y_{1j}\ y_{1j} < 0$			$\dfrac{-7/3}{-1/3} = 7$	—	—	$\dfrac{-1/3}{-1/3} = 1$ $\uparrow$	
Y_4	0	1	1	0	-3	1	All $x_{Br} > 0$
Y_2	-1	1	1	1	-1	0	
$Z_P = -1$		Δ_j	-2	0	-1	0	

From the last table $x_1 = 0,\ x_2 = 1,\ x_3 = 0,\ x_4 = 1$ which is feasible and optimal as $x_j \ge 0$ and $\Delta_j \le 0$ for all $j = 1, 2, 3, 4$.

Hence the optimal solution of the given L.P.P. is $x_1 = 0,\ x_2 = 0$, Mini. $Z = -\text{Max } Z_P = 1$.

Example 14: *Solve the following L.P.P. by the simplex algorithm :*

$$\textit{Min. } Z = 3x_1 + 2x_2 + x_3 + 4x_4$$

subject to

$$2x_1 + 4x_2 + 5x_3 + x_4 \ge 10$$
$$3x_1 - x_2 + 7x_3 - 2x_4 \ge 2$$
$$5x_1 + 2x_2 + x_3 + 6x_4 \ge 15$$

and

$$x_1, x_2, x_3, x_4 \ge 0.$$

Solution: **Step I :** The given L.P.P. in the standard primal form is

$$\text{Max. } Z_P = -3x_1 - 2x_2 - x_3 - 4x_4$$

subject to $\quad -2x_1 - 4x_2 - 5x_3 - x_4 \le -10$

$$-3x_1 + x_2 - 7x_3 + 2x_4 \le -2$$

$$-5x_1 - 2x_2 - x_3 - 6x_4 \le -15$$

and $\quad x_1, x_2, x_3, x_4 \ge 0.$

Since objective function is of maximization and all $c_j < 0$, we can solve this L.P.P. by dual simplex algorithm.

Step II : Introducing the slack variables x_5, x_6 and x_7 the constraints of the above problem reduce to the following equalities :

$$-2x_1 - 4x_2 - 5x_3 - x_4 + x_5 = -10$$

$$-3x_1 + x_2 - 7x_3 + 2x_4 + x_6 = -2$$

$$-5x_1 - 2x_2 - x_3 - 6x_4 + x_7 = -15$$

Taking $x_1 = 0 = x_2 = x_3 = x_4$, we get $x_5 = -10$, $x_6 = -2$, $x_7 = -15$ which is the starting basic solution to the primal and is infeasible.

The solution to the problem using dual simplex algorithm is given in the following table :

B	c_B	c_j	-3	-2	-1	-4	0	0	0	Mini x_{Br}
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	$x_{Br} < 0$
Y_5	0	-10	-2	-4	-5	-1	1	0	0	
Y_6	0	-2	-3	1	-7	2	0	1	0	$-15\,(x_{B3})$ →
Y_7	0	-15	$\boxed{-5}$	-2	-1	-6	0	0	1	
$Z = c_B x_B = 0$		Δ_j	-3	-2	-1	-4	0	0	0	
Mini ratio Δ_j / y_{3j}, $Y_{3j} < 0$			$-3/-5 = 3/5$ ↑ min	$\frac{-2}{-2} = 1$	$\frac{-1}{-1} = 1$	$\frac{-4}{-6} = \frac{2}{3}$				
Y_5	0	-4	0	$-16/5$	$\boxed{-23/5}$	$7/5$	1	0	$-2/5$	$-4\,(x_{B1})$ →
Y_6	0	7	0	$11/5$	$-32/5$	$28/5$	0	1	$-3/5$	
Y_1	-3	3	1	$2/5$	$1/5$	$6/5$	0	0	$-1/5$	
$Z = -9$		Δ_j	0	$-4/5$	$-2/5$	$-2/5$	0	0	$-3/5$	
Mini ratio Δ_j / y_{1j}, $Y_{1j} < 0$			—	$\frac{-4/5}{-16/5} = 1/4$	$\frac{-2/5}{-23/5} = 2/23$ min↑	—	—	—	$\frac{-3/5}{-2/5} = 3/2$	
Y_3	-1	$20/23$	0	$16/23$	1	$-7/23$	$-5/23$	0	$2/23$	All $x_{Br} > 0$
Y_6	0	$289/23$	0	$153/23$	0	$84/23$	$-32/23$	1	$-1/23$	
Y_1	-3	$65/23$	1	$6/23$	0	$29/23$	$1/23$	0	$-5/23$	
$Z = -215/23$		Δ_j	0	$-12/23$	0	$-12/23$	$-2/23$	0	$-13/23$	

The solution given in the last table is

$$x_1 = 65/23,\ x_2 = 0,\ x_3 = 20/23,\ x_4 = 0 = x_5,\ x_6 = 289/23,\ x_7 = 0$$

which is feasible and optimal (since all $\Delta_j \le 0$).

Hence the optimal feasible solution of the given L.P.P. is

$$x_1 = 65/23,\ x_2 = 0,\ x_3 = 20/23,\ x_4 = 0$$

and $\quad$ Min. $Z = -\text{Max. } Z_P = 215/23$

Example 15: *Use dual simplex method to solve the following L.P.P. :*

$$\text{Min. } Z = 6x_1 + 7x_2 + 3x_3 + 5x_4$$

subject to,
$$5x_1 + 6x_2 - 3x_3 + 4x_4 \ge 12$$
$$x_2 + 5x_3 - 6x_4 \ge 10$$
$$2x_1 + 5x_2 + x_3 + x_4 \ge 8$$

and
$$x_1, x_2, x_3, x_4 = 0.$$

Solution: **Step I :** The given L.P.P. in the standard primal form is

$$\text{Max. } Z_P = -6x_1 - 7x_2 - 3x_3 - 5x_4$$

subject to
$$-5x_1 - 6x_2 + 3x_3 - 4x_4 \le -12$$
$$-x_2 - 5x_3 + 6x_4 \le -10$$
$$-2x_1 - 5x_2 - x_3 - x_4 \le -8$$

and
$$x_1, x_2, x_3, x_4 \ge 0$$

Since objective function is of maximization and all $c_j < 0$,we can solve this L.P.P. by dual simplex algorithm.

Step II : Introducing the slack variables x_5, x_6 and x_7 the constraints of the above problem reduce to the following equalities :

$$-5x_1 - 6x_2 + 3x_3 - 4x_4 + x_5 = -12$$
$$-x_2 - 5x_3 + 6x_4 + x_6 = -10$$
$$-2x_1 - 5x_2 - x_3 - x_4 + x_7 = -8$$

Taking $x_1 = 0 = x_2 = x_3 = x_4$, we get $x_5 = -12,\ x_6 = -10,\ x_7 = -8$ is the starting basic solution to the primal and is infeasible.

The solution to the problem using dual simplex algorithm is given in the following table :

B	c_B	c_j / x_B	-6 / Y_1	-7 / Y_2	-3 / Y_3	-5 / Y_4	0 / Y_5	0 / Y_6	0 / Y_7	Min. x_{Br}, $x_{Br} < 0$
Y_5	0	-12	-5	$\boxed{-6}$	3	-4	1	0	0	$-12 (x_{B1})$
Y_6	0	-10	0	-1	-5	6	0	1	0	—
Y_7	0	-8	-2	-5	-1	-1	0	0	1	—
$Z_P = C_B X_B = 0$		Δ_j	-6	-7	-3	-5	0	0	0	
Mini. Ratio Δ_j / y_{1j}, $y_{1j} < 0$			$-6/(-5) = 6/5$	$-7/(-6) = 7/6$ (min) ↑	—	$(-5)/(-4) = 5/4$				
Y_2	-7	2	5/6	1	$-1/2$	2/3	$-1/6$	0	0	—
Y_6	0	-8	5/6	0	$\boxed{-11/2}$	20/3	$-1/6$	1	0	$-8 (x_{B2})$
Y_7	0	2	13/6	0	$-7/2$	7/3	$-5/6$	0	1	—
$Z_P = -14$		Δ_j	$-1/6$	0	$-13/2$	$-1/3$	$-7/6$	0	0	
Mini Ratio Δ_j / y_{2j}, $y_{2j} < 0$			—	—	$\frac{-13/2}{-11/2} = \frac{13}{11}$ (min) ↑	—	$\frac{-7/6}{-1/6} = 7$	—	—	
Y_2	-7	30/11	25/33	1	0	2/33	$-5/33$	$-1/11$	0	All $x_{Br} > 0$
Y_3	-3	16/11	$-5/33$	0	1	$-40/33$	1/33	$-2/11$	0	
Y_7	0	78/11	18/11	0	0	$-21/11$	$-8/11$	$-7/11$	1	
$Z_P = -258/11$		Δ_j	$-38/33$	0	0	$-271/33$	$-32/33$	$-13/11$	0	

The solution given in the last table is

$$x_1 = 0 = x_4 = x_5 = x_6,\ x_2 = 30/11,\ x_3 = 16/11 \text{ and } x_7 = 78/11,$$

which is feasible and optimal. (since all $\Delta_j \le 0$)

Hence, the optimal feasible solution of the given L.P.P. is

$$x_1 = 0,\ x_2 = 30/11,\ x_3 = 16/11,\ x_4 = 0$$

and Min. $Z = -\text{Max. } Z_P = 258/11$

4.12 Primal Dual Algorithm

When artificial variables are introduced (necessarily) in a L.P.P. then we use either two phase method or Charne's M-method for the removal of the artificial vectors from the basis *i.e.*, for driving the artificial variables to zero. In the process of driving the artificial variables to zero, we are hardly concerned with the optimality of the problem because the criterion used to select the vector to enter the basis is only concerned with driving the artificial variable to zero. Thus at the end of phase 1 or when all the artificial variables are zero in Carnes M-method, the resulting feasible solution may not be any where near optimal. Hence we require a large number of iterations (improvements) to get the optimal solution in phase II. It would be very much beneficial if both the objectives of optimality and of driving the artificial variables to zero could be attained in phase I, this will reduce considerably the total number of iterations.

The **primal-dual algorithm** is developed for this purpose.

We have seen that the dual simplex algorithm can be used in special circumstances to remove the necessity of introducing artificial variables. Thus dual simplex algorithm cannot be used in general because it is not always easy to find a basic solution with all $c_j - Z_j \leq 0$.

In primal dual algorithm we introduce the artificial variables and hence require phase I. But here the computational procedure is such that at the end of phase I we found an optimal as well as feasible solution of the given L.P.P.

In this method we work simultaneously on the primal and dual problems. This method is symmetrically followed as follows :

1. We start with a solution to the dual problem *i.e.* first of all we determine an initial dual solution.
2. Then we proceed to find an infeasible solution to the primal such that the theorem of complementary slackness (article 4.6) remain satisfied. For this we proceeed as follows :

   (a) We associate with the initial dual solution a restricted primal problem which is deduced from the original primal by zeroing certain variables to satisfy the theorem of complementary slackness and by replacing the original linear form by the sum of artifical variables.

   (b) Then we maximize the sum of the artificial variables by using simplex method.

   (c) In case the optimal solution thus obtained is not the primal solution of the original primal then we determine a new dual solution and apply steps (a) and (b).

We proceed in this way again and again till we obtain an optimal solution of the original primal.

Thus when a feasible solution to the primal is found it is also optimal.

4.13 To determine the Initial Dual Solution

Let the given L.P.P. be as follows.

Primal Problem.

$$\left.\begin{array}{l}\text{Max. } Z_P = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n = \mathbf{c}\ \mathbf{x} \\ \text{s.t. } \ A\mathbf{x} = \mathbf{b} \\ \text{and } \ \mathbf{x} \geq \mathbf{0}\end{array}\right] \quad \ldots(1)$$

Dual Problem. The dual of the above primal is given by

$$\left.\begin{array}{l}\text{Max.} Z_D = b_1 w_1 + b_2 w_2 + \ldots + b_m w_m = \mathbf{b}' \mathbf{w} \\ \text{s.t. } \ A' \mathbf{x} \geq \mathbf{c}'.\end{array}\right] \quad \ldots(2)$$

In the discussion of the dual simplex algorithm we have seen that it is not easy to find an immediate solution to the dual (2). It was **Beale** who suggested a very simple modification to the primal (1), which provide us an immediate solution of the dual. The procedure is as follows.

Introducing an additional constraint

$$x_0 + \mathbf{x} = x_0 + x_1 + x_2 + \ldots + x_n = b_0$$

in the given primal (1), we obtain the following new primal called the **modified primal problem**.

$$\text{Max. } Z_P = \mathbf{0} \cdot x_0 + c_1 x_1 + c_2 x_2 + \ldots + c_n x_n = (0, \mathbf{c}) \cdot [\mathbf{x}_0, \mathbf{x}]$$

$$\left.\begin{array}{l}\text{s.t. } \ \mathbf{x}_0 + \mathbf{x} = b_o. \\ \qquad A\mathbf{x} = \mathbf{b} \\ \text{and } [x_0, \mathbf{x}] \geq \mathbf{0}.\end{array}\right] \quad \ldots(3)$$

Here it is assumed that the new constant b_0 has arbitrarily large value.

(3) can also be written as Max. $Z_P = \mathbf{cx}$

$$\text{s.t.} \begin{bmatrix} 1 & 1 \\ \mathbf{0} & A \end{bmatrix} \begin{bmatrix} x_0 \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} b_0 \\ \mathbf{b} \end{bmatrix}, [x_0, \mathbf{x}] \geq 0 \quad \ldots(4)$$

The dual of the modified primal (3) or (4), called the modified dual is as follows.

$$\text{Min. } Z_D = b_0 w_0 + \mathbf{b}' \mathbf{w} = b_0 w_0 + b_1 w_1 + \ldots + b_m w_m$$

$$\text{s.t.} \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{1} & A' \end{bmatrix} \begin{bmatrix} w_0 \\ \mathbf{w} \end{bmatrix} \geq \begin{bmatrix} 0 \\ \mathbf{c}' \end{bmatrix}$$

which can also be written as

$$\left.\begin{array}{llllll} 1\cdot w_0 & +0\cdot w_1 & +0\cdot w_2 & +\ldots+0\cdot w_m & \geq 0 \\ 1\cdot w_0 & +a_{11}w_1 & +a_{21}w_2 & +\ldots+a_{m1}\;w_m & \geq c_1 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 1\cdot w_0 & +a_{1n}w_1 & +a_{2n}w_2 & +\ldots+a_{mn}w_m & \geq c_n \end{array}\right] \quad \ldots(5)$$

where $w_1, w_2, \ldots, w_m$ are unrestricted in sign and by virtue of first constraint in (5),

$w_0 \geq 0$, *i.e.*, w_0 is non-negative.

In case $w_0 = 0$ then (5) is converted to the dual (2) of primal (1). We can immediately obtain a solution of this dual (5) which is

$$\left.\begin{array}{ll} w_0 = \text{Max.}\,\{0, c_j\} & \mathbf{w} = \mathbf{0} \\ j = 1, \ldots, n & i.e.,\; w_1 = 0 = w_2 = \ldots = w_m \end{array}\right] \quad \ldots(6)$$

To solve (5), we add surplus variables $w_{sj}, j = 0, 1, 2, \ldots, n$ to the constraints of (5), and then the resulting problem is

$$\text{Min. } Z_D = b_0 w_0 + b_1 w_1 + \ldots + b_m w_m + 0\cdot w_{s0} + 0\cdot w_{s1} + \ldots + 0\cdot w_{sn}$$

$$\left.\begin{array}{llllllll} \text{s.t.} & 1\cdot w_0 + 0\cdot w_1 + 0\cdot w_2 + \ldots + 0\cdot w_m & - w_{s0} & & & & & = 0 = c_0 \\ & 1\cdot w_0 + a_{11}w_1 + a_{21}w_2 + \ldots + a_{m1}w_m & & - w_{st} & & & & = c_1 \\ & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ & 1\cdot w_0 + a_{1n}w_1 + a_{2n}w_2 + \ldots + a_{mn}w_n & & & - w_{sn} & & & = c_n \end{array}\right] \quad \ldots(7)$$

and $\quad w_{s0}, w_{sj} \geq 0$

i.e., $\quad w_{s0}, w_{s1}, \ldots, w_{sn} \geq 0$

For the solution (6), from (7) have

$$\left.\begin{array}{l} w_{s0} = w_0 - c_0 \\ w_{s1} = w_0 - c_1 \\ \ldots \quad \ldots \quad \ldots \\ \ldots \quad \ldots \quad \ldots \\ w_{sn} = w_0 - c_n \end{array}\right] \quad \ldots(8)$$

The solution (6) is the initial dual solution with which we start in primal dual method. Since $w_0 = \text{Max.}\{0, c_j\}\; j = 1, \ldots, n$,

it follows that at least one of $w_{s0}, w_{s1}, \ldots, w_{sn}$ is zero.

From (5), we know that if $\mathbf{x}'\cdot\mathbf{w}_s = 0$ or equivalently for dual problem (4) and (7), if

$$\left.\begin{aligned} \begin{bmatrix} x_0 \\ \mathbf{x}' \end{bmatrix}' \begin{bmatrix} w_{s0} \\ \mathbf{w}_s \end{bmatrix} &= \mathbf{0}, \\ \text{or} \quad x_0 \cdot w_{s0} + \mathbf{x}' \cdot \mathbf{w}_s &= 0 \end{aligned}\right\} \qquad \ldots(9)$$

then $Z_P = Z_D$ where $\mathbf{x}$ is any solution (not necessarily feasible) to the primal.

If along with the condition (9), $[x_0, \mathbf{x}]$ is a feasible solution to the modified primal (3) or (4) then $[x_0, \mathbf{x}]$ is an optimal feasible solution to the modified primal and $[w_0, \mathbf{w}]$ is an optimal solution to the modified dual.

If in addition $w_0 = 0$, then $Z_P = \mathbf{c}\ \mathbf{x} = Z_D = \mathbf{b}'\mathbf{w}$ and from (9) $\mathbf{x}' \cdot \mathbf{w}_s = 0$. Consequently $\mathbf{x}$ is an optimal solution to the given primal and $\mathbf{w}$ is an optimal solution to its dual. If we get a solution to the modified dual with $w_0 \neq 0$ and a feasible solution to the modified primal such that (9) is satisfied, then it following that $x_0 = 0$, and then

$$Z_P = Z_D = b_0 w_0 + \mathbf{b}'\mathbf{w}.$$

4.14 To Determine the Restricted Primal (or Extended Primal Problem)

To solve the modifed primal problem (3) or (4) of article 6.13 we add artificial variables to each of the constraints and assign price -1 to each artificial variables and zero prices corresponding to every legitimate variable. Thus we get following new primal, called the **restricted primal** or **extended primal**.

$$\text{Max. } Z^* = -1 \cdot x_{a0} - 1 \cdot x_{a1} - x_{a2} \ldots - 1 \cdot x_{am} = -1 \cdot x_{a0} - 1 \cdot \mathbf{x}_a$$

s.t. $$\begin{bmatrix} 1 & 1 \\ 0 & A \end{bmatrix} \begin{bmatrix} x_0 \\ \mathbf{x} \end{bmatrix} + \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & I_m \end{bmatrix} \begin{bmatrix} x_{a0} \\ \mathbf{x}_a \end{bmatrix} = \begin{bmatrix} b_0 \\ \mathbf{b} \end{bmatrix}$$

and $[x_0, \mathbf{x}] \geq 0$, $[x_{a0}, \mathbf{x}_a] \geq \mathbf{0}$.

The constraints can be written as

$$\begin{aligned} x_0 + x_1 + x_2 + \ldots + x_n + x_{a0} &= b \\ a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n + x_{a1} &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n + x_{a2} &= b_2 \\ &\ldots\ldots\ldots\ldots \\ &\ldots\ldots\ldots\ldots \\ a_{m1}x_1 + a_{m2}x_2 + .. + a_{mn}x_n + x_{am} &= b_m \end{aligned}$$

and $x_0, x_1, \ldots, x_n \geq 0, x_{a0}, x_{a1}, \ldots, x_{am} \geq 0$

Now we shall apply simplex method on the above restricted primal and go on improving Z^* by entering some vector in place of some leaving vector in each iteration till we obtain $Z^* = 0$

which follows that

$$x_{a0} = 0 = a_{a1} = \ldots = x_{am}.$$

The criterion for the selection of the entering and leaving vector will be different from that in the regular simplex method.

Here we shall use the following notations :

1. $(m+1)$ component legitimate activity vectors will be denoted by $Y_j^{(1)}, j = 0, 1, \ldots, n.$
2. A basis matrix for the constrains will be denoted by B_1.
3. The vector containing the prices c_j^* of the variables in the basis will be denoted by C_{B1}^* ,
4. $\Delta_j^* = c_j^* - Z_j^* = c_j^* - C_{B1}^* \cdot B_1^{-1} \alpha_j^{(1)} = c_j^* - C_{B1}^* \cdot Y_j^{(1)}.$

4.15 To Find the Entering and Leaving Vectors

Entering vector: In article 6.13, we obtained the solution of the modifed dual as follows

$$w_0 = \text{Max.} \{ 0, c_j \}, w_1 = 0 = w_2 = \ldots = w_m$$
$$j = 1, 2, \ldots n.$$

Which implies that at least one component w_{sj} will be zero.

The vector $Y_j^{(1)}$ for which the corresponding $w_{sj} = 0$ is taken as the vector entering the restricted primal basis. In this way all x_j for which w_{sj} are strictly positive remain zero.

Leaving vector: The vector leaving the basis corresponding to the entering vector $Y_j^{(1)}$ is selected one by one in the same manner as in Phase 1 of ordinary simplex method.

4.16 Method to Obtain New Dual Solution $(\hat{w}_{sj})$ and the Improved value of $\hat{Z}_D$ and $\hat{Z}^*$.

(i) The new dual solution $\hat{w}_{sj}$ is obtained by using the formula

$$\hat{w}_{sj} = w_{sj} - \theta \Delta_j^*$$

where $$\theta = \text{Mini}\left\{\frac{w_{sj}}{\Delta_j^*}, \Delta_j^* > 0\right\} \quad \ldots(1)$$

Here also at least one of $\hat{w}_{sj}$ will be zero and hence the corresponding $Y_j^{(1)}$ will be selected to enter the basis at the next iteration.

(ii) The improved value of Z^* will be obtained by the formula

$$Z^* = \mathbf{c}_{B1}{}^* \mathbf{x}_B{}^{(1)} \quad \ldots(2)$$

(iii) The value of $\hat{Z}_D$, will be obtained by the formula

$$\hat{Z}_D = Z_D + \theta Z^*$$

where θ and Z^* are given by (1) and (2) respectively.

4.17 Test of Optimality

The primal dual method will terminate (come to end) with one of the following results :

(i) When $Z^* = 0$, we get an optimal solution to the primal and $Z_P = Z_D$, provided $w_{s0} = 0 = w_0$.

(ii) When $Z^* = 0$ abut $w_{s0} \neq 0$, the primal problem has an unbounded solution.

(iii) When $Z^* < 0, \Delta_j{}^* \leq 0$ for all j (at any interation), the primal have no solution.

4.18 Computational Procedure of Primal-Dual Algorithm

The primal-dual algorithm is applied in the following order :

1. First we construct the restricted primal problem (see article 6.14).
2. Then we determine the initial dual solution by

$$w_0 = \text{Max.}\ \{0, c_j\}, c_j > 0, w_1 = 0 = w_2 = \ldots = w_m$$
$$j = 1, \ldots, n$$

and $w_{s0} = w_0 - c_0, w_{sj} = w_0 - c_j, j = 1, 2, \ldots, n.$

3. Then we select the entering vector $Y_j^{(1)}$ corresponding to that j for which $w_{sj} = \mathbf{0}$.

4. Then we select the leaving vector in the similar manner as in Phase I of regular simplex method.
5. For the vectors which are not in the primal basis we compute Δ_j^* by the formula

$$\Delta_j^* = c_j^* - C_{B1} Y_j^{*(1)} \; .$$

Also we compute $Z^* = C_{B1}^* \mathbf{x}_B^{(1)}$

and $$Z_w = b_0 w_0 + b_1 w_1 + \ldots + b_m w_m .$$

6. Then we construct the starting simplex table as follows.

B_1	C_{B1}^*	c_j^*	c_0^* c_1^* c_2^* ... c_n^* ... c_{n+m}^*	Mini $\left\{\dfrac{\mathbf{x}_B^{(1)}}{Y_k^{(1)}}\right\}$
		$x_B^{(1)}$	$Y_0^{(1)}$ $Y_1^{(1)}$ $Y_2^{(1)}$ ... $Y_n^{(1)}$ ... $y_{n+m}^{(1)}$	
$Y_{n+1}^{(1)}$	-1	b_0		
$Y_{n+2}^{(1)}$	-1			
$\vdots$	$\vdots$			
$Y_{n+m}^{(1)}$	-1			
$Z^* = C_{B1} \cdot \mathbf{x}_B^{(1)}$		x_j	... ...	
		Δ_j^*	... ...	
$Z_w = \sum_{j=0}^{m} b_j w_j$		w_{sj}		

7. If $Z^* = 0$ with $\Delta_j^* \le 0$, the solution under test is optimal. If the solution is not optimal, we proceed to the next iteration.
8. The solution of the new dual is obtained by using the formula

$$\hat{w}_{sj} = w_{sj} - \theta_0 \, \Delta_j^*$$

where $$\theta_0 = \text{Mini}\left\{\frac{w_{sj}}{\Delta_j^*} \; \Delta_j^* > 0\right\}.$$

9. We again find the entering vector $Y_j^{(1)}$ for which $\hat{w}_{sj} = 0$ and repeat steps 5 to 8 again till we get the optimal solution.
10. The optimality of solution of the primal is decided as shown in article 6.17.

Illustrative Examples

Examaple 12: *Solve the following L.P.P. by primal dual method.*

Max. $Z = x_1 + 6x_2$

s.t. $x_1 + x_2 \geq 2$

$x_1 + 3x_2 \leq 3$

and $x_1, x_2 \geq 0.$

Solution: 1. Adding the slack and surplus variables x_3, x_4, introducing the additional constraint and introducing the artificial variables x_5, x_6, x_7 we obtain the following **restricted primal**.

$$\text{Max. } Z^* = 0 \cdot x_0 + 0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 0 \cdot x_4 - 1 \cdot x_5 - 1 \cdot x_6 - 1 \cdot x_7.$$

$$\begin{aligned} \text{s.t.} \quad x_0 + x_1 + x_2 + x_3 + x_4 + x_5 &= b_0 \\ x_1 + x_2 - x_3 + x_6 &= 2 \\ x_1 + 3x_2 + x_4 + x_7 &= 3 \end{aligned}$$

2. To find the **initial dual solution**.

For initial solution to the dual, $w_0 = \text{Max.} \{ 0, c_j \}, c_j > 0$

$$j = 1, \ldots, 4.$$

$$= \text{Max} \{0, 1, 6\} = 6$$

$$w_0 = 6, w_1 = w_2 = w_3 = w_4 = 0$$

$\therefore$ $w_{s0} = w_0 - c_0 = 6 - 0 = 6, w_{s1} = w_0 - c_1 = 6 - 1 = 5$

$w_{s2} = w_0 - c_2 = 6 - 6 = 0, w_{s3} = w_0 - c_3 = 6 - 0 = 6.$

$w_{s4} = w_0 - c_4 = 6 - 0 = 6.$

Also $Z_w = \sum_{j=0}^{4} b_j w_j = b_0 w_0 + b_1 w_1 + b_2 w_2 + b_3 w_3 + b_4 w_4$

$= 6b_0 + 0 = 6b_0.$

The starting simplex table is as follows.

Table 1

B_1	C_{B1}^*	c_j^*	0	0	0	0	0	−1	−1	−1	Mini Ratio $\mathbf{x}_B/Y_2$
		$X_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$	$Y_2^{(1)}$	$Y_3^{(1)}$	$Y_4^{(1)}$	$Y_5^{(1)}$ (β_1)	$Y_6^{(1)}$ (β_2)	$Y_7^{(1)}$ (β_3)	
$Y_5^{(1)}$	−1	b_0	1	1	1	1	1	1	0	0	b_0 /1
$Y_6^{(1)}$	−1	2	0	1	1	−1	0	0	1	0	2/1
$Y_7^{(1)}$	−1	3	0	1	$\boxed{3}$	0	1	0	0	1	3/3 → Min
$Z^* = C_{B1}^* X_B^{(1)}$		x_j	0	0	0	0	0	b_0	2	3	
$= -b_0 - 5$		Δ_j^*	1	3	5	0	2	0	0	0	
$Z_w = 6\,b_0$		w_{sj}	6	5	0	6	6	0	0	0	

↑ entering vector ↓ out going

$$\Delta_0^* = c_0^* - C^*_{B1}\ Y_0^{(1)} = 0 - (-1, -1, -1)\ (1, 0, 0) = 1$$

$$\Delta_1^* = c_1^* - C^*_{B1}\ Y_1^{(1)} = 0 - (-1, -1, -1)\ (1, 1, 1) = 3$$

$$\Delta_2^* = c_2^* - C^*_{B1}\ Y_2^{(1)} = 0 - (-1, -1, -1)\ (1, 1, 3) = 5$$

$$\Delta_3^* = c_3^* - C^*_{B1} Y_3^{(1)} = 0 - (-1, -1, -1)\ (1, -1, 0) = 0$$

$$\Delta_4^* = c_4^* - C^*_{B1}\ Y_4^{(1)} = 0 - (-1, -1, -1)\ (1, 0, 1) = 2$$

Since $\Delta_j^* > 0$, so the solution is not optimal.

Therefore we proceed to improve the solution.

To find entering and outgoing vectors.

Since $w_{s2} = 0 \therefore Y_2^{(1)}$ is taken as the entering vector. By mini. ratio rule the outgoing (leaving) vector is taken as $Y_7^{(1)}$ (β_3).

$\therefore$ key element = $y_{32} = 3$.

Proceeding as usual and leaving the column $Y_7^{(1)}$ (outgoing artificial vector), we get the following table.

Table 2

B_1	C^*_{B1}	c^*_j	0	0	0	0	0	-1	-1	Mini Ratio X_B/Y_1
		$X_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$	$Y_2^{(1)}$ (β_3)	$Y_3^{(1)}$	$Y_4^{(1)}$	$Y_5^{(1)}$ (β_2)	$Y_6^{(1)}$ (β_2)	
$Y_5^{(1)}$	-1	b_0-1	1	2/3	0	1	2/3	1	0	$(b_0-1)/\frac{2}{3}$
$Y_6^{(1)}$	-1	1	0	$\boxed{2/3}$	0	-1	$-1/3$	0	1	$1/\frac{2}{3}\rightarrow$ (Mini)
$Y_2^{(1)}$	0	1	0	1/3	1	0	1/3	0	0	$1/\frac{1}{3}$
$Z^* = C^*_{B1}\cdot X_B^{(1)}$		x_j	0	0	1	0	0	b_0-1	1	
$=-b_0$		Δ_j^*	1	4/3	0	0	1/3	0	0	
$\hat{Z}^*_w = Z_w + \theta_0 Z^*$		w_{sj}	6	5	0	6	6	0	0	
$=6b_0 + 15/4\cdot(-b_0)$ $=9b_0/4$		$\hat{w}_{sj}$	9/4	0	0	6	19/4	0	0	
				↑ Entering vector					↓ Outgoing	

$$\Delta_0^* = c_0^* - C^*_{B1}\ Y_0^{(1)} = 0-(-1,-1,0)\,(1,0,0) = 1$$

$$\Delta_1^* = c_1^* - C^*_{B1}\ Y_1^{(1)} = 0-(-1,-1,0)\,(2/3, 2/3, 1/3) = 4/3$$

$$\Delta_3^* = c_3^* - C^*_{B1}\ Y_3^{(1)} = 0-(-1,-1,0)\,(1,-1,0) = 0$$

$$\Delta_4^* = c_4^* - C^*_{B1}\ Y_4^{(1)} = 0-(-1,-1,0)\cdot(2/3,-1/3,1/3) = 1/3$$

Sine $\Delta_j^* > 0$, so solution is not optimal.

$\therefore$ We compute the solution $\hat{w}_{sj}$ of the new dual

$$\theta_0 = \text{Mini}\left\{\frac{w_{sj}}{\Delta_j^*}\ \Delta_j^* > 0\right\} = \text{Mini}\left\{\frac{w_{s0}}{\Delta_0^*}, \frac{w_{s1}}{\Delta_1^*}, \frac{w_{s4}}{\Delta_4^*}\right\}$$

$$= \text{Mini}\left\{\frac{6}{1}, \frac{5}{4/3}, \frac{6}{1/3}\right\} = \frac{15}{4} = \frac{w_{s1}}{\Delta_1^*}$$

∴ the new dual solution is

$$\hat{w}_{s0} = w_{s0} - \theta_0 \cdot \Delta_0^* = 6 - \frac{15}{4} \cdot 1 = \frac{9}{4}$$

$$\hat{w}_{s1} = w_{s1} - \theta_0 \cdot \Delta_1^* = 5 - \frac{15}{4} \cdot \frac{4}{3} = 0$$

$$\hat{w}_{s2} = w_{s2} - \theta_0 \cdot \Delta_2^* = 0 - \frac{15}{4} \cdot 0 = 0$$

$$\hat{w}_{s3} = w_{s3} - \theta_0 \cdot \Delta_3^* = 6 - \frac{15}{4} \cdot 0 = 6$$

$$\hat{w}_{s4} = w_{s4} - \theta_0 \cdot \Delta_4^* = 6 - \frac{15}{4} \cdot \frac{1}{3} = \frac{19}{4}.$$

To find the improved solution we select the vector $Y_1^{(1)}$ as the entering vector, since $\hat{w}_{s1} = 0$,[$Y_2^{(1)}$ is already in the basic so it cannot be selected, even $\hat{w}_{s2} = 0$].

By minimum ratio rule we find the vector $Y_6^{(1)}$ (β_2) as the outgoing vector.

∴ key element $y_{22} = 2/3$.

Proceeding as usual and leaving the column $Y_6^{(1)}$ (outgoing artificial vector), we get the following table. 3.

$$\Delta_0^* = c_0^* - C_{B1}^* Y_0^{(1)} = 0 - (-1, 0, 0) \cdot (1, 0, 0) = 1$$

$$\Delta_3^* = c_3^* - C_{B1}^* Y_3^{(1)} = 0 - (-1, 0, 0) \cdot (2, -3/2, 1/2) = 2$$

$$\Delta_4^* = c_4^* - C_{B1}^* Y_4^{(1)} = 0 - (-1, 0, 0) \cdot (1, -1/2, 1/2) = 1$$

Since $\Delta_j^* > 0$, so the solution is not optimal.

∴ we compute the solution $\hat{w}'_{sj}$ of the new dual.

Table 3

B_1	C^*_{B1}	c^*_j	0	0	0	0	0	−1	Mini Ratio
		$Y_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$ (β_2)	$Y_2^{(1)}$ (β_3)	$Y_3^{(1)}$	$Y_4^{(1)}$	$Y_5^{(1)}$ (β_1)	X_B/Y_0
$Y_5^{(1)}$	−1	$b_0 - 2$	1	0	0	2	1	1	$(b_0 - 2)/1 \rightarrow$
$Y_1^{(1)}$	0	3/2	0	1	0	−3/2	−1/2	0	(Mini)
$Y_2^{(1)}$	0	1/2	0	0	1	1/2	1/2	0	

$Z^* = C_B^* \cdot \mathbf{x}_B^{(1)}$ $= -b_0 + 2$	x_j Δ_j^*	0 1	3/2 0	1/2 0	0 2	0 1	−1 0	
$= \hat{Z}_w + \theta_0 Z^*$ $= 9b_0/4 + 9/4$	$\hat{w}_{sj}$	9/4	0	0	6	19/4	0	
$(b_0 + 2) = 9/2$	$\hat{w}_{sj}$	0	0	0	3/2	5/2	0	
		↑ Entering vector					↓ Outgoing	

$$\theta_0 = \text{Mini}\left\{\frac{\hat{w}_{sj}}{\Delta_j^*}, \Delta_j^* > 0\right\} = \text{Mini}\left\{\frac{\hat{w}_{s0}}{\Delta_0^*}, \frac{\hat{w}_{s3}}{\Delta_3^*}, \frac{\hat{w}_{s4}}{\Delta_4^*}\right\}$$

$$= \text{Mini}\left\{\frac{9/4}{1}, \frac{6}{2}, \frac{19/4}{1}\right\} = \frac{9}{4} = \frac{\hat{w}_{s0}}{\Delta_0^*}$$

∴ the new dual solution is

$$\hat{w}'_{s0} = \hat{w}_{s0} - \theta_0 \cdot \Delta_0^* = 9/4 - 9/4 \cdot 1 = 0$$

$$\hat{w}'_{s1} = \hat{w}_{s1} - \theta_0 \cdot \Delta_1^* = 0 - 9/4 \cdot 0 = 0$$

$$\hat{w}'_{s2} = \hat{w}_{s2} - \theta_0 \cdot \Delta_2^* = 0 - 9/4 \cdot 0 = 0$$

$$\hat{w}'_{s3} = \hat{w}_{s3} - \theta_0 \cdot \Delta_3^* = 6 - 9/4 \cdot 2 = 3/2$$

$$\hat{w}'_{s4} = \hat{w}_{s4} - \theta_0 \cdot \Delta_4^* = 19/4 - 9/4 \cdot 1 = 5/2.$$

To find the improved solution we select the vector $Y_0^{(1)}$ as the entering vector, since $w'_{s0} = 0, [Y_1^{(1)}, Y_2^{(1)}$, are already in basis].

By minimum ratio rule we find the vector $Y_5^{(1)}$ (β_1) as the **outgoing vector**.

∴ key element $= y_{11} = 1$.

Proceeding as usual and leaving the column $Y_5^{(1)}$ (outgoing artificial vector), we get the following table.

Table 4

B_1	C^*_{B1}	c^*_j	0	0	0	0	0
		$Y_B^{(1)}$	$Y_0^{(1)}$	$Y_1^{(1)}$	$Y_2^{(1)}$	$Y_3^{(1)}$	$Y_4^{(1)}$
$Y_0^{(1)}$	0	$b_0 - 2$	1	0	0	2	1
$Y_1^{(1)}$	0	3/2	0	1	0	–3/2	–1/2
$Y_2^{(1)}$	0	1/2	0	0	1	1/2	1/2
$Z^* = C^*_{B1} \cdot x_B^{(1)}$		x_j	$b_0 - 2$	3/2	1/2	0	0
$= 0$		Δ_j^*	0	0	0	0	0
$Z''_w = Z'_w + \theta_0 \cdot Z^*$ $= 9/2 + \theta_0 \cdot 0$ $= 9/2$		$\hat{w}_{sj}$	0	0	0	3/2	5/2

$$\Delta_3^* = c_3^* - C^*_{B1} \cdot Y_3^{(1)} = 0 - (0, 0, 0) \cdot (2, -3/2, 1/2) = 0$$

$$\Delta_4^* = c_4^* - C^*_{B1} \cdot Y_4^{(1)} = 0 - (0, 0, 0) \cdot (1, -1/2, 1/2) = 0.$$

Here all $\Delta_j^* \leq 0$ $\therefore$ The solution under test is optimal. $Z^* = 0$. Since $w_0 = 0 = \hat{w}_{s0} \cdot$ $\therefore$ The sol. of the given primal problem is also optimal. Thus the optimal sol. of the given primal is $x_1 = 3/2, x_2 = 1/2, Z = \hat{Z}_w'' = 9/2$.

Comprehensive Exercise 3

1. Define the dual of a L.P.P.
2. Write a short note on duality theory.
3. Prove that the dual of the dual of a L.P.P. is the problem itself.
4. What are the advantages of duality method ?
5. What are the advantages of dual simplex algorithm ?
6. Write short note on “Reading the solution to the dual from the final simplex table of the primal”.

Solve the following L.P.P. by the dual simplex algorithm :

7. Min. $Z = x_1 + 2x_2$

subject to $2x_1 + x_2 \geq 4$

$x_1 + 7x_2 \geq 7$

and $x_1, x_2 \geq 0$

8. Min. $Z = x_1 + 2x_2 + 3x_3$

subject to $2x_1 - x_2 + x_3 \geq 4$

$x_1 + x_2 + 2x_3 \leq 8$

$x_2 - x_3 \geq 2$

and $x_1, x_2, x_3 \geq 0$

9. Max. $Z = -2x_1 - 2x_2 - 4x_3$

subject to $2x_1 + 3x_2 + 5x_3 \geq 2$

$3x_1 + x_2 + 7x_3 \leq 3$

$x_1 + 4x_2 + 6x_3 \leq 5$

and $x_1, x_2, x_3 \geq 0$

10. Max. $Z = -2x_1 - x_3$

subject to $x_1 - 2x_2 + 4x_3 \geq 8$

$x_1 + x_2 - x_3 \geq 5$

and $x_1, x_2, x_3 \geq 0$

11. Can you apply Dual Simplex Algorithm to the following L.P.P. ?

Max. $Z = 5x_1 + 3x_2$

subject to $3x_1 + 5x_2 \leq 15$, $5x_1 + 2x_2 \leq 10$ and $x_1, x_2 \geq 0$.

Answers 3

7. $x_1 = 21/13$, $x_2 = 10/13$, Min. $Z = 41/13$
8. $x_1 = 3$, $x_2 = 2$, $x_3 = 0$ Min. $Z = 7$
9. $x_1 = 0$, $x_2 = 2/3$, $x_3 = 0$, Max. $Z = -4/3$
10. $x_1 = 0$, $x_2 = 14$, $x_3 = 9$, Max. $Z = -9$
11. No

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. In standard primal form if the problem is of maximization, all the constraints involve the sign :

(a) $\geq$ (b) $\leq$

(c) $=$ (d) unrestricted

2. If the standard primal problem is of minimization, all the constraints involve the sign :
   (a) $\geq$ (b) $\leq$
   (c) $=$ (d) none of these
3. If a finite optimal feasible solution exists for the primal then the dual has :
   (a) unbounded solution (b) no solution
   (c) a finite feasible optimal solution (d) none of these
4. If the primal problem has an unbounded solution, the dual problem has :
   (a) a finite optimal feasible solution
   (b) no solution
   (c) either no solution or an unbounded solution
   (d) none of these
5. If the *i*-th slack variable of the primal is positive, then the *i*-th variable of the dual is :
   (a) + ive (b) – ive
   (c) zero (d) unrestricted
6. If both the primal and dual problems have finite optimal solutions and Z_P, Z_D are the optimal values of the objective functions of the primal and dual respectively then we have :
   (a) $Z_P > Z_D$ (b) $Z_P < Z_D$
   (c) $Z_P = Z_D$ (d) none of these

Fill in the Blank

Fill in the blanks "……" so that the following statements are complete and correct.

1. If the primal problem is a maximization problem, its dual will be a .... problem.
2. If any constraint in the primal is a perfect equality, the corresponding dual variable is .... in sign.
3. If the primal problem has an unbounded solution, the dual has either no solution or an .... solution.
4. If the primal has a finite optimal solution then the values of the objective functions of the primal and dual are ....
5. If **x** is any feasible solution to the primal and **w** is any feasible solution to the dual problem then Z_p .... Z_D where Z_P and Z_D are the objective functions of the primal and dual respectively.

True or False

Write 'T' for true and 'F' for false statement.

1. The dual of a dual is the primal itself.
2. If the primal problem has a finite feasible solution, the dual problem has no solution.

3. The coefficient matrix of the dual is obtained by transposing the coefficient matrix of the primal.
4. If the primal is a maximization problem, the dual is also a maximization problem.
5. Requirement vector of the primal is the price vector of the dual.
6. In a standard primal problem, if all the constraints have the sign ≥, it is a maximization problem.

Answers

Multiple Choice Questions

1. (b) 2. (a) 3. (c) 4. (c) 5. (c)
6. (c)

Fill in the Blank

1. minimization
2. unrestricted
3. unbounded
4. equal
5. ≤

True or False

1. *T* 2. *F* 3. *T* 4. *F* 5. *T*
6. *F*

□

Annihilators and Transpose of Linear Transformation

1. Annihilators

Definition: *If V is a vector space over the field F and S is a subset of V, the* **annihilator** *of S is the set S^0 of all linear functionals f on V such that*

$$f(\alpha) = 0 \;\; \forall \;\; \alpha \in S.$$

Sometimes $A(S)$ is also used to denote the annihilator of S.

Thus $S^0 = \{ f \in V' : f(\alpha) = 0 \;\; \forall \, \alpha \in S \}$.

It should be noted that we have defined the annihilator of S which is simply a subset of V. S should not necessarily be a subspace of V.

If $S =$ zero subspace of V, then $S^0 = V'$.

If $S = V$, then $S^0 = V^0 =$ zero subspace of V'.

If V is finite dimensional and S contains a non-zero vector, then $S^0 \neq V'$. If $\mathbf{0} \neq \alpha \in S$, then there is a linear functional f on V such that $f(\alpha) \neq 0$. Thus there is $f \in V'$ such that $f \notin S^0$. Therefore $S^0 \neq V'$.

Theorem 1: *If S is any subset of a vector space $V(F)$, then S^0 is a subspace of V'.*

Proof: First we see that S^0 is a non-empty subset of V' because at least $\hat{\mathbf{0}} \in S^0$. We have

$$\hat{\mathbf{0}}(\alpha) = 0 \; \forall \, \alpha \in S.$$

Let $f, g \in S^0$. Then $f(\alpha) = 0 \; \forall \, \alpha \in S$, and $g(\alpha) = 0 \; \forall \, \alpha \in S$.

If $a, b \in F$, then

$$(af + bg)(\alpha) = (af)(\alpha) + (bg)(\alpha) = af(\alpha) + bg(\alpha) = a0 + b0 = 0.$$

$\therefore \quad af + bg \in S^0.$

Thus $a, b \in F$ and $f, g \in S^0 \quad \Rightarrow \quad af + bg \in S^0$.

$\therefore \quad S^0$ is a subspace of V'.

Dimension of annihilator.

Theorem 2: *Let V be a finite dimensional vector space over the field F, and let W be a subspace of V. Then*

$$\dim W + \dim W^0 = \dim V.$$

Proof: If W is zero subspace of V, then $W^0 = V'$.

$$\therefore \quad \dim W^0 = \dim V' = \dim V.$$

Also in this case $\dim W = 0$. Hence the result.

Let us now suppose that W is a proper subspace of V.

Let $\dim V = n$, and $\dim W = m$ where $0 < m < n$.

Let $B_1 = \{\alpha_1, \ldots, \alpha_m\}$ be a basis for W. Since B_1 is a linearly independent subset of V also, therefore it can be extended to form a basis for V. Let $B = \{\alpha_1, \ldots, \alpha_m, \alpha_{m+1}, \ldots, \alpha_n\}$ be a basis for V.

Let $B' = \{f_1, \ldots, f_m, f_{m+1}, \ldots, f_n\}$ be the dual basis of B. Then B' is a basis for V' such that $f_i(\alpha_j) = \delta_{ij}$.

We claim that $S = \{f_{m+1}, \ldots, f_n\}$ is a basis for W^0.

Since $S \subset B'$, therefore S is linearly independent because B' is linearly independent. So S will be a basis for W^0, if W^0 is equal to the subspace of V' spanned by S *i.e.*, if $W^0 = L(S)$.

First we shall show that $W^0 \subseteq L(S)$. Let $f \in W^0$. Then $f \in V'$. So let

$$f = \sum_{i=1}^{n} x_i f_i. \qquad \ldots(1)$$

Now $f \in W^0 \Rightarrow f(\alpha) = 0 \ \forall \alpha \in W$

$\Rightarrow \quad f(\alpha_j) = 0$ for each $j = 1, \ldots, m$ $\qquad [\because \alpha_1, \ldots, \alpha_m$ are in $W]$

$\Rightarrow \quad \left(\sum_{i=1}^{n} x_i f_i\right)(\alpha_j) = 0$ $\qquad$ [From (1)]

$\Rightarrow \quad \sum_{i=1}^{n} x_i f_i(\alpha_j) = 0 \Rightarrow \sum_{i=1}^{n} x_i \delta_{ij} = 0$

$\Rightarrow \quad x_j = 0$ for each $j = 1, \ldots, m$.

Putting $x_1 = 0, x_2 = 0, \ldots, x_m = 0$ in (1), we get

$$f = x_{m+1} f_{m+1} + \ldots + x_n f_n$$

$=$ a linear combination of the elements of S.

$\therefore \quad f \in L(S)$.

Thus $f \in W^0 \Rightarrow f \in L(S)$.

$\therefore \quad W^0 \subseteq L(S)$.

Now we shall show that $L(S) \subseteq W^0$.

Let $g \in L(S)$. Then g is a linear combination of $f_{m+1}, \ldots, f_n$.

Let $$g = \sum_{k=m+1}^{n} y_k f_k. \qquad \ldots(2)$$

Let $\alpha \in W$. Then α is a linear combination of $\alpha_1, \ldots, \alpha_m$. Let

$$\alpha = \sum_{j=1}^{m} c_j \alpha_j. \qquad \ldots(3)$$

We have $g(\alpha) = g\left(\sum_{j=1} c_j \alpha_j\right)$ [From (3)]

$$= \sum_{j=1}^{m} c_j \, g(\alpha_j) \qquad [\because \; g \text{ is linear functional}]$$

$$= \sum_{j=1}^{m} c_j \left(\sum_{k=m+1}^{n} y_k \, f_k\right)(\alpha_j) \qquad \text{[From (2)]}$$

$$= \sum_{j=1}^{m} c_j \sum_{k=m+1}^{n} y_k \, f_k \,(\alpha_j) = \sum_{j=1}^{m} c_j \sum_{k=m+1}^{n} y_k \, \delta_{kj}$$

$$= \sum_{j=1}^{m} c_j \, 0 \qquad [\because \delta_{kj} = 0 \text{ if } k \neq j \text{ which is so for each } k = m+1, \ldots, n \text{ and for each } j = 1, \ldots, m]$$

$$= 0.$$

Thus $g(\alpha) = 0 \;\forall\; \alpha \in W$. Therefore $g \in W^0$.

Thus $g \in L(S) \;\Rightarrow\; g \in W^0$.

$\therefore \qquad L(S) \subseteq W^0$.

Hence $W^0 = L(S)$ and S is a basis for W^0.

$\therefore \qquad \dim W^0 = n - m = \dim V - \dim W$

or $\qquad \dim V = \dim W + \dim W^0$.

Corollary: *If V is finite-dimensional and W is a subspace of V, then W' is isomorphic to V'/W^0.*

Proof: Let $\dim V = n$ and $\dim W = m$. W' is dual space of W,

so $\qquad \dim W' = \dim W = m.$

Now $\dim V'/W^0 = \dim V' - \dim W^0$

$$= \dim V - (\dim V - \dim W) = \dim W = m.$$

Since $\dim W' = \dim V'/W^0$, therefore $W' \cong V'/W^0$.

Annihilator of an annihilator. Let V be a vector space over the field F. If S is any subset of V, then S^0 is a subspace of V'. By definition of an annihilator, we have

$$(S^0)^0 = S^{00} = \{L \in V'' : \; L(f) = 0 \;\; \forall \; f \in S^0\}.$$

Obviously S^{00} is a subspace of V''. But if V is finite dimensional, then we have identified V'' with V through the natural isomorphism $\alpha \leftrightarrow L_\alpha$. Therefore we may regard S^{00} as a subspace of V. Thus

$$S^{00} = \{\alpha \in V : f(\alpha) = 0 \;\forall\; f \in S^0\}.$$

Theorem 3: *Let V be a finite dimensional vector space over the field F and let W be a subspace of V. Then $W^{00} = W$.*

Proof: We have

and $\qquad W^{00} = \{\alpha \in V : f(\alpha) = 0 \ \forall \ f \in W^0\}.$ ...(2)

Let $\alpha \in W$. Then from (1), $f(\alpha) = 0 \ \ \forall \ f \in W^0$ and so from (2), $\alpha \in W^{00}$.

$\therefore \qquad \alpha \in W \ \Rightarrow \ \alpha \in W^{00}.$

Thus $W \subseteq W^{00}$. Now W is a subspace of V and W^{00} is also a subspace of V. Since $W \subseteq W^{00}$, therefore W is a subspace of W^{00}.

Now $\dim W + \dim W^0 = \dim V$. [by theorem (2)]

Applying the same theorem for vector space V' and its subspace W^0, we get

$$\dim W^0 + \dim W^{00} = \dim V' = \dim V.$$

$$\therefore \qquad \dim W = \dim V - \dim W^0 = \dim V - [\dim V - \dim W^{00}]$$

$$= \dim W^{00}.$$

Since W is a subspace of W^{00} and $\dim W = \dim W^{00}$, therefore $W = W^{00}$.

Illustrative Examples

Example 1: *If S_1 and S_2 are two subsets of a vector space V such that $S_1 \subseteq S_2$, then show that $S_2^0 \subseteq S_1^0$.*

Solution: Let $f \in S_2^0$. Then $\quad f(\alpha) = 0 \ \ \forall \ \alpha \in S_2$

$\Rightarrow \qquad f(\alpha) = 0 \ \forall \ \alpha \in S_1$ $\qquad [\because \ S_1 \subseteq S_2]$

$\Rightarrow \qquad f \in S_1^0.$

$\therefore \qquad S_2^0 \subseteq S_1^0.$

Example 2: *Let V be a vector space over the field F. If S is any subset of V, then show that $S^0 = [L(S)]^0$.*

Solution: We know that $S \subseteq L(S)$.

$\therefore \qquad [L(S)]^0 \subseteq S^0.$ ...(1)

Now let $f \in S^0$. Then $f(\alpha) = 0 \ \forall \ \alpha \in S$.

If β is any element of $L(S)$, then

$$\beta = \sum_{i=1}^{n} x_i \, \alpha_i \text{ where each } \alpha_i \in S.$$

We have $\quad f(\beta) = \sum_{i=1}^{n} x_i \, f(\alpha_i) = 0$, since each $f(\alpha_i) = 0$.

Thus $f(\beta) = 0 \ \forall \ \beta \in L(S)$.

$\therefore \qquad f \in (L(S))^0.$

From (1) and (2), we conclude that $S^0 = (L(S))^0$.

Example 3: *Let V be a finite-dimensional vector space over the field F. If S is any subset of V, then $S^{00} = L(S)$.*

Solution: *We have* $S^0 = (L(S))^0$. [See Example 2]

$\therefore$ $$S^{00} = (L(S))^{00}. \quad \text{...(1)}$$

But V is finite-dimensional and $L(S)$ is a subspace of V. Therefore by theorem 3,

$$(L(S))^{00} = L(S).$$

$\therefore$ from (1), we have

$$S^{00} = L(S).$$

Example 4: *Let W_1 and W_2 be subspaces of a finite dimensional vector space V.*

(a) Prove that $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$.

(b) Prove that $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$.

Solution: *(a) First we shall prove that*

$$W_1^0 \cap W_2^0 \subseteq (W_1 + W_2)^0.$$

Let $f \in W_1^0 \cap W_2^0$. Then $f \in W_1^0, f \in W_2^0$.

Suppose α is any vector in $W_1 + W_2$. Then

$$\alpha = \alpha_1 + \alpha_2 \text{ where } \alpha_1 \in W_1,\ \alpha_2 \in W_2.$$

We have $f(\alpha) = f(\alpha_1 + \alpha_2) = f(\alpha_1) + f(\alpha_2)$

$= 0 + 0$ [$\because$ $\alpha_1 \in W_1$ and $f \in W_1^0 \Rightarrow f(\alpha_1) = 0$ and similarly $f(\alpha_2) = 0$]

$= 0.$

Thus $f(\alpha) = 0 \;\forall\, \alpha \in W_1 + W_2$.

$\therefore$ $$f \in (W_1 + W_2)^0.$$

$\therefore$ $$W_1^0 \cap W_2^0 \subseteq (W_1 + W_2)^0. \quad \text{...(1)}$$

Now we shall prove that

$$(W_1 + W_2)^0 \subseteq W_1^0 \cap W_2^0.$$

We have $W_1 \subseteq W_1 + W_2$.

$\therefore$ $$(W_1 + W_2)^0 \subseteq W_1^0. \quad \text{...(2)}$$

Similarly, $W_2 \subseteq W_1 + W_2$.

$\therefore$ $$(W_1 + W_2)^0 \subseteq W_2^0. \quad \text{...(3)}$$

From (2) and (3), we have

$$(W_1 + W_2)^0 \subseteq W_1^0 \cap W_2^0. \quad \text{...(4)}$$

From (1) and (4), we have

(b) Let us use the result (a) for the vector space V' in place of the vector space V. Thus replacing W_1 by W_1^0 and W_2 by W_2^0 in (a), we get

$$(W_1^0 + W_2^0)^0 = W_1^{00} \cap W_2^{00}$$

$$\Rightarrow \quad (W_1^0 + W_2^0)^0 = W_1 \cap W_2 \qquad [\because W_1^{00} = W_1 \text{ etc.}]$$

$$\Rightarrow \quad (W_1^0 + W_2^0)^{00} = (W_1 \cap W_2)^0$$

$$\Rightarrow \quad W_1^0 + W_2^0 = (W_1 \cap W_2)^0.$$

Example 5: *If W_1 and W_2 are subspaces of a vector space V, and if $V = W_1 \oplus W_2$, then $V' = W_1^0 \oplus W_2^0$.*

Solution: To prove that $V' = W_1^0 \oplus W_2^0$, we are to prove that

(i) $W_1^0 \cap W_2^0 = \{\hat{\mathbf{0}}\}$

and (ii) $V' = W_1^0 + W_2^0$ *i.e.*, each $f \in V'$ can be written as $f_1 + f_2$

where $f_1 \in W_1^0$, $f_2 \in W_2^0$.

(i) First to prove that $W_1^0 \cap W_2^0 = \{\hat{\mathbf{0}}\}$.

Let $f \in W_1^0 \cap W_2^0$. Then $f \in W_1^0$ and $f \in W_2^0$.

If α is any vector in V, then, V being the direct sum of W_1 and W_2, we can write

$$\alpha = \alpha_1 + \alpha_2 \text{ where } \alpha_1 \in W_1,\ \alpha_2 \in W_2.$$

We have

$$\begin{aligned} f(\alpha) &= f(\alpha_1 + \alpha_2) \\ &= f(\alpha_1) + f(\alpha_2) && [\because f \text{ is linear functional}] \\ &= 0 + 0 && [\because f \in W_1^0 \text{ and } \alpha_1 \in W_1 \Rightarrow f(\alpha_1) = 0 \\ & && \text{and similarly } f(\alpha_2) = 0] \\ &= 0. \end{aligned}$$

Thus $f(\alpha) = 0 \ \forall \alpha \in V$.

$$\therefore \quad f = \hat{\mathbf{0}}.$$

$$\therefore \quad W_1^0 \cap W_2^0 = \{\hat{\mathbf{0}}\}.$$

(ii) Now to prove that $V' = W_1^0 + W_2^0$.

Let $f \in V'$.

If $\alpha \in V$, then α can be uniquely written as

$$\alpha = \alpha_1 + \alpha_2 \text{ where } \alpha_1 \in W_1, \alpha_2 \in W_2.$$

For each f, let us define two functions f_1 and f_2 from V into F such that

$$f_1(\alpha) = f_1(\alpha_1 + \alpha_2) = f(\alpha_2) \qquad \ldots(1)$$

and

$$f_2(\alpha) = f_2(\alpha_1 + \alpha_2) = f(\alpha_1). \qquad \ldots(2)$$

First we shall show that f_1 is a linear functional on V. Let $a, b \in F$ and $\alpha = \alpha_1 + \alpha_2, \beta = \beta_1 + \beta_2 \in V$ where $\alpha_1, \beta_1 \in W_1$ and $\alpha_2, \beta_2 \in W_2$. Then

$$= f_1 \left[(a\alpha_1 + b\beta_1) + (a\alpha_2 + b\beta_2) \right]$$

$$= f\,(a\alpha_2 + b\beta_2)$$

$$[\because \; a\alpha_1 + b\beta_1 \in W_1, a\alpha_2 + b\beta_2 \in W_2]$$

$$= af\,(\alpha_2) + bf\,(\beta_2) \qquad [\because \; f \text{ is linear functional}]$$

$$= af_1\,(\alpha) + bf_1\,(\beta) \qquad [\text{From (1)}]$$

$\therefore$ f_1 is linear functional on V *i.e.*, $f_1 \in V'$.

Now we shall show that $f_1 \in W_1^0$.

Let α_1 be any vector in W_1. Then α_1 is also in V. We can write

$$\alpha_1 = \alpha_1 + \mathbf{0}, \text{ where } \alpha_1 \in W_1, \mathbf{0} \in W_2 .$$

$\therefore$ from (1), we have

$$f_1\,(\alpha_1) = f_1\,(\alpha_1 + \mathbf{0}) = f\,(\mathbf{0}) = 0.$$

Thus $f_1\,(\alpha_1) = 0 \;\forall\; \alpha_1 \in W_1$.

$\therefore$ $$f_1 \in W_1^0.$$

Similarly we can show that f_2 is a linear functional on V and $f_2 \in W_2^0$.

Now we claim that $f = f_1 + f_2$.

Let α be any element in V. Let

$$\alpha = \alpha_1 + \alpha_2 , \text{ where } \alpha_1 \in W_1, \alpha_2 \in W_2 .$$

Then $(f_1 + f_2)\,(\alpha) = f_1\,(\alpha) + f_2\,(\alpha) = f\,(\alpha_2) + f\,(\alpha_1)$ [From (1) and (2)]

$$= f\,(\alpha_1) + f\,(\alpha_2) = f\,(\alpha_1 + \alpha_2)$$

$$[\because f \text{ is linear functional}]$$

$$= f\,(\alpha).$$

Thus $(f_1 + f_2)\,(\alpha) = f\,(\alpha) \;\forall\; \alpha \in V$.

$\therefore$ $$f = f_1 + f_2.$$

Thus $f \in V' \;\Rightarrow\; f = f_1 + f_2$

where $$f_1 \in W_1^0, f_2 \in W_2^0.$$

$\therefore$ $$V' = W_1^0 + W_2^0.$$

Hence $$V' = W_1^0 \oplus W_2^0$$

2. The Adjoint or the Transpose of a Linear Transformation

In order to bring some simplicity in our work we shall introduce a few changes in our notation of writing the image of an element of a vector space under a linear transformation and that under a linear functional. If T is a linear transformation on a vector space V and $\alpha \in V$, then in place of writing $T\,(\alpha)$ we shall simply write $T\alpha$ *i.e.*, we shall omit the brackets. Thus $T\alpha$ will mean the image of α under T. If will stand for $T_1 \, [T_2 \, (\alpha)]$.

Let f be a linear functional on V. If $\alpha \in V$, then in place of writing $f(\alpha)$ we shall write $[\alpha, f]$. This is the square brackets notation to write the image of a vector under a linear functional. Thus $[\alpha, f]$ will stand for $f(\alpha)$. If $a, b \in F$ and $\alpha, \beta \in V$, then in this new notation the linearity property of f

i.e., $$f(a\alpha + b\beta) = af(\alpha) + bf(\beta)$$

will be written as

$$[a\alpha + b\beta, f] = a[\alpha, f] + b[\beta, f].$$

Also if f and g are two linear functionals on V and $a, b \in F$, then the property defining addition and scalar multiplication of linear functionals *i.e.*, the property

$$(af + bg)(\alpha) = af(\alpha) + bg(\alpha)$$

will be written as

$$[\alpha, af + bg] = a[\alpha, f] + b[\alpha, g].$$

Note that in this new notation, we get

$$[\alpha, f] = f(\alpha), [\alpha, g] = g(\alpha).$$

Theorem 1. *Let U and V be vector spaces over the field F. For each linear transformation T from U into V, there is a unique linear transformation T' from V' into U' such that*

$$[T'(g)](\alpha) = g[T(\alpha)] \qquad \textit{(in old notation)}$$

or $$[\alpha, T'g] = [T\alpha, g] \qquad \textit{(in new notation)}$$

for every g in V' and α in U.

The linear transformation T' is called the **adjoint** or the **transpose** or the **dual** of T. In some books it is denoted by T^t or by T^*.

Proof : T is a linear transformation from U to V. U' is the dual space of U and V' is the dual space of V. Suppose $g \in V'$ *i.e.*, g is a linear functional on V. Let us define

$$f(\alpha) = g[T(\alpha)] \;\; \forall \;\; \alpha \in U \qquad \text{...(1)}$$

Then f is a function from U into F. We see that f is nothing but the product or composite of the two functions T and g where $T : U \to V$ and $g : V \to F$. Since both T and g are linear therefore f is also linear. Thus f is a linear functional on U *i.e.*, $f \in U'$. In this way T provides us with a rule T' which associates with each functional g on V a linear functional $f = T'(g)$ on U, defined by (1).

Thus $\quad T' : V' \to U'$ such that

$$T'(g) = f \;\; \forall \;\; g \in V' \text{ where } \; f(\alpha) = g[T(\alpha)] \;\; \forall \alpha \in U.$$

Putting $f = T'(g)$ in (1), we see that T' is a function from V' into U' such that

$$[T'(g)](\alpha) = g[T(\alpha)]$$

or in square brackets notation

$$[\alpha, T'g] = [T\alpha, g] \qquad \text{...(2)}$$

$$\forall \;\; g \in V' \text{ and } \forall \;\; \alpha \in U.$$

and $a, b \in F$.

Then we are to prove that

$$T'(ag_1 + bg_2) = aT'g_1 + bT'g_2 \qquad \ldots(3)$$

where $T'g_1$ stands for $T'(g_1)$ and $T'g_2$ stands for $T'(g_2)$.

We see that both the sides of (3) are elements of U' *i.e.*, both are linear functionals on U. So if α is any element of U, we have

$$[\alpha, T'(ag_1 + bg_2)] = [T\alpha, ag_1 + bg_2]$$

[From (2) because $ag_1 + bg_2 \in V'$]

$$= [T\alpha, ag_1] + [T\alpha, bg_2] \quad \text{[by def. of addition in } V'\text{]}$$

$$= a[T\alpha, g_1] + b[Ta, g_2]$$

[by def. of scalar multiplication in V']

$$= a[\alpha, T'g_1] + b[\alpha_1, T'g_2] \qquad \text{[From (2)]}$$

$$= [\alpha, aT'g_1] + [\alpha, bT'g_2]$$

[by def. of scalar multiplication in U'. Note that $T'g_1, T'g_2 \in U'$]

$$= [\alpha, aT'g_1 + bT'g_2] \qquad \text{[By addition in } U'\text{]}$$

Thus $\forall \ \alpha \in U$, we have

$$[\alpha, T'(ag_1 + bg_2)] = [\alpha, aT'g_1 + bT'g_2].$$

$\therefore$ $$T'(ag_1 + bg_2) = aT'g_1 + bT'g_2$$

[by def. of equality of two functions]

Hence T' is a linear transformation from V' into U'.

Now let us show that T' is uniquely determined for a given T. If possible, let T_1 be a linear transformation from V' into U' such that

$$[\alpha, T_1 g] = [T\alpha, g] \ \forall \ g \in V' \text{ and } \alpha \in U. \qquad \ldots(4)$$

Then from (2) and (4), we get

$$[\alpha, T_1 g] = [\alpha, T'g] \ \forall \ \alpha \in U, \forall \ g \in V'$$

$\Rightarrow$ $$T_1 g = T'g \ \forall \ g \in V' \ \Rightarrow \ T_1 = T'.$$

$\therefore$ T' is uniquely determined for each T. Hence the theorem.

Note. If T is a linear transformation on the vector space V, then in the proof of the above theorem we should simply replace U by V.

Theorem 2 : *If T is a linear transformation from a vector space U into a vector space V, then*

(i) the annihilator of the range of T is equal to the null space of T' i.e.,

$$[R(T)]^0 = N(T').$$

If in addition U and V are finite dimensional, then

(ii) $\rho(T') = \rho(T)$

and (iii) the range of T' is the annihilator of the null space of T i.e.,

$$R(T') = [N(T)]^0.$$

$$[\alpha, T' g] = [T\alpha, g] \ \forall \alpha \in U. \qquad \ldots(1)$$

Let $g \in N(T')$ which is a subspace of V'. Then

$T' g = \hat{\mathbf{0}}$ where $\hat{\mathbf{0}}$ is zero element of U' *i.e.*, $\hat{\mathbf{0}}$ is zero functional on U. Therefore from (1), we get

$$[T\alpha, g] = [\alpha, \hat{\mathbf{0}}] \ \forall \alpha \in U$$

$\Rightarrow$ $[T\alpha, g] = 0 \ \forall \alpha \in U$ $\quad [\because \hat{\mathbf{0}}(\alpha) = 0 \ \forall \alpha \in U]$

$\Rightarrow$ $g(\beta) = 0 \ \forall \beta \in R(T)$ $\quad [\because R(T) = \{\beta \in V : \beta = T(\alpha) \text{ for some } \alpha \in U\}]$

$\Rightarrow$ $g \in [R(T)]^0$.

$\therefore$ $N(T') \subseteq [R(T)]^0$.

Now let $g \in [R(T)]^0$ which is a subspace of V'. Then

$$g(\beta) = 0 \ \forall \beta \in R(T)$$

$\Rightarrow$ $[T\alpha, g] = 0 \ \forall \alpha \in U$ $\quad [\because \forall \alpha \in U, T\alpha \in R(T)]$

$\Rightarrow$ $[\alpha, T' g] = 0 \ \forall \alpha \in U$ $\quad$ [From (1)]

$\Rightarrow$ $T' g = \hat{\mathbf{0}}$ (zero functional on U) $\Rightarrow$ $g \in N(T')$.

$\therefore$ $[R(T)]^0 \subseteq N(T')$.

Hence $[R(T)]^0 = N(T')$.

(ii) Suppose U and V are finite dimensional. Let $\dim U = n$, $\dim V = m$. Let $r = \rho(T) =$ the dimension of $R(T)$.

Now $R(T)$ is a subspace of V. Therefore

$$\dim R(T) + \dim [R(T)]^0 = \dim V. \qquad \text{[See Th. 2 of article 1]}$$

$$\therefore \quad \dim [R(T)]^0 = \dim V - \dim R(T)$$

$$= \dim V - r = m - r.$$

By part (i) of this theorem $[R(T)]^0 = N(T')$.

$\therefore$ $\dim N(T') = m - r$ $\Rightarrow$ nullity of $T' = m - r$.

But T' is a linear transformation from V' into U'.

$\therefore$ $\rho(T') + \nu(T') = \dim V'$

or $\rho(T') = \dim V' - \nu(T')$

$= \dim V -$ nullity of $T' = m - (m - r) = r$.

$\therefore$ $\rho(T) = \rho(T') = r$.

(iii) T' is a linear transformation from V' into U'. Therefore $R(T')$ is a subspace of U'. Also $[N(T)]^0$ is a subspace of U' because $N(T)$ is a subspace of U. First we shall show that

$$R(T') \subseteq [N(T)]^0.$$

Let $f \in R(T')$.

If α is any vector in $N(T)$, then $T\alpha = \mathbf{0}$. We have

$$[\alpha, f] = [\alpha, T' g] = [T\alpha, g] = [\mathbf{0}, g] = 0.$$

Thus $\quad f(\alpha) = 0 \ \forall \ \alpha \in N(T)$.

Therefore $\quad f \in [N(T)]^0$.

$\therefore \quad R(T') \subseteq [N(T)]^0$

$\Rightarrow \quad R(T')$ is a subspace of $[N(T)]^0$.

Now $\quad \dim N(T) + \dim [N(T)]^0 = \dim U$. [Theorem 2 of article 1]

$\therefore \quad \dim [N(T)]^0 = \dim U - \dim N(T)$

$= \dim R(T)$ $\quad [\because \dim U = \dim R(T) + \dim N(T)]$

$= \rho(T)$

$= \rho(T') = \dim R(T')$.

Thus $\quad \dim R(T') = \dim [N(T)]^0$

and $\quad R(T') \subseteq [N(T)]^0$.

$\therefore \quad R(T') = [N(T)]^0$.

Note. If T is a linear transformation on a vector space V, then in the proof of the above theorem we should replace U by V and m by n.

Theorem 3 : *Let U and V be finite-dimensional vector spaces over the field F. Let B be an ordered basis for U with dual basis B', and let B_1 be an ordered basis for V with dual basis B_1'. Let T be a linear transformation from U into V. Let A be the matrix of T relative to B, B_1 and let C be the matrix of T' relative to B_1', B'. Then $C = A'$ i.e., the matrix C is the transpose of the matrix A.*

Proof : Let $\dim U = n$, $\dim V = m$.

Let $\quad B = \{\alpha_1, \ldots, \alpha_n\}, B' = \{f_1, \ldots, f_n\}$,

$B_1 = \{\beta_1, \ldots, \beta_m\}, B_1' = \{g_1, \ldots, g\}$.

Now T is a linear transformation from U into V and T' is that from V' into U'. The matrix A of T relative to B, B_1 will be of the type $m \times n$. If $A = [a_{ij}]_{m \times n}$, then by definition

$$T(\alpha_j) \text{ or simply } T\alpha_j = \sum_{i=1}^{m} a_{ij} \beta_i, j = 1, 2 \ldots, n. \qquad \ldots(1)$$

The matrix C of T' relative to B_1', B' will be of the type $n \times m$. If $C = [c_{ji}]_{n \times m}$, then by definition

$$T'(g_i) \text{ or simply } T' g_i = \sum_{j=1}^{n} c_{ji} f_j, i = 1, 2, \ldots, m \qquad \ldots(2)$$

Now $T' g_i$ is an element of U' *i.e.,* $T' g_i$ is a linear functional on U. If f is any linear functional on U, then we know that

$$f = \sum_{j=1}^{n} f(\alpha_j) f_j. \qquad \text{[See theorem 3 of Chapter 7, article 7.4]}$$

$$T' g_i = \sum_{j=1}^{n} \{(T' g_i)(\alpha_j)\} f_j . \qquad \ldots(3)$$

Now let us find $(T' g_i)(\alpha_j)$. We have

$$(T' g_i)(\alpha_j) = g_i T(\alpha_j) \qquad \text{[by def. of } T'\text{]}$$

$$= g_i \left(\sum_{k=1}^{m} a_{kj} \beta_k \right) \qquad \text{[From (1), replacing the suffix } i \text{ by } k \text{ which is immaterial]}$$

$$= \sum_{k=1}^{m} a_{kj} g_i(\beta_k) \qquad [\because g_i \text{ is linear}]$$

$$= \sum_{k=1}^{m} a_{kj} \delta_{ik} \qquad [\because g_i \in B_1' \text{ which is dual basis of } B_1]$$

$$= a_{ij} \qquad \text{[On summing with respect to } k \text{ and remembering that } \delta_{ik} = 1 \text{ when } k = i \text{ and } \delta_{ik} = 0 \text{ when } k \neq i]$$

Putting this value of $(T' g_i)(\alpha_j)$ in (3), we get

$$T' g_i = \sum_{j=1}^{m} a_{ij} f_j . \qquad \ldots(4)$$

Since $f_1, \ldots, f_n$ are linearly independent, therefore from (2) and (4), we get

$$c_{ji} = a_{ij} .$$

Hence by definition of transpose of a matrix, we have

$$C = A' .$$

Note. If T is a linear transformation on a finite-dimensional vector space V, then in the above theorem we put $U = V$ and $m = n$. Also according to our convention we take $B_1 = B$. The students should write the complete proof themselves.

Theorem 4 : *Let A be any m × n matrix over the field F. Then the row rank of A is equal to the column rank of A.*

Proof : Let $A = [a_{ij}]_{m \times n}$. Let

$$B = \{\alpha_1, \ldots, \alpha_n\} \text{ and } B_1 = \{\beta_1, \ldots, \beta_m\}$$

be the standard ordered bases for $V_n(F)$ and $V_m(F)$ respectively. Let T be the linear transformation from $V_n(F)$ into $V_m(F)$ whose matrix is A relative to ordered bases B and B_1. Then obviously the vectors $T(\alpha_1), \ldots, T(a_n)$ are nothing but the column vectors of the matrix A. Also these vectors span the range of T because $\alpha_1, \ldots, \alpha_n$ form a basis for the domain of T *i.e.,* $V_n(F)$.

$\therefore$ the range of T = the column space of A

$\Rightarrow$ the dimension of the range of T = the dimension of the column space of A

$\Rightarrow$ $\rho(T)$ = the column rank of A. ...(1)

If T' is the adjoint of the linear transformation T, then the matrix of T' relative to the dual bases B_1' and B' is the matrix A' which is the transpose of the matrix A. The reasoning as given in proving the result (1), we have

$$\rho(T') = \text{the column rank of } A'$$
$$= \text{the row rank of } A \qquad \ldots(2)$$

Since $\rho(T) = \rho(T')$, therefore from (1) and (2), we get the result that the column rank of A = the row rank of A.

Theorem 5 : *Prove the following properties of adjoints of linear operators on a vector space $V(F)$:*

(i) $\hat{\mathbf{0}}' = \hat{\mathbf{0}}$;

(ii) $I' = I$;

(iii) $(T_1 + T_2)' = T_1' + T_2'$;

(iv) $(T_1 T_2)' = T_2' T_1'$;

(v) $(aT)' = aT'$ *where* $a \in F$;

(vi) $(T^{-1})' = (T')^{-1}$ *if* T *is invertible ;*

(vii) $(T')' = T'' = T$ *if* V *is finite-dimensional.*

Proof : (i) If $\hat{\mathbf{0}}$ is the zero transformation on V, then by the definition of the adjoint of a linear transformation, we have

$$[\alpha, \hat{\mathbf{0}}' g] = [\hat{\mathbf{0}}\alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$= [\mathbf{0}, g] \text{ for every } g \text{ in } V' \qquad [\because \hat{\mathbf{0}}(\alpha) = \mathbf{0} \ \forall \ \alpha \in V]$$

$$= 0 \qquad [\because g(\mathbf{0}) = 0]$$

$$= [\alpha, \hat{\mathbf{0}}] \ \forall \ \alpha \in V \qquad [\text{Here } \hat{\mathbf{0}} \in V' \text{ and } \hat{\mathbf{0}}(\alpha) = 0]$$

$$= [\alpha, \hat{\mathbf{0}} g] \ \forall \ g \in V' \text{ and } \alpha \in V$$

[Here $\hat{\mathbf{0}}$ is the zero transformation on V']

Thus we have

$$[\alpha, \hat{\mathbf{0}}' g] = [\alpha, \hat{\mathbf{0}} g] \text{ for all } g \text{ in } V' \text{ and } \alpha \text{ in } V.$$

$$\therefore \qquad \hat{\mathbf{0}}' = \hat{\mathbf{0}}.$$

(ii) If I is the identity transformation on V, then by the definition of the adjoint of a linear transformation, we have

$$[\alpha, I' g] = [I\alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$= [\alpha, g] \quad \text{for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$[\because I(\alpha) = \alpha \ \forall \ \alpha \in V]$$

$$= [\alpha, Ig] \quad \text{for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

[Here I is the identity operator on V']

$$\therefore \qquad I' = I.$$

definition of adjoint, we have

$$[\alpha, (T_1 + T_2)' g] = [(T_1 + T_2) \alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$= [T_1 \alpha + T_2 \alpha, g]$$

[by def. of addition of linear transformations]

$$= [T_1 \alpha, g] + [T_2 \alpha, g] \qquad \text{[by linearity property of } g]$$

$$= [\alpha, T_1' g] + [\alpha, T_2' g] \qquad \text{[by def. of adjoint]}$$

$$= [\alpha, T_1' g + T_2' g]$$

[by def. of addition of linear functionals. Note that $T_1' g, T_2' g$ are elements of V']

$$= [\alpha, (T_1' + T_2') g].$$

Thus we have

$$[\alpha, (T_1 + T_2)' g] = [a, (T_1' + T_2') g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V.$$

$\therefore \qquad (T_1 + T_2)' g = (T_1' + T_2') g \; \forall \; g \in V'.$

$\therefore \qquad (T_1 + T_2)' = T_1' + T_2'.$

(iv) If T_1, T_2 are linear operators on V, then $T_1 T_2$ is also a linear operator on V. By the definition of adjoint, we have

$$[\alpha, (T_1 T_2)' g] = [(T_1 T_2) \alpha, g] \qquad \text{for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$= [(T_1) T_2 \alpha, g]$$

[by def. of product of linear transformations]

$$= [T_2 \alpha, T_1' g] \qquad \text{[by def. of adjoint]}$$

$$= [\alpha, T_2' T_1' g] \qquad \text{[by def. of adjoint]}$$

Thus we have

$$[\alpha, (T_1 T_2)' g] = [a, T_2' T_1' g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V.$$

$\therefore \qquad (T_1 T_2)' = T_2' T_1'.$

Note. This is called the reversal law for the adjoint of the product of two linear transformations.

(v) If T is a linear operator on V and $a \in F$, then aT is also a linear operator on V. By the definition of the adjoint, we have

$$[\alpha, (aT)' g] = [(aT) \alpha, g] \text{ for every } g \text{ in } V' \text{ and } \alpha \text{ in } V$$

$$= [a (T\alpha), g]$$

[by def. of scalar multiplication of a linear transformation]

$$= a [T\alpha, g] \qquad [\because g \text{ is linear}]$$

$$= a [\alpha, T' g] \qquad \text{[by def. of adjoint]}$$

$$= [\alpha, a (T' g)]$$

[by def. of scalar multiplication in V'. Note that $T' g \in V'$]

$$= [\alpha, (aT') g]$$

[by def. of scalar multiplication of T' by a]

$\therefore \qquad (aT)' = aT'.$

$T^{-1}T = I = TT^{-1}$

$\Rightarrow \qquad (T^{-1}T)' = I' = (TT^{-1})'$

$\Rightarrow \qquad T'(T^{-1})' = I = (T^{-1})'T'$ [Using results (ii) and (iv)]

$\therefore \qquad T'$ is invertible and $(T')^{-1} = (T^{-1})'$.

(vii) V is a finite dimensional vector space. T is a linear operator on V, T' is a linear operator on V' and $(T')'$ or T'' is a linear operator on V''. We have identified V'' with V through natural isomorphism $\alpha \leftrightarrow L_\alpha$ where $\alpha \in V$ and $L_\alpha \in V''$. Here L_α is a linear functional on V' and is such that

$$L_\alpha(g) = g(\alpha) \; \forall \; g \in V'. \qquad \ldots(1)$$

Through this natural isomorphism we shall take $\alpha = L_\alpha$ and thus T'' will be regarded as a linear operator on V.

Now T' is a linear operator on V'. Therefore by the definition of adjoint, we have

$$[g, T'' L_\alpha] = [g, (T')' L_\alpha] = [T' g, L_\alpha] \text{ for every } g \in V' \text{ and } \alpha \in V.$$

Now $T' g$ is an element of V'. Therefore from (1), we have

$$[T' g, L_\alpha] = [\alpha, T' g]$$

[Note that from (1), $L_\alpha(T' g) = (T' g)(\alpha)$]

$$= [T\alpha, g]. \qquad \text{[by def. of adjoint]}$$

Again $T'' L_\alpha$ is an element of V''. Therefore from (1), we have

$[g, T'' L_\alpha] = [\beta, g]$ where $\beta \in V$ and $\beta \leftrightarrow T'' L_\alpha$ under natural isomorphism

$= [T''\alpha, g]$ [$\because \beta = T'' L_\alpha = T''\alpha$ when we regard T'' as linear operator on V in place of V'']

Thus, we have

$[T\alpha, g] = [T''\alpha, g]$ for every g in V' and α in V

$\Rightarrow \qquad g(T\alpha) = g(T''\alpha)$ for every g in V' and α in V

$\Rightarrow \qquad g(T\alpha - T''\alpha) = 0$ for every g in V' and α in V

$\Rightarrow \qquad T\alpha - T''\alpha = \mathbf{0}$ for every α in V

$\Rightarrow \qquad (T - T'')\alpha = \mathbf{0}$ for every α in V

$\Rightarrow \qquad T - T'' = \hat{\mathbf{0}}$

$\Rightarrow \qquad T = T''.$

Illustrative Examples

Example 6 : *Let f be the linear functional on $\mathbf{R}^2$ defined by $f(x, y) = 2x - 5y$. For each linear mapping*

$$T : \mathbf{R}^3 \to \mathbf{R}^2 \text{ find } [T'(f)](x, y, z), \textit{ where}$$

(ii) $T(x, y, z) = (x + y + 2z, 2x + y)$,

(iii) $T(x, y, z) = (x + y, 0)$.

Solution : By definition of the transpose mapping,

$$T'(f) = f \circ T \quad i.e., \; [T'(f)]\alpha = f[T(\alpha)] \text{ for every } \alpha \in \mathbf{R}^3.$$

(i) $[T'(f)](x, y, z) = f[T(x, y, z)]$

$$= f(x - y, y + z) = 2(x - y) - 5(y + z) = 2x - 7y - 5z.$$

(ii) $[T'(f)](x, y, z) = f[T(x, y, z)]$

$$= f(x + y + 2z, 2x + y)$$

$$= 2(x + y + 2z) - 5(2x + y) = -8x - 3y + 4z.$$

(iii) $[T'(f)](x, y, z) = f[T(x, y, z)]$

$$= f(x + y, 0) = 2(x + y) - 5.0 = 2x + 2y.$$

Example 7 : *If A and B are similar linear transformations on a vector space V, then so also are A′ and B′.*

Solution : A is similar to B means that there exists an invertible linear transformation C on V such that

$$A = CBC^{-1} \quad \Rightarrow \quad A = (CBC^{-1})' \quad \Rightarrow \quad A' = (C^{-1})' B' C'.$$

Now C is invertible implies that C' is also invertible and

$$(C')^{-1} = (C^{-1})'.$$

$\therefore$ $$A' = (C')^{-1} B' C' \quad \Rightarrow \quad C' A' (C')^{-1} = B'$$

[Multiplying on right by $(C')^{-1}$ and on left by C']

$\Rightarrow$ B' is similar to A' $\Rightarrow$ A' and B' are similar.

Example 8 : *Let V be a finite dimensional vector space over the field F. Show that $T \to T'$ is an isomorphism of $L(V, V)$ onto $L(V', V')$.*

Solution : Let $\dim V = n$. Then $\dim V' = n$.

Also $\dim L(V, V) = n^2, \dim L(V', V') = n^2$.

Let $\psi : L(V, V) \to L(V', V')$ such that

$\psi(T) = T' \; \forall \; T \in L(V, V)$.

(i) ψ is linear transformation.

Let $a, b \in F$ and $T_1, T_2 \in L(V, V)$. Then

$$\psi(aT_1 + bT_2) = (aT_1 + bT_2)' \qquad \text{[by def. of } \psi]$$

$$= (aT_1)' + (bT_2)' \qquad [\because (A + B)' = A' + B']$$

$$= aT_1' + bT_2' \qquad [\because (aA)' = aA']$$

$$= a\psi(T_1) + b\psi(T_2) \qquad \text{[by def. of } \psi]$$

$\therefore$ ψ is a linear transformation from $L(V, V)$ into $L(V', V')$.

(ii) ψ is one-one.

Let $T_1, T_2 \in L(V, V)$.

$\Rightarrow \quad T_1'' = T_2'' \quad \Rightarrow \quad T_1 = T_2 \quad [\because V \text{ is finite-dimensional}]$

$\therefore \quad \psi$ is one-one.

(iii) ψ is onto.

We have $\dim L(V, V) = \dim L(V', V') = n^2$.

Since ψ is a linear transformation from $L(V, V)$ into $L(V', V')$ therefore ψ is one-one implies that ψ must be onto.

Hence ψ is an isomorphism of $L(V, V)$ onto $L(V', V')$.

Example 9 : *If A and B are linear transformations on a finite-dimensional vector space V, then prove that*

(i) $\rho(A + B) \leq \rho(A) + \rho(B)$

(ii) $\rho(AB) \leq \min\{\rho(A), \rho(B)\}$.

(iii) If B is invertible, then $\rho(AB) = \rho(BA) = \rho(A)$.

Solution : If A is a linear transformation on V, let $R(A)$ denote the range of A. We know that $R(A)$, $R(B)$, and $R(A + B)$ are all subspaces of V. First we shall prove that $R(A + B)$ is a subspace of $R(A) + R(B)$.

Let $\quad \alpha \in R(A + B)$.

Then $\quad \alpha = (A + B)(\beta)$ for some $\beta \in V$

$\quad = A(\beta) + B(\beta)$.

But $\quad A(\beta) \in R(A)$ and $B(\beta) \in R(B)$.

$\therefore \quad \alpha \in R(A) + R(B)$.

Thus $\quad R(A + B) \subseteq R(A) + R(B)$.

$\therefore \quad R(A + B)$ is a subspace of $R(A) + R(B)$.

$\therefore \quad \dim R(A + B) \leq \dim\{R(A) + R(B)\}$

i.e., $\quad \rho(A + B) \leq \dim \{R(A) + R(B)\} \quad$ ...(1)

Now if W_1 and W_2 are subspaces of a finite dimensional vector space V, then

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$$

$\Rightarrow \quad \dim(W_1 + W_2) \leq \dim W_1 + \dim W_2$

i.e., the dimension of the sum $\leq$ sum of the dimensions.

$\therefore \quad \dim\{R(A) + R(B)\} \leq \dim R(A) + \dim R(B)$,

i.e., $\quad \dim\{R(A) + R(B)\} \leq \rho(A) + \rho(B). \quad$ ...(2)

From (1) and (2), we get

$$\rho(A + B) \leq \rho(A) + \rho(B).$$

(ii) For every α in V, we have

$$(AB)(\alpha) = A[B(\alpha)].$$

$\therefore \quad$ the range of AB is a subset of the range of A *i.e.,*

$$R(AB) \subseteq R(A).$$

$\therefore \quad R(AB)$ is a subspace of $R(A)$.

$\therefore \quad \dim R(AB) \leq \dim R(A)$

Applying the result (1) for the transformations B' and A', we get

$$\rho(B'A') \leq \rho(B')$$

$$\Rightarrow \quad \rho[(AB)'] \leq \rho(B') \qquad [\because (AB)' = B'A']$$

$$\Rightarrow \quad \rho(AB) \leq \rho(B) \qquad [\because \rho(B) = \rho(B') \text{ and } \rho[(AB)'] = \rho(AB)]$$

Thus $\quad \rho(AB) \leq \rho(A)$, and $\rho(AB) \leq \rho(B)$. ...(2)

$\therefore \quad \rho(AB) \leq \text{minimum } \{\rho(A), \rho(B)\}$.

(iii) Let B be invertible. Then we can write

$$A = (AB)\, B^{-1}.$$

$$\therefore \quad \rho(A) = \rho[(AB)\, B^{-1}]$$

$$\leq \rho(AB) \qquad \text{[by result (1) proved in the proof of (ii)]}$$

Similarly, we can write

$$A = B^{-1}(BA).$$

$$\therefore \quad \rho(A) = \rho[B^{-1}(BA)]$$

$$\leq \rho(BA). \qquad \text{[by result (2) proved in the proof of (ii)]}$$

Now $\quad \rho(A) \leq \rho(AB)$, and $\rho(AB) \leq \rho(A)$ implies that

$$\rho(A) = \rho(AB).$$

Similarly $\quad \rho(A) \leq \rho(BA)$, and $\rho(BA) \leq \rho(A)$ implies that

$$\rho(A) = \rho(BA).$$

Example 10 : *If A and B are linear transformations on an n-dimensional vector space V, then prove that*

(i) $\quad \rho(AB) \geq \rho(A) + \rho(B) - n.$

(ii) $\quad \nu(AB) \leq \nu(A) + \nu(B).$ **(Sylvester's law of nullity)**

Solution : (i) First we shall prove, that if T is a linear transformation on V and W_1 is an h-dimensional subspace of V, then the dimension of $T(W_1)$ is $\geq h - \nu(T)$.

Since V is finite-dimensional, therefore the subspace W_1 will possess complement. Let $V = W_1 \oplus W_2$. Then

$$\dim W_2 = n - h = k \text{ (say).}$$

Since $V = W_1 + W_2$, therefore

$$T(V) = T(W_1) + T(W_2), \text{ as can be easily seen.}$$

$$\therefore \quad \dim T(V) = \dim [T(W_1) + T(W_2)]$$

$$\leq \dim T(W_1) + \dim T(W_2)$$

$$[\because \text{ the dimension of a sum is} \leq \text{the sum of the dimensions}]$$

But $\quad T(V) = \text{the range of } T.$

$\therefore \quad \dim T(V) = \rho(T).$

Thus $\quad \dim T(W_1) + \dim T(W_2) \geq \rho(T).$ ...(1)

Now $T(W_2)$ is a subspace of W_2. Therefore

$$\dim W_2 \geq \dim T(W_2). \qquad ...(2)$$

$$\dim T(W_1) + \dim W_2 \geq \rho(T)$$

$\Rightarrow$ $\dim T(W_1) \geq \rho(T) - \dim W_2$

$\Rightarrow$ $\dim T(W_1) \geq n - \nu(T) - k$ $\quad [\because \rho(T) + \nu(T) = n]$

$\Rightarrow$ $\dim T(W_1) \geq n - k - \nu(T)$

$\Rightarrow$ $\dim T(W_1) \geq h - \nu(T).$ $\quad \ldots(3)$

Now taking $T = A$ and $W_1 = B(V)$ in (3), we get

$$\dim A[B(V)] \geq \dim B(V) - \nu(A)$$

$\Rightarrow$ $\dim (AB)(V) \geq \rho(B) - \nu(A)$ $\quad [\because B(V) = \text{the range of } B]$

$\Rightarrow$ $\rho(AB) \geq \rho(B) - [n - \rho(A)]$

$\Rightarrow$ $\rho(AB) \geq \rho(A) + \rho(B) - n.$

(ii) We have $\rho(AB) + \nu(AB) = n.$

$\therefore$ $\rho(AB) = n - \nu(AB).$

But $\rho(AB) \geq \rho(A) + \rho(B) - n.$

$\therefore$ $n - v(AB) \geq \rho(A) + \rho(B) - n$

$\Rightarrow$ $\nu(AB) \leq [n - \rho(A)] + [n - \rho(B)]$

$\Rightarrow$ $\nu(AB) \leq \nu(A) + \nu(B)$ $\quad [\because \rho(A) + \nu(A) = n]$

Comprehensive Exercise 1

1. Let V be a finite dimensional vector space over the field F. If W_1 and W_2 are subspaces of V, then $W_1^0 = W_2^0$ iff $W_1 = W_2$.

2. If W_1 and W_2 are subspaces of a finite-dimensional vector space V and if $V = W_1 \oplus W_2$, then
   (i) W_1' is isomorphic to W_2^0.
   (ii) W_2' is isomorphic to W_1^0.

3. Let W be the subspace of $\mathbf{R}^3$ spanned by $(1, 1, 0)$ and $(0, 1, 1)$. Find a basis of the annihilator of W.

4. Let W be the subspace of $\mathbf{R}^4$ spanned by $(1, 2, -3, 4), (1, 3, -2, 6)$ and $(1, 4, -1, 8)$. Find a basis of the annihilator of W.

Answers 1

3. Basis is $\{f(x, y, z) = x - y + z\}$
4. Basis is $\{f_1, f_2\}$ where $f_1(x, y, z, w) = 5x - y + z$, $f_2(x, y, z, w) = 2y - w$

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. If S is any subset of a vector space $V(F)$, then S^0 is a subspace of

(a) V (b) V'

(c) V'' (d) None of these.

2. Let V be a finite dimensional vector space over the field F. If S is any subset of V, then $S^{00} =$

(a) S (b) $L(S)$

(c) $[L(S)]^0$ (d) None of these.

Fill in the Blank

Fill in the blanks "......" so that the following statements are complete and correct.

1. Let V be a finite dimensional vector space over the field F, and let W be a subspace of V. Then

$$\dim W + \dim W^0 = \ldots\ldots$$

2. Let V be a finite dimensional vector space over the field F and let W be a subspace of V. Then $W^{00} = \ldots\ldots$

True or False

Write 'T' for true and 'F' for false statement.

1. If V is finite-dimensional and W is a subspace of V then W' is isomorphic to V'/W^0.

2. If T is a linear transformation from a vector space U into a vector space V then $\rho(T') \neq \rho(T)$.

Answers

Multiple Choice Questions

1. (b) 2. (b)

Fill in the Blank

1. $\dim V$ 2. W

True or False

1. T 2. F

APPENDIX - II

Bilinear, Quadratic and Hermitian Forms

1 External Direct Product of Two Vector Spaces

Suppose U and V are two vector spaces over the same field F. Let

$$W = U \times V \quad i.e., \quad W = \{(\alpha, \beta) : \alpha \in U, \beta \in V\}.$$

If (α_1, β_1) and (α_2, β_2) are two elements in W, then we define their equality as follows :

$$(\alpha_1, \beta_1) = (\alpha_2, \beta_2) \text{ if } \alpha_1 = \alpha_2 \text{ and } \beta_1 = \beta_2 .$$

Also we define the sum of (α_1, β_1) and (α_2, β_2) as follows :

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1 + \beta_2).$$

If c is any element in F and (α, β) is any element in W, then we define scalar multiplication in W as follows :

$$c\,(\alpha, \beta) = (c\alpha, c\beta).$$

It can be easily shown that with respect to addition and scalar multiplication as defined above, W is a vector space over the field F. We call W as the external direct product of the vector spaces U and V and we shall write $W = U \oplus V$.

Now we shall consider some special type of scalar-valued functions on W known as *bilinear forms.*

2 Bilinear Forms

Definition: *Let U and V be two vector spaces over the same field F. A bilinear form on $W = U \oplus V$ is a function f from W into F, which assigns to each element (α, β) in W a scalar $f(\alpha, \beta)$ in such a way that*

$$f(a\alpha_1 + b\alpha_2, \beta) = a f(\alpha_1, \beta) + b f(\alpha_2, \beta)$$

and

$$f(\alpha, a\beta_1 + b\beta_2) = a f(\alpha, \beta_1) + b f(\alpha, \beta_2).$$

Here $f(\alpha, \beta)$ is an element of F. It denotes the image of (α, β) under the function f. Thus a bilinear form on W is a function from W into F which is linear as a function of either of its arguments when the other is fixed.

If $U = V$, then in place of saying that f is a bilinear form on $W = U \oplus V$, we shall simply say that f is a bilinear form on V.

Thus if V is a vector space over the field F, then a bilinear form on V is a function f, which assigns to each ordered pair of vectors α, β in V a scalar $f(\alpha, \beta)$ in F, and which satisfies

$$f(a\alpha_1 + b\alpha_2, \beta) = a f(\alpha_1, \beta) + b f(\alpha_2, \beta)$$

and

$$f(\alpha, a\beta_1 + b\beta_2) = a f(\alpha, \beta_1) + b f(\alpha, \beta_2).$$

Example 1: *Suppose V is a vector space over the field F. Let L_1, L_2 be linear functionals on V. Let f be a function from $V \times V$ into F defined as*

$$f(\alpha, \beta) = L_1(\alpha) L_2(\beta).$$

Then f is a bilinear form on V.

Solution: If $\alpha, \beta \in V$, then $L_1(\alpha), L_2(\beta)$ are scalars. We have

$$\begin{aligned} f(a\alpha_1 + b\alpha_2, \beta) &= L_1(a\alpha_1 + b\alpha_2) L_2(\beta) \\ &= [a L_1(\alpha_1) + b L_1(\alpha_2)] L_2(\beta) \\ &= aL_1(\alpha_1) L_2(\beta) + bL_1(\alpha_2) L_2(\beta) \\ &= af(\alpha_1, \beta) + bf(\alpha_2, \beta). \end{aligned}$$

Also
$$\begin{aligned} f(\alpha, a\beta_1 + b\beta_2) &= L_1(\alpha) L_2(a\beta_1 + b\beta_2) \\ &= L_1(\alpha)[a L_2(\beta_1) + b L_2(\beta_2)] \\ &= aL_1(\alpha) L_2(\beta_1) + b L_1(\alpha) L_2(\beta_2) \\ &= a f(\alpha, \beta_1) + b f(\alpha, \beta_2). \end{aligned}$$

Hence f is a bilinear form on V.

Example 2: *Suppose V is a vector space over the field F. Let T be a linear operator on V and f a bilinear form on V. Suppose g is a function from $V \times V$ into F defined as*

$$g(\alpha, \beta) = f(T\alpha, T\beta).$$

Then g is a bilinear form on V.

Solution: We have
$$\begin{aligned} g(a\alpha_1 + b\alpha_2, \beta) &= f(T(a\alpha_1 + b\alpha_2), T\beta) \\ &= f(aT\alpha_1 + bT\alpha_2, T\beta) \\ &= a f(T\alpha_1, T\beta) + b f(T\alpha_2, T\beta) \\ &= a g(\alpha_1, \beta) + b g(\alpha_2, \beta). \end{aligned}$$

Also
$$\begin{aligned} g(\alpha, a\beta_1 + b\beta_2) &= f(T\alpha, T(a\beta_1 + b\beta_2)) \\ &= f(T\alpha, aT\beta_1 + bT\beta_2) \\ &= a f(T\alpha, T\beta_1) + b f(T\alpha, T\beta_2) \\ &= a g(\alpha, \beta_1) + b g(\alpha, \beta_2). \end{aligned}$$

Hence g is a bilinear form on V.

3 Matrix of a Bilinear Form

Theorem 1: *If U is an n-dimensional vector space with basis $\{\alpha_1, \ldots, \alpha_n\}$, if V is an m-dimensional vector space with basis $\{\beta_1, \ldots, \beta_m\}$, and if $\{a_{ij}\}$ is any set of nm scalars $(i = 1, \ldots, n;\ j = 1, \ldots, m)$, then there is one and only one bilinear form f on $U \oplus V$ such that $f(\alpha_i, \beta_j) = a_{ij}$ for all i and j.*

Proof: Let $\alpha = \sum_{i=1}^{n} x_i \alpha_i \in U$ and $\beta = \sum_{j=1}^{m} y_j \beta_j \in V$.

$$f(\alpha, \beta) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i \, y_j \, a_{ij} . \qquad \ldots(1)$$

We shall show that f is a bilinear form on $U \times V$.

Let $a, b \in F$ and let $\alpha_1, \alpha_2 \in U$.

Let $\qquad \alpha_1 = \sum_{i=1}^{n} a_i \, \alpha_i, \; \alpha_2 = \sum_{i=1}^{n} b_i \, \alpha_i$.

Then $\qquad f(\alpha_1, \beta) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \, y_j \, a_{ij}$

and $\qquad f(\alpha_2, \beta) = \sum_{i=1}^{n} \sum_{j=1}^{m} b_i \, y_j \, a_{ij}$.

Also $\quad a\alpha_1 + b\alpha_2 = a \sum_{i=1}^{n} a_i \, \alpha_i + b \sum_{i=1}^{n} b_i \, \alpha_i = \sum_{i=1}^{n} (aa_i + bb_i) \, \alpha_i$.

$$\therefore \qquad f(a\alpha_1 + b\alpha_2, \beta) = \sum_{i=1}^{n} \sum_{j=1}^{m} (aa_i + bb_i) \, y_j \, a_{ij}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} aa_i \, y_j \, a_{ij} + \sum_{i=1}^{n} \sum_{j=1}^{m} bb_i \, y_j \, a_{ij}$$

$$= a \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \, y_j \, a_{ij} + b \sum_{i=1}^{n} \sum_{j=1}^{m} b_i \, y_j \, a_{ij}$$

$$= af(\alpha_1, \beta) + bf(\alpha_2, \beta).$$

Similarly, we can prove that if $a, b \in F$, and $\beta_1, \beta_2 \in V$, then

$$f(\alpha, a\beta_1 + b\beta_2) = a f(\alpha, \beta_1) + b f(\alpha, \beta_2).$$

Therefore f is a bilinear form on $U \times V$.

Now $\qquad \alpha_i = 0\alpha_1 + \ldots + 0\alpha_{i-1} + 1\alpha_i + 0\alpha_{i+1} + \ldots + 0\alpha_n$

and $\qquad \beta_j = 0\beta_1 + \ldots + 0\beta_{j-1} + 1\beta_j + 0\beta_{j+1} + \ldots + 0\beta_m$.

Therefore from (1), we have

$$f(\alpha_i, \beta_j) = a_{ij} .$$

Thus there exists a bilinear form f on $U \times V$ such that

$$f(\alpha_i, \beta_j) = a_{ij} .$$

Now to show that f is unique.

Let g be a bilinear form on $U \times V$ such that $\quad g(\alpha_i, \beta_j) = a_{ij}$. $\qquad \ldots(2)$

If $\alpha = \sum_{i=1}^{n} x_i \, \alpha_i$ be in U and $\beta = \sum_{j=1}^{m} y_j \, \beta_j$ be in V, then

$$g(\alpha, \beta) = g\left(\sum_{i=1}^{n} x_i \, \alpha_i, \sum_{j=1}^{m} y_j \, \beta_j \right)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} x_i \, y_j \, g(\alpha_i, \beta_j) \qquad [\because \; g \text{ is a bilinear form}]$$

$$= \sum_{i=1} \sum_{j=1} x_i\, y_j\, a_{ij} \qquad \text{[from (2)]}$$

$$= f\,(\alpha\,,\beta). \qquad \text{[from (1)]}$$

∴ By the equality of two functions, we have

$$g = f.$$

Thus f is unique.

Matrix of a bilinear form. Definition : *Let V be a finite-dimensional vector space and let $B = \{\alpha_1, \ldots, \alpha_n\}$ be an ordered basis for V. If f is a bilinear form on V, the* **matrix of f in the ordered basis** *B is the $n \times n$ matrix $\mathbf{A} = [a_{ij}]_{n \times n}$ such that*

$$f\,(\alpha_i\,, \alpha_j) = a_{ij}\,,\ i = 1, \ldots, n\,;\ j = 1, \ldots, n.$$

We shall denote this matrix $\mathbf{A}$ by $[\,f\,]_B$.

Rank of a bilinear form. Definition: *The rank of a bilinear form is defined as the rank of the matrix of the form in any ordered basis.*

Let us describe all bilinear forms on a finite-dimensional vector space V of dimension n.

If $\qquad \alpha = \sum_{i=1}^{n} x_i\, \alpha_i$, and $\beta = \sum_{j=1}^{n} y_j\, \alpha_j$ are vectors in V, then

$$f\,(\alpha, \beta) = f\left(\sum_{i=1}^{n} x_i\, \alpha_i\,,\ \sum_{j=1}^{n} y_j\, \alpha_j\right) = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i\, y_j\, f\,(\alpha_i\,, \alpha_j)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} x_i\, y_j\, a_{ij} = \mathbf{X}'\,\mathbf{AY},$$

where $\mathbf{X}$ and $\mathbf{Y}$ are coordinate matrices of α and β in the ordered basis $\mathbf{B}$ and $\mathbf{X}'$ is the transpose of the matrix $\mathbf{X}$. Thus

$$f\,(\alpha, \beta) = [\,\alpha\,]_B{}'\,\mathbf{A}\,[\,\beta\,]_B\,.$$

From the definition of the matrix of a bilinear form, we note that if f is a bilinear form on an n-dimensional vector space V over the field F and B is an ordered basis of V, then there exists a unique $n \times n$ matrix $\mathbf{A} = [\,a_{ij}\,]_{n \times n}$ over the field F such that

$$\mathbf{A} = [\,f\,]_B\,.$$

Conversely, if $\mathbf{A} = [\,a_{ij}\,]_{n \times n}$ be an $n \times n$ matrix over the field F, then from theorem 1, we see that there exists a unique bilinear form f on V such that

$$[\,f\,]_B = [\,a_{ij}\,]_{n \times n}\,.$$

If $\alpha = \sum_{i=1}^{n} x_i\, \alpha_i$, $\beta = \sum_{j=1}^{n} y_j\, \alpha_j$ are vectors in V, then the bilinear form f is defined as

$$f\,(\alpha, \beta) = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i\, y_j\, a_{ij} = \mathrm{X}'\,\mathrm{AY}, \qquad \ldots(1)$$

where $\mathbf{X}, \mathbf{Y}$ are the coordinate matrices of α, β in the ordered basis B. Hence the bilinear forms on V are precisely those obtained from an $n \times n$ matrix as in (1).

Example 3: *Let f be the bilinear form on V_2 (**R**) defined by*

$$f\ ((x_1, y_1), (x_2, y_2)) = x_1\ y_1 + x_2\ y_2\ .$$

Find the matrix of f in the ordered basis

$$B = \{(1, -1), (1, 1)\} \text{ of } V_2\ (\mathbf{R}).$$

Solution: Let $B = \{\alpha_1, \alpha_2\}$

where $\alpha_1 = (1, -1), \alpha_2 = (1, 1)$.

We have
$$f\ (\alpha_1, \alpha_1) = f\ ((1, -1), (1, -1)) = -1 - 1 = -2\ ,$$
$$f\ (\alpha_1, \alpha_2) = f\ ((1, -1), (1, 1)) = -1 + 1 = 0,$$
$$f\ (\alpha_2, \alpha_1) = f\ ((1, 1), (1, -1)) = 1 - 1 = 0,$$
$$f\ (\alpha_2, \alpha_2) = f\ ((1, 1), (1, 1)) = 1 + 1 = 2\ .$$

$\therefore$
$$[f]_B = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}.$$

4 Bilinear Form Corresponding to a Given Matrix

Bilinear Form. Definition: *Let V_m and V_n be two vector spaces over the same field F and let $\mathbf{A} = [a_{ij}]_{m \times n}$ be an $m \times n$ matrix over the field F.*

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_m \end{bmatrix}$ *and* $\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ \dots \\ y_n \end{bmatrix}$ *be any two elements of V_m and V_n respectively so that*

$\mathbf{X}^T$ = *the transpose of the column matrix* $\mathbf{X}$

$$= [x_1 \quad x_2 \quad \dots \quad \dots \quad x_m]$$

and $\mathbf{Y}^T = [y_1 \quad y_2 \quad \dots \quad \dots \quad y_n]$.

Then an expression of the form

$$b\ (\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y} = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}\ x_i\ y_j \qquad \dots(1)$$

is called a bilinear form over the field F corresponding to the matrix $\mathbf{A}$.

It should be noted that b (**X**, **Y**) is an element of the field F and b **(X, Y)** is a mapping from

$$V_m \times V_n \rightarrow F.$$

The matrix **A** is called the **matrix** of the bilinear form (1) and the rank of the matrix **A** is called the rank of the bilinear form (1).

the matrix $\mathbf{A}$ which occurs in the i th row and the j th column.

Symmetric bilinear form. The bilinear form (1) is said to be **symmetric** if its matrix $\mathbf{A}$ is a symmetric matrix.

If the field F is the real field $\mathbf{R}$, then the bilinear form (1) is said to be a **real bilinear form.** Thus in a real bilinear form $b(\mathbf{X}, \mathbf{Y})$ assumes real values.

If the vectors $\mathbf{X}$ and $\mathbf{Y}$ belong to the same vector space V_n over a field F, then $\mathbf{A}$ is a square matrix and $\mathbf{X}^T \mathbf{AY}$ is called a bilinear form on the vector space V_n over the field F.

In order to show that $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{AY}$ given by (1) is a bilinear form, first we show that the mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_m \to F$.

Let $\mathbf{X}_1, \mathbf{X}_2$ be any two elements of V_m and α, β be any two elements of the field F and let the vector $\mathbf{Y}$ be fixed. Then

$$\begin{aligned} b(\alpha\mathbf{X}_1 + \beta\mathbf{X}_2, Y) &= (\alpha\, \mathbf{X}_1 + \beta\mathbf{X}_2)_T\ \mathbf{AY} \\ &= (\alpha\, \mathbf{X}_1{}^T + \beta\mathbf{X}_2{}^T)\, \mathbf{AY} \\ &= \alpha\, \mathbf{X}_1{}^T\ \mathbf{AY} + \beta\mathbf{X}_2{}^T\ \mathbf{AY} \\ &= \alpha b(\mathbf{X}_1, \mathbf{Y}) + \beta b(\mathbf{X}_2, \mathbf{Y}). \end{aligned}$$

$\therefore$ The mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_m \to F$.

Now we show that the mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_n \to F$.

Let $\mathbf{Y}_1, \mathbf{Y}_2$ be any two elements of V_n and α, β be any two elements of the field F and let the vector $\mathbf{X}$ be fixed. Then

$$\begin{aligned} b(\mathbf{X}, \alpha\mathbf{Y}_1 + \beta\mathbf{Y}_2) &= \mathbf{X}^T\mathbf{A}\,(\alpha\mathbf{Y}_1 + \beta\mathbf{Y}_2) \\ &= \alpha\, \mathbf{X}^T\mathbf{AY}_1 + \beta\mathbf{X}^T\mathbf{AY}_2 \\ &= \alpha\, b(\mathbf{X}, \mathbf{Y}_1) + \beta b(\mathbf{X}, \mathbf{Y}_2). \end{aligned}$$

$\therefore$ The mapping $b(\mathbf{X}, \mathbf{Y})$ is a linear mapping from $V_n \to F$.

Hence, $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T\mathbf{AY} = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}\, x_i\, y_j$ is a bilinear form.

Example of a real bilinear form

$$\text{Let } b(\mathbf{X}, \mathbf{Y}) = [x_1\ x_2]\begin{bmatrix} 5 & 3 & 1 \\ 7 & 4 & 9 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= 5x_1\, y_1 + 3x_1\, y_2 + x_1\, y_3 + 7x_2\, y_1 + 4x_2\, y_2 + 9x_2\, y_3\,.$$

Then $b(\mathbf{X}, \mathbf{Y})$ is an example of a real bilinear form.

The matrix $\begin{bmatrix} 5 & 3 & 1 \\ 7 & 4 & 9 \end{bmatrix}$ is the matrix of this bilinear form. The rank of this matrix is 2 and so the rank of this bilinear form is 2.

F. The matrix $\mathbf{A}$ *is said to be equivalent to the matrix* $\mathbf{B}$ *if there exist non-singular matrices* $\mathbf{P}$ *and* $\mathbf{Q}$ *of orders m and n respectively such that*

$$\mathbf{B} = \mathbf{P}^T \mathbf{A} \mathbf{Q}.$$

Equivalent Bilinear Forms. Definition: *Let* $\mathbf{V}_m$ *and* $\mathbf{V}_n$ *be two vector spaces over the same field F and let* $\mathbf{A}$ *and* $\mathbf{D}$ *be two* $m \times n$ *matrices over the field F.*

The bilinear form $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y}$ *is said to be equivalent to the bilinear form* $b(\mathbf{U}, V) = \mathbf{U}^T \mathbf{D} \mathbf{V}$, *where* $\mathbf{X}$ *and* $\mathbf{U}$ *are in* $\mathbf{V}_m$, $\mathbf{Y}$ *and* $\mathbf{V}$ *are in* V_n *if there exist non-singular matrices* $\mathbf{B}$ *and* $\mathbf{C}$ *of orders m and n respectively such that*

$$\mathbf{D} = \mathbf{B}^T \mathbf{A} \mathbf{C}$$

i.e., the matrices $\mathbf{A}$ *and* $\mathbf{D}$ *are equivalent.*

Thus, the bilinear form $b(\mathbf{U}, \mathbf{V})$ equivalent to the bilinear $b(\mathbf{X}, \mathbf{Y})$ is given by

$$b(\mathbf{U}, \mathbf{V}) = \mathbf{U}^T (\mathbf{B}^T \mathbf{A} \mathbf{C}) \mathbf{V} = \mathbf{U}^T \mathbf{D} \mathbf{V},$$

where $\mathbf{D} = \mathbf{B}^T \mathbf{A} \mathbf{C}$ and $\mathbf{B}$ and $\mathbf{C}$ are non-singular matrices of orders m and n respectively.

The transformations of vectors yielding these equivalent bilinear forms are

$$\mathbf{X} = \mathbf{B} \mathbf{U}, \quad \mathbf{Y} = \mathbf{C} \mathbf{V}.$$

The matrices $\mathbf{B}$ and $\mathbf{C}$ are called the transformation matrices yielding these equivalent bilinear forms.

Equivalent Canonical form of a given bilinear form.

Let V_m and V_n be two vector spaces over the same field F and let A be an $m \times n$ matrix over the field F.

Let $$b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{A} \mathbf{Y},$$

where $$\mathbf{X}^T = [x_1 \quad x_2 \quad \ldots \quad x_m],$$

$$\mathbf{Y}^T = [y_1 \quad y_2 \quad \ldots \quad y_n]$$

be a given bilinear form over the field F.

If the matrix $\mathbf{A}$ is of rank r, then there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ of orders m and n respectively such that

$$\mathbf{P}^T \mathbf{A} \mathbf{Q} = \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$$

is in the normal form.

If we transform the vectors $\mathbf{X}$ and $\mathbf{Y}$ to $\mathbf{U}$ and $\mathbf{V}$ by the transformations

$$\mathbf{X} = \mathbf{P} \mathbf{U}, \quad \mathbf{Y} = \mathbf{Q} \mathbf{V},$$

then the bilinear form $b(\mathbf{U}, \mathbf{V})$ equivalent to the bilinear form $b(\mathbf{X}, \mathbf{Y})$ is given by

$$b(\mathbf{U}, \mathbf{V}) = \mathbf{U}^T (\mathbf{P}^T \mathbf{A} \mathbf{Q}) \mathbf{V} = \mathbf{U}^T \begin{bmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \mathbf{V}$$

$$= u_1 v_1 + u_2 v_2 + \ldots + u_r v_r,$$

where $$\mathbf{U} = [u_1 \ u_2 \ \ldots \ u_m]^T \text{ and } \mathbf{V} = [v_1 \ v_2 \ \ldots \ v_n]^T.$$

the *equivalent normal form* of the bilinear form

$$b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{AY}.$$

Congruent Matrices. Definition: *A square matrix* $\mathbf{B}$ *of order n over a field F is said to be congruent to another square matrix* $\mathbf{A}$ *of order n over F, if there exists a non-singular matrix* $\mathbf{P}$ *over F such that*

$$\mathbf{B} = \mathbf{P}^T \mathbf{AP}.$$

Cogradient Transformations. Definition : *Let* $\mathbf{X}$ *and* $\mathbf{Y}$ *be vectors belonging to the same vector space* V_n *over a field F and let* $\mathbf{A}$ *be a square matrix of order n over F. Let*

$$b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{AY}$$

be a bilinear form on the vector space V_n *over F.*

Let $\mathbf{B}$ *be a non-singular matrix of order n and let the vectors* $\mathbf{X}$ *and* $\mathbf{Y}$ *be transformed to the vectors* $\mathbf{U}$ *and* $\mathbf{V}$ *by the transformations*

$$\mathbf{X} = \mathbf{BU}, \quad \mathbf{Y} = \mathbf{BV}.$$

Then the bilinear form $b(\mathbf{X}, \mathbf{Y})$ *transforms to the equivalent bilinear form*

$$b(\mathbf{U}, \mathbf{V}) = \mathbf{U}^T (\mathbf{B}^T \mathbf{AB}) \mathbf{V} = \mathbf{U}^T \mathbf{DV},$$

where $\mathbf{D} = \mathbf{B}^T \mathbf{AB}$ *and* $\mathbf{U}$ *and* $\mathbf{V}$ *are both n-vectors.*

The bilinear form $\mathbf{U}^T \mathbf{DV}$ *is said to be congruent to the bilinear form* $\mathbf{X}^T \mathbf{AY}$*. Under such circumstances when* $\mathbf{X}$ *and* $\mathbf{Y}$ *are subjected to the same transformation* $\mathbf{X} = \mathbf{BU}$ *and* $\mathbf{Y} = \mathbf{BV}$*, we say that* $\mathbf{X}$ *and* $\mathbf{Y}$ *are transformed cogradiently.*

Here, the matrix $\mathbf{A}$ is congruent to $\mathbf{B}$ because $\mathbf{D} = \mathbf{B}^T \mathbf{AB}$, where $\mathbf{B}$ is non- singular.

Illustrative Examples

Example 4: *Find the matrix* $\mathbf{A}$ *of each of the following bilinear forms* $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{AY}$.

(i) $3x_1 y_1 + x_1 y_2 - 2x_2 y_1 + 3x_2 y_2 - 3x_1 y_3$

(ii) $2x_1 y_1 + x_1 y_2 + x_1 y_3 + 3x_2 y_1 - 2x_2 y_3 + x_3 y_2 - 5x_3 y_3$

Which of the above forms is symmetric ?

Solution: (i) The element a_{ij} of the matrix $\mathbf{A}$ is the coefficient of $x_i y_j$ in the given bilinear form.

$\therefore$ $$\mathbf{A} = \begin{bmatrix} 3 & 1 & -3 \\ -2 & 3 & 0 \end{bmatrix}.$$

The matrix $\mathbf{A}$ is not a symmetric matrix. So the given bilinear form is not symmetric.

(ii) The element a_{ij} of the matrix $\mathbf{A}$ which occurs in the ith row and the jth column is the coefficient of $x_i y_j$ in the given bilinear form.

$$\therefore \qquad \mathbf{A} = \begin{bmatrix} 3 & 0 & -2 \\ 0 & 1 & -5 \end{bmatrix}.$$

The matrix **A** is not symmetric. So the given bilinear form is not symmetric.

Example 5: *Transform the bilinear form* $\mathbf{X}^T\mathbf{AY}$ *to the equivalent canonical form where*

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}.$$

Solution: We write $\mathbf{A} = \mathbf{I}_3\ \mathbf{AI}_3$ *i.e.*,

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_1 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 4 & 2 & 2 \\ 2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_2 \to R_2 - 4R_1, R_3 \to R_3 - 2R_1$, we get

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -6 & -6 \\ 0 & -3 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -4 \\ 1 & 0 & -2 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $C_2 \to C_2 - 2C_1, C_3 \to C_3 - 2C_1$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -6 & -6 \\ 0 & -3 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -4 \\ 1 & 0 & -2 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_2 \to -\dfrac{1}{6} R_2, R_3 \to -\dfrac{1}{3} R_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} A \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \to R_3 - R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & \frac{1}{6} & 0 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $C_3 \to C_3 - C_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & \frac{1}{6} & 0 \end{bmatrix} A \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\therefore \qquad \mathbf{P}^T\mathbf{AQ} = \begin{bmatrix} \mathbf{I}_2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}, \text{where } \mathbf{P} = \begin{bmatrix} \vdots & \vdots & \vdots \\ 0 & -\frac{1}{6} & \frac{1}{6} \\ 1 & \frac{2}{3} & 0 \end{bmatrix}$$

and
$$\mathbf{Q} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence if the vectors $\mathbf{X} = [x_1 \quad x_2 \quad x_3]^T$ and $\mathbf{Y} = [y_1 \quad y_2 \quad y_3]^T$ are transformed to the vectors $\mathbf{U} = [u_1 \quad u_2 \quad u_3]^T$ and $\mathbf{V} = [v_1 \quad v_2 \quad v_3]^T$ respectively by the transformations

$$\mathbf{X} = \mathbf{PU}, \ \mathbf{Y} = \mathbf{QV},$$

then the bilinear form

$$[x_1 \quad x_2 \quad x_3] \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

transforms to the equivalent canonical form

$$[u_1 \quad u_2 \quad u_3] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = u_1 v_1 + u_2 v_2.$$

5 Quadratic Forms

Definition: *An expression of the form* $\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j$, *where* a_{ij}*'s are elements of a field F, is called a quadratic form in the n variables* $x_1, x_2, \ldots\ldots, x_n$ *over the field F.*

Real Quadratic Form. Definition : *An expression of the form*

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j,$$

where a_{ij}*'s are all real numbers, is called a real quadratic form in the n variables* $x_1, x_2, \ldots, x_n$.

For example,

(i) $2x^2 + 7xy + 5y^2$ is a real quadratic form in the two variables x and y.

(ii) $2x^2 - y^2 + 2z^2 - 2yz - 4zx + 6xy$ is a real quadratic form in the three variables x, y and z.

(iii) $x_1^2 - 2x_2^2 + 4x_3^2 - 4x_4^2 - 2x_1 x_2 + 3x_1 x_4 + 4x_2 x_3 - 5x_3 x_4$ is a real quadratic form in the four variables x_1, x_2, x_3 and x_4.

Theorem 1: *Every quadratic form over a field F in n variables* $x_1, x_2, \ldots\ldots, x_n$ *can be expressed in the form* $\mathbf{X'BX}$ *where* *is a column vector and* **B** *is a symmetric matrix of order n over the field F.*

Proof: Let $\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j,$ ...(1)

be a quadratic form over the field F in the n variables $x_1, x_2, \ldots\ldots, x_n$.

In (1) it is assumed that $x_i\, x_j = x_j\, x_i$. Then the total coefficient of $x_i\, x_j$ in (1) is $a_{ij} + a_{ji}$. Let us assign half of this coefficient to x_{ij} and half to x_{ji}. Thus we define another set of scalars b_{ij}, such that $b_{ii} = a_{ii}$ and $b_{ij} = b_{ji} = \frac{1}{2}(a_{ij} + a_{ji}), i \neq j$. Then we have

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j = \sum_{i=1}^{n} \sum_{j=1}^{n} b_{ij}\, x_i\, x_j .$$

Let $\mathbf{B} = [b_{ij}]_{n \times n}$. Then $\mathbf{B}$ is a symmetric matrix of order n over the field F since $b_{ij} = b_{ji}$.

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ \ldots \\ x_n \end{bmatrix}$. Then $\mathbf{X}^T$ or $\mathbf{X}' = [\,x_1 \quad x_2 \quad \ldots\ldots \quad x_n\,]$.

Now $\mathbf{X}^T\, \mathbf{BX}$ is a matrix of the type 1×1. It can be easily seen that the single element of this matrix is $\sum_{i=1}^{n} \sum_{j=1}^{n} b_{ij}\, x_i\, x_j$. If we identify a 1×1 matrix with its single element *i.e.*, if we regard a 1×1 matrix equal to its single element, then we have

$$\mathbf{X}^T \mathbf{BX} = \sum_{i=1}^{n} \sum_{j=1}^{n} b_{ij}\, x_i\, x_j = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j.$$

Hence the result.

6 Matrix of a Quadratic Form

Definition: *If* $\phi = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j$ *is a quadratic form in n variables* $x_1, x_2, \ldots\ldots, x_n$, *then there exists a unique symmetric matrix* **B** *of order n such that* $\phi = \mathbf{X}^T \mathbf{BX}$ *where* $\mathbf{X} = [x_1 \quad x_2 \quad \ldots\ldots \quad x_n]^T$. *The symmetric matrix* **B** *is called the matrix of the quadratic form*

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i\, x_j .$$

Since every quadratic form can always be so written that matrix of its coefficients is a symmetric matrix, therefore we shall be considering quadratic forms which are so adjusted that the coefficient matrix is symmetric.

7 Quadratic form Corresponding to a Symmetric Matrix

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ be a symmetric matrix over the field F and let

$$\mathbf{X} = [x_1 \quad x_2 \quad \ldots\ldots \quad x_n]^T$$

be a column vector. Then $\mathbf{X}^T \mathbf{AX}$ determines a unique quadratic form $\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$ in n variables $x_1, x_2, \ldots\ldots, x_n$ over the field F.

Thus we have seen that there exists a one-to-one correspondence between the set of all quadratic forms in n variables over a field F and the set of all n-rowed symmetric matrices over F.

8 Quadratic Form Corresponding to any Given Square Matrix

Definition: *Let* $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ \ldots \\ x_n \end{bmatrix}$ *be any n-vector in the vector space* V_n *over a field F so that*

$$\mathbf{X}^T = [x_1 \quad x_2 \quad \ldots \quad x_n].$$

Let $\mathbf{A} = [a_{ij}]_{n \times n}$ *be any given square matrix of order n over the field F. Then any polynomial of the form*

$$q\,(x_1, x_2, \ldots, x_n) = \mathbf{X}^T \mathbf{AX} = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$$

is called a quadratic form of order n over F in the n variables $x_1, x_2, \ldots, x_n$.

We can always find a unique symmetric matrix $B = [b_{ij}]_{n \times n}$ of order n such that $\mathbf{X}^T \mathbf{AX} = \mathbf{X}^T \mathbf{BX}$. We have $b_{ij} = b_{ji} = \frac{1}{2}(a_{ij} + a_{ji})$.

Discriminant of a quadratic form. Singular and Non-singular Quadratic forms. By the *discriminant* of a quadratic form $\mathbf{X}^T \mathbf{AX}$, we mean det $\mathbf{A}$. The quadratic form $\mathbf{X}^T \mathbf{AX}$ is said to be *non-singular* if det $A \neq 0$, and it is said to be *singular* if det $\mathbf{A} = 0$.

Illustrative Examples

Example 6: *Write down the matrix of each of the following quadratic forms and verify that they can be written as matrix products* $\mathbf{X}^T \mathbf{AX}$ *:*

(ii) $x_1^2 + 2x_2^2 - 5x_3^2 - x_1 x_2 + 4x_2 x_3 - 3x_3 x_1$.

Solution: (i) The given quadratic form can be written as

$$x_1 x_1 - 9x_1 x_2 - 9x_2 x_1 + 5x_2 x_2 .$$

Let **A** be the matrix of this quadratic form. Then $\mathbf{A} = \begin{bmatrix} 1 & -9 \\ -9 & 5 \end{bmatrix}$.

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Then $\mathbf{X}' = [x_1 \quad x_2]$.

We have $\mathbf{X}'\mathbf{A} = [x_1 \quad x_2] \begin{bmatrix} 1 & -9 \\ -9 & 5 \end{bmatrix} = [x_1 - 9x_2 \quad -9x_1 + 5x_2]$.

$\therefore \quad \mathbf{X}'\mathbf{A}\mathbf{X} = [x_1 - 9x_2 \quad -9x_1 + 5x_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$= x_1 (x_1 - 9x_2) + x_2 (-9x_1 + 5x_2) = x_1^2 - 9x_1 x_2 - 9x_2 x_1 + 5x_2^2$$

$$= x_1^2 - 18x_1 x_2 + 5x_2^2.$$

(ii) The given quadratic form can be written as

$$x_1 x_1 - \frac{1}{2} x_1 x_2 - \frac{3}{2} x_1 x_3 - \frac{1}{2} x_2 x_1 + 2x_2 x_2 + 2x_2 x_3 - \frac{3}{2} x_3 x_1 + 2x_3 x_2 - 5x_3 x_3 .$$

Let **A** be the matrix of this quadratic form. Then

$$\mathbf{A} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} \\ -\frac{1}{2} & 2 & 2 \\ -\frac{3}{2} & 2 & -5 \end{bmatrix}.$$

Obviously **A** is a symmetric matrix.

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Then $\mathbf{X}' = [x_1 \quad x_2 \quad x_3]$.

We have $\mathbf{X}'\mathbf{A} = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} \\ -\frac{1}{2} & 2 & 2 \\ -\frac{3}{2} & 2 & -5 \end{bmatrix}$

$$= [x_1 - \frac{1}{2} x_2 - \frac{3}{2} x_3 \qquad -\frac{1}{2} x_1 + 2x_2 + 2x_3 \qquad -\frac{3}{2} x_1 + 2x_2 - 5x_3].$$

$\therefore \quad \mathbf{X}'\mathbf{A}\mathbf{X} = [x_1 - \frac{1}{2} x_2 - \frac{3}{2} x_3 \quad -\frac{1}{2} x_1 + 2x_2 + 2x_3 \quad -\frac{3}{2} x_1 + 2x_2 - 5x_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

$$= x_1 (x_1 - \frac{1}{2} x_2 - \frac{3}{2} x_3) + x_2 (-\frac{1}{2} x_1 + 2x_2 + 2x_3) + x_3 (-\frac{3}{2} x_1 + 2x_2 - 5x_3)$$

$$= x_1^2 - \frac{1}{2} x_1 x_2 - \frac{3}{2} x_1 x_3 - \frac{1}{2} x_2 x_1 + 2x_2^2 + 2x_2 x_3 - \frac{3}{2} x_3 x_1 + 2x_3 x_2 - 5x_3^2$$

Example 7: *Obtain the matrices corresponding to the following quadratic forms :*

(i) $x^2 + 2y^2 + 3z^2 + 4xy + 5yz + 6zx.$

(ii) $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy.$

Solution: (i) The given quadratic form can be written as

$$x^2 + 2xy + 3xz + 2yx + 2y^2 + \frac{5}{2}yz + 3zx + \frac{5}{2}zy + 3z^2.$$

$\therefore$ if **A** is the matrix of this quadratic form, then

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & \frac{5}{2} \\ 3 & \frac{5}{2} & 3 \end{bmatrix}, \text{which is a symmetric matrix of order 3.}$$

(ii) The given quadratic form can be written as

$$ax^2 + hxy + gxz + hyx + by^2 + fyz + gzx + fzy + cz^2.$$

$\therefore$ if **A** is the matrix of this quadratic form, then

$$\mathbf{A} = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}.$$

Example 8: *Write down the quadratic forms corresponding to the following matrices :*

(i) $\begin{bmatrix} 0 & 5 & -1 \\ 5 & 1 & 6 \\ -1 & 6 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$.

Solution: (i) Let $\mathbf{X} = [x_1 \quad x_2 \quad x_3]^T$ and **A** denote the given symmetric matrix. Then $\mathbf{X^T A X}$ is the quadratic form corresponding to this matrix. We have

$$\mathbf{X^T A} = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 0 & 5 & -1 \\ 5 & 1 & 6 \\ -1 & 6 & 2 \end{bmatrix}$$

$$= [5x_2 - x_3 \quad 5x_1 + x_2 + 6x_3 \quad -x_1 + 6x_2 + 2x_3].$$

$$\therefore \quad \mathbf{X^T A X} = x_1(5x_2 - x_3) + x_2(5x_1 + x_2 + 6x_3) + x_3(-x_1 + 6x_2 + 2x_3)$$

$$= x_2^2 + 2x_3^2 + 10x_1x_2 - 2x_1x_3 + 12x_2x_3.$$

(ii) Let $\mathbf{X} = [x_1 \quad x_2 \quad x_3 \quad x_4]^T$ and **A** denote the given symmetric matrix. Then $\mathbf{X^T A X}$ is the quadratic form corresponding to this matrix. We have

$$\mathbf{X}^{\mathrm{T}}\mathbf{A} = [x_1 \ \ x_2 \ \ x_3 \ \ x_4]\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

$$= [x_2 + 2x_3 + 3x_4 \quad x_1 + 2x_2 + 3x_3 + 4x_4 \quad 2x_1 + 3x_2 + 4x_3 + 5x_4 \quad 3x_1 + 4x_2 + 5x_3 + 6x_4]$$

$\therefore$
$$\mathbf{X}^T\mathbf{AX} = x_1 (x_2 + 2x_3 + 3x_4) + x_2 (x_1 + 2x_2 + 3x_3 + 4x_4) + x_3 (2x_1 + 3x_2 + 4x_3 + 5x_4) + x_4 (3x_1 + 4x_2 + 5x_3 + 6x_4)$$

$$= 2x_2^2 + 4x_3^2 + 6x_4^2 + 2x_1 x_2 + 4x_1 x_3 + 6x_1 x_4 + 6x_2 x_3 + 8x_2 x_4 + 10 x_3 x_4.$$

Comprehensive Exercise 1

1. Which of the following functions f, defined on vectors $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ in $\mathbf{R}^2$, are bilinear forms ?

(i) $f(\alpha, \beta) = x_1 y_2 - x_2 y_1$

(ii) $f(\alpha, \beta) = (x_1 - y_1)^2 + x_2 y_2$.

2. Find the matrix $\mathbf{A}$ of each of the following bilinear forms $b(\mathbf{X}, \mathbf{Y}) = \mathbf{X}^T \mathbf{AY}$.

(i) $-5x_1 y_1 - x_1 y_2 + 2x_2 y_1 - x_3 y_1 + 3x_3 y_2$

(ii) $4x_1 y_1 + x_1 y_2 + x_2 y_1 - 2x_2 y_2 - 4x_2 y_3 - 4x_3 y_2 + 7x_3 y_3$.

Which of the above forms is symmetric ?

3. Determine the transformation matrices $\mathbf{P}$ and $\mathbf{Q}$ so that the bilinear form

$$\mathbf{X}^T\mathbf{AY} = x_1 y_1 + x_1 y_2 + 2x_1 y_3 + x_2 y_1 + 2x_2 y_2 + 3x_2 y_3 - x_3 y_2 - x_3 y_3$$

is equivalent to a canonical form.

4. Obtain the matrices corresponding to the following quadratic forms

(i) $ax^2 + 2hxy + by^2$ (ii) $2x_1 x_2 + 6x_1 x_3 - 4x_2 x_3$.

(iii) $x_1^2 + 5x_2^2 - 7x_3^2$ (iv) $2x_1^2 - 7x_3^2 + 4x_1 x_2 - 6x_2 x_3$.

5. Obtain the matrices corresponding to the following quadratic forms :

(i) $a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + 2a_{12}x_1 x_2 + 2a_{23}x_2 x_3 + 2a_{31}x_3 x_1$.

(ii) $x_1^2 - 2x_2^2 + 4x_3^2 - 4x_4^2 - 2x_1 x_2 + 3x_1 x_3 + 4x_2 x_3 - 5x_3 x_4$.

(iii) $x_1 x_2 + x_2 x_3 + x_3 x_1 + x_1 x_4 + x_2 x_4 + x_3 x_4$.

(iv) $x_1^2 - 2x_2 x_3 - x_3 x_4$.

(v) $d_1 x_1^2 + d_2 x_2^2 + d_3 x_3^2 + d_4 x_4^2 + d_5 x_5^2$.

6. Find the matrix of the quadratic form

$x_1^2 - 2x_2^2 - 3x_3^2 + 4x_1 x_2 + 6x_1 x_3 - 8x_2 x_3$.

and verify that it can be written as a matrix product $\mathbf{X}'\mathbf{AX}$.

matrices :

(i) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 3 \\ 3 & 3 & 1 \end{bmatrix}$ (ii) $diag.[\lambda_1, \lambda_2, \ldots\ldots, \lambda_n]$

8. Write down the quadratic form corresponding to the matrix

$$\begin{bmatrix} 0 & a & b & c \\ a & 0 & l & m \\ b & l & 0 & p \\ c & m & p & 0 \end{bmatrix}.$$

9. Write down the quadratic form associated with the matrix

$$\mathbf{B} = \begin{bmatrix} 2 & 1 & 2 \\ -3 & -3 & -1 \\ 4 & 1 & 3 \end{bmatrix}.$$

Rewrite the matrix A of the form so that it is symmetric.

Answers 1

1. (i) f is a bilinear form on $\mathbf{R}^2$

(ii) f is not a bilinear form on $\mathbf{R}^2$

2. (i) $\mathbf{A} = \begin{bmatrix} -5 & -1 \\ 2 & 0 \\ -1 & 3 \end{bmatrix}$ · The given bilinear form is not symmetric

(ii) $\mathbf{A} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & -2 & -4 \\ 0 & -4 & 7 \end{bmatrix}$ · The given bilinear form is symmetric

3. $\mathbf{P} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, $\mathbf{Q} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

4. (i) $\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & -2 \\ 3 & -2 & 0 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -7 \end{bmatrix}$ (iv) $\begin{bmatrix} 2 & 2 & 0 \\ 2 & 0 & -3 \\ 0 & -3 & -7 \end{bmatrix}$

5. (i) $\begin{bmatrix} a_{11} & a_{12} & a_{31} \\ a_{12} & a_{22} & a_{23} \\ a_{31} & a_{23} & a_{33} \end{bmatrix}$. (ii) $\begin{bmatrix} -1 & -2 & 2 & 0 \\ 0 & 2 & 4 & -\frac{5}{2} \\ \frac{3}{2} & 0 & -\frac{5}{2} & -4 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 0 \end{bmatrix}$

(v) diag. $[d_1, d_2, d_3, d_4, d_5]$

6. $\begin{bmatrix} 1 & 2 & 3 \\ 2 & -2 & -4 \\ 3 & -4 & -3 \end{bmatrix}$

7. (i) $x_1^2 + x_3^2 + 4x_1x_2 + 6x_1x_3 + 6x_2x_3$

(ii) $\lambda_1 x_1^2 + \lambda_2 x_2^2 + \ldots.. + \lambda_n x_n^2$

8. $2ax_1x_2 + 2bx_1x_3 + 2cx_1 x_4 + 2lx_2x_3 + 2mx_2x_4 + 2px_3x_4$

9. $2x_1^2 - 3x_2^2 + 3x_3^2 - 2x_1x_2 + 6x_1x_3$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ -1 & -3 & 0 \\ 3 & 0 & 3 \end{bmatrix}$$

9 Linear Transformations

Suppose V_n is the vector space of all ordered n-tuples of the elements of a field F and let the vectors in V_n be written as column vectors. Let **P** be a matrix of order n over the field F. If $\mathbf{Y} = [y_1, y_2, \ldots, y_n]^T$ is a vector in V_n, then **Y** is a matrix of the type $n \times 1$. Obviously **PY** is a matrix of the type $n \times 1$. Thus **PY** is also a vector in V_n. Let

$$\mathbf{PY} = \mathbf{X} = [x_1, x_2, \ldots, x_n]^T.$$

The relation **PY = X** thus gives a mapping form V_n into V_n.

Since $\mathbf{P}(a\mathbf{Y}_1 + b\mathbf{Y}_2) = a\,(\mathbf{PY}_1) + b\,(\mathbf{PY}_2)$, therefore this mapping is a linear transformation. If the matrix **P** is non-singular, then the linear transformation is also said to be non-singular. Also if the matrix **P** is non-singular, then the mapping **PY** = **X** is one-one onto as shown below :

Mapping **P** is **one-one.** We have $\mathbf{PY_1} = \mathbf{PY_2}$

$\Rightarrow$ $$\mathbf{P}^{-1}(\mathbf{PY}_1) = \mathbf{P}^{-1}(\mathbf{PY}_2) \Rightarrow \mathbf{Y}_1 = \mathbf{Y}_2.$$

Mapping $\mathbf{P}$ is **onto**. Let $\mathbf{Z}$ be any vector in V_n. Then $\mathbf{P}^{-1}\mathbf{Z}$ is also a vector of V_n and we have $\mathbf{P}(\mathbf{P}^{-1}\mathbf{Z}) = (\mathbf{P}\mathbf{P}^{-1})\mathbf{Z} = \mathbf{I}\mathbf{Z} = \mathbf{Z}.$ Therefore the mapping $\mathbf{P}$ is onto.

If the linear transformation $\mathbf{PY} = \mathbf{X}$ is non-singular, then $\mathbf{PY} = \mathbf{O}$ if and only if $\mathbf{Y} = \mathbf{O}.$
If $\mathbf{Y} = \mathbf{O}$ then obviously $\mathbf{PY} = \mathbf{O}.$

Conversely, $\mathbf{PY} = \mathbf{O} \Rightarrow \mathbf{P}^{-1}(\mathbf{PY}) = \mathbf{P}^{-1}\mathbf{O} \Rightarrow \mathbf{Y} = \mathbf{O}.$

10 Congruence of Matrices

Definition: *A square matrix* $\mathbf{B}$ *of order* n *over a field* F *is said to be congruent to another square matrix* $\mathbf{A}$ *of order* n *over* F, *if there exists a non-singular matrix* $\mathbf{P}$ *over* F *such that*

$$\mathbf{B} = \mathbf{P}^T \mathbf{AP}.$$

Theorem 1: *The relation of 'congruence of matrices' is an equivalence relation in the set of all* $n \times n$ *matrices over a field* F.

Proof: **Reflexivity.** Let $\mathbf{A}$ be any $n \times n$ matrix over a field F. Then $\mathbf{A} = \mathbf{I}^T \mathbf{AI}$, where $\mathbf{I}$ is unit matrix of order n over F. Since $\mathbf{I}$ is non-singular, therefore $\mathbf{A}$ is congruent to itself.

Symmetry: Suppose $\mathbf{A}$ is congruent to $\mathbf{B}$. Then $\mathbf{A} = \mathbf{P}'\mathbf{BP}$, where $\mathbf{P}$ is non-singular.

$\therefore \qquad (\mathbf{P}')^{-1}\mathbf{AP}^{-1} = (\mathbf{P}')^{-1}\mathbf{P}'\mathbf{BPP}^{-1} = \mathbf{B}$

$\Rightarrow \qquad (\mathbf{P}^{-1})'\mathbf{AP}^{-1} = \mathbf{B} \qquad [\because (\mathbf{P}')^{-1} = (\mathbf{P}^{-1})']$

$\Rightarrow \qquad \mathbf{B}$ is congruent to $\mathbf{A}$.

Transitivity. Suppose $\mathbf{A}$ is congruent to $\mathbf{B}$ and $\mathbf{B}$ is congruent to $\mathbf{C}$. Then $\mathbf{A} = \mathbf{P}'\mathbf{BP}$, $\mathbf{B} = \mathbf{Q}'\mathbf{CQ}$, where $\mathbf{P}$ and $\mathbf{Q}$ are non-singular. Therefore

$$\mathbf{A} = \mathbf{P}'(\mathbf{Q}'\mathbf{CQ})\mathbf{P} = (\mathbf{P}'\mathbf{Q}')\mathbf{CQP} = (\mathbf{QP})'\mathbf{C}(\mathbf{QP}).$$

Since $\mathbf{QP}$ is also a non-singular matrix, therefore $\mathbf{A}$ is congruent to $\mathbf{C}$.

Thus the relation of 'congruence of matrices' is reflexive, symmetric and transitive. So it is an equivalence relation.

Theorem 2 : *Every matrix congruent to a symmetric matrix is a symmetric matrix.*

Proof : Let a matrix $\mathbf{B}$ be congruent to a symmetric matrix $\mathbf{A}$.

Then there exists a non-singular matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{P}'\mathbf{AP}.$

We have $\qquad \mathbf{B}' = (\mathbf{P}'\mathbf{AP})' = \mathbf{P}'\mathbf{A}'(\mathbf{P}')' = \mathbf{P}'\mathbf{A}'\mathbf{P}$

$\qquad = \mathbf{P}'\mathbf{AP} \qquad [\because \mathbf{A}$ is symmetric $\Rightarrow \mathbf{A}' = \mathbf{A}]$

$\qquad = \mathbf{B}.$

$\therefore \qquad \mathbf{B}$ is also a symmetric matrix.

Congruence operation on a square matrix or Congruence transformations of a square matrix.

three types :

(i) Interchange of the ith and the jth rows as well as of the ith and the jth columns. Both should be applied simultaneously. Thus the operation $R_i \leftrightarrow R_j$ followed by $C_i \leftrightarrow C_j$ is a congruence operation.

(ii) Multiplication of the ith row as well as the ith column by a non-zero number c *i.e.*, $R_i \to cR_i$ followed by $C_i \to cC_i$.

(iii) $R_i \to R_i + kR_j$ followed by $C_i \to C_i + kC_j$.

Now we shall show that *each congruent transformation of a matrix consists of a pair of elementary transformations, one row and the other column, such that of the corresponding elementary matrices each is the transpose of the other.*

(a) Let $\mathbf{E}^*, \mathbf{E}$ be the elementary matrices corresponding to the elementary transformations $R_i \leftrightarrow R_j$ and $C_i \leftrightarrow C_j$ respectively. Then

$$\mathbf{E}^* = \mathbf{E} = \mathbf{E}'.$$

(b) Let $\mathbf{E}^*, \mathbf{E}$ be the elementary matrices corresponding to the elementary transformations $R_i \to cR_i$ and $C_i \to cC_i$ respectively where $c \neq 0$. Then

$$\mathbf{E}^* = \mathbf{E} = \mathbf{E}'.$$

(c) Let $\mathbf{E}^*, \mathbf{E}$ be the elementary matrices corresponding to the elementary transformations $R_i \to R_i + kR_j$ and $C_i \to C_i + kC_j$ respectively. Then

$$\mathbf{E}^* = \mathbf{E}'.$$

Now we know that every elementary row (column) transformation of a matrix can be brought about by pre-multiplication (post-multiplication) with the corresponding elementary matrix. Therefore if a matrix **B** has been obtained from **A** by a finite chain of congruent operations applied on **A**, then there exist elementary matrices $\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_s$ such that

$$\begin{aligned}\mathbf{B} &= \mathbf{E}_s' \ldots \mathbf{E}_2' \mathbf{E}_1' \mathbf{A}\mathbf{E}_1 \mathbf{E}_2 \ldots \mathbf{E}_s \\ &= (\mathbf{E}_1 \mathbf{E}_2 \ldots \mathbf{E}_s)' \mathbf{A} (\mathbf{E}_1 \mathbf{E}_2 \ldots \mathbf{E}_s) \\ &= \mathbf{P}'\mathbf{A}\mathbf{P}, \text{ where } \mathbf{P} = \mathbf{E}_1 \mathbf{E}_2 \ldots \mathbf{E}_s \text{ is a non-singular matrix.}\end{aligned}$$

Therefore **B** is congruent to **A.** *Thus every matrix* **B** *obtained from any given matrix* **A** *by subjecting* **A** *to a finite chain of congruent operations is congruent to* **B.**

The converse is also true. If **B** is congruent to **A,** then

$$\mathbf{B} = \mathbf{P}'\mathbf{A}\mathbf{P}$$

where **P** is a non-singular matrix. Now every non-singular matrix can be expressed as the product of elementary matrices. Therefore we can write $\mathbf{P} = \mathbf{E}_1 \mathbf{E}_2 \ldots \mathbf{E}_s$ where $\mathbf{E}_1, \ldots, \mathbf{E}_s$ are elementary matrices. Then $\mathbf{B} = \mathbf{E}_s' \ldots \mathbf{E}_2' \mathbf{E}_1' \mathbf{A}\mathbf{E}_1 \mathbf{E}_2 \ldots \mathbf{E}_s$. Therefore **B** is obtained from **A** by a finite chain of congruent operations applied on **A.**

11 Congruence of Quadratic Forms or Equivalence of Quadratic Forms

Definition: *Two quadratic forms* $\mathbf{X}^T\mathbf{AX}$ *and* $\mathbf{Y}^T\mathbf{BY}$ *over a field F are said to be congruent or equivalent over F if their respective matrices* **A** *and* **B** *are congruent over F. Thus* $\mathbf{X}^T\mathbf{AX}$ *is equivalent to* $\mathbf{Y}^T\mathbf{BY}$ *if there exists a non-singular matrix* **P** *over F such that* $\mathbf{P}^T\mathbf{AP} = \mathbf{B}.$

Since congruence of matrices is an equivalence relation, therefore equivalence of quadratic forms is also an equivalence relation.

Equivalence of Real Quadratic Forms:

Definition: *Two real quadratic forms* $\mathbf{X}^T\mathbf{AX}$ *and* $\mathbf{Y}^T\mathbf{BY}$ *are said to be real equivalent, orthogonally equivalent, or complex equivalent according as there exists a non-singular real, orthogonal, or a non-singular complex matrix* **P** *such that*

$$\mathbf{B} = \mathbf{P}^T\mathbf{AP}.$$

12 The Linear Transformation of a Quadratic Form

Consider a quadratic form

$$\mathbf{X}^T\mathbf{AX} \qquad \ldots(1)$$

and a non-singular linear transformation

$$\mathbf{X} = \mathbf{PY} \qquad \ldots(2)$$

so that **P** is a non-singular matrix.

Putting $\mathbf{X} = \mathbf{PY}$ in (1), we get

$$\mathbf{X}^T\mathbf{AX} = (\mathbf{PY})^T\mathbf{A}(\mathbf{PY}) = \mathbf{Y}^T\mathbf{P}^T\mathbf{APY}$$

$$= \mathbf{Y}^T\mathbf{BY}, \quad \text{where } \mathbf{B} = \mathbf{P}^T\mathbf{AP}.$$

Since **B** is congruent to a symmetric matrix **A,** therefore **B** is also a symmetric matrix. Thus $\mathbf{Y}^T\mathbf{BY}$ is a quadratic form. It is called a linear transform of the form $\mathbf{X}^T\mathbf{AX}$ by the non-singular matrix **P.** The matrix of the quadratic form $\mathbf{Y}^T\mathbf{BY}$ is

$$\mathbf{B} = \mathbf{P}^T\mathbf{AP}.$$

Thus the quadratic form $\mathbf{Y}^T\mathbf{BY}$ is congruent to $\mathbf{X}^T\mathbf{AX}.$

Theorem 1: *The ranges of values of two congruent quadratic forms are the same.*

Proof: Let $\phi = \mathbf{X}'\mathbf{AX}$ and $\psi = \mathbf{Y}'\mathbf{BY}$ be two congruent quadratic forms. Then there exists a non-singular matrix **P** such that

$$\mathbf{B} = \mathbf{P}^T\mathbf{AP}.$$

Consider the linear transformation $\mathbf{X} = \mathbf{PY}.$

Let $\phi = p$ when $\mathbf{X} = \mathbf{X}_1$. Then $p = \mathbf{X}_1'\mathbf{AX}_1$. The value of ψ when $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}_1$ is

$$= (\mathbf{P}^{-1}\mathbf{X}_1)'\mathbf{B}(\mathbf{P}^{-1}\mathbf{X}_1) = \mathbf{X}_1'(\mathbf{P}^{-1})'\mathbf{P}'\mathbf{APP}^{-1}\mathbf{X}_1$$

Thus each value of ϕ is equal to some value of ψ.

Conversely let $\psi = q$ when $\mathbf{Y} = \mathbf{Y_1}$. Then $q = \mathbf{Y_1}' \mathbf{B} \mathbf{Y_1}$. The value of ϕ when $\mathbf{X} = \mathbf{P}\mathbf{Y_1}$ is

$$= (\mathbf{P}\mathbf{Y_1})' \mathbf{A} (\mathbf{P}\mathbf{Y_1}) = \mathbf{Y_1}' \mathbf{P}' \mathbf{A}\mathbf{P}\mathbf{Y_1}$$
$$= \mathbf{Y_1}' \mathbf{B}\mathbf{Y_1} = q.$$

Thus each value of ψ is equal to some value of ϕ.

Hence ϕ and ψ have the same ranges of values.

Corollary: *If the quadratic form* $\mathbf{Y}' \mathbf{B}\mathbf{Y}$ *is a linear transform of the quadratic form* $\mathbf{X}' \mathbf{A}\mathbf{X}$ *by a non-singular matrix* **P**, *then the two forms are congruent and so have the same ranges of values.*

13 Congruent Reduction of a Symmetric Matrix

Theorem 1: *If* **A** *be any n-rowed non-zero symmetric marix of rank r, over a field F, then there exists an n-rowed non-singular matrix* **P** *over F such that*

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \begin{bmatrix} \mathbf{A_1} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$$

where $\mathbf{A_1}$ *is a non-singular diagonal matrix of order r over F and each* **O** *is a null matrix of suitable size.*

Or

Every symmetric matrix of rank r is congruent to a diagonal matrix, r, of whose diagonal elements only are non-zero.

Proof : We shall prove the theorem by induction on n, the order of the given matrix. If $n = 1$, the theorem is obviously true. Let us suppose that the theorem is true for all symmetric matrices of order $n - 1$. Then we shall show that it is also true for an $n \times n$ symmetric matrix **A**.

Let $\mathbf{A} = [\, a_{ij} \,]_{n \times n}$ be a symmetric matrix of rank r over a field F. First we shall show that there exists a matrix $\mathbf{B} = [\, b_{ij} \,]_{n \times n}$ over F congruent to **A** such that $b_{11} \neq 0$.

Case 1. If $a_{11} \neq 0$, then we take $\mathbf{B} = \mathbf{A}$.

Case 2. If $a_{11} = 0$, but some diagonal element of **A**, say, $a_{ii} \neq 0$, then applying the congruent operation $R_i \leftrightarrow R_1, C_i \leftrightarrow C_1$ to **A**, we obtain a matrix **B** congruent to **A** such that

$$b_{11} = a_{ii} \neq 0.$$

Case 3. Suppose that each diagonal element of **A** is 0. Since **A** is a non-zero matrix, let a_{ij} be a non-zero element of **A**. Then

$$a_{ij} = a_{ji} \neq 0.$$

Applying the congruent operation $R_i \to R_i + R_j, C_i \to C_i + C_j$ to **A**, we obtain a matrix $\mathbf{D} = [\, d_{ij} \,]_{n \times n}$ congruent to **A** such that

$$d_{ii} = a_{ij} + a_{ji} = 2\, a_{ij} \neq 0.$$

Now applying the congruent operation $R_i \leftrightarrow R_1, C_i \leftrightarrow C_1$ to **D** we obtain a matrix congruent to **D** and, therefore, also congruent to **A** such that

$$b_{11} = d_{ii} \neq 0.$$

Thus there always exists a matrix

$$\mathbf{B} = [\, b_{ij} \,]_{n \times n}$$

congruent to a symmetric matrix, such that the leading element of **B** is not zero. Since **B** is congruent to a symmetric matrix, therefore **B** itself is a symmetric matrix. Since $b_{11} \neq 0$, therefore all elements in the first row and first column of **B,** except the leading element, can be made 0 by suitable congruent operations. We thus have a matrix

$$\mathbf{C} = \begin{bmatrix} a_{11} & 0 & \ldots & 0 \\ 0 & & & \\ \vdots & \mathbf{B}_1 & & \\ 0 & & & \end{bmatrix},$$

congruent to **B** and, therefore, also congruent to **A** such that $\mathbf{B_1}$ is a square matrix of order $n - 1$. Since **C** is congruent to a symmetric matrix **A,** therefore **C** is also a symmetric matrix and consequently $\mathbf{B_1}$ is also a symmetric matrix. Thus $\mathbf{B_1}$ is a symmetric matrix of order $n - 1$. Therefore by our induction hypothesis it can be reduced to a diagonal matrix by congruent operations. If the congruent operations applied to $\mathbf{B_1}$ for this purpose be applied to **C,** they will not affect the first row and the first column of **C.** So **C** can be reduced to a diagonal matrix by congruent operations. Thus **A** is congruent to a diagonal matrix, say diag $[\lambda_1, \lambda_2, \ldots, \lambda_k, 0, 0, \ldots, 0]$. Thus there exists a non-singular matrix **P** such that

$$\mathbf{P'AP} = \text{diag}\,[\lambda_1, \ldots, \lambda_k, 0, \ldots, 0].$$

Since rank $\mathbf{A} = r$ and the rank of a matrix does not change on multiplication by a non-singular matrix, therefore rank of the matrix $\mathbf{P'AP} = \text{diag.}\,[\lambda_1, \ldots, \lambda_k, 0, \ldots, 0]$ is also r. So precisely r elements of diag. $[\lambda_1, \ldots, \lambda_k, 0, \ldots, 0]$ are non-zero. Therefore $k = r$ and thus $\mathbf{P'AP} = \text{diag.}\,[\lambda_1, \ldots, \lambda_r, 0, \ldots, 0]$.

Thus **A** can be reduced to diagonal form by congruent operations.

The proof is now complete by induction.

Corollary: *Corresponding to every quadratic form* $\mathbf{X'AX}$ *over a field* F*, there exists a non-singular linear transformation*

$$\mathbf{X} = \mathbf{PY}$$

over F*, such that the form* $\mathbf{X'AX}$ *transforms to a sum of* r*, square terms*

$$\lambda_1 y_1^2 + \ldots + \lambda_r y_r^2,$$

where $\lambda_1, \ldots, \lambda_r$ *belong to the field* F *and* r *is the rank of the matrix* **A.**

Rank of a quadratic form. Definition : *Let* $\mathbf{X'AX}$ *be a quadratic form over a field* F*. The rank of the matrix* **A** *is called the rank of the quadratic form* $\mathbf{X'AX}$.

If $\mathbf{X'AX}$ is a quadratic form of rank r, then there exists a non-singular matrix **P** which will reduce the form $\mathbf{X'AX}$ to a sum of r square terms.

We should transform the given symmetric matrix **A** to diagonal form by applying congruent operations. Then the application of corresponding column operations to the unit matrix $\mathbf{I}_n$ will give us a non-singular matrix **P** such that

$$\mathbf{P}'\mathbf{AP} = \text{a diagonal matrix.}$$

The whole process will be clear from the following examples.

Illustrative Examples

Example 9: *Determine a non-singular matrix* **P** *such that* **P′AP** *is a diagonal matrix, where*

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

Interpret the result in terms of quadratic forms.

Solution: We write **A = IAI**

i.e.,
$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We shall reduce **A** to diagonal form by applying congruent operations. On the right hand side **IAI,** the corresponding row operations will be applied on prefactor **I** and the column operations will be applied on the post-factor **I.** There is no need of actually applying column operations on post-factor **I** because at any stage the matrix obtained by applying column operations on post-factor **I** will be the transpose of the matrix obtained by applying row operations on pre-factor **I.**

Performing the congruent operations

$$R_2 \to R_2 + \frac{1}{3} R_1, C_2 \to C_2 + \frac{1}{3} C_1$$

and
$$R_3 \to R_3 - \frac{1}{3} R_1, C_3 \to C_3 - \frac{1}{3} C_1,$$

we have

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & \frac{7}{3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now performing the congruent operation

$$R_3 \to R_3 + \frac{1}{7} R_2, C_3 \to C_3 + \frac{1}{7} C_2, \text{we get}$$

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & 0 \\ 0 & 0 & \frac{16}{7} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{7} & \frac{1}{7} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & \frac{1}{3} & -\frac{2}{7} \\ 0 & 1 & \frac{1}{7} \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\mathbf{P} = \begin{bmatrix} 1 & \frac{1}{3} & -\frac{2}{7} \\ 0 & 1 & \frac{1}{7} \\ 0 & 0 & 1 \end{bmatrix} \text{ such that } \mathbf{P'AP} = \text{diag.} \left[6, \frac{7}{3}, \frac{16}{7}\right].$$

The quadratic form corresponding to the matrix **A** is

$$\mathbf{X'AX} = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4\, x_1\, x_2 - 2\, x_2\, x_3 + 4\, x_3\, x_1 . \qquad \ldots(1)$$

The non-singular transformation corresponding to the matrix **P** is given by **X = PY** *i.e.*,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{3} & -\frac{2}{7} \\ 0 & 1 & \frac{1}{7} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

which is equivalent to

$$\left.\begin{aligned} x_1 &= y_1 + \frac{1}{3} y_2 - \frac{2}{7} y_3 \\ x_2 &= y_2 + \frac{1}{7} y_3 \\ x_3 &= y_3 . \end{aligned}\right\} \qquad \ldots(2)$$

The transformation (2) will reduce the quadratic form (1) to the diagonal form

$$\mathbf{Y'\,P'APY} = 6\,y_1^2 + \frac{7}{3}\, y_2^2 + \frac{16}{7}\, y_3^2 .$$

The rank of the quadratic form $\mathbf{X'AX}$ is 3. So it has been reduced to a form which is a sum of three squares.

14 Reduction of a Real Quadratic Form

Definition: Diagonal and Unit Quadratic Forms. *If the matrix of a quadratic form is diagonal, then it is called a diagonal quadratic form.*

For example, X^T diag. $[a_1, a_2, \ldots, a_n]\, X = a_1\, x_1^2 + a_2\, x_2^2 + \ldots + a_n\, x_n^2$ is a diagonal quadratic form. In the diagonal quadratic form, some of the a_i's may be zero.

If the matrix of a quadratic form is unit, then it is called a unit quadratic form.

For example, $X^T\, I_n\, X = x_1^2 + x_2^2 + \ldots + x_n^2$ is a unit quadratic form.

Theorem 1: *If* **A** *be any n-rowed real symmetric matrix of rank r, then there exists a real non-singular matrix* **P** *such that*

$$\mathbf{P'\,AP} = \textit{diag}. [1, 1, \ldots, 1, -1, -1, \ldots, -1, 0, \ldots, 0]$$

so that 1, *appears p times and* -1 *appears* $r - p$ *times.*

Proof: **A** is a real symmetric matrix of rank r. Therefore there exists a non-singular real matrix **Q** such that **Q′AQ** is a diagonal matrix **D** with precisely r non-zero diagonal elements. Let

$$\mathbf{Q'AQ} = \mathbf{D} = \text{diag.} [\lambda_1, \lambda_2, \ldots, \lambda_r, 0, \ldots, 0].$$

Suppose that p of the non-zero diagonal elements are positive. Then $r - p$ are negative.

j^{th} rows can be interchanged by applying the congruent operation $R_i \leftrightarrow R_j, C_i \leftrightarrow C_j$, therefore without any loss of generality we can take $\lambda_1, \ldots, \lambda_p$ to be positive and $\lambda_{p+1}, \ldots, \lambda_r$ to be negative.

Let **S** be the $n \times n$ (real) diagonal matrix with diagonal elements

$$\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_p}}, \frac{1}{\sqrt{(-\lambda_{p+1})}}, \ldots, \frac{1}{\sqrt{(-\lambda_r)}}, 1, \ldots, 1.$$

Then $$\mathbf{S} = \text{diag}\left[\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_p}}, \frac{1}{\sqrt{(-\lambda_{p+1})}}, \ldots, \frac{1}{\sqrt{(-\lambda_r)}}, 1, \ldots, 1\right]$$

is a real non-singular diagonal matrix and $\mathbf{S}' = \mathbf{S}$.

If we take $\mathbf{P} = \mathbf{QS}$, then **P** is also real non-singular matrix and we have

$$\mathbf{P}' \mathbf{AP} = (\mathbf{QS})' \mathbf{A} (\mathbf{QS}) = \mathbf{S}'\mathbf{Q}' \mathbf{AQS} = \mathbf{S}' \mathbf{DS} = \mathbf{SDS}$$
$$= \text{diag.}\, [1, \ldots, 1, -1, \ldots, -1, 0, \ldots, 0]$$

so that 1 and -1 appear p and $r-p$ times respectively.

Corollary: *If* $\mathbf{X'AX}$ *is a real quadratic form of rank r in n variables, then there exists a real non-singular linear transformation* $\mathbf{X} = \mathbf{PY}$ *which transforms* $\mathbf{X'AX}$ *to the form*

$$\mathbf{Y'}\,\mathbf{P'}\,\mathbf{APY} = y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2.$$

Canonical or Normal form of a real quadratic form. **Definition:** *If* $\mathbf{X'AX}$ *is a real quadratic form in n variables, then there exists real non-singular linear transformation* $\mathbf{X} = \mathbf{PY}$ *which transforms* $\mathbf{X'AX}$ *to the form*

$$y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2.$$

In the new form the given quadratic form has been expressed as a sum and difference of the squares of new variables. This latter expression is called the canonical form or normal form of the given quadratic form.

If $\phi = \mathbf{X'AX}$ is a real quadratic form of rank r, then **A** is a matrix of rank r. If the real non-singular linear transformation $\mathbf{X} = \mathbf{PY}$ reduces ϕ to normal form, then $\mathbf{P'AP}$ is a diagonal matrix having 1 and -1 as its non-zero diagonal elements. Since $\mathbf{P'AP}$ is also of rank r, therefore it will have precisely r non-zero diagonal elements. Thus the number of terms in each normal form of a given real quadratic form is the same. Now we shall prove that the number of positive terms in any two normal reductions of a real quadratic form is the same.

Theorem 2: *The number of positive terms in any two normal reductions of a real quadratic form is the same.*

Proof: Let $\phi = \mathbf{X'AX}$ be a real quadratic form of rank r in n variables. Suppose the real non-singular linear transformations

$$\mathbf{X} = \mathbf{PY} \quad \text{and} \quad \mathbf{X} = \mathbf{QZ}$$

transform ϕ to the normal forms

$$y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2 \qquad \ldots(1)$$

respectively.

To prove that $p = q$.

Let $p < q$. Obviously $y_1, \ldots, y_n, z_1, \ldots, z_n$ are linear homogeneous functions of $x_1, \ldots, x_n$. Since $q > p$, therefore $q - p > 0$. So $n - (q - p)$ is less than n. Therefore $(n - q) + p$ is less than n.

Now $y_1 = 0, y_2 = 0, \ldots, y_p = 0, z_{q+1} = 0, z_{q+2} = 0, \ldots, z_n = 0$ are $(n - q) + p$ linear homogeneous equations in n unknowns $x_1, \ldots, x_n$. Since the number of equations is less than the number of unknowns n, therefore these equations must possess a non-zero solution. Let $x_1 = a_1, \ldots, x_n = a_n$ be a non-zero solution of these equations and let $\mathbf{X}_1 = [a_1, \ldots, a_n]'$. Let $\mathbf{Y} = [b_1, \ldots, b_n]' = \mathbf{Y}_1$ and $\mathbf{Z} = [c_1, \ldots, c_n]'$ when $\mathbf{X} = \mathbf{X}_1$. Then $b_1 = 0, \ldots, b_p = 0$ and $c_{q+1} = 0, \ldots, c_n = 0$. Putting $\mathbf{Y} = [b_1, \ldots, b_n]'$ in (1) and $\mathbf{Z} = [c_1, \ldots, c_n]'$ in (2), we get two values of ϕ when $\mathbf{X} = \mathbf{X}_1$.

These must be equal. Therefore we have

$$- b_{p+1}^2 - \ldots - b_r^2 = c_1^2 + \ldots + c_q^2$$

$$\Rightarrow \quad b_{p+1} = 0, \ldots, b_r = 0$$

$$\Rightarrow \quad \mathbf{Y}_1 = \mathbf{O}$$

$$\Rightarrow \quad \mathbf{P}^{-1} \mathbf{X}_1 = \mathbf{O} \qquad [\because \mathbf{X}_1 = \mathbf{P}\mathbf{Y}_1]$$

$$\Rightarrow \quad \mathbf{X}_1 = \mathbf{O},$$

which is a contradiction since $\mathbf{X}_1$ is a non-zero vector.

Thus we cannot have $p < q$. Similarly, we cannot have $q < p$. Hence we must have $p = q$.

Corollary: *The number of negative terms in any two normal reductions of a real quadratic form is the same. Also the excess of the number of positive terms over the number of negative terms in any two normal reductions of a real quadratic form is the same.*

Signature and index of a real quadratic form.

Definition: *Let $y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2$ be a normal form of a real quadratic form* $\mathbf{X}'\mathbf{A}\mathbf{X}$ *of rank r. The number p of positive terms in a normal form of* $\mathbf{X}'\mathbf{A}\mathbf{X}$ *is called the index of the quadratic form. The excess of the number of positive terms over the number of negative terms in a normal form of*

$$\mathbf{X}'\mathbf{A}\mathbf{X} \quad i.e., \quad p - (r - p) = 2p - r$$

is called the signature of the quadratic form and is usually denoted by s.

Thus $s = 2p - r$.

In terms of signature theorem 2 may be stated as follows :

Theorem 3: Sylvester's Law of Inertia. *The signature of a real quadratic form is invariant for all normal reductions.*

Proof: For its proof give definition of signature and the proof of theorem 2.

Theorem 4: *Two real quadratic forms in n variables are real equivalent if and only if they have the same rank and index (or signature).*

variables.

Let us first assume that the two forms are equivalent. Then there exists a real non-singular linear transformation $\mathbf{X} = \mathbf{PY}$ which transforms $\mathbf{X'AX}$ to $\mathbf{Y'BY}$ *i.e.*, $\mathbf{B} = \mathbf{P'AP}$.

Now suppose the real non-singular linear transformation $\mathbf{Y} = \mathbf{QZ}$ transforms $\mathbf{Y'BY}$ to normal form $\mathbf{Z'CZ}$. Then $\mathbf{C} = \mathbf{Q'BQ}$. Since $\mathbf{P}$ and $\mathbf{Q}$ are real non-singular matrices, therefore $\mathbf{PQ}$ is also a real non-singular matrix. The linear transformation $\mathbf{X} = (\mathbf{PQ})\mathbf{Z}$ will transform $\mathbf{X'AX}$ to the form

$$(\mathbf{PQZ})'\,\mathbf{A}\,(\mathbf{PQZ}) = \mathbf{Z'Q'P'APQZ} = \mathbf{Z'Q'BQZ}.$$

Thus the two given quadratic forms have a common normal form. Hence they have the same rank and same index (or signature).

Conversely, suppose that the two forms have the same rank r and the same signature s. Then they have the same index p where $2p - r = s$. So they can be reduced to the same normal form

$$\mathbf{Z'CZ} = z_1^2 + \ldots + z_p^2 - z_{p+1}^2 - \ldots - z_r^2$$

by real non-singular linear transformations, say, $\mathbf{X} = \mathbf{P\,Z}$ and $\mathbf{Y} = \mathbf{QZ}$ respectively. Then $\mathbf{P'AP} = \mathbf{C}$ and $\mathbf{Q'BQ} = \mathbf{C}$.

Therefore $\mathbf{Q'BQ} = \mathbf{P'AP}$. This gives

$$\mathbf{B} = (\mathbf{Q}')^{-1}\,\mathbf{P'APQ}^{-1} = (\mathbf{Q}^{-1})'\,\mathbf{P'APQ}^{-1} = (\mathbf{PQ}^{-1})'\,\mathbf{A}\,(\mathbf{PQ}^{-1}).$$

Therefore the real non-singular linear transformation $\mathbf{X} = (\mathbf{PQ}^{-1})\,\mathbf{Y}$ transforms $\mathbf{X'AX}$ to $\mathbf{Y'BY}$. Hence the two given quadratic forms are real equivalent.

Reduction of a real quadratic form in the complex field.

Theorem 5: *If* $\mathbf{A}$ *be any n-rowed real symmetric matrix of rank* r, *there exists a non-singular matrix* $\mathbf{P}$ *whose elements may be any complex numbers such that*

$$\mathbf{P'AP} = \textit{diag.}\,[1, 1, \ldots, 1, 0, \ldots, 0] \text{ where } 1 \text{ appears } r \text{ times.}$$

Proof: $\mathbf{A}$ is a real symmetric matrix of rank r. Therefore there exists a non-singular real matrix $\mathbf{Q}$ such that $\mathbf{Q'AQ}$ is a diagonal matrix $\mathbf{D}$ with precisely r non-zero diagonal elements. Let

$$\mathbf{Q'AQ} = \mathbf{D} = \text{diag. } [\lambda_1, \ldots, \lambda_r, 0, \ldots, 0].$$

The real numbers $\lambda_1, \ldots, \lambda_r$ may be positive or negative or both.

Let $\mathbf{S}$ be the $n \times n$ (complex) diagonal matrix with diagonal elements

$$\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_r}}, 1, \ldots, 1.$$

Then $\mathbf{S} = \text{diag.} \left[\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_r}}, 1, \ldots, 1\right]$ is a complex non-singular diagonal matrix and $\mathbf{S'} = \mathbf{S}$.

$\mathbf{P'AP} = \mathbf{(QS)'\ A\ (QS)} = \mathbf{S'Q'\ AQS} = \mathbf{S'\ DS} = \mathbf{SDS} = \text{diag. } [1, 1, \ldots, 1, 0, \ldots, 0]$ so that 1 appears r times. Hence the result.

Corollary 1: *Every real quadratic form* $\mathbf{X'AX}$ *is complex-equivalent to the form* $z_1^2 + \ldots + z_r^2$ *where r is the rank of* $\mathbf{A}$.

Corollary 2: *Two real quadratic forms in n variables are complex equivalent if and only if they have the same rank.*

Orthogonal reduction of a real quadratic form.

Theorem 6: *If* $\phi = \mathbf{X'AX}$ *be a real quadratic form of rank r in n variables, then there exists a real orthogonal transformation* $\mathbf{X} = \mathbf{PY}$ *which transforms* ϕ *to the diagonal form*

$$\lambda_1 y_1^2 + \ldots + \lambda_r y_r^2,$$

where $\lambda_1, \ldots, \lambda_r$ *are the, r, non-zero eigenvalues of* $\mathbf{A}$, $n - r$ *eigenvalues of* $\mathbf{A}$ *being equal to zero.*

Proof: Since $\mathbf{A}$ is a real symmetric matrix, therefore there exists a real orthogonal matrix $\mathbf{P}$ such that $\mathbf{P^{-1}AP} = \mathbf{D},$ where $\mathbf{D}$ is a diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$.

Since $\mathbf{A}$ is of rank r, therefore $\mathbf{P^{-1}AP} = \mathbf{D}$ is also of rank r. So $\mathbf{D}$ has precisely r non-zero diagonal elements. Consequently $\mathbf{A}$ has exactly r non-zero eigenvalues, the remaining $n - r$ eigenvalues of $\mathbf{A}$ being zero. Let $\mathbf{D} = \text{diag. } [\lambda_1, \ldots, \lambda_r, \ldots 0, \ldots 0]$.

Since $\mathbf{P^{-1}} = \mathbf{P'}$, therefore $\mathbf{P^{-1}AP} = \mathbf{D}$

$$\Rightarrow \qquad \mathbf{P'AP} = \mathbf{D} \Rightarrow \mathbf{A} \text{ is congruent to } \mathbf{D}.$$

Now consider the real orthogonal transformation $\mathbf{X} = \mathbf{PY}.$ We have

$$\mathbf{X'AX} = \mathbf{(PY)'\ A\ (PY)} = \mathbf{Y'P'APY} = \mathbf{Y'\ DY}$$
$$= \lambda_1 y_1^2 + \ldots + \lambda_r y_r^2.$$

Hence the result.

Theorem 7: *Every real quadratic form* $\mathbf{X'AX}$ *in n variables is real equivalent to the form*

$$y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2,$$

where r is the rank of $\mathbf{A}$ *and p is the number of positive eigenvalues of* $\mathbf{A}$.

Proof: $\mathbf{A}$ is a real symmetric matrix. Therefore there exists a real orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{Q^{-1}AQ} = \mathbf{Q'\ AQ} = \mathbf{D},$$

where $\mathbf{D}$ is a diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$. Since $\mathbf{A}$ is of rank r, therefore $\mathbf{D}$ is also of rank r. So $\mathbf{D}$ has exactly r non-zero diagonal elements. Consequently $\mathbf{A}$ has exactly r non-zero eigenvalues, the remaining $n - r$ eigenvalues of $\mathbf{A}$ being zero. Let $\mathbf{D} = \text{diag. } [\lambda_1, \lambda_2 \ldots, \lambda_r, 0, \ldots 0]$.

Let $\lambda_1, \ldots, \lambda_p$ be positive and $\lambda_{p+1}, \ldots \lambda_r$ be negative. Let $\mathbf{S}$ be the $n \times n$ real diagonal matrix with diagonal elements

$\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_p}, \sqrt{(-\lambda_{p+1})}, \ldots, \sqrt{(-\lambda_r)}, 1, \ldots, 1$

Then **S** is non-singular and **S′ = S.** If we take **P = QS,** then **P** is also a real non-singular matrix and we have

$$\mathbf{P'AP} = \mathbf{(QS)'A(QS)} = \mathbf{S'Q'AQS} = \mathbf{SDS}$$
$$= \text{diag. } [1, \ldots, 1, -1, \ldots, -1, 0, \ldots, 0]$$

so that 1 and -1 appear p and $r-p$ times respectively.

Now the real non-singular linear transformation **X = PY** reduces **X′AX** to the form **Y′P′APY** *i.e.*,

$$y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2.$$

Hence the result.

Corollary: *Two real quadratic forms* **X′AX** *and* **Y′BY** *in the same number of variables are real equivalent if and only if* **A** *and* **B** *have the same number of positive and negative eigenvalues.*

Important Note. If **X′AX** is a real quadratic form, then the number of non-zero eigenvalues of **A** is equal to the rank of **X′AX** and the number of positive eigenvalues of **A** is equal to the index of **X′AX.**

Theorem 8 : *Two real quadratic forms* **X′AX** *and* **Y′BY** *are orthogonally equivalent if and only if* **A** *and* **B** *have the same eigenvalues and these occur with the same multiplicities.*

Proof: If **A** and **B** have eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ and **D** is a diagonal matrix with $\lambda_1, \lambda_2, \ldots, \lambda_n$ as diagonal elements, then there exist orthogonal matrices **P** and **Q** such that **P′AP = D = Q′BQ.**

Now $$\mathbf{Q'BQ} = \mathbf{P'AP}$$

$$\Rightarrow \quad \mathbf{B} = \mathbf{(Q')^{-1} P'APQ^{-1}}$$
$$= \mathbf{(Q^{-1})' P'APQ^{-1}}$$
$$= \mathbf{(PQ^{-1})' A (PQ^{-1})}.$$

Since $\mathbf{PQ^{-1}}$ is an orthogonal matrix, therefore **Y′BY** is orthogonally equivalent to **X′AX.**

Conversely, if the two forms are orthogonally equivalent, then there exists an orthogonal matrix **P** such that $\mathbf{B = P'AP = P^{-1}AP}$**.** Therefore **A** and **B** are similar matrices and so have the same eigenvalues with the same multiplicities.

Illustrative Examples

Example 10: *Reduce each of the following quadratic forms in three variables to real canonical form and find its rank and signature. Also write in each case the linear transformation which brings about the normal reduction.*

(i) $2x_1^2 + x_2^2 - 3x_3^2 - 8x_2x_3 - 4x_3x_1 + 12x_1x_2$.

Solution: (i) The matrix **A** of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 2 & 6 & -2 \\ 6 & 1 & -4 \\ -2 & -4 & -3 \end{bmatrix}.$$

We write $\mathbf{A} = \mathbf{IAI}$ *i.e.*,

$$\begin{bmatrix} 2 & 6 & -2 \\ 6 & 1 & -4 \\ -2 & -4 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now we shall reduce **A** to diagonal form by applying congruence operations on it. Performing $R_2 \to R_2 - 3R_1, C_2 \to C_2 - 3C_1$ and $R_3 \to R_3 + R_1, C_3 \to C_3 + C_1$, we get

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -17 & 2 \\ 0 & 2 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -3 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

[Note that we apply the row and column operations on **A** in two separate steps. But in order to save labour we should apply them in one step. For this we should not first write the first row. After changing R_2 and R_3 with the corresponding row operations we should simply write 0 in the second and third places of the first row and the first element of the first row should be kept unchanged].

Now performing $R_3 \to R_3 + \dfrac{2}{17} R_2$, $C_3 \to C_3 + \dfrac{2}{17} C_2$, we get

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -17 & 0 \\ 0 & 0 & -\frac{81}{17} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ \frac{11}{17} & \frac{2}{17} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -3 & \frac{11}{17} \\ 0 & 1 & \frac{2}{17} \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_1 \to \dfrac{1}{\sqrt{2}} R_1$, $C_1 \to \dfrac{1}{\sqrt{2}} C_1$; $R_2 \to \dfrac{1}{\sqrt{17}} R_2$, $C_2 \to \dfrac{1}{\sqrt{17}} C_2$;

and $R_3 \to \sqrt{(17/81)}\, R_3$, $C_3 \to \sqrt{(17/81)}\, C_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ -3b & b & 0 \\ \frac{11}{17}c & \frac{2}{17}c & c \end{bmatrix} \mathbf{A} \begin{bmatrix} a & -3b & \frac{11}{17}c \\ 0 & b & \frac{2}{17}c \\ 0 & 0 & c \end{bmatrix},$$

where $a = 1/\sqrt{2}, b = 1/\sqrt{17}, c = \sqrt{(17/81)}$.

Thus the linear transformation $\mathbf{X} = \mathbf{PY}$

where $\mathbf{P} = \begin{bmatrix} a & -3b & \frac{11}{17}c \\ 0 & b & \frac{2}{17}c \\ 0 & 0 & c \end{bmatrix}$, $\mathbf{X} = [x_1 \quad x_2 \quad x_3]'$, $\mathbf{Y} = [y_1 \quad y_2 \quad y_3]'$,

transforms the given quadratic form to the normal form

The rank r of the given quadratic form = the number of non-zero terms in its normal form (1) = 3.

The signature of the given quadratic form = the excess of the number of positive terms over the number of negative terms in its normal form = 1 – 2 = – 1.

The index of the given quadratic form = the number of positive terms in its normal form = 1.

The linear transformation **X = PY** which brings about this normal reduction is given by

$$x_1 = ay_1 - 3by_2 + \frac{11}{7}cy_3\,,\, x_2 = by_2 + \frac{2}{17}cy_3\,,\, x_3 = cy_3\,.$$

(ii) The matrix of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & 2 & 9 \\ 2 & 3 & 2 \\ 9 & 2 & 14 \end{bmatrix}.$$

We write $\begin{bmatrix} 6 & 2 & 9 \\ 2 & 3 & 2 \\ 9 & 2 & 14 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

Performing congruence operations

$$R_2 \to R_2 - \frac{1}{3}R_1, C_2 \to C_2 - \frac{1}{3}C_1\,;$$

and $$R_3 \to R_3 - \frac{3}{2}R_1, C_3 \to C_3 - \frac{3}{2}C_1,$$

we get

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & -1 \\ 0 & -1 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ -\frac{3}{2} & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -\frac{1}{3} & -\frac{3}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \to R_3 + \frac{3}{7}R_2,\ C_3 \to C_3 + \frac{3}{7}C_2$, we get

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & 0 \\ 0 & 0 & \frac{1}{14} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ \frac{23}{14} & \frac{3}{7} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -\frac{1}{3} & \frac{23}{14} \\ 0 & 1 & \frac{3}{7} \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_1 \to (1/\sqrt{6})R_1, C_1 \to (1/\sqrt{6})C_1\,; R_2 \to \sqrt{(3/7)}\,R_2, C_2 \to \sqrt{(3/7)}\,C_2;$ $R_3 \to (1/14)\,R_3, C_3 \to (1/\sqrt{14})\,C_3$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ -\frac{1}{3}b & b & 0 \\ \frac{23}{14}c & \frac{3}{7}c & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} a & -\frac{1}{3}b & \frac{23}{14}c \\ 0 & b & \frac{3}{7}c \\ 0 & 0 & 1 \end{bmatrix}$$

where $$a = \frac{1}{\sqrt{6}},\ b = \sqrt{(3/7)},\ c = \frac{1}{\sqrt{14}}.$$

$$\mathbf{P} = \begin{bmatrix} a & -\frac{1}{3}b & \frac{23}{14}c \\ 0 & b & \frac{3}{7}c \\ 0 & 0 & 1 \end{bmatrix}$$

transforms the given quadratic form to the normal form

$$y_1^2 + y_2^2 + y_3^2.$$

The rank of the given quadratic form is 3 and its signature is

$$3 - 0 = 3.$$

Example 11: *Find an orthogonal matrix* **P** *that will diagonalize the real symmetric matrix*

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Interpret the result in terms of quadratic forms.

Solution: The characteristic equation of the given matrix is

$$|\mathbf{A} - \lambda\mathbf{I}| = 0$$

i.e., $$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & -1 \\ 1 & -1 & -\lambda \end{vmatrix} = 0 \text{ } i.e., (\lambda - 1)^2 (\lambda + 2) = 0.$$

$\therefore$ the eigenvalues of **A** are $1, 1, -2$.

Corresponding to the eigenvalue 1 we can find two mutually orthogonal eigenvectors of **A** by solving

$$(\mathbf{A} - \mathbf{I})\,\mathbf{X} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

or $$-x_1 + x_2 + x_3 = 0.$$

Two orthogonal solutions are

$$\mathbf{X}_1 = [1, \ 0, \ 1]' \text{ and } \mathrm{X}_2 = [1, \ 2, \ -1]'.$$

An eigenvector corresponding to the eigenvalue -2, is found by solving

$$2x_1 + x_2 + x_3 = 0, \ x_1 + 2x_2 - x_3 = 0$$

to be $\mathbf{X}_3 = [-1, \ 1, \ 1]'$. The required matrix **P** is therefore a matrix whose columns are unit vectors which are scalar multiples of $\mathbf{X}_1, \mathbf{X}_2$ and $\mathbf{X}_3$.

$\therefore$ $$\mathbf{P} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

We have $\mathbf{P}'\,\mathbf{AP} =$ diag. $[1, 1, -2]$

The quadratic form corresponding to the symmetric matrix **A** is

The orthogonal linear transformation $\mathbf{X} = \mathbf{PY}$ will transform it to the diagonal form

$$y_1^2 + y_2^2 - 2y_3^2 .$$

The rank of the quadratic form ϕ = the number of non-zero eigenvalues of its matrix $\mathbf{A} = 3$.

The signature of the quadratic form ϕ = the number of positive eigenvalues of $\mathbf{A}$ = the number of negative eigenvalues of $\mathbf{A} = 2 - 1 = 1$. The normal form is $z_1^2 + z_2^2 - z_3^2$.

Example 12: *Reduce the quadratic form*

$$6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 + 4x_1x_3 - 2x_2x_3$$

to the canonical form by an orthogonal transformation and hence find the rank and signature of the given quadratic form.

Solution: The matrix $\mathbf{A}$ of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

The characteristic equation of $\mathbf{A}$ is $|\mathbf{A} - \lambda I| = 0$

or $$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

or $$\begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 3-\lambda & 2-\lambda \\ 2 & -1 & 2-\lambda \end{vmatrix} = 0, C_3 \to C_3 + C_2$$

or $$(2-\lambda)\begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 3-\lambda & 1 \\ 2 & -1 & 1 \end{vmatrix} = 0$$

or $$(2-\lambda)\begin{vmatrix} 6-\lambda & -2 & 0 \\ -4 & 4-\lambda & 0 \\ 2 & -1 & 1 \end{vmatrix} = 0, R_2 \to R_2 - R_3$$

or $$(2-\lambda)[(6-\lambda)(4-\lambda) - 8] = 0$$

or $$(2-\lambda)(\lambda^2 - 10\lambda + 16) = 0$$

or $$(2-\lambda)(\lambda - 2)(\lambda - 8) = 0.$$

$\therefore$ the eigenvalues of $\mathbf{A}$ are $2, 2, 8$.

The eigenvalue 8 is of algebraic multiplicity 1. So there will be only one linearly independent eigenvector corresponding to this value. The eigenvectors corresponding to the eigenvalue 8 are given by the equation

$$(\mathbf{A} - 8\mathbf{I})\,\mathbf{X} = O$$

or $$\begin{bmatrix} & & \\ -2 & -5 & -1 \\ 2 & -1 & -6 \end{bmatrix}\begin{bmatrix} \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} \\ 0 \\ 0 \end{bmatrix}.$$

Since these equations have only one linearly independent solution, therefore the coefficient matrix of these equations is of rank 2 and its third row can be made zero by elementary row operations. So in order to find an eigenvector $\mathbf{X}=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ corresponding to the eigenvalue 8, it is sufficient to find x_1, x_2, x_3 satisfying the equations

$$-2x_1-2x_2+2x_3=0 \qquad \text{...(1)}$$

and $$-2x_1-5x_2-x_3=0 \qquad \text{...(2)}$$

Subtracting (2) from (1), we get

$$3x_2+3x_3=0.$$

$\therefore$ $\quad x_2=1, x_3=-1, x_1=-2$ is a solution

Thus $\mathbf{X}_1=\begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue 8.

The eigenvalue 2 is of algebraic multiplicity 2. So we are to find two mutually orthogonal eigenvectors corresponding to it.

The eigenvectors $\mathbf{X}$ corresponding to the eigenvalue 2 are given by the equation

$$(\mathbf{A}-2\mathbf{I})\mathbf{X}=\mathbf{O}$$

or $$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since these equations have two linearly independent solutions, therefore their coefficient matrix is of rank 1 and its second and third rows can be made zero by elementary row operations. So we should find two orthogonal solutions of the equation

$$4x_1-2x_2+2x_3=0$$

i.e., $$2x_1-x_2+x_3=0. \qquad \text{...(3)}$$

Obviously $x_1=0, x_2=1, x_3=1$ is a solution of (3).

$\therefore$ $\quad \mathbf{X}_2=\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to the eigenvalue 2.

Let $\mathbf{X}_3=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be another eigenvector of $\mathbf{A}$ corresponding to the eigenvalue 2 and let $\mathbf{X}_3$ be orthogonal to $\mathbf{X}_2$.

Then $\quad 2x-y+z=0 \qquad [\because \mathbf{X}_3$ is a solution of (3)]

Obviously $y = 1, z = -1, x = 1$ is a solution.

$$\therefore \qquad \mathbf{X}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Lengths of the vectors $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ are $\sqrt{6}, \sqrt{2}, \sqrt{3}$ respectively.

$$\therefore \qquad \mathbf{P} = \left[\frac{1}{\sqrt{6}}\mathbf{X}_1, \frac{1}{\sqrt{2}}\mathbf{X}_2, \frac{1}{\sqrt{3}}\mathbf{X}_3\right] = \begin{bmatrix} -2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix}$$

is the required orthogonal matrix that will diagonalize $\mathbf{A}$. We have $\mathbf{P}^{-1} = \mathbf{P}^T$ and

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{P}^T\mathbf{A}\mathbf{P} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The orthogonal linear transformation $\mathbf{X} = \mathbf{PY}$ will transform the given quadratic form to the diagonal form

$$8y_1^2 + 2y_2^2 + 2y_3^2.$$

The canonical form of the given quadratic form is

$$z_1^2 + z_2^2 + z_3^2.$$

The rank of the given quadratic form = the number of non-zero eigenvalues of its matrix $\mathbf{A} = 3$.

The signature of the given quadratic form = the number of positive eigenvalues of A – the number of negative eigenvalues of $\mathbf{A}$

$$= 3 - 0 = 3.$$

15 Lagrange's Reduction of a Real Quadratic Form

This method proceeds by forming squares successively by taking lesser and lesser number of variables among $x_1, x_2, \ldots, x_n$. We shall illustrate this method by the following examples.

Illustrative Examples

Example 13: *Reduce the quadratic form*

$$q(x_1, x_2, x_3) = 4x_1^2 + 10x_2^2 + 11x_3^2 - 4x_1x_2 + 12x_1x_3 - 12x_2x_3$$

to diagonal form by using Lagrange's reduction.

Solution: We have

$$q(x_1, x_2, x_3) = 4x_1^2 + 10x_2^2 + 11x_3^2 - 4x_1x_2 + 12x_1x_3 - 12x_2x_3.$$

$x_1 x_2$ and $x_1 x_3$, we have

$$q = 4\{x_1^2 - x_1(x_2 - 3x_3)\} + 10x_2^2 + 11x_3^2 - 12x_2x_3 .$$

Completing a square on the term x_1 by adding and subtracting suitable terms as needed, we get

$$q = 4\left\{x_1 - \frac{1}{2}(x_2 - 3x_3)\right\}^2 - (x_2 - 3x_3)^2 + 10x_2^2 + 11x_3^2 - 12x_2x_3$$

$$= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9x_2^2 - 6x_2x_3 + 2x_3^2$$

$$= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9\left(x_2^2 - \frac{2}{3}x_2 x_3\right) + 2x_3^2.$$

Now completing square on x_2, we have

$$q = 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9\left(x_2 - \frac{1}{3}x_3\right)^2 - x_3^2 + 2x_3^2$$

$$= 4\left(x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3\right)^2 + 9\left(x_2 - \frac{1}{3}x_3\right)^2 + x_3^2.$$

Now we put

$$y_1 = x_1 - \frac{1}{2}x_2 + \frac{3}{2}x_3$$

$$y_2 = x_2 - \frac{1}{3}x_3$$

$$y_3 = x_3$$

which is of the form $\mathbf{Y} = \mathbf{PX}$ with the transformation matrix

$$\mathbf{P} = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix}$$

which is non-singular with determinant equal to 1.

With this transformation, the given quadratic form reduces to the diagonal form

$$q = 4y_1^2 + 9y_2^2 + y_3^2.$$

If we choose $z_1 = 2y_1, z_2 = 3y_2, z_3 = y_3$, then we have $q = z_1^2 + z_2^2 + z_3^2$, so that the given quadratic form is reduced to a unit quadratic form.

Example 14: *Using Lagrange's reduction reduce the quadratic form*

$$q(x_1, x_2, x_3) = x_1 x_2 + 2x_1 x_3 + 3x_2 x_3$$

to a diagonal form.

Solution: Set

$$x_1 = y_1,\ x_2 = y_1 + y_2,\ x_3 = y_3 .$$

The transformation $\mathbf{X} = \mathbf{PY} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{Y}$ is non-singular because

det $P = 1 \neq 0$.

The quadratic form q now becomes

$$q = y_1 (y_1 + y_2) + 2 y_1 y_3 + 3 (y_1 + y_2) y_3$$
$$= y_1^2 + y_1 y_2 + 5 y_1 y_3 + 3 y_2 y_3$$
$$= \{ y_1^2 + y_1 (y_2 + 5 y_3) \} + 3 y_2 y_3$$
$$= \{ y_1 + \frac{1}{2} (y_2 + 5 y_3) \}^2 - \frac{1}{4} (y_2^2 + 10 y_2 y_3 + 25 y_3^2) + 3 y_2 y_3$$
$$= \left(y_1 + \frac{1}{2} y_2 + \frac{5}{2} y_3 \right)^2 - \frac{1}{4} (y_2^2 + 10 y_2 y_3 - 12 y_2 y_3) - \frac{25}{4} y_3^2$$
$$= \left(y_1 + \frac{1}{2} y_2 + \frac{5}{2} y_3 \right)^2 - \frac{1}{4} (y_2^2 - 2 y_2 y_3) - \frac{25}{4} y_3^2$$
$$= \left(y_1 + \frac{1}{2} y_2 + \frac{5}{2} y_3 \right)^2 - \frac{1}{4} (y_2 - y_3)^2 + \frac{1}{4} y_3^2 - \frac{25}{4} y_3^2$$
$$= \left(y_1 + \frac{1}{2} y_2 + \frac{5}{2} y_3 \right)^2 - \frac{1}{4} (y_2 - y_3)^2 - 6 y_3^2.$$

Now put $\quad z_1 = y_1 + \frac{1}{2} y_2 + \frac{5}{2} y_3 = \frac{1}{2} x_1 + \frac{1}{2} x_2 + \frac{5}{2} x_3$

$$z_2 = y_2 - y_3 = x_2 - x_1 - x_3$$
$$z_3 = y_3 = x_3$$

so that the non-singular transformation $\mathbf{Z} = \mathbf{QX}$ *i.e.*,

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{5}{2} \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

with det $\mathbf{Q} = 1$ will reduce the given quadratic form to the diagonal form

$$q = z_1^2 - \frac{1}{4} z_2^2 - 6 z_3^2.$$

Comprehensive Exercise 2

1. Write the matrix and find the rank of each of the following quadratic forms :

(i) $x_1^2 - 2 x_1 x_2 + 2 x_2^2$

(ii) $4 x_1^2 + x_2^2 - 8 x_3^2 + 4 x_1 x_2 - 4 x_1 x_3 + 8 x_2 x_3$.

2. Determine a non-singular matrix **P** such that **P′ AP** is a diagonal matrix, where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 3 & 0 \end{bmatrix}.$$

form and find its rank and signature. Also write in each case the linear transformation which brings about the normal reduction.

(i) $x^2 - 2y^2 + 3z^2 - 4yz + 6zx.$

(ii) $x^2 + 2y^2 + 2z^2 - 2xy - 2yz + zx.$

4. Reduce the following quadratic form to canonical form and find its rank and signature :

$$x^2 + 4y^2 + 9z^2 + t^2 - 12yz + 6zx - 4xy - 2xt - 6zt.$$

5. Using Lagrange's reduction reduce the quadratic form

$$[x_1 \quad x_2 \quad x_3]\begin{bmatrix} 5 & -2 & 0 \\ -2 & 6 & 2 \\ 0 & 2 & 7 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

to a diagonal form.

6. Using Lagrange's reduction reduce the quadratic form

$$[x_1 \quad x_2 \quad x_3 \quad x_4]\begin{bmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

to a diagonal form.

7. Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = (x_1 + x_2 + x_3)\, x_2$$

to a diagonal form.

8. Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1$$

to a diagonal form.

9. Using Lagrange's reduction reduce the quadratic form

$$q(x_1, x_2, x_3) = x_1^2 + 4x_2^2 + 16x_3^2 + 4x_1x_2 + 8x_1x_3 + 17x_2x_3$$

to a diagonal form.

Answers 2

1. (i) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$, rank 2 (ii) $\begin{bmatrix} 4 & 2 & -2 \\ 2 & 1 & 4 \\ -2 & 4 & -8 \end{bmatrix}$, rank 3

2. $\mathbf{P} = \begin{bmatrix} 1 & -\frac{1}{2} & -3 \\ 1 & \frac{1}{2} & -2 \\ 0 & 0 & 1 \end{bmatrix}$

3. (i) The linear transformation $\mathbf{X} = \mathbf{PY}$ where

$$\mathbf{P} = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \mathbf{X} = [x \quad y \quad z]', \mathbf{Y} = [y_1 \quad y_2 \quad y_3]'$$

transforms the given quadratic form to the normal form $y_1^2 - y_2^2 - y_3^2$

The rank of the given quadratic form is 3 and its signature is -1

(ii) The linear transformation $\mathbf{X} = \mathbf{PY}$ where

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1/\sqrt{6} \\ 0 & 0 & \sqrt{(2/3)} \end{bmatrix}$$

transforms the given quadratic form to the normal form $y_1^2 + y_2^2 + y_3^2$

Rank of the given quadratic form is 3 and its signature is 3

4. $y_1^2 - y_2^2 + y_4^2$

Rank of the given quadratic form is 3 and its signature is 1

5. $\frac{1}{5} y_1^2 + \frac{10}{13} y_2^2 + \frac{81}{13} y_3^2$

6. $\frac{1}{3} y_1^2 + \frac{8}{3} y_2^2 + \frac{1}{3} y_3^2 + \frac{8}{3} y_4^2$

7. $\frac{1}{4} (y_1^2 - y_2^2)$

8. $\frac{1}{4} (z_1^2 - z_2^2) - z_3^2$

9. $z_1^2 + \frac{1}{4} (z_2^2 - z_3^2)$

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16 Value Class of a Real Quadratic Form. Definite, Semi-definite and Indefinite Real Quadratic Forms

Definitions: *Let* $\phi = \mathbf{X}' \mathbf{A} \mathbf{X}$ *be a real quadratic form in n variables* $x_1, \ldots, x_n$. *The form* ϕ *is said to be*

(i) Positive Definite (PD) *if* $\phi \geq 0$ *for all real values of the variables* $x_1, \ldots, x_n$ *and* $\phi = 0$ *only if* $\mathbf{X} = \mathbf{O}$ *i.e.*, $\phi = 0 \Rightarrow x_1 = x_2 = \ldots = x_n = 0$.

For example the quadratic form $x_1^2 - 4x_1 x_2 + 5x_2^2$ in two variables is positive definite because it can be written as

$$(x_1 - 2x_2)^2 + x_2^2,$$

which is ≥ 0 for all real values of x_1, x_2 and

$$(x_1 - 2x_2)^2 + x_2^2 = 0 \Rightarrow x_1 - 2x_2 = 0, x_2 = 0$$

$$\Rightarrow \qquad x_1 = 0, x_2 = 0.$$

Similarly the quadratic form $x_1^2 + x_2^2 + x_3^2$ in three variables is a positive definite form.

only if $x_1 = x_2 = \ldots = x_n = 0$.

For example $-x_1^2 - x_2^2 - x_3^2$ is a negative definite form in three variables.

(iii) Positive semi-definite (PSD) *if* $\phi \geq 0$ *for all real values of the variables* $x_1, \ldots, x_n$ *and* $\phi = 0$ *for some non-zero real vector* $\mathbf{X}$ *i.e.,* $\phi = 0$ *for some real values of the variables* $x_1, x_2, \ldots, x_n$ *not all zero.*

For example the quadratic form

$$x_1^2 + x_2^2 + 2x_3^2 - 2x_1x_3 - 2x_2x_3$$

is positive semi-definite because it can be written in the form

$$(x_1 - x_3)^2 + (x_2 - x_3)^2,$$

which is ≥ 0 for all real values of x_1, x_2, x_3 but is zero for non-zero values also, for example, $x_1 = x_2 = x_3 = 1$.

Similarly the quadratic form $x_1^2 + x_2^2 + 0x_3^2$ in three variables x_1, x_2, x_3 is positive semi-definite. It is non-negative for all real values of x_1, x_2, x_3 and it is zero for values $x_1 = 0, x_2 = 0, x_3 = 2$ which are not all zero.

(iv) Negative semi-definite (NSD) *if* $\phi \leq 0$ *for all real values of the variables* $x_1, \ldots, x_n$ *and* $\phi = 0$ *for some values of the variables* $x_1, \ldots, x_n$ *not all zero.*

For example the quadratic form $-x_1^2 - x_2^2 - 0x_3^2$ in three variables x_1, x_2, x_3 is negative semi-definite.

(v) Indefinite (I) *if* ϕ *takes positive as well as negative values for real values of the variables* $x_1, \ldots, x_n$.

For example the quadratic form $x_1^2 - x_2^2 + x_3^2$ in three variables is indefinite. It takes positive value 1 when $x_1 = 1, x_2 = 1, x_3 = 1$ and it takes negative value -1 when $x_1 = 0, x_2 = 1, x_3 = 0$.

Note 1: The above five classes of real quadratic forms are mutually exclusive and are called **value classes of real quadratic forms.** *Every real quadratic form must belong to one and only one value class.*

Note 2: A form which is positive definite or negative definite is called *definite* and a form which is positive semi-definite or negative semi-definite is called *semi-definite.*

Non-negative definite quadratic form.

Definition: *A real quadratic form* $\phi = \mathbf{X'AX}$ *in* n *variables* $x_1, \ldots, x_n$, *is said to be non-negative definite if it takes only non-negative values for all real values of* $x_1, \ldots, x_n$.

Thus ϕ is non-negative definite if $\phi \geq 0$ for all real values of $x_1, \ldots, x_n$. A non-negative definite quadratic form may be positive definite or positive semi-definite. It is positive definite if it takes the value 0 only when $x_1 = x_2 = \ldots = x_n = 0$.

Classification of real-symmetric matrices.

Definite, semi-definite and indefinite real symmetric matrices.

Definition: *A real symmetric matrix* $\mathbf{A}$ *is said to be definite, semi-definite or indefinite if the corresponding quadratic form* $\mathbf{X'AX}$ *is definite, semi-definite or indefinite respectively.*

Positive definite real symmetric matrix.

Definition: *A real symmetric matrix* $\mathbf{A}$ *is said to be positive definite if the corresponding form* $\mathbf{X'}$ $\mathbf{AX}$ *is positive definite.*

Definition: *A real symmetric matrix* $\mathbf{A}$ *is said to be non-negative definite if the associated quadratic form* $\mathbf{X'AX}$ *is non-negative definite.*

Theorem 1: *All real equivalent real quadratic forms have the same value class.*

Proof: Let $\phi = \mathbf{X'AX}$ and $\psi = \mathbf{Y'BY}$ be two real equivalent real quadratic forms. Then there exists a real non-singular matrix $\mathbf{P}$ such that $\mathbf{P'AP} = \mathbf{B}$ and $(\mathbf{P}^{-1})'\mathbf{BP}^{-1} = \mathbf{A}.$

The real non-singular linear transformation $\mathbf{X} = \mathbf{PY}$ transforms the quadratic form ϕ into the quadratic form ψ and the inverse transformation $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$ transforms the quadratic form ψ into the quadratic form ϕ. The two quadratic forms have the same ranges of values. The vectors $\mathbf{X}$ and $\mathbf{Y}$ for which ϕ and ψ have the same value are connected by the relations $\mathbf{X} = \mathbf{PY}$ and $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}.$ Thus the vector $\mathbf{Y}$ for which ψ has the same value as ϕ has for the vector $\mathbf{X}$ is given by $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}.$ Similarly the vector $\mathbf{X}$ for which ϕ has the same value as ψ has for the vector $\mathbf{Y}$ is given by $\mathbf{X} = \mathbf{PY}.$

Now we shall discuss the five cases separately.

Case I. ϕ *is positive definite if and only if* ψ *is positive definite.*

Suppose ϕ is positive definite.

Then $\phi \geq 0$ and $\phi = 0 \Rightarrow \mathbf{X} = \mathbf{O}.$

Since ϕ and ψ have same ranges of values, therefore

$$\phi \geq 0 \Rightarrow \psi \geq 0.$$

Also $\psi = 0 \Rightarrow \mathbf{Y'BY} = 0$

$\Rightarrow \quad (\mathbf{PY})'\mathbf{A}(\mathbf{PY}) = 0$ [$\because$ ϕ has the same value for the vector $\mathbf{PY}$ as ψ has for the vector $\mathbf{Y}$]

$\Rightarrow \quad \mathbf{PY} = \mathbf{O}$ [$\because$ ϕ is positive definite means $\mathbf{X'AX} = 0 \Rightarrow \mathbf{X} = \mathbf{O}$]

$\Rightarrow \quad \mathbf{P}^{-1}(\mathbf{PY}) = \mathbf{P}^{-1}\mathbf{O}$

$\Rightarrow \quad \mathbf{Y} = \mathbf{O}.$

Thus ψ is also positive definite.

Conversely suppose that ψ is positive definite.

Then $\psi \geq 0$ and $\psi = 0 \Rightarrow \mathbf{Y} = \mathbf{O}.$

Since ϕ and ψ have the same ranges of values, therefore

$$\psi \geq 0 \Rightarrow \phi \geq 0.$$

Also $\phi = 0 \Rightarrow \mathbf{X'AX} = 0$

$\Rightarrow \quad (\mathbf{P}^{-1}\mathbf{X})'\mathbf{B}(\mathbf{P}^{-1}\mathbf{X}) = 0$ [$\because$ ψ has the same value for the vector $\mathbf{P}^{-1}\mathbf{X}$ as ϕ has for the vector $\mathbf{X}$]

$\Rightarrow \quad \mathbf{P}^{-1}\mathbf{X} = \mathbf{O}$ [$\because$ ψ is positive definite]

$\Rightarrow \quad \mathbf{P}(\mathbf{P}^{-1}\mathbf{X}) = \mathbf{PO}$

$\Rightarrow \quad \mathbf{X} = \mathbf{O}.$

Thus ϕ is also positive definite.

Case II. ϕ *is negative definite if and only if* ψ *is negative definite.*

The proof is the same as in case **I.**

The only difference is that we are to replace the expressions $\phi \geq 0, \psi \geq 0$ by the expressions $\phi \leq 0, \psi \leq 0.$

Since ϕ and ψ have the same ranges of values, therefore $\phi \geq 0$ if and only if $\psi \geq 0$.
Further since **P** is non-singular, therefore

$$\mathbf{X} \neq \mathbf{O} \Rightarrow \mathbf{Y} = \mathbf{P}^{-1}\mathbf{X} \neq \mathbf{O}$$

and $$\mathbf{Y} \neq \mathbf{O} \Rightarrow \mathbf{X} = \mathbf{PY} \neq \mathbf{O}.$$

Also the vectors **X** and **Y** for which ϕ and ψ have the same values are connected by the relations $\mathbf{X} = \mathbf{PY}$ and $\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$. Therefore $\phi = 0$ for some non-zero vector **X** if and only if $\psi = 0$ for some non-zero vector **Y**.
Hence ϕ is positive semi-definite if and only if ψ is positive semi-definite.
Case IV. *ϕ is negative semi-definite if and only if ψ is negative semi-definite.*
For proof replace the expressions $\phi \geq 0, \psi \geq 0$ in case III by the expressions

$$\phi \leq 0, \psi \leq 0.$$

Case V. *ϕ is indefinite if and only if ψ is indefinite*. Since ϕ and ψ have the same ranges of values, therefore the result follows immediately.
Thus the proof of the theorem is complete.

Criterion for the value of a real quadratic form in terms of its rank and signature.

Theorem 2: *Suppose r is the rank and s is the signature of a real quadratic form $\phi = \mathbf{X'AX}$ in n variables. Then ϕ is (i) positive definite if and only if $s = r = n$, (ii) negative definite if and only if $-s = r = n$, (iii) positive semi-definite if and only if $s = r < n$, (iv) negative semi-definite if and only if $-s = r < n$; and (v) indefinite if and only if $|s| \neq r$.*

Proof: Let $\psi = y_1^2 + \ldots + y_p^2 - y_{p+1}^2 - \ldots - y_r^2$ ...(1)

be the real canonical form of the real **quadratic** form ϕ of rank r and signature s. Then $s = 2p - r$. Since ϕ and ψ are real equivalent real quadratic forms, therefore they have the same value class.

(i) Suppose $s = r = n$. Then $p = n$ and the real canonical form of ϕ becomes $y_1^2 + \ldots + y_n^2$. But this is a positive definite quadratic form. So ϕ is also positive definite.
Conversely suppose that ϕ is positive definite. Then ψ is also a positive definite form in n variables. So we must have

$$\psi = y_1^2 + \ldots + y_n^2.$$

Hence $r = n, p = n,$
$2p - r = s = n.$

(ii) Suppose $-s = r = n$. Then $s = 2p - r$ gives $p = 0$. The real canonical form of ϕ becomes $-y_1^2 - \ldots - y_n^2$ which is negative definite and so ϕ is also negative definite.
Conversely if ϕ is negative definite, then ψ is also negative definite and so we must have $\psi = -y_1^2 - \ldots - y_n^2$. Hence $r = n, p = 0, 2p - r = s = -n$ *i.e.*, $-s = n$.

(iii) Suppose $s = r < n$. Then $s = 2p - r$ gives $p = r$ and the real canonical form of ϕ becomes $y_1^2 + \ldots + y_r^2$ where $r < n$. But this is a positive semi-definite form in n variables. So ϕ is also positive semi-definite.

form in n variables. So we must have $\psi = y_1^2 + \ldots + y_r^2$ where $r < n$. Therefore $p = r < n$ and $s = 2p - r = r$. Thus $s = r < n$.

(iv) Suppose $-s = r < n$. Then $s = 2p - r$ gives $p = 0$ and the real canonical form of ϕ becomes $-y_1^2 - \ldots - y_r^2$ where $r < n$. This is a negative semi-definite form in n variables. So ϕ is also negative semi-definite.

Conversely if ϕ is negative semi-definite, then ψ is also a negative semi-definite form in n variables. So we must have

$$\psi = -y_1^2 - \ldots - y_r^2, \text{ where } r < n.$$

Therefore $p = 0$ and $s = 2p - r = -r$. Thus $-s = r < n$.

(v) Suppose $|s| \neq r$. Then $|2p - r| \neq r$. Therefore $p \neq 0$ and $p \neq r$ and so $0 < p < r$. Then in this case the canonical form of ϕ has positive as well as negative terms and so it is an indefinite form. Consequently ϕ is also indefinite.

Conversely if ϕ is indefinite, then ψ is also indefinite. So there must be positive as well as negative terms in ψ. Therefore

$$|s| \neq r.$$

Criterion for the value class of a real quadratic form in terms of the eigenvalues of its matrix.

Suppose $\phi = \mathbf{X'AX}$ is a real quadratic form in n variables. Then $\mathbf{A}$ is a real symmetric matrix of order n. Suppose r is the number of non-zero eigenvalues of $\mathbf{A}$. Then r = rank of the quadratic form ϕ. Further if $s =$ the number of positive eigenvalues of $\mathbf{A}$ – the number of negative eigenvalues of $\mathbf{A}$, then s = signature of ϕ. Hence with the help of theorem 2, we arrive at the following conclusion.

Theorem 3: *A real quadratic form $\phi = \mathbf{X'AX}$ in n variables is*

(i) positive definite if and only if all the eigenvalues of $\mathbf{A}$ *are positive.*

(ii) negative definite if and only if all the eigenvalues of $\mathbf{A}$ *are negative.*

(iii) positive semi-definite if and only if all the eigenvalues of $\mathbf{A}$ *are* ≥ 0 *and at least one eigenvalue is* 0.

(iv) negative semi-definite if and only if all the eigenvalues of $\mathbf{A}$ *are* ≤ 0 *and at least one eigenvalue of* $\mathbf{A}$ *is* 0.

(v) indefinite if and only if $\mathbf{A}$ *has positive as well as negative eigenvalues.*

On account of its importance we shall give an independent proof of case (i).

Theorem 4: *A real symmetric matrix is positive definite if and only if all its eigenvalues are positive.*

Proof: Let $\mathbf{A}$ be a real symmetric matrix of order n. Then there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P^{-1}AP = P'AP = D} = \text{diag } [\lambda_1, \lambda_2, \ldots, \lambda_n]$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of $\mathbf{A}$.

Let $\mathbf{X'AX}$ be the real quadratic form corresponding to the matrix $\mathbf{A}$. Let us transform this quadratic form by the real non-singular linear transformation $\mathbf{X = PY}$ where $\mathbf{Y} = [y_1, y_2, \ldots, y_n]'$. Then

$$\mathbf{X'AX = (PY)'A\,(PY) = Y'\,P'APY = Y'\,DY.}$$

Now suppose that $\lambda_1, \lambda_2, \ldots, \lambda_n$ are all positive. Then the right hand side of (1) ensures that $\mathbf{X'AX} \geq 0$ for all real vectors $\mathbf{X}$. Also

$$\mathbf{X'AX} = 0$$

$\Rightarrow$ $\lambda_1 y_1^2 + \ldots + \lambda_n y_n^2 = 0$

$\Rightarrow$ $y_1 = y_2 = \ldots = y_n = 0$ [$\because \lambda_1, \ldots, \lambda_n$ are all positive]

$\Rightarrow$ $\mathbf{Y} = \mathbf{O}$

$\Rightarrow$ $\mathbf{P}^{-1}\mathbf{X} = \mathbf{O}$ [$\because \mathbf{X} = \mathbf{PY} \Rightarrow \mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}$]

$\Rightarrow$ $\mathbf{P}(\mathbf{P}^{-1}\mathbf{X}) = \mathbf{PO}$

$\Rightarrow$ $\mathbf{X} = \mathbf{O}.$

Thus if $\lambda_1, \lambda_2, \ldots, \lambda_n$ are all positive, then $\mathbf{X'AX}$ is positive definite and so the matrix $\mathbf{A}$ is positive definite.

Conversely suppose that $\mathbf{A}$ is a positive definite matrix. Then the quadratic form $\mathbf{X'AX}$ is positive definite. So

$\mathbf{X'AX} \geq 0$ for all real vectors $\mathbf{X}$

$\Rightarrow$ $\lambda_1 y_1^2 + \ldots + \lambda_n y_n^2 \geq 0$ for all real vectors $\mathbf{Y}$

$\Rightarrow$ $\lambda_1, \ldots, \lambda_n$ are all ≥ 0.

Also $\mathbf{X'AX} = 0$ only if $\mathbf{X} = \mathbf{O}$

$\Rightarrow$ $\lambda_1 y_1^2 + \ldots + \lambda_n y_n^2 = 0$ only if $\mathbf{PY} = \mathbf{O}$

$\Rightarrow$ $\lambda_1 y_1^2 + \ldots + \lambda_n y_n^2 = 0$ only if $\mathbf{Y} = \mathbf{O}$

[$\because$ $\mathbf{P}$ is non-singular means $\mathbf{PY} = \mathbf{O}$ only if $\mathbf{Y} = \mathbf{O}$]

$\Rightarrow$ $\lambda_1, \ldots, \lambda_n$ are all not equal to zero.

Therefore if $\mathbf{A}$ is positive definite, then $\lambda_1, \lambda_2, \ldots, \lambda_n$ are all > 0.

This completes the proof of the theorem.

Corollary 1: *A positive definite real symmetric matrix is non-singular.*

Proof: Suppose $\mathbf{A}$ is a positive definite real symmetric matrix. Then the eigenvalues of $\mathbf{A}$ are all positive. Also there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D},$$

where $\mathbf{D}$ is a diagonal matrix having the eigenvalues of $\mathbf{A}$ as its diagonal elements. So all diagonal elements of $\mathbf{D}$ are positive and thus $\mathbf{D}$ is non-singular.

Now $\mathbf{A} = \mathbf{PDP}^{-1} \Rightarrow \mathbf{A}$ is non-singular.

Corollary 2: *If the real quadratic form* $\mathbf{X}^T\mathbf{AX}$ *is positive definite, then there exists a non-singular transformation* $\mathbf{X} = \mathbf{PY}$ *such that*

$$\mathbf{X}^T\mathbf{AX} = \mathbf{Y}^T\mathbf{Y}.$$

Theorem 5: *A real matrix* $\mathbf{A}$ *is symmetric and positive definite if and only if* $\mathbf{P}^T\mathbf{AP}$ *is symmetric and positive definite for any real and non-singular* $\mathbf{P}$.

Proof: The condition is necessary. Suppose that $\mathbf{A}$ is real, symmetric and positive definite.

This implies that the real quadratic form $\mathbf{X}^T\mathbf{AX}$ is positive definite. Put $\mathbf{X} = \mathbf{PY}$, where $\mathbf{Y} \neq \mathbf{O}$ if $\mathbf{X} \neq \mathbf{O}$ because $\mathbf{P}$ is non-singular. Hence from we conclude that the right hand side is positive for all non-zero values of **Y.**

So the quadratic form $\mathbf{Y}^T\,(\mathbf{P}^T\mathbf{AP})\,\mathbf{Y}$ is positive definite and hence the matrix $\mathbf{P}^T\mathbf{AP}$ is positive definite.

Also $\quad \mathbf{P}^T\mathbf{AP} = \mathbf{P}^T\mathbf{A}^T\mathbf{P}$ $\quad$ [$\because$ **A** is symmetric $\Rightarrow \mathbf{A}^T = \mathbf{A}$]

$$= (\mathbf{P}^T\mathbf{AP})^T.$$

$\therefore \quad \mathbf{P}^T\mathbf{AP}$ is symmetric.

Hence $\mathbf{P}^T\mathbf{AP}$ is symmetric and positive definite.

The condition is sufficient. Now suppose that $\mathbf{P}^T\mathbf{AP}$ is symmetric and positive definite for any real and non-singular **P**. Putting $\mathbf{P} = \mathbf{I}$, we get

$$\mathbf{P}^T\mathbf{AP} = \mathbf{I}^T\mathbf{AI} = \mathbf{IAI} = \mathbf{A}.$$

Since $\mathbf{P}^T\mathbf{AP}$ is symmetric and positive definite, therefore **A** is symmetric and positive definite.

Theorem 6: *A real symmetric matrix* **A** *is positive definite if and only if there exists a non-singular matrix* **Q** *such that*

$$\mathbf{A} = \mathbf{Q}'\,\mathbf{Q}.$$

Proof: Suppose **A** is positive definite. Then all the eigenvalues of **A** are positive and we can find an orthogonal matrix **P** such that

$$\mathbf{P}^{-1}\mathbf{AP} = \mathbf{P}'\mathbf{AP} = \mathbf{D} = \text{diag.}\,[\lambda_1, \ldots, \lambda_n]$$

where each $\lambda_i > 0$. Let $\mathbf{D_1} = \text{diag.}\,[\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}]$. Then $\mathbf{D_1}^2 = \mathbf{D}$ and $\mathbf{D_1}' = \mathbf{D_1}$. We have

$$\mathbf{A} = \mathbf{PDP}^{-1} = \mathbf{PD_1}^2\,\mathbf{P}^{-1} = \mathbf{PD_1}\,\mathbf{D_1P}'$$

$$= (\mathbf{PD_1})\,(\mathbf{PD_1})' = \mathbf{Q}'\mathbf{Q} \text{ where } \mathbf{Q} = (\mathbf{PD_1})'.$$

Clearly, **Q** is non-singular since **P** and $\mathbf{D_1}$ are non-singular.

Conversely suppose that $\mathbf{A} = \mathbf{Q}'\mathbf{Q}$ where **Q** is non-singular. We have for all real vectors **X,**

$$\mathbf{X}'\mathbf{AX} = \mathbf{X}'\mathbf{Q}'\mathbf{QX} = (\mathbf{QX})'\,(\mathbf{QX})$$

$$= \mathbf{Y}'\mathbf{Y}, \quad \text{where} \quad \mathbf{Y} = \mathbf{QX} \text{ is a real } n\text{-vector}$$

$$\geq 0.$$

Also $\quad \mathbf{X}'\mathbf{AX} = 0 \;\Rightarrow\; \mathbf{Y}'\mathbf{Y} = 0$

$\Rightarrow \quad \mathbf{Y} = \mathbf{O} \;\Rightarrow\; \mathbf{QX} = \mathbf{O}$ $\quad$ [$\because \mathbf{Y} = \mathbf{QX}$]

$\Rightarrow \quad \mathbf{Q}^{-1}(\mathbf{QX}) = \mathbf{Q}^{-1}\mathbf{O} \;\Rightarrow\; \mathbf{X} = \mathbf{O}.$

$\therefore$ $\mathbf{X}'\mathbf{AX}$ is a positive definite real quadratic form and so the symmetric matrix **A** is positive definite.

Theorem 7: *Every real non-singular matrix* **A** *can be written as a product* $\mathbf{A} = \mathbf{PS}$, *where* **S** *is a positive definite symmetric matrix and* **P** *is orthogonal.*

Proof: Since **A** is non-singular, therefore by theorem 6, $\mathbf{A}'\mathbf{A}$ is a positive definite real symmetric matrix. Let **Q** be an orthogonal matrix such that

$$\mathbf{Q}^{-1}\,(\mathbf{A}'\mathbf{A})\,\mathbf{Q} = \mathbf{Q}'\,(\mathbf{A}'\mathbf{A})\,\mathbf{Q} = \mathbf{D} = \text{diag.}\,[\lambda_1, \ldots, \lambda_n],$$

where $\lambda_1, \ldots, \lambda_n$ are the positive real eigenvalues of $\mathbf{A}'\mathbf{A}$. Let

$$\mathbf{D_1} = \text{diag.}\,[\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}].$$

Now let $\mathbf{S} = \mathbf{Q}\mathbf{D}_1\mathbf{Q}'$. Clearly $\mathbf{S}' = \mathbf{S}$ and so $\mathbf{S}$ is symmetric. Moreover $\mathbf{S}$ is positive definite because it is similar to $\mathbf{D}_1$ which has positive eigenvalues.

Also $\mathbf{S}^2 = \mathbf{Q}\mathbf{D}_1\mathbf{Q}'\mathbf{Q}\mathbf{D}_1\mathbf{Q}' = \mathbf{Q}\mathbf{D}_1\mathbf{Q}^{-1}\mathbf{Q}\mathbf{D}_1\mathbf{Q}'$

$$= \mathbf{Q}\mathbf{D}_1{}^2\mathbf{Q}^{-1} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} = \mathbf{A}'\mathbf{A}.$$

Now let $\mathbf{P} = \mathbf{A}\mathbf{S}^{-1}$. Then $\mathbf{P}$ is orthogonal because

$$\begin{aligned}
\mathbf{P}'\mathbf{P} &= (\mathbf{A}\mathbf{S}^{-1})'\mathbf{A}\mathbf{S}^{-1} = (\mathbf{S}^{-1})'\mathbf{A}'\mathbf{A}\mathbf{S}^{-1} \\
&= (\mathbf{S}^{-1})'\mathbf{S}^2\mathbf{S}^{-1} && [\because\ \mathbf{A}'\mathbf{A} = \mathbf{S}^2] \\
&= (\mathbf{S}^{-1})'\mathbf{S}\mathbf{S}\mathbf{S}^{-1} = (\mathbf{S}^{-1})'\mathbf{S} \\
&= (\mathbf{S}')^{-1}\mathbf{S} = \mathbf{S}^{-1}\mathbf{S} && [\because\ \mathbf{S} \text{ is symmetric}] \\
&= \mathbf{I}.
\end{aligned}$$

Thus $\mathbf{S} = \mathbf{Q}\mathbf{D}_1\mathbf{Q}'$ is a positive definite real symmetric matrix and $\mathbf{P} = \mathbf{A}\mathbf{S}^{-1}$ is an orthogonal matrix and we have

$$\mathbf{P}\mathbf{S} = \mathbf{A}\mathbf{S}^{-1}\mathbf{S} = \mathbf{A}.$$

Hence the result.

Note: The decomposition $\mathbf{A} = \mathbf{P}\mathbf{S}$ obtained in theorem 7 is called the *polar factorization of* $\mathbf{A}$.

17 Criterion for Positive-Definiteness of a Quadratic Form in Terms of Leading Principal Minors of its Matrix

Leading principal minors of a matrix. Definition : *Let*

$$\mathbf{A} = [\,a_{ij}\,]_{n \times n}$$

be a square matrix of order n. Then

$$A_1 = a_{11},\ A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$A_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \ldots, A_n = \begin{vmatrix} a_{11} & \ldots & a_{1n} \\ \ldots & \ldots & \ldots \\ a_{n1} & \ldots & a_{nn} \end{vmatrix}$$

are called the leading principal minors of $\mathbf{A}$.

Before stating the main theorem we shall prove the following Lemma.

Lemma : *If* $\mathbf{A}$ *is the matrix of a positive definite form, then*

$$|\mathbf{A}| > 0.$$

Proof: If $\mathbf{X}'\mathbf{A}\mathbf{X}$ is a positive definite real quadratic form, then there exists a real non-singular matrix $\mathbf{P}$ such that

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{I}.$$

$$\therefore \qquad |\mathbf{P}'\mathbf{A}\mathbf{P}| = |\mathbf{I}| = 1$$

or $\qquad |\mathbf{P}'|\,|\mathbf{A}|\,|\mathbf{P}| = 1$ or $|\mathbf{A}| = 1/|\mathbf{P}|^2 \qquad [\because\ |\mathbf{P}| = |\mathbf{P}'| \neq 0]$

Therefore $|\mathbf{A}|$ is positive.

Theorem 1: Sylvester's Criterion for Positive Definiteness: *A necessary and sufficient condition for a real quadratic form* $\mathbf{X'}\,\mathbf{AX}$ *to be positive definite is, that the leading principal minors of* $\mathbf{A}$ *are all positive.*

Proof: The condition is necessary. Suppose $\mathbf{X'AX}$ is a positive definite quadratic form in n variables. Let k be any natural number such that $k \le n$. Putting $x_{k+1} = 0, \dots, x_n = 0$ in the positive definite form $\mathbf{X'AX}$, we get a positive definite form in k variables $x_1, \dots, x_k$. The determinant of the matrix of this new quadratic form is the leading principal minor of order k of $\mathbf{A}$ and is positive by virtue of the lemma we have just proved. Thus every leading principal minor of the matrix of a positive definite quadratic form is positive.

The condition is sufficient. Now it is given that the leading principal minors of $\mathbf{A}$ are all positive and we are to prove that the form $\mathbf{X'AX}$ is positive definite. Here we shall use the principle of mathematical induction.

The result is true for quadratic forms in one variable since $a_{11}\, x^2$ is positive definite when a_{11} is positive.

Assume as our induction hypothesis that the theorem is true for quadratic forms in m variables. Then we shall prove that it is also true for quadratic forms in $(m+1)$ variables.

Now let $\mathbf{S}$ be any real symmetric matrix of order $(m+1)$ and let the leading principal minors of $\mathbf{S}$ be all positive. We partition $\mathbf{S}$ as follows :

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{B_1} \\ \mathbf{B_1}' & \lambda \end{bmatrix},$$

where $\mathbf{B}$ is a real symmetric matrix of order m and $\mathbf{B_1}$ is an $m \times 1$ column matrix.

By hypothesis the leading principal minors of $\mathbf{S}$ are all positive. Therefore $|\mathbf{S}|$ and the leading principal minors of $\mathbf{B}$ are all positive. Thus $\mathbf{B}$ is a real symmetric matrix of order m having all its leading principal minors positive. So by induction hypothesis the quadratic form corresponding to $\mathbf{B}$ is positive definite. Therefore there exists a non-singular matrix $\mathbf{P}$ of order m such that $\mathbf{P'\,BP} = \mathbf{I_m}$.

Since $|\mathbf{B}| > 0$, therefore $\mathbf{B}$ is non-singular. Let $\mathbf{C} = -\,\mathbf{B}^{-1}\,\mathbf{B_1}$. Then $\mathbf{C}$ is an $m \times 1$ column matrix. Also

$$\begin{aligned} \mathbf{C}' &= -\,(\mathbf{B}^{-1}\,\mathbf{B_1})' = -\,\mathbf{B_1}'\,(\mathbf{B}^{-1})' \\ &= -\,\mathbf{B_1}'\,(\mathbf{B}')^{-1} = -\,\mathbf{B_1}'\,\mathbf{B}^{-1}, \end{aligned}$$

since $\mathbf{B'} = \mathbf{B}$, $\mathbf{B}$ being symmetric. We have

$$\begin{aligned} \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} S \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} &= \begin{bmatrix} \mathbf{P}' & \mathbf{O} \\ \mathbf{C}' & 1 \end{bmatrix} \begin{bmatrix} \mathbf{B} & \mathbf{B_1} \\ \mathbf{B_1}' & \lambda \end{bmatrix} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{P'\,B} & \mathbf{P'\,B_1} \\ \mathbf{C'\,B} + \mathbf{B_1}' & \mathbf{C'\,B_1} + \lambda \end{bmatrix} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{P'\,BP} & \mathbf{P'\,BC} + \mathbf{P'\,B_1} \\ \mathbf{C'\,BP} + \mathbf{B_1}'\,\mathbf{P} & \mathbf{C'\,BC} + \mathbf{B_1}'\,\mathbf{C} + \mathbf{C'\,B_1} + \lambda \end{bmatrix} \end{aligned}$$

$\mathbf{O} \quad \mathbf{B_1'C} + \lambda$ $[\because \mathbf{P'BP} = \mathbf{I}_m, \mathbf{C} = -\mathbf{B}^{-1}\mathbf{B_1}, \mathbf{C'} = -\mathbf{B_1'}\mathbf{B}^{-1}]$

Thus $\begin{bmatrix} \mathbf{P'} & \mathbf{O} \\ \mathbf{C'} & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \mathbf{B_1'C} + \lambda \end{bmatrix}.$

Taking determinants of both sides, we get

$$|\mathbf{P'}|.|\mathbf{S}|.|\mathbf{P}| = |\mathbf{I_m}|.|\mathbf{B_1'C} + \lambda| = \mathbf{B_1'C} + \lambda$$

because $\mathbf{B_1'C} + \lambda$ is an 1×1 matrix.

$$\therefore \qquad |\mathbf{P}|^2 .|\mathbf{S}| = \mathbf{B_1'C} + \lambda \qquad [\because |\mathbf{P}| = |\mathbf{P'}|].$$

Since $|\mathbf{S}| > 0$ and $|\mathbf{P}| \neq 0$, therefore $\mathbf{B_1'C} + \lambda$ is positive. Let $\mathbf{B_1'C} + \lambda = \alpha^2$, where α is real. Then

$$\begin{bmatrix} \mathbf{P'} & \mathbf{O} \\ \mathbf{C'} & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \alpha^2 \end{bmatrix}.$$

Pre-multiplying and post-multiplying both sides with

$\begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix}$, we get

$$\begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{P'} & \mathbf{O} \\ \mathbf{C'} & 1 \end{bmatrix} \mathbf{S} \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix} = \mathbf{I_{m+1}}.$$

Now let $\mathbf{Q} = \begin{bmatrix} \mathbf{P} & \mathbf{C} \\ \mathbf{O} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix}.$

Then $\mathbf{Q}$ is non-singular as it is the product of two non-singular matrices. Also

$$\mathbf{Q'} = \begin{bmatrix} \mathbf{I_m} & \mathbf{O} \\ \mathbf{O} & \alpha^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{P'} & \mathbf{O} \\ \mathbf{C'} & 1 \end{bmatrix}.$$

Therefore, we have

$$\mathbf{Q'SQ} = \mathbf{I}_{m+1}.$$

Thus the real symmetric matrix $\mathbf{S}$ of order $m+1$ is congruent to $\mathbf{I_{m+1}}$. So the quadratic form corresponding to $\mathbf{S}$ is positive definite.

The proof is now complete by induction.

Corollary: *The real quadratic form*

$$q(x_1, \ldots, x_n) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$$

is negative definite if and only if

$$a_{11} < 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} < 0, \ldots.$$

Proof: For $q(x_1, \ldots, x_n)$ to be negative definite, $-q(x_1, \ldots, x_n)$ must be positive definite. Applying Sylvester's criterion for positive definiteness of $-q$, we get successively

$$-a_{11}>0,\begin{vmatrix} a_{11} & a_{12} \\ -a_{21} & -a_{22}\end{vmatrix}>0,\begin{vmatrix} -a_{21} & -a_{22} & -a_{23} \\ -a_{31} & -a_{32} & -a_{33}\end{vmatrix}>0,\dots.$$

From this follows the condition stated in the statement of the above corollary.

18 Some More Criteria to Check the Value Class of a Real Quadratic Form

Principal minors of a matrix.

Definition: *Let $A=[a_{ij}]_{n\times n}$ be a square matrix of order n. Any minors of A obtained by deleting the corresponding rows and columns of A are called principal minors of A.*

Consider the square matrix

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{bmatrix}.$$

The principal minors of A of order 1 are a_{11}, a_{22}, a_{33}.

The principal minors of A of order 2 are

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}\end{vmatrix},\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33}\end{vmatrix},\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{vmatrix}.$$

These have obtained respectively by deleting the third row and the third column, the second row and the second column and the first row and the first column.

The principal minor of A of order 3 is

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{vmatrix}.$$

Now we give some theorems without proof.

Theorem 1: *A real quadratic form $\mathbf{X}^T A\,\mathbf{X}$ is positive definite if and only if all the principal minors of A are positive.*

Theorem 2: *A real quadratic form $\mathbf{X}^T A\,\mathbf{X}$ is negative definite if and only if all the principal minors of A of even order are positive and those of odd order are negative.*

Theorem 3: *A real quadratic form $\mathbf{X}^T A\,\mathbf{X}$ is positive semi-definite if and only if A is singular and all its principal minors are non-negative.*

Theorem 4: *A real quadratic form $\mathbf{X}^T A\,\mathbf{X}$ is negative semi-definite if and only if A is singular and all principal minors of even order of A are non-negative while those of odd order are non-positive.*

Theorem 5: *A real quadratic form $\mathbf{X}^T A\,\mathbf{X}$ is positive semi-definite if the leading principal minors $A_1,\dots,A_{n-1}$ of A are positive and det $A=0$.*

Theorem 6: *A real symmetric matrix A is indefinite if and only if at least one of the following conditions is satisfied :*

(b) *A has a positive principal minor of odd order and a negative principal minor of odd order.*

Illustrative Examples

Example 15: *Prove that the quadratic form*

$$6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$$

in three variables is positive definite.

Solution: The matrix **A** of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

The leading principal minors of **A** are

$$A_1 = 6, A_2 = \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix} = 18 - 4 = 14,$$

$$A_3 = \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 0 & 1 & -7 \\ 0 & 2 & 2 \\ 2 & -1 & 3 \end{vmatrix}, \text{by } R_2 + R_3, R_1 - 3R_3$$

$$= 2\,(2 + 14) = \text{positive}.$$

Since the leading principal minors of **A** are all positive, therefore the given quadratic form is positive definite.

Example 16: *Write the matrix* **A** *of the quadratic form*

$$6x^2 + 35\,y^2 + 11z^2 + 4\,zx.$$

Find the eigenvalues of **A** *and hence determine the value class of the given quadratic form.*

Solution: The matrix **A** of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 6 & 0 & 2 \\ 0 & 35 & 0 \\ 2 & 0 & 11 \end{bmatrix}.$$

The characteristic equation of **A** is

$$\begin{vmatrix} 6-\lambda & 0 & 2 \\ 0 & 35-\lambda & 0 \\ 2 & 0 & 11-\lambda \end{vmatrix} = 0$$

or $(6-\lambda)(35-\lambda)(11-\lambda) - 2 \times 2\,(35-\lambda) = 0$

or $(35-\lambda)\,[(6-\lambda)(11-\lambda) - 4] = 0$

or $(35-\lambda)\,[\lambda^2 - 17\lambda + 62] = 0$

or $(35-\lambda) = 0, (\lambda^2 - 17\lambda + 62) = 0.$

$\therefore$ the eigenvalues of **A** are 35, $\dfrac{17 \pm \sqrt{(41)}}{2}$.

Since the eigenvalues of **A** are all positive, therefore the quadratic form is positive definite.

$$5x_1^2 + 26x_2^2 + 10x_3^2 + 4\,x_2x_3 + 14x_3x_1 + 6x_1x_2$$

in three variables is positive semi-definite and find a non-zero set of values of x_1, x_2, x_3 *which makes the form zero.*

Solution: The matrix of the given form is

$$A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix}.$$

We write $\begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

Now we shall reduce **A** to diagonal form by applying congruence operations on it. Performing

$$R_2 \to R_2 - \frac{3}{5} R_1, C_2 \to C_2 - \frac{3}{5} C_1 \,;\, R_3 \to R_3 - \frac{7}{5} R_1,$$

$$C_3 \to C_3 - \frac{7}{5} C_1,$$

we get $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 121/5 & -11/5 \\ 0 & -11/5 & 1/5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3/5 & 1 & 0 \\ -7/5 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -\frac{3}{5} & -\frac{7}{5} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

Performing $R_3 \to R_3 + \frac{1}{11} R_2 \,, C_3 \to C_3 + \frac{1}{11} C_2$, we get

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & \frac{121}{5} & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{5} & 1 & 0 \\ -\frac{16}{11} & \frac{1}{11} & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -\frac{3}{5} & -\frac{16}{11} \\ 0 & 1 & \frac{1}{11} \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore the linear transformation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{3}{5} & -\frac{16}{11} \\ 0 & 1 & \frac{1}{11} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

i.e., $x_1 = y_1 - \frac{3}{5} y_2 - \frac{16}{11} y_3 \,, x_2 = y_2 + \frac{1}{11} y_3 \,, x_3 = y_3$

transforms the given form to the diagonal form

$$5\,y_1^2 + \frac{121}{5}\,y_2^2 + 0\,y_3^2. \qquad \ldots(1)$$

But the quadratic form (1) in 3 variables is positive semi-definite and equivalent quadratic forms have the same value class. Therefore the given quadratic form is positive semi-definite.

The set of values $y_1 = 0, y_2 = 0, y_3 = 1$ makes (1) zero. Corresponding to this set of values, we have $x_3 = 1, x_2 = \frac{1}{11}, x_1 = -\frac{16}{11}$. This is a non-zero set of values of x_1, x_2, x_3 which makes the given quadratic form zero.

$$x_1^2 + 2x_2^2 + 3x_3^2 + 2x_2x_3 - 2x_3x_1 + 2x_1x_2$$

in three variables is indefinite and find two sets of values of x_1, x_2, x_3 for which the form assumes positive and negative values.

Solution: The matrix of the given quadratic form is

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

We write $\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

Now we shall reduce **A** to diagonal form by applying congruent operations on it. Performing

$R_2 \to R_2 - R_1, C_2 \to C_2 - C_1$; and $R_3 \to R_3 + R_1, C_3 \to C_3 + C_1$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Performing $R_3 \to R_3 - 2R_2$, $C_3 \to C_3 - 2C_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \mathbf{A} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

$\therefore$ the linear transformation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

i.e.,

$$\left.\begin{aligned} x_1 &= y_1 - y_2 + 3y_3 \\ x_2 &= y_2 - 2y_3 \\ x_3 &= y_3 \end{aligned}\right\} \qquad \ldots(A)$$

transforms the given form to the diagonal form

$$y_1^2 + y_2^2 - 2y_3^2. \qquad \ldots(1)$$

The form (1) is indefinite and so the given quadratic form is also indefinite. Obviously $y_1 = 0, y_2 = 0, y_3 = 1$ makes the form (1) negative and $y_1 = 0, y_2 = 1, y_3 = 0$ makes the form (1) positive. Substituting these values in the relations (A), we see that the sets of values $x_1 = 3, x_2 = -2, x_3 = 1$; $x_1 = -1, x_2 = 1, x_3 = 0$ respectively make the given form negative and positive.

Example 19: *Show that every real non-singular matrix* **A** *can be expressed as*

$$\mathbf{A} = \mathbf{QDR},$$

where **Q** *and* **R** *are orthogonal and* **D** *is real diagonal.*

real symmetric matrix. Let **P** be an orthogonal matrix such that

$$\mathbf{P}^{-1}\,(\mathbf{A}'\mathbf{A})\,\mathbf{P} = \mathbf{P}'(\mathbf{A}'\mathbf{A})\,\mathbf{P} = \text{diag.}\,[\lambda_1,\ldots,\lambda_n],$$

where $\lambda_1,\ldots,\lambda_n$ are the positive real eigenvalues of the positive definite matrix $\mathbf{A}'\mathbf{A}$. Let $\mathbf{D} = \text{diag.}\,[\sqrt{\lambda_1},\ldots,\sqrt{\lambda_n}]$. Then **D** is a real diagonal matrix and $\mathbf{D}' = \mathbf{D}$.

We have $\mathbf{D}'\mathbf{D} = \mathbf{D}^2 = \text{diag.}\,[\lambda_1,\ldots,\lambda_n]$

$\Rightarrow \qquad \mathbf{D}'\mathbf{D} = \mathbf{P}'\mathbf{A}'\mathbf{A}\mathbf{P}$

$\Rightarrow \qquad (\mathbf{P}')^{-1}\,\mathbf{D}'\,\mathbf{D}\mathbf{P}^{-1} = (\mathbf{P}')^{-1}\,\mathbf{P}'\mathbf{A}'\mathbf{A}\mathbf{P}\mathbf{P}^{-1}$

$\Rightarrow \qquad (\mathbf{P}')^{-1}\,\mathbf{D}'\,\mathbf{D}\mathbf{P}^{-1} = \mathbf{A}'\mathbf{A}$

$\Rightarrow \qquad (\mathbf{A}')^{-1}\,(\mathbf{P}')^{-1}\,\mathbf{D}'\,\mathbf{D}\,\mathbf{P}^{-1}\,\mathbf{A}^{-1} = \mathbf{I}$

$\Rightarrow \qquad (\mathbf{A}^{-1})'\,\mathbf{P}\mathbf{D}'\,\mathbf{D}\mathbf{P}'\,\mathbf{A}^{-1} = \mathbf{I}$ $\qquad [\because$ **P** is orthogonal $\rightarrow \mathbf{P}' = \mathbf{P}^{-1}]$

$\Rightarrow \qquad (\mathbf{D}\mathbf{P}'\mathbf{A}^{-1})'\,(\mathbf{D}\mathbf{P}'\mathbf{A}^{-1}) = \mathbf{I}$

$\Rightarrow \qquad \mathbf{D}\mathbf{P}'\mathbf{A}^{-1}$ is orthogonal.

Let $\quad \mathbf{S} = \mathbf{D}\mathbf{P}'\,\mathbf{A}^{-1}$.

Then **S** is an orthogonal matrix.

Now let $\mathbf{Q} = \mathbf{S}^{-1}$; then **Q** is an orthogonal matrix. Also let $\mathbf{R} = \mathbf{P}'$. Then **R** is an orthogonal matrix.

We have $\mathbf{Q}\mathbf{D}\mathbf{R} = (\mathbf{D}\mathbf{P}'\mathbf{A}^{-1})^{-1}\,\mathbf{D}\mathbf{P}' = \mathbf{A}\,(\mathbf{P}')^{-1}\,\mathbf{D}^{-1}\,\mathbf{D}\mathbf{P}' = \mathbf{A}\,(\mathbf{P}')^{-1}\,\mathbf{P}' = \mathbf{A}.$

Hence the result.

Example 20: *If* **A** *is a positive definite real symmetric matrix, show that there exists a positive definite real symmetric matrix* **B** *such that*

$$\mathbf{B}^2 = \mathbf{A}.$$

Solution: *Since* **A** *is a positive definite real symmetric matrix, therefore the eigenvalues* $\lambda_1,\ldots,\lambda_n$ *of* **A** *are all real and positive. Also there exists an orthogonal matrix* **P** *such that*

$$\mathbf{P}^{-1}\,\mathbf{A}\mathbf{P} = \mathbf{D} = \text{diag.}\,[\lambda_1,\ldots,\lambda_n].$$

Let $\mathbf{D}_1 = \text{diag.}\,[\sqrt{\lambda_1},\ldots,\sqrt{\lambda_n}]$. Then $\mathbf{D}_1^2 = \mathbf{D}$, $\mathbf{D}_1' = \mathbf{D}_1$, and the eigenvalues of $\mathbf{D}_1$ are all positive.

Now suppose that $\mathbf{B} = \mathbf{P}\,\mathbf{D}_1\mathbf{P}^{-1} = \mathbf{P}\,\mathbf{D}_1\mathbf{P}'$.

We have $\mathbf{B}' = (\mathbf{P}\mathbf{D}_1\mathbf{P}')' = \mathbf{P}\mathbf{D}_1'\,\mathbf{P}' = \mathbf{P}\mathbf{D}_1\mathbf{P}' = \mathbf{B}.$

$\therefore \qquad$ **B** is a real symmetric matrix.

Also $\mathbf{B} = \mathbf{P}\mathbf{D}_1\,\mathbf{P}^{-1} \Rightarrow$ **B** is similar to $\mathbf{D}_1$. So **B** and $\mathbf{D}_1$ have the same eigenvalues. Therefore the eigenvalues of **B** are all positive. So **B** is a positive definite real symmetric matrix.

Finally, we have

$$\mathbf{B}^2 = (\mathbf{P}\mathbf{D}_1\mathbf{P}^{-1})^2 = \mathbf{P}\mathbf{D}_1\mathbf{P}^{-1}\,\mathbf{P}\mathbf{D}_1\,\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}_1^{\,2}\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{A}.$$

Hence the result.

$$q(x_1, x_2, x_3) = [x_1, x_2, x_3]\begin{bmatrix} 2 & 0 & -2 \\ 1 & 5 & 2 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Solution: The given quadratic form is

$$q = 2x_1^2 - 2x_1x_3 + x_2x_1 + 5x_2^2 + 2x_2x_3 - x_3x_1 + x_3x_2 + x_3^2$$

$$= 2x_1x_1 + \frac{1}{2}x_1x_2 - \frac{3}{2}x_1x_3 + \frac{1}{2}x_2x_1 + 5x_2x_2$$

$$+ \frac{3}{2}x_2x_3 - \frac{3}{2}x_3x_1 + \frac{3}{2}x_3x_2 + x_3x_3.$$

The real symmetric matrix associated with the above quadratic form is

$$\mathbf{A} = \begin{bmatrix} 2 & \frac{1}{2} & -\frac{3}{2} \\ \frac{1}{2} & 5 & \frac{3}{2} \\ -\frac{3}{2} & \frac{3}{2} & 1 \end{bmatrix}.$$

The principal minors of **A** of order 1 are $2, 5, 1$, those of order 2 are $\frac{39}{4}, -\frac{1}{4}$ and $\frac{11}{4}$; the principal minor of order three is $-\frac{33}{4}$.

We see that **A** has a negative principal minor of even order, namely of order 2. Hence, by theorem 6 of 18, the matrix **A** is indefinite. Therefore, the given quadratic form is indefinite.

Example 22: *Determine the value class of the real quadratic form*

$$-x_1^2 - 2x_2^2 - 2x_3^2 + 2x_1x_2 + 2x_2x_3$$

in three variables.

Solution: The given quadratic form is

$$q(x_1, x_2, x_3) = -x_1^2 - 2x_2^2 - 2x_3^2 + 2x_1x_2 + 2x_2x_3$$

$$= -x_1x_1 + x_1x_2 + 0\,x_1x_3 + x_2x_1 - 2x_2x_2 + x_2x_3$$

$$+ 0\,x_3x_1 + x_3x_2 - 2x_3x_3$$

$$= \mathbf{X}^T A\mathbf{X}, \text{ where } A = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

is a symmetric matrix.

The leading principal minors of A are

$$A_1 = -1,\ A_2 = \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix} = 2 - 1 = 1,$$

$$A_3 = \begin{vmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{vmatrix} = \begin{vmatrix} -1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{vmatrix}, C_2 \to C_2 + C_1$$

$$= -1(2 - 1) = -1.$$

We see that $A_1 < 0, A_2 > 0, A_3 < 0$.

So the given quadratic form is negative definite.

Comprehensive Exercise 3

1. Prove that the quadratic form

$$6x_1^2 + 3x_2^2 + 14x_3^2 + 4x_2x_3 + 18x_3x_1 + 4x_1x_2$$

in three variables is positive definite.

2. Prove that the quadratic form

$$2x_1^2 + x_2^2 - 3x_3^2 - 8x_2x_3 - 4x_3x_1 + 12x_1x_2$$

in three variables is indefinite.

3. Prove that the quadratic form

$$6x^2 + 49y^2 + 51z^2 - 82yz + 20zx - 4xy$$

in three variables is positive definite.

4. Show that the quadratic form

$$6x^2 + 17y^2 + 3z^2 - 20xy - 14yz + 8zx$$

in three variables is positive semi-definite and find a non-zero set of values of x, y, z which makes the form zero.

5. Classify the following forms in three variables as definite, semi-definite and indefinite

(i) $2x^2 + 2y^2 + 3z^2 - 4yz - 4zx + 2xy$

(ii) $26x^2 + 20y^2 + 10z^2 - 4yz - 16zx - 36xy$

(iii) $x_1^2 + 4x_2^2 + x_3^2 - 4x_2x_3 + 2x_3x_1 - 4x_1x_2$.

6. Show that a real symmetric matrix is positive definite if and only if all its characteristic roots are positive.

7. Show that a real symmetric matrix $\boldsymbol{A}$ is positive definite iff $\mathbf{A}^{-1}$ exists and is positive definite.

8. Show that the quadratic form

$$2x^2 - 4xy + 3xz + 6y^2 + 6zy + 8z^2$$

in three variables is positive definite.

9. Show that the quadratic form

$$y^2 + 2z^2 - 2yz + 2zx - 2xy$$

in three variables is indefinite.

10. Determine the value class of the real quadratic form in three variables

$$x_1^2 + 3x_2^2 + 6x_3^2 - 2x_1x_2 + 4x_1x_3.$$

11. Determine the value class of the real quadratic form in three variables

$$-2x_1^2 - 2x_2^2 - 2x_3^2 + 4x_2x_3.$$

12. Determine the definiteness of the real quadratic form in three variables

$$4x_1^2 + x_2^2 + 9x_3^2 - 4x_1x_2 + 12x_1x_3.$$

Hermitian form. Definition.

Let $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix}$ *be any n-vector in the unitary space* $\mathbf{C}^n$ *and let* $H = [h_{ij}]_{n \times n}$ *be a Hermitian matrix of order n over the complex field* **C**. *Then the expression*

$$h(x_1, x_2, \ldots, x_n) = \mathbf{X}^\theta H \mathbf{X} = \sum_{i=1}^{n} \sum_{j=1}^{n} h_{ij} \bar{x}_i x_j$$

is called a Hermitian form of order n in the n complex variables $x_1, x_2, \ldots, x_n$. *The Hermitian matrix H is called the matrix of this Hermitian form.*

If H is real, then a Hermitian form is called the *real Hermitian form*. Also, det H is defined as the *discriminant* of the Hermitian form, and a Hermitian form is called *singular* if its determinant is zero, otherwise it is called *non-singular.*

Note. *A matrix H over the complex field* **C** *is called a Hermitian matrix if* $H^\theta = H$, *where* H^θ *denotes the conjugate transpose of the matrix H i.e.,*

$$H^\theta = (\overline{H})' = (\overline{H})^T = \overline{(H')}.$$

Some authors denote the conjugate transpose of a matrix H by the symbol H^*. A real symmetric matrix A is always a Hermitian matrix.

Theorem 1: *A Hermitian form* $\mathbf{X}^\theta H \mathbf{X}$ *assumes only real values for all complex n-vectors* X.

Proof : Suppose $\mathbf{X}^\theta H \mathbf{X}$ is a Hermitian form. Then H is a Hermitian matrix and so $H^\theta = H$.

Since $\mathbf{X}^\theta H \mathbf{X}$ is a 1×1 matrix, therefore it is symmetric and so

$$(\mathbf{X}^\theta H \mathbf{X})' = \mathbf{X}^\theta H \mathbf{X}.$$

Now
$$\overline{(\mathbf{X}^\theta H \mathbf{X})} = \overline{(\mathbf{X}^\theta H \mathbf{X})'} = (\mathbf{X}^\theta H \mathbf{X})^\theta$$
$$= \mathbf{X}^\theta H^\theta (\mathbf{X}^\theta)^\theta = \mathbf{X}^\theta H \mathbf{X}.$$

Thus $\mathbf{X}^\theta H \mathbf{X}$ and its conjugate are equal.

$\therefore$ $\mathbf{X}^\theta H \mathbf{X}$ is a real 1×1 matrix.

Hence, $\mathbf{X}^\theta H \mathbf{X}$ has only real values.

Theorem 2 : *The determinant and every leading principal minor of a Hermitian matrix H are real.*

Proof : We have $\overline{|H|} = |\overline{H}| = |(\overline{H})'|$

$$= |H^\theta| = |H|. \qquad [\because H^\theta = H, H \text{ being Hermitian}]$$

$\therefore$ $|H|$ is real.

Since every leading principal sub-matrix of a Hermitian matrix is Hermitian, we conclude that every leading principal minor of a Hermitian matrix is real.

This completes the proof of the theorem.

transformation of coordinates defined by

$$\mathbf{X} = P\,\mathbf{Y}$$

where P is a non-singular matrix over the complex field **C.**

Proof : Substituting $\mathbf{X} = P\,\mathbf{Y}$ in $h = \mathbf{X}^{\theta}\, H\, \mathbf{X}$, we get

$$h = (P\,\mathbf{Y})^{\theta}\, H\,(P\,\mathbf{Y}) = \mathbf{Y}^{\theta}\,(P^{\theta}\, HP)\,\mathbf{Y} = \mathbf{Y}^{\theta}\, Q\,\mathbf{Y},$$

where $\qquad Q = P^{\theta}\, HP.$

We have $\qquad Q^{\theta} = (P^{\theta}\, HP)^{\theta} = P^{\theta}\, H^{\theta}\,(P^{\theta})^{\theta} = P^{\theta}\, HP = Q.$

$\therefore \qquad Q$ is Hermitian.

Hence, the transformed form $\mathbf{Y}^{\theta}\, Q\,\mathbf{Y}$ is Hermitian.

Remark. It can be easily seen that under this non-singular transformation, the rank of a Hermitian form remains invariant. We know that the rank of a matrix does not alter by pre-multiplication or post-multiplication by a non-singular matrix. Since P and P^{θ} are both non-singular, therefore rank H = rank $(P^{\theta}\, HP)$ = rank Q.

Hence rank $\mathbf{X}^{\theta}\, H\,\mathbf{X}$ = rank $\mathbf{Y}^{\theta}\, Q\,\mathbf{Y}$.

20. Unitary Reduction of a Hermitian Form.

Diagonal and Unit Hermitian forms.

Definition : A Hermitian form represented by

$$h = c_1\, \bar{x}_1\, x_1 + c_2\, \bar{x}_2\, x_2 + \ldots + c_n\, \bar{x}_n\, x_n$$

where the coefficients c_i's are real as h is real, is referred to as diagonal Hermitian form.

If all c_i's can be reduced to one, then a Hermitian form given by

$$h = \bar{x}_1\, x_1 + \bar{x}_2\, x_2 + \ldots + \bar{x}_n\, x_n$$

is called a *unit Hermitian form.*

Theorem 1 : *Every Hermitian form* $\mathbf{X}^{\theta}\, H\,\mathbf{X}$ *of order n is unitarily equivalent (under a transformation* $\mathbf{X} = P\,\mathbf{Y}$, *P unitary) to the diagonal form*

$$\lambda_1\, \bar{y}_1\, y_1 + \ldots + \lambda_n\, \bar{y}_n\, y_n$$

where $\lambda_1, \ldots, \lambda_n$ are eigenvalues of H.

Proof. Since H is a Hermitian matrix, therefore there exists a unitary matrix P such that

$$P^{-1}\, HP = P^{\theta}\, HP = D = \text{diag.}\,[\lambda_1, \ldots, \lambda_n]$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of H.

Consider the unitary linear transformation $\mathbf{X} = P\,\mathbf{Y}$. We have

$$\mathbf{X}^{\theta}\, H\,\mathbf{X} = (P\,\mathbf{Y})^{\theta}\, H\,(P\,\mathbf{Y}) = \mathbf{Y}^{\theta}\, P^{\theta}\, HP\,\mathbf{Y} = \mathbf{Y}^{\theta}\, D\,\mathbf{Y}$$

$$= \lambda_1\, \bar{y}_1\, y_1 + \ldots + \lambda_n\, \bar{y}_n\, y_n\,.$$

Hence, the result.

Remark. If the rank of the Hermitian form $\mathbf{X}^{\theta}\, H\,\mathbf{X}$ is r, then only r of its n eigenvalues $\lambda_1, \ldots, \lambda_n$ will be non-zero and consequently only r terms in the above diagonal form will be non-zero.

21. Value Class of a Hermitian Form. Definite, Semi-definite and Indefinite Hermitian Forms.

Definitions : A Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is said to be

(i) **Positive definite** if $\mathbf{X}^{\theta} H \mathbf{X} \geq 0$ for all complex n-vectors $\mathbf{X}$ and $\mathbf{X}^{\theta} H \mathbf{X} = 0$ only if $\mathbf{X} = \mathbf{O}$ *i.e.*, $x_1 = x_2 = \ldots = x_n = 0$.

(ii) **Negative definite** if $\mathbf{X}^{\theta} H \mathbf{X} \leq 0$ for all complex n-vectors $\mathbf{X}$ and $\mathbf{X}^{\theta} H \mathbf{X} = 0$ only if $\mathbf{X} = \mathbf{O}$ *i.e.*, $x_1 = x_2 = \ldots = x_n = 0$.

(iii) **Positive semi-definite** if $\mathbf{X}^{\theta} H \mathbf{X} \geq 0$ for all complex n-vectors $\mathbf{X}$ and $\mathbf{X}^{\theta} H \mathbf{X} = 0$ for some non-zero complex n-vector $\mathbf{X}$ *i.e.*, $\mathbf{X}^{\theta} H \mathbf{X} = 0$ for some complex values of the variables $x_1, x_2, \ldots, x_n$ not all zero.

(iv) **Negative semi-definite** if $\mathbf{X}^{\theta} H \mathbf{X} \leq 0$ for all complex n-vectors $\mathbf{X}$ and $\mathbf{X}^{\theta} H \mathbf{X} = 0$ for some non-zero complex n-vector $\mathbf{X}$ *i.e.*, $\mathbf{X}^{\theta} H \mathbf{X} = 0$ for some complex values of the variables $x_1, x_2, \ldots, x_n$ not all zero.

(v) **Indefinite** if $\mathbf{X}^{\theta} H \mathbf{X}$ takes positive as well as negative values for some complex values of the variables $x_1, x_2, \ldots, x_n$.

(vi) **Non-negative definite** if $\mathbf{X}^{\theta} H \mathbf{X} \geq 0$ for all complex n-vectors $\mathbf{X}$.

Non-negative definite and positive definite Hermitian matrices. Definition. *A Hermitian matrix H is called non-negative definite if the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is non-negative definite . A Hermitian matrix H is called positive definite if the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is positive definite.*

Similar definitions are for other cases.

22. Some Results for Hermitian Forms Parallel to the Corresponding Results for Real Quadratic Forms.

Index and Signature of a Hermitian form.

Definition : Let $\mathbf{X}^{\theta} H \mathbf{X}$ be a Hermitian form of order n. Let

$$\lambda_1 \bar{z}_1 z_1 + \lambda_2 \bar{z}_2 z_2 + \ldots + \lambda_p \bar{z}_p z_p - \lambda_{p+1} \bar{z}_{p+1} z_{p+1} - \ldots - \lambda_r \bar{z}_r z_r$$

be its diagonal form where $\lambda_1, \lambda_2, \ldots, \lambda_p, \lambda_{p+1}, \ldots, \lambda_r$ are all positive.

Here, r is the rank of the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$.

The number p of positive terms in this diagonal form is called the *index* of the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$. The excess of the number of positive terms over the number of negative terms in this diagonal form *i.e.*, $p - (r - p) = 2p - r$ is called the *signature* of the Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ and is denoted by s.

Under all non-singular linear transformations the rank r and the index p are invariant of a Hermitian form and consequently the signature s of a Hermitian form is also invariant under all non-singular linear transformations.

Equivalent Hermitian forms. Definition. Two Hermitian forms $\mathbf{X}^\theta H \mathbf{X}$ and $\mathbf{Y}^\theta Q \mathbf{Y}$ are said to be *equivalent* over the complex field **C** if one can be obtained from the other by a non-singular transformation over **C.** Thus $\mathbf{X}^\theta H \mathbf{X}$ and $\mathbf{Y}^\theta Q \mathbf{Y}$ are equivalent if and only if there exists a non-singular transformation $\mathbf{X} = P\,\mathbf{Y}$ such that

$$\mathbf{X}^\theta H \mathbf{X} = (P\mathbf{Y})^\theta H (P\mathbf{Y}) = \mathbf{Y}^\theta (P^\theta HP)\,\mathbf{Y}$$
$$= \mathbf{Y}^\theta Q\,\mathbf{Y}, \text{ where } Q = P^\theta HP.$$

Theorem 2 : Equivalence theorem of Hermitian forms. *Two Hermitian forms are equivalent if and only if they have the same rank and the same index.*

Theorem 3 : *Let $\mathbf{X}^\theta H \mathbf{X}$ be a Hermitian form of order n, rank r, and signature s. Then $\mathbf{X}^\theta H \mathbf{X}$ is*

(i) positive definite if and only if $s = r = n$.
(ii) negative definite if and only if $-s = r = n$.
(iii) positive semi-definite if and only if $s = r < n$.
(iv) negative semi-definite if and only if $-s = r < n$.
(v) indefinite if and only if $|s| \neq r$ i.e., if and only if $|s| < r$.

Theorem 4 : *If the Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ is positive definite, then there exists a non-singular transformation $\mathbf{X} = P\,\mathbf{Y}$ such that $\mathbf{X}^\theta H \mathbf{X} = \mathbf{Y}^\theta \mathbf{Y}$.*

Theorem 5 : Sylvester's criterion for positive definiteness of Hermitian forms. *A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$, where $H = [h_{ij}]_{n\times n}$, is positive definite if and only if all the leading principal minors of H are positive i.e., iff*

$$h_{11}, \begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix}, \dots, \begin{vmatrix} h_{11} & \dots & h_{1n} \\ \dots & \dots & \dots \\ \dots & \dots & \dots \\ h_{n1} & \dots & h_{nn} \end{vmatrix} \text{ are all positive.}$$

Theorem 6 : *A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ is positive definite if and only if all the principal minors of H are positive.*

Theorem 7 : *The Hermitian form*

$$h(x_1, \dots, x_n) = \mathbf{X}^\theta H \mathbf{X} = \sum_{i=1}^{n} \sum_{j=1}^{n} h_{ij}\,\bar{x}_i\,x_j$$

is negative definite if and only if the leading principal minors of H are alternately negative and positive i.e., iff

$$h_{11} < 0, \begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix} > 0, \begin{vmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{vmatrix} < 0, \dots.$$

minors of H of even order are positive and those of odd order are negative.

Theorem 9 : *A Hermitian form* $\mathbf{X}^{\theta} H \mathbf{X}$ *is*

(i) Positive definite if and only if all the eigenvalues of H are positive.

(ii) Negative definite if and only if all the eigenvalues of H are negative.

(iii) Positive semi-definite iff all the eigenvalues of H are ≥ 0 *with at least one zero eigenvalue.*

(iv) Negative semi-definite if and only if all the eigenvalues of H are ≤ 0 *with at least one zero eigenvalue.*

(v) Indefinite iff H has at least one positive eigenvalue and one negative eigenvalue.

(vi) Non-negative definite iff the eigenvalues of H are all non-negative.

Theorem 10 : *A Hermitian form* $\mathbf{X}^{\theta} H \mathbf{X}$ *is positive semi-definite iff H is singular and all its principal minors are non-negative.*

Theorem 11 : *A Hermitian form* $\mathbf{X}^{\theta} H \mathbf{X}$ *is negative semi-definite iff H is singular and the principal minors of even order of H are non-negative while those of odd order are non-positive.*

Theorem 12 : *A Hermitian matrix H is indefinite iff at least one of the following conditions is satisfied :*

(i) H has a negative principal minor of even order.

(ii) H has a positive principal minor of odd order and a negative principal minor of odd order.

Illustrative Examples

Example 23 : *Determine the definiteness of the following Hermitian forms* $\mathbf{X}^{\theta} H \mathbf{X}$ *in* $\mathbf{C}^3$ *where*

(i) $H = \begin{bmatrix} 0 & -i & -i \\ i & 0 & -i \\ i & i & 0 \end{bmatrix}$ *(ii)* $H = \begin{bmatrix} -2 & 1+2i & 0 \\ 1-2i & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Solution. (i) The characteristic equation of the matrix H is

$$|H - \lambda I| = 0$$

i.e., $\begin{vmatrix} 0-\lambda & -i & -i \\ i & 0-\lambda & -i \\ i & i & 0-\lambda \end{vmatrix} = 0$ or $\begin{vmatrix} -\lambda & -i & -i \\ i & -\lambda & -i \\ i & i & -\lambda \end{vmatrix} = 0$

or $-\lambda(\lambda^2 + i^2) + i(-\lambda i + i^2) - i(i^2 + \lambda i) = 0$

or $-\lambda(\lambda^2 - 1) - \lambda i^2 + i^3 - i^3 - \lambda i^2 = 0$

or $-\lambda(\lambda^2 - 1) + \lambda + \lambda = 0$

or $\lambda[-(\lambda^2 - 1) + 2] = 0$ or $\lambda(\lambda^2 - 3) = 0.$

$\therefore$ the eigenvalues of H are $0, \sqrt{3}, -\sqrt{3}$.

Since the given Hermitian matrix H has a positive eigenvalue as well as a negative eigenvalue, therefore the given Hermitian form $\mathbf{X}^{\theta} H \mathbf{X}$ is indefinite.

$$\sqrt{3}\,\bar{y}_1\, y_1 - \sqrt{3}\,\bar{y}_2\, y_2\,.$$

Its canonical form is $\bar{z}_1\, z_1 - \bar{z}_2\, z_2$.

The order of the given Hermitian form is 3. The rank r of the given Hermitian form = the number of non-zero terms in its diagonal form = 2.

The index p of the given Hermitian form = the number of positive terms in its diagonal form = 1.

The signature s of the given Hermitian form = the number of positive terms in its diagonal form – the number of negative terms in its diagonal form = 1 – 1 = 0.

(ii) The characteristic equation of the matrix H of the given Hermitian form $\mathbf{X}^{\theta}\, H\, \mathbf{X}$ is $|\, H - \lambda I\,| = 0$

or $$\begin{vmatrix} -2-\lambda & 1+2\,i & 0 \\ 1-2\,i & -4-\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = 0$$

or $$-\lambda\,[(-2-\lambda)(-4-\lambda) - (1-2\,i)(1+2\,i)] = 0$$

or $$-\lambda\,[(\lambda+2)(\lambda+4) - (1+4)] = 0$$

or $$-\lambda\,(\lambda^2 + 6\,\lambda + 8 - 5) = 0 \quad \text{or} \quad \lambda\,(\lambda^2 + 6\,\lambda + 3) = 0.$$

$\therefore$ $$\lambda = 0, \frac{-6 \pm \sqrt{(36-12)}}{2} \quad i.e., \quad \frac{-6 \pm \sqrt{24}}{2} \quad i.e., \ -3 \pm \sqrt{6}.$$

The eigenvalues $-3 + \sqrt{6}$ and $-3 - \sqrt{6}$ are both negative.

Thus the eigenvalues of the matrix H are all ≤ 0 and at least one of them is zero. Hence the given Hermitian form $\mathbf{X}^{\theta}\, H\, \mathbf{X}$ is negative semi-definite.

Remark. Order of the given Hermitian form = order of the matrix $H = 3$.

Rank r of the given Hermitian form = 2.

Index p of the given Hermitian form = 0.

Signature s of the given Hermitian form = 0 – 2 = – 2.

A diagonal form of the given Hermitian form

$$= -(3 - \sqrt{6})\,\bar{y}_1\, y_1 - (3 + \sqrt{6})\,\bar{y}_2\, y_2\,.$$

Its canonical form or normal form is

$$-\bar{z}_1\, z_1 - \bar{z}_2\, z_2\,.$$

Example 24 : *Determine whether or not the following Hermitian forms in* $\mathbf{C}^2$ *are equivalent.*

(i) $2\,\bar{x}_1\, x_1 + 3i\,\bar{x}_2\, x_1 - 3i\,\bar{x}_1\, x_2 - \bar{x}_2\, x_2\,, (2+i)\,\bar{x}_2\, x_1 + (2-i)\,\bar{x}_1\, x_2$

(ii) $\bar{x}_1\, x_1 + 2\,\bar{x}_2\, x_2 + (1+2\,i)\,\bar{x}_1\, x_2 + (1-2\,i)\,\bar{x}_2\, x_1\,, i\,\bar{x}_1\, x_2 - i\,\bar{x}_2\, x_1$.

Solution. (i) Out of the two given Hermitian forms the first Hermitian form is

$$2\,\bar{x}_1\, x_1 - 3i\,\bar{x}_1\, x_2 + 3i\,\bar{x}_2\, x_1 - \bar{x}_2\, x_2$$

$$= \mathbf{X}^{\theta}\, H_1\, \mathbf{X}, \text{ where } H_1 = \begin{bmatrix} 2 & -3i \\ 3i & -1 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_1 is $|\, H_1 - \lambda I\,| = 0$

i.e., $$\begin{vmatrix} 2-\lambda & -3i \\ 3i & -1-\lambda \end{vmatrix} = 0$$

or $\qquad -2-\lambda+\lambda^2-9=0$

or $\qquad \lambda^2-\lambda-11=0.$

$\therefore$ the eigenvalues of H_1 are

$$\frac{1\pm\sqrt{1+44}}{2} \quad i.e., \quad \frac{1\pm\sqrt{45}}{2} \quad i.e., \quad \frac{1\pm 3\sqrt{5}}{2}.$$

The eigenvalue $\dfrac{1+3\sqrt{5}}{2}$ is positive and the eigenvalue $\dfrac{1-3\sqrt{5}}{2}$ is negative.

The rank r of the Hermitian form $\mathbf{X}^\theta H_1 \mathbf{X}$ = the number of non-zero eigenvalues of $H_1 = 2$, the index p of the Hermitian form $\mathbf{X}^\theta H_1 \mathbf{X}$ = the number of positive eigenvalues of $H_1 = 1$.

The second Hermitian form is

$$0\,\bar{x}_1 x_1 + (2-i)\,\bar{x}_1 x_2 + (2+i)\,\bar{x}_2 x_1 + 0\,\bar{x}_2 x_2$$

$$= \mathbf{X}^\theta H_2 \mathbf{X}, \text{ where } H_2 = \begin{bmatrix} 0 & 2-i \\ 2+i & 0 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_2 is

$$|H_2-\lambda I|=0 \qquad i.e., \qquad \begin{vmatrix} -\lambda & 2-i \\ 2+i & -\lambda \end{vmatrix}=0$$

or $\qquad \lambda^2-(2+i)(2-i)=0 \qquad$ or $\qquad \lambda^2-5=0$

$\therefore \qquad \lambda^2=5 \quad$ or $\quad \lambda=\pm\sqrt{5}.$

$\therefore$ the eigenvalues of H_2 are $+\sqrt{5}, -\sqrt{5}$.

The rank r of the Hermitian form $\mathbf{X}^\theta H_2 \mathbf{X}$ = the number of non-zero eigenvalues of $H_2 = 2$, the index p of the Hermitian form $\mathbf{X}^\theta H_2 \mathbf{X}$ = the number of positive eigenvalues of $H_2 = 1$.

Thus the Hermitian forms $\mathbf{X}^\theta H_1 \mathbf{X}$ and $\mathbf{X}^\theta H_2 \mathbf{X}$ have the same rank and the same index. Hence, they are equivalent.

(ii) Out of the two given Hermitian forms the first Hermitian form is

$$\bar{x}_1 x_1 + (1+2i)\,\bar{x}_1 x_2 + (1-2i)\,\bar{x}_2 x_1 + 2\,\bar{x}_2 x_2$$

$$= \mathbf{X}^\theta H_1 \mathbf{X}, \text{ where } H_1 = \begin{bmatrix} 1 & 1+2i \\ 1-2i & 2 \end{bmatrix} \text{ is a Hermitian matrix.}$$

The characteristic equation of H_1 is $|H_1-\lambda I|=0$

i.e., $$\begin{vmatrix} 1-\lambda & 1+2i \\ 1-2i & 2-\lambda \end{vmatrix}=0$$

or $\qquad (1-\lambda)(2-\lambda)-(1-2i)(1+2i)=0$

or $\qquad \lambda^2-3\lambda+2-5=0 \quad$ or $\quad \lambda^2-3\lambda-3=0.$

$$\therefore \qquad \lambda=\frac{3\pm\sqrt{9+12}}{2}=\frac{3\pm\sqrt{21}}{2}.$$

$\therefore$ the eigenvalues of H_1 are $\dfrac{3+\sqrt{21}}{2}, \dfrac{3-\sqrt{21}}{2}$.

The eigenvalue $\dfrac{\quad}{2}$ is positive and the eigenvalue $\dfrac{\quad}{2}$ is negative.

The rank r of the Hermitian form $\mathbf{X}^{\theta} H_1 \mathbf{X}$ = the number of non-zero eigenvalues of $H_1 = 2$,

the index p of this Hermitian form = the number of positive eigenvalues of $H_1 = 1$.

The second Hermitian form is

$$0\, \bar{x}_1 x_1 + i\, \bar{x}_1 x_2 - i\, \bar{x}_2 x_1 + 0\, \bar{x}_2 x_2$$

$$= \mathbf{X}^{\theta} H_2 \mathbf{X}, \text{ where } H_2 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

is a Hermitian matrix.

The characteristic equation of H_2 is $| H_2 - \lambda I | = 0$

i.e., $$\begin{vmatrix} -\lambda & i \\ -i & -\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^2 + i^2 = 0 \quad \text{or} \quad \lambda^2 - 1 = 0.$$

$\therefore$ $$\lambda^2 = 1 \quad \text{or} \quad \lambda = \pm 1.$$

$\therefore$ the eigenvalues of H_2 are $1, -1$.

The rank r of the Hermitian form $\mathbf{X}^{\theta} H_2 \mathbf{X}$ = the number of non-zero eigenvalues of $H_2 = 2$ and its index p = the number of positive eigenvalues of $H_2 = 1$.

Thus the Hermitian forms $\mathbf{X}^{\theta} H_1 \mathbf{X}$ and $\mathbf{X}^{\theta} H_2 \mathbf{X}$ have the same rank and the same index. Hence, they are equivalent.

Example 25 : *Reduce the Hermitian form* $\mathbf{X}^{\theta} H \mathbf{X}$, *where*

$$H = \begin{bmatrix} 2 & 1 - 2i \\ 1 + 2i & -2 \end{bmatrix},$$

to the canonical form by a unitary transformation and hence find the rank and signature of the given Hermitian form.

Solution : The characteristic equation of H is

$$| H - \lambda I | = 0 \qquad i.e., \qquad \begin{vmatrix} 2 - \lambda & 1 - 2i \\ 1 + 2i & -2 - \lambda \end{vmatrix} = 0$$

or $$(\lambda - 2)(\lambda + 2) - (1 + 2i)(1 - 2i) = 0$$

or $$\lambda^2 - 4 - (1 + 4) = 0 \quad \text{or} \quad \lambda^2 - 9 = 0.$$

$\therefore$ the eigenvalues of H are $-3, 3$.

The eigenvector $\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ corresponding to the eigenvalue -3 is given by the equation

$$(H + 3I)\mathbf{X} = \mathbf{O} \qquad \text{or} \qquad \begin{bmatrix} 5 & 1 - 2i \\ 1 + 2i & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

i.e., $$5x_1 + (1 - 2i)\, x_2 = 0, \text{ and } (1 + 2i)\, x_1 + x_2 = 0.$$

Obviously $x_1 = 1 - 2i, x_2 = -5$ is a solution.

$\therefore$ $\mathbf{X}_1 = \begin{bmatrix} 1 - 2i \\ -5 \end{bmatrix}$ is an eigenvector corresponding to the eigenvalue $\lambda = -3$.

Corresponding to the eigenvalue 3, the eigenvector is given by the equation or $$\begin{bmatrix} -1 & 1-2i \\ 1+2i & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

i.e., $-x_1 + (1-2i)\,x_2 = 0$, and $(1+2i)\,x_1 - 5x_2 = 0$.

Obviously $x_1 = 5, x_2 = 1+2i$ is a solution.

$\therefore$ $\mathbf{X}_2 = \begin{bmatrix} 5 \\ 1+2i \end{bmatrix}$ is an eigenvector corresponding to $\lambda = 3$.

Length of the vector $\mathbf{X}_1 = \sqrt{[\,|1-2i|^2 + |-5|^2\,]} = \sqrt{(5+25)} = \sqrt{30}$

and the length of the vector $\mathbf{X}_2 = \sqrt{[\,|5|^2 + |1+2i|^2\,]}$

$$= \sqrt{25+5} = \sqrt{30}.$$

$\therefore$ the unitary matrix P that will transform H to diagonal form is

$$P = \left[\frac{1}{\sqrt{30}}\mathbf{X}_1, \frac{1}{\sqrt{30}}\mathbf{X}_2\right] = \begin{bmatrix} (1-2i)/\sqrt{30} & 5/\sqrt{30} \\ -5/\sqrt{30} & (1+2i)/\sqrt{30} \end{bmatrix}.$$

Also $P^{-1} = P^{\theta}$

and $P^{-1} HP = P^{\theta} HP = \begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix} = \text{diag.}\,[-3, 3].$

The unitary transformation $\mathbf{X} = P\,\mathbf{Y}$ will transform the given Hermitian form to the equivalent diagonal form

$$\mathbf{Y}^{\theta}\,\text{diag.}\begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix}\mathbf{Y} = -3\,\bar{y}_1\,y_1 + 3\,\bar{y}_2\,y_2\,.$$

The canonical form of the given Hermitian form is

$$-\bar{z}_1\,z_1 + \bar{z}_2\,z_2\,.$$

The rank of the given Hermitian form = the number of non-zero eigenvalues of its matrix $H = 2$.

The signature of the given Hermitian form = the number of positive eigenvalues of H – the number of negative eigenvalues of $H = 1 - 1 = 0$.

Comprehensive Exercise 4

1. Determine the definiteness of the following Hermitian forms $\mathbf{X}^{\theta}\,H\,\mathbf{X}$ in $\mathbf{C}^3$ where

(i) $H = \begin{bmatrix} 2 & 1-i & 2 \\ 1+i & 2 & 1+i \\ 2 & 1-i & 2 \end{bmatrix}$ (ii) $H = \begin{bmatrix} -2 & 1-i & i \\ 1+i & -2 & 2+i \\ -i & 2-i & -6 \end{bmatrix}$

(iii) $H = \begin{bmatrix} 2 & 1-i & i \\ 1+i & 4 & 2+i \\ -i & 2-i & 8 \end{bmatrix}.$

$$H = \begin{bmatrix} 2 & i & 0 \\ -i & 1 & -i \\ 0 & i & 2 \end{bmatrix},$$

to the equivalent diagonal form by a unitary transformation and hence find the rank and signature of the given Hermitian form.

3. Show that a Hermitian matrix H is positive definite if and only if its eigenvalues are all positive and is non-negative definite if and only if its eigenvalues are all non-negative.
4. Show that a Hermitian matrix H is positive definite if and only if there exists a non-singular matrix Q such that $H = Q^{\theta} Q$.
5. If H is a positive definite or a positive semi-definite Hermitian matrix, show that there exists a Hermitian matrix B such that $B^2 = H$.

Answers 4

1. (i) Positive semi-definite. (ii) Negative definite.
   (iii) Positive definite.
2. The unitary matrix P that will transform H to diagonal form is

$$P = \begin{bmatrix} -i/\sqrt{6} & 1/\sqrt{2} & i/\sqrt{3} \\ 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -i/\sqrt{6} & -1/\sqrt{2} & i/\sqrt{3} \end{bmatrix}.$$

The unitary transformation $\mathbf{X} = P\,\mathbf{Y}$ will transform the given Hermitian form to the equivalent diagonal form $2\,\bar{y}_2\,y_2 + 3\,\bar{y}_3\,y_3$.
The rank of the given Hermitian form is 2 and its signature is 2.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) *and* (d).

1. A real quadratic form $\mathbf{X}^T A\,\mathbf{X}$ in three variables is equivalent to the diagonal form $3y_1^2 - 4y_2^2 + 5y_3^2$. Then the quadratic form $\mathbf{X}^T A\,\mathbf{X}$ is
   (a) Positive definite (b) Negative definite
   (c) Indefinite (d) Positive semi-definite.
2. A real quadratic form $\mathbf{X}^{\mathbf{T}} A\,\mathbf{X}$ in three variables is equivalent to the normal form $y_1^2 - y_2^2 - y_3^2$. If s is the signature of this quadratic form, then

(c) $s = 2$ (d) $s = 3$.

3. A real quadratic form $\mathbf{X}^{\mathbf{T}} A \, \mathbf{X}$ is positive semi-definite if and only if

(a) the eigenvalues of the matrix A are all ≤ 0

(b) the eigenvalues of the matrix A are all > 0

(c) the eigenvalues of the matrix A are all negative

(d) the eigenvalues of the matrix A are all ≥ 0 and at least one eigenvalue of A is zero.

4. If n is the order, r is the rank and s is the signature of a real quadratic form in n variables, then the quadratic form is negative semi-definite if

(a) $s = r = n$ (b) $-s = r = n$

(c) $s = r < n$ (d) $-s = r < n$.

5. A real quadratic form in three variables is equivalent to the diagonal form

$$6 y_1^2 + 3 y_2^2 + 0 y_3^2.$$

Then the quadratic form is

(a) Positive definite (b) Indefinite

(c) Positive semi-definite (d) Negative.

Fill in the Blank

Fill in the blanks "......" so that the following statements are complete and correct.

1. If the bilinear form

$$b(X, Y) = X^T AY = 2 x_1 y_1 + x_1 y_2 - 2 x_2 y_1 + 3 x_2 y_2 - 3 x_1 y_3,$$

then the matrix $A = \ldots\ldots$

2. The bilinear form $b(X, Y) = X^T AY$ is said to be a symmetric bilinear form if the matrix A is a ...... matrix.

3. An expression of the form $\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \, x_i \, x_j$, where a_{ij}'s are elements of a field F, is called a ...... in the n variables $x_1, x_2, \ldots, x_n$ over the field F.

4. There exists a one-to-one correspondence between the set of all quadratic forms in n variables over a field F and the set of all n-rowed ...... matrices over F.

5. The matrix A corresponding to the quadratic form

$$d_1 \, x_1^2 + d_2 \, x_2^2 + d_3 \, x_3^2 + d_4 \, x_4^2 + d_5 \, x_5^2$$

in five variables $x_1, x_2, \ldots, x_5$ is $A = \ldots\ldots$

6. The quadratic form corresponding to the symmetric matrix diag. $[\lambda_1, \lambda_2, \ldots, \lambda_n]$ is ....

7. The quadratic form corresponding to the matrix

$$\begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & -2 \\ 5 & -2 & 4 \end{bmatrix} \text{ is } \ldots\ldots .$$

8. Two real quadratic forms in n variables are equivalent if and only if they have the same ...... and the same index.

9. The number of positive terms in any two normal reductions of a real quadratic form is the ....

10. The rank of a real quadratic form $\mathbf{X}^T A \mathbf{X}$ is equal to the number of ...... eigenvalues of the symmetric matrix A.

11. The real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive definite if and only if the eigenvalues of the matrix A are all ....

12. The real quadratic form $\mathbf{X}^T A \mathbf{X}$ is positive definite if $\mathbf{X}^T A \mathbf{X}$......0, for all non-zero vectors $\mathbf{X}$.

13. The eigenvalues of a Hermitian matrix are always ....

14. A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ always takes ...... values for all complex n-vectors $\mathbf{X}$.

15. A Hermitian form $\mathbf{X}^\theta H \mathbf{X}$ is negative definite iff the eigenvalues of the matrix H are all ....

True or False

Write 'T' for true and 'F' for false statement.

1. The bilinear form

$$3x_1 y_1 + x_1 y_2 + x_2 y_1 - 2\, x_2 y_2 - 4\, x_2 y_3 - 4x_3 y_2 + 3x_3 y_3$$

is symmetric.

2. The matrix corresponding to the quadratic form

$$d_1 x_1^2 + d_2 x_2^2 + d_3 x_3^2 + d_4 x_4^2$$

is a diagonal matrix.

3. Every real quadratic form over a field F in n variables $x_1, x_2, \dots, x_n$ can be expressed in the form $\mathbf{X}'B\,\mathbf{X}$ where $\mathbf{X}' = [x_1, x_2, \dots, x_n]'$ is a column vector and B is a skew-symmetric matrix of order n over the field F.

4. A real quadratic form in three variables is equivalent to the diagonal form

$$6\, y_1^2 + \frac{7}{3}\, y_2^2 - \frac{16}{7}\, y_3^2.$$

Then the rank of the quadratic form is 2.

5. A real quadratic form $\mathbf{X}^T A\mathbf{X}$ is positive definite iff the leading principal minors of the matrix A are all positive.

6. A real quadratic form $\mathbf{X}^T A \mathbf{X}$ is negative definite iff the leading principal minors of the matrix A are all negative.

7. A real quadratic form in three variables is equivalent to the diagonal form

$$4\, y_1^2 + 3\, y_2^2.$$

Then the quadratic form is positive definite.

8. A real quadratic form in four variables is equivalent to the normal form

$$y_1^2 + y_2^2 - y_3^2 - y_4^2.$$

Then the signature of the quadratic form is 0.

$$4\,\bar{y}_1\,y_1 + 5\,\bar{y}_2\,y_2 + 7\,\bar{y}_3\,y_3\,.$$

Then the Hermitian form is positive definite.

10. A Hermitian form $\mathbf{X}^\theta\ H\ \mathbf{X}$ in three variables is equivalent to the diagonal form

$$\frac{3}{2}\,\bar{y}_1\,y_1 + 5\,\bar{y}_2\,y_2\,.$$

Then the Hermitian form is positive definite.

Answers

Multiple Choice Questions

1. (c) **2.** (b) **3.** (d)
4. (d) **5.** (c)

Fill in the Blank

1. $\begin{bmatrix} 2 & 1 & -3 \\ -2 & 3 & 0 \end{bmatrix}$

2. symmetric **3.** quadratic form
4. symmetric **5.** diag. $[d_1, d_2, d_3, d_4, d_5]$
6. $\lambda_1\,x_1^2 + \lambda_2\,x_2^2 + \ldots + \lambda_n\,x_n^2$
7. $2\,x_1^2 + 3x_2^2 + 4\,x_3^2 + 2\,x_1\,x_2 + 10\,x_1\,x_3 - 4\,x_2\,x_3$
8. rank **9.** same **10.** non-zero
11. positive **12.** > **13.** real
14. real **15.** negative

True or False

1. T **2.** T **3.** F
4. F **5.** T **6.** F
7. F **8.** T **9.** T
10. F